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Discrete Mathematical Structures
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1 Sets and Subsets

Sets

A set is any well-de ned collection of objects called the elements or members of the set. For example, the collection of all wooden chairs, the collection of all one-legged black birds, or the collection of real numbers between zero and one are all sets. Well-de ned just means that it is possible to decide if a given object belongs to the collection or not. Almost all mathematical objects are rst of all sets, regardless of any additional properties they may possess. Thus set theory is, in a sense, the foundation on which virtually all of mathematics is constructed. In spite of this, set theory (at least the informal brand we need) is quite easy to learn and use.

One way of describing a set that has a nite number of elements is by listing the elements of the set between braces. Thus the set of all positive integers that are less than 4 can be written as

$$\{1, 2, 3\}. \tag{1}$$

The order in which the elements of a set are listed is not important. Thus $\{1, 3, 2\}$, $\{3, 2, 1\}$, $\{3, 1, 2\}$, $\{2, 1, 3\}$, and $\{2, 3, 1\}$ are all representations of the set given in (1). Moreover, repeated elements in the listing of the elements of a set can be ignored. Thus $\{1, 3, 2, 3, 1\}$ is another representation of the set given in (1).

We use uppercase letters such as A, B, C to denote sets, and lowercase letters such as a, b, c, x, y, z, t to denote the members (or elements) of sets.

We indicate the fact that x is an element of the set A by writing $x \in A$, and we indicate the fact that x is not an element of A by writing $x \notin A$.

Example 1 Let $A = \{1, 3, 5, 7\}$. Then $1 \in A$, $3 \in A$, but $2 \notin A$.

Sometimes it is inconvenient or impossible to describe a set by listing all of its elements. Another useful way to de ne a set is by specifying a property that the elements of the set have in common. We use the notation $P(x)$ to denote a sentence or statement P concerning the variable object x. The set de ned by $P(x)$, written $\{x \mid P(x)\}$, is just the collection of all objects for which P is sensible and true. $\{x \mid P(x)\}$ is read, "the set of all x such that $P(x)$." For example, $\{x \mid x \text{ is a positive integer less than } 4\}$ is the set $\{1, 2, 3\}$ described in (1) by listing its elements.

Example 2 The set consisting of all the letters in the word "byte" can be denoted by $\{b, y, t, e\}$ or by $\{x \mid x \text{ is a letter in the word "byte"}\}$.

Example 3 We introduce here several sets and their notations.

(a) $Z^+ = \{x \mid x \text{ is a positive integer}\}$.
Thus Z^+ consists of the numbers used for counting: $1, 2, 3, \ldots$.

(b) $N = \{x \mid x \text{ is a positive integer or zero}\} = \{x \mid x \text{ is a natural number}\}$.
Thus N consists of the positive integers and zero: $0, 1, 2, \ldots$.

(c) $Z = \{x \mid x \text{ is an integer}\}$.
Thus Z consists of all the integers: $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$.

(d) $Q = \{x \mid x \text{ is a rational number}\}$.
Thus Q consists of numbers that can be written as $\frac{a}{b}$, where a and b are integers and b is not 0.

(e) $R = \{ x \mid x \text{ is a real number} \}$.

(f) The set that has no elements in it is denoted either by $\{\ \}$ or by the symbol $\emptyset$ and is called the *empty set*.

Example 4 Since the square of a real number is always nonnegative,

$$\{x \mid x \text{ is a real number and } x^2 = -1\} = \emptyset.$$

Sets are completely known when their members are all known. Thus we say two sets A and B are *equal* if they have the same elements, and we write $A = B$.

Example 5 If $A = \{1, 2, 3\}$ and $B = \{ x \mid x \text{ is a positive integer and } x^2 < 12 \}$, then $A = B$.

Example 6 If $A = \{$ JAVA, PASCAL, C++ $\}$ and $B = \{$ C++, JAVA, PASCAL $\}$, then $A = B$.

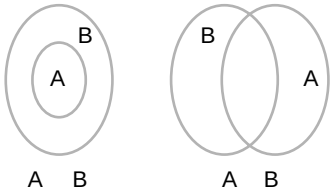


Figure 1

Subsets

If every element of A is also an element of B, that is, if whenever $x \in A$ then $x \in B$, we say that A is a *subset* of B or that A is *contained* in B, and we write $A \subseteq B$. If A is not a subset of B, we write $A \not\subseteq B$. (See Figure 1.)

Diagrams, such as those in Figure 1, which are used to show relationships between sets, are called *Venn diagrams* after the British logician John Venn. Venn diagrams will be used extensively in Section 2.

Example 7 We have $\mathbb{Z}^+ \subseteq \mathbb{Z}$. Moreover, if $\mathbb{Q}$ denotes the set of rational numbers, then $\mathbb{Z} \subseteq \mathbb{Q}$.

Example 8 Let $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 5\}$, and $C = \{1, 2, 3, 4, 5\}$. Then $B \subseteq A$, $B \subseteq C$, and $C \subseteq A$. However, $A \not\subseteq B$, $A \not\subseteq C$, and $C \not\subseteq B$.

Example 9 If A is any set, then $A \subseteq A$. That is, every set is a subset of itself.

Example 10 Let A be a set and let $B = \{A, \{A\}\}$. Then, since $A \in B$ and $\{A\} \in B$, we have $A \subseteq B$ and $\{A\} \subseteq B$. It follows that $\{A\} \subseteq B$ and $\{\{A\}\} \subseteq B$. However, it is not true that $A \subseteq B$.

For any set A, since there are no elements of $\emptyset$ that are not in A, we have $\emptyset \subseteq A$.

It is easy to see that $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. This simple statement is the basis for proofs of many statements about sets.

The collection of everything, it turns out, cannot be considered a set without presenting serious logical difficulties. To avoid this and other problems, which need not concern us here, we will assume that for each discussion there is a "universal set" U (which will vary with the discussion) containing all objects for which the discussion is meaningful. Any other set mentioned in the discussion will automatically be assumed to be a subset of U. Thus, if we are discussing real numbers and we mention sets A and B, then A and B must (we assume) be sets of real numbers, not matrices, electronic circuits, or rhesus monkeys. In most problems, a universal set will be apparent from the setting of the problem. In Venn diagrams, the universal set U will be denoted by a rectangle, while sets within U will be denoted by circles as shown in Figure 2.

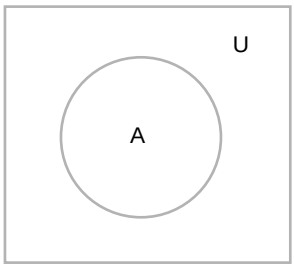


Figure 2

A set A is called **finite** if it has n distinct elements, where $n \in \mathbb{N}$. In this case, n is called the **cardinality** of A and is denoted by $|A|$. Thus, the sets of Examples 1, 2, 4, 5, and 6 are finite. A set that is not finite is called **infinite**. The sets introduced in Example 3 (except $\mathbb{N}$) are infinite sets.

If A is a set, then the set of all subsets of A is called the **power set** of A and is denoted by $\mathcal{P}(A)$. (Be sure to distinguish between $A \in \mathcal{P}(A)$, a statement about A, and $\mathcal{P}(A)$, the power set of A.)

Example 11 Let $A = \{1, 2, 3\}$. Then $\mathcal{P}(A)$ consists of the following subsets of A: $\{\ \}$, $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$, and $\{1, 2, 3\}$ (or A). In a later section, we will count the number of subsets that a set can have.

1 Exercises

1. Let $A = \{1, 2, 4, a, b, c\}$. Identify each of the following as true or false.
   (a) $2 \in A$ (b) $3 \in A$ (c) $c \notin A$
   (d) $\in A$ (e) $\{\ \} \notin A$ (f) $A \in A$
2. Let $A = \{x \mid x \text{ is a real number and } x < 6\}$. Identify each of the following as true or false.
   (a) $3 \in A$ (b) $6 \in A$ (c) $5 \notin A$
   (d) $8 \notin A$ (e) $-8 \in A$ (f) $3.4 \notin A$
3. In each part, give the set of letters in each word by listing the elements of the set.
   (a) AARDVARK (b) BOOK
   (c) MISSISSIPPI
4. Give the set by listing its elements.
   (a) The set of all positive integers that are less than ten.
   (b) $\{x \mid x \in \mathbb{Z} \text{ and } x^2 < 12\}$
5. Let $A = \{1, \{2, 3\}, 4\}$. Identify each of the following as true or false.
   (a) $3 \in A$ (b) $\{1, 4\} \in A$ (c) $\{2, 3\} \in A$
   (d) $\{2, 3\} \subseteq A$ (e) $\{4\} \in A$ (f) $\{1, 2, 3\} \subseteq A$

In Exercises 6 through 9, write the set in the form $\{x \mid P(x)\}$, where $P(x)$ is a property that describes the elements of the set.

6. $\{2, 4, 6, 8, 10\}$
7. $\{a, e, i, o, u\}$
8. $\{1, 8, 27, 64, 125\}$
9. $\{-2, -1, 0, 1, 2\}$
10. Let $A = \{1, 2, 3, 4, 5\}$. Which of the following sets are equal to A?
    (a) $\{4, 1, 2, 3, 5\}$ (b) $\{2, 3, 4\}$ (c) $\{1, 2, 3, 4, 5, 6\}$
    (d) $\{x \mid x \text{ is an integer and } x^2 \quad 25\}$
    (e) $\{x \mid x \text{ is a positive integer and } x \quad 5\}$
    (f) $\{x \mid x \text{ is a positive rational number and } x \quad 5\}$
11. Which of the following sets are the empty set?
    (a) $\{x \mid x \text{ is a real number and } x^2 - 1 = 0\}$
    (b) $\{x \mid x \text{ is a real number and } x^2 + 1 = 0\}$
    (c) $\{x \mid x \text{ is a real number and } x^2 = -9\}$
    (d) $\{x \mid x \text{ is a real number and } x = 2x + 1\}$
    (e) $\{x \mid x \text{ is a real number and } x = x + 1\}$
12. List all the subsets of $\{a, b\}$.
13. List all the subsets of $\{\text{JAVA, PASCAL, C++}\}$.
14. List all the subsets of $\{\ \}$.
15. Let $A = \{1, 2, 5, 8, 11\}$. Identify each of the following as true or false.
    (a) $\{5, 1\} \subseteq A$ (b) $\{8, 1\} \subseteq A$
    (c) $\{1, 8, 2, 11, 5\} \subseteq A$ (d) $\in A$
    (e) $\{1, 6\} \subseteq A$ (f) $\{2\} \subseteq A$
    (g) $\{3\} \notin A$ (h) $A \subseteq \{11, 2, 5, 1, 8, 4\}$
16. Let $A = \{x \mid x \text{ is an integer and } x^2 < 16\}$. Identify each of the following as true or false.
    (a) $\{0, 1, 2, 3\} \subseteq A$ (b) $\{-3, -2, -1\} \subseteq A$
    (c) $\{\ \} \subseteq A$
    (d) $\{x \mid x \text{ is an integer and } |x| < 4\} \subseteq A$
    (e) $A \subseteq \{-3, -2, -1, 0, 1, 2, 3\}$
17. Let $A = \{1\}$, $B = \{1, a, 2, b, c\}$, $C = \{b, c\}$, $D = \{a, b\}$, and $E = \{1, a, 2, b, c, d\}$. For each part, replace the symbol with either $\in$ or $\notin$ to give a true statement.
    (a) $A \quad B$ (b) $\quad A$ (c) $B \quad C$
    (d) $C \quad E$ (e) $D \quad C$ (f) $B \quad E$

In Exercises 18 through 20, find the set of smallest cardinality that contains the given sets as subsets.

18. $\{a, b, c\}, \{a, d, e, f\}, \{b, c, e, g\}$
19. $\{1, 2\}, \{1, 3\},$
20. $\{2, 4, 6, \ldots, 20\}, \{3, 6, 9, \ldots, 21\}$
21. Is it possible to have two different (appropriate) universal sets for a collection of sets? Would having different universal sets create any problems? Explain.
22. Use the Venn diagram in Figure 3 to identify each of the following as true or false.
    (a) $A \subseteq B$ (b) $B \subseteq A$ (c) $C \subseteq B$
    (d) $x \in B$ (e) $x \in A$ (f) $y \in B$

36. Choose $w \in D \cup (E \cap F)$.
    (a) $w \in D$ (b) $w \in E$ (c) $w \in F$
    (d) $w \notin D$ (e) $w \in F \setminus E$
    (f) $w \in (D \cup E) \cap (D \cup F)$
37. Choose $t \notin \overline{D \cup E}$.
    (a) $t \notin D$ (b) $t \in E$ (c) $t \notin D$
    (d) $t \in E$ (e) $t \notin D \cup E$
38. Choose $x \in \overline{A} \cap (B \cup C)$.
    (a) $x \in A$ (b) $x \in B$ (c) $x \in C$
    (d) $x \in A \cap B$ (e) $x \in (\overline{A} \cap B) \cup (\overline{A} \cap C)$
39. Complete the following proof that $A \cap B \subseteq A$. Suppose $x \in A \cap B$. Then x belongs to _____. Thus $A \cap B \subseteq A$.
40. (a) Draw a Venn diagram to represent the situation $C \subseteq A \cup B$ and $C \not\subseteq A$ and $C \not\subseteq B$.
    (b) To prove $C \not\subseteq A \cup B$, we should choose an element from which set?
    (c) Prove that if $C \not\subseteq A$ and $C \not\subseteq B$, then $C \not\subseteq A \cup B$.
41. (a) Draw a Venn diagram to represent the situation $A \not\subseteq C$ and $B \not\subseteq C$.
    (b) To prove $A \cup B \not\subseteq C$, we should choose an element from which set?
    (c) Prove that if $A \not\subseteq C$ and $B \not\subseteq C$, then $A \cup B \not\subseteq C$.
42. Prove that $A \cap (A \cap B) \subseteq B$.
43. Suppose that $A \cup B = A \cup C$. Does this guarantee that $B = C$? Justify your conclusion.
44. Prove that $A \cap B = A \setminus \overline{B}$.
45. If $A \cup B = A \cup C$, must $B = C$? Explain.
46. If $A \cap B = A \cap C$, must $B = C$? Explain.
47. Prove that if $A \subseteq B$ and $C \subseteq D$, then $A \cup C \subseteq B \cup D$ and $A \cap C \subseteq B \cap D$.
48. When is $A \cap B = B \cap A$? Explain.
49. Explain the last term in the sum in Theorem 3. Why is $|A \cap B \cap C|$ added and $|B \cap C|$ subtracted?
50. Write the four-set version of Theorem 3; that is, $|A \cup B \cup C \cup D| = \cdots$.
51. Describe in words the n-set version of Theorem 3.

3 Sequences

Some of the most important sets arise in connection with sequences. A sequence is simply a list of objects arranged in a definite order; a first element, second element, third element, and so on. The list may stop after n steps, $n \in \mathbb{N}$, or it may go on forever. In the first case we say that the sequence is *finite*, and in the second case we say that it is *infinite*. The elements may all be different, or some may be repeated.

Example 1 The sequence 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 1 is a finite sequence with repeated items. The digit zero, for example, occurs as the second, third, fifth, seventh, and eighth elements of the sequence.

Example 2 The list 3, 8, 13, 18, 23, . . . is an infinite sequence. The three dots in the expression mean "and so on," that is, continue the pattern established by the first few elements.

Example 3 Another infinite sequence is 1, 4, 9, 16, 25, . . . , the list of the squares of all positive integers.

It may happen that how a sequence is to continue is not clear from the first few terms. Also, it may be useful to have a compact notation to describe a sequence. Two kinds of formulas are commonly used to describe sequences. In Example 2, a natural description of the sequence is that successive terms are produced by adding 5 to the previous term. If we use a subscript to indicate a term's position in the sequence, we can describe the sequence in Example 2 as $a_1 = 3$, $a_n = a_{n-1} + 5$, $2 \le n$. A formula, like this one, that refers to previous terms to define the next term is called *recursive*. Every recursive formula must include a starting place.

On the other hand, in Example 3 it is easy to describe a term using only its position number. In the nth position is the square of n, $b_n = n^2$, $1 \le n$. This type of formula is called *explicit*, because it tells us exactly what value any particular term has.

Example 4 The recursive formula $c_1 = 5, c_n = 2c_{n-1}, 2 \leq n \leq 6$, defines the finite sequence 5, 10, 20, 40, 80, 160.

Example 5 The infinite sequence 3, 7, 11, 15, 19, 23, ... can be defined by the recursive formula $d_1 = 3, d_n = d_{n-1} + 4$.

Example 6 The explicit formula $s_n = (-4)^n$, $1 \leq n$, describes the infinite sequence $-4, 16, -64, 256, \ldots$.

Example 7 The finite sequence 87, 82, 77, 72, 67 can be defined by the explicit formula $t_n = 92 - 5n$, $1 \leq n \leq 5$.

Example 8 An ordinary English word such as "sturdy" can be viewed as the finite sequence

s, t, u, r, d, y

composed of letters from the ordinary English alphabet.

In examples such as Example 8, it is common to omit the commas and write the word in the usual way, if no confusion results. Similarly, even a meaningless word such as "abacabcd" may be regarded as a finite sequence of length 8. Sequences of letters or other symbols, written without the commas, are also referred to as strings.

Example 9 An infinite string such as ababababab... may be regarded as the infinite sequence a, b, a, b, a, b, ...

Example 10 The sentence "now is the time for the test" can be regarded as a finite sequence of English words: now, is, the, time, for, the, test. Here the elements of the sequence are themselves words of varying length, so we would not be able simply to omit the commas. The custom is to use spaces instead of commas in this case.

The set corresponding to a sequence is simply the set of all distinct elements in the sequence. Note that an essential feature of a sequence is the order in which the elements are listed. However, the order in which the elements of a set are listed is of no significance at all.

Example 11 (a) The set corresponding to the sequence in Example 3 is $\{1, 4, 9, 16, 25, \ldots\}$.
(b) The set corresponding to the sequence in Example 9 is simply $\{a, b\}$.

The idea of a sequence is important in computer science, where a sequence is sometimes called a linear array or list. We will make a slight but useful distinction between a sequence and an array, and use a slightly different notation. If we have a sequence S: $s_1, s_2, s_3, \ldots$, we think of all the elements of S as completely determined. The element s_4, for example, is some fixed element of S, located in position four. Moreover, if we change any of the elements, we have a new sequence and will probably name it something other than S. Thus if we begin with the finite sequence S: 0, 1, 2, 3, 2, 1, 1 and we change the 3 to a 4, getting 0, 1, 2, 4, 2, 1, 1, we would think of this as a different sequence, say

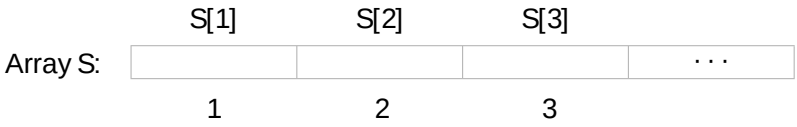


Figure 16

An array, on the other hand, may be viewed as a sequence of positions, which we represent in Figure 16 as boxes. The positions form a finite or infinite list, depending on the desired size of the array. Elements from some set may be assigned to the positions of the array S. The element assigned to position n will be denoted by $S[n]$, and the sequence $S[1], S[2], S[3], \ldots$ will be called the sequence of values of the array S. The point is that S is considered to be a well-defined object, even if some of the positions have not been assigned values, or if some values are changed during the discussion. The following shows one use of arrays.

Characteristic Functions

A very useful concept for sets is the characteristic function. Think of a function on a set as a rule that assigns some "value" to each element of the set. If A is a subset of a universal set U, the characteristic function f_A of A is defined for each $x \in U$ as follows:

$$f_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A. \end{cases}$$

We may add and multiply characteristic functions, since their values are numbers, and these operations sometimes help us prove theorems about properties of subsets.

THEOREM 1 Characteristic functions of subsets satisfy the following properties:

(a) $f_{A \cap B} = f_A f_B$; that is, $f_{A \cap B}(x) = f_A(x) f_B(x)$ for all x.

(b) $f_{A \cup B} = f_A + f_B - f_A f_B$; that is, $f_{A \cup B}(x) = f_A(x) + f_B(x) - f_A(x) f_B(x)$ for all x.

(c) $f_{A \oplus B} = f_A + f_B - 2 f_A f_B$; that is, $f_{A \oplus B}(x) = f_A(x) + f_B(x) - 2 f_A(x) f_B(x)$ for all x.

Proof

(a) $f_A(x) f_B(x)$ equals 1 if and only if both $f_A(x)$ and $f_B(x)$ are equal to 1, and this happens if and only if x is in A and x is in B, that is, x is in $A \cap B$. Since $f_A f_B$ is 1 on $A \cap B$ and 0 otherwise, it must be $f_{A \cap B}$.

(b) If $x \in A$, then $f_A(x) = 1$, so $f_A(x) + f_B(x) - f_A(x) f_B(x) = 1 + f_B(x) - f_B(x) = 1$. Similarly, when $x \in B$, $f_A(x) + f_B(x) - f_A(x) f_B(x) = 1$. If x is not in A or B, then $f_A(x)$ and $f_B(x)$ are 0, so $f_A(x) + f_B(x) - f_A(x) f_B(x) = 0$. Thus $f_A + f_B - f_A f_B$ is 1 on $A \cup B$ and 0 otherwise, so it must be $f_{A \cup B}$.

(c) We leave the proof of (c) as an exercise.

Note that the proof of Theorem 1 proceeds by direct application of the definition of the characteristic function.

Computer Representation of Sets and Subsets

Another use of characteristic functions is in representing sets in a computer. To represent a set in a computer, the elements of the set must be arranged in a sequence. The particular sequence selected is of no importance. When we list the set $A = \{a, b, c, \ldots, r\}$ we normally assume no particular ordering of the elements in A. Let us identify for now the set A with the sequence $a, b, c, \ldots, r$.

When a universal set U is finite, say $U = \{x_1, x_2, \ldots, x_n\}$, and A is a subset of U, then the characteristic function assigns 1 to an element that belongs to A and 0 to an element that does not belong to A. Thus f_A can be represented by a sequence of 0's and 1's of length n.

Example 12 Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2\}$, $B = \{2, 4, 6\}$, and $C = \{4, 5, 6\}$. Then $f_A(x)$ has value 1 when x is 1 or 2, and otherwise is 0. Hence f_A corresponds to the sequence 1, 1, 0, 0, 0, 0. In a similar way, the finite sequence 0, 1, 0, 1, 0, 1 represents f_B and 0, 0, 0, 1, 1, 1 represents f_C.

Any set with n elements can be arranged in a sequence of length n, so each of its subsets corresponds to a sequence of zeros and ones of length n, representing the characteristic function of that subset. This fact allows us to represent a universal set in a computer as an array A of length n. Assignment of a zero or one to each location $A[k]$ of the array specifies a unique subset of U.

Example 13 Let $U = \{a, b, e, g, h, r, s, w\}$. The array of length 8 shown in Figure 17 represents U, since $A[k] = 1$ for $1 \leq k \leq 8$.

1	1	1	1	1	1	1	1

Figure 17

If $S = \{a, e, r, w\}$, then

$$f_S(x) = \begin{cases} 1 & \text{if } x = a, e, r, w \\ 0 & \text{if } x = b, g, h, s. \end{cases}$$

Hence the array in Figure 18 represents the subset S.

1	0	1	0	0	1	0	1

Figure 18

A set is called *countable* if it is the set corresponding to some sequence. Informally, this means that the members of the set can be arranged in a list, with a first, second, third, . . , element, and the set can therefore be "counted." Not all infinite sets are countable. A set that is not countable is called *uncountable*.

The most accessible example of an uncountable set is the set of all real numbers that can be represented by an infinite decimal of the form $.a_1a_2a_3\ldots$, where a_i is an integer and $0 \leq a_i \leq 9$. We shall now show that this set is uncountable. We will prove this result by contradiction; that is, we will show the countability of this set implies an impossible situation.

Assume that the set of all decimals $0.a_1a_2a_3\ldots$ is countable. Then we could form the following list (sequence), containing all such decimals:

$$\begin{aligned} d_1 &= 0.a_1a_2a_3\ldots \\ d_2 &= 0.b_1b_2b_3\ldots \\ d_3 &= 0.c_1c_2c_3\ldots \\ &\vdots \end{aligned}$$

Each of our infinite decimals must appear somewhere on this list. We shall establish a contradiction by constructing an infinite decimal of this type that is not on the list. Now construct a number $x = 0.x_1x_2x_3\ldots$, where x_1 is 1 if $a_1 = 2$, otherwise x_1 is 2; $x_2 = 1$ if $b_2 = 2$, otherwise $x_2 = 2$; $x_3 = 1$ if $c_3 = 2$, otherwise $x_3 = 2$. This process can clearly be continued indefinitely. The resulting number is an infinite decimal consisting of 1's and 2's, but by its construction x differs from each number in the list at some position. Thus x is not on the list, a contradiction to our assumption. Hence no matter how the list is constructed, there is some real number of the form $0.x_1x_2x_3\ldots$ that is not in the list. On the other hand, it can be shown that the set of rational numbers is countable.

Strings and Regular Expressions

Given a set A, we can construct the set A^* consisting of all finite sequences of elements of A. Often, the set A is not a set of numbers, but some set of symbols. In this case, A is called an *alphabet*, and the finite sequences in A^* are called *words* from A, or sometimes strings from A. For this case in particular, the sequences in A^* are *not* written with commas. We assume that A^* contains the *empty sequence* or empty string, containing no symbols, and we denote this string by λ.

Example 14 Let $A = \{a, b, c, \ldots, z\}$, the usual English alphabet. Then A^* consists of ordinary words, such as ape, sequence, antidisestablishmentarianism, and so on, as well as "words" such as yxaloble, zigadongdong, esy, and pqrst. All finite sequences from A are in A^*, whether they have meaning or not.

If $w_1 = s_1s_2s_3\ldots s_n$ and $w_2 = t_1t_2t_3\ldots t_k$ are elements of A^* for some set A, we define the *catenation* of w_1 and w_2 as the sequence $s_1s_2s_3\ldots s_nt_1t_2t_3\ldots t_k$. The catenation of w_1 with w_2 is written as $w_1 \cdot w_2$ or w_1w_2, and is another element of A^*. Note that if w belongs to A^*, then $w \cdot \lambda = w$ and $\lambda \cdot w = w$. This property is convenient and is one of the main reasons for defining the empty string.

Example 15 Let $A = \{$John, Sam, Jane, swims, runs, well, quickly, slowly$\}$. Then A^* contains real sentences such as "Jane swims quickly" and "Sam runs well," as well as nonsense sentences such as "Well swims Jane slowly John." Here we separate the elements in each sequence with spaces. This is often done when the elements of A are words.

The idea of a recursive formula for a sequence is useful in more general settings as well. Regular expressions are defined recursively. A regular expression over A is a string constructed from the elements of A and the symbols $\emptyset$, $($, $)$, $\cup$, $\cdot$, *, according to the following definition.

RE1. The symbol $\emptyset$ is a regular expression.

RE2. If $x \in A$, the symbol x is a regular expression.

RE3. If r and s are regular expressions, then the expression rs is regular.

RE4. If r and s are regular expressions, then the expression $(r \cup s)$ is regular.

RE5. If r is a regular expression, then the expression $(r)^*$ is regular.

Note here that RE1 and RE2 provide initial regular expressions. The other parts of the definition are used repetitively to define successively larger sets of regular expressions from those already defined. Thus the definition is recursive.

By convention, if the regular expression r consists of a single symbol x, where $x \in A$, or if r begins and ends with parentheses, then we will write r simply as r. When no confusion results, we will refer to a regular expression r over A simply as a regular expression (omitting reference to A).

Example 16 Let $A = \{0, 1\}$. Show that the following expressions are all regular expressions over A.

(a) $0(0 \cup 1)^*$ (b) $00(0 \cup 1)^*1$ (c) $(01)^*(01 \cup 1)^*$

Solution

(a) By RE2, 0 and 1 are regular expressions. Thus $(0 \cup 1)$ is regular by RE4, and so 0 and $(0 \cup 1)^*$ are regular by RE5 (and the convention mentioned previously). Finally, we see that $0(0 \cup 1)^*$ is regular by RE3.

(b) We know that 0, 1, and $(0 \cup 1)^*$ are all regular. Thus, using RE3 twice, $00(0 \cup 1)^*1$ must be regular.

(c) By RE3, 01 is a regular expression. Since 1 is regular, $(01 \cup 1)^*$ is regular by RE4, and $(01)^*$ is regular by RE5. Then the regularity of $(01)^*(01 \cup 1)^*$ follows from RE3.

Associated with each regular expression over A, there is a corresponding subset of A^*. Such sets are called *regular subsets* of A^* or just *regular sets* if no reference to A is needed. To compute the regular set corresponding to a regular expression, we use the following correspondence rules.

1. The expression $\emptyset$ corresponds to the set $\{\varepsilon\}$, where ε is the empty string in A^*.
2. If $x \in A$, then the regular expression x corresponds to the set $\{x\}$.
3. If r and s are regular expressions corresponding to the subsets M and N of A^*, then rs corresponds to $M \cdot N = \{s \cdot t \mid s \in M \text{ and } t \in N\}$. Thus $M \cdot N$ is the set of all catenations of strings in M with strings in N.
4. If the regular expressions r and s correspond to the subsets M and N of A^*, then $(r \cup s)$ corresponds to $M \cup N$.
5. If the regular expression r corresponds to the subset M of A^*, then $(r)^*$ corresponds to the set M^*. Note that M is a set of strings from A^*. Elements from M^* are finite sequences of such strings, and thus may themselves be interpreted as strings from A^*. Note also that we always have $\varepsilon \in M^*$.

Example 17 Let $A = \{a, b, c\}$. Then the regular expression a^* corresponds to the set of all finite sequences of a's, such as aaa, $aaaaaaa$, and so on. The regular expression $a(b \cup c)$ corresponds to the set $\{ab, ac\} \subseteq A^*$. Finally, the regular expression $ab(bc)^*$ corresponds to the set of all strings that begin with ab and then repeat the symbols bc n times, where $n \geq 0$. This set includes the strings ab, $abbcbc$, $abbcbcbcbc$, and so on.

Example 18 Let $A = \{0, 1\}$. Find regular sets corresponding to the three regular expressions in Example 16.

Solution

(a) The set corresponding to $(0 \cup 1)$ consists of all sequences of 0's and 1's. Thus, the set is A.

(b) The expression $00(0 \cup 1)^*1$ corresponds to the set of all sequences of 0's and 1's that begin with at least one 0 and end with at least one 1.

(c) The expression $(01)^* \cup (01 \cup 1)^*$ corresponds to the set of all sequences of 0's and 1's that either repeat the string 01 a total of $n \geq 0$ times, or begin with a total of $n \geq 0$ repetitions of 01 and end with some number $k \geq 0$ of 1's. This set includes, for example, the strings 111, 01, 010101, 01010111111, and 011.

3 Exercises

In Exercises 1 through 4, give the set corresponding to the sequence.

1. 1, 2, 1, 2, 1, 2, 1, 2, 1
2. 0, 2, 4, 6, 8, 10, . . .
3. aabbccddee . . . zz
4. abbcccdddd
5. Give three different sequences that have $\{x, y, z\}$ as a corresponding set.
6. Give three different sequences that have $\{1, 2, 3, \ldots\}$ as a corresponding set.

In Exercises 7 through 14, write out the first four terms (begin with $n = 1$) of the sequence whose general term is given.

7. $a_n = 5^n$
8. $b_n = 3n^2 + 2n - 6$
9. $g_n = 1 \cdot 2 \cdot 3 \cdot \cdots \cdot n$
10. $h_n = \dfrac{a^n - 1}{a - 1}, a \neq 1$
11. $c_1 = 2.5, c_n = c_{n-1} + 1.5$
12. $d_1 = -3, d_n = -2d_{n-1} + 1$
13. $e_1 = 0, e_n = e_{n-1} - 2$
14. $f_1 = 4, f_n = n \cdot f_{n-1}$

In Exercises 15 through 20, write a formula for the nth term of the sequence. Identify your formula as recursive or explicit.

15. 1, 3, 5, 7, . . .
16. 0, 3, 8, 15, 24, 35, . . .
17. $1, -1, 1, -1, 1, -1, \ldots$
18. 0, 2, 0, 2, 0, 2, . . .
19. 1, 4, 7, 10, 13, 16
20. $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \ldots$
21. Write an explicit formula for the sequence 2, 5, 8, 11, 14, 17, ...
22. Write a recursive formula for the sequence 2, 5, 7, 12, 19, 31, ...
23. Let $A = \{x \mid x$ is a real number and $0 \leq x < 1\}$, $B = \{x \mid x$ is a real number and $x^2 + 1 = 0\}$, $C = \{x \mid x = 4m, m \in Z\}$, $D = \{(x, 3) \mid x$ is an English word whose length is 3$\}$, and $E = \{x \mid x \in Z$ and $x^2 \leq 100\}$. Identify each set as finite, countable, or uncountable.
24. Let $A = \mathcal{W}$ for $W = \{a, b\}$, $B = \{x \mid x \in R$ and $x^2 + 41x + 41 = 0\}$, $C = \{x \mid x = \frac{m}{n}, m, n \in Z^+, n > 4\}$, $D = \{x \mid x \in R$ and $x^2 + 3x + 2 = 0\}$, and $E = \{(x, y, z) \mid x \in Z, y \in R^+, z \in Z^+\}$. Identify each set as finite, countable, or uncountable.
25. Let $A = \{ab, bc, ba\}$. In each part, tell whether the string belongs to A.
    (a) ababab (b) abc (c) abba
    (d) abbcbaba (e) bcabbab (f) abbbcba
26. Let $U = \{$arithmetic, algebra, calculus, geometry, trigonometry, analysis, topology, statistics$\}$, $B = \{$analysis, topology, calculus$\}$, $C = \{$algebra, calculus, trigonometry, analysis$\}$, $D = \{$arithmetic, algebra, calculus, topology, statistics$\}$, $E = \{$algebra, calculus, geometry, trigonometry, analysis$\}$. In each of the following, represent the given set by an array of zeros and ones.
    (a) $B \cap C$ (b) $C \cap D$
    (c) $B \cap (D \cup E)$ (d) $\overline{B} \cap E$
    (e) $\overline{C} \cap (B \cup E)$
27. Let $U = \{b, d, e, g, h, k, m, n\}$, $B = \{b\}$, $C = \{d, g, m, n\}$, and $D = \{d, k, n\}$.
    (a) What is $f_B(b)$? (b) What is $f_C(e)$?
    (c) Find the sequences of length 8 that correspond to f_B, f_C, and f_D.
    (d) Represent $B \cup C$, $C \cap D$, and $C \cup D$ by arrays of zeros and ones.
28. Complete the proof that $f_{A \cap B} = f_A + f_B - 2f_A f_B$ [Theorem 1(c)]. Suppose $x \in A$ and $x \notin B$. Then $f_A(x) =$ _____, $f_B(x) =$ _____, and $f_A(x) f_B(x) =$ _____, so $f_A(x) + f_B(x) - 2f_A(x) f_B(x) =$ _____.

Now suppose $x \in A$ and $x \notin B$. Then $f_A(x) =$ _____, $f_B(x) =$ _____, and $f_A(x)f_B(x) =$ _____, so $f_A(x) + f_B(x) - 2f_A(x)f_B(x) =$ _____. The remaining case to check is $x \notin A \cup B$. If $x \notin A \cup B$, then $x \notin$ _____ and $f_A(x) + f_B(x) - 2f_A(x)f_B(x) =$ _____. Explain how these steps prove Theorem 1(c).

29. Using characteristic functions, prove that $(A \cap B) \cap C = A \cap (B \cap C)$.

30. Let $A = \{+, \times, a, b\}$. Show that the following expressions are regular over A.
    (a) $a + b(ab)^*(a \times b^* a)^*$
    (b) $a + b \times (a^* b)^*$
    (c) $(a^* b^* +)^* \times b^*$

In Exercises 31 and 32, let $A = \{a, b, c\}$. In each exercise a string in A^* is listed and a regular expression over A is given. In each case, tell whether or not the string on the left belongs to the regular set corresponding to the regular expression on the right.

31. (a) ac $a^* b^* c^*$ (b) $abcc$ $(abc)^* c^*$
    (c) $aaabc$ $((a^* b^*)^* c^*)^*$

32. (a) ac $(a^* b^* c^*)^*$ (b) $abab$ $(ab)^* c^*$
    (c) $aaccc$ $(a^* b)^* c^*$

33. Give three expressions that are not regular over A that are given for Exercises 31 and 32.

34. Let $A = \{p, q, r\}$. Give the regular set corresponding to the regular expression given.
    (a) $(p^* q)^* rq^*$ (b) $p^*(qq^*)^* r^*$

35. Let $S = \{0, 1\}$. Give the regular expression corresponding to the regular set given.
    (a) $\{00, 010, 0110, 011110, \ldots\}$
    (b) $\{0, 001, 000, 00001, 00000, 0000001, \ldots\}$

36. We define T-numbers recursively as follows:
    1. 0 is a T-number.
    2. If X is a T-number, $X + 3$ is a T-number.

    Write a description of the set of T-numbers.

37. Define an S-number by
    1. 8 is an S-number.
    2. If X is an S-number and Y is a multiple of X, then Y is an S-number.
    3. If X is an S-number and X is a multiple of Y, then Y is an S-number.

    Describe the set of S-numbers.

38. Let F be a function defined for all nonnegative integers by the following recursive definition.

$$F(0) = 0, \qquad F(1) = 1$$
$$F(n + 2) = 2F(n) + F(n + 1), \quad n \geq 0$$

Compute the first six values of F; that is, write the values of $F(n)$ for $n = 0, 1, 2, 3, 4, 5$.

39. Let G be a function defined for all nonnegative integers by the following recursive definition.

$$G(0) = 1, \qquad G(1) = 2$$
$$G(n + 2) = G(n)^2 + G(n + 1), \quad n \geq 0$$

Compute the first five values of G.

4 Properties of Integers

We shall now discuss some results needed later about division and factoring in the integers. If m is an integer and n is a positive integer, we can plot the integer multiples of n on a line, and locate m as in Figure 19. If m is a multiple of n, say $m = qn$, then we can write $m = qn + r$, where r is 0. On the other hand (as shown in Figure 19), if m is not a multiple of n, we let qn be the first multiple of n lying to the left of m and let r be $m - qn$. Then r is the distance from qn to m, so clearly $0 < r < n$, and again we have $m = qn + r$. We state these observations as a theorem.

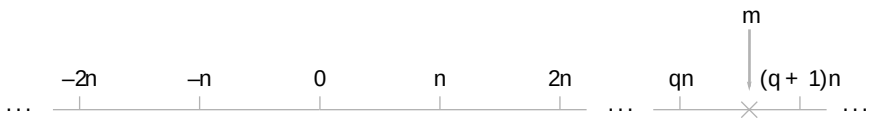


Figure 19

THEOREM 1 If n and m are integers and $n > 0$, we can write $m = qn + r$ for integers q and r with $0 \leq r < n$. Moreover, there is just one way to do this.

Example 1

(a) If n is 3 and m is 16, then $16 = 5(3) + 1$ so q is 5 and r is 1.
(b) If n is 10 and m is 3, then $3 = 0(10) + 3$ so q is 0 and r is 3.
(c) If n is 5 and m is -11, then $-11 = -3(5) + 4$ so q is -3 and r is 4.

If the r in Theorem 1 is zero, so that m is a multiple of n, we write $n \mid m$, which is read "n divides m." If $n \mid m$, then $m = qn$ and $n \nmid m$. If m is not a multiple of n, we write $n \nmid m$, which is read "n does not divide m." We now prove some simple properties of divisibility.

THEOREM 2 Let a, b, and c be integers.

(a) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
(b) If $a \mid b$ and $a \mid c$, where $b > c$, then $a \mid (b - c)$.
(c) If $a \mid b$ or $a \mid c$, then $a \mid bc$.
(d) If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof

(a) If $a \mid b$ and $a \mid c$, then $b = k_1 a$ and $c = k_2 a$ for integers k_1 and k_2. So $b + c = (k_1 + k_2)a$ and $a \mid (b + c)$.
(b) This can be proved in exactly the same way as (a).
(c) As in (a), we have $b = k_1 a$ or $c = k_2 a$. Then either $bc = k_1 ac$ or $bc = k_2 ab$, so in either case bc is a multiple of a and $a \mid bc$.
(d) If $a \mid b$ and $b \mid c$, we have $b = k_1 a$ and $c = k_2 b$, so $c = k_2 b = k_2(k_1 a) = (k_2 k_1)a$ and hence $a \mid c$.

Note that again we have a proof that proceeds directly from a de nition by restating the original conditions. As a consequence of Theorem 2, we have that if $a \mid b$ and $a \mid c$, then $a \mid (mb + nc)$, for any integers m and n. (See Exercises 23 and 24.)

A number $p > 1$ in Z^+ is called *prime* if the only positive integers that divide p are p and 1.

Example 2 The numbers 2, 3, 5, 7, 11, and 13 are prime, while 4, 10, 16, and 21 are not prime.

It is easy to write a set of steps, or an *algorithm*,[1] to determine if a positive integer $n > 1$ is a prime number. First we check to see if n is 2. If $n > 2$, we could divide by every integer from 2 to $n - 1$, and if none of these is a divisor of n, then n is prime. To make the process more ef cient, we note that if $n = rk$, then either r or k is less than or equal to $\sqrt{n}$. This means that if n is not prime, it has a divisor k satisfying the inequality $1 < k \leq \sqrt{n}$, so we need only test for divisors in this range. Also, if n has any even number as a divisor, it must have 2 as a divisor. Thus after checking for divisibility by 2, we may skip all even integers.

Algorithm

To test whether an integer $N > 1$ is prime:

Step 1 Check whether N is 2. If so, N is prime. If not, proceed to
Step 2 Check whether $2 \mid N$. If so, N is not prime; otherwise, proceed to
Step 3 Compute the largest integer k for $\sqrt{N}$. Then

[1] Algorithms are discussed in Appendix A.

Step 4 Check whether $D \mid N$, where D is any odd number such that $1 < D \leq K$. If $D \mid N$, then N is not prime; otherwise, N is prime.

Testing whether an integer is prime is a common task for computers. The algorithm given here is too inefficient for testing very large numbers, but there are many other algorithms for testing whether an integer is prime.

THEOREM 3 Every positive integer $n > 1$ can be written uniquely as $p_1^{k_1} p_2^{k_2} \cdots p_s^{k_s}$, where $p_1 < p_2 < \cdots < p_s$ are distinct primes that divide n and the k's are positive integers giving the number of times each prime occurs as a factor of n.

Example 3
(a) $9 = 3 \cdot 3 = 3^2$
(b) $24 = 8 \cdot 3 = 2 \cdot 2 \cdot 2 \cdot 3 = 2^3 \cdot 3$
(c) $30 = 2 \cdot 3 \cdot 5$

Greatest Common Divisor

If a, b, and k are in Z^+, and $k \mid a$ and $k \mid b$, we say that k is a common divisor of a and b. If d is the largest such k, d is called the greatest common divisor, or GCD, of a and b, and we write $d = \text{GCD}(a, b)$. This number has some interesting properties. It can be written as a combination of a and b, and it is not only larger than all the other common divisors, it is a multiple of each of them.

THEOREM 4 If d is $\text{GCD}(a, b)$, then

(a) $d = sa + tb$ for some integers s and t. (These are not necessarily positive.)
(b) If c is any other common divisor of a and b, then $c \mid d$.

Proof
Let x be the smallest positive integer that can be written as $sa + tb$ for some integers s and t, and let c be a common divisor of a and b. Since $c \mid a$ and $c \mid b$, it follows from Theorem 2 that $c \mid x$, so $c \leq x$. If we can show that x is a common divisor of a and b, it will then be the greatest common divisor of a and b and both parts of the theorem will have been proved. By Theorem 1.1, $a = qx + r$ with $0 \leq r < x$. Solving for r, we have

$$r = a - qx = a - q(sa + tb) = a - qsa - qtb = (1 - qs)a + (-qt)b.$$

If r is not zero, then since $r < x$ and r is the sum of a multiple of a and a multiple of b, we will have a contradiction to the fact that x is the smallest positive number that is a sum of multiples of a and b. Thus r must be 0 and $x \mid a$. In the same way we can show that $x \mid b$, and this completes the proof.

This proof is more complex than the earlier ones. At this stage you should focus on understanding the details of each step.

From the definition of greatest common divisor and Theorem 4(b), we have the following result: Let a, b, and d be in Z^+. The integer d is the greatest common divisor of a and b if and only if

(a) $d \mid a$ and $d \mid b$.
(b) Whenever $c \mid a$ and $c \mid b$, then $c \mid d$.

Example 4 (a) The common divisors of 12 and 30 are 1, 2, 3, and 6, so that

$$\text{GCD}(12, 30) = 6 \quad \text{and} \quad 6 = 1 \cdot 30 + (-2) \cdot 12.$$

(b) It is clear that $\text{GCD}(17, 95) = 1$ since 17 is prime and 17 ∤ 95, and the reader may verify that $1 = 28 \cdot 17 + (-5) \cdot 95$.

If $\text{GCD}(a, b) = 1$, as in Example 4(b), we say a and b are **relatively prime**.

One remaining question is that of how to compute the GCD conveniently in general. Repeated application of Theorem 1 provides the key to doing this.

We now present a procedure, called the **Euclidean algorithm**, for finding $\text{GCD}(a, b)$. Suppose that $a > b > 0$ (otherwise interchange a and b). Then by Theorem 1, we may write

$$a = k_1 b + r_1, \qquad \text{where } k_1 \text{ is in } \mathbb{Z}^+ \text{ and } 0 \le r_1 < b. \tag{1}$$

Now Theorem 2 tells us that if n divides a and b, then it must divide r_1, since $r_1 = a - k_1 b$. Similarly, if n divides b and r_1, then it must divide a. We see that the common divisors of a and b are the same as the common divisors of b and r_1, so $\text{GCD}(a, b) = \text{GCD}(b, r_1)$.

We now continue using Theorem 1 as follows:

$$\begin{array}{rrll}
\text{divide } b \text{ by } r_1: & b = k_2 r_1 + r_2 & 0 & r_2 < r_1 \\
\text{divide } r_1 \text{ by } r_2: & r_1 = k_3 r_2 + r_3 & 0 & r_3 < r_2 \\
\text{divide } r_2 \text{ by } r_3: & r_2 = k_4 r_3 + r_4 & 0 & r_4 < r_3 \\
\vdots & \vdots & & \vdots \\
\text{divide } r_{n-2} \text{ by } r_{n-1}: & r_{n-2} = k_n r_{n-1} + r_n & 0 & r_n < r_{n-1} \\
\text{divide } r_{n-1} \text{ by } r_n: & r_{n-1} = k_{n+1} r_n + r_{n+1} & 0 & r_{n+1} < r_n.
\end{array} \tag{2}$$

Since $a > b > r_1 > r_2 > r_3 > r_4 > \cdots$, the remainder will eventually become zero, so at some point we have $r_{n+1} = 0$.

We now show that $r_n = \text{GCD}(a, b)$. We saw previously that

$$\text{GCD}(a, b) = \text{GCD}(b, r_1).$$

Repeating this argument with b and r_1, we see that

$$\text{GCD}(b, r_1) = \text{GCD}(r_1, r_2).$$

Upon continuing, we have

$$\text{GCD}(a, b) = \text{GCD}(b, r_1) = \text{GCD}(r_1, r_2) = \cdots = \text{GCD}(r_{n-1}, r_n).$$

Since $r_{n-1} = k_{n+1} r_n$, we see that $\text{GCD}(r_{n-1}, r_n) = r_n$. Hence $r_n = \text{GCD}(a, b)$.

Example 5 Compute $\text{GCD}(273, 98)$. Let a be 273 and b be 98. Using the Euclidean algorithm,

$$\begin{array}{ll}
\text{divide 273 by 98:} & 273 = 2 \cdot 98 + 77 \\
\text{divide 98 by 77:} & 98 = 1 \cdot 77 + 21 \\
\text{divide 77 by 21:} & 77 = 3 \cdot 21 + 14 \\
\text{divide 21 by 14:} & 21 = 1 \cdot 14 + 7 \\
\text{divide 14 by 7:} & 14 = 2 \cdot 7 + 0
\end{array}$$

so $\text{GCD}(273, 98) = 7$, the last of the nonzero divisors.

In Theorem 4(a), we observed that if $d = \text{GCD}(a, b)$, we can find integers s and t such that $d = sa + tb$. The integers s and t can be found as follows. Solve the next-to-last equation in (2) for r_n:

$$r_n = r_{n-2} - k_n r_{n-1}. \tag{3}$$

Now solve the second-to-last equation in (2), $r_{n-3} = k_{n-1} r_{n-2} + r_{n-1}$, for r_{n-1}:

$$r_{n-1} = r_{n-3} - k_{n-1} r_{n-2}$$

and substitute this expression in (3):

$$r_n = r_{n-2} - k_n [r_{n-3} - k_{n-1} r_{n-2}].$$

Continue to work up through the equations in (2) and (1), replacing r_i by an expression involving r_{i-1} and r_{i-2}, and finally arriving at an expression involving only a and b.

Example 6 (a) Let a be 273 and b be 98, as in Example 5. Then

$$\begin{aligned}
\text{GCD}(273, 98) &= 7 = 21 - 1 \cdot 14 \\
&= 21 - 1(77 - 3 \cdot 21) && 14 = 77 - 3 \cdot 21 \\
&= 4 \cdot 21 - 1 \cdot 77 \\
&= 4(98 - 1 \cdot 77) - 1 \cdot 77 && 21 = 98 - 1 \cdot 77 \\
&= 4 \cdot 98 - 5 \cdot 77 \\
&= 4 \cdot 98 - 5(273 - 2 \cdot 98) && 77 = 273 - 2 \cdot 98 \\
&= 14(98) - 5(273)
\end{aligned}$$

Hence $s = -5$ and $t = 14$. Note that the key is to carry out the arithmetic only partially.

(b) Let $a = 108$ and $b = 60$. Then

$$\begin{aligned}
\text{GCD}(108, 60) &= 12 = 60 - 1(48) \\
&= 60 - 1[108 - 1(60)] && 48 = 108 - 1 \cdot 60 \\
&= 2(60) - 108.
\end{aligned}$$

Hence $s = -1$ and $t = 2$.

THEOREM 5 If a and b are in Z^+, $b > a$, then $\text{GCD}(a, b) = \text{GCD}(b, b - a)$.

Proof
If c divides a and b, it divides $b - a$, by Theorem 2. Since $a = b - (b - a) = -b + (b + a)$, we see, also by Theorem 2, that a common divisor of a and $b - a$ also divides a and b. Since a and b have the same common divisors as b and $b - a$, they must have the same greatest common divisor.

This is another direct proof, but one that uses a previous theorem as well as definitions.

Least Common Multiple

If a, b, and k are in Z^+, and $a \mid k$, $b \mid k$, we say k is a common multiple of a and b. The smallest such k, call it c, is called the least common multiple, or LCM, of a and b, and we write $c = \text{LCM}(a, b)$. The following result shows that we can obtain the least common multiple from the greatest common divisor, so we do not need a separate procedure for finding the least common multiple.

THEOREM 6 If a and b are two positive integers, then $\text{GCD}(a, b) \cdot \text{LCM}(a, b) = ab$.

Proof
Let $p_1, p_2, \ldots, p_k$ be all the prime factors of either a or b. Then we can write

$$a = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k} \quad \text{and} \quad b = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}$$

where some of the a_i and b_i may be zero. It then follows that

$$\text{GCD}(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_k^{\min(a_k, b_k)}$$

and

$$\text{LCM}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_k^{\max(a_k, b_k)}.$$

Hence

$$\begin{aligned} \text{GCD}(a, b) \cdot \text{LCM}(a, b) &= p_1^{a_1 + b_1} p_2^{a_2 + b_2} \cdots p_k^{a_k + b_k} \\ &= (p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}) \cdot (p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}) \\ &= ab. \end{aligned}$$

Example 7 Let $a = 540$ and $b = 504$. Factoring a and b into primes, we obtain

$$a = 540 = 2^2 \cdot 3^3 \cdot 5 \quad \text{and} \quad b = 504 = 2^3 \cdot 3^2 \cdot 7.$$

Thus all the prime numbers that are factors of either a or b are $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, and $p_4 = 7$. Then $a = 2^2 \cdot 3^3 \cdot 5^1 \cdot 7^0$ and $b = 2^3 \cdot 3^2 \cdot 5^0 \cdot 7^1$. We then have

$$\begin{aligned} \text{GCD}(540, 504) &= 2^{\min(2,3)} \cdot 3^{\min(3,2)} \cdot 5^{\min(1,0)} \cdot 7^{\min(0,1)} \\ &= 2^2 \cdot 3^2 \cdot 5^0 \cdot 7^0 \\ &= 2^2 \cdot 3^2 \text{ or } 36. \end{aligned}$$

Also,

$$\begin{aligned} \text{LCM}(540, 504) &= 2^{\max(2,3)} \cdot 3^{\max(3,2)} \cdot 5^{\max(1,0)} \cdot 7^{\max(0,1)} \\ &= 2^3 \cdot 3^3 \cdot 5^1 \cdot 7^1 \text{ or } 7560. \end{aligned}$$

Then

$$\text{GCD}(540, 504) \cdot \text{LCM}(540, 504) = 36 \cdot 7560 = 272{,}160 = 540 \cdot 504.$$

As a verification, we can also compute $\text{GCD}(540, 504)$ by the Euclidean algorithm and obtain the same result.

If n and m are integers and $n > 1$, Theorem 1 tells us we can write $m = qn + r$, $0 \le r < n$. Sometimes the remainder r is more important than the quotient q.

Example 8 If the time is now 4 o'clock, what time will it be 101 hours from now?

Solution

Let $n = 12$ and $m = 4 + 101$ or 105. Then we have $105 = 8 \cdot 12 + 9$. The remainder 9 answers the question. In 101 hours it will be 9 o'clock.

For each $n \in Z^+$, we define a function f_n, the mod n function, as follows: If z is a nonnegative integer, $f_n(z) = r$, the remainder when z is divided by n. (As in Section 3, we need only think of a function as a rule that assigns some "value" to each member of a set.)

Example 9 (a) $f_3(14) = 2$, because $14 = 4 \cdot 3 + 2$.
(b) $f_7(153) = 6$.

Pseudocode Versions

An alternative to expressing an algorithm in ordinary English as we did in this section is to express it in something like a computer language. Here we give pseudocode versions for an algorithm that determines if an integer is prime and for an algorithm that calculates the greatest common divisor of two integers.

In the pseudocode for the algorithm to determine if an integer is prime, we assume the existence of functions SQR and INT, where SQR(N) returns the greatest integer not exceeding $\sqrt{N}$, and INT(X) returns the greatest integer not exceeding X. For example, SQR(10) = 3, SQR(25) = 5, INT(7.124) = 7, and INT(8) = 8.

```
SUBROUTINE  PRIME( N )
1.  IF  ( N = 2)  THEN
      a.  PRINT  (•PRIME•)
      b.  RETURN
2.  ELSE
      a.  IF  ( N/2 = INT(  N/2))  THEN
            1.  PRINT  (•NOT PRIME•)
            2.  RETURN
      b.  ELSE
            1.  FOR  D = 3  THRU  SQR( N)  BY  2
                  a.  IF  ( N/ D = INT(  N/ D))  THEN
                        1.  PRINT  (•NOT PRIME•)
                        2.  RETURN
            2.  PRINT  (•PRIME•)
            3.  RETURN
END OF SUBROUTINE PRIME
```

The following gives a pseudocode program for finding the greatest common divisor of two positive integers. This procedure is different from the Euclidean algorithm.

```
FUNCTION  GCD( X, Y)
1.  WHILE  ( X = Y)
      a.  IF  ( X > Y)  THEN
            1.  X  X Š Y
```

b. ELSE
1. Y Y Š X
2. RETURN (X)
END OF FUNCTION GCD

Example 10 Use the pseudocode for GCD to calculate the greatest common divisor of 190 and 34.

Solution

The following table gives the values of X, Y, X Š Y, or Y Š X as we go through the program.

X	Y	X Š Y	Y Š X
190	34	156	
156	34	122	
122	34	88	
88	34	54	
54	34	20	
20	34		14
20	14	6	
6	14		8
6	8		2
6	2	4	
4	2	2	
2	2		

Since the last value of X is 2, the greatest common divisor of 190 and 34 is 2.

Representations of Integers

The decimal representation of an integer is so familiar that we sometimes regard it as the symbol, or name for that integer. For example, when we write the integer 3264, we are saying that the number is the result of adding 3 times 10^3, 2 times 10^2, 6 times 10^1, and 4, or 4 times 10^0. We say 3264 is the *base 10 expansion* of the integer n or the *decimal expansion* of n; 10 is called the *base* of this expansion.

There is nothing special about using the number 10 as a base, and it is likely the result of our having 10 fingers on which to count. Any positive integer b > 1 can be used in a similar way, and these expansions are often of much greater use than the standard decimal expansion. The bases 2, 8, and 16 are frequently used in computer science, and the base 26 is sometimes used in *cryptology* (the science of producing and deciphering secret codes).

THEOREM 7 If b > 1 is an integer, then every positive integer n can be uniquely expressed in the form

$$n = d_k b^k + d_{k-1} b^{k-1} + \cdots + d_1 b + d_0, \tag{4}$$

where $0 \le d_i < b$, $i = 0, 1, \ldots, k$, and $d_k \ne 0$. The sequence $d_k d_{k-1} \ldots d_1 d_0$ is called the *base b expansion* of n. If we need to explicitly indicate the base b, we will write the above sequence as $(d_k d_{k-1} \ldots d_1 d_0)_b$.

Proof
Suppose that k is the largest nonnegative integer so that $b^k \le n$ (k could be 0). By Theorem 1 we can uniquely write $n = qb^k + r$, where $0 \le r < b^k$. Let $d_k = q$. If $k = 0$, then $r = 0$ and we are done ($n = d_0$). Otherwise we have

$$n = d_k b^k + r.$$

Repeat this process, using r in place of n. Let s be the largest nonnegative integer so that $b^s \le r$, write $r = qb^s + r_1$, where $0 \le r_1 < b^s$, and define d_s to be q. If $s < k$, define $d_{s+1}, \ldots, d_{k-1}$ to be 0. Then

$$n = d_k b^k + d_{k-1} b^{k-1} + \cdots + d_s b^s + r_1.$$

Continuing this process using $r_1, r_2, \ldots$, we will eventually arrive at (4).

Now that we know that the base b expansion exists, we can find the digits $d_k, d_{k-1}, \ldots, d_1, d_0$ by a more direct method, easily implemented on a computer. Note that

$$n = (d_k b^{k-1} + d_{k-1} b^{k-2} + \cdots + d_1)b + d_0$$

so that d_0 is the remainder after dividing n by b, and the quotient is $d_k b^{k-1} + d_{k-1} b^{k-2} + \cdots + d_1$. Similarly, if this quotient is divided by b, the remainder is d_1. By repeatedly dividing the quotients by b and saving the remainders, we produce the digits of the base b representation of n from right to left.

Example 11 Find the base 4 representation of 158.

Solution
We repeatedly divide by 4 and save the remainder:

$$\begin{array}{r|l} 4 & 158 \quad 2 \\ 4 & 39 \quad 3 \\ 4 & 9 \quad 1 \\ 4 & 2 \quad 2 \\ & 0 \end{array}$$

Thus $158 = (2132)_4$.

The following pseudocode algorithm returns the digits of the base b expansion of an integer. We use the expression $m \bmod n$ to denote the mod n function value for m, that is, the remainder after dividing m by n. These are the functions defined after Example 8. The mod functions are commonly implemented in most programming languages.

```
SUBROUTINE  EXPANSION( N )
1.  Q     N
2.  K     0
3.  WHILE  ( Q  =  0)
      a.  Dk     Q  mod  B
      b.  Q      INT(  Q/ B)
      c.  K      K  +  1
4.  RETURN
END  OF  SUBROUTINE  EXPANSION
```

When this subroutine ends, the base B expansion of N consists of the integers D_i, which can then be further processed.

No matter what base is used to represent integers, the elementary rules of addition and multiplication are still valid. Only the appearance of the numbers changes.

Example 12 Let $m = (313)_4$ and $n = (322)_4$. Find the base 4 expansion of $m + n$.

Solution

Adding the digits in the last column, we have $3 + 2 = (11)_4$, so we record a 1 and carry a 1 to the next column.

$$\begin{array}{r} {\scriptstyle 1} \\ 313 \\ +\ 322 \\ \hline 1 \end{array}$$

Adding the digits in the second column, we have $1 + 1 + 2 = (10)_4$, so we record a 0 and carry the 1.

$$\begin{array}{r} {\scriptstyle 11} \\ 313 \\ +\ 322 \\ \hline 01 \end{array}$$

Finally, adding the digits in the first column, we obtain $3 + 3 + 1 = (13)_4$, so the answer is $(1301)_4$.

The most common expansion used in computer work is the base 2 or binary expansion. Since the only remainders of division by 2 are 0 and 1, the binary expansion of every number needs only the digits 0 and 1 and so is easily implemented and manipulated by the on-off digital circuits of a computer.

Example 13 (a) The binary expansion of 39 is $(100111)_2$.
(b) $(110101)_2 = 32 + 16 + 4 + 1 = 53$.

Binary addition and multiplication are usually implemented in computer circuitry rather than by software. Another common base in computer science is base 16, or hexadecimal (or hex) representation. This representation requires six additional symbols to use with 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 to represent the digits usually written 10, 11, 12, 13, 14, 15. It is customary to choose the letters A, B, C, D, E, F for this purpose.

Example 14 To use 26 as a base, we can use the letters A, B, ..., Z of the English alphabet to represent the digits 0, 1, . . , 25. In this way we can interpret any text as the base 26 representation of an integer. Thus we can interpret "TWO" as the base 26 representation of $(T \times 26^2) + (W \times 26) + O = (19 \times 676) + (22 \times 26) + 14 = 13430$. The ability to "add" words can be made the basis of cryptographic encoding.

Example 15 As an example of cryptology, we consider a remarkable code due to Sir Francis Bacon, the English writer and philosopher. Suppose that we wish to encode a message, say FLEE NOW. We first write the base 2 representation of the position of each letter of the message in the English alphabet, starting with A in position 0, and ending with Z in position 25. Thus we have the following table.

F	L	E	E	N	O	W
00101	01011	00100	00100	01101	01110	10110

Since F is in position 5, its binary representation is $(00101)_2$, and so on. Now choose an unrelated "dummy" message exactly five times as long (padded with a few extra letters if necessary). Place the dummy message in a third row. The letters of the dummy message correspond exactly to the string of 0's and 1's in the second row of the above table. We agree to write each letter in one font (say Times Roman) if it corresponds to a 0 in the table, and another font (say Times Roman Italic) if it corresponds to a 1 (Bacon used fonts that were even more similar). Thus if the dummy message is ONCE UPON A TIME IN THE WEST THERE WAS A TOWN, we would write that message as ONCE UPON A TIME IN THE WEST THERE WAS A TOWN. Note that when the letters in this message are arranged in a third row of the table, the patterns of nonitalic for 0 and italic for 1 allow us to decode the message.

F	L	E	E	N	O	W
00101	01011	00100	00100	01101	01110	10110
ONCEU	PONAT	IMEIN	THEWE	STTHE	REWAS	ATOWN

Example 16 Suppose that we wish to decode the following dummy message, using the Bacon code

NOW IS THE TIME FOR ALL GOOD MEN TO AID THE COUNTRY

Since there are 40 letters, the true message must have 8 letters. Arrange the dummy message, 5 letters at a time in the following table, and then list the corresponding binary digits, using 1 for italic and 0 for plain text.

NOWIS	THETI	MEFOR	ALLGO	ODMEN	TOAID	THECO	UNTRY
10010	10011	00000	10001	10011	01001	01110	00001

The binary representations in the second row correspond respectively to the numbers 18, 19, 0, 17, 19, 9, 14, and 1, and therefore represent the letters STARTJOB. Thus the decoded message is START JOB.

If the fonts used are quite similar, it will not be obvious that the text is a coded message. Thus this example also illustrates *steganography*, the science of concealment of coded information. Modern versions include hiding information in stray bits of a digital photograph. One use of this is to watermark copyrighted artistic material.

4 Exercises

In Exercises 1 through 4, for the given integers m and n, write $m = qn + r$, with $0 \le r < n$.

1. $m = 20, n = 3$
2. $m = 64, n = 37$
3. $m = 3, n = 22$
4. $m = 48, n = 12$
5. Write each integer as a product of powers of primes (as in Theorem 3).
   (a) 828 (b) 1666 (c) 1781
   (d) 1125 (e) 107

In Exercises 6 through 9, find the greatest common divisor of the integers a and b, and write it as $sa + tb$.

6. $a = 60, b = 100$
7. $a = 45, b = 33$
8. $a = 34, b = 58$
9. $a = 77, b = 128$

In Exercises 10 through 13, find the least common multiple of the integers.

10. 72, 108
11. 150, 70
12. 175, 245
13. 32, 27

14. If f is the mod-7 function, compute each of the following.
    (a) $f(17)$ (b) $f(48)$ (c) $f(1207)$
    (d) $f(130)$ (e) $f(93)$ (f) $f(169)$

15. If f is the mod-11 function, compute each of the following.
    (a) $f(39)$ (b) $f(386)$ (c) $f(1232)$
    (d) $f(573)$ (e) $2f(87)$ (f) $f(175) + 4$

16. If f is the mod-7 function, compute each of the following.
    (a) $f(752 + 793)$ (b) $f(752) + f(793)$
    (c) $f(3 \cdot 1759)$ (d) $3 \cdot f(1759)$

17. If f is the mod-12 function, compute each of the following.
    (a) $f(1259 + 743)$ (b) $f(1259) + f(743)$
    (c) $f(2 \cdot 319)$ (d) $2 \cdot f(319)$

18. Let f be the mod-n function for a fixed n. What do the results of Exercises 16 and 17 suggest about the relationship between $k \cdot f(a)$ and $f(k \cdot a)$?

19. Let f be the mod-n function for a fixed n. Based on the results of Exercises 16 and 17, explain why $f(a + b)$ does not always equal $f(a) + f(b)$.

20. Let f be the mod-n function for a fixed n. Explain when $f(a + b) = f(a) + f(b)$ is true.

21. If g is the mod-5 function, solve each of the following.
    (a) $g(n) = 2$ (b) $g(n) = 4$

22. If g is the mod-6 function, solve each of the following.
    (a) $g(n) = 3$ (b) $g(n) = 1$

23. Prove that if $a \mid b$, then $a \mid mb$, for any $m \in Z$.

24. Prove that if $a \mid b$ and $a \mid c$, then $a \mid mb + nc$, for any $m, n \in Z$.

25. Complete the following proof. Let a and b be integers. If p is a prime and $p \mid ab$, then $p \mid a$ or $p \mid b$. We need to show that if $p \nmid a$, then p must divide b. If $p \nmid a$, then $GCD(a, p) = 1$, because____. By Theorem 4, we can write $1 = sa + tp$ for some integers s and t. Then $b = sab + tpb$. (Why?) Then p must divides $sab + tpb$, because____. So $p \mid b$. (Why?)

26. Show that if $GCD(a, c) = 1$ and $c \mid ab$, then $c \mid b$. (Hint: Model the proof on the one in Exercise 25.)

27. Show that if $GCD(a, c) = 1$, $a \mid m$, and $c \mid m$, then $ac \mid m$. (Hint: Use Exercise 26.)

28. Show that if $d = GCD(a, b)$, $a \mid b$, and $c \mid b$, then $ac \mid bd$.

29. Show that $GCD(ca, cb) = c\,GCD(a, b)$.

30. Show that $LCM(a, ab) = ab$.

31. Show that if $GCD(a, b) = 1$, then $LCM(a, b) = ab$.

32. Let $c = LCM(a, b)$. Show that if $a \mid k$ and $b \mid k$, then $c \mid k$.

33. Prove that if a and b are positive integers such that $a \mid b$ and $b \mid a$, then $a = b$.

34. Let a be an integer and let p be a positive integer. Prove that if $p \nmid a$, then $p = GCD(a, p)$.

35. Theorem 2(c) says that if $a \mid b$ or $a \mid c$, then $a \mid bc$. Is the converse true; that is, if $a \mid bc$, then $a \mid b$ or $a \mid c$? Justify your conclusion.

36. Prove that if m and n are relatively prime and mn is a perfect square, then m and n are each perfect squares.

37. Is the statement in Exercise 36 true for cubes? For any fixed power? Justify your conclusion.

In Exercises 38 through 40, let $U = \{1, 2, 3, \ldots, 1683\}$, $A = \{x \mid x \in U \text{ and } 3 \mid x\}$, $B = \{y \mid y \in U \text{ and } 5 \mid y\}$, and $C = \{z \mid z \in U \text{ and } 11 \mid z\}$. Compute each of the following.

38. (a) $|A|$ (b) $|B|$ (c) $|C|$

39. (a) The number of elements of U that are divisible by 15
    (b) The number of elements of U that are divisible by 165
    (c) The number of elements of U that are divisible by 55

40. Use the results of Exercises 38 and 39 to compute each of the following.
    (a) $|A \cap B|$ (b) $|A \cup B \cup C|$

41. (a) Write the expansion in base 5 of each of the following numbers.
        (i) 29 (ii) 73 (iii) 215 (iv) 732
    (b) Write the expansion in base 10 of each of the following numbers.
        (i) $(144)_5$ (ii) $(320)_5$
        (iii) $(1242)_5$ (iv) $(11231)_5$

42. (a) Write the expansion in base 7 of each of the following numbers.
        (i) 29 (ii) 73 (iii) 215 (iv) 732
    (b) Write the expansion in base 10 of each of the following numbers.
        (i) $(102)_7$ (ii) $(161)_7$
        (iii) $(460)_7$ (iv) $(1613)_7$

43. For each of the following, write the expansion in the specified base.
    (i) 29 (ii) 73 (iii) 215 (iv) 732
    (a) 2 (b) 4 (c) 16

44. (a) How are the numbers 2, 4, and 16 related?
    (b) Because of the way 2, 4, and 16 are related, it is possible to change the expansion of a number relative to one of these numbers to the expansion relative to another directly without using the number's base 10 expansion. Examine the results of Exercise 43 (and other examples, if needed) and describe how to change from
        (i) a base 2 expansion to a base 4 expansion.
        (ii) a base 16 expansion to a base 2 expansion.
        (iii) a base 4 expansion to a base 16 expansion.

45. Use Bacon's code as given in Example 15
    (a) to create a dummy message for COME BACK
    (b) to decode WHEN THE MOON COMES OVER THE MOUNTAIN, THE OWLS FLY HIGH.
46. (a) For Bacon's code as given in Example 15, why should the dummy message be five times as long as the true message?
    (b) Modify Bacon's code so that it handles the spaces between words and the digits 0, 1, 2, ..., 9.
47. If ONE and TWO are the base 26 representations of two integers, then what is the base 26 representation of the sum ONE + TWO?
48. Use Bacon's code, as given in Example 15, to decode DO YOU KNOW THAT THE NUMBER PI IS NOW KNOWN TO MORE THAN FOUR HUNDRED MILLION DECIMAL PLACES.

5 Matrices

A matrix is a rectangular array of numbers arranged in m horizontal rows and n vertical columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \tag{1}$$

The ith row of A is $a_{i1}\ a_{i2}\ \cdots\ a_{in}$, $1 \le i \le m$, and the jth column of A is $\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$, $1 \le j \le n$. We say that A is m by n, written $m \times n$. If $m = n$, we say A is a square matrix of order n and that the numbers $a_{11}, a_{22}, \ldots, a_{nn}$ form the main diagonal of A. We refer to the number a_{ij}, which is in the ith row and jth column of A as the i, j th element of A or as the (i, j) entry of A, and we often write (1) as $A = [a_{ij}]$. Note that first the row is named and then the column.

Example 1 Let

$$A = \begin{bmatrix} 2 & 3 & 5 \\ 0 & -1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -1 & 3 & 4 \end{bmatrix}$$

$$D = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \quad \text{and} \quad E = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 2 & 3 \\ 2 & 4 & 5 \end{bmatrix}.$$

Then A is 2×3 with $a_{12} = 3$ and $a_{23} = 2$, B is 2×2 with $b_{21} = 4$, C is 1×4, D is 3×1, and E is 3×3.

A square matrix $A = [a_{ij}]$ for which every entry off the main diagonal is zero, that is, $a_{ij} = 0$ for $i \ne j$, is called a diagonal matrix.

Example 2 Each of the following is a diagonal matrix.

$$F = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}, \quad G = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 5 \end{bmatrix}, \quad \text{and} \quad H = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

Matrices are used in many applications in computer science. We present the following simple application showing how matrices can be used to display data in a tabular form.

Example 3 The following matrix gives the airline distances between the cities indicated.

	London	Madrid	New York	Tokyo
London	0	785	3469	5959
Madrid	785	0	3593	6706
New York	3469	3593	0	6757
Tokyo	5959	6706	6757	0

Two $m \times n$ matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be **equal** if $a_{ij} = b_{ij}$, $1 \le i \le m$, $1 \le j \le n$; that is, if corresponding elements are the same. Notice how easy it is to state the definition using generic elements a_{ij}, b_{ij}.

Example 4 If

$$A = \begin{bmatrix} 2 & -3 & -1 \\ 0 & 5 & 2 \\ 4 & -4 & 6 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & x & -1 \\ y & 5 & 2 \\ 4 & -4 & z \end{bmatrix},$$

then $A = B$ if and only if $x = -3$, $y = 0$, and $z = 6$.

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are $m \times n$ matrices, then the **sum** of A and B is the matrix $C = [c_{ij}]$ defined by $c_{ij} = a_{ij} + b_{ij}$, $1 \le i \le m$, $1 \le j \le n$. That is, C is obtained by adding the corresponding elements of A and B. Once again the use of generic elements makes it easy to state the definition.

Example 5 Let $A = \begin{bmatrix} 3 & 4 & -1 \\ 5 & 0 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 5 & 3 \\ 0 & -3 & 2 \end{bmatrix}$. Then

$$A + B = \begin{bmatrix} 3+4 & 4+5 & -1+3 \\ 5+0 & 0+(-3) & -2+2 \end{bmatrix} = \begin{bmatrix} 7 & 9 & 2 \\ 5 & -3 & 0 \end{bmatrix}.$$

Observe that the sum of the matrices A and B is defined only when A and B have the same number of rows and the same number of columns. We agree to write $A + B$ only when the sum is defined.

A matrix all of whose entries are zero is called a **zero matrix** and is denoted by 0.

Example 6 Each of the following is a zero matrix.

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The following theorem gives some basic properties of matrix addition; the proofs are omitted.

THEOREM 1
(a) $A + B = B + A$
(b) $(A + B) + C = A + (B + C)$
(c) $A + 0 = 0 + A = A$

If $A = [a_{ij}]$ is an $m \times p$ matrix and $B = [b_{ij}]$ is a $p \times n$ matrix, then the **product** of A and B, denoted AB, is the $m \times n$ matrix $C = [c_{ij}]$ defined by

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj} \qquad 1 \le i \le m, \ 1 \le j \le n. \qquad (2)$$

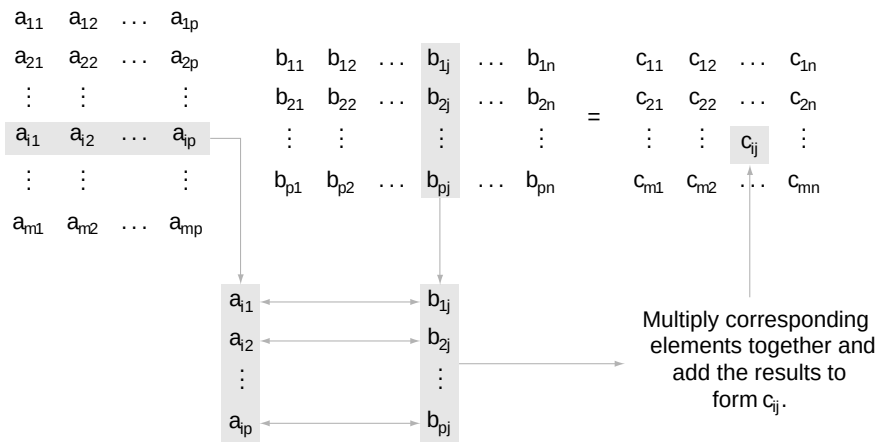


Figure 20

Let us explain (2) in more detail. The elements $a_{i1}, a_{i2}, \ldots, a_{ip}$ form the ith row of A, and the elements $b_{1j}, b_{2j}, \ldots, b_{pj}$ form the jth column of B. Then (2) states that for any i and j, the element c_{ij} of $C = AB$ can be computed in the following way, illustrated in Figure 20.

1. Select row i of A and column j of B, and place them side by side.
2. Multiply corresponding entries and add all the products.

Example 7 Let $A = \begin{bmatrix} 2 & 3 & -4 \\ 1 & 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ -2 & 2 \\ 5 & -3 \end{bmatrix}$. Then

$$AB = \begin{bmatrix} (2)(3) + (3)(-2) + (-4)(5) & (2)(1) + (3)(2) + (-4)(-3) \\ (1)(3) + (2)(-2) + (3)(5) & (1)(1) + (2)(2) + (3)(-3) \end{bmatrix}$$

$$= \begin{bmatrix} -20 & 20 \\ 14 & -4 \end{bmatrix}.$$

An array of dimension two is a modification of the idea of a matrix, in the same way that a linear array is a modification of the idea of a sequence. By an $m \times n$ array A we will mean an $m \times n$ matrix A of mn positions. We may assign numbers to these positions later, make further changes in these assignments, and still refer to the array as A. This is a model for two-dimensional storage of information in a computer. The number assigned to row i and column j of an array A will be denoted $A[i, j]$.

As we have seen, the properties of matrix addition resemble the familiar properties for the addition of real numbers. However, some of the properties of matrix multiplication do not resemble those of real number multiplication. First, observe that if A is an $m \times p$ matrix and B is a $p \times n$ matrix, then AB can be computed and is an $m \times n$ matrix. As for BA, we have the following four possibilities:

1. BA may not be defined; we may have $n \neq m$.
2. BA may be defined and then BA is $p \times p$, while AB is $m \times m$ and $p = m$. Thus AB and BA are not equal.
3. AB and BA may both be the same size, but not be equal as matrices.
4. $AB = BA$.

We agree as before to write AB only when the product is defined.

Example 8 Let $A = \begin{bmatrix} 2 & 1 \\ 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 2 & -3 \end{bmatrix}$. Then $AB = \begin{bmatrix} 4 & -5 \\ -1 & 3 \end{bmatrix}$ and $BA = \begin{bmatrix} -1 & 3 \\ -5 & 8 \end{bmatrix}$.

The basic properties of matrix multiplication are given by the following theorem.

THEOREM 2
(a) $A(BC) = (AB)C$
(b) $A(B + C) = AB + AC$
(c) $(A + B)C = AC + BC$

The $n \times n$ diagonal matrix

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix},$$

all of whose diagonal elements are 1, is called the *identity matrix* of order n. If A is an $m \times n$ matrix, it is easy to verify that $I_m A = A I_n = A$. If A is an $n \times n$ matrix and p is a positive integer, we define

$$A^p = \underbrace{A \cdot A \cdots A}_{p \text{ factors}} \quad \text{and} \quad A^0 = I_n.$$

If p and q are nonnegative integers, we can prove the following laws of exponents for matrices:

$$A^p A^q = A^{p+q} \quad \text{and} \quad (A^p)^q = A^{pq}.$$

Observe that the rule $(AB)^p = A^p B^p$ does not hold for all square matrices. However, if $AB = BA$, then $(AB)^p = A^p B^p$.

If $A = [a_{ij}]$ is an $m \times n$ matrix, then the $n \times m$ matrix $A^T = [a_{ij}^T]$, where $a_{ij}^T = a_{ji}$, $1 \le i \le m$, $1 \le j \le n$, is called the *transpose* of A. Thus the transpose of A is obtained by interchanging the rows and columns of A.

Example 9 Let $A = \begin{bmatrix} 2 & -3 & 5 \\ 6 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 & 5 \\ 2 & -1 & 0 \\ 1 & 6 & -2 \end{bmatrix}$. Then

$$A^T = \begin{bmatrix} 2 & 6 \\ -3 & 1 \\ 5 & 3 \end{bmatrix} \quad \text{and} \quad B^T = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -1 & 6 \\ 5 & 0 & -2 \end{bmatrix}.$$

The following theorem summarizes the basic properties of the transpose operation.

THEOREM 3 If A and B are matrices, then
(a) $(A^T)^T = A$
(b) $(A + B)^T = A^T + B^T$
(c) $(AB)^T = B^T A^T$

A matrix $A = [a_{ij}]$ is called **symmetric** if $A^T = A$. Thus, if A is symmetric, it must be a square matrix. It is easy to show that A is symmetric if and only if $a_{ij} = a_{ji}$. That is, A is symmetric if and only if the entries of A are symmetric with respect to the main diagonal of A.

Example 10 If $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & 5 \\ -3 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & 0 \\ 3 & 2 & 1 \end{bmatrix}$, then A is symmetric and B is not symmetric.

If x is a nonzero number, there is a number y such that $xy = 1$. The number y is called the **multiplicative inverse** of x. When we have a multiplication for objects other than numbers, it is reasonable to ask if a multiplicative inverse exists. One example is matrix multiplication, described in this section.

If A and B are $n \times n$ matrices, then we say that B is an **inverse** of A if $AB = I_n$ and $BA = I_n$, where I_n is the $n \times n$ identity matrix defined earlier. The identity matrix behaves like the number 1, in that $AI_n = A$ for any $n \times n$ matrix A, so the matrix inverse is analogous to the reciprocal of a nonzero number. However, it is not clear how to construct inverses, or even when they exist.

Example 11 An inverse of the matrix $\begin{bmatrix} 1 & 3 & 0 \\ 2 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ is the matrix $\begin{bmatrix} -2 & 3 & -3 \\ 1 & -1 & 1 \\ 2 & -3 & 4 \end{bmatrix}$. This can be verified by checking that

$$\begin{bmatrix} 1 & 3 & 0 \\ 2 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 & 3 & -3 \\ 1 & -1 & 1 \\ 2 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and

$$\begin{bmatrix} -2 & 3 & -3 \\ 1 & -1 & 1 \\ 2 & -3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 \\ 2 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

There are tests to determine if an $n \times n$ matrix has an inverse, and formulas for computing an inverse if it exists. Many of these are programmed into calculators and computer algebra systems. We will be content to give the results for the case of 2×2 matrices.

Suppose that the 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has an inverse $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$. Then

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and we have two pairs of equations

$$\begin{aligned} ae + bg &= 1 \\ ce + dg &= 0 \end{aligned} \quad \text{and} \quad \begin{aligned} af + bh &= 0 \\ cf + dh &= 1. \end{aligned}$$

When we solve the first pair for e and g, we find that we must divide by $ad - bc$. This can only be done if $ad - bc = 0$. The same condition is needed when solving the second pair for f and h. The results of solving for e, f, g, and h are summarized in Theorem 4.

THEOREM 4 A matrix $A = \begin{matrix} a & b \\ c & d \end{matrix}$ has a unique inverse if and only if $ad - bc \neq 0$. In this case we have

$$A^{-1} = \begin{matrix} \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{matrix}.$$

Boolean Matrix Operations

A Boolean matrix (also called a bit matrix) is an $m \times n$ matrix whose entries are either zero or one. We shall now define three operations on Boolean matrices.

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be $m \times n$ Boolean matrices. We define $A \vee B = C = [c_{ij}]$, the join of A and B, by

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0 & \text{if } a_{ij} \text{ and } b_{ij} \text{ are both } 0 \end{cases}$$

and $A \wedge B = D = [d_{ij}]$, the meet of A and B, by

$$d_{ij} = \begin{cases} 1 & \text{if } a_{ij} \text{ and } b_{ij} \text{ are both } 1 \\ 0 & \text{if } a_{ij} = 0 \text{ or } b_{ij} = 0. \end{cases}$$

Note that these operations are only possible when A and B have the same size, just as in the case of matrix addition. Instead of adding corresponding elements in A and B, to compute the entries of the result, we simply examine the corresponding elements for particular patterns.

Example 12 Let $A = \begin{matrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{matrix}$ and $B = \begin{matrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{matrix}$.

(a) Compute $A \vee B$. (b) Compute $A \wedge B$.

Solution

(a) Let $A \vee B = [c_{ij}]$. Then, since a_{43} and b_{43} are both 0, we see that $c_{43} = 0$. In all other cases, either a_{ij} or b_{ij} is 1, so c_{ij} is also 1. Thus

$$A \vee B = \begin{matrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{matrix}.$$

(b) Let $A \wedge B = [d_{ij}]$. Then, since a_{11} and b_{11} are both 1, $d_{11} = 1$, and since a_{23} and b_{23} are both 1, $d_{23} = 1$. In all other cases, either a_{ij} or b_{ij} is 0, so $d_{ij} = 0$. Thus

$$A \wedge B = \begin{matrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{matrix}.$$

Fundamentals

- Size of a matrix: A is $m \times n$ if it has m rows and n columns
- Diagonal matrix: a square matrix with zero entries off the main diagonal
- Equal matrices: matrices of the same size whose corresponding entries are equal
- $A + B$: the matrix obtained by adding corresponding entries of A and B
- Zero matrix: a matrix all of whose entries are zero
- AB
- I_n (identity matrix): a square matrix with ones on the diagonal and zeros elsewhere
- Array of dimension 2
- A^T: the matrix obtained from A by interchanging the rows and columns of A
- Symmetric matrix: $A^T = A$
- Inverse of a matrix
- Boolean matrix: a matrix whose entries are either one or zero
- $A \quad B$
- $A \quad B$
- $A \quad B$
- Properties of Boolean matrix operations
- Mathematical structure: a collection of objects with operations de ned on them and the accompanying properties
- Binary operation: an operation that combines two objects
- Unary operation: an operation that requires only one object
- Closure property: each application of the operation produces another object in the collection
- Associative property: $(x \quad y) \quad z = x \quad (y \quad z)$
- Distributive property: $x \quad (y \quad z) = (x \quad y) \quad (x \quad z)$
- De Morgan's laws: $(x \quad y) = x \quad y$ and $(x \quad y) = x \quad y$
- Identity for : an element e such that $x \quad e = e \quad x = x$ for all x in the structure
- -inverse for x: an element y such that $x \quad y = y \quad x = e$, where e is the identity for

Chapter Self-Test

1. What kind of mathematical object is $P(A)$?
2. What kind of mathematical object is $|P(A)|$?
3. What kind of mathematical object is LCM(a, b)?
4. What kind of mathematical object is A?
5. What are the components of a mathematical structure?
6. Let $A = \{ x \mid x$ is a real number and $0 < x < 1 \}$,
   $B = \{ x \mid x$ is a real number and $x^2 + 1 = 0 \}$,
   $C = \{ x \mid x = 4m, m \quad Z \}$, $D = \{ 0, 2, 4, 6, \ldots \}$, and
   $E = \{ x \mid x \quad Z$ and $x^2 \quad 100 \}$.
   (a) Identify the following as true or false.
   (i) $C \quad D$ (ii) $\{4, 16\} \quad C$
   (iii) $\{4, 16\} \quad E$ (iv) $D \quad D$
   (v) B
   (b) Identify the following as true or false.
   (i) $C \quad E \quad (C \quad E)$
   (ii) $(A \quad B)$ (iii) $C \quad D = D$
   (iv) $C \quad E \quad D$
   (v) $A \quad D \quad A \quad C$
7. Let $A = \{ x \mid x = 2n, n \quad Z^+ \}$,
   $B = \{ x \mid x = 2n + 1, n \quad Z^+ \}$, $C = \{ x \mid x = 4n, n \quad Z^+ \}$,
   and $D = \{ x \mid x(x^2 - 6x + 8) = 0, x \quad Z \}$. Use Z as the universal set and nd
   (a) $A \quad B$ (b) $\overline{A}$
   (c) $(A \quad D) \quad (A \quad B)$ (d) $A \quad C$
   (e) $A \quad C$
8. Draw a Venn diagram to represent (a) $A \quad \overline{B}$ and (b) $\overline{A} \quad \overline{B}$.
9. Under what conditions will $A \quad B = A \quad B$?
10. Suppose that 109 of the 150 mathematics students at Verysmall College take at least one of the following computer languages: PASCAL, BASIC, C++. Suppose 45 study BASIC, 61 study PASCAL, 53 study C++, 18 study BASIC and PASCAL, 15 study BASIC and C++, and 23 study PASCAL and C++.
    (a) How many students study all three languages?
    (b) How many students study only BASIC?
    (c) How many students do not study any of the languages?
11. De ne a sequence as follows: $a_0 = 0, a_1 = 0$, $a_n = 1 - 3a_{n-1} + 2a_{n-2}$. Compute the rst six terms of this sequence.
12. Let $U = \{ a, b, c, d, e, f, g, h, i, j \}$, $A = \{ a, b, d, f \}$, $B = \{ a, b, c, h, j \}$, $C = \{ b, c, f, h, i \}$, and $D = \{ g, h \}$. Represent each of the following sets by an array of zeros and ones.
    (a) $A \quad B$ (b) $A \quad B$
    (c) $A \quad (B \quad C)$ (d) $(\overline{A} \quad B) \quad D$
13. Let $I = \{ a, b, c \}$. In each part that follows is listed a string in I and a regular expression over I. For each, state whether the string belongs to the regular set corresponding to the expression.
    (a) ab a bc (b) acbb ((acb) b)
    (c) bc ((ab) c) (d) abaca (ab) ac
14. Use the Euclidean algorithm to compute GCD(4389, 7293) and write it as $s(7293) + t(4389)$.

15. Let $A = \begin{bmatrix} 2 & 6 & 4 \\ -1 & 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ -3 & 1 \end{bmatrix}$. Compute, if possible, each of the following.
    (a) AB (b) BA (c) B^T
    (d) $A + B$ (e) $A^T B$ (f) B^{-1}
    (g) $B^{-1}A$

16. Let $C = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$.
    Compute each of the following.
    (a) $C \vee D$ (b) $C \wedge D$ (c) $C \oplus D$

17. Let $S = (2 \times 2$ Boolean matrices, $\vee$, $\wedge$, $')$ and A be a 2×2 Boolean matrix. Describe the $\vee$-inverse of A in S.

Experiment 1

In many voting procedures the rules are one person, one vote, and a simple majority is required to elect a candidate or to pass a motion. But it is not unusual to have a procedure where individual voters have more than one vote or where something other than a simple majority is required. An example of such a situation is when not all shareholders in a company own the same number of shares, and each shareholder has as many votes as shares. Does a shareholder with twice as many shares as another have twice as much control, or power, over the company? In this experiment you will investigate this question and some related ones. First, we begin with some definitions. The number of votes that a voter has is called the voter's *weight*. Here only counting numbers can be weights. The total number of votes needed to elect a candidate or to pass a motion is the *quota*. The collection of the quota and the individual weights for all voters is called a *weighted voting system*. If the voters are designated $v_1, v_2, \ldots, v_k$ with corresponding weights $w_1, w_2, \ldots, w_k$ and q is the quota, then the weighted voting system may be conveniently represented by $[q : w_1, w_2, \ldots, w_k]$. For ease of computations, the weights are usually listed from largest to smallest.

1. For the weighted voting system $[9 : 9, 4, 2, 1]$, what is the quota? How many voters are there? What is the total number of votes available?
2. In a weighted voting system $[q : w_1, w_2, \ldots, w_k]$, what are the restrictions on the possible values of q? Explain each restriction.
3. For the weighted voting system $[9 : 9, 4, 2, 1]$, describe how much power voter v_1 has. Such a voter is called a *dictator*. Why is this appropriate? Could a system have two dictators? Explain why or why not.
4. For $[8 : 5, 3, 2, 1]$, is v_1 a dictator? Describe v_1's power relative to the other voters.

More interesting cases arise when the power of each voter is not so obvious as in these first examples. One way to measure a voter's power was developed by John Banzhaf in 1965. A *coalition* is a subset of the voters in a weighted voting system. If the total number of votes controlled by the members of the coalition equals or exceeds the quota, we call the coalition a winning coalition. If not, this is a losing coalition.

5. (a) List all the coalitions for $[9 : 9, 4, 2, 1]$. Which of these are winning coalitions?
   (b) List all the winning coalitions for $[8 : 5, 3, 2, 1]$.

Banzhaf's idea is to measure a voter's power by examining how many times removal of that voter from a coalition would change the coalition from winning to losing. Consider the system $[7 : 5, 4, 3]$. The winning coalitions are $\{v_1, v_2\}$, $\{v_1, v_3\}$, $\{v_2, v_3\}$, and $\{v_1, v_2, v_3\}$. Each member of the first three coalitions has the power to change it from winning to losing, but none have this power in the last

coalition. All together there are six opportunities for change. Each of v_1, v_2, v_3 has two of these opportunities. We record this information as the Banzhaf power distribution for the system: $v_1 : \frac{2}{6}$, $v_2 : \frac{2}{6}$, $v_3 : \frac{2}{6}$. According to this analysis, all three voters have the same amount of power despite having different weights. The fraction of power assigned to a voter is the voter's Banzhaf power index

6. Here is a test for Banzhaf's de nition of power. Calculate the Banzhaf power distribution for $[9 : 9, 4, 2, 1]$. Explain how the results are consistent with the designation of v_1 as a dictator.
7. Calculate the Banzhaf power distribution for $[8 : 5, 3, 2, 1]$. A voter like v_1 that must belong to every winning coalition has veto power in the system.
8. Let $[q : 6, 3, 1]$ be a weighted voting system.
   (a) Give values for q for which the system has a dictator and identify that voter.
   (b) Give values for q for which one or more voters have veto power and identify these voters.
   (c) Give values for q for which at least one player is powerless, but there is no dictator. Which player is powerless?

Banzhaf's idea is adaptable to cases where each voter has one vote, but special rules for voting apply.

9. The four partners in a company agree that each partner has one vote and a simple majority passes a motion. In the case of a tie the coalition containing the senior partner is the winning coalition. Give the Banzhaf power distribution for this system. Would the distribution change if the tie-breaking rule were changed to the coalition containing the most junior member is the losing coalition? Explain.
10. Suppose you are the voter with weight one in $[8 : 5, 3, 2, 1]$.
    (a) What is your Banzhaf power index?
    (b) Unhappy with this situation, you offer to buy a vote from one of the other voters. If each is willing to sell and each asks the same price, from whom should you buy a vote and why? Give the Banzhaf power distribution for this system for the resulting weighted voting system.
11. Here is another feature of Banzhaf's way of measuring power. Let $[q : w_1, w_2, \ldots, w_k]$ be a weighted voting system and n be a positive integer. Prove that the Banzhaf power distributions for $[q : w_1, w_2, \ldots, w_k]$ and $[nq : nw_1, nw_2, \ldots, nw_k]$ are the same.
12. We now return to the original question about power. Suppose we have a weighted voting system in which v_1 has weight w_1, v_2 has weight w_2, and $w_1 = 2w_2$. Construct such a system where the Banzhaf power index of v_1 is
    (a) the same as that of v_2
    (b) twice that of v_2
    (c) more than twice that of v_2.

Coding Exercises

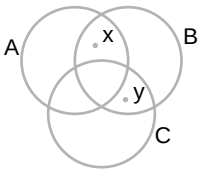
In Exercises 1 through 3, assume that A and B are Þnite sets of integers. Write a subroutine to compute the speciÞed set.

1. $A \cup B$ 2. $A \cap B$ 3. $A \setminus B$

4. Consider the sequence recursively de ned by $g(0) = 1, g(1) = -1, g(n) = 3g(n-1) - 2g(n-2)$.
   (a) Write a subroutine that will print the rst 20 terms of the sequence.
   (b) Write a subroutine that will print the rst n terms of the sequence. The user should be able to supply the value of n at runtime.
5. Write a subroutine to nd the least common multiple of two positive integers.

Answers to Odd-Numbered Exercises

Exercise Set 1

1. (a) True. (b) False. (c) False. (d) False. (e) True. (f) False.
3. (a) {A, R, D, V, K}. (b) {B, O, K}. (c) {M, I, S, P}.
5. (a) False. (b) True. (c) False. (d) True. (e) False. (f) False.
7. {x | x is a vowel}.
9. $\{x \mid x \in Z \text{ and } x^2 < 5\}$.
11. (b), (c), (e).
13. { }, {JAVA}, {PASCAL}, {C++}, {JAVA, PASCAL}, {JAVA, C++}, {PASCAL, C++}, {JAVA, PASCAL, C++}.
15. (a) True. (b) False. (c) False. (d) True. (e) True. (f) True. (g) True. (h) True.
17. (a) $\in$. (b) $\in$. (c) $\in$. (d) $\in$. (e) $\in$. (f) $\in$.
19. {1, 2, 3}
21. Yes, Yes, the complement of a set would not be de ned unambiguously.
23. (a) False. (b) False. (c) Insuf cient information. (d) False. (e) True. (f) True.
25. Eight. There are three parts that represent what is left of each set when common parts are removed, three regions that each represent the part shared by one of the three pairs of sets, a region that represents what all three sets have in common, and a region outside all three sets.
27. B = { m, n}.
29. B = { a, b, c}.
31. 

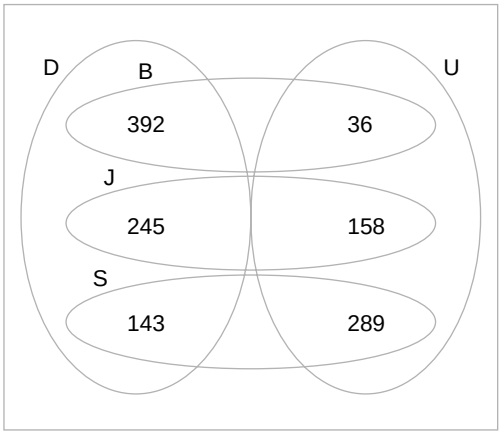
is one solution.
33. $Z^+ \subseteq N \subseteq Z \subseteq Q \subseteq R$.
35. B; B.
37. 4; 8.

Exercise Set 2

1. (a) {a, b, c, d, e, f, g }. (b) {a, c, d, e, f, g }. (c) {a, c}. (d) {f }. (e) {b, g, d, e}. (f) {a, b, c}. (g) {d, e, f, h, k }. (h) {a, b, c, d, e, f }. (i) {b, g, f }. (j) {g}.
3. (a) {a, b, c, d, e, f, g }. (b) { }. (c) {a, c, g}. (d) {a, c, f }. (e) {h, k}. (f) {a, b, c, d, e, f, h, k }.
5. (a) {1, 2, 4, 5, 6, 8, 9}. (b) {1, 2, 3, 4, 6, 8}. (c) {1, 2, 4, 6, 7, 8}. (d) {1, 2, 3, 4, 5, 9}. (e) {1, 2, 4}. (f) {8}. (g) {2, 4}. (h) { }.
7. (a) {1, 2, 3, 4, 5, 6, 8, 9}. (b) {2, 4}. (c) {1, 2, 4}. (d) {8}. (e) {3, 7}. (f) {1, 3, 5, 6, 7, 8, 9}.
9. (a) {b, d, e, h}. (b) {b, c, d, f, g, h }. (c) {b, d, h}. (d) {b, c, d, e, f, g, h }. (e) { }. (f) {c, f, g }.
11. (a) All real numbers except -1 and 1.
    (b) All real numbers except -1 and 4.
    (c) All real numbers except -1, 1, and 4.
    (d) All real numbers except -1.
13. (a) True. (b) True. (c) False. (d) False.
15. 1.
17. (a) $|A \cup B| = 10, |A| = 6, |B| = 7, |A \cap B| = 3$. Hence $|A \cup B| = |A| + |B| - |A \cap B|$.
    (b) $|A \cup B| = 11, |A| = 5, |B| = 6, |A \cap B| = 0$. Hence $|A \cup B| = |A| + |B| - |A \cap B|$.
19. B must be the empty set.
21. The complement of the intersection of two sets is the union of the complements of the two sets.
23. $|A| = 6, |B| = 5, |C| = 6, |A \cap B| = 2, |A \cap C| = 3, |B \cap C| = 3, |A \cap B \cap C| = 2, |A \cup B \cup C| = 11$. Hence $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.
25. (a) 106. (b) 60.

27. 16; 23.

29. (a) 162. (b) 118. (c) 236.
    (d) 290. (e) 264.

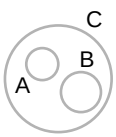
31. 

33. (a) True. (b) True. (c) False. (d) False.

35. (a) Not possible to decide. (b) Not possible to decide.
    (c) Not possible to decide. (d) Not possible to decide.
    (e) Not possible to decide. (f) Not possible to decide.

37. (a) Not possible to decide. (b) Not possible to decide.
    (c) Not possible to decide. (d) Not possible to decide.
    (e) Not possible to decide.

39. A and to B.

41. (a) 

    (b) A B.
    (c) Let x A B. Then x A or x B. Since A C and B C, x A or x B means that x C. Hence A B C.

43. Yes. Suppose x B. Either x A or x / A. If x A, then x / A B = A C. But then x must be in C. If x / A, then x A B = A C, and again x must be in C. So B C. A similar argument shows that C B, so B = C.

45. No. Let A = {1, 2, 3}, B = {4}, and C = {3, 4}. Then A B = A C and B = C.

47. (a) Let x A C. Then x A or x C, so x B or x D and x B D. Hence A C B D.
    (b) Let x A C. Then x A and x C, so x B and x D. Hence x B D. Thus A C B D.

49. We must subtract |B C|, because each element B C has been counted twice in |A| + |B| + |C|. But when we subtract both |B C| and |A C|, we have "uncounted" all the elements of C that also belong to B and A. These (and the similar elements of A and B) are counted again by adding |A B C|.

51. The cardinality of the union of n sets is the sum of the cardinalities of each of the n sets minus the sum of the cardinalities of the $_nC_2$ different intersections of two of the sets plus the sum of the $_nC_3$ different intersections of three of the sets and so on, alternating plus and minus the sum of the $_nC_k$ k-set intersections, k = 4, . . . , n.

Exercise Set 3

1. {1, 2}.

3. {a, b, c, . . . , z}.

5. Possible answers include xyzxyz. . . , xxyyzzxxyyzz, and yzxyzx. . .

7. 5, 25, 125, 625.

9. 1, 2, 6, 24.

11. 2.5, 4, 5.5, 7.

13. 0, Š2, Š4, Š6.

15. $a_n = a_{n\check{S}1} + 2, a_1 = 1$, recursive; $a_n = 2n \check{S} 1$, explicit.

17. $c_n = (\check{S}1)^{n+1}$, explicit.

19. $e_n = e_{n\check{S}1} + 3, e_1 = 1$, recursive.

21. $a_n = 2 + 3(n \check{S} 1)$.

23. A, uncountable; B, finite; C, countable; D, finite; E, finite.

25. (a) Yes. (b) No. (c) Yes.
    (d) Yes. (e) No. (f) No.

27. (a) 1. (b) 0.
    (c) f_B: 10000000; f_C: 01010011; f_D: 01000101.
    (d) 11010011, 01010111, 01000001.

29. $f_{(A\ B)\ C} = f_{A\ B} + f_C \check{S} 2f_{A\ B}f_C$ by Theorem 4
    $= (f_A + f_B \check{S} 2f_Af_B) + f_C \check{S} 2(f_A + f_B \check{S} 2f_Af_B)f_C$
    $= f_A + (f_B + f_C \check{S} 2f_Bf_C) \check{S} 2f_A(f_B + f_C \check{S} 2f_Bf_C)$
    $= f_A + f_{B\ C} \check{S} 2f_Af_{B\ C}$
    $= f_{A\ (B\ C)}$
    Since the characteristic functions are the same, the sets must be the same.

31. (a) Yes. (b) Yes. (c) Yes.

33. Possible answers include ac a ab(.

35. (a) 01 0. (b) 0(00) (00) 1.

37. By (1), 8 is an S-number. By (3), 1 is an S-number. By (2), all multiples of 1, that is, all integers are S-numbers.

39. 1, 2, 3, 7, 16.

Exercise Set 4

1. $20 = 6 \cdot 3 + 2$.

3. $3 = 0 \cdot 22 + 3$.

5. (a) $828 = 2^2 \cdot 3^2 \cdot 23$. (b) $1666 = 2 \cdot 7^2 \cdot 17$.
   (c) $1781 = 13 \cdot 137$. (d) $1125 = 3^2 \cdot 5^3$.
   (e) 107.

7. $d = 3$; $3 = 3 \cdot 45 - 4 \cdot 33$.

9. $d = 1$; $1 = 5 \cdot 77 - 3 \cdot 128$.

11. 1050. 13. 864.

15. (a) 6. (b) 1. (c) 0.
   (d) 1. (e) 20. (f) 14.

17. (a) 10. (b) 22. (c) 2. (d) 14.

19. $f(a) + f(b)$ may be greater than m.

21. (a) $\{2, 7, 12, 17, \ldots\}$. (b) $\{1, 6, 11, 16, \ldots\}$.

23. If $a \mid b$, then $b = ka$, for some $k \in Z$. Thus, $mb = m(ka) = (mk)a$ and mb is a multiple of a.

25. The only divisors of p are $\pm p$ and ± 1, but p does not divide a. (Multiply both sides by b.) $p \mid sab$ and $p \mid tpb$. If p divides the right side of the equation, then it must divide the left side also.

27. Because $a \mid m$, $ac \mid mc$ and because $c \mid m$, $ac \mid am$. If $GCD(a, c) = 1$, there are integers s, t such that $1 = sa + tc$. Thus $m = sam + tcm$. But ac divides each term on the left so $ac \mid m$.

29. Let $d = GCD(a, b)$. Then $cd \mid ca$ and $cd \mid cb$; that is, cd is a common divisor of ca and cb. Let $e = GCD(ca, cd)$. Then $cd \mid e$ and $e = cdk$ is a divisor of ca and cb. But then $dk \mid a$ and $dk \mid b$. Because d is the greatest common divisor of a and b, k must be 1 and $e = cd$.

31. By Theorem 6, $GCD(a, b) \cdot LCM(a, b) = ab$. Since $GCD(a, b) = 1$, we have $LCM(a, b) = ab$.

33. $a \mid b$ means $b = am$. $b \mid a$ means $a = bn$. Thus $b = am = bnm$. Hence $nm = 1$ and $n = m = 1$, because a and b are positive.

35. No; consider $a = 6, b = 4, c = 3$.

37. Yes. Using the same reasoning as in Exercise 36, m and n share no prime factors, so for mn to be a perfect tth power, each prime in the factorizations of m and n must appear a multiple of t times. But this means each of m and n are also perfect tth powers.

39. (a) 112. (b) 10. (c) 30.

41. (a) (i) $(104)_5$. (ii) $(243)_5$. (iii) $(1330)_5$. (iv) $(10412)_5$.
   (b) (i) 49. (ii) 85. (iii) 197. (iv) 816.

43. (a) $(11101)_2$. $(1001001)_2$. $(11010111)_2$. $(1011011100)_2$.
   (b) $(131)_4$. $(1021)_4$. $(3113)_4$. $(23130)_4$.
   (c) $(1D)_{16}$. $(49)_{16}$. $(D7)_{16}$. $(2DC)_{16}$.

45. (a) Answers will vary, but the pattern of italicized and non-italicized letters should match
   AAA AAA *AAA* AAA *AAAAAAA* AAA *AAAAAA* AA *AAAAAAAAA*.
   (b) STUDY WELL.

47. BIJS.

Exercise Set 5

1. (a) $(-2, 1, 2)$. (b) $(3, 4)$. (c) $(4, -1, 8)$.
   (d) $(2, 6, 8)$.

3. a is 3, b is 1, c is 8, and d is -2.

5. (a) $\begin{bmatrix} 4 & 0 & 2 \\ 9 & 6 & 2 \\ 3 & 2 & 4 \end{bmatrix}$.

   (b) $AB = \begin{bmatrix} 7 & 13 \\ -3 & 0 \end{bmatrix}$.

   (c) Not possible. (d) $\begin{bmatrix} 21 & 14 \\ -7 & 17 \end{bmatrix}$.

7. (a) EB is 3×2 and FA is 2×3; the sum is undefined.
   (b) $B + D$ does not exist.
   (c) $\begin{bmatrix} 10 & 0 & -25 \\ 40 & 14 & 12 \end{bmatrix}$. (d) DE does not exist.

9. (a) $\begin{bmatrix} 22 & 34 \\ 3 & 11 \\ -31 & 3 \end{bmatrix}$. (b) BC is not defined.

   (c) $\begin{bmatrix} 25 & 5 & 26 \\ 20 & -3 & 32 \end{bmatrix}$.

   (d) $D^T + E$ is not defined.

11. Let $B = [b_{jk}] = I_m A$. Then $b_{jk} = \sum_{l=1}^{m} i_{jl} a_{lk}$, for $1 \le j \le m$ and $1 \le k \le n$. But $i_{jj} = 1$ and $i_{jl} = 0$ if $j = l$. Hence $b_{jk} = i_{jj} a_{jk}$, $1 \le j \le m$, $1 \le k \le n$. This means $B = I_m A = A$. Similarly, if $C = AI_n = [c_{jk}]$, $c_{jk} = \sum_{l=1}^{n} a_{jl} i_{lk} = a_{jk} i_{kk} = a_{jk}$ for $1 \le j \le m$, $1 \le k \le n$.

13. $A^3 = \begin{bmatrix} 27 & 0 & 0 \\ 0 & -8 & 0 \\ 0 & 0 & 64 \end{bmatrix}$ or $\begin{bmatrix} 3^3 & 0 & 0 \\ 0 & (-2)^3 & 0 \\ 0 & 0 & 4^3 \end{bmatrix}$, $A^k = \begin{bmatrix} 3^k & 0 & 0 \\ 0 & (-2)^k & 0 \\ 0 & 0 & 4^k \end{bmatrix}$.

15. The entries of I_n^T satisfy $i_{kj} = i_{jk}$. But $i_{jk} = 1$ if $j = k$ and is 0 otherwise. Thus $i_{kj} = 1$ if $k = j$ and is 0 if $k \ne j$ for $1 \le j \le n$, $1 \le k \le n$.

17. The jth column of AB has entries $c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$. Let $D = [d_{ij}] = AB_j$, where B_j is the jth column of B. Then $d_{ij} = \sum_{m=1}^{n} a_{im} b_{mj} = c_{ij}$.

19. $A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$.

21. (a) $(A + B)^T = A^T + B^T$ by Theorem 3. Since A and B are symmetric $A^T + B^T = A + B$ and $A + B$ is also symmetric.

(b) $(AB)^T = B^TA^T = BA$, but this may not be AB, so AB may not be symmetric. Let $A = \begin{matrix} 1 & 2 \\ 2 & 4 \end{matrix}$ and $B = \begin{matrix} 3 & -1 \\ -1 & 2 \end{matrix}$. Then AB is not symmetric.

23. (a) The i, j th element of $(A^T)^T$ is the j, i th element of A^T. But the j, i th element of A^T is the i, j th element of A. Thus $(A^T)^T = A$.

(b) The i, j th element of $(A + B)^T$ is the j, i th element of $A + B$, $a_{ji} + b_{ji}$. But this is the sum of the j, ith entry of A and the j, ith entry of B. It is also the sum of the i, j th entry of A^T and the i, j th entry of B^T. Thus $(A + B)^T = A^T + B^T$.

(c) Let $C = c_{ij} = (AB)^T$. Then $c_{ij} = \sum_{k=1}^{n} a_{jk} b_{ki}$, the j, ith entry of AB. Let $D = d_{ij} = B^TA^T$, then

$$d_{ij} = \sum_{k=1}^{n} b_{ik} a_{kj} = \sum_{k=1}^{n} b_{ki} a_{jk} = \sum_{k=1}^{n} a_{jk} b_{ki} = c_{ij}.$$

Hence $(AB)^T = B^TA^T$.

25. (a) $\begin{matrix} \frac{7}{9} & -\frac{1}{9} \\ -\frac{5}{9} & \frac{2}{9} \end{matrix}$. (b) $\begin{matrix} -\frac{9}{27} & \frac{4}{27} \\ \frac{0}{27} & \frac{3}{27} \end{matrix}$. (c) $\begin{matrix} \frac{2}{32} & \frac{5}{32} \\ \frac{4}{32} & -\frac{6}{32} \end{matrix}$.

27. (a) $C = A^{-1}$ since $CA = I_3$.

(b) $D = B^{-1}$ since $DB = I_3$.

29. Since $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n$, $(B^{-1}A^{-1}) = (AB)^{-1}$.

31. (a) $A \quad B = \begin{matrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{matrix}$; $A \quad B = \begin{matrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{matrix}$;

$A \quad B = \begin{matrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{matrix}$.

(b) $A \quad B = \begin{matrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{matrix}$; $A \quad B = \begin{matrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{matrix}$;

$A \quad B = \begin{matrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{matrix}$.

(c) $A \quad B = \begin{matrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{matrix}$; $A \quad B = \begin{matrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{matrix}$;

$A \quad B = \begin{matrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{matrix}$.

33. (a) f_{p1}. (b) f_{11}. (c) f_{pq}.

(d) f_{1q}. (e) f_{ir}. (f) f_{ii} if $p = q$.

35. (a) $p_{i1}m_{1j} + p_{i2}m_{2j} + \cdots + p_{it}m_{tj}$.

(b) $m_{i1}p_{1j} + m_{i2}p_{2j} + \cdots + m_{it}p_{tj}$.

(c) $p_{i1}(m_{1j} + n_{1j}) + p_{i2}(m_{2j} + n_{2j}) + \cdots + p_{it}(m_{tj} + n_{tj})$.

(d) $(m_{i1}n_{1j} + m_{i2}n_{2j} + \cdots + m_{it}n_{tj}) + (p_{i1}n_{1j} + p_{i2}n_{2j} + \cdots + p_{it}n_{tj})$.

37. (a) $\begin{cases} 1 & \text{if } m_{ik} = p_{kj} = 1 \text{ for some } k \\ 0 & \text{otherwise.} \end{cases}$

(b) $\begin{cases} 1 & \text{if } p_{ik} = m_{kj} = 1 \text{ for some } k \\ 0 & \text{otherwise.} \end{cases}$

(c) $\begin{cases} 1 & \text{if } m_{ik} = n_{kl} = p_{lj} = 1 \text{ for some } k, l \\ 0 & \text{otherwise.} \end{cases}$

39. Let $C = c_{ij} = A \quad B$ and $D = d_{ij} = B \quad A$.

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0 & \text{if } a_{ij} = 0 = b_{ij} \end{cases} = d_{ij}$$

Hence $C = D$.

41. Let $d_{ij} = B \quad C$, $e_{ij} = A \quad (B \quad C)$, $f_{ij} = A \quad B$, and $g_{ij} = (A \quad B) \quad C$. Then

$$d_{ij} = \begin{cases} 1 & \text{if } b_{ij} = 1 \text{ or } c_{ij} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$e_{ij} = \begin{cases} 1 & \text{if } a_{ij} = 1 \text{ or } d_{ij} = 1 \\ 0 & \text{otherwise.} \end{cases}$$

But this means $e_{ij} = 1$ if $a_{ij} = 1$ or $b_{ij} = 1$ or $c_{ij} = 1$ and $d_{ij} = 0$ otherwise.

$$f_{ij} = \begin{cases} 1 & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$g_{ij} = \begin{cases} 1 & \text{if } f_{ij} = 1 \text{ or } c_{ij} = 1 \\ 0 & \text{otherwise.} \end{cases}$$

But this means $g_{ij} = 1$ if $a_{ij} = 1$ or $b_{ij} = 1$ or $c_{ij} = 1$ and $g_{ij} = 0$ otherwise. Hence $A \quad (B \quad C) = (A \quad B) \quad C$.

43. Let $d_{ij} = B \quad C$, $e_{ij} = A \quad (B \quad C)$, $f_{ij} = A \quad B$, and $g_{ij} = (A \quad B) \quad C$. Then

$$d_{ij} = \begin{cases} 1 & \text{if } b_{ik} = c_{kj} \text{ for some } k \\ 0 & \text{otherwise} \end{cases}$$

and

$$e_{ij} = \begin{cases} 1 & \text{if } a_{il} = 1 = d_{lj} \text{ for some } l \\ 0 & \text{otherwise.} \end{cases}$$

But this means

$$e_{ij} = \begin{cases} 1 & \text{if } a_{il} = 1 = b_{lk} = c_{kj} \text{ for some } k, l \\ 0 & \text{otherwise.} \end{cases}$$

$$f_{ij} = \begin{cases} 1 & \text{if } a_{ik} = 1 = b_{kj} \text{ for some } k \\ 0 & \text{otherwise} \end{cases}$$

and

$$g_{ij} = \begin{cases} 1 & \text{if } f_{il} = 1 = c_{lj} \text{ for some } l \\ 0 & \text{otherwise.} \end{cases}$$

But then

$$g_{ij} = \begin{cases} 1 & \text{if } a_{ik} = 1 = b_{kl} = c_{ij} \text{ for some } k, l \\ 0 & \text{otherwise} \end{cases}$$

and $A \cap (B \cap C) = (A \cap B) \cap C$.

45. An argument similar to that in Exercise 44 shows that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

47. Since $c_{ij} = \sum_{t=1}^{n} a_{it} b_{tj}$ and $k \mid a_{it}$ for any i and t, k divides each term in c_{ij}, and thus $k \mid c_{ij}$ for all i and j.

49. (a) $\begin{bmatrix} 6 & -9 & -3 \\ 0 & 15 & 6 \\ 12 & -12 & 18 \end{bmatrix}$.

(b) $\begin{bmatrix} 10 & 20 & -30 \\ 20 & 0 & 45 \\ 35 & -5 & 15 \end{bmatrix}$.

(c) $\begin{bmatrix} -4 & 0 \\ -3 & -1 \\ 2 & -5 \end{bmatrix}$.

51. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two $m \times n$ matrices. Then $k(A + B) = k[a_{ij} + b_{ij}] = k(a_{ij} + b_{ij}) = ka_{ij} + kb_{ij} = ka_{ij} + kb_{ij} = kA + kB$.

53. Let K be the $m \times m$ diagonal matrix with each diagonal entry equal to k. Then $KA = kA$.

Exercise Set 6

1. (a) Yes. (b) Yes.

3. (a) No. (b) Yes.

5. $A \cup B = \{x \mid (x \in A \cup B) \text{ and } (x \notin A \cap B)\} = \{x \mid (x \in B \cup A) \text{ and } (x \notin B \cap A)\} = B \cup A$.

7.

x	y	z	$y \cup z$	$x \cap (y \cup z)$	$x \cap y$	$x \cap z$	$(x \cap y) \cup (x \cap z)$
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	0	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	0	0	1	1	0
				(A)			(B)

Since columns (A) and (B) are identical, the distributive property $x \cap (y \cup z) = (x \cap y) \cup (x \cap z)$ holds.

9. 5×5 zero matrix for $\cap$; 5×5 matrix of 1's for $\cup$; I_5 for $\cdot$.

11. Let A, B be $n \times n$ diagonal matrices. Let $c_{ij} = AB$. Then $c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$, but $a_{ik} = 0$ if $i = k$. Hence $c_{ij} = a_{ii} b_{ij}$. But $b_{ij} = 0$ if $i = j$. Thus $c_{ij} = 0$ if $i = j$ and AB is an $n \times n$ diagonal matrix.

13. Yes, the $n \times n$ zero matrix, which is a diagonal matrix.

15. $-A$ is the diagonal matrix with ith entry $-a_{ii}$.

17. $\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix}$ or $\begin{bmatrix} a+b & 0 \\ 0 & 0 \end{bmatrix}$ belongs to M.

19. $\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}^T$ or $\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$ belongs to M.

21. Yes, $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

23. If $a = 0$, then $A^{-1} = \begin{bmatrix} \frac{1}{a} & 0 \\ 0 & 0 \end{bmatrix}$.

25. Yes. 27. No.

29. (a) Yes. (b) Yes. (c) Yes.
(d) Yes, $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$. (e) Yes, $\begin{bmatrix} -x \\ -y-2 \end{bmatrix}$.

31. (a) Yes. (b) Yes. (c) No.

33. (a) $k \begin{bmatrix} x & w \\ y & z \end{bmatrix} = k \begin{bmatrix} x \\ y \end{bmatrix} \cup k \begin{bmatrix} w \\ z \end{bmatrix}$.
(b) No.

35. Let $C = [c_{ij}] = \text{comp}(A \cup B)$ and $D = [d_{ij}] = \text{comp}(A) \cap \text{comp}(B)$. Then

$$c_{ij} = \begin{cases} 0 & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 1 & \text{if } a_{ij} = 0 = b_{ij} \end{cases}$$

and

$$d_{ij} = \begin{cases} 0 & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 1 & \text{if } a_{ij} = 0 = b_{ij}. \end{cases}$$

Hence $C = D$. Similarly, we can show that $\text{comp}(A \cap B) = \text{comp}(A) \cup \text{comp}(B)$.

37. $A \cap \overline{B}$. 39. $\{\}$.

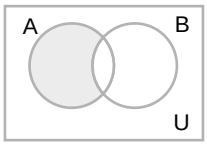
Answers to Chapter Self-Tests

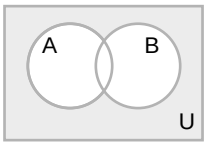
1. $P(A)$ is a set of sets.
2. $|P(A)|$ is a counting number or infinity.
3. LCM(a, b) is a positive integer.
4. kA is a matrix of the same size as A.
5. A mathematical structure consists of a set of objects, operations on those objects, and the properties of those operations.

6. (a) (i) False. (ii) True. (iii) False. (iv) True. (v) True.
(b) (i) False. (ii) True. (iii) False. (iv) False. (v) True.

7. (a) $\{1, 2, 3, 4, \ldots, \}$. (b) $\{\ldots, -3, -2, -1, 0\}$ B.
(c) $\{2, 4\}$. (d) A.

(e) $\{2, 6, 10, 14, \ldots\}$.

8. (a) 

(b) 

9. $A \cup B$ is always a subset of $A \cap B$. $A \cap B = A \cup B$ if and only if $A = B$.

10. (a) 6. (b) 18. (c) 41.

11. $0, 0, 1, -2, 9, -30$.

12. (a)

1	1	1	1	0	1	0	1	0	1

(b)

1	1	0	0	0	0	0	0	0	0

(c)

1	1	0	0	0	1	0	0	0	0

(d)

0	0	1	0	0	0	1	1	0	1

13. (a) Yes. (b) Yes. (c) No. (d) No.

14. $33 = 65(7293) - 108(4389)$.

15. (a) AB does not exist.

(b) $BA = \begin{matrix} 4 & 12 & 8 \\ -7 & -15 & -10 \end{matrix}$.

(c) $\begin{matrix} 2 & -3 \\ 0 & 1 \end{matrix}$.

(d) $A + B$ does not exist.

(e) $\begin{matrix} 7 & -1 \\ 3 & 3 \\ 2 & 2 \end{matrix}$.

(f) $\begin{matrix} \frac{1}{2} & 0 \\ \frac{3}{2} & 1 \end{matrix}$.

(g) $\begin{matrix} 1 & 3 & 2 \\ 2 & 12 & 8 \end{matrix}$.

16. (a) $\begin{matrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{matrix}$.

(b) $\begin{matrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{matrix}$.

(c) $\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{matrix}$.

17. A has a $\{1\}$-inverse if and only if $A = \begin{matrix} 1 & 1 \\ 1 & 1 \end{matrix}$, the $\{1\}$-identity.

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Logic

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Logic

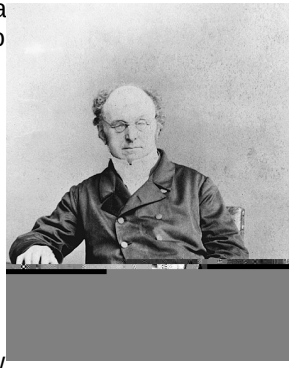
Logic is the discipline that deals with the methods of reasoning. On an elementary level, logic provides rules and techniques for determining whether a given argument is valid. Logical reasoning is used in mathematics to prove theorems, in computer science to verify the correctness of programs and to prove theorems, in the natural and physical sciences to draw conclusions from experiments, in the social sciences, and in our everyday lives to solve a multitude of problems. Indeed, we are constantly using logical reasoning. In this chapter we discuss a few of the basic ideas.

Looking Back

In the 1840s Augustus De Morgan, a British mathematician, set out to extend the logic developed by the early Greeks and others and to correct some of the weaknesses in these ideas. De Morgan (1806–1871) was born in India but was educated in England. He taught at London University for many years and was the first to use the word "induction" for a method of proof that had been used in a rather informal manner and put it on a firm rigorous foundation. In 1847, a few years after De Morgan's work on an extended system of logic had appeared, his countryman George Boole published the book entitled *The Mathematical Analysis of Logic* and followed it up a few years later by the book *An Investigation of the Laws of Thought*. Boole's objective in these books was

> to investigate the fundamental laws of those operations of the mind by which reasoning is performed; to give expression to them in the symbolical language of a Calculus; and upon this foundation to establish the science of Logic and construct its method.

Boole's work in this area firmly established the point of view that logic should use symbols and that algebraic properties should be studied in logic. George Boole (1815–1864) taught at Queen's College in Ireland for many years. Thus, De Morgan started and Boole completed the task of folding a large part of the study of logic into mathematics. We shall briefly study the work of De Morgan and Boole in logic in this chapter.



Augustus De Morgan
CORBIS BETTMANN

George Boole
Mary Evans Picture Library Ltd.

Quoted in Victor J. Katz, *A History of Mathematics, An Introduction*, New York: HarperCollins, 1993, p. 619.

1 Propositions and Logical Operations

A statement or proposition is a declarative sentence that is either true or false, but not both.

Example 1 Which of the following are statements?

(a) The earth is round.
(b) $2 + 3 = 5$
(c) Do you speak English?
(d) $3 - x = 5$
(e) Take two aspirins.
(f) The temperature on the surface of the planet Venus is 800°F.
(g) The sun will come out tomorrow.

Solution

(a) and (b) are statements that happen to be true.
(c) is a question, so it is not a statement.
(d) is a declarative sentence, but not a statement, since it is true or false depending on the value of x.
(e) is not a statement; it is a command.
(f) is a declarative sentence whose truth or falsity we do not know at this time; however, we can in principle determine if it is true or false, so it is a statement.
(g) is a statement since it is either true or false, but not both, although we would have to wait until tomorrow to find out if it is true or false.

Logical Connectives and Compound Statements

In mathematics, the letters $x, y, z, \ldots$ often denote variables that can be replaced by real numbers, and these variables can be combined with the familiar operations $+, \times, -$, and $\div$. In logic, the letters $p, q, r, \ldots$ denote propositional variables; that is, variables that can be replaced by statements. Thus we can write p: The sun is shining today. q: It is cold. Statements or propositional variables can be combined by logical connectives to obtain compound statements. For example, we may combine the preceding statements by the connective *and* to form the compound statement p and q: The sun is shining today, and it is cold. The truth value of a compound statement depends only on the truth values of the statements being combined and on the types of connectives being used. We shall look at the most important connectives.

If p is a statement, the *negation* of p is the statement "not p", denoted by $\neg p$. Thus $\neg p$ is the statement "it is not the case that p." From this definition, it follows that if p is true, then $\neg p$ is false, and if p is false, then $\neg p$ is true. The truth value of $\neg p$ relative to p is given in Table 1. Such a table, giving the truth values of a compound statement in terms of its component parts, is called a *truth table*. Strictly speaking, *not* is not a connective, since it does not join two statements, and $\neg p$ is not really a compound statement. However, *not* is a unary operation for the collection of statements and $\neg p$ is a statement if p is.

TABLE 1

p	$\neg p$
T	F
F	T

Example 2 Give the negation of the following statements:

(a) p: $2 + 3 > 1$ (b) q: It is cold.

Solution

(a) p: $2 + 3$ is not greater than 1. That is, p: $2 + 3 \not> 1$. Since p is true in this case, $\sim p$ is false.

(b) q: It is not the case that it is cold. More simply, q: It is not cold.

TABLE 2

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

If p and q are statements, the **conjunction** of p and q is the compound statement "p and q," denoted by $p \wedge q$. The connective **and** is denoted by the symbol $\wedge$. The compound statement $p \wedge q$ is true when both p and q are true; otherwise, it is false. The truth values of $p \wedge q$ in terms of the truth values of p and q are given in the truth table shown in Table 2. Observe that in giving the truth table of $p \wedge q$ we need to look at four possible cases. This follows from the fact that each of p and q can be true or false.

Example 3 Form the conjunction of p and q for each of the following.

(a) p: It is snowing. q: I am cold.

(b) p: $2 < 3$ q: $\sim 5 > \sim 8$

(c) p: It is snowing. q: $3 < 5$

Solution

(a) $p \wedge q$: It is snowing and I am cold.

(b) $p \wedge q$: $2 < 3$ and $\sim 5 > \sim 8$

(c) $p \wedge q$: It is snowing and $3 < 5$.

Example 3(c) shows that in logic, unlike in everyday English, we may join two totally unrelated statements by the connective **and**.

If p and q are statements, the **disjunction** of p and q is the compound statement "p or q," denoted by $p \vee q$. The connective **or** is denoted by the symbol $\vee$. The compound statement $p \vee q$ is true if at least one of p or q is true; it is false when both p and q are false. The truth values of $p \vee q$ are given in the truth table shown in Table 3.

Example 4 Form the disjunction of p and q for each of the following.

(a) p: 2 is a positive integer q: $\sqrt{2}$ is a rational number.

(b) p: $2 + 3 = 5$ q: London is the capital of France.

TABLE 3

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Solution

(a) $p \vee q$: 2 is a positive integer or $\sqrt{2}$ is a rational number. Since q is true, the disjunction $p \vee q$ is true, even though p is false.

(b) $p \vee q$: $2 + 3 = 5$ or London is the capital of France. Since both p and q are false $p \vee q$ is false.

Example 4(b) shows that in logic, unlike in ordinary English, we may join two totally unrelated statements by the connective **or**.

The connective **or** is more complicated than the connective **and** because it is used in two different ways in English. Suppose that we say "I left for Spain on Monday or I left for Spain on Friday." In this compound statement we have the disjunction of the statements p: I left for Spain on Monday and q: I left for Spain on Friday. Of course, exactly one of the two possibilities could have occurred. Both could not have occurred, so the connective **or** is being used in an **exclusive** sense. On the other hand, consider the disjunction "I passed mathematics or I failed French." In this case, at least one of the two possibilities occurred. However, both

could have occurred, so the connective *or* is being used in an *inclusive* sense. In mathematics and computer science we agree to use the connective *or* always in the inclusive manner.

In general, a compound statement may have many component parts, each of which is itself a statement, represented by some propositional variable. The statement s: $p \wedge (q \vee (p \wedge r))$ involves three propositions p, q, and r, each of which may independently be true or false. There are altogether 2^3 or 8 possible combinations of truth values for p, q, and r, and a truth table for s must give the truth or falsity of s in all these cases. If a compound statement s contains n component statements, there will need to be 2^n rows in the truth table for s. Such a truth table may be systematically constructed in the following way.

Step 1 The first n columns of the table are labeled by the component propositional variables. Further columns are included for all intermediate combinations of the variables, culminating in a column for the full statement.

Step 2 Under each of the first n headings, we list the 2^n possible n-tuples of truth values for the n component statements.

Step 3 For each of the remaining columns, we compute, in sequence, the remaining truth values.

Example 5 Make a truth table for the statement $(p \vee q) \rightarrow (\sim p)$.

Solution

Because two propositions are involved, the truth table will have 2^2 or 4 rows. In the first two columns we list all possible pairs of truth values for p and q. The numbers below the remaining columns show the order in which the columns were filled.

p	q	$p \vee q$		$\sim p$
T	T	T	T	F
T	F	F	F	F
F	T	F	T	T
F	F	F	T	T
		(1)	(3)	(2)

Quantifiers

Sets are defined by specifying a property $P(x)$ that elements of the set have in common. Thus, an element of $\{x \mid P(x)\}$ is an object x for which the statement $P(x)$ is true. Such a sentence $P(x)$ is called a *predicate*, because in English the property is grammatically a predicate. $P(x)$ is also called a *propositional function*, because each choice of x produces a proposition $P(x)$ that is either true or false. Another use of predicates is in programming. Two common constructions are "if $P(x)$, then execute certain steps" and "while $Q(x)$, do specified actions." The predicates $P(x)$ and $Q(x)$ are called the *guards* for the block of programming code. Often the guard for a block is a conjunction or disjunction.

Example 6 Let $A = \{x \mid x \text{ is an integer less than } 8\}$. Here $P(x)$ is the sentence "x is an integer less than 8." The common property is "is an integer less than 8," and $P(1)$ is the statement "1 is an integer less than 8." Since $P(1)$ is true, $1 \in A$.

Example 7 (a) Consider the following program fragment,

```
1.  IF  N < 10  THEN
      a.  Replace  N  with  N + 1
      b.  RETURN
```

Here the statement $N < 10$ is the guard.

(b) Consider the following program fragment,

```
1.  WHILE  t      T  and  s      S
      a.  PRINT  t  +  s
      b.  RETURN
```

Here the compound statement t T and s S is the guard.

The universal quantification of a predicate P(x) is the statement "For all values of x, P(x) is true." We assume here that only values of x that make sense in P(x) are considered. The universal quantification of P(x) is denoted x P(x). The symbol is called the universal quantifier.

Example 8

(a) The sentence P(x): Š(Šx) = x is a predicate that makes sense for real numbers x. The universal quantification of P(x), x P(x), is a true statement, because for all real numbers Š(Šx) = x.

(b) Let Q(x): x + 1 < 4. Then x Q(x) is a false statement, because Q(5) is not true.

Universal quantification can also be stated in English as "for every x," "every x," or "for any x."

A predicate may contain several variables. Universal quantification may be applied to each of the variables. For example, a commutative property can be expressed as x y x y = y x. The order in which the universal quantifiers are considered does not change the truth value. Often mathematical statements contain implied universal quantifications.

In some situations we only require that there be at least one value for which the predicate is true. The existential quantification of a predicate P(x) is the statement "There exists a value of x for which P(x) is true." The existential quantification of P(x) is denoted x P(x). The symbol is called the existential quantifier.

Example 9

(a) Let Q(x): x + 1 < 4. The existential quantification of Q(x), x Q(x), is a true statement, because Q(2) is a true statement.

(b) The statement y y + 2 = y is false. There is no value of y for which the propositional function y + 2 = y produces a true statement.

In English x can also be read "there is a x," "there is some x," "there exists an x," or "there is at least one x." Occasionally a variation of a universal or existential quantifier is useful. The possible values of the variable may be restricted as in the following examples. The statement x A P(x) represents "for all x in the set A, P(x) is true," and the statement k n Q(k) represents "there is a k greater than or equal to n such that Q(k) is true."

Example 10

(a) Let D be the set of n × n diagonal matrices. Consider the statement M D such that $M^{Š1}$ does not exist. To determine if this statement is true or false, we must decide if there is an n × n diagonal matrix that does not have an inverse.

(b) For the statement n 6, $2^n > 2n$, only integers greater than or equal to 6 need to be tested.

Existential quantification may be applied to several variables in a predicate and the order in which the quantifications are considered does not affect the truth value. For a predicate with several variables we may apply both universal and existential quantification. In this case the order does matter.

Example 11 Let A and B be $n \times n$ matrices.

(a) The statement $\exists B \; A + B = I_n$ is read "for every A there is a B such that $A + B = I_n$." For a given $A = [a_{ij}]$, define $B = [b_{ij}]$ as follows: $b_{ii} = 1 - a_{ii}$, $1 \le i \le n$ and $b_{ij} = -a_{ij}$, $i \ne j$, $1 \le i, j \le n$. Then $A + B = I_n$ and we have shown that $\forall A \; \exists B \; A + B = I_n$ is a true statement.

(b) $\exists B \; \forall A \; A + B = I_n$ is the statement "there is a B such that for all A, $A + B = I_n$." This statement is false; no single B has this property for all A's.

(c) $\exists B \; \forall A \; A + B = A$ is true. What is the value for B that makes the statement true?

Let $p: \forall x P(x)$. The negation of p is false when p is true, and true when p is false. For p to be false there must be at least one value of x for which $P(x)$ is false. Thus, $\neg p$ is false if $\exists x \neg P(x)$ is true. On the other hand, if $\exists x \neg P(x)$ is false, then for every x, $\neg P(x)$ is false; that is, $\forall x P(x)$ is true.

Example 12 (a) Let p: For all positive integers n, $n^2 + 41n + 41$ is a prime number. Then $\neg p$ is There is at least one positive integer n for which $n^2 + 41n + 41$ is not prime.

(b) Let q: There is some integer k for which $12 = 3k$. Then $\neg q$: For all integers k, $12 \ne 3k$.

Example 13 Let p: The empty set is a subset of any set A. For p to be false, there must be an element of $\emptyset$ that is not in A, but this is impossible. Thus p is true.

1 Exercises

1. Which of the following are statements?
   (a) Is 2 a positive number?
   (b) $x^2 + x + 1 = 0$
   (c) Study logic.
   (d) There will be snow in January.
   (e) If stock prices fall, then I will lose money.
2. Give the negation of each of the following statements.
   (a) $2 + 7 \quad 11$
   (b) 2 is an even integer and 8 is an odd integer.
3. Give the negation of each of the following statements.
   (a) It will rain tomorrow or it will snow tomorrow.
   (b) If you drive, then I will walk.
4. In each of the following, form the conjunction and the disjunction of p and q.
   (a) p: $3 + 1 < 5$ q: $7 = 3 \times 6$
   (b) p: I am rich. q: I am happy.
5. In each of the following, form the conjunction and the disjunction of p and q.
   (a) p: I will drive my car. q: I will be late.
   (b) p: NUM > 10 q: NUM 15
6. Determine the truth or falsity of each of the following statements.
   (a) $2 < 3$ and 3 is a positive integer.
   (b) $2 \quad 3$ and 3 is a positive integer.
   (c) $2 < 3$ and 3 is not a positive integer.
   (d) $2 \quad 3$ and 3 is not a positive integer.
7. Determine the truth or falsity of each of the following statements.
   (a) $2 < 3$ or 3 is a positive integer.
   (b) $2 \quad 3$ or 3 is a positive integer.
   (c) $2 < 3$ or 3 is not a positive integer.
   (d) $2 \quad 3$ or 3 is not a positive integer.

In Exercises 8 and 9, find the truth value of each proposition if p and r are true and q is false.

8. (a) $p \quad q$ (b) $(p \quad q) \quad r$
   (c) $p \quad q \quad r$ (d) $(p \quad q) \quad r$
9. (a) $p \quad (q \quad r)$ (b) $p \quad (\quad (q \quad r))$
   (c) $(r \quad q) \quad (p \quad r)$ (d) $(q \quad r) \quad (p \quad r)$
10. Which of the following statements is the negation of the statement "2 is even and -3 is negative"?
    (a) 2 is even and -3 is not negative.
    (b) 2 is odd and -3 is not negative.
    (c) 2 is even or -3 is not negative.
    (d) 2 is odd or -3 is not negative.

11. Which of the following statements is the negation of the statement "2 is even or 3 is negative"?
    (a) 2 is even or 3 is not negative.
    (b) 2 is odd or 3 is not negative.
    (c) 2 is even and 3 is not negative.
    (d) 2 is odd and 3 is not negative.

In Exercises 12 and 13, use p: Today is Monday; q: The grass is wet; and r: The dish ran away with the spoon.

12. Write each of the following in terms of p, q, r, and logical connectives.
    (a) Today is Monday and the dish did not run away with the spoon.
    (b) Either the grass is wet or today is Monday.
    (c) Today is not Monday and the grass is dry.
    (d) The dish ran away with the spoon, but the grass is wet.
13. Write an English sentence that corresponds to each of the following.
    (a) $r \quad q$ (b) $q \quad r$
    (c) $(p \quad q)$ (d) $p \quad r$

In Exercises 14 through 19, use $P(x)$: x is even; $Q(x)$: x is a prime number; $R(x, y)$: $x + y$ is even. The variables x and y represent integers.

14. Write an English sentence corresponding to each of the following.
    (a) $x \; P(x)$ (b) $x \; Q(x)$
15. Write an English sentence corresponding to each of the following.
    (a) $x \; y \; R(x, y)$ (b) $x \; y \; R(x, y)$
16. Write an English sentence corresponding to each of the following.
    (a) $x \; (\quad Q(x))$ (b) $y \; (\quad P(y))$
17. Write an English sentence corresponding to each of the following.
    (a) $(\quad x \; P(x))$ (b) $(\quad x \; Q(x))$
18. Write each of the following in terms of $P(x)$, $Q(x)$, $R(x, y)$, logical connectives, and quanti ers.
    (a) Every integer is an odd integer.
    (b) The sum of any two integers is an even number.
    (c) There are no even prime numbers.
    (d) Every integer is even or a prime.
19. Determine the truth value of each statement given in Exercises 14 through 18.
20. If $P(x)$: $x^2 < 12$, then
    (a) $P(4)$ is the statement ____.
    (b) $P(1.5)$ is the statement ____.
21. If $Q(n)$: $n + 3 = 6$, then
    (a) $Q(5)$ is the statement ____.
    (b) $Q(m)$ is the statement ____.
22. If $P(y)$: $1 + 2 + \cdots + y = 0$, then
    (a) $P(1)$ is the statement ____.
    (b) $P(5)$ is the statement ____.
    (c) $P(k)$ is the statement ____.
23. If $Q(m)$: $m \quad 3^m$, then
    (a) $Q(0)$ is the statement ____.
    (b) $Q(2)$ is the statement ____.
    (c) $Q(k)$ is the statement ____.
24. Give a symbolic statement of the commutative property for addition of real numbers using appropriate quanti ers.
25. Give a symbolic statement of De Morgan's laws for sets using appropriate quanti ers.
26. Give a symbolic statement of the multiplicative inverse property for real numbers using appropriate quanti ers.

In Exercises 27 through 30, make a truth table for the statement.

27. $(\quad p \quad q) \quad p$ 28. $(p \quad q) \quad q$
29. $(p \quad q) \quad r$ 30. $(\quad p \quad q) \quad r$

For Exercises 31 through 33, define $p \quad q$ to be a true statement if neither p nor q is true.

p	q	$p \quad q$
T	T	F
T	F	F
F	T	F
F	F	T

31. Make a truth table for $(p \quad q) \quad r$.
32. Make a truth table for $(p \quad q) \quad (p \quad r)$.
33. Make a truth table for $(p \quad q) \quad (p \quad r)$.

For Exercises 34 through 36, define $p \quad q$ to be true if either p or q, but not both, is true. Make a truth table for the statement.

34. (a) $p \quad q$ (b) $p \quad p$
35. $(p \quad q) \quad p$
36. $(p \quad q) \quad (q \quad r)$

In Exercises 37 through 40, revision of the given programming block is needed. Replace the guard $P(x)$ with $P(x)$.

37. IF ($x =$ max and $y > 4$) THEN take action
38. WHILE (key = "open" or $t <$ limit) take action
39. WHILE (item = sought and index $<$ 101) take action
40. IF (cell > 0 or found) THEN take action

2 Conditional Statements

If p and q are statements, the compound statement "if p then q," denoted $p \to q$, is called a **conditional statement**, or **implication**. The statement p is called the **antecedent** or **hypothesis**, and the statement q is called the **consequent** or **conclusion**. The connective *if . . . then* is denoted by the symbol $\to$.

Example 1 Form the implication $p \to q$ for each of the following.

(a) p: I am hungry. q: I will eat.
(b) p: It is snowing. q: $3 + 5 = 8$.

Solution

(a) If I am hungry, then I will eat.
(b) If it is snowing, then $3 + 5 = 8$.

Example 1(b) shows that in logic we use conditional statements in a more general sense than is customary. Thus in English, when we say "if p, then q," we are tacitly assuming there is a cause-and-effect relationship between p and q. That is, we would never use the statement in Example 1(b) in ordinary English, since there is no way statement p can have any effect on statement q.

In logic, implication is used in a much weaker sense. To say the compound statement $p \to q$ is true simply asserts that if p is true, then q will also be found to be true. In other words, $p \to q$ says only that we will not have p true and q false at the same time. It does not say that "p caused" q in the usual sense. Table 4 describes the truth values of $p \to q$ in terms of the truth of p and q. Notice that $p \to q$ is considered false only if p is true and q is false. In particular, if p is false, then $p \to q$ is true for any q. This fact is sometimes described by the statement "A false hypothesis implies any conclusion." This statement is misleading, since it seems to say that if the hypothesis is false, the conclusion must be true, an obviously silly statement. Similarly, if q is true, then $p \to q$ will be true for any statement p. The implication "If $2 + 2 = 5$, then I am the king of England" is true, simply because p: $2 + 2 = 5$ is false, so it is not the case that p is true and q is false simultaneously.

TABLE 4

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

In the English language, and in mathematics, each of the following expressions is an equivalent form of the conditional statement q: p implies q; q, if p; p only if q; p is a sufficient condition for q; q is a necessary condition for p.

If $p \to q$ is an implication, then the **converse** of $p \to q$ is the implication $q \to p$, and the **contrapositive** of $p \to q$ is the implication $\sim q \to \sim p$.

Example 2 Give the converse and the contrapositive of the implication "If it is raining, then I get wet."

Solution

We have p: It is raining; and q: I get wet. The converse is $q \to p$: If I get wet, then it is raining. The contrapositive is $\sim q \to \sim p$: If I do not get wet, then it is not raining.

If p and q are statements, the compound statement p if and only if q, denoted by $p \leftrightarrow q$, is called an **equivalence** or **biconditional**. The connective *if and only if* is denoted by the symbol $\leftrightarrow$. The truth values of $p \leftrightarrow q$ are given in Table 5. Observe that $p \leftrightarrow q$ is true only when both p and q are true or when both p and q are false. The equivalence $p \leftrightarrow q$ can also be stated as "p is a necessary and sufficient condition for q."

TABLE 5

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example 3 Is the following equivalence a true statement? $2 < 3$ if and only if $0 < 3 < 2$.

Solution

Let p be the statement $2 < 3$ and let q be the statement $0 < 3 < 2$. Since both p and q are true, we conclude that $p \leftrightarrow q$ is true.

Example 4 Compute the truth table of the statement $(p \rightarrow q) \wedge (q \rightarrow p)$.

Solution

The following table is constructed using steps 1, 2, and 3 as given in Section 1. The numbers below the columns show the order in which they were constructed.

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T
		(1)	(2)	(3)

A statement that is true for all possible values of its propositional variables is called a **tautology**. A statement that is always false is called a **contradiction** or an **absurdity**, and a statement that can be either true or false, depending on the truth values of its propositional variables, is called a **contingency**.

Example 5 (a) The statement in Example 4 is a tautology.

(b) The statement $p \wedge \neg p$ is an absurdity. (Verify this.)

(c) The statement $(p \rightarrow q) \rightarrow (p \wedge q)$ is a contingency.

We have now defined a new mathematical structure with two binary operations and one unary operation, (propositions, $\wedge$, $\vee$, $\neg$). It makes no sense to say two propositions are equal; instead we say p and q are **logically equivalent**, or simply **equivalent**, if $p \leftrightarrow q$ is a tautology. When an equivalence is shown to be a tautology, this means its two component parts are always either both true or both false, for any values of the propositional variables. Thus the two sides are simply different ways of making the same statement and can be regarded as "equal." We denote that p is equivalent to q by $p \equiv q$. Now we can adapt our properties for operations to say this structure has a property if using equivalent in place of equal gives a true statement.

Example 6 The binary operation $\wedge$ has the commutative property; that is, $q \wedge p \equiv p \wedge q$. The truth table for $(p \wedge q) \leftrightarrow (q \wedge p)$ shows the statement is a tautology.

p	q	$p \wedge q$	$q \wedge p$	$(p \wedge q) \leftrightarrow (q \wedge p)$
T	T	T	T	T
T	F	F	F	T
F	T	F	F	T
F	F	F	F	T

Another way to use a truth table to determine if two statements are equivalent is to construct a column for each statement and compare these to see if they are identical. In Example 6 the third and fourth columns are identical, and this will guarantee that the statements they represent are equivalent.

Forming $p \quad q$ from p and q is another binary operation for statements, but we can express it in terms of the operations in Section 1.

Example 7 The conditional statement $p \quad q$ is equivalent to $(\quad p) \quad q$. Columns 1 and 3 in the following table show that for any truth values of p and q, $p \quad q$ and $(\quad p) \quad q$ have the same truth values.

p	q	$p \quad q$	$\quad p$	$(\quad p) \quad q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T
		(1)	(2)	(3)

The structure (propositions, , ,) has many of the same properties as the structure (sets, , , $^{-}$).

THEOREM 1 The operations for propositions have the following properties.

Commutative Properties

1. $p \quad q \quad q \quad p$
2. $p \quad q \quad q \quad p$

Associative Properties

3. $p \quad (q \quad r) \quad (p \quad q) \quad r$
4. $p \quad (q \quad r) \quad (p \quad q) \quad r$

Distributive Properties

5. $p \quad (q \quad r) \quad (p \quad q) \quad (p \quad r)$
6. $p \quad (q \quad r) \quad (p \quad q) \quad (p \quad r)$

Idempotent Properties

7. $p \quad p \quad p$
8. $p \quad p \quad p$

Properties of Negation

9. $(\quad p) \quad p$
10. $(p \quad q) \quad (\quad p) \quad (\quad q)$ Properties 10 and 11
11. $(p \quad q) \quad (\quad p) \quad (\quad q)$ are De Morgan's laws.

Proof

We have proved Property 1 in Example 6. The remaining properties may be proved the same way and are left for the reader as exercises.

Truth tables can be used to prove statements about propositions, because in a truth table all possible cases are examined.

The implication operation also has a number of important properties.

THEOREM 2

(a) $(p \quad q) \quad ((\quad p) \quad q)$
(b) $(p \quad q) \quad (\quad q \quad p)$
(c) $(p \quad q) \quad ((p \quad q) \quad (q \quad p))$
(d) $(p \quad q) \quad (p \quad q)$
(e) $(p \quad q) \quad ((p \quad q) \quad (q \quad p))$

20. Fill the grid so that each row, column, and marked 3×3 block contains 1, 2, 3, 4, 5, 6, 7, 8, 9, with no repeats.

		4						7
	9			1		5		
6		1	7		4	8	9	
	4		1		7	9		
9			8	2	3			4
			4		5		2	
	8	2			6	1		5
		9		7				
5			2			6		

In Exercises 21 and 22, find the truth value of each statement if p and q are true and r, s, and t are false.

21. (a) $(p \vee q)$ (b) $(\sim p) \wedge r$
    (c) $(p \wedge s) \vee (s \wedge t)$ (d) $t \vee q$
22. (a) $(\sim q) \vee (r \wedge (r \vee (p \wedge s)))$
    (b) $p \vee (r \wedge q)$
    (c) $(q \vee (r \wedge s)) \wedge ((p \vee s) \wedge (\sim t))$
    (d) $(r \wedge s \wedge t) \vee (p \wedge q)$
23. Use the definition of $p \veebar q$ given for Exercise 31 in Section 1 and show that $((p \veebar p) \veebar (q \veebar q))$ is equivalent to $p \wedge q$.
24. Write the negation of each of the following in good English.
    (a) The weather is bad and I will not go to work.
    (b) If Carol is not sick, then if she goes to the picnic, she will have a good time.
    (c) I will not win the game or I will not enter the contest.
25. Write the negation of each of the following in good English.
    (a) Jack did not eat fat, but he did eat broccoli.
    (b) Mary lost her lamb or the wolf ate the lamb.
    (c) If Tom stole a pie and ran away, then the three pigs do not have any supper.
26. Consider the following conditional statement:

    p: If the flood destroys my house or the fire destroys my house, then my insurance company will pay me.

    (a) Which of the following is the converse of p?
    (b) Which of the following is the contrapositive of p?
        (i) If my insurance company pays me, then the flood destroys my house or the fire destroys my house.
        (ii) If my insurance company pays me, then the flood destroys my house and the fire destroys my house.
        (iii) If my insurance company does not pay me, then the flood does not destroy my house or the fire does not destroy my house.
        (iv) If my insurance company does not pay me, then the flood does not destroy my house and the fire does not destroy my house.
27. Prove Theorem 1 part 6.
28. Prove Theorem 1 part 11.
29. Prove Theorem 2 part (e).
30. Prove Theorem 3 part (d).
31. Prove Theorem 3 part (e).
32. Prove Theorem 4 part (a).
33. Prove Theorem 4 part (d).
34. Prove Theorem 4 part (g).
35. Prove Theorem 4 part (j).
36. Explain why proving part (e) of Theorem 4 provides a one-line proof of part (f) of Theorem 4.
37. Explain why proving part (a) of Theorem 4 provides a one-line proof of part (b) of Theorem 4.

3 Methods of Proof

Some methods of proof we have already used are direct proofs using generic elements, definitions, and previously proven facts, and proofs by cases, such as examining all possible truth value situations in a truth table. Here we look at proofs in more detail.

If an implication $p \rightarrow q$ is a tautology, where p and q may be compound statements involving any number of propositional variables, we say that q logically follows from p. Suppose that an implication of the form $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \rightarrow q$ is a tautology. Then this implication is true regardless of the truth values of any of its components. In this case, we say that q logically follows from $p_1, p_2, \ldots, p_n$.

When q logically follows from $p_1, p_2, \ldots, p_n$, we write

$$\begin{array}{l} p_1 \\ p_2 \\ \vdots \\ \underline{p_n} \\ q \end{array}$$

where the symbol $\therefore$ means therefore. This means if we know that p_1 is true, p_2 is true, $\ldots$, and p_n is true, then we know q is true.

Virtually all mathematical theorems are composed of implications of the type

$$(p_1 \quad p_2 \quad \cdots \quad p_n) \quad q.$$

The p_i's are called the *hypotheses* or *premises*, and q is called the *conclusion*. To "prove the theorem" means to show that the implication is a tautology. Note that we are not trying to show that q (the conclusion) is true, but only that q will be true if all the p_i are true. For this reason, mathematical proofs often begin with the statement "suppose that $p_1, p_2, \ldots,$ and p_n are true" and conclude with the statement "therefore, q is true." The proof does not show that q is true, but simply shows if the p_i are all true, then q has to be true.

Arguments based on tautologies represent universally correct methods of reasoning. Their validity depends only on the form of the statements involved and not on the truth values of the variables they contain. Such arguments are called *rules of inference*. The various steps in a mathematical proof of a theorem must follow from the use of various rules of inference, and a mathematical proof of a theorem must begin with the hypotheses, proceed through various steps, each justified by some rule of inference, and arrive at the conclusion.

Example 1 According to Theorem 4(j) of the last section, $((p \quad q) \quad (q \quad r)) \quad (p \quad r)$ is a tautology. Thus the argument

$$\begin{array}{l} p \quad q \\ \underline{q \quad r} \\ p \quad r \end{array}$$

is universally valid, and so is a rule of inference.

Example 2 Is the following argument valid?

If you invest in the stock market, then you will get rich.
If you get rich, then you will be happy.
If you invest in the stock market, then you will be happy.

Solution

The argument is of the form given in Example 1, hence the argument is valid, although the conclusion may be false.

Example 3 The tautology $(p \quad q) \quad ((p \quad q) \quad (q \quad p))$ is Theorem 2(c), Section 2. Thus both of the following arguments are valid.

$$\begin{array}{l} p \quad q \\ \underline{\qquad\qquad} \\ (p \quad q) \quad (q \quad p) \end{array} \qquad\qquad \begin{array}{l} p \quad q \\ \underline{q \quad p} \\ p \quad q \end{array}$$

Some mathematical theorems are equivalences; that is, they are of the form $p \leftrightarrow q$. They are usually stated "p if and only if q." By Example 3, the proof of such a theorem is logically equivalent with proving both $p \rightarrow q$ and $q \rightarrow p$, and this is almost always the way in which equivalences are proved. We wish to show that both $p \rightarrow q$ and $q \rightarrow p$ are true. To do so, we first assume that p is true, and show that q must then be true; next we assume that q is true and show that p must then be true.

A very important rule of inference is

$$\begin{array}{l} p \\ \underline{p \rightarrow q} \\ q. \end{array}$$

That is, "p is true, and $p \rightarrow q$ is true, so q is true." This follows from Theorem 4(g), Section 2.

Some rules of inference were given Latin names by classical scholars. Theorem 4(g) is referred to as *modus ponens*, or loosely, the method of asserting.

Example 4 Is the following argument valid?

Smoking is healthy.
If smoking is healthy, then cigarettes are prescribed by physicians.
Cigarettes are prescribed by physicians.

Solution

The argument is valid since it is of the form modus ponens. However, the conclusion is false. Observe that the first premise "smoking is healthy" is false. The second premise $p \rightarrow q$ is then true and $(p \wedge (p \rightarrow q))$, the conjunction of the two premises, is false.

Example 5 Is the following argument valid?

If taxes are lowered, then income rises.
Income rises.
Taxes are lowered.

Solution

Let p: taxes are lowered and q: income rises. Then the argument is of the form

$$\begin{array}{l} p \rightarrow q \\ \underline{q} \\ p. \end{array}$$

Assume that $p \rightarrow q$ and q are both true. Now $p \rightarrow q$ may be true with p being false. Then the conclusion p is false. Hence the argument is not valid. Another approach to answering this question is to verify whether the statement $((p \rightarrow q) \wedge q) \rightarrow p$ logically implies the statement p. A truth table shows this is not the case. (Verify this.)

An important proof technique, which is an example of an *indirect* method of proof, follows from the tautology $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$. This states that an implication is equivalent to its contrapositive. Thus to prove q indirectly, we assume q is false (the statement $\neg q$) and show that p is then false (the statement $\neg p$).

Example 6 Let n be an integer. Prove that if n^2 is odd, then n is odd.

Solution

Let p: n^2 is odd and q: n is odd. We have to prove that $p \rightarrow q$ is true. Instead, we prove the contrapositive $\neg q \rightarrow \neg p$. Thus suppose that n is not odd, so that n is even. Then $n = 2k$, where k is an integer. We have $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, so n^2 is even. We thus show that if n is even, then n^2 is even, which is the contrapositive of the given statement. Hence the given statement has been proved.

Another important indirect proof technique is proof by contradiction. This method is based on the tautology $((p \rightarrow q) \wedge (\neg q)) \rightarrow (\neg p)$. Thus the rule of inference

$$\begin{array}{l} p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

is valid. Informally, this states that if a statement p implies a false statement q, then p must be false. This is often applied to the case where q is an absurdity or contradiction, that is, a statement that is always false. An example is given by taking q as the contradiction $r \wedge (\neg r)$. Thus any statement that implies a contradiction must be false. In order to use proof by contradiction, suppose we wish to show that a statement q logically follows from statements $p_1, p_2, \ldots, p_n$. Assume that q is true (that is, $\neg q$ is false) as an extra hypothesis, and that $p_1, p_2, \ldots, p_n$ are also true. If this enlarged hypothesis $p_1 \wedge p_2 \wedge \cdots \wedge p_n \wedge (\neg q)$ implies a contradiction, then at least one of the statements $p_1, p_2, \ldots, p_n, \neg q$ must be false. This means that if all the p_i's are true, then $\neg q$ must be false, so q must be true. Thus q follows from $p_1, p_2, \ldots, p_n$. This is proof by contradiction.

Example 7 Prove there is no rational number p/q whose square is 2. In other words, show $\sqrt{2}$ is irrational.

Solution

This statement is a good candidate for proof by contradiction, because we could not check all possible rational numbers to demonstrate that none had a square equal to 2. Assume $(p/q)^2 = 2$ for some integers p and q, which have no common factors. If the original choice of p/q is not in lowest terms, we can replace it with its equivalent lowest-term form. Then $p^2 = 2q^2$, so p^2 is even. This implies p is even, since the square of an odd number is odd. Thus $p = 2n$ for some integer n. We see that $2q^2 = p^2 = (2n)^2 = 4n^2$, so $q^2 = 2n^2$. Thus q^2 is even, and so q is even. We now have that both p and q are even, and therefore have a common factor 2. This is a contradiction to the assumption. Thus the assumption must be false.

We have presented several rules of inference and logical equivalences that correspond to valid proof techniques. In order to prove a theorem of the (typical) form $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \rightarrow q$, we begin with the hypothesis $p_1, p_2, \ldots, p_n$ and show that some result r_1 logically follows. Then, using $p_1, p_2, \ldots, p_n, r_1$, we show that some other statement r_2 logically follows. We continue this process, producing intermediate statements $r_1, r_2, \ldots, r_k$, called *steps* in the proof, until we can finally show that the conclusion q logically follows from $p_1, p_2, \ldots, p_n, r_1, r_2, \ldots, r_k$. Each logical step must be justified by some valid proof technique, based on the rules of inference we have developed, or on some other rules that come from tautological implications we have not discussed. At any stage, we can replace a statement that needs to be derived by its contrapositive statement, or any other equivalent form.

In practice, the construction of proofs is an art and must be learned in part from observation and experience. The choice of intermediate steps and methods of deriving them is a creative activity that cannot be precisely described. But a few simple techniques are applicable to a wide variety of settings. We will focus on these techniques. The "Tips for Proofs" notes at the end of this chapter highlight the methods most useful for the chapter's material.

Example 8 Let m and n be integers. Prove that $n^2 = m^2$ if and only if n is m or n is $-m$.

Solution

Let us analyze the proof as we present it. Suppose p is the statement $n^2 = m^2$, q is the statement n is m, and r is the statement n is $-m$. Then we wish to prove the theorem $p \leftrightarrow (q \vee r)$. We know from previous discussion that we may instead prove $s: p \rightarrow (q \vee r)$ and $t: (q \vee r) \rightarrow p$ are true. First, we assume that either $q: n$ is m or $r: n$ is $-m$ is true. If q is true, then $n^2 = m^2$, and if r is true, then $n^2 = (-m)^2 = m^2$, so in either case p is true. We have therefore shown that the implication $t: (q \vee r) \rightarrow p$ is true.

Now we must prove that $s: p \rightarrow (q \vee r)$ is true; that is, we assume p and try to prove either q or r. If p is true, then $n^2 = m^2$, so $n^2 - m^2 = 0$. But $n^2 - m^2 = (n - m)(n + m)$. If r_1 is the intermediate statement $(n - m)(n + m) = 0$, we have shown $p \rightarrow r_1$ is true. We now show that $r_1 \rightarrow (q \vee r)$ is true, by showing that the contrapositive $(q \vee r) \rightarrow (\sim r_1)$ is true. Now $(q \vee r)$ is equivalent to $(\sim q) \rightarrow (\sim r)$, so we show that $(\sim q) \rightarrow (\sim r) \rightarrow (\sim r_1)$. Thus, if $(\sim q): n$ is not m and $(\sim r): n$ is not $-m$ are true, then $(n - m) \neq 0$ and $(n + m) \neq 0$, so $(n - m)(n + m) \neq 0$ and r_1 is false. We have therefore shown that $r_1 \rightarrow (q \vee r)$ is true. Finally, from the truth of $p \rightarrow r_1$ and $r_1 \rightarrow (q \vee r)$, we can conclude that $p \rightarrow (q \vee r)$ is true, and we are done.

We do not usually analyze proofs in this detailed manner. We have done so only to illustrate that proofs are devised by piecing together equivalences and valid steps resulting from rules of inference. The amount of detail given in a proof depends on who the reader is likely to be.

As a final remark, we remind the reader that many mathematical theorems actually mean that the statement is true for all objects of a certain type. Sometimes this is not evident. Thus the theorem in Example 8 really states that for all integers m and n, $n^2 = m^2$ if and only if n is m or n is $-m$. Similarly, the statement "If x and y are real numbers, and $x = y$, then $x < y$ or $y < x$" is a statement about all real numbers x and y. To prove such a theorem, we must make sure that the steps in the proof are valid for every real number. We could not assume, for example, that x is 2, or that y is -3 or $\sqrt{3}$. This is why proofs often begin by selecting a generic element, denoted by a variable. On the other hand, we know from Section 2 that the negation of a statement of the form $\forall x P(x)$ is $\exists x \sim P(x)$, so we need only find a single example where the statement is false to disprove it.

Example 9 Prove or disprove the statement that if x and y are real numbers, $(x^2 = y^2) \rightarrow (x = y)$.

Solution

The statement can be restated in the form $\forall x \forall y R(x, y)$. Thus, to prove this result, we would need to provide steps, each of which would be true for all x and y. To disprove the result, we need only find one example for which the implication is false.

Since $(-3)^2 = 3^2$, but $-3 \neq 3$, the result is false. Our example is called a counterexample, and any other counterexample would do just as well.

In summary, if a statement claims that a property holds for all objects of a certain type, then to prove it, we must use steps that are valid for all objects of that type and that do not make references to any particular object. To disprove such a statement, we need only show one counterexample, that is, one particular object or set of objects for which the claim fails.

3 Exercises

In Exercises 1 through 11, state whether the argument given is valid or not. If it is valid, identify the tautology or tautologies on which it is based.

1. If I drive to work, then I will arrive tired.
   I am not tired when I arrive at work.

   I do not drive to work.

2. If I drive to work, then I will arrive tired.
   I arrive at work tired.

   I drive to work.

3. If I drive to work, then I will arrive tired.
   I do not drive to work.

   I will not arrive tired.

4. If I drive to work, then I will arrive tired.
   I drive to work.

   I will arrive tired.

5. I will become famous or I will not become a writer.
   I will become a writer.

   I will become famous.

6. I will become famous or I will be a writer.
   I will not be a writer.

   I will become famous.

7. If I try hard and I have talent, then I will become a musician.
   If I become a musician, then I will be happy.

   If I will not be happy, then I did not try hard or I do not have talent.

8. If I graduate this semester, then I will have passed the physics course.
   If I do not study physics for 10 hours a week, then I will not pass physics.
   If I study physics for 10 hours a week, then I cannot play volleyball.

   If I play volleyball, I will not graduate this semester.

9. If my plumbing plans do not meet the construction code, then I cannot build my house.
   If I hire a licensed contractor, then my plumbing plans will meet the construction code.
   I hire a licensed contractor.

   I can build my house.

10. (a) $p \quad q$ / q / $\therefore p$ (b) $p \quad q$ / p / $\therefore q$

11. Write each argument in Exercise 10 as a single compound statement.

12. (a) $(p \quad q) \quad (q \quad r)$ / $(\quad q) \quad r$ / $\therefore p$ (b) $(p \quad q)$ / p / $\therefore q$

13. Write each argument in Exercise 12 as a single compound statement.

14. Prove that the sum of two even numbers is even.

15. Prove that the sum of two odd numbers is even.

16. Prove that the structure (even integers,) is closed with respect to .

17. Prove that the structure (odd integers,) is closed with respect to .

18. Prove that n^2 is even if and only if n is even.

19. Prove that $A = B$ if and only if $A \quad B$ and $B \quad A$.

20. Let A and B be subsets of a universal set U. Prove that $A \quad B$ if and only if $\overline{B} \quad \overline{A}$.

21. Show that
    (a) $A \quad B$ is a necessary and sufficient condition for $A \quad B = B$.
    (b) $A \quad B$ is a necessary and sufficient condition for $A \quad B = A$.

22. Show that k is odd is a necessary and sufficient condition for k^3 to be odd.

23. Prove or disprove $n^2 + 41n + 41$ is a prime number for every integer n.

24. Prove or disprove: The sum of any five consecutive integers is divisible by 5.

25. Prove or disprove that $3 \mid (n^3 - n)$ for every positive integer n.

26. Prove or disprove: $1 + 2^n > 3^n$, for all $n \quad Z^+$.

27. Determine if the following is a valid argument. Explain your conclusion.

    Prove: $x\ x^3 > x^2$.
    Proof: $x\ x^2 > 0$ so $x\ x^2(x - 1) > 0(x - 1)$ and $x\ x^3 - x^2 > 0$. Hence $x\ x^3 > x^2$.

28. Determine if the following is a valid argument. Explain your conclusion.

5 Exercises

1. Prove or disprove that the cube of an even number is even.
2. Prove or disprove that the cube of an odd number is odd.
3. (a) Prove or disprove that the sum of three consecutive integers is divisible by 3.
   (b) Prove or disprove that the sum of four consecutive integers is divisible by 4.
4. Prove that the product of a nonzero rational number and an irrational number is a(n) ____ number.
5. Prove that the quotient of a nonzero rational number and an irrational number is a(n) ____ number.
6. State and prove the extension of $\overline{A \cap B} = \overline{A} \cup \overline{B}$ for three sets.
7. State and prove the extension of $\overline{A \cup B} = \overline{A} \cap \overline{B}$ for three sets.
8. Let A, B, C, and D be $n \times n$ Boolean matrices. State and prove the extension of $A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C)$ to four matrices.
9. Modify the conjecture and proof in Example 4 to give the general case for $n + 1$ Boolean matrices.
10. Let p and q be propositions. State and prove an extension of $(p \vee q) \wedge p \wedge q$ to the case of
    (a) 3 propositions (b) n propositions.
11. Let p and q be propositions. State and prove an extension of $(p \wedge q) \vee p \vee q$ to the case of
    (a) 3 propositions (b) n propositions.
12. Prove that the commutative property holds for the mathematical structure (3 × 3 diagonal matrices, matrix multiplication).
13. Prove that the commutative property holds for the mathematical structure ($r \times r$ diagonal matrices, matrix multiplication).
14. State and prove a conjecture about the sum of the n first positive odd integers.

For Exercises 15 through 17, use the sequence 3, 9, 15, 21, 27, 33, ...

15. Give both a recursive and an explicit formula for this sequence.
16. Experiment with the sums of the terms of the sequence to produce a reasonable conjecture.
17. Prove the conjecture from Exercise 16.
18. State and prove a reasonable conjecture about the sum of the first n terms of the sequence 6, 10, 14, 18, 22, . . .

For Exercises 19 through 22, use the recursively defined sequence $g_1 = 1, g_2 = 3, g_n = g_{n-1} + g_{n-2}$.

19. Write the first ten terms of the sequence.
20. Experiment with the sums of terms in the sequence that occupy even-numbered positions to produce a reasonable conjecture.
21. Experiment with the sums of terms of the sequence to produce a reasonable conjecture about the sum of the n first terms of the sequence.
22. Experiment with the sums of terms in the sequence that occupy odd-numbered positions to produce a reasonable conjecture.
23. Prove the conjecture from Exercise 20.
24. Prove the conjecture from Exercise 21.
25. Prove the conjecture from Exercise 22.
26. Produce and prove a conjecture about the sum of the powers of 3 beginning with 3^0.
27. Build on the work done in Exercise 3 to produce a conjecture about when the sum of k consecutive integers is divisible by k.
28. Prove the conjecture of Exercise 27.

6 Logic and Problem Solving

In previous sections, we investigated the use of logic to prove mathematical theorems and to verify the correctness of computational algorithms. However, logic is also valuable in less formal settings. Logic is used every day to decide between alternatives and investigate consequences. It is, in many situations, essentially the same as precise and careful thinking, particularly in cases where the details of the situation are complex. One of the most important uses of logic is to develop correct and efficient algorithms to solve problems. Part of this process may be to express the solution in terms of a computation, even if the problem does not at first seem computational. In Section 5, we noted that one common way to create mathematical conjectures is to apply old ideas in new settings. This approach is also a powerful problem-solving technique. To demonstrate these two ideas, we present a variety of seemingly unrelated problems that can be solved by a single method.

We begin with an easily stated mathematical question. Given a set A and a finite number of subsets of A, say $A_1, A_2, \ldots, A_k$, is it possible to find a collection S of the A_i's with the following two properties:

1. Any two sets in S are disjoint.
2. The union of all sets in S is A?

The collection S is called an *exact cover* of A with respect to the A_i's, because together the A_i's "cover" A with no overlaps.

Example 1 Let $A = \{a, b, c, d, e, f, g, h, i, j\}$ and $A_1 = \{a, c, d\}$, $A_2 = \{a, b, e, f\}$, $A_3 = \{b, f, g\}$, $A_4 = \{d, h, i\}$, $A_5 = \{a, h, j\}$, $A_6 = \{e, h\}$, $A_7 = \{c, i, j\}$, $A_8 = \{i, j\}$. Is there an exact cover of A with respect to A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8?

Solution

We start with A_1 and consider each of the other sets in turn. We reject A_2, since $A_1 \cap A_2 = \emptyset$, but A_3 is suitable. A_1 and A_3 together contain the elements a, b, c, d, f, and g of A. The sets A_4 and A_5 contain d and a respectively, and so they are rejected. At this point we can see that A_6 and A_8 will complete the search, and A_1, A_3, A_6, A_8 form an exact cover of A.

If we did not find an exact cover that contained A_1, we would have started the search over with A_2 as the first choice. This method would be continued until either we found an exact cover or failed for each choice of a starting set. A systematic way to try all possibilities, such as we just described, is a key step in writing algorithms and computer programs to solve many types of problems involving only a finite number of possible solutions. The technique is called *backtracking*, and is an essential part of many algorithms that search all possible outcomes of a process. This method is a way of implementing a brute force search of possibilities and will always produce a solution if one exists.

Example 2 Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and let $A_1 = \{2, 4, 6, 7\}$, $A_2 = \{1, 2, 5\}$, $A_3 = \{9, 10\}$, $A_4 = \{5, 8, 1\}$, $A_5 = \{1, 3, 5\}$. Find an exact cover of U with respect to the given subsets.

Solution

A_1 is not disjoint from A_2, so A_2 cannot be used. A_1, A_3, and A_4 are a collection of disjoint sets, but A_4 has elements in common with A_5, so A_5 is eliminated. However, $A_1 \cup A_3 \cup A_4 = A$ and we begin the search again with A_2. A_2 is only disjoint from A_3. Again, this is not enough for a cover: $A_2 \cup A_3 = A$. Continuing to check, starting with A_3, A_4, and A_5 in turn, we find there is no exact cover of U with respect to these subsets.

The exact cover problem can be restated in a way that leads to an algorithm that can be implemented by a computer program. If U is a set with n elements, then each subset of U can be represented by a unique sequence of 0's and 1's of length n. We can use this representation to restate the exact cover problem. If U is a set with n elements, and if $S_1, S_2, \ldots, S_k$ are a finite number of subsets of U, we form a matrix M whose rows correspond, in order, to the representations of $S_1, S_2, \ldots, S_k$ as sequences of 0's and 1's.

Example 3 Represent the subsets $A_1, A_2, \ldots, A_8$ of Example 1 by a matrix M.

Solution

The set A of Example 1 has 10 elements. The eight subsets can be represented in order by the rows of the following matrix

$$M = \begin{matrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{matrix}.$$

We can now use M to find an exact cover of A with respect to A_1, A_2, A_3, A_4, A_5, A_6, A_7, and A_8. Let

$$0 = \begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$$

and

$$1 = \begin{matrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{matrix}.$$

In matrix terms the problem becomes: Find a collection $r_1, r_2, \ldots, r_k$ of rows of M having the following properties:

(a) $r_i \quad r_j = 0$ for $1 \quad i, j \quad k$ and

(b) $r_1 \quad r_2 \quad \cdots \quad r_k = 1$

It is easy to see that one solution is given by rows 1, 3, 6, and 8. This is the solution found before.

Once it is reduced to searching a Boolean matrix, the exact cover problem can be more easily tackled with computer-implemented algorithms. The best-known algorithm for solving the exact cover problem is Donald Knuth's Algorithm X.

Algorithm ALGORITHM X

Label the rows of the original matrix A and retain these labels throughout the application of the algorithm. Begin with $M = A$ and $L = \{\ \}$.

Step 1 If M has a column of 0's, then there is no solution, since by de nition the rows representing an exact cover must together have 1's in every position.

Step 2 Otherwise:

(a) Choose the column c of M with the fewest 1's.

(b) Choose a row r with a 1 in column c, and place the number r in L.

(c) Eliminate any row r_1 having the property that $r \quad r_1 = 0$ (where 0 represents a row with all 0 entries).

(d) Eliminate all columns in which r has a 1.

(e) Eliminate row r.

(f) Let M be the resulting matrix and repeat steps 1 and 2.

The logic behind (c) is that any such row r_1 represents a subset that is not disjoint from the subset of row r. The logic of (d) is that the deleted columns record elements that are already in the subset of row r, and so cannot play any further role in additional rows chosen.

We repeat Steps 1 and 2 producing ever smaller matrices, choosing a row each time and accumulating corresponding row labels from A, until the process terminates. Let us call such a succession of choices and computations a path. In general, there may be many paths to follow, since we may have several rows available at each successive stage in the algorithm. A path is successful if the final matrix produced is empty (has no rows or columns), in which case L will contain the row labels of an exact cover. A path fails if the final matrix produced has a column of 0's.

Note that if all paths that start with row r fail, there may still be a solution. In this case, we remove r from L (and from any further consideration) and choose (if possible) another row with a 1 in column c.

Example 4 Apply Algorithm X to find an exact cover for $A = \{a, b, c, d, e, f, g\}$ with respect to $A_1 = \{a, b, e\}$, $A_2 = \{a, b\}$, $A_3 = \{a, e, f\}$, $A_4 = \{d, f, g\}$, $A_5 = \{c, d, e, g\}$, $A_6 = \{c, e\}$.

Solution

The matrix A represents the original situation.

$$A = \begin{array}{c|ccccccc} 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 4 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 5 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 6 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{array}$$

The rows of A have been numbered. Let $M = A$ and $L = \{\ \}$.

Step 1 M does not have a column of zeros.

Step 2 The least number of 1's in any column is 2, so we may choose the first such column, column 2. We need to choose a row with a 1 in column 2, say $r = 1$. Now $L = \{1\}$. We perform actions (c) through (e). Rows 2, 3, 5, and 6 are eliminated, since they each have a 1 in a common column with the first row. We then eliminate columns 1, 2, and 5, and row 1. The new matrix is what remains of row 4 of A.

$$4 \quad 0 \quad 1 \quad 1 \quad 1$$

Note: The original row label is still attached. However, this matrix has a column of 0's, and so this path fails.

That means that we backtrack, removing row 1 from further consideration. So $L =$ again, and we return to the original matrix A. Since row 1 failed to start a successful path, we try row 2, the only other row to have a 1 in column 2. If this also fails, then no solution is possible.

Perform Steps 1 and 2 on M, based on row 2. This first gives $L = \{2\}$, and the following matrix:

$$\begin{array}{c|ccccc} 4 & 0 & 1 & 0 & 1 & 1 \\ 5 & 1 & 1 & 1 & 0 & 1 \\ 6 & 1 & 0 & 1 & 0 & 0 \end{array}.$$

Column 4 of this matrix has the least number of 1's, and the top row has a 1 in column 4, so we choose to work with it. Since that row corresponds to row 4 of A, we have $L = \{2, 4\}$. Actions (c) through (e) on the new matrix, using its top row, produce

$$6 \quad 1 \quad 1 \ .$$

Now we can only choose the last remaining row and add 6 to L, so that $L = \{2, 4, 6\}$. Actions (c) through (e), applied to this matrix, eliminate all rows and columns, so the path is successful, and an exact cover consists of rows 2, 4, and 6 of A, as is easily verified.

Many other problems and recreational puzzles are equivalent to the exact cover problem.

Example 5 Figure 1(a) shows a 3×3 matrix with two positions filled.

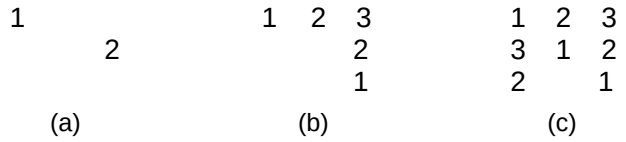


Figure 1

Fill the remaining positions with 1, 2, and 3, so that each row and each column has exactly one 1, one 2, and one 3.

Solution

We reason as follows: Position $(1, 3)$ must contain a 3, since there is already a 1 in the first row and a 2 in the third column, and duplicates are not permitted. The rest of row 1 and column 3 must then be as shown in Figure 1(b).

Similar reasoning shows that position $(2, 1)$ can only contain a 3, and so row 2 and column 1 must be as shown in Figure 1(c). Finally, we see that position $(3, 2)$ must contain a 3, and we have found the unique solution.

Example 6 Fill the remaining positions of Figure 2 with 1, 2, and 3, so that each row and each column has exactly one 1, one 2, and one 3.

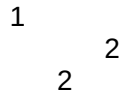


Figure 2

Solution

As before, we reason that position $(1, 3)$ must contain the number 3. However, this forces position $(1, 2)$ to contain a 2. On the other hand, this is impossible, since there would then be two 2's in the second column. Thus we see that no solution is possible.

Consider again the puzzle given in Figure 1. Recall that we are trying to put the digits 1, 2, and 3 in a 3×3 matrix so that each row and column contains these digits exactly once. Suppose that we are free to choose nine digits, each of which is either 1, 2, or 3 (without restrictions on the number of each), and place them in the matrix at any location (where more than one number can occupy the same location, and some locations can be empty). In order for such a placement to constitute an allowable solution to the puzzle, three sets of constraints must be true.

1. Each cell in the matrix must contain a digit. (9 constraints)
2. Each digit must occur in each column. (9 constraints)

3. Each digit must occur in each row. (9 constraints)

Thus, there are 27 constraints in all.

With the complete freedom of placement described above, we could easily fill the array so that any two of the three conditions were true, but the third was not, even if we always had three of each number. Some cases are shown in Figure 3.

1, 2, 3			1	2	3	1	1	1
	1, 2, 3		1	2	3	2	2	2
		1, 2, 3	1	2	3	3	3	3

Figure 3

Thus, we need all 27 constraints satisfied in order to have a solution. We now construct a Boolean matrix A with 27 columns, one for each constraint. The first 9 columns of A represent respectively the constraints that the positions $(1, 1)$, $(1, 2)$, $(1, 3)$, $(2, 1)$, $\ldots$, $(3, 3)$ contain one digit. The columns from 10 to 18 represent respectively the constraints that (a 1 is in row 1), (a 1 is in row 2), $\ldots$, (a 3 is in row 3). Finally, the columns from 19 to 27 represent respectively the constraints that (a 1 is in column 1), (a 1 is in column 2), $\ldots$, (a 3 is in column 3).

Let us say that a "move" is the placement of one of the digits 1, 2, 3 in a particular location of A. Since there are three digits and nine cells, there are 27 moves. Hence, let matrix A have 27 rows, one for each possible move. The rows of A contain 0's and 1's and are constructed in the following way. If a row corresponds to putting a 2 in position $(3, 1)$, then three constraints are satisfied—there is a digit in position $(3, 1)$, there is a 2 in row 3, and there is a 2 in column 1. We place 1's in the columns of that row to correspond to these three constraints, namely columns 7, 15, and 22, and place 0's elsewhere. Then the row of A that corresponds to the placement of a 2 in position $(3, 1)$ is

$$0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0.$$

Each row of A corresponds to a move, thus satisfying three constraints: one in the range 1–9, one in the range 10–18, and one in the range 19–27. It therefore has exactly three 1's in the positions corresponding to satisfied constraints. Thus, the row

$$0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1$$

corresponds to placing a 3 in position $(2, 3)$.

Example 7 (a) Give the row of A for the move that places a 1 in position $(2, 2)$.

(b) Describe the move represented by this row of A:

$$0\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0.$$

Solution

(a) $0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1$
$0\ 0\ 0\ 0\ 0\ 0\ 0$

(b) The move consists of placing a 2 in the position $(1, 2)$.

If we now solve the exact cover problem for a set represented by A, then the rows in the solution correspond to placements that are a legitimate solution to the

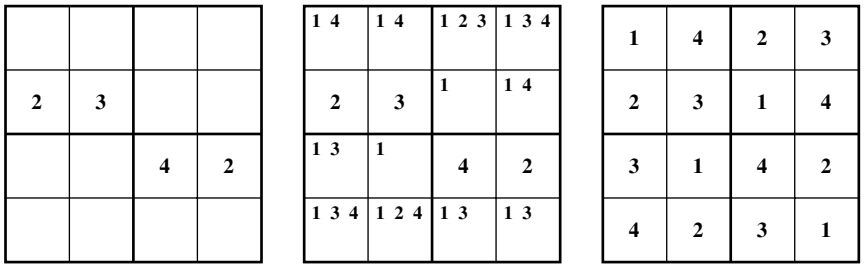
puzzle, since each constraint is satisfied in exactly one way. If there is an initial placement of digits, then the corresponding rows must belong to the solution. The solution to the puzzle shown in Figure 1(a) is then represented by the following matrix.

1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0
0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Rows 1 and 6 correspond to the initial placement of digits.

Sudoku

A generalization of the simple puzzle shown in Example 5 is one of the most popular puzzles of recent times, Sudoku. We begin with a simplified version of Sudoku shown in Figure 4(a). It has 16 small squares, organized into 4 larger 2×2 squares, outlined more thickly.



(a) (b) (c)

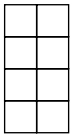
Figure 4

The goal of the puzzle is to fill the grid with 1, 2, 3, and 4 in such a way that each row, each column, and each of the outlined 2×2 squares will contain these numbers exactly once. Often there is an initial assignment, as shown in Figure 4(a).

The main hand techniques for solving a Sudoku puzzle are variations of what may be described as "guess and check the consequences." The ideas are similar in nature to those for Example 1, but may involve more complex logical deductions. We could, for example, temporarily place in each cell all the numbers that might go there; that is, numbers not obviously ruled out by the initial placements. Thus in cell $(1, 1)$, the top left cell, we would place 1 and 4, since both 2 and 3 are elsewhere in the top left large block. The resulting matrix is shown in Figure 4(b). Then we test these possible choices by investigating the logical consequences, as we did in the simpler 3×3 version.

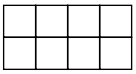
In Figure 4(b) we see that positions $(2, 3)$ and $(3, 2)$ can only contain 1. Having used 1's in columns 2 and 3, we see that positions $(1, 2)$ and $(4, 3)$ must contain 4 and 3 respectively. In turn, we deduce that positions $(1, 1)$ and $(4, 4)$ must contain 1's to fill in the top left and bottom right 2×2 squares. We continue this

21. (a) How many ways can tile D be placed in rectangle A?
    (b) How many ways can tile L be placed in rectangle A?



A

22. (a) How many ways can tile U be placed in rectangle B?
    (b) How many ways can tile R be placed in rectangle B?



B

23. Using tiles U, D, R, L exactly once each and rectangle A, what information is required to de ne a move?
24. Using tiles U, D, R, L exactly once each and rectangle B, what information is required to de ne a move?
25. Give a matrix that records the initial placement shown in Figure 12. Include the column names.

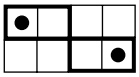


Figure 12

26. Give a matrix that records the initial placement shown in Figure 13. Include the column names.

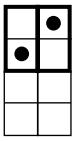


Figure 13

27. Figure 12 cannot be covered with U, D, R, L. Use Algorithm X to prove this.
28. Figure 13 can be covered in two different ways using U, D, R, L exactly once each. Use Algorithm X to prove this.

Tips for Proofs

This chapter provides the formal basis for our proofs, although most proofs are not so formal as the patterns given in Section 3. Two new types of proofs are presented: indirect proofs and induction proofs. Indirect proofs are based either on the pattern (p q) q (proof by contradiction) or on the fact that (p q) (q p) (prove the contrapositive). There are no hard and fast rules about when to use a direct or indirect proof. One strategy is to proceed optimistically with a direct proof. If that does not lead to anything useful, you may be able to identify a counterexample if the statement is in fact false or start a new proof based on one of the indirect models. Where the dif culty occurs in the attempted direct proof can often point the way to go next. Remember that a certain amount of creativity is required for any proof.

Conjectures that are good candidates for proof by induction are ones that involve the counting of whole numbers in some way, either to count something or to describe a pattern. Examples of these are in Section 4, Exercises 11 and 15. Notice that for most of the induction proofs in Section 4, (k) is used early and then properties of operations and arithmetic are used, but in proving loop invariants, the "arithmetic" comes rst, then the use of (k).

In proving conjectures about propositions, try to use the properties of logical operations (see Section 2, Theorem 1 for some of these). Building truth tables should be your second-choice strategy. The proof of a conjecture that is the extension of a known fact usually requires the application of the original fact.

Key Ideas for Review

- Statement: declarative sentence that is either true or false, but not both
- Propositional variable: letter denoting a statement
- Compound statement: statement obtained by combining two or more statements by a logical connective
- Logical connectives: not (), and ( ), or ( ), if then ( ), if and only if ( )
- Conjunction: p q (p and q)

- Disjunction: $p \vee q$ (p or q)
- Predicate (propositional function): a sentence of the form $P(x)$
- Universal quantification: $\forall x\, P(x)$ [For all values of x, $P(x)$ is true.]
- Existential quantification: $\exists x\, P(x)$ [There exists an x such that $P(x)$ is true.]
- Conditional statement or implication: $p \to q$ (if p then q); p is the antecedent or hypothesis and q is the consequent or conclusion
- Converse of $p \to q$: $q \to p$
- Contrapositive of $p \to q$: $\neg q \to \neg p$
- Equivalence: $p \leftrightarrow q$
- Tautology: a statement that is true for all possible values of its propositional variables
- Absurdity: a statement that is false for all possible values of its propositional variables
- Contingency: a statement that may be true or false, depending on the truth values of its propositional variables
- $p \equiv q$ (Logically equivalent statements p and q): $p \leftrightarrow q$ is a tautology
- Methods of proof:
  - q logically follows from p
  - Rules of inference
  - Modus ponens
  - Indirect method
  - Proof by contradiction
- Counterexample: single instance that disproves a theorem or proposition
- Principle of mathematical induction: Let n_0 be a fixed integer. Suppose that for each integer $n \geq n_0$ we have a proposition $P(n)$. Suppose that (a) $P(n_0)$ is true and (b) If $P(k)$, then $P(k+1)$ is a tautology for every $k \geq n_0$. Then the principle of mathematical induction states that $(\forall n)P$ is true for all $n \geq n_0$.
- Loop invariant: a statement that is true before and after every pass through a programming loop
- Strong form of mathematical induction
- Conjecture: a mathematical statement that has not been proved or disproved
- Exact cover
- Algorithm X

Chapter Self-Test

1. Why is it important to recognize the converse and the contrapositive of a conditional statement?
2. How does the strong form of induction differ from basic mathematical induction?
3. What mathematical structure previously studied has the same properties as (logical statements, $\wedge$, $\vee$, $\neg$)?
4. How does an indirect proof technique differ from a direct proof?
5. What is the structure of a proof by contradiction?
6. Determine the truth value of the given statements if p is true and q is false.
   (a) $p \vee q$ (b) $p \wedge q$
7. Determine the truth value for each of the following statements. Assume $x, y \in Z$.
   (a) $\forall x, y$ $x + y$ is even.
   (b) $\forall x \exists y$ $x + y$ is even.
8. Make a truth table for $(p \wedge p) \vee (\neg(q \vee r))$.

For Problems 9 through 11, let p: $1 < -1$, q: $|2| = |-2|$, r: $-3 < -1$, and s: $1 < 3$.

9. Write the symbolic version of the converse and of the contrapositive for each of the following propositions.
   (a) $p \to q$ (b) $(\neg r) \wedge (\neg s) \to q$
   (c) $q \to p \vee s$
10. Write the converse and the contrapositive of the propositions in Problem 4 as English sentences.
11. Give the truth value of each proposition in Problem 9.
12. The English word "or" is sometimes used in the exclusive sense meaning that either p or q, but not both, is true. Make a truth table for this exclusive or, $\veebar$.
13. Let p: An Internet business is cheaper to start, q: I will start an Internet business, and r: An Internet business makes less money. For each of the following write the argument in English sentences and also determine the validity of the argument.

    (a) $\dfrac{r \to (q \to p) \quad p}{(\neg r) \wedge (\neg q)}$ (b) $\dfrac{p \to q \quad q \to r \quad p}{r}$

14. Suppose that m and n are integers such that $m \mid n$ and $m \mid n$. Are these hypotheses sufficient to prove that $m = n$? If so, give a proof. If not, supply a simple additional hypothesis that will guarantee $m = n$ and provide a proof.
15. Prove or disprove by giving a counterexample that the sum of any three consecutive odd integers is divisible by 6.
16. Use mathematical induction to prove that $3^n - 1$ is divisible by 3.
17. Use mathematical induction to prove that

$$1 + 2 + 3 + \cdots + n < \frac{(n+1)^2}{2}.$$

Experiment 2

Many games and puzzles use strategies based on the rules of mathematical logic developed in this chapter. We begin here with a simple puzzle situation: Construct an object from beads and wires that satisfies some given conditions. After investigating this object, you will prove that it satisfies certain properties.

Part I. Here are the conditions for the first object.

(a) You must use exactly three beads.

(b) There is exactly one wire between every pair of beads.

(c) Not all beads can be on the same wire.

(d) Any pair of wires has at least one bead in common.

1. Draw a picture of the object.
2. Your object might not be the only one possible, so the following statements are to be proved referring only to the conditions and not to your object.

   T1. Any two wires have at most one bead in common.
   T2. There are exactly three wires.
   T3. No bead is on all the wires.

Part II. Here are the conditions for the second object.

(a) You must use at least one bead.

(b) Every wire has exactly two beads on it.

(c) Every bead is on exactly two wires.

(d) Given a wire, there are exactly three other distinct wires that have no beads in common with the given wire.

1. Draw a picture of the object.
2. Your object might not be the only one possible, so the following statements are to be proved referring only to the conditions and not to your object.

   T1. There is at least one wire.
   T2. Given a wire, there are exactly two other wires that have a bead in common with the given wire.
   T3. There are exactly ____ wires.
   T4. There are exactly ____ beads.

Part III. The two objects you created in Parts I and II can be viewed in a number of ways. Instead of beads and wires, consider players and two-person teams, or substitute the words point and line for bead and wire.

1. Translate the statements T1, T2, and T3 in Part I into statements about players and two-person teams.
2. Translate the conditions (a)–(d) given in Part II into statements about points and lines.
3. The type of object created here is often called a finite geometry, because each has a finite number of points and lines. What common geometric concept is described in condition (d) of Part II?
4. The Acian Bolex Tournament will be played soon. Determine the number of players needed and the number of teams that will be formed according to these ancient rules for bolex.

   (a) A team must consist of exactly three players.

(b) Two players may be on at most one team in common.
(c) Each player must be on at least three teams.
(d) Not all the players can be on the same team.
(e) At least one team must be formed.
(f) If a player is not on a given team, then the player must be on exactly one team that has no members in common with the given team.

Coding Exercises

1. Write a program that will print a truth table for $p \vee q$.
2. Write a program that will print a truth table for $(p \wedge q) \rightarrow r$.
3. Write a program that will print a truth table for any two-variable propositional function.
4. Write a subroutine EQUIVALENT that determines if two logical expressions are equivalent.
5. Write a subroutine that determines if a logical expression is a tautology, a contingency, or an absurdity.

Answers to Odd-Numbered Exercises

Exercise Set 1

1. (b), (d), and (e) are statements.
3. (a) It will not rain tomorrow and it will not snow tomorrow.
   (b) It is not the case that if you drive, I will walk.
5. (a) I will drive my car and I will be late.
   I will drive my car or I will be late.
   (b) $10 < NUM \leq 15$. $NUM > 10$ or $NUM \leq 15$.
7. (a) True. (b) True. (c) True. (d) False.
9. (a) False. (b) True. (c) True. (d) False.
11. (d) is the negation.
13. (a) The dish did not run away with the spoon and the grass is wet.
   (b) The grass is dry or the dish ran away with the spoon.
   (c) It is not true that today is Monday or the grass is wet.
   (d) Today is Monday or the dish did not run away with the spoon.
15. (a) For all x there exists a y such that $x + y$ is even.
   (b) There exists an x such that, for all y, $x + y$ is even.
17. (a) It is not true that there is an x such that x is even.
   (b) It is not true that, for all x, x is a prime number.
19. 14: (a) False. (b) True.
   15: (a) True. (b) False.
   16: (a) False. (b) True.
   17: (a) False. (b) True.
   18: (a) False. (b) False. (c) False. (d) False.
21. (a) $5 + 3 = 6$. (b) $m + 3 = 6$.
23. (a) $0 \quad 3^0$. (b) $2 \quad 3^2$. (c) $k \quad 3^k$.
25. $A \quad B(\overline{A \quad B}) = \overline{A} \quad \overline{B}$; $A \quad B(\overline{A \quad B}) = \overline{A} \quad \overline{B}$.
27.

p	q	(p	q)	p
T	T	F	F	T
T	F	F	F	T
F	T	T	T	T
F	F	T	F	F
		(1)	(2)	

29.

p	q	r	(p	q)	r
T	T	T	T		T
T	T	F	T		F
T	F	T	T		T
T	F	F	T		F
F	T	T	T		T
F	T	F	T		F
F	F	T	F		F
F	F	F	F		F
			(1)		

31.

p	q	r	(p q)	r
T	T	T	F	F
T	T	F	F	T
T	F	T	F	F
T	F	F	F	T
F	T	T	F	F
F	T	F	F	T
F	F	T	T	F
F	F	F	T	F

(1)

33.

p	q	r	(p q)		(p r)
T	T	T	F	T	F
T	T	F	F	T	F
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	F	T	F
F	T	F	F	F	T
F	F	T	T	F	F
F	F	F	T	F	T

(1) (2)

35.

p	q	(p q)	p
T	T	T	F
T	F	F	T
F	T	T	T
F	F	F	F

(1)

37. (x = max) or (y 4).

39. WHILE (item = sought or index 101) take action

Exercise Set 2

1. (a) p q. (b) r p.
   (c) q p. (d) r p.

3. (a) If I am not the Queen of England, then 2 2 = 4.
   (b) If I walk to work, then I am not the President of the United States.
   (c) If I did not take the train to work, then I am late.
   (d) If I go to the store, then I have time and I am not too tired.
   (e) If I buy a car and I buy a house, then I have enough money.

5. (a) True. (b) False.
   (c) True. (d) True.

7. (a) If I do not study discrete structures and I go to a movie, then I am in a good mood.
   (b) If I am in a good mood, then I will study discrete structures or I will go to a movie.
   (c) If I am not in a good mood, then I will not go to a movie or I will study discrete structures.
   (d) I will go to a movie and I will not study discrete structures if and only if I am in a good mood.

9. (a) If 4 > 1 and 2> 2, then 4< 5.
   (b) It is not true that 3 3 and 4< 5.
   (c) If 3 > 3, then 4> 1.

11. (a)

p	q	p	(q p)
T	T	T	T
T	F	T	T
F	T	T	F
F	F	T	T

tautology

(b)

p	q	q	(q p)
T	T	T	T
T	F	T	T
F	T	F	F
F	F	T	T

contingency

13. Yes. If p q is false, then p is true and q is false. Hence p q is false, (p q) is true, and ((p q)) q is false.

15. No, because if p q is true, it may be that both p and q are true, so (p q) q is false. But it could also be that p and q are both false, and then (p q) q is true.

17. The solultions are self-checking.

19. The solution is self-checking.

21. (a) False. (b) True. (c) False. (d) True.

23.

p	q	(p q)	(p p)		(q q)
T	T	T	F	T	F
T	F	F	F	F	T
F	T	F	T	F	F
F	F	F	T	F	T

(A) (B)

Since columns (A) and (B) are the same, the statements are equivalent.

25. (a) Jack did eat fat or he did not eat broccoli.
   (b) Mary did not lose her lamb and the wolf did not eat the lamb.
   (c) Tom stole a pie and ran away and the three pigs have some supper.

27.

p	q	r	p	(q r)	(p q)		(p r)
T	T	T	T	T	T	T	T
T	T	F	T	T	T	T	F
T	F	T	T	T	F	T	T
T	F	F	F	F	F	F	F
F	T	T	F	T	F	F	F
F	T	F	F	T	F	F	F
F	F	T	F	T	F	F	F
F	F	F	F	F	F	F	F

(A) (B)

Because (A) and (B) are the same, the statements are equivalent.

29.

p	q	(p ∨ q)	(p ∨ q)	(q ∨ p)
T	T	T	T	T
T	F	T	T	T
F	T	T	T	T
F	F	F	F	F
		(A)	(B)	

Because columns (A) and (B) are the same, the statements are equivalent.

31. The statement $\forall x(P(x) \wedge Q(x))$ is true if and only if x both $P(x)$ and $Q(x)$ are true, but this means $\forall x P(x)$ is true and $\forall x Q(x)$ is true. This holds if and only if $\forall x P(x) \wedge \forall x Q(x)$ is true.

33.

p	q	q → (p ∧ q)
T	T	T
T	F	T
F	T	T
F	F	T

35.

p	q	r	((p → q) ∧ (q → r))	(p → r)
T	T	T	T	T
T	T	F	F	F
T	F	T	F	T
T	F	F	F	F
F	T	T	T	T
F	T	F	F	T
F	F	T	T	T
F	F	F	T	T
			(1) (3) (2)	(4)

37. Because $(p \to q) \equiv (\sim q \to \sim p)$, parts (a) and (b) of Theorem 4 say the same thing; that is, either p or q can be considered the first statement.

Exercise Set 3

1. Valid: $((d \to t) \wedge \sim t) \to \sim d$.
3. Invalid.
5. Valid: $((f \to w) \wedge \sim w) \to \sim f$.
7. Valid: $[(h \to t) \wedge (m \to h) \wedge p] \to [h \wedge p \wedge t]$.
9. Invalid.
11. (a) $((p \to q) \wedge \sim q) \to \sim p$.
    (b) $((p \to q) \wedge \sim p) \to \sim q$.
13. (a) $(((p \to q) \wedge (q \to r)) \wedge (\sim (\sim q) \vee r)) \to p$.
    (b) $(\sim (\sim p \to q) \wedge \sim p) \to q$.
15. Suppose m and n are odd. Then there exist integers j and k such that $m = 2j + 1$ and $n = 2k + 1$. $m + n = (2j + 1) + (2k + 1) = 2j + 2k + 2 = 2(j + k + 1)$. Since $j + k + 1$ is an integer, $m + n$ is even.
17. Suppose that m and n are odd. Then there exist integers j and k such that $m = 2j + 1$ and $n = 2k + 1$. $m \cdot n = 2j \cdot 2k + 2j + 2k + 1 = 2(2jk + j + k) + 1$. Since $2jk + j + k$ is an integer, $m \cdot n$ is odd and the system is closed with respect to multiplication.
19. If $A = B$, then, clearly, $A \subseteq B$ and $B \subseteq A$. If $A \subseteq B$ and $B \subseteq A$, then $A \subseteq A$ and B must be the same as A.
21. (a) If $A \subseteq B$, then $A \cup B \subseteq B$. But $B \subseteq A \cup B$. Hence $A \cup B = B$. If $A \cup B = B$, then since $A \subseteq A \cup B$, we have $A \subseteq B$.
    (b) If $A \subseteq B$, then $A \cap A \subseteq A \cap B$. But $A \cap B \subseteq A$. Hence $A \cap B = A$. If $A \cap B = A$, then since $A \subseteq B$, we have $A \subseteq B$.
23. For $n = 41$, we have a counterexample. $41^2 + 41 + 41 = 41 \cdot 41 + 41$ is $41(41 + 1) + 1$ or $41 \cdot 83$.
25. $n^3 - n = n(n - 1)(n + 1)$, the product of three consecutive integers. One of these must be a multiple of 3, so $3 \mid n^3 - n$.
27. Invalid. Multiplying by $x - 1$ may or may not preserve the order of the inequality.
29. Valid.
31. Let x and y be prime numbers, each larger than 2. Then x and y are odd and their sum is even (Exercise 15). The only even prime is 2, so $x + y$ is not a prime.
33. Suppose $x + y$ is rational. Then there are integers a and b such that $x + y = \frac{a}{b}$. Since x is rational, we can write $x = \frac{c}{d}$ with integers c and d. But now $y = x + y - x = \frac{a}{b} - \frac{c}{d}$. $y = \frac{ad - bc}{bd}$. Both $ad - bc$ and bd are integers. This is a contradiction since y is an irrational number and cannot be expressed as the quotient of two integers.

Exercise Set 4

Note: Only the outlines of the induction proofs are given. These are not complete proofs.

1. Basis step $n = 1$: $P(1)$: $2(1) = 1(1 + 1)$ is true.
   Induction step: $P(k)$: $2 + 4 + \cdots + 2k = k(k + 1)$.
   $P(k + 1)$: $2 + 4 + \cdots + 2(k + 1) = (k + 1)(k + 2)$.
   LHS of $P(k + 1)$: $2 + 4 + \cdots + 2k + 2(k + 1) = k(k + 1) + 2(k + 1) = (k + 1)(k + 2)$
   RHS of $P(k + 1)$.
3. Basis step $n = 0$: $P(0)$: $2^0 = 2^{0+1} - 1$ is true.
   Induction step: LHS of $P(k + 1)$: $1 + 2^1 + 2^2 + \cdots + 2^k + 2^{k+1} = (2^{k+1} - 1) + 2^{k+1} = 2 \cdot 2^{k+1} - 1 = 2^{k+2} - 1$.
   RHS of $P(k + 1)$.
5. Basis step $n = 1$: $P(1)$: $1^2 = \dfrac{1(1 + 1)(2 + 1)}{6}$ is true.
   Induction step: LHS of $P(k + 1)$:

$$\begin{aligned} 1^2 + 2^2 + \cdots + k^2 + (k + 1)^2 &= \frac{k(k + 1)(2k + 1)}{6} + (k + 1)^2 \\ &= (k + 1)\left[\frac{k(2k + 1)}{6} + (k + 1)\right] \\ &= \frac{k + 1}{6}(2k^2 + k + 6(k + 1)) \\ &= \frac{k + 1}{6}(2k^2 + 7k + 6) \end{aligned}$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$
$$= \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}.$$
RHS of $P(k+1)$.

7. Basis step: $n = 1$. $P(1)$: $a = \frac{a(1-r^1)}{1-r}$ is true.
Induction step: LHS of $P(k+1)$: $a + ar + \cdots + ar^{k-1} + ar^k = \frac{a(1-r^k)}{1-r} + ar^k = \frac{a-ar^k+ar^k-ar^{k+1}}{1-r} = \frac{a(1-r^{k+1})}{1-r}$.
RHS of $P(k+1)$.

9. (a) LHS of $P(k+1)$: $1+5+9+\cdots+(4(k+1)-3) = (2k+1)(k-1)+4(k+1)-3 = 2k^2+3k = (2k+3)(k) = (2(k+1)+1)((k+1)-1)$.
RHS of $P(k+1)$.
(b) No; $P(1)$: $1 = (2\cdot 1+1)(1-1)$ is false.

11. Basis step: $n = 2$. $P(2)$: $2 < 2^2$ is true.
Induction step: LHS of $P(k+1)$: $k+1 < 2^k+1 < 2^k+2^k = 2\cdot 2^k = 2^{k+1}$. RHS of $P(k+1)$.

13. Basis step: $n = 5$. $P(5)$: $1+5^2 < 2^5$ is true.
Induction step: LHS of $P(k+1)$: $1+(k+1)^2 = k^2+1+2k+1 < 2^k+2k+1 < 2^k+k^2+1 < 2^k+2^k = 2\cdot 2^k = 2^{k+1}$.
RHS of $P(k+1)$.

15. Basis step: $n = 0$. $A = \{\}$ and $P(A) = \{\{\}\}$, so $|P(A)| = 2^0$ and $P(0)$ is true.
Induction step: Use $P(k)$: If $|A| = k$, then $|P(A)| = 2^k$ to show $P(k+1)$: If $|A| = k+1$, then $|P(A)| = 2^{k+1}$. Suppose that $|A| = k+1$. Set aside one element x of A. Then $|A-\{x\}| = k$ and $A-\{x\}$ has 2^k subsets. These subsets are also subsets of A. We can form another 2^k subsets of A by forming the union of $\{x\}$ with each subset of $A-\{x\}$. None of these subsets are duplicates. Now A has 2^k+2^k, or 2^{k+1}, subsets.

17. Basis step: $n = 1$. $P(1)$: $\overline{A_1} = \overline{A_1}$ is true.
Induction step: LHS of $P(k+1)$:
$$\overline{\bigcup_{i=1}^{k+1} A_i} = \overline{\bigcup_{i=1}^{k} A_i \cup A_{k+1}}$$
$$= \overline{\bigcup_{i=1}^{k} A_i} \cap \overline{A_{k+1}} \quad \text{(De Morgan's laws)}$$
$$= \bigcap_{i=1}^{k} \overline{A_i} \cap \overline{A_{k+1}}$$
$$= \bigcap_{i=1}^{k+1} \overline{A_i}. \quad \text{RHS of } P(k+1).$$

19. Basis step: $n = 1$. $P(1)$: $A_1 \cap B = A_1 \cap B$ is true.
Induction step: LHS of $P(k+1)$:
$$\bigcup_{i=1}^{k+1} A_i \cap B = \bigcup_{i=1}^{k} A_i \cup A_{k+1} \cap B$$
$$= \bigcup_{i=1}^{k} A_i \cap B \cup (A_{k+1} \cap B)$$
(distributive property)
$$= \bigcup_{i=1}^{k} (A_i \cap B) \cup (A_{k+1} \cap B)$$
$$= \bigcup_{i=1}^{k+1} (A_i \cap B). \quad \text{RHS of } P(k+1).$$

21. (a) $(k+1)^2+(k+1) = k^2+2k+1+k+1 = k^2+k+2(k+1)$.
Using $P(k)$, k^2+k is odd; $2(k+1)$ is clearly even. Hence their sum is odd.
(b) No, $P(1)$ is false.

23. The flaw is that to carry out the procedure in the induction step, you must have at least three trucks, but the basis step was done for one truck.

25. Basis step: $n = 1$. $P(1)$: $A^2 \cdot A = A^{2+1}$ is true.
Induction step: LHS of $P(k+1)$: $A^2 \cdot A^{k+1} = A^2(A^k \cdot A) = (A^2 \cdot A^k) \cdot A = A^{2+k} \cdot A = A^{2+k+1}$. RHS of $P(k+1)$.

27. Basis step: $n = 5$. $P(5)$: A restaurant bill of \$5 can be paid exactly with a \$5 bill.
Induction step: We use $P(j)$: A restaurant bill of j can be paid exactly using \$2 and \$5 bills for $j = 5, 6, \ldots, k$ to show $P(k+1)$. Write $k+1$ as $2m+r$, $0 \le r < 2$. If $r = 0$, then the bill can be paid with m \$2 bills. If $r = 1$, then $2m+r = 2m+1 = 2(m-2)+5$. Since $k+1 \ge 7$, $m \ge 2$, and $m-2 \ge 0$. Thus a bill of $k+1$ dollars can be paid exactly with $(m-2)$ \$2 bills and a \$5 bill.

29. Basis step: $n = 1$. $P(1)$: If p is prime and $p \mid a^1$, then $p \mid a$ is true.
Induction step: If $p \mid a^{k+1}$, then $p \mid a^k \cdot a$ and either $p \mid a^k$ or $p \mid a$. If $p \mid a^k$, then (using $P(k)$), $p \mid a$.

31. (a) 5.
(b) Basis step: $n = 5$. $P(5)$: $2^5 > 5^2$ is true.
Induction step: LHS of $P(k+1)$: $2^{k+1} = 2\cdot 2^k = 2^k+2^k > k^2+k^2 > k^2+3k > k^2+2k+1$ (since $k \ge 5$) $= (k+1)^2$. RHS of $P(k+1)$.

33. Basis step: $n = 1$. $P(1)$: $x-y$ divides $x-y$ is true.
Induction step: $x^{k+1}-y^{k+1} = x\cdot x^k - y\cdot y^k = x\cdot x^k - x\cdot y^k + x\cdot y^k - y\cdot y^k = x(x^k-y^k)+y^k(x-y)$. This rewriting gives an expression where each term is divisible by $x-y$ and so the sum is as well.

35. Loop invariant check:
Basis step: $n = 0$. $P(0)$: $X-Z_0+W_0 = Y$ is true, because $Z_0 = X$ and $W_0 = Y$.
Induction step: LHS of $P(k+1)$: $X-Z_{k+1}+W_{k+1} = X-(Z_k-1)+(W_k-1) = X-Z_k+W_k = Y$.
RHS of $P(k+1)$.
Exit condition check: $W = 0$. $X-Z+W = Y$ yields $X-Z = Y$ or $Z = X-Y$.

37. Loop invariant check:
Basis step: $n = 0$. $P(0)$: $Z_0+(X\times W_0) = X\times Y$ is true, because $Z_0 = 0$ and $W_0 = Y$.
Induction step: LHS of $P(k+1)$: $Z_{k+1}+(X\times W_{k+1}) = Z_k+X+(X\times(W_k-1)) = Z_k+(X\times W_k) = X\times Y$.
RHS of $P(k+1)$.
Exit condition check: $W = 0$. $Z+(X\times W) = X\times Y$ yields $Z = X\times Y$.

Loop invariant check:
Basis step: $n = 0$ $P(0)$: $Z_0 + (X \times Y \times W_0) = X + Y^2$ is true, because $Z_0 = X \times Y$ and $W_0 = Y - 1$.
Induction step: LHS of $P(k + 1)$: $Z_{k+1} + (X \times Y \times W_{k+1}) = Z_k + X \times Y + (X \times Y \times (W_k - 1)) = Z_k + X \times Y \times W_k = X + Y^2$. RHS of $P(k + 1)$.
Exit condition check: $W = 0$ $Z + X \times Y \times W = X + Y^2$ yields $Z = X + Y^2$.

39. Loop invariant check:
Basis step: $n = 0$ $P(0)$: $Z_0 + (X \times W_0) = Y + X^2$ is true, because $Z_0 = Y$ and $W_0 = X$.
Induction step: LHS of $P(k + 1)$: $Z_{k+1} + (X \times W_{k+1}) = (Z_k + X) + (X \times (W_k - 1)) = Z_k + X \times W_k = Y + X^2$ RHS of $P(k + 1)$.
Exit condition check: $W = 0$ $Z + (X \times W) = Y + X^2$ yields $Z = Y + X^2$.
Loop invariant check:
Basis step: $n = 0$ $P(0)$: $Z_0 + (Y \times W_0) = X^2 + Y^2$ is true, because $Z_0 = Y + X^2$ and $W_0 = Y - 1$.
Induction step: LHS of $P(k + 1)$: $Z_{k+1} + (Y \times W_{k+1}) = Z_k + Y + Y \times (W_k - 1) = Z_k + Y \times W_k = X^2 + Y^2$.
RHS of $P(k + 1)$.
Exit condition check: $W = 0$ $Z + Y \times W = X^2 + Y^2$ yields $Z = X^2 + Y^2$.

Exercise Set 5

1. Let $2k$ be any even number. $(2k)^3 = 8k^3 = 2(4k^3)$ is clearly an even number.

3. (a) Let n, $n + 1$, and $n + 2$ be any three consecutive integers. $n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1)$, which is clearly divisible by 3.

   (b) Let n, $n + 1$, $n + 2$, and $n + 3$ be any four consecutive integers. $n + (n + 1) + (n + 2) + (n + 3) = 4n + 6$, which is not divisible by 4.

5. The quotient of a nonzero rational number and an irrational number is irrational.
Proof: Let $r = \frac{a}{b}$, $a \neq 0$, represent a rational number and j, an irrational number. Suppose that $\frac{r}{j}$ is rational, then $\exists\, c, d \in \mathbb{Z}$ such that $\frac{r}{j} = \frac{c}{d} = \frac{a}{b}/j$. But then $\frac{ad}{bc} = j$; this is a contradiction.

7. $\overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}$
Proof: $\overline{A \cap B \cap C}$
$= \overline{A} \cup \overline{(B \cap C)}$ by the associative property of $\cap$.
$= \overline{A} \cup (\overline{B} \cup \overline{C})$ by De Morgan's law
$= \overline{A} \cup (\overline{B} \cup \overline{C})$ by De Morgan's law
$= \overline{A} \cup \overline{B} \cup \overline{C}$ by the associative property of $\cup$

9. Let $A_1, A_2, \ldots, A_m$ be m Boolean matrices of the same size. Then
$$A_1 \cdot (A_2 \cdot \cdots \cdot A_m) = (A_1 \cdot A_2) \cdot (A_1 \cdot A_3) \cdot \cdots \cdot (A_1 \cdot A_m).$$
Proof (by induction):
Basis Step: $n = 3$
Induction Step: LHS of $P(k + 1)$:
$$\begin{aligned} &A_1 \cdot (A_2 \cdot \cdots \cdot A_k \cdot A_{k+1}) \\ &= A_1 \cdot ((A_2 \cdot \cdots \cdot A_k) \cdot A_{k+1}) \\ &= ((A_1 \cdot A_2) \cdot (A_1 \cdot A_3) \cdot \cdots \\ &\qquad (A_1 \cdot A_k)) \cdot (A_1 \cdot A_{k+1}) \\ &= (A_1 \cdot A_2) \cdot (A_1 \cdot A_3) \cdot \cdots \cdot (A_1 \cdot A_{k+1}). \end{aligned}$$
RHS of $P(k + 1)$.

11. (a) $(p \wedge q \wedge r) \equiv p \wedge q \wedge r$.
Proof: $(p \wedge q \wedge r) \equiv (p \wedge (q \wedge r))$ since $\wedge$ is associative
$\equiv p \wedge (q \wedge r)$
$\equiv p \wedge (q \wedge r)$
$\equiv p \wedge q \wedge r$

(b) $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \equiv p_1 \wedge p_2 \wedge \cdots \wedge p_n$
Basis step: $n = 2$ $(p \wedge q) \equiv p \wedge q$ by Theorem 1, part 11, Section 2.
Induction step: LHS of $P(k + 1)$:
$(p_1 \wedge p_2 \wedge \cdots \wedge p_k \wedge p_{k+1})$
$\equiv ((p_1 \wedge p_2 \wedge \cdots \wedge p_k) \wedge p_{k+1})$
$\equiv (p_1 \wedge p_2 \wedge \cdots \wedge p_k) \wedge p_{k+1}$
$\equiv (p_1 \wedge p_2 \wedge \cdots \wedge p_k) \wedge p_{k+1}$
$\equiv p_1 \wedge p_2 \wedge \cdots \wedge p_{k+1}$. RHS of $P(k + 1)$.

13. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two $r \times r$ diagonal matrices; thus $a_{ij} = 0$ if $i \neq j$ and $b_{ij} = 0$ if $i \neq j$. The ij-entry in AB is $a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ir}b_{rj}$, but this sum is $a_{ii}b_{ii}$. Similarly, the ij-entry in BA is $b_{ii}a_{ii}$. Hence, $AB = BA$.

15. recursive $d_n = d_{n-1} + 6$, $d_1 = 3$; explicit $d_n = 3(2n - 1)$

17. Basis step: $n = 1$ $3 = 3(2(1) - 1) = 3 \cdot 1^2$ is true.
Induction step: LHS of $P(k + 1)$: $3 + 9 + 15 + \cdots + 3(2(k + 1) - 1) = 3 + 9 + 15 + \cdots + 3(2k - 1) + 3(2k + 1) = 3k^2 + 6k + 3 = 3(k + 1)^2$ RHS of $P(k + 1)$.

19. 1, 3, 4, 7, 11, 18, 29, 47, 76, 123.

21. The sums are 1, 4, 8, 15, 26, 44, 73, 120. A reasonable conjecture would be $g_1 + g_2 + \cdots + g_n = g_{n+2} - 3$.

23. Proof: Basis step: $n = 1$ $g_2 = 3 = g_3 - 1 = g_{2(1)+1} - 1$.
Induction step: LHS of $P(k + 1)$:
$g_2 + g_4 + \cdots + g_{2k} + g_{2(k+1)}$
$= g_{2k+1} - 1 + g_{2(k+1)} = g_{2k+1} + g_{2k+2} - 1$
$= g_{2k+3} - 1 = g_{2(k+1)+1} - 1$. RHS of $P(k + 1)$.

25. Proof: Basis step: $n = 1$ $g_1 = 1 = g_2 - 2 = g_{2(1)} - 2$
Induction step: LHS of $P(k + 1)$:
$g_1 + g_3 + \cdots + g_{2k-1} + g_{2(k+1)-1}$
$= g_{2k} - 2 + g_{2(k+1)-1} = g_{2k} + g_{2k+1} - 2$
$= g_{2k+2} - 2 = g_{2(k+1)} - 2$. RHS of $P(k + 1)$.

27. Further experimentation suggests the conjecture that the sum of k consecutive integers is divisible by k if and only if k is odd.

Exercise Set 6

1. A_2, A_6, A_4.

3.
$$\begin{matrix} 1&1&0&0&1&0&0&0&0&0\\ 0&0&1&0&1&0&0&1&1&0\\ 1&0&0&0&0&0&0&1&0&1\\ 0&0&0&1&0&0&1&0&0&1\\ 0&0&1&0&0&0&1&0&0&0\\ 1&1&0&0&0&1&0&0&0&0 \end{matrix}.$$

5. Pick A_4.
$$\begin{matrix} 1&1&0&1&0&0&0&A_1\\ 0&0&1&1&0&1&1&A_2\\ 1&1&0&0&1&0&0&A_6 \end{matrix}$$
Pick A_2. $1\ \ 1\ \ 1\ \ A_6$
An exact cover is A_4, A_2, A_6.

7.
$$\begin{matrix} 1&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0&0&0&1&0&0& \end{matrix}$$

9.
$$\begin{matrix} 0&0&0&0&0&0&0&1&0&0&0&1&0&0\\ 0&0&0&0&0&1&0&0&0&0&0&0&0& \end{matrix}$$

11. One possible solution is

1	2	3	4
2	3	4	1
3	4	1	2
4	1	2	3

13. (a) 1. Each of the 81 cells must contain a digit.
2. Each digit 1 through 9 must appear in each of the nine columns.
3. Each digit 1 through 9 must appear in each of the nine rows.
4. Each digit 1 through 9 must appear in each of the nine 3×3 blocks.

(b) 81; 9×9; 9×9; 9×9 for a total of 324 columns.

(c) 1. $(1,1), (1,2), (1,3), \ldots, (1,9), (2,1), \ldots, (9,9)$
2. a 1 in column 1, a 1 in column 2, . . . , a 1 in column 9, a 2 in column 1, . . . , a 9 in column 9
3. a 1 in row 1, a 1 in row 2, . . . , a 1 in row 9, a 2 in row 1, . . . , a 9 in row 9
4. a 1 in block 1, a 1 in block 2, . . . , a 1 in block 9, a 2 in block 1, . . . , a 9 in block 9

15. (a) (5, 7); 4 in column 7; 4 in row 5; 4 in block 6 (using the numbering of Exercise 13(c) 4)

(b) (4, 4); 6 in column 4; 6 in row 4; 6 in block 5

17. (a) A move consists of placing F, I, L, N, P, T, U, V, W, X, Y, or Z in one specified squares. Squares may be labelled by their position $(1,1), (1,2), \ldots, (1,6), (2,1), \ldots, (5,6)$.

(b) One solution is F, I, L, N, P, T, U, V, W, X, Y, Z, $(1,1), (1,2), \ldots, (1,6), (2,1), \ldots, (5,6)$ for a total of 42 columns.

19. 2 by 4 21. (a) 6 (b) 4

23. A move is placing U, D, R, or L in two specified squares. Squares may be labelled by position $(1,1), (1,2), (2,1), \ldots, (4,2)$.

25.
$$\begin{matrix} & U&D&R&L&(1,1)&(1,2)&(1,3)&(1,4)&(2,1)&(2,2)&(2,3)&(2,4)\\ 1&1&0&0&0&1&0&0&0&1&0&0&0\\ 2&1&0&0&0&0&1&0&0&0&1&0&0\\ 3&1&0&0&0&0&0&1&0&0&0&1&0\\ 4&1&0&0&0&0&0&0&1&0&0&0&1\\ 5&0&1&0&0&1&0&0&0&1&0&0&0\\ 6&0&1&0&0&0&1&0&0&0&1&0&0\\ 7&0&1&0&0&0&0&1&0&0&0&1&0\\ 8&0&1&0&0&0&0&0&1&0&0&0&1\\ 9&0&0&1&0&0&0&0&0&0&0&1&1\\ 10&0&0&0&1&1&1&0&0&0&0&0&0 \end{matrix}.$$

27. Use the matrix from Exercise 25 and choose row 9 for Algorithm X.

$$\begin{matrix} 1&1&0&0&1&0&0&0&1&0\\ 2&1&0&0&0&1&0&0&0&1\\ 5&0&1&0&1&0&0&0&1&0\\ 6&0&1&0&0&1&0&0&0&1\\ 10&0&0&1&1&1&0&0&0&0 \end{matrix}$$

The two columns of zeros signal that there is no solution using row 9, but row 9 represents an initial placement. Hence, there is no solultion with this initial placement.

Answers to Chapter Self-Tests

1. The converse is not equivalent to the original statement, but the contrapositive is. In some cases, the contrapositive may be easier to prove than the original statement.

2. In the strong form of induction all statements $P(n_0), P(n_0+1), \ldots, P(k)$ may be used to show $P(k+1)$, not just $P(k)$.

3. The mathematical structure of sets, union, intersection, and complement has the same properties as (logical statements, , ,).

4. An indirect proof proves the contrapositive of the statement or assumes the statement is false and derives a contradiction.

5. A proof by contradiction proceeds by assuming the negation of the conclusion of the statement to be proved. Then definitions, previous theorems, and commonly known facts are used to derive a contradiction.

6. (a) False. (b) True.

7. (a) False. (b) True.

8.

p	q	r	(p p)	((q r))
T	T	T	F	T
T	T	F	F	F
T	F	T	F	F
T	F	F	F	F
F	T	T	F	T
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F
			(1)	(3) (2)

9. (a) q p. q p.

(b) q (r) (s). q (r s).

(c) (p s) q. (p s) q.

10. (a) If |2| = |Š2|, then 1< Š1.
If |2| = |Š2|, then 1 Š 1.

(b) If |2| = |Š2|, then either Š3 Š 1 or 1 3.
If |2| = |Š2|, then Š3 < Š1 and 1< 3.

(c) If 1 < Š1 or 1< 3, then |2| = |Š2|.
If 1 Š 1 and 1 3, then |2| = |Š2|.

11. (a) False; True.

(b) False; True.

(c) True; True.

12.

p	q	p xor q
T	T	F
T	F	T
F	T	T
F	F	F

13. (a) If an Internet business makes less money, then if I start an Internet business, then an Internet business is cheaper to start. An Internet business is not cheaper to start. Therefore, either an Internet business does not make less money or I do not start an Internet business. Valid.

(b) If an Internet business is cheaper to start, then I will start an Internet business. If I start an Internet business, then an Internet business will make less money. An Internet business is cheaper to start. Therefore, an Internet business will make less money.
Valid.

14. No, consider 6 | Š6 and Š6 | 6. If both m and n are positive (or negative), then m | m and m | n guarantees that n = m.

15. Consider 7, 9, and 11. These are three consecutive odd integers whose sum is not divisible by 6.

16. Basis step: n = 0. P(0): 4^0 Š 1 is divisible by 3 is true since 3| 0.
Induction step: We use P(k): 3 divides 4^k Š 1 to show P(k + 1): 3 divides 4^{k+1} Š 1. Consider 4^{k+1} Š 1 = $4(4^k$ Š 1) + 3. By P(k), 3 | (4^k Š 1) and we have 4^{k+1} Š 1 = 3(a + 1) where a = 4^k Š 1. So 3| (4^{k+1} Š 1).

17. Basis step n = 1. P(1): $1 < \frac{(1+1)^2}{2}$ is true.
Induction step: We use P(k):

$$1 + 2 + 3 + \cdots + k < \frac{(k+1)^2}{2}$$

to show P(k + 1):

$$1 + 2 + \cdots + (k+1) < \frac{(k+2)^2}{2}.$$

LHS of P(k + 1):

$$\begin{aligned} 1 + 2 + 3 + \cdots + k + (k+1) &< \frac{(k+1)^2}{2} + (k+1) \\ &= \frac{k^2 + 4k + 3}{2} \\ &< \frac{k^2 + 4k + 4}{2} \\ &= \frac{(k+2)^2}{2}. \end{aligned}$$

RHS of P(k + 1).

Counting

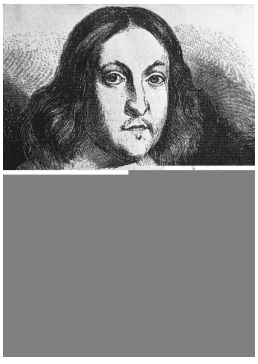
Techniques for counting are important in mathematics and in computer science, especially in the analysis of algorithms. In this chapter, we present other counting techniques, in particular those for permutations and combinations, and we look at two applications of counting, the pigeonhole principle and probability. In addition, recurrence relations, another tool for the analysis of computer programs, are discussed.

Looking Back

An early contributor to the study of combinations was Abraham ben Meir ibn Ezra (1092–1167), who was born and died in Spain. Rabbi ben Ezra also worked in astrology, astronomy, philosophy, and medicine. Another early researcher in the area of permutations and combinations was the French mathematician, astronomer, and philosopher Levi ben Gerson (1288–1344).

In 1654 Blaise Pascal (1623–1662), a French mathematician who had been a child prodigy, exchanged a small series of letters with Pierre de Fermat (1601–1665), a French lawyer for whom mathematics was a hobby. This series of letters laid the foundation for the theory of probability. The correspondence between Pascal and Fermat developed when Pascal's friend, the Chevalier de Méré, asked him to solve several dice problems. In addition to making many other important contributions in mathematics and hydrostatics, Pascal invented a mechanical calculator that was very similar to the mechanical calculators used in the 1940s, just before the development of the digital electronic computer.

Blaise Pascal
Académie des Sciences



Pierre de Fermat
Photo Reserchers, Inc.

From Discrete Mathematical Structures, Sixth Edition, Bernard Kolman, Robert C. Busby, Sharon Cutler Ross.
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Counting

1 Permutations

We begin with a simple but general result we will use frequently in this section.

THEOREM 1 Suppose that two tasks T_1 and T_2 are to be performed in sequence. If T_1 can be performed in n_1 ways, and for each of these ways T_2 can be performed in n_2 ways, then the sequence T_1T_2 can be performed in n_1n_2 ways.

Proof
Each choice of a method of performing T_1 will result in a different way of performing the task sequence. There are n_1 such methods, and for each of these we may choose n_2 ways of performing T_2. Thus, in all, there will be n_1n_2 ways of performing the sequence T_1T_2. See Figure 1 for the case where n_1 is 3 and n_2 is 4.

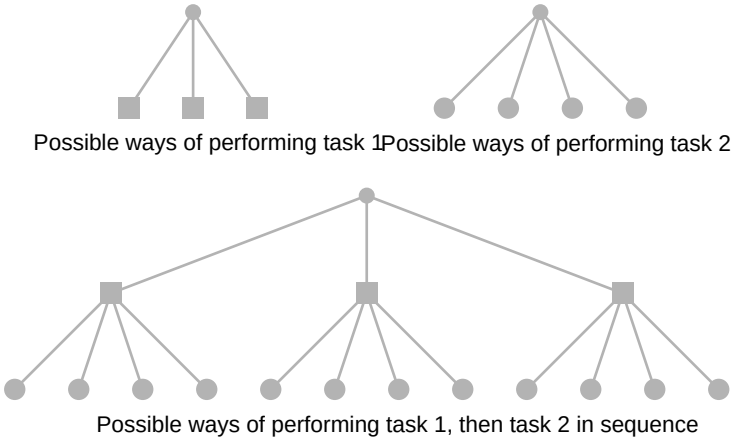


Figure 1

Theorem 1 is sometimes called the *multiplication principle of counting*. It is an easy matter to extend the multiplication principle as follows.

THEOREM 2 Suppose that tasks $T_1, T_2, \ldots, T_k$ are to be performed in sequence. If T_1 can be performed in n_1 ways, and for each of these ways T_2 can be performed in n_2 ways, and for each of these n_1n_2 ways of performing T_1T_2 in sequence T_3 can be performed in n_3 ways, and so on, then the sequence $T_1T_2 \cdots T_k$ can be performed in exactly $n_1n_2 \cdots n_k$ ways.

Proof
This result can be proved by using the principle of mathematical induction on k.

Example 1 A label identifier, for a computer system, consists of one letter followed by three digits. If repetitions are allowed, how many distinct label identifiers are possible?

Solution

There are 26 possibilities for the beginning letter and there are 10 possibilities for each of the three digits. Thus, by the extended multiplication principle, there are $26 \times 10 \times 10 \times 10$ or 26,000 possible label identifiers.

Example 2 Let A be a set with n elements. How many subsets does A have?

Solution

Each subset of A is determined by its characteristic function, and if A has n elements, this function may be described as an array of 0's and 1's having length n. The first element of the array can be filled in two ways (with a 0 or a 1), and this is true for all succeeding elements as well. Thus, by the extended multiplication principle, there are

$$\underbrace{2 \cdot 2 \cdot \cdots \cdot 2}_{n \text{ factors}} = 2^n$$

ways of filling the array, and therefore 2^n subsets of A.

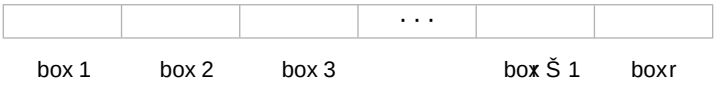
We now turn our attention to the following counting problem. Let A be any set with n elements, and suppose that $1 \le r \le n$.

Problem 1 How many different sequences, each of length r, can be formed using elements from A if

(a) elements in the sequence may be repeated?

(b) all elements in the sequence must be distinct?

First we note that any sequence of length r can be formed by filling r boxes in order from left to right with elements of A. In case (a) we may use copies of elements of A.



Let T_1 be the task "fill box 1," let T_2 be the task "fill box 2," and so on. Then the combined task $T_1 T_2 \cdots T_r$ represents the formation of the sequence.

Case (a). T_1 can be accomplished in n ways, since we may copy any element of A for the first position of the sequence. The same is true for each of the tasks $T_2, T_3, \ldots, T_r$. Then by the extended multiplication principle, the number of sequences that can be formed is

$$\underbrace{n \cdot n \cdot \cdots \cdot n}_{r \text{ factors}} = n^r.$$

We have therefore proved the following result.

THEOREM 3 Let A be a set with n elements and $1 \le r \le n$. Then the number of sequences of length r that can be formed from elements of A, allowing repetitions, is n^r.

Example 3 How many three-letter "words" can be formed from letters in the set $\{x, y, z\}$ if repeated letters are allowed?

Solution

Here n is 4 and r is 3, so the number of such words is $4^3 = 64$, by Theorem 3.

Now we consider case (b) of Problem 1. Here also T_1 can be performed in n ways, since any element of A can be chosen for the first position. Whichever element is chosen, only $(n - 1)$ elements remain, so that T_2 can be performed in $(n - 1)$ ways, and so on, until finally T_r can be performed in $n - (r - 1)$ or

$(n - r + 1)$ ways. Thus, by the extended principle of multiplication, a sequence of r distinct elements from A can be formed in $n(n - 1)(n - 2) \cdots (n - r + 1)$ ways.

A sequence of distinct elements of A is often called a permutation of A taken r at a time. This terminology is standard, and therefore we adopt it, but it is confusing. A better terminology might be a "permutation of elements chosen from A." Many sequences of interest are permutations of some set of n objects taken r at a time. The preceding discussion shows that the number of such sequences depends only on n and r, not on A. This number is often written $_nP_r$ and is called the number of permutations of n objects taken r at a time. We have just proved the following result.

THEOREM 4 If $1 \leq r \leq n$, then $_nP_r$, the number of permutations of n objects taken r at a time, is $n \cdot (n - 1) \cdot (n - 2) \cdot \cdots \cdot (n - r + 1)$.

Example 4 Let A be $\{1, 2, 3, 4\}$. Then the sequences 124, 421, 341, and 243 are some permutations of A taken 3 at a time. The sequences 12, 43, 31, 24, and 21 are examples of different permutations of A taken two at a time. By Theorem 4, the total number of permutations of A taken three at a time is $_4P_3$ or $4 \cdot 3 \cdot 2$ or 24. The total number of permutations of A taken two at a time is $_4P_2$ or $4 \cdot 3$ or 12.

When $r = n$, we are counting the distinct arrangements of the elements of A, with $|A| = n$, into sequences of length n. Such a sequence is simply called a permutation of A. The number of permutations of A is thus $_nP_n$ or $n \cdot (n - 1) \cdot (n - 2) \cdot \cdots \cdot 2 \cdot 1$, if $n \geq 1$. This number is also written $n!$ and is read n factorial. Both $_nP_r$ and $n!$ are built-in functions on many calculators.

Example 5 Let A be $\{a, b, c\}$. Then the possible permutations of A are the sequences abc, acb, bac, bca, cab, and cba.

For convenience, we define $0!$ to be 1. Then for every $n \geq 0$ the number of permutations of n objects is $n!$. If $n \geq 0$ and $0 \leq r \leq n$, we can now give a more compact form for $_nP_r$ as follows:

$$
\begin{aligned}
{}_nP_r &= n \cdot (n - 1) \cdot (n - 2) \cdot \cdots \cdot (n - r + 1) \\
&= \frac{n \cdot (n - 1) \cdot \cdots \cdot (n - r + 1) \cdot (n - r) \cdot (n - r - 1) \cdot \cdots \cdot 2 \cdot 1}{(n - r) \cdot (n - r - 1) \cdot \cdots \cdot 2 \cdot 1} \\
&= \frac{n!}{(n - r)!}.
\end{aligned}
$$

Example 6 Let A consist of all 52 cards in an ordinary deck of playing cards. Suppose that this deck is shuffled and a hand of five cards is dealt. A list of cards in this hand, in the order in which they were dealt, is a permutation of A taken five at a time. Examples would include AH, 3D, 5C, 2H, JS; 2H, 3H, 5H, QH, KD; JH, JD, JS, 4H, 4C; and 3D, 2H, AH, JS, 5C. Note that the first and last hands are the same, but they represent different permutations since they were dealt in a different order. The number of permutations of A taken five at a time is $_{52}P_5 = \frac{52!}{47!}$ or $52 \cdot 51 \cdot 50 \cdot 49 \cdot 48$ or 311,875,200. This is the number of five-card hands that can be dealt if we consider the order in which they were dealt.

Example 7 If A is the set in Example 5, then n is 3 and the number of permutations of A is $3!$ or 6. Thus, all the permutations of A are listed in Example 5, as claimed.

Example 8 How many "words" of three distinct letters can be formed from the letters of the word MAST?

Solution

The number is $_4P_3 = \frac{4!}{(4-3)!}$ or $\frac{4!}{1!}$ or 24.

In Example 8 if the word had been MASS, $_4P_3$ would count as distinct some permutations that cannot be distinguished. For example, if we tag the two S's as S_1 and S_2, then SAS_2 and S_2AS_1 are two of the 24 permutations counted, but without the tags, these are the same "word." We have one more case to consider, permutations with limited repeats.

Example 9 How many distinguishable permutations of the letters in the word BANANA are there?

Solution

We begin by tagging the A's and N's in order to distinguish between them temporarily. For the letters B, A_1, N_1, A_2, N_2, A_3, there are 6! or 720 permutations. Some of these permutations are identical except for the order in which the N's appear; for example, $A_1A_2A_3BN_1N_2$ and $A_1A_2A_3BN_2N_1$. In fact, the 720 permutations can be listed in pairs whose members differ only in the order of the two N's. This means that if the tags are dropped from the N's only $\frac{720}{2}$ or 360 distinguishable permutations remain. Reasoning in a similar way we see that these can be grouped in groups of 3! or 6 that differ only in the order of the three A's. For example, one group of 6 consists of $BNNA_1A_2A_3$, $BNNA_1A_3A_2$, $BNNA_2A_1A_3$, $BNNA_2A_3A_1$, $BNNA_3A_1A_2$, $BNNA_3A_2A_1$. Dropping the tags would change these 6 into the single permutation BNNAAA. Thus, there are $\frac{360}{6}$ or 60 distinguishable permutations of the letters of BANANA.

The following theorem describes the general situation for permutations with limited repeats.

THEOREM 5 The number of distinguishable permutations that can be formed from a collection of n objects where the first object appears k_1 times, the second object k_2 times, and so on, is

$$\frac{n!}{k_1!\,k_2!\cdots k_t!}, \quad \text{where } k_1 + k_2 + \cdots + k_t = n.$$

Example 10 The number of distinguishable "words" that can be formed from the letters of MISSISSIPPI is $\frac{11!}{1!\,4!\,4!\,2!}$ or 34,650.

1 Exercises

1. A bank password consists of two letters of the English alphabet followed by two digits. How many different passwords are there?
2. In a psychological experiment, a person must arrange a square, a cube, a circle, a triangle, and a pentagon in a row. How many different arrangements are possible?
3. A coin is tossed four times and the result of each toss is recorded. How many different sequences of heads and tails are possible?
4. A catered menu is to include a soup, a main course, a dessert, and a beverage. Suppose a customer can select from four soups, five main courses, three desserts, and two beverages. How many different menus can be selected?
5. A fair six-sided die is tossed four times and the numbers shown are recorded in a sequence. How many different sequences are there?

6. Let $A = \{0, 1\}$. A^* denotes the set of all finite sequences of elements of A as defined in Section 3.
   (a) How many strings of length three are there in A^*?
   (b) How many strings of length seven are there in A^*?
7. (a) Compute the number of strings of length four in the set corresponding to the regular expression $(01)^*1$.
   (b) Compute the number of strings of length five in the set corresponding to the regular expression $(01)^*1$.
8. Compute each of the following.
   (a) ${}_4P_4$ (b) ${}_6P_5$ (c) ${}_7P_2$
9. Compute each of the following.
   (a) ${}_nP_{n-1}$ (b) ${}_nP_{n-2}$ (c) ${}_{n+1}P_{n-1}$

In Exercises 10 through 13, compute the number of permutations of the given set.

10. $\{r, s, t, u\}$
11. $\{1, 2, 3, 4, 5\}$
12. $\{a, b, 1, 2, 3, c\}$
13. $\{4, 7, 10, 13\}$

In Exercises 14 through 16, find the number of permutations of A taken r at a time.

14. $A = \{1, 2, 3, 4, 5, 6, 7\}$, $r = 3$
15. $A = \{a, b, c, d, e, f\}$, $r = 2$
16. $A = \{x \mid x \text{ is an integer and } x^2 < 16\}$, $r = 4$
17. In how many ways can six men and six women be seated in a row if
   (a) any person may sit next to any other?
   (b) men and women must occupy alternate seats?
18. Find the number of different permutations of the letters in the word GROUP.
19. How many different arrangements of the letters in the word BOUGHT can be formed if the vowels must be kept next to each other?
20. Find the number of distinguishable permutations of the letters in BOOLEAN.
21. Find the number of distinguishable permutations of the letters in PASCAL.
22. Find the number of distinguishable permutations of the letters in ASSOCIATIVE.
23. Find the number of distinguishable permutations of the letters in REQUIREMENTS.
24. In how many ways can seven people be seated in a circle?
25. How many different ways can n people be seated around a circular table?
26. Give a proof of your result for Exercise 25.
27. A bookshelf is to be used to display six new books. Suppose there are eight computer science books and five French books from which to choose. If we decide to show four computer science books and two French books and we are required to keep the books in each subject together, how many different displays are possible?
28. Three fair six-sided dice are tossed and the numbers showing on the top faces are recorded as a triple. How many different records are possible?
29. Prove that $n \cdot {}_{n-1}P_{n-1} = {}_nP_n$.
30. Most versions of Pascal allow variable names to consist of eight letters or digits with the requirement that the first character must be a letter. How many eight-character variable names are possible?
31. Until recently, U.S. telephone area codes were three-digit numbers whose middle digit was 0 or 1. Codes whose last two digits are 1's are used for other purposes (for example, 911). With these conditions how many area codes were available?
32. How many Social Security numbers can be assigned at any one time? Identify any assumptions you have made.
33. How many zeros are there at the end of 12!? at the end of 26!? at the end of 52!?
34. Give a procedure for determining the number of zeros at the end of $n!$. Justify your procedure.

In Exercises 35 through 37, use the following information. There are three routes from Atlanta to Athens, four routes from Athens to Augusta, and two routes from Atlanta to Augusta.

35. (a) How many ways are there to travel from Atlanta to Augusta?
   (b) How many ways are there to travel from Athens to Atlanta?
36. (a) How many different ways can the round trip between Atlanta and Augusta be made?
   (b) How many different ways can the round trip between Atlanta and Augusta be made if each route is used only once?
37. (a) How many different ways can the round trip between Augusta and Athens be made if the trip does not go through Atlanta?
   (b) How many different ways can the round trip between Augusta and Athens be made if the trip does not go through Atlanta and each route is used only once?

2 Combinations

The multiplication principle and the counting methods for permutations all apply to situations where order matters. In this section we look at some counting problems where order does not matter.

Problem 2 Let A be any set with n elements and $0 \le r \le n$. How many different subsets of A are there, each with r elements?

The traditional name for an r-element subset of an n-element set A is a combination of A, taken r at a time.

Example 1 Let $A = \{1, 2, 3, 4\}$. The following are all distinct combinations of A, taken three at a time: $A_1 = \{1, 2, 3\}$, $A_2 = \{1, 2, 4\}$, $A_3 = \{1, 3, 4\}$, $A_4 = \{2, 3, 4\}$. Note that these are subsets, not sequences. Thus $A_1 = \{2, 1, 3\} = \{2, 3, 1\} = \{1, 3, 2\} = \{3, 1, 2\} = \{3, 2, 1\}$. In other words, when it comes to combinations, unlike permutations, the order of the elements is irrelevant.

Example 2 Let A be the set of all 52 cards in an ordinary deck of playing cards. Then a combination of A, taken five at a time, is just a hand of five cards regardless of how these cards were dealt.

We now want to count the number of r-element subsets of an n-element set A. This is most easily accomplished by using what we already know about permutations. Observe that each permutation of the elements of A, taken r at a time, can be produced by performing the following two tasks in sequence.

Task 1: Choose a subset B of A containing r elements.
Task 2: Choose a particular permutation of B.

We are trying to compute the number of ways to choose B. Call this number C. Then task 1 can be performed C ways, and task 2 can be performed $r!$ ways. Thus the total number of ways of performing both tasks is, by the multiplication principle, $C \cdot r!$. But it is also ${}_nP_r$. Hence,

$$C \cdot r! = {}_nP_r = \frac{n!}{(n - r)!}.$$

Therefore,

$$C = \frac{n!}{r!\,(n - r)!}.$$

We have proved the following result.

THEOREM 1 Let A be a set with $|A| = n$, and let $0 \le r \le n$. Then the number of combinations of the elements of A, taken r at a time, that is, the number of r-element subsets of A, is

$$\frac{n!}{r!\,(n - r)!}.$$

Note again that the number of combinations of A, taken r at a time, does not depend on A, but only on n and r. This number is often written ${}_nC_r$ and is called the number of combinations of n objects taken r at a time. We have

$${}_nC_r = \frac{n!}{r!\,(n - r)!}.$$

This computation is a built-in function on many calculators.

Example 3 Compute the number of distinct five-card hands that can be dealt from a deck of 52 cards.

Solution

This number is $_{52}C_5$ because the order in which the cards were dealt is irrelevant. $_{52}C_5 = \frac{52!}{5!\,47!}$ or 2,598,960. Compare this number with the number computed in Example 6, Section 1.

In the discussion of permutations, we considered cases where repetitions are allowed. We now look at one such case for combinations.

Consider the following situation. A radio station offers a prize of three CDs from the Top Ten list. The choice of CDs is left to the winner, and repeats are allowed. The order in which the choices are made is irrelevant. To determine the number of ways in which prize winners can make their choices, we use a problem-solving technique we have used before; we model the situation with one we already know how to handle.

Suppose choices are recorded by the station's voice mail system. After properly identifying herself, a winner is asked to press 1 if she wants CD number n and to press 2 if she does not. If 1 is pressed, the system asks again about CD number n. When 2 is pressed, the system asks about the next CD on the list. When three 1's have been recorded, the system tells the caller the selected CDs will be shipped. A record must be created for each of these calls. A record will be a sequence of 1's and 2's. Clearly there will be three 1's in the sequence. A sequence may contain as many as nine 2's, for example, if the winner refuses the first nine CDs and chooses three copies of CD number 10. Our model for counting the number of ways a prize winner can choose her three CDs is the following. Each three-CD selection can be represented by an array containing three 1's and nine 2's or blanks, or a total of twelve cells. Some possible records are 222122122221 (selecting numbers 4, 6, 10), 121bbbbbbbbb (selecting number 1 and two copies of number 2), and 222222222111 (selecting three copies of number 10). The number of ways to select three cells of the array to hold 1's is $_{12}C_3$ since the array has 3 + 9 or 12 cells and the order in which this selection is made does not matter. The following theorem generalizes this discussion.

THEOREM 2 Suppose k selections are to be made from n items without regard to order and repeats are allowed, assuming at least k copies of each of the n items. The number of ways these selections can be made is $_{(n+k-1)}C_k$.

Example 4 In how many ways can a prize winner choose three CDs from the Top Ten list if repeats are allowed?

Solution

Here n is 10 and k is 3. By Theorem 2, there are $_{10+3-1}C_3$ or $_{12}C_3$ ways to make the selections. The prize winner can make the selection in 220 ways.

In general, when order matters, we count the number of sequences or permutations; when order does not matter, we count the number of subsets or combinations.

Some problems require that the counting of permutations and combinations be combined or supplemented by the direct use of the addition or the multiplication principle.

Example 5 Suppose that a valid computer password consists of seven characters, the first of which is a letter chosen from the set {A, B, C, D, E, F, G} and the remaining six characters are letters chosen from the English alphabet or a digit. How many different passwords are possible?

Solution

A password can be constructed by performing the tasks T_1 and T_2 in sequence.

Task 1: Choose a starting letter from the set given.

Task 2: Choose a sequence of letter and digits. Repeats are allowed.

Task T_1 can be performed in ${}_7C_1$ or 7 ways. Since there are 26 letters and 10 digits that can be chosen for each of the remaining six characters, and since repeats are allowed, task T_2 can be performed in 36^6 or 2,176,782,336 ways. By the multiplication principle, there are $7 \cdot 2176782336$ or 15,237,476,352 different passwords.

Example 6 How many different seven-person committees can be formed each containing three women from an available set of 20 women and four men from an available set of 30 men?

Solution

In this case a committee can be formed by performing the following two tasks in succession:

Task 1: Choose three women from the set of 20 women.

Task 2: Choose four men from the set of 30 men.

Here order does not matter in the individual choices, so we are merely counting the number of possible subsets. Thus task 1 can be performed in ${}_{20}C_3$ or 1140 ways and task 2 can be performed in ${}_{30}C_4$ or 27,405 ways. By the multiplication principle, there are $(1140)(27405)$ or 31,241,700 different committees.

2 Exercises

1. Compute each of the following.
   (a) ${}_7C_7$ (b) ${}_7C_4$ (c) ${}_{16}C_5$
2. Compute each of the following.
   (a) ${}_nC_{n-1}$ (b) ${}_nC_{n-2}$ (c) ${}_{n+1}C_{n-1}$
3. Show that ${}_nC_r = {}_nC_{n-r}$.
4. In how many ways can a committee of three faculty members and two students be selected from seven faculty members and eight students?
5. In how many ways can a six-card hand be dealt from a deck of 52 cards?
6. At a certain college, the housing office has decided to appoint, for each floor, one male and one female residential advisor. How many different pairs of advisors can be selected for a seven-story building from 12 male candidates and 15 female candidates?
7. A microcomputer manufacturer who is designing an advertising campaign is considering six magazines, three newspapers, two television stations, and four radio stations. In how many ways can six advertisements be run if
   (a) all six are to be in magazines?
   (b) two are to be in magazines, two are to be in newspapers, one is to be on television, and one is to be on radio?
8. How many different eight-card hands with five red cards and three black cards can be dealt from a deck of 52 cards?
9. (a) Find the number of subsets of each possible size for a set containing four elements.
   (b) Find the number of subsets of each possible size for a set containing n elements.

For Exercises 10 through 13, suppose that an urn contains 15 balls, of which eight are red and seven are black.

10. In how many ways can five balls be chosen so that
    (a) all five are red?
    (b) all five are black?
11. In how many ways can five balls be chosen so that
    (a) two are red and three are black?
    (b) three are red and two are black?
12. In how many ways can five balls be chosen so that at most three are black?
13. In how many ways can five balls be chosen so that at least two are red?
14. Give a model in terms of combinations to count the number of strings of length 6 in $\{0, 1\}$ that have exactly four ones.

15. Give a model in terms of combinations to count the number of ways to arrange three of seven people in order from youngest to oldest.
16. A committee of six people with one person designated as chair of the committee is to be chosen. How many different committees of this type can be chosen from a group of 10 people?
17. A gift certi cate at a local bookstore allows the recipient to choose six books from the combined list of ten best-selling ction books and ten bestselling non ction books. In how many different ways can the selection of six books be made?
18. The college food plan allows a student to chose three pieces of fruit each day. The fruits available are apples, bananas, peaches, pears, and plums. For how many days can a student make a different selection?
19. Show that $_{n+1}C_r = {}_nC_{r-1} + {}_nC_r$.
20. (a) How many ways can a student choose eight out of ten questions to answer on an exam?
    (b) How many ways can a student choose eight out of ten questions to answer on an exam if the rst three questions must be answered?
21. Five fair coins are tossed and the results are recorded.
    (a) How many different sequences of heads and tails are possible?
    (b) How many of the sequences in part (a) have exactly one head recorded?
    (c) How many of the sequences in part (a) have exactly three heads recorded?
22. Three fair six-sided dice are tossed and the numbers showing on top are recorded.
    (a) How many different record sequences are possible?
    (b) How many of the records in part (a) contain exactly one six?
    (c) How many of the records in part (a) contain exactly two fours?
23. If n fair coins are tossed and the results recorded, how many
    (a) record sequences are possible?
    (b) sequences contain exactly three tails, assuming 3?
    (c) sequences contain exactly heads, assuming k?
24. If n fair six-sided dice are tossed and the numbers showing on top are recorded, how many
    (a) record sequences are possible?
    (b) sequences contain exactly one six?
    (c) sequences contain exactly four twos, assuming 4?
25. How many ways can you choose three of seven ction books and two of six non ction books to take with you on your vacation?
26. For the driving part of your vacation you will take 6 of the 35 rock cassettes in your collection, 3 of the 22 classical cassettes, and 1 of the 8 comedy cassettes. In how many ways can you make your choice(s)?
27. The array commonly called Pascal's triangle can be de ned by giving enough information to establish its pattern.

```
                1
              1   1
            1   2   1
          1   3   3   1
        1   4   6   4   1
```

    (a) Write the next three rows of Pascal's triangle.
    (b) Give a rule for building the next row from the previous row(s).
28. Pascal's triangle can also be de ned by an explicit pattern. Use the results of Exercises 9 and 27 to give an explicit rule for building the nth row of Pascal's triangle.
29. Explain the connections between Exercises 19, 27, and 28.
30. The list of numbers in any row of Pascal's triangle reads the same from left to right as it does from right to left. Such a sequence is called a palindrome Use the results of Exercise 28 to prove that each row of Pascal's triangle is a palindrome.
31. (a) The sum of the entries in the second row of Pascal's triangle is____.
    (b) The sum of the entries in the third row of Pascal's triangle is____.
    (c) The sum of the entries in the fourth row of Pascal's triangle is____.
32. Make a conjecture about the sum of the entries in the row of Pascal's triangle and prove it.
33. Marcy wants to buy a book of poems. If she wants to read a different set of three poems every day for a year (365 days), what is the minimum number of poems the book should contain?

3 Pigeonhole Principle

In this section we introduce another proof technique, one that often makes use of the counting methods we have discussed.

THEOREM 1
The Pigeonhole Principle

If n pigeons are assigned to m pigeonholes, and $m < n$, then at least one pigeonhole contains two or more pigeons.

Proof
Suppose each pigeonhole contains at most 1 pigeon. Then at most m pigeons have been assigned. But since $m < n$, not all pigeons have been assigned pigeonholes. This is a contradiction. At least one pigeonhole contains two or more pigeons.

This informal and almost trivial-sounding theorem is easy to use and has unexpected power in proving interesting consequences.

Example 1

If eight people are chosen in any way from some group, at least two of them will have been born on the same day of the week. Here each person (pigeon) is assigned to the day of the week (pigeonhole) on which he or she was born. Since there are eight people and only seven days of the week, the pigeonhole principle tells us that at least two people must be assigned to the same day of the week.

Note that the pigeonhole principle provides an *existence proof*; there must be an object or objects with a certain characteristic. In Example 1, this characteristic is having been born on the same day of the week. The pigeonhole principle guarantees that there are at least two people with this characteristic but gives no information on identifying these people. Only their existence is guaranteed. In contrast, a constructive proof guarantees the existence of an object or objects with a certain characteristic by actually constructing such an object or objects. For example, we could prove that given two rational numbers p and q there is a rational number between them by showing that $\frac{p+q}{2}$ is between p and q.

In order to use the pigeonhole principle we must identify pigeons (objects) and pigeonholes (categories of the desired characteristic) and be able to count the number of pigeons and the number of pigeonholes.

Example 2

Show that if any five numbers from 1 to 8 are chosen, then two of them will add to 9.

Solution

Construct four different sets, each containing two numbers that add up to 9 as follows: $A_1 = \{1, 8\}$, $A_2 = \{2, 7\}$, $A_3 = \{3, 6\}$, $A_4 = \{4, 5\}$. Each of the five numbers chosen must belong to one of these sets. Since there are only four sets, the pigeonhole principle tells us that two of the chosen numbers belong to the same set. These numbers add up to 9.

Example 3

Show that if any 11 numbers are chosen from the set $\{1, 2, \ldots, 20\}$, then one of them will be a multiple of another.

Solution

The key to solving this problem is to create 10 or fewer pigeonholes in such a way that each number chosen can be assigned to only one pigeonhole, and when x and y are assigned to the same pigeonhole we are guaranteed that either $x \mid y$ or $y \mid x$. Factors are a natural feature to explore. There are eight prime numbers between 1 and 20, but knowing that x and y are multiples of the same prime will not guarantee that either $x \mid y$ or $y \mid x$. We try again. There are ten odd numbers between 1 and 20. Every positive integer n can be written as $n = 2^k m$, where m is odd and $k \geq 0$. This can be seen by simply factoring all powers of 2 (if any) out of n. In this case let us call m the odd part of n. If 11 numbers are chosen from the set

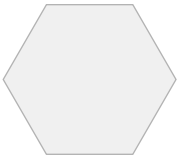


Figure 2

{1, 2, . . . , 20}, then two of them must have the same odd part. This follows from the pigeonhole principle since there are 11 numbers (pigeons), but only 10 odd numbers between 1 and 20 (pigeonholes) that can be odd parts of these numbers.

Let n_1 and n_2 be two chosen numbers with the same odd part. We must have $n_1 = 2^{k_1}m$ and $n_2 = 2^{k_2}m$, for some k_1 and k_2. If $k_1 \quad k_2$, then n_1 is a multiple of n_2; otherwise, n_2 is a multiple of n_1.

Example 4

Consider the region shown in Figure 2. It is bounded by a regular hexagon whose sides are of length 1 unit. Show that if any seven points are chosen in this region, then two of them must be no farther apart than 1 unit.

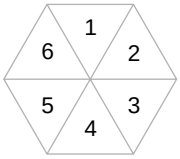


Figure 3

Solution

Divide the region into six equilateral triangles, as shown in Figure 3. If seven points are chosen in the region, we can assign each of them to a triangle that contains it. If the point belongs to several triangles, arbitrarily assign it to one of them. Then the seven points are assigned to six triangular regions, so by the pigeonhole principle, at least two points must belong to the same region. These two cannot be more than one unit apart. (Why?)

Example 5

Shirts numbered consecutively from 1 to 20 are worn by the 20 members of a bowling league. When any three of these members are chosen to be a team, the league proposes to use the sum of their shirt numbers as a code number for the team. Show that if any eight of the 20 are selected, then from these eight one may form at least two different teams having the same code number.

Solution

From the eight selected bowlers, we can form a total of $C(8,3)$ or 56 different teams. These will play the role of pigeons. The largest possible team code number is $18 + 19 + 20$ or 57, and the smallest possible is $1 + 2 + 3$ or 6. Thus only the 52 code numbers (pigeonholes) between 6 and 57 inclusive are available for the 56 possible teams. By the pigeonhole principle, at least two teams will have the same code number. The league should use another way to assign team numbers.

The Extended Pigeonhole Principle

Note that if there are n pigeonholes and more than n pigeons, two or more pigeons will have to be assigned to at least one of the pigeonholes. (Consider the most even distribution of pigeons you can make.) In general, if the number of pigeons is much larger than the number of pigeonholes, Theorem 1 can be restated to give a stronger conclusion.

First a word about notation. If n and m are positive integers, then n/m stands for the largest integer less than or equal to the rational number n/m. Thus $3/2$ is 1, $9/4$ is 2, and $6/3$ is 2.

THEOREM 2

The Extended Pigeonhole Principle

If n pigeons are assigned to m pigeonholes, then one of the pigeonholes must contain at least $(n - 1)/m + 1$ pigeons.

Proof (by contradiction)

If each pigeonhole contains no more than $(n - 1)/m$ pigeons, then there are at most $m \cdot (n - 1)/m$ $m \cdot (n - 1)/m = n - 1$ pigeons in all. This contradicts our hypothesis, so one of the pigeonholes must contain at least $(n - 1)/m + 1$ pigeons.

This proof by contradiction uses the fact that there are two ways to count the total number of pigeons, the original count n and as the product of the number of pigeonholes times the number of pigeons per pigeonhole.

Example 6 We give an extension of Example 1. Show that if any 30 people are selected, then one may choose a subset of five so that all five were born on the same day of the week.

Solution

Assign each person to the day of the week on which she or he was born. Then 30 pigeons are being assigned to 7 pigeonholes. By the extended pigeonhole principle with $n = 30$ and $m = 7$, at least $(30 - 1)/7 + 1$ or 5 of the people must have been born on the same day of the week.

Example 7 Show that if 30 dictionaries in a library contain a total of 61,327 pages, then one of the dictionaries must have at least 2045 pages.

Solution

Let the pages be the pigeons and the dictionaries the pigeonholes. Assign each page to the dictionary in which it appears. Then by the extended pigeonhole principle, one dictionary must contain at least $61,326/30 + 1$ or 2045 pages.

3 Exercises

1. If thirteen people are assembled in a room, show that at least two of them must have their birthday in the same month.
2. Show that if seven integers from 1 to 12 are chosen, then two of them will add up to 13.
3. Let T be an equilateral triangle whose sides are of length 1 unit. Show that if any five points are chosen lying on or inside the triangle, then two of them must be no more than $\frac{1}{2}$ unit apart.
4. Show that if any eight positive integers are chosen, two of them will have the same remainder when divided by 7.
5. Show that if seven colors are used to paint 50 bicycles, at least eight bicycles will be the same color.
6. Ten people volunteer for a three-person committee. Every possible committee of three that can be formed from these ten names is written on a slip of paper, one slip for each possible committee, and the slips are put in ten hats. Show that at least one hat contains 12 or more slips of paper.
7. Six friends discover that they have a total of $21.61 with them on a trip to the movies. Show that one or more of them must have at least $3.61.
8. A store has an introductory sale on 12 types of candy bars. A customer may choose one bar of any five different types and will be charged no more than $1.75. Show that although different choices may cost different amounts, there must be at least two different ways to choose so that the cost will be the same for both choices.
9. If the store in Exercise 8 allows repetitions in the choices, show that there must be at least ten ways to make different choices that have the same cost.
10. Show that there must be at least 90 ways to choose six integers from 1 to 15 so that all the choices have the same sum.
11. How many friends must you have to guarantee at least five of them will have birthdays in the same month?
12. Show that if five points are selected in a square whose sides have length 1 inch, at least two of the points must be no more than $\frac{\sqrt{2}}{2}$ inches apart.
13. Let A be an 8×8 Boolean matrix. If the sum of the entries in A is 51, prove that there is a row i and a column j in A such that the entries in row i and in column j add up to more than 13.
14. Write an exercise similar to Exercise 13 for a 12×12 Boolean matrix.
15. Prove that if any 14 integers from 1 to 25 are chosen, then one of them is a multiple of another.
16. How large a subset of the integers from 1 to 50 must be chosen to guarantee that one of the numbers in the subset is a multiple of another number in the subset?
17. How large a subset of the integers from 1 to n must be chosen to guarantee that one of the numbers in the subset is a multiple of another number in the subset?
18. Twenty disks numbered 1 through 20 are placed face down on a table. Disks are selected one at a time and turned over until 10 disks have been chosen. If two of the

disks add up to 21, the player loses. Is it possible to win this game?

19. Suppose the game in Exercise 18 has been changed so that 12 disks are chosen. Is it possible to win this game?

20. Complete the following proof. It is not possible to arrange the numbers 1, 2, 3, . . . , 10 in a circle so that every triple of consecutively placed numbers has a sum less than 15. Proof: In any arrangement of 1, 2, 3, . . . , 10 in a circle, there are ____ triples of consecutively placed numbers, because ____. Each number appears in ____ of these triples. If the sum of each triple were less than 15, then the total sum of all triples would be less than ____ times 15 or ____. But 1 + 2 + 3 + · · · + 10 is 55 and since each number appears in ____ triples, the total sum should be ____ times 55. This is a contradiction so not all triples can have a sum less than 15.

21. Prove that any sequence of six integers must contain a subsequence whose sum is divisible by six. (Hint: Consider the sums $c_1, c_1 + c_2, c_1 + c_2 + c_3, \ldots$ and the possible remainders when dividing by six.)

22. Prove that any sequence of n integers must contain a subsequence whose sum is divisible by n.

23. Show that any set of six positive integers whose sum is 13 must contain a subset whose sum is three.

24. Use the pigeonhole principle to prove that any rational number can be expressed as a number with a finite or repeating decimal part.

25. The computer classroom has 12 PCs and 5 printers. What is the minimum number of connections that must be made to guarantee that any set of 5 or fewer PCs can access printers at the same time?

4 Elements of Probability

Another area where counting techniques are important is probability theory. In this section we present a brief introduction to probability.

Many experiments do not yield exactly the same results when performed repeatedly. For example, if we toss a coin, we are not sure if we will get heads or tails, and if we toss a die, we have no way of knowing which of the six possible numbers will turn up. Experiments of this type are called *probabilistic*, in contrast to deterministic experiments, whose outcome is always the same.

Sample Spaces

A set A consisting of all the outcomes of an experiment is called a *sample space* of the experiment. With a given experiment, we can often associate more than one sample space, depending on what the observer chooses to record as an outcome.

Example 1 Suppose that a nickel and a quarter are tossed in the air. We describe three possible sample spaces that can be associated with this experiment.

1. If the observer decides to record as an outcome the number of heads observed, the sample space A is $A = \{0, 1, 2\}$.
2. If the observer decides to record the sequence of heads (H) and tails (T) observed, listing the condition of the nickel first and then that of the quarter, then the sample space A is $A = \{HH, HT, TH, TT\}$.
3. If the observer decides to record the fact that the coins match (M) (both heads or both tails) or do not match (N), then the sample space A is $A = \{M, N\}$.

We thus see that in addition to describing the experiment, we must indicate exactly what the observer wishes to record. Then the set of all outcomes of this type becomes the sample space for the experiment.

A sample space may contain a finite or an infinite number of outcomes, but in this chapter, we need only finite sample spaces.

Example 2 Determine the sample space for an experiment consisting of tossing a six-sided die twice and recording the sequence of numbers showing on the top face of the die after each toss.

Solution

An outcome of the experiment can be represented by an ordered pair of numbers (n, m), where n and m can be 1, 2, 3, 4, 5, or 6. Thus the sample space A contains 6×6 or 36 elements (by the multiplication principle).

Example 3 An experiment consists of drawing three coins in succession from a box containing four pennies and five dimes, and recording the sequence of results. Determine the sample space of this experiment.

Solution

An outcome can be recorded as a sequence of length 3 constructed from the letters P (penny) and D (dime). Thus the sample space A is {PPP, PPD, PDP, PDD, DPP, DPD, DDP, DDD}.

Events

A statement about the outcome of an experiment, which for a particular outcome will be either true or false, is said to describe an *event*. Thus for Example 2, the statements, "Each of the numbers recorded is less than 3" and "The sum of the numbers recorded is 4" would describe events. The event described by a statement is taken to be the set of all outcomes for which the statement is true. With this interpretation, any event can be considered a subset of the sample space. Thus the event E described by the first statement is $E = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$. Similarly, the event F described by the second statement is $F = \{(1, 3), (2, 2), (3, 1)\}$.

Example 4 Consider the experiment in Example 2. Determine the events described by each of the following statements.

(a) The sum of the numbers showing on the top faces is 8.

(b) The sum of the numbers showing on the top faces is at least 10.

Solution

(a) The event consists of all ordered pairs whose sum is 8. Thus the event is $\{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$.

(b) The event consists of all ordered pairs whose sum is 10, 11, or 12. Thus the event is $\{(4, 6), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)\}$.

If A is a sample space of an experiment, then A itself is an event called the *certain event* and the empty subset $\emptyset$ of A is called the *impossible event*.

Since events are sets, we can combine them by applying the operations of union, intersection, and complementation to form new events. The sample space A is the universal set for these events. Thus if E and F are events, we can form the new events $E \cup F$, $E \cap F$, and $\overline{E}$. What do these new events mean in terms of the experiment? An outcome of the experiment belongs to $E \cup F$ when it belongs to E or F (or both). In other words, the event $E \cup F$ occurs exactly when E or F occurs. Similarly, the event $E \cap F$ occurs if and only if both E and F occur. Finally, $\overline{E}$ occurs if and only if E does not occur.

Example 5 Consider the experiment of tossing a die and recording the number on the top face. Let E be the event that the number is even and F be the event that the number is prime. Then $E = \{2, 4, 6\}$ and $F = \{2, 3, 5\}$. The event that the number showing is either even or prime is $E \cup F = \{2, 3, 4, 5, 6\}$. The event that the number showing is an even prime is $E \cap F = \{2\}$. Finally, the event that the number

showing is not even is $\overline{E} = \{1, 3, 5\}$ and the event that the number showing is not prime is $\overline{F} = \{1, 4, 6\}$.

Events E and F are said to be **mutually exclusive** or **disjoint** if $E \cap F = \{\}$. If E and F are mutually exclusive events, then E and F cannot both occur at the same time; if E occurs, then F does not occur, and if F occurs, then E does not. If $E_1, E_2, \ldots, E_n$ are all events, then we say that these sets are **mutually exclusive**, or **disjoint**, if each pair of them is mutually exclusive. Again, this means that at most one of the events can occur on any given outcome of the experiment.

Assigning Probabilities to Events

In probability theory, we assume that each event E has been assigned a number $p(E)$ called the **probability** of the event E. We now look at probabilities. We will investigate ways in which they can be assigned, properties they must satisfy, and the meaning that can be given to them.

The number $p(E)$ reflects our assessment of the likelihood that the event E will occur. More precisely, suppose the underlying experiment is performed repeatedly, and that after n such performances, the event E has occurred n_E times. Then the fraction $f_E = n_E/n$, called the **frequency of occurrence** of E in n trials, is a measure of the likelihood that E will occur. When we assign the probability $p(E)$ to the event E, it means that in our judgment or experience, we believe that the fraction f_E will tend ever closer to a certain number as n becomes larger, and that $p(E)$ is this number. Thus probabilities can be thought of as idealized frequencies of occurrence of events, to which actual frequencies of occurrence will tend when the experiment is performed repeatedly.

Example 6 Suppose an experiment is performed 2000 times, and the frequency of occurrence f_E of an event E is recorded after 100, 500, 1000, and 2000 trials. Table 1 summarizes the results.

TABLE 1

Number of Repetitions of the Experiment	n_E	$f_E = n_E/n$
100	48	0.48
500	259	0.518
1000	496	0.496
2000	1002	0.501

Based on this table, it appears that the frequency f_E approaches $\frac{1}{2}$ as n becomes larger. It could therefore be argued that $p(E)$ should be set equal to $\frac{1}{2}$. On the other hand, one might require more extensive evidence before assigning the value of $p(E)$. In any case, this sort of evidence can never "prove" that $p(E)$ is $\frac{1}{2}$. It only serves to make this a plausible assumption.

If probabilities assigned to various events are to represent frequencies of occurrence of the events meaningfully, as explained previously, then they cannot be assigned in a totally arbitrary way. They must satisfy certain conditions. In the first place, since every frequency f_E must satisfy the inequalities $0 \le f_E \le 1$, it is only reasonable to assume that

P1: $0 \le p(E) \le 1$ for every event E in A.

Also, since the event A must occur every time (every outcome belongs to A), and the event $\emptyset$ cannot occur, we assume that

$$\text{P2:}\quad p(A) = 1 \text{ and } p(\emptyset) = 0.$$

Finally, if $E_1, E_2, \ldots, E_k$ are mutually exclusive events, then

$$n_{(E_1 \text{ or } E_2 \text{ or } \cdots \text{ or } E_k)} = n_{E_1} + n_{E_2} + \cdots + n_{E_k},$$

since only one of these events can occur at a time. If we divide both sides of this equation by n, we see that the frequencies of occurrence must satisfy a similar equation. We therefore assume

$$\text{P3:}\quad p(E_1 \text{ or } E_2 \text{ or } \cdots \text{ or } E_k) = p(E_1) + p(E_2) + \cdots + p(E_k)$$

whenever the events are mutually exclusive. If the probabilities are assigned to all events in such a way that P1, P2, and P3 are always satisfied, then we have a probability space. We call P1, P2, and P3 the axioms for a probability space.

It is important to realize that mathematically, no demands are made on a probability space except those given by the probability axioms P1, P2, and P3. Probability theory begins with all probabilities assigned, and then investigates consequences of any relations between these probabilities. No mention is made of how the probabilities were assigned. However, the mathematical conclusions will be useful in an actual situation only if the probabilities assigned reflect what actually occurs in that situation.

Experimentation is not the only way to determine reasonable probabilities for events. The probability axioms can sometimes provide logical arguments for choosing certain probabilities.

Example 7

Consider the experiment of tossing a coin and recording whether heads or tails results. Consider the events E: heads turns up and F: tails turns up. The mechanics of the toss are not controllable in detail. Thus in the absence of any defect in the coin that might unbalance it, one may argue that E and F are equally likely to occur. There is a symmetry in the situation that makes it impossible to prefer one outcome over the other. This argument lets us compute what the probabilities of E and F must be.

We have assumed that $p(E) = p(F)$, and it is clear that E and F are mutually exclusive events and $A = E \text{ or } F$. Thus, using the properties P2 and P3, we see that $1 = p(A) = p(E) + p(F) = 2p(E)$ since $p(E) = p(F)$. This shows that $p(E) = \frac{1}{2} = p(F)$. One may often assign appropriate probabilities to events by combining the symmetry of situations with the axioms of probability.

Finally, we will show that the problem of assigning probabilities to events can be reduced to the consideration of the simplest cases. Let A be a probability space. We assume that A is finite, that is, $A = \{x_1, x_2, \ldots, x_n\}$. Then each event $\{x_k\}$, consisting of just one outcome, is called an elementary event. For simplicity, let us write p_k for $p(\{x_k\})$. Then p_k is called the elementary probability corresponding to the outcome x_k. Since the elementary events are mutually exclusive and their union is A, the axioms of probability tell us that

$$\text{EP1:}\quad 0 \le p_k \le 1 \text{ for all } k$$

$$\text{EP2:}\quad p_1 + p_2 + \cdots + p_n = 1.$$

If E is any event in A, say $E = \{x_{i_1}, x_{i_2}, \ldots, x_{i_m}\}$, then we can write $E = \{x_{i_1}\} \text{ or } \{x_{i_2}\} \text{ or } \cdots \text{ or } \{x_{i_m}\}$. This means, by axiom P2, that $p(E) = p_{i_1} + p_{i_2} + \cdots + p_{i_m}$. Thus if we know the elementary probabilities, then we can compute the probability of any event E.

Example 8 Suppose that an experiment has a sample space $A = \{1, 2, 3, 4, 5, 6\}$ and that the elementary probabilities have been determined as follows:

$$p_1 = \frac{1}{12}, \quad p_2 = \frac{1}{12}, \quad p_3 = \frac{1}{3}, \quad p_4 = \frac{1}{6}, \quad p_5 = \frac{1}{4}, \quad p_6 = \frac{1}{12}.$$

Let E be the event "The outcome is an even number." Compute $p(E)$.

Solution

Since $E = \{2, 4, 6\}$, we see that

$$p(E) = p_2 + p_4 + p_6 = \frac{1}{12} + \frac{1}{6} + \frac{1}{12} \quad \text{or} \quad \frac{1}{3}.$$

In a similar way we can determine the probability of any event in A.

Thus we see that the problem of assigning probabilities to all events in a consistent way can be reduced to the problem of finding numbers $p_1, p_2, \ldots, p_n$ that satisfy EP1 and EP2. Again, mathematically speaking, there are no other restrictions on the p_k's. However, if the mathematical structure that results is to be useful in a particular situation, then the p_k's must reflect the actual behavior occurring in that situation.

Equally Likely Outcomes

Let us assume that all outcomes in a sample space A are equally likely to occur. This is, of course, an assumption, and so cannot be proved. We would make such an assumption if experimental evidence or symmetry indicated that it was appropriate in a particular situation (see Example 7). Actually these situations arise commonly. One additional piece of terminology is customary. Sometimes experiments involve choosing an object, in a nondeterministic way, from some collection. If the selection is made in such a way that all objects have an equal probability of being chosen, we say that we have made a *random selection* or chosen an object *at random* from the collection. We will often use this terminology to specify examples of experiments with equally likely outcomes.

Suppose that $|A| = n$ and these n outcomes are equally likely. Then the elementary probabilities are all equal, and since they must add up to 1, this means that each elementary probability is $1/n$. Now let E be an event that contains k outcomes, say $E = \{x_1, x_2, \ldots, x_k\}$. Since all elementary probabilities are $1/n$, we must have

$$p(E) = \underbrace{\frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n}}_{k \text{ summands}} = \frac{k}{n}.$$

Since $k = |E|$, we have the following principle: If all outcomes are equally likely, then for every event E

$$p(E) = \frac{|E|}{|A|} = \frac{\text{total number of outcomes in } E}{\text{total number of outcomes}}.$$

In this case, the computation of probabilities reduces to counting numbers of elements in sets. For this reason, the methods of counting discussed in the earlier sections of this chapter are quite useful.

Example 9 Choose four cards at random from a standard 52-card deck. What is the probability that four kings will be chosen?

29. Solve the recurrence relation of Example 2.

30. Using the argument in Example 3 for P_r would produce ${}_nP_r = r \cdot {}_nP_{r-1}$. But this is easily shown to be false for nearly all choices of n and r. Explain why the argument is not valid.

31. For the Fibonacci sequence, prove that for $n \geq 2$, $f_{n+1}^2 - f_n^2 = f_{n-1} f_{n+2}$.

32. Solve the recurrence relation of Exercise 7.

33. Solve the recurrence relation of Exercise 9.

Theorem 1 can be extended to a linear homogeneous relation of degree k, $a_n = r_1 a_{n-1} + r_2 a_{n-2} + \cdots + r_k a_{n-k}$. If the characteristic equation has k distinct roots $s_1, s_2, \ldots, s_k$, then $a_n = u_1 s_1^n + u_2 s_2^n + \cdots + u_k s_k^n$, where $u_1, u_2, \ldots, u_k$ depend on the initial conditions.

34. Let $a_n = 7a_{n-2} + 6a_{n-3}$, $a_1 = 3, a_2 = 6, a_3 = 10$.
    (a) What is the degree of this linear homogeneous relation?
    (b) Solve the recurrence relation.

35. Solve the recurrence relation $a_n = -2a_{n-1} + 2a_{n-2} + 4a_{n-3}$, $a_1 = 0, a_2 = 2, a_3 = 8$.

36. Use mathematical induction to prove that for the recurrence relation $b_n = b_{n-1} + 2b_{n-2}$, $b_1 = 1$, $b_2 = 3$, $b_n < \left(\frac{5}{2}\right)^n$.

37. Use mathematical induction to prove that for the recurrence relation $a_n = 2a_{n-1} + a_{n-2}$, $a_1 = 10$, $a_2 = 12$, $5 \mid a_{3n+1}$, $n \geq 0$.

38. Let $A_1, A_2, A_3, \ldots, A_{n+1}$ each be a $k \times k$ matrix. Let C_n be the number of ways to evaluate the product $A_1 \times A_2 \times A_3 \times \cdots \times A_{n+1}$ by choosing different orders in which to do then multiplications. Compute C_1, C_2, C_3, C_4, C_5.

39. Give a recurrence relation for C_n (defined in Exercise 38).

40. Verify that $C_n = \frac{2nC_n}{n+1}$ is a possible solution to the recurrence relation of Exercise 39 by showing that this formula produces the first five values as found in Exercise 38. (The terms of this sequence are called the Catalan numbers.)

Tips for Proofs

Proofs based on the pigeonhole principle are introduced in this chapter. Two situations are possible; the pigeons and pigeonholes are implicitly defined in the statement of the problem (Section 3, Exercise 5) or you must create pigeons and pigeonholes by defining categories into which the objects must fall (Section 3, Exercises 12 and 13). In the first case, the phrases "at least k objects" "have the same property" identify the pigeons (objects) and the labels on the pigeonholes (possible properties).

Proofs of statements about ${}_nC_r$ and ${}_nP_r$ are usually direct proofs based on the definitions and elementary algebra. Remember that a direct proof is generally the first approach to try.

Key Ideas for Review

- Theorem (The Multiplication Principle): Suppose two tasks T_1 and T_2 are to be performed in sequence. If T_1 can be performed in n_1 ways and for each of these ways T_2 can be performed in n_2 ways, then the sequence $T_1 T_2$ can be performed in $n_1 n_2$ ways.
- Theorem (The Extended Multiplication Principle)
- Theorem: Let A be a set with n elements and $1 \leq r \leq n$. Then the number of sequences of length r that can be formed from elements of A, allowing repetitions, is n^r.
- Permutation of n objects taken r at a time ($1 \leq r \leq n$): a sequence of length r formed from distinct elements
- Theorem: If $1 \leq r \leq n$, then ${}_nP_r$, the number of permutations of n objects taken r at a time, is $n \cdot (n-1) \cdot (n-2) \cdot \cdots \cdot (n-r+1)$ or $\frac{n!}{(n-r)!}$.
- Permutation: an arrangement of elements of a set A into a sequence of length n
- Theorem: The number of distinguishable permutations that can be formed from a collection of n objects where the first object appears k_1 times, the second object k_2 times, and so on, is $\frac{n!}{k_1! k_2! \cdots k_t!}$, where $k_1 + k_2 + \cdots + k_t = n$.
- Combination of n objects taken r at a time: a subset of r elements taken from a set with n elements
- Theorem: Let A be a set with $|A| = n$ and let $0 \leq r \leq n$. Then ${}_nC_r$, the number of combinations of the elements of A, taken r at a time, is $\frac{n!}{r!(n-r)!}$.
- Theorem: Suppose k selections are to be made from n items without regard to order and that repeats are allowed, assuming at least k copies of each of the n items. The number of ways these selections can be made is ${}_{(n+k-1)}C_k$.
- The pigeonhole principle
- The extended pigeonhole principle
- Sample space: the set of all outcomes of an experiment
- Event: a subset of the sample space
- Certain event: an event certain to occur

- Impossible event: the empty subset of the sample space
- Mutually exclusive events: any two events E and F with $E \cap F = \{\}$
- f_E: the frequency of occurrence of the event E in n trials
- $p(E)$: the probability of event E
- Probability space
- Elementary event: an event consisting of just one outcome
- Random selection
- Expected value: the sum of the products (value a_i)(p(a_i)) for all outcomes a_i of an experiment
- Recurrence relation: a recursive formula for a sequence
- Initial conditions: information about the beginning of a recursively defined sequence
- Linear homogeneous relation of degree k: a recurrence relation of the form $a_n = r_1 a_{n-1} + r_2 a_{n-2} + \cdots + r_k a_{n-k}$ with the r_i's constants
- Characteristic equation
- Catalan numbers

Chapter Self-Test

1. How can you decide whether combinations or permutations or neither should be counted in a particular counting problem?
2. What are some clues for deciding how to define pigeons and pigeonholes?
3. What are some of the advantages and disadvantages of the recursive form of a recurrence relation?
4. What are some of the advantages and disadvantages of the solution form of a recurrence relation?
5. What is our primary tool (from previous work) for answering probability questions?
6. Compute the number of
   (a) five-digit binary numbers.
   (b) five-card hands from a deck of 52 cards.
   (c) distinct arrangements of the letters of DISCRETE.
7. A computer program is used to generate all possible six-letter names for a new medication. Suppose that all the letters of the English alphabet may be used. How many possible names can be formed
   (a) if the letters are to be distinct?
   (b) if exactly two letters are repeated?
8. A fair six-sided die is rolled five times. If the results of each roll are recorded, how many
   (a) record sequences are possible?
   (b) record sequences begin 1, 2?
9. The IC Shoppe has 14 flavors of ice cream today. If you allow repeats, how many different triple-scoop ice cream cones can be chosen? (The order in which the scooping is done does not matter.)
10. The Spring Dance Committee must have 3 freshman and 5 sophomore members. If there are 23 eligible freshmen and 18 eligible sophomores, how many different committees are possible?
11. Show that $_{2n}C_2 = 2 \cdot {}_nC_2 + n^2$.
12. Pizza Quik always puts 50 pepperoni slices on a pepperoni pizza. If you cut a pepperoni pizza into eight equal size pieces, at least one piece must have ___ pepperoni slices. Justify your answer.
13. Complete and prove the following statement. At least ____ months of the year must begin on the same day of the week.
14. What is the probability that exactly two coins will land heads up when five fair coins are tossed?
15. Let $p(A) = 0.29, p(B) = 0.41$, and $p(A \cap B) = 0.65$. Are A and B mutually exclusive events? Justify your answer.
16. Solve the recurrence relation $b_n = 7b_{n-1} - 12b_{n-2}, b_1 = 1, b_2 = 7$.
17. Develop a formula for the solution of a recurrence relation of the form $a_n = ma_{n-1} - 1, a_1 = m$.

Experiment 3

The purpose of this experiment is to introduce the concept of a Markov chain. The investigations will use your knowledge of probability and matrices.

Suppose that the weather in Acia is either rainy or dry. We say that the weather has two possible states. As a result of extensive record keeping, it has been determined that the probability of a rainy day following a dry day is $\frac{1}{3}$, and the probability of a rainy day following a rainy day is $\frac{1}{2}$. If we know the weather today, then we can predict the probability that it will be rainy tomorrow. In fact, if we know the state in which the weather is today, then we can predict the probability for each possible state tomorrow. A Markov chain is a process in which the probability of

a system's being in a particular state at a given observation period depends only on its state at the immediately preceding observation period t_{ij}. Let the probability that if the system is in state j at a certain observation period, it will be in state i at the next period t_{ij} is called a *transition probability*. It is convenient to arrange the transition probabilities for a system with n possible states as an $n \times n$ transition matrix. A transition matrix for Acia's weather is

$$T = \begin{array}{cc} \text{D} & \text{R} \\ \begin{bmatrix} \frac{2}{3} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} \end{bmatrix} & \begin{array}{l} \text{D} \\ \text{R} \end{array} \end{array}$$

1. What is the sum of the entries in each column of T? Explain why this must be the same for each column of any transition matrix.

The transition matrix of a Markov chain can be used to determine the probability of the system being in any of its n possible states at future times. Let

$$P^{(k)} = \begin{bmatrix} p_1^{(k)} \\ p_2^{(k)} \\ \vdots \\ p_n^{(k)} \end{bmatrix}$$

denote the *state vector* of the Markov chain at the observation period k, where $p_j^{(k)}$ is the probability that the system is in state j at the observation period k. The state vector $P^{(0)}$ is called the initial state vector.

2. Suppose today, a Wednesday, is dry in Acia and this is observation period 0.
   (a) Give the initial state vector for the system.
   (b) What is the probability that it will be dry tomorrow? What is the probability that it will be rainy tomorrow? Give $P^{(1)}$.
   (c) Compute $TP^{(0)}$. What is the relationship between $TP^{(0)}$ and $P^{(1)}$?

It can be shown that, in general, $P^{(k)} = T^k P^{(0)}$. Thus the transition matrix and the initial state vector completely determine every other state vector.

3. Using the initial state vector from part 2, what is the state vector for next
   (a) Friday?
   (b) Sunday?
   (c) Monday?
   (d) What appears to be the long-term behavior of this system?

In some cases the Markov chain reaches an equilibrium state, because the state vectors converge to a fixed vector. This vector is called the *steady-state vector*. The most common use of Markov chains is to determine long-term behavior, so it is important to know if a particular Markov chain has a steady-state vector.

4. Let

$$T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad P^{(0)} = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}.$$

   Compute enough state vectors to determine the long-term behavior of this Markov chain.

A transition matrix T of a Markov chain is called *regular* if all the entries in some power of T are positive. If a Markov chain has a regular transition matrix, then the process has a steady-state vector. One way to find the steady-state vector, if it exists, is to proceed as in question 3; that is, calculate enough successive state vectors to identify the vector to which they are converging. Another method requires the solution of a system of linear equations. The steady-state vector U must be a solution of the matrix equation $TU = U$, and the entries of U have a sum equal to 1.

5. Verify that the transition matrix for the weather in Acia is regular and that the transition matrix in part 4 is not regular.
6. Solve
$$\begin{bmatrix} \frac{2}{3} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$
with the condition that $x + y = 1$. Compare your solution with the results of part 3.
7. Consider a plant that can have red (R), pink (P), or white (W) flowers depending on the genotypes RR, RW, and WW. When we cross each of these genotypes with genotype RW, we have the following transition matrix.

		Flowers of parent plant		
		R	P	W
Flowers of offspring plant	R	0.5	0.25	0.0
	P	0.5	0.50	0.5
	W	0.0	0.25	0.5

Suppose that each successive generation is produced by crossing only with plants of RW genotype.

(a) Will the process reach an equilibrium state? Why or why not?
(b) If there is a steady-state vector for this Markov chain, what are the long-term percentages of plants with red, pink, and white flowers?

8. In Acia there are two companies that produce widgets, Widgets, Inc., and Acia Widgets. Each year Widgets, Inc., keeps one-fourth of its customers while three-fourths switch to Acia Widgets. Each year Acia Widgets keeps two-thirds of its customers and one-third switch to Widgets, Inc. Both companies began business the same year and in that first year Widgets, Inc., had three-fifths of the market and Acia Widgets had the other two-fifths of the market. Under these conditions, will Acia Widgets ever run Widgets, Inc., out of business? Justify your answer.

Coding Exercises

1. Write a subroutine that accepts two positive integers n and r, and if $r \le n$, returns the number of permutations of n objects taken r at a time.
2. Write a program that has as input positive integers n and r and, if $r \le n$, prints the permutations of $1, 2, 3, \ldots, n$ taken r at a time.
3. Write a subroutine that accepts two positive integers n and r and, if $r \le n$, returns the number of combinations of n objects taken r at a time.
4. Write a program that has as input positive integers n and r and, if $r \le n$, prints the combinations of $1, 2, 3, \ldots, n$ taken r at a time.

5. (a) Write a recursive subroutine that with input k prints the first k Fibonacci numbers.
   (b) Write a nonrecursive subroutine that with input k prints the kth Fibonacci number.

Answers to Odd-Numbered Exercises

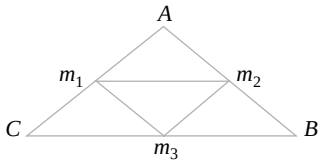
Exercise Set 1

1. 67,600. 3. 16. 5. 1296.
7. (a) 0. (b) 1.
9. (a) $n!$. (b) $\frac{n!}{2}$. (c) $\frac{(n+1)!}{2}$.
11. 120. 13. 4! or 24. 15. 30.
17. (a) 479,001,600. (b) 1,036,800.
19. 240. 21. 360.
23. 39,916,800. 25. $(n-1)!$. 27. 67,200.
29. $n \cdot {}_{n-1}P_{n-1} = n \cdot (n-1)(n-2)\cdots 2 \cdot 1 = n! = {}_nP_n$.
31. 190. 33. 2; 6; 12.
35. (a) 14. (b) 11. 37. 16. (b) 12.

Exercise Set 2

1. (a) 1. (b) 35. (c) 4368.
3. ${}_nC_r = \dfrac{n!}{r!(n-r)!} = \dfrac{n!}{(n-(n-r))!(n-r)!} = {}_nC_{n-r}$.
5. 20,358,520.
7. (a) 1. (b) 360.
9. (a) One of size 0, four of size 1, six of size 2, four of size 3, and one of size 4.
   (b) For each r, $0 \le r \le n$, there are ${}_nC_r$ subsets of size r.
11. (a) 980. (b) 1176.
13. 2702.
15. Because three people can be arranged in only one way from youngest to oldest, the problem is to count the number of ways to choose three people from seven.
17. 177,100 (repeats are allowed).
19. ${}_nC_{r-1} + {}_nC_r = \dfrac{n!}{(r-1)!(n-(r-1))!} + \dfrac{n!}{r!(n-r)!}$
$= \dfrac{n!r + n!(n-r+1)}{r!(n-r+1)!} = \dfrac{n!(n+1)}{r!(n+1-r)!}$
$= \dfrac{(n+1)!}{r!(n+1-r)!} = {}_{n+1}C_r$.
21. (a) 32. (b) 5. (c) 10.
23. (a) 2^n. (b) ${}_nC_3$. (c) ${}_nC_k$.
25. 525.
27. (a)
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1
   (b) Begin the row with a 1; write the sum of each consecutive pair of numbers in the previous row, moving left to right; end the row with a 1.
29. Exercise 19 shows another way to express the results of Exercise 27(b) and Exercise 28.
31. (a) 2. (b) 4. (c) 8. 33. 15.

Exercise Set 3

1. Let the birth months play the role of the pigeons and the calendar months, the pigeonholes. Then there are 13 pigeons and 12 pigeonholes. By the pigeonhole principle, at least two people were born in the same month.
3. 

m_1, m_2, m_3 are the midpoints of sides AC, AB, and BC, respectively. Let the four small triangles created be the pigeonholes. For any five points in or on triangle ABC, at least two must be in or on the same small triangle and thus are no more than $\frac{1}{2}$ unit apart.
5. By the extended pigeonhole principle, at least $\lfloor (50-1)/7 \rfloor + 1$ or 8 will be the same color.
7. Let 2161 cents be the pigeons and the six friends, the pigeonholes. Then at least one friend has $\lfloor (2161-1)/6 \rfloor + 1$ or 361 cents.
9. If repetitions are allowed, there are ${}_{15}C_5$ or 4368 choices. At least $\lfloor 4367/175 \rfloor + 1$, or 25, choices have the same cost.
11. You must have at least 49 friends.
13. Consider the first eight rows; one row must have at least 7 ones since there are 51 ones in all. Similarly, there is a column with at least 7 ones. The sum of the entries in this row and this column is at least 14.
15. Label the pigeonholes with $1, 3, 5, \ldots, 25$, the odd numbers between 1 and 25 inclusive. Assign each of the selected 14 numbers to the pigeonhole labeled with its odd part. There are only 13 pigeonholes, so two numbers must have the same odd part. One is a multiple of the other.
17. Using an argument similar to that for Exercise 16, the subset must contain at least $\frac{n-1}{2} + 1$ elements.
19. No. At least one pair of the 12 disks must add up to 21.

21. Consider the six sums c_1, $c_1 + c_2$, $c_1 + c_2 + c_3$, . . . , $c_1 + c_2 + c_3 + c_4 + c_5 + c_6$. If one of these has remainder 0 when divided by 6, then we are done. If none have remainder 0 when divided by 6, then two of them must give the same remainder. The positive difference of these two is a subsequence whose sum is divisible by 6.

23. We consider the cases of 3 or more ones, 2 ones, 1 one, and no ones. If there are at least 3 ones, we are done. If there are 2 ones and no two, then the sum is at least $1 + 4 \cdot 3$, but this is not possible. So if there are 2 ones, there is at least 1 two and we are done. If there is 1 one and no twos, then the sum is at least $1 + 5 \cdot 3$. This is impossible. If there are no ones and no three, then the sum is 26 or at least $5 \cdot 2 + 5$. But again, neither of these is possible.

25. There need to be at least 40 connections. If one printer has seven or fewer connections to PCs, then there can be a set of 5 PCs requesting printer access, but only four printers are available.

Exercise Set 4

1. {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}.
3. {sb, sr, sg, cb, cr, cg}.
5. (a) { }, {1}, {2}, {3}, {2, 3}, {1, 2}, {1, 3}, {1, 2, 3}.
   (b) 2^n.
7. (a) The card is a red ace.
   (b) The card is black or a diamond or an ace.
9. (a) {(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (1, 5), (2, 5), (3, 5), (4, 5), (6, 5)}.
   (b) { }.
11. (a) No, 3 satisfies both descriptions.
   (b) No, 2 satisfies both descriptions.
   (c) Yes, $E \cap F = \{3, 4, 5, 1, 2, 3\}$.
   (d) No, $E \cap F = \{3\}$.
13. (a) No. (b) No. (c) No. (d) Yes.
15. (a) {dls, dln, dms, dmn, dus, dun, als, aln}.
   (b) {als, aln}.
   (c) {dls, dln, als}.
17. $\overline{E} \cap F = \{2, 6, 3\}$, $\overline{F} \cap G = \{4\}$.
19. {club}, {spade}, {diamond}, {heart}.
21. (a) $\frac{10}{18}$. (b) $\frac{11}{18}$. (c) $\frac{12}{18}$. (d) $\frac{9}{18}$.
23. (a) 0.7. (b) 0. (c) 0.7. (d) 1.
25. $p(A) = \frac{6}{11}$, $p(B) = \frac{3}{11}$, $p(C) = \frac{1}{11}$, $p(D) = \frac{1}{11}$.
27. $\frac{10}{32}$.
29. (a) $\frac{1}{6}$. (b) 1.
31. (a) $\frac{11}{36}$. (b) $\frac{27}{36}$. (c) $\frac{6}{36}$. (d) $\frac{21}{36}$.
33. (a) $\frac{35}{220}$. (b) $\frac{80}{220}$. (c) $\frac{210}{220}$. (d) $\frac{210}{220}$.
35. (a) $\frac{4}{24}$. (b) $\frac{20}{24}$.
37. $\frac{n+1}{2}$.
39. $\$\frac{11}{9}$ dollars.
41. (a) $\frac{8}{52} \cdot \frac{7}{51}$. (b) $\frac{8}{52} \cdot \frac{8}{52}$.

Exercise Set 5

1. 4, 10, 25, 62.5. Yes, degree 1.
3. 3, 12, 24, 48. No.
5. 1, 8, 43, 216. No.
7. $s_1 = 2, s_2 = 3, s_n = s_{n-1} + 1$.
9. $A_1 = 100\left(1 + \frac{0.06}{12}\right)$, $A_n = \left(1 + \frac{0.06}{12}\right)(A_{n-1} + 100)$.
11. $c_n = c_{n-2} + c_{n-3}, c_1 = 0, c_2 = 1, c_3 = 1$.
13. $b_n = 3 \cdot 5^{n-1} + \frac{3}{4}(5^{n-1} - 1)$.
15. $d_n = 5(1.1)^{n-1}$.
17. $g_n = n! \cdot 6$.
19. $b_n = (-2)^n$.
21. $d_n = -\frac{3}{4} \cdot 2^n + \frac{5}{4} \cdot n \cdot 2^n$.
23. $g_n = \frac{1-2i}{2}(1+i)^n + \frac{1+2i}{2}(1-i)^n$.
27. (a) $a_n = 2a_{n-1} + 1, a_1 = 0$. (b) $a_n = 2^{n-1} - 1$.
29. $c_1 = 2, c_2 = 3, c_n = u\left(\frac{1+\sqrt{5}}{2}\right)^n + v\left(\frac{1-\sqrt{5}}{2}\right)^n$, $u = \frac{5+3\sqrt{5}}{10}$, $v = \frac{5-3\sqrt{5}}{10}$.
31. For $n \geq 2$, $f_{n+1}^2 - f_n^2 = (f_{n+1} - f_n)(f_{n+1} + f_n) = f_{n-1}f_{n+2}$, by the definition of f_k.
33. $A_1 = 100.5$.
    $A_n = (1.005)^{n-1}(100.5) + 20{,}100[(1.005)^{n-1} - 1]$.
35. $a_n = -2(-2)^n + \frac{5}{2}\left(-\sqrt{2}\right)^n + \frac{5}{2} + \sqrt{2} - \sqrt{2}^n$.
37. Basis step: $n = 0$. $P(0)$: $5 \mid a_1$ is clearly true.
    Induction step: We use $P(k)$: $5 \mid a_{3k+1}$ to show $P(k+1)$: $5 \mid a_{3(k+1)+1}$. Consider $a_{3(k+1)+1} = 2a_{3(k+1)} + a_{3k+2} = 2(2a_{3k+2} + a_{3k+1}) + a_{3k+2} = 5a_{3k+2} + 2a_{3k+1}$. Clearly $5 \mid 5a_{3k+2}$ and $P(k)$ guarantees $5 \mid 2a_{3k+1}$.
39. $C_n = C_1C_{n-1} + C_2C_{n-2} + \cdots + C_{n-1}C_1$, $C_1 = 1$.

Answers to Chapter Self-Tests

1. If all items to be chosen are not from the same set, then the problem is not a simple combination or permutation problem. If the items to be chosen are from the same set, and if the order in which they are chosen matters, permutations should be counted; otherwise, combinations.
2. Pigeons can often be identified by the phrases "at least k items have the property P" and the corresponding pigeonholes are the possible values for $P(x)$.
3. The recursive form may be easier to find or to justify, but this form may be difficult to evaluate for large n.

4. The solution form may be more efficient to evaluate for large n, but this form may be difficult to find.

5. The counting rules from Sections 1 and 2 are the primary tools needed to answer the probability questions presented in this chapter.

6. (a) 32. (b) 2,598,960. (c) 20,160.

7. (a) 165,765,600. (b) 118,404,000.

8. (a) 7776. (b) 216.

9. 560.

10. 15,173,928.

11. $2 \cdot \dfrac{n!}{(n-2)!\,2!} + n^2 = \dfrac{2 \cdot n! + 2!\,(n-2)!\,n^2}{(n-2)!\,2!}$

$$= \frac{(n-2)!\,n(n-1+n)}{(n-2)!}$$

$$= n(2n-1) \cdot \frac{(2n-2)!}{(2n-2)!} \cdot \frac{2}{2}$$

$$= \frac{(2n)!}{2!\,(2n-2)!}$$

$$= {}_{2n}C_2.$$

12. $\frac{50}{8} + 1 = 7$ pieces, assuming the pepperoni slices are not cut.

13. At least two months must begin on the same day of the week. Let the seven days of the week be the pigeonholes and the twelve months of the year, the pigeons. Then by the pigeonhole principle, at least $\frac{11}{7} + 1$, or 2, months begin on the same day of the week.

14. $\frac{10}{32}$.

15. No, $p(A \quad B) = p(A) + p(B) - p(A \quad B) = 0.29 + 0.41 - 0.65 = 0.05$.

16. $b_n = -3^n + 4^n$.

17. $a_n = m^{n-1}a_1 - m^{n-2} - m^{n-3} - \cdots - m^2 - 1 = m^n - \dfrac{m^{n-1} - 1}{m - 1}$.

Relations and Digraphs

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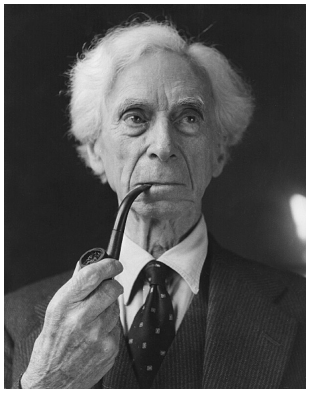
Relations and Digraphs

Relationships between people, numbers, sets, and many other entities can be formalized in the idea of a binary relation. In this chapter we develop the concept of binary relation, and we give several different methods of representing such objects. We also discuss a variety of different properties that a binary relation may possess, and we introduce important examples such as equivalence relations. Finally, we introduce several useful types of operations that may be performed on binary relations. We discuss these operations from both a theoretical and computational point of view.

Looking Back

According to John N. WarÞeld, the theory of relations was developed by Augustus De Morgan. WarÞeld also notes that the earliest illustration of a digraph was given by the great British philosopher and mathematician Bertrand Russell in 1919.



Bertrand Russell
Getty Images Inc.-Hulton Archive Photos

1 Product Sets and Partitions

Product Sets

An ordered pair (a, b) is a listing of the objects a and b in a prescribed order, with a appearing first and b appearing second. Thus an ordered pair is merely a sequence of length 2. It follows that the ordered pairs (a_1, b_1) and (a_2, b_2) are equal if and only if $a_1 = a_2$ and $b_1 = b_2$.

If A and B are two nonempty sets, we define the product set or Cartesian product $A \times B$ as the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$. Thus

$$A \times B = \{ (a, b) \mid a \in A \text{ and } b \in B \}.$$

A \ B	r	s
1	(1, r)	(1, s)
2	(2, r)	(2, s)
3	(3, r)	(3, s)

Figure 1

Example 1 Let

$$A = \{ 1, 2, 3 \} \quad \text{and} \quad B = \{ r, s \};$$

then

$$A \times B = \{ (1, r), (1, s), (2, r), (2, s), (3, r), (3, s) \}.$$

Observe that the elements of $A \times B$ can be arranged in a convenient tabular array as shown in Figure 1.

Example 2 If A and B are as in Example 1, then

$$B \times A = \{ (r, 1), (s, 1), (r, 2), (s, 2), (r, 3), (s, 3) \}.$$

THEOREM 1 For any two finite, nonempty sets A and B, $|A \times B| = |A| \, |B|$.

Proof
Suppose that $|A| = m$ and $|B| = n$. To form an ordered pair (a, b), $a \in A$ and $b \in B$, we must perform two successive tasks. Task 1 is to choose a first element from A, and task 2 is to choose a second element from B. There are m ways to perform task 1 and n ways to perform task 2; so, by the multiplication principle, there are $m \times n$ ways to form an ordered pair (a, b). In other words, $|A \times B| = m \cdot n = |A| \cdot |B|$.

Example 3 If $A = B = R$, the set of all real numbers, then $R \times R$, also denoted by R^2, is the set of all points in the plane. The ordered pair (a, b) gives the coordinates of a point in the plane.

Example 4 A marketing research firm classifies a person according to the following two criteria:

Gender: male (m); female (f)
Highest level of education completed: elementary school (e); high school (h); college (c); graduate school (g)

Let $S = \{ m, f \}$ and $L = \{ e, h, c, g \}$. The product set $S \times L$ contains all the categories into which the population is classified. Thus the classification (f, g) represents a female who has completed graduate school. There are eight categories in this classification scheme.

We now define the Cartesian product of three or more nonempty sets by generalizing the earlier definition of the Cartesian product of two sets. That is, the

Cartesian product $A_1 \times A_2 \times \cdots \times A_m$ of the nonempty sets $A_1, A_2, \ldots, A_m$ is the set of all ordered m-tuples $(a_1, a_2, \ldots, a_m)$, where $a_i \in A_i$, $i = 1, 2, \ldots, m$. Thus

$$A_1 \times A_2 \times \cdots \times A_m = \{(a_1, a_2, \ldots, a_m) \mid a_i \in A_i,\ i = 1, 2, \ldots, m\}.$$

Example 5 A manufacturer offers the following options for its refrigerators:

Doors: side-by-side (s), over-under (u), three (t)
Icemaker freezer (f), door (d)
Finish: standard (r), metallic (m), custom (c)

Let $D = \{s, u, t\}$, $I = \{f, d\}$, and $F = \{r, m, c\}$. Then the Cartesian product $D \times I \times F$ contains all the categories that describe refrigerator options. There are $3 \cdot 2 \cdot 3$ or 18 categories.

Proceeding in a manner similar to that used to prove Theorem 1, using the extended multiplication principle, we can show that if A_1 has n_1 elements, A_2 has n_2 elements, ..., and A_m has n_m elements, then $A_1 \times A_2 \times \cdots \times A_m$ has $n_1 \cdot n_2 \cdot \cdots \cdot n_m$ elements.

The remainder of this chapter uses only ordered pairs, but there are many applications for ordered n-tuples. One application is to represent records in a database, a collection of data usually stored in a computer. A relational database D is a subset of $A_1 \times A_2 \times \cdots \times A_n$, where each A_i designates a characteristic or attribute of the data. Each n-tuple in D is a single record of related information. In a relational database there must be a single attribute (or set of attributes) whose value(s) in a record uniquely identifies the record. This attribute (or set of attributes) is called a key.

Table 1 shows part of a relational database, Employees, that contains information about a company's employees. The attributes are Employee ID, Last Name, Department, and Years with Company. The attribute Employee ID is the key for this database. For ease in reading the information, the 4-tuples are displayed without parentheses in this table.

TABLE 1 Employees

Employee ID	Last Name	Department	Years with Company
8341	Croft	Front office	2
7984	Cottongim	Sales	2
2086	King	Human Resources	4
0340	Boswell	Research	3
7182	Chin	Human Resources	3
1748	Harris	Sales	1
4039	Gonzalez	Public Relations	6
4596	Greene	Human Resources	1
2914	Salamat	Sales	5
5703	Sahni	Research	7
3465	Harris	Sales	4

Digraphs are nothing but geometrical representations of relations, and any statement made about a digraph is actually a statement about the corresponding relation. This is especially important for theorems and their proofs. In some cases, it is easier or clearer to state a result in graphical terms, but a proof will always refer to the underlying relation. The reader should be aware that some authors allow more general objects as digraphs; for example, by permitting several edges in the same direction between the same vertices.

An important concept for relations is inspired by the visual form of digraphs. If R is a relation on a set A and $a \in A$, then the **in-degree** of a (relative to the relation R) is the number of $b \in A$ such that $(b, a) \in R$. The **out-degree** of a is the number of $b \in A$ such that $(a, b) \in R$.

What this means, in terms of the digraph of R, is that the in-degree of a vertex is the number of edges terminating at the vertex. The out-degree of a vertex is the number of edges leaving the vertex. Note that the out-degree is $|R(a)|$.

Example 21 Consider the digraph of Figure 4. Vertex 1 has in-degree 3 and out-degree 2. Also consider the digraph shown in Figure 5. Vertex 3 has in-degree 4 and out-degree 2, while vertex 4 has in-degree 0 and out-degree 1.

Example 22 Let $A = \{a, b, c, d\}$, and let R be the relation on A that has the matrix

$$M_R = \begin{matrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{matrix} .$$

Construct the digraph of R, and list in-degrees and out-degrees of all vertices.

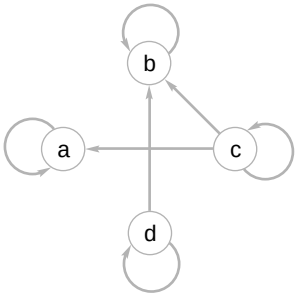


Figure 6

Solution

The digraph of R is shown in Figure 6. The following table gives the in-degrees and out-degrees of all vertices. Note that the sum of all in-degrees must equal the sum of all out-degrees.

	a	b	c	d
In-degree	2	3	1	1
Out-degree	1	1	3	2

Example 23 Let $A = \{1, 4, 5\}$, and let R be given by the digraph shown in Figure 7. Find M_R and R.

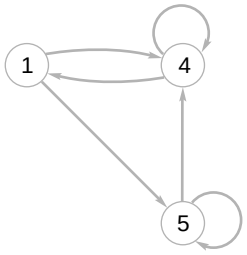


Figure 7

Solution

$$M_R = \begin{matrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{matrix}, \quad R = \{(1,4),(1,5),(4,1),(4,4),(5,4),(5,5)\}$$

If R is a relation on a set A, and B is a subset of A, the restriction of R to B is $R \cap (B \times B)$.

Example 24 Let $A = \{a, b, c, d, e, f\}$ and $R = \{(a,a),(a,c),(b,c),(a,e),(b,e),(c,e)\}$. Let $B = \{a, b, c\}$. Then

$$B \times B = \{(a,a),(a,b),(a,c),(b,a),(b,b),(b,c),(c,a),(c,b),(c,c)\}$$

and the restriction of R to B is $\{(a,a),(a,c),(b,c)\}$.

2 Exercises

1. For the relation R defined in Example 4, which of the following ordered pairs belong to R?
   (a) (2, 3) (b) (0, 8) (c) (1, 3)
   (d) (6, 18) (e) (−6, 24) (f) (8, 0)
2. For the relation R defined in Example 6, which of the following ordered pairs belong to R?
   (a) (2, 0) (b) (0, 2) (c) (0, 3)
   (d) (0, 0) (e) $1, \frac{3}{2}$ (f) (1, 1)
3. Let $A = Z^+$, the positive integers, and R be the relation defined by $a\ R\ b$ if and only if $2a = b + 1$. Which of the following ordered pairs belong to R?
   (a) (2, 2) (b) (3, 2) (c) (6, 15)
   (d) (1, 1) (e) (15, 6) (f) (n, n)

In Exercises 4 through 12, find the domain, range, matrix, and, when $A = B$, the digraph of the relation R.

4. $A = \{a, b, c, d\}$, $B = \{1, 2, 3\}$,
   $R = \{(a, 1), (a, 2), (b, 1), (c, 2), (d, 1)\}$
5. $A = \{$daisy, rose, violet, daffodil, peony$\}$,
   $B = \{$red, white, purple, yellow, blue, pink, orange$\}$
   $R = \{$(daisy, red), (violet, pink), (rose, purple),
   (daffodil, white)$\}$
6. $A = \{1, 2, 3, 4\}$, $B = \{1, 4, 6, 8, 9\}$; $a\ R\ b$ if and only if $b = a^2$.
7. $A = \{1, 2, 3, 4, 8\} = B$; $a\ R\ b$ if and only if $a = b$.
8. $A = \{1, 2, 3, 4, 8\}$, $B = \{1, 4, 6, 9\}$; $a\ R\ b$ if and only if $a \mid b$.
9. $A = \{1, 2, 3, 4, 6\} = B$; $a\ R\ b$ if and only if a is a multiple of b.
10. $A = \{1, 2, 3, 4, 5\} = B$; $a\ R\ b$ if and only if $a \quad b$.
11. $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 4, 6, 8\}$; $a\ R\ b$ if and only if $b < a$.
12. $A = \{1, 2, 3, 4, 8\} = B$; $a\ R\ b$ if and only if $a + b \quad 9$.
13. Let $A = Z^+$, the positive integers, and R be the relation defined by $a\ R\ b$ if and only if there exists k in Z^+ so that $a = b^k$ (k depends on a and b). Which of the following belong to R?
   (a) (4, 16) (b) (1, 7) (c) (8, 2)
   (d) (3, 3) (e) (2, 8) (f) (2, 32)
14. Let $A = R$. Consider the following relation R on A: $a\ R\ b$ if and only if $2a + 3b = 6$. Find Dom(R) and Ran(R).
15. Let $A = R$. Consider the following relation R on A: $a\ R\ b$ if and only if $a^2 + b^2 = 25$. Find Dom(R) and Ran(R).
16. Let R be the relation defined in Example 6. Find $R(A_1)$ for each of the following.
   (a) $A_1 = \{1, 8\}$ (b) $A_1 = \{3, 4, 5\}$
   (c) $A_1 = \{\ \}$
17. Let R be the relation defined in Exercise 9. Find each of the following.
   (a) $R(3)$ (b) $R(6)$ (c) $R(\{2, 4, 6\})$
18. Let R be the relation defined in Exercise 11. Find each of the following.
   (a) $R(\{3, 7\})$ (b) $R(9)$ (c) $R(\{1, 3\})$
19. Let R be the relation defined in Exercise 13. Find each of the following.
   (a) $R(3)$ (b) $R(4)$
   (c) $R(\{4, 3\})$ (d) $R(\{2, 4\})$
20. Let R be a relation from A to B. Prove that for all subsets A_1 and A_2 of A

$$R(A_1 \quad A_2) = R(A_1) \quad R(A_2) \text{ if and only if}$$
$$R(a) \quad R(b) = \{\ \} \text{ for any distinct } a, b \text{ in } A.$$

21. Let $A = R$. Give a description of the relation R specified by the shaded region in Figure 8.

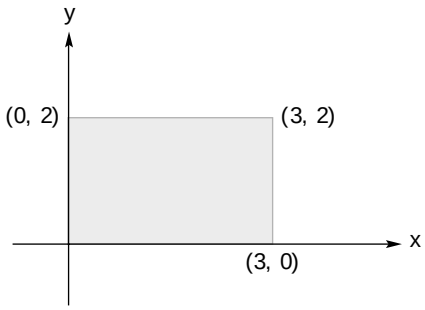


Figure 8

22. If A has n elements and B has m elements, how many different relations are there from A to B?

In Exercises 23 and 24, give the relation R defined on A and its digraph.

23. Let $A = \{1, 2, 3, 4\}$ and $M_R = \begin{matrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{matrix}$.

24. Let $A = \{a, b, c, d, e\}$ and $M_R = \begin{matrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{matrix}$.

In Exercises 25 and 26, find the relation determined by the digraph and give its matrix.

25.

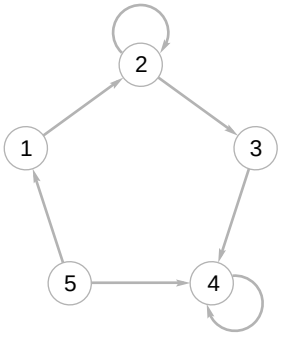


Figure 9

26.

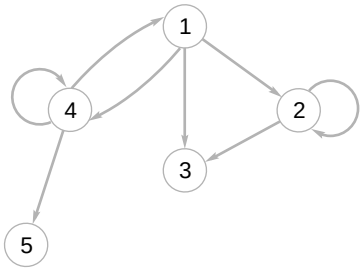


Figure 10

27. For the digraph in Exercise 25, give the in-degree and the out-degree of each vertex.

28. For the digraph in Exercise 26, give the in-degree and the out-degree of each vertex.

29. Describe how to find the in-degree and the out-degree of a vertex directly from the matrix of a relation.

In Exercises 30 and 31, let $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $R = \{(1, 2), (1, 4), (2, 3), (2, 5), (3, 6), (4, 7)\}$. Compute the restriction of R to B for the given subset of A.

30. $B = \{1, 2, 4, 5\}$

31. $B = \{2, 3, 4, 6\}$

32. Let R be a relation on a set A and $B \subseteq A$. Describe how to create the matrix of the restriction of R to B from M_R.

33. Let R be a relation on a set A and $B \subseteq A$. Describe how to create the digraph for the restriction of R to B from the digraph of R.

34. Let R be a relation on $A = \{a_1, a_2, \ldots, a_n\}$ given by the matrix M_R. Give a procedure using M_R directly to find each of the following.

    (a) $R(a_k)$ (b) $R(\{a_1, a_j, a_n\})$

35. Let R be a relation on $A = \{a_1, a_2, \ldots, a_n\}$ given by its digraph. Give a procedure using the digraph directly to find each of the following.

    (a) $R(a_k)$ (b) $R(\{a_1, a_j, a_n\})$

36. Let S be the product set $\{1, 2, 3\} \times \{a, b\}$. How many relations are there on S?

37. Let S be the product set $A \times B$. If $|A| = m$ and $|B| = n$, then how many relations are there on S?

3 Paths in Relations and Digraphs

Suppose that R is a relation on a set A. A path of length n in R from a to b is a finite sequence $a, x_1, x_2, \ldots, x_{n-1}, b$, beginning with a and ending with b, such that

$$a \; R \; x_1, \; x_1 \; R \; x_2, \; \ldots, \; x_{n-1} \; R \; b.$$

Note that a path of length n involves $n + 1$ elements of A, although they are not necessarily distinct.

A path is most easily visualized with the aid of the digraph of the relation. It appears as a geometric path or succession of edges in such a digraph, where the indicated directions of the edges are followed, and in fact a path derives its name from this representation. Thus the length of a path is the number of edges in the path, where the vertices need not all be distinct.

Example 1

Consider the digraph in Figure 11. Then μ_1: 1, 2, 5, 4, 3 is a path of length 4 from vertex 1 to vertex 3, μ_2: 1, 2, 5, 1 is a path of length 3 from vertex 1 to itself, and μ_3: 2, 2 is a path of length 1 from vertex 2 to itself.

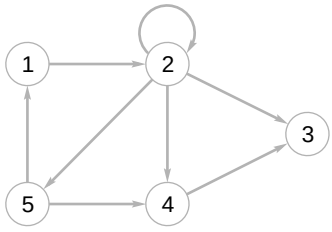


Figure 11

A path that begins and ends at the same vertex is called a cycle. In Example 1, μ_2 and μ_3 are cycles of lengths 3 and 1, respectively. It is clear that the paths of length 1 can be identified with the ordered pairs (x, y) that belong to R. Paths in a relation R can be used to define new relations that are quite useful. If n is a fixed positive integer, we define a relation R^n on A as follows: $x\ R^n\ y$ means that there is a path of length n from x to y in R. We may also define a relation R^* on A, by letting $x\ R^*\ y$ mean that there is some path in R from x to y. The length of such a path will depend, in general, on x and y. The relation R^* is sometimes called the connectivity relation for R.

Note that $R^n(x)$ consists of all vertices that can be reached from x by means of a path in R of length n. The set $R^*(x)$ consists of all vertices that can be reached from x by some path in R.

Example 2

Let A be the set of all living human beings, and let R be the relation of mutual acquaintance. That is, $a\ R\ b$ means that a and b know one another. Then $a\ R^2\ b$ means that a and b have an acquaintance in common. In general, $a\ R^n\ b$ if a knows someone x_1, who knows x_2, ..., who knows x_{n-1}, who knows b. Finally, $a\ R^*\ b$ means that some chain of acquaintances exists that begins at a and ends at b. It is interesting (and unknown) whether every two Americans, say, are related by R^*.

Example 3

Let A be a set of U.S. cities, and let $x\ R\ y$ if there is a direct flight from x to y on at least one airline. Then x and y are related by R^n if one can book a flight from x to y having exactly $n-1$ intermediate stops, and $x\ R^*\ y$ if one can get from x to y by plane.

Example 4

Let $A = \{1, 2, 3, 4, 5, 6\}$. Let R be the relation whose digraph is shown in Figure 12. Figure 13 shows the digraph of the relation R^2 on A. A line connects two vertices in Figure 13 if and only if they are R^2-related, that is, if and only if there is a path of length two connecting those vertices in Figure 12. Thus

$$
\begin{array}{llllll}
1\ R^2\ 2 & \text{since} & 1R2 & \text{and} & 2R2 \\
1\ R^2\ 4 & \text{since} & 1R2 & \text{and} & 2R4 \\
1\ R^2\ 5 & \text{since} & 1R2 & \text{and} & 2R5 \\
2\ R^2\ 2 & \text{since} & 2R2 & \text{and} & 2R2 \\
2\ R^2\ 4 & \text{since} & 2R2 & \text{and} & 2R4 \\
2\ R^2\ 5 & \text{since} & 2R2 & \text{and} & 2R5 \\
2\ R^2\ 6 & \text{since} & 2R5 & \text{and} & 5R6 \\
3\ R^2\ 5 & \text{since} & 3R4 & \text{and} & 4R5 \\
4\ R^2\ 6 & \text{since} & 4R5 & \text{and} & 5R6.
\end{array}
$$

In a similar way, we can construct the digraph of R^n for any n.

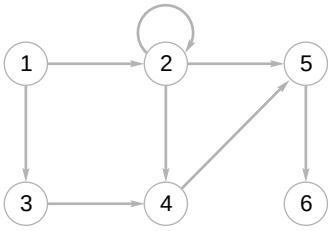


Figure 12

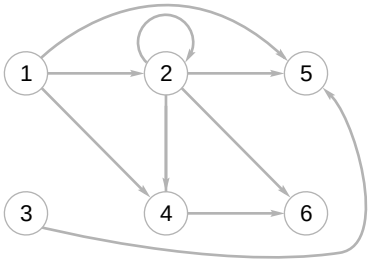


Figure 13

Example 5 Let $A = \{a, b, c, d, e\}$ and

$$R = \{(a, a), (a, b), (b, c), (c, e), (c, d), (d, e)\}.$$

Compute (a) R^2; (b) R^*.

Solution

(a) The digraph of R is shown in Figure 14.

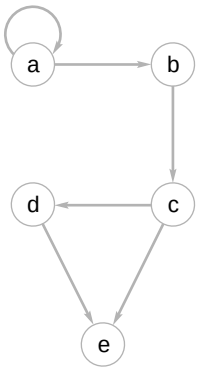


Figure 14

$a\ R^2\ a$ since $a\ R\ a$ and $a\ R\ a$.
$a\ R^2\ b$ since $a\ R\ a$ and $a\ R\ b$.
$a\ R^2\ c$ since $a\ R\ b$ and $b\ R\ c$.
$b\ R^2\ e$ since $b\ R\ c$ and $c\ R\ e$.
$b\ R^2\ d$ since $b\ R\ c$ and $c\ R\ d$.
$c\ R^2\ e$ since $c\ R\ d$ and $d\ R\ e$.

Hence

$$R^2 = \{(a, a), (a, b), (a, c), (b, e), (b, d), (c, e)\}.$$

(b) To compute R^*, we need all ordered pairs of vertices for which there is a path of any length from the first vertex to the second. From Figure 14 we see that

$$R^* = \{(a, a), (a, b), (a, c), (a, d), (a, e), (b, c), (b, d), (b, e), (c, d), (c, e), (d, e)\}.$$

For example, $(a, d) \in R^*$, since there is a path of length 3 from a to d: a, b, c, d. Similarly, $(a, e) \in R^*$, since there is a path of length 3 from a to e: a, b, c, e as well as a path of length 4 from a to e: a, b, c, d, e.

If $|R|$ is large, it can be tedious and perhaps difficult to compute R^*, or even R^2, from the set representation of R. However, M_R can be used to accomplish these tasks more efficiently.

Let R be a relation on a finite set $A = \{a_1, a_2, \ldots, a_n\}$, and let M_R be the $n \times n$ matrix representing R. We will show how the matrix M_{R^2}, of R^2, can be computed from M_R.

THEOREM 1 If R is a relation on $A = \{a_1, a_2, \ldots, a_n\}$, then $M_{R^2} = M_R \cdot M_R$.

Proof

Let $M_R = [m_{ij}]$ and $M_{R^2} = [n_{ij}]$. By definition, the i, j th element of $M_R \cdot M_R$ is equal to 1 if and only if row i of M_R and column j of M_R have a 1 in the same relative position, say position k. This means that $m_{ik} = 1$ and $m_{kj} = 1$ for some

k, $1 \le k \le n$. By definition of the matrix M_R, the preceding conditions mean that $a_i \mathrel{R} a_k$ and $a_k \mathrel{R} a_j$. Thus $a_i \mathrel{R^2} a_j$, and so $m_{ij} = 1$. We have therefore shown that position i, j of $M_R \odot M_R$ is equal to 1 if and only if $m_{ij} = 1$. This means that $M_R \odot M_R = M_{R^2}$.

For brevity, we will usually denote $M_R \odot M_R$ simply as $(M_R)^2$ (the symbol $\odot$ reminds us that this is not the usual matrix product).

Example 6 Let A and R be as in Example 5. Then

$$M_R = \begin{matrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{matrix}.$$

From the preceding discussion, we see that

$$M_{R^2} = M_R \odot M_R = \begin{matrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{matrix} \odot \begin{matrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{matrix} = \begin{matrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{matrix}.$$

Computing M_{R^2} directly from R^2, we obtain the same result.

We can see from Examples 5 and 6 that it is often easier to compute R^2 by computing $M_R \odot M_R$ instead of searching the digraph of R for all vertices that can be joined by a path of length 2. Similarly, we can show that $M_{R^3} = M_R \odot (M_R \odot M_R) = (M_R)^3$. In fact, we now show by induction that these two results can be generalized.

THEOREM 2 For $n \ge 2$ and R a relation on a finite set A, we have

$$M_{R^n} = M_R \odot M_R \odot \cdots \odot M_R \quad (n \text{ factors})$$

Proof
Let $P(n)$ be the assertion that the statement holds for an integer $n \ge 2$.

Basis Step

$P(2)$ is true by Theorem 1.

Induction Step

We use $P(k)$ to show $P(k + 1)$. Consider the matrix $M_{R^{k+1}}$. Let $M_{R^{k+1}} = [x_{ij}]$, $M_{R^k} = [y_{ij}]$, and $M_R = [m_{ij}]$. If $x_{ij} = 1$, we must have a path of length $k + 1$ from a_i to a_j. If we let a_s be the vertex that this path reaches just before the last vertex a_j, then there is a path of length k from a_i to a_s and a path of length 1 from a_s to a_j. Thus $y_{is} = 1$ and $m_{sj} = 1$, so $M_{R^k} \odot M_R$ has a 1 in position i, j. We can

see, similarly, that if M_{R^k} M_R has a 1 in position i, j, then $x_{ij} = 1$. This means that $M_{R^{k+1}} = M_{R^k}$ M_R.

Using

$$P(k): M_{R^k} = M_R \cdots M_R \quad (k \text{ factors}),$$

we have

$$M_{R^{k+1}} = M_{R^k} \; M_R = (M_R \; M_R \cdots M_R) \; M_R$$

and hence

$$P(k+1): M_{R^{k+1}} = M_R \cdots M_R \; M_R \quad (k+1 \text{ factors})$$

is true. Thus, by the principle of mathematical induction, $P(n)$ is true for all $n \geq 2$. This proves the theorem. As before, we write $M_R \cdots M_R$ (n factors) as $(M_R)^n$.

Note that the key to an induction step is finding a useful connection between $P(k)$ and $P(k+1)$.

Now that we know how to compute the matrix of the relation R^n from the matrix of R, we would like to see how to compute the matrix of R^*. We proceed as follows. Suppose that R is a relation on a finite set A, and $x \in A, y \in A$. We know that $x R^* y$ means that x and y are connected by a path in R of length n for some n. In general, n will depend on x and y, but, clearly, $x R^* y$ if and only if $x R y$ or $x R^2 y$ or $x R^3 y$ or .... Thus the preceding statement tells us that $R^* = R \cup R^2 \cup R^3 \cup \cdots = \bigcup_{n=1}^{\infty} R^n$. If R and S are relations on A, the relation $R \circ S$ is defined by $x (R \circ S) y$ if and only if $x R y$ or $x S y$. (The relation $R \circ S$ will be discussed in more detail in Section 7.) The reader may verify that $M_{R \cup S} = M_R \vee M_S$, and we will show this in Section 7. Thus

$$M_{R^*} = M_R \vee M_{R^2} \vee M_{R^3} \vee \cdots = M_R \vee (M_R)^2 \vee (M_R)^3 \vee \cdots .$$

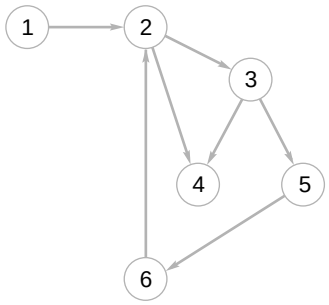


Figure 15

The reachability relation R^* of a relation R on a set A that has n elements is defined as follows: $x R^* y$ means that $x = y$ or $x R y$. The idea is that y is reachable from x if either y is x or there is some path from x to y. It is easily seen that $M_{R^*} = M_R \vee I_n$, where I_n is the $n \times n$ identity matrix. Thus our discussion shows that

$$M_{R^*} = I_n \vee M_R \vee (M_R)^2 \vee (M_R)^3 \vee \cdots .$$

Let $\alpha_1: a, x_1, x_2, \ldots, x_{n-1}, b$ be a path in a relation R of length n from a to b, and let $\alpha_2: b, y_1, y_2, \ldots, y_{m-1}, c$ be a path in R of length m from b to c. Then the composition of α_1 and α_2 is the path $a, x_1, x_2, \ldots, b, y_1, y_2, \ldots, y_{m-1}, c$ of length $n + m$, which is denoted by $\alpha_2 \circ \alpha_1$. This is a path from a to c.

Example 7 Consider the relation whose digraph is given in Figure 15 and the paths

$$\alpha_1: 1, 2, 3 \quad \text{and} \quad \alpha_2: 3, 5, 6, 2, 4.$$

Then the composition of α_1 and α_2 is the path $\alpha_2 \circ \alpha_1: 1, 2, 3, 5, 6, 2, 4$ from 1 to 4 of length 6.

3 Exercises

For Exercises 1 through 8, let R be the relation whose digraph is given in Figure 16.

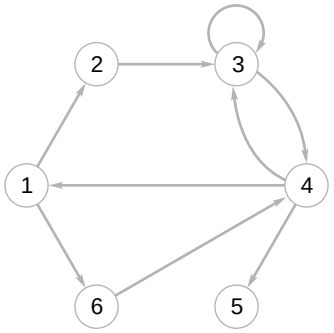


Figure 16

1. List all paths of length 1.
2. (a) List all paths of length 2 starting from vertex 2.
   (b) List all paths of length 2.
3. (a) List all paths of length 3 starting from vertex 3.
   (b) List all paths of length 3.
4. Find a cycle starting at vertex 2.
5. Find a cycle starting at vertex 6.
6. Draw the digraph of R^2.
7. Find M_{R^2}.
8. (a) Find R^{-1}.
   (b) Find $M_{R^{-1}}$.

For Exercises 9 through 16, let R be the relation whose digraph is given in Figure 17.

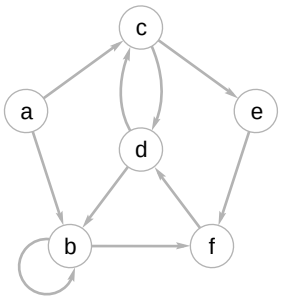


Figure 17

9. List all paths of length 1.
10. (a) List all paths of length 2 starting from vertex c.
    (b) Find all paths of length 2.
11. (a) List all paths of length 3 starting from vertex a.
    (b) Find all paths of length 3.
12. Find a cycle starting at vertex c.
13. Find a cycle starting at vertex d.
14. Find a cycle starting at vertex e.
15. Draw the digraph of R^2.
16. Find M_{R^2}. Is this result consistent with the result of Exercise 15?
17. (a) Find $M_{R^{-1}}$.
    (b) Find R^{-1}.
18. Let R and S be relations on a set A. Show that

$$M_{R \cap S} = M_R \wedge M_S.$$

19. Let R be a relation on a set A that has n elements. Show that $M_{R^2} = M_R \wedge I_n$, where I_n is the $n \times n$ identity matrix.

In Exercises 20 through 25, let R be the relation whose digraph is given in Figure 18.

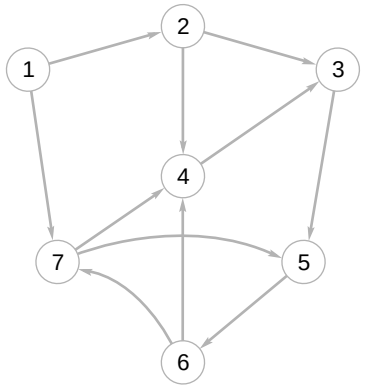


Figure 18

20. If $f_1 = \{(1, 2), (4, 3)\}$ and $f_2 = \{(3, 5), (6, 4)\}$, find the composition $f_2 \circ f_1$.
21. If $f_1 = \{(1, 7), (5, 6)\}$ and $f_2 = \{(5, 6), (7, 4), (3, 3)\}$, find the composition $f_2 \circ f_1$.
22. If $f_1 = \{(3, 4), (5, 6)\}$ and $f_2 = \{(6, 7), (4, 3), (5, 5)\}$, find the composition $f_2 \circ f_1$.
23. If $f_1 = \{(2, 3), (5, 6), (7, 7)\}$ and $f_2 = \{(7, 5), (6, 4), (4, 4)\}$, find the composition $f_2 \circ f_1$.
24. Find two cycles of length at least 3 in the relation R.
25. Find a cycle with maximum length in the relation R.
26. Let $A = \{1, 2, 3, 4, 5\}$ and R be the relation defined by $a \; R \; b$ if and only if $a < b$.
    (a) Compute R^2 and R^3.
    (b) Complete the following statement: $a \; R^2 \; b$ if and only if _____.
    (c) Complete the following statement: $a \; R^3 \; b$ if and only if _____.

27. By Theorem 1, $M_R \odot M_R = M_{R^2}$ so that $M_R \odot M_R$ shows where there are paths of length 2 in the digraph of R. Let

$$M_R = \begin{matrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{matrix}.$$

What does $M_R \cdot M_R$ show? Justify your conclusion.

28. Is it possible to generalize the results of Exercise 27? For example, does $(M_R)^3$ tell us anything useful about R?

29. Complete the following. The proof of Theorem 1 is ___ proof based on ____ of matrices.

30. (a) What about the statement of Theorem 2 indicates that an induction proof is appropriate?

    (b) What is the central idea of the induction step in the proof of Theorem 2?

31. Let D be the digraph of a finite relation R. Show that if there are no cycles in D, then at least one vertex has out-degree 0.

32. Draw a digraph with six vertices that has exactly one path of length 6 and exactly six paths of length 1.

33. Juan and Nils have each drawn a digraph to represent the relation R. The digraphs do not "look" alike. How would you determine if the digraphs both represent R correctly?

4 Properties of Relations

In many applications to computer science and applied mathematics, we deal with relations on a set A rather than relations from A to B. Moreover, these relations often satisfy certain properties that will be discussed in this section.

Reflexive and Irreflexive Relations

A relation R on a set A is **reflexive** if $(a, a) \in R$ for all $a \in A$, that is, if $a \mathrel{R} a$ for all $a \in A$. A relation R on a set A is **irreflexive** if $a \not\mathrel{R} a$ for every $a \in A$.

Thus R is reflexive if every element of A is related to itself and it is irreflexive if no element is related to itself.

Example 1

(a) Let $\Delta = \{(a, a) \mid a \in A\}$, so that Δ is the relation of equality on the set A. Then Δ is reflexive, since $(a, a) \in \Delta$ for all $a \in A$.

(b) Let $R = \{(a, b) \in A \times A \mid a \neq b\}$, so that R is the relation of inequality on the set A. Then R is irreflexive, since $(a, a) \notin R$ for all $a \in A$.

(c) Let $A = \{1, 2, 3\}$, and let $R = \{(1, 1), (1, 2)\}$. Then R is not reflexive since $(2, 2) \notin R$ and $(3, 3) \notin R$. Also, R is not irreflexive, since $(1, 1) \in R$.

(d) Let A be a nonempty set. Let $R = \emptyset \subseteq A \times A$, the empty relation. Then R is not reflexive, since $(a, a) \notin R$ for all $a \in A$ (the empty set has no elements). However, R is irreflexive.

We can identify a reflexive or irreflexive relation by its matrix as follows. The matrix of a reflexive relation must have all 1's on its main diagonal, while the matrix of an irreflexive relation must have all 0's on its main diagonal.

Similarly, we can characterize the digraph of a reflexive or irreflexive relation as follows. A reflexive relation has a cycle of length 1 at every vertex, while an irreflexive relation has no cycles of length 1. Another useful way of saying the same thing uses the equality relation Δ on a set A: R is reflexive if and only if $\Delta \subseteq R$, and R is irreflexive if and only if $R \cap \Delta = \emptyset$.

Finally, we may note that if R is reflexive on a set A, then $\mathrm{Dom}(R) = \mathrm{Ran}(R) = A$.

Symmetric, Asymmetric, and Antisymmetric Relations

A relation R on a set A is **symmetric** if whenever $a \mathrel{R} b$, then $b \mathrel{R} a$. It then follows that R is not symmetric if we have some a and $b \in A$ with $a \mathrel{R} b$, but $b \not\mathrel{R} a$. A relation R on a set A is **asymmetric** if whenever $a \mathrel{R} b$, then $b \not\mathrel{R} a$. It

then follows that R is not asymmetric if we have some a and b A with both $a \ R \ b$ and $b \ R \ a$.

A relation R on a set A is antisymmetric if whenever $a \ R \ b$ and $b \ R \ a$, then $a = b$. The contrapositive of this definition is that R is antisymmetric if whenever $a = b$, then $a \not R \ b$ or $b \not R \ a$. It follows that R is not antisymmetric if we have a and b in A, $a = b$, and both $a \ R \ b$ and $b \ R \ a$.

Given a relation R, we shall want to determine which properties hold for R. Keep in mind the following remark: A property fails to hold in general if we can find one situation where the property does not hold.

Example 2 Let $A = Z$, the set of integers, and let

$$R = \{ (a, b) \quad A \times A \mid a < b \}$$

so that R is the relation less than. Is R symmetric, asymmetric, or antisymmetric?

Solution

Symmetry. If $a < b$, then it is not true that $b < a$, so R is not symmetric.
Asymmetry. If $a < b$, then $b < a$ (b is not less than a), so R is asymmetric.
Antisymmetry. If $a = b$, then either $a < b$ or $b < a$, so that R is antisymmetric.

Example 3 Let A be a set of people and let $R = \{ (x, y) \quad A \times A \mid x \text{ is a cousin of } y \}$. Then R is a symmetric relation (verify).

Example 4 Let $A = \{ 1, 2, 3, 4 \}$ and let

$$R = \{ (1, 2), (2, 2), (3, 4), (4, 1) \}.$$

Then R is not symmetric, since $(1, 2) \quad R$, but $(2, 1) \ / \ R$. Also, R is not asymmetric, since $(2, 2) \quad R$. Finally, R is antisymmetric, since if $a = b$, either $(a, b) \ / \ R$ or $(b, a) \ / \ R$.

Example 5 Let $A = Z^+$, the set of positive integers, and let

$$R = \{ (a, b) \quad A \times A \mid a \text{ divides } b \}.$$

Is R symmetric, asymmetric, or antisymmetric?

Solution

If $a \mid b$, it does not follow that $b \mid a$, so R is not symmetric. For example, $|24$, but $4 \quad 2$.

If $a = b = 3$, say, then $a \ R \ b$ and $b \ R \ a$, so R is not asymmetric.

If $a \mid b$ and $b \mid a$, then $a = b$, so R is antisymmetric.

We now relate symmetric, asymmetric, and antisymmetric properties of a relation to properties of its matrix. The matrix $M_R = \quad m_{ij} \quad$ of a symmetric relation satisfies the property that

$$\text{if} \quad m_{ij} = 1, \quad \text{then} \quad m_{ji} = 1.$$

Moreover, if $m_{ji} = 0$, then $m_{ij} = 0$. Thus M_R is a matrix such that each pair of entries, symmetrically placed about the main diagonal, are either both 0 or both 1. It follows that $M_R = M_R^T$, so that M_R is a symmetric matrix.

The matrix $M_R = [m_{ij}]$ of an asymmetric relation R satisfies the property that

$$\text{if } m_{ij} = 1, \text{ then } m_{ji} = 0.$$

If R is asymmetric, it follows that $m_{ii} = 0$ for all i; that is, the main diagonal of the matrix M_R consists entirely of 0's. This must be true since the asymmetric property implies that if $m_{ii} = 1$, then $m_{ii} = 0$, which is a contradiction.

Finally, the matrix $M_R = [m_{ij}]$ of an antisymmetric relation R satisfies the property that if $i \neq j$, then $m_{ij} = 0$ or $m_{ji} = 0$.

Example 6 Consider the matrices in Figure 19, each of which is the matrix of a relation, as indicated.

Relations R_1 and R_2 are symmetric since the matrices M_{R_1} and M_{R_2} are symmetric matrices. Relation R_3 is antisymmetric, since no symmetrically situated, off-diagonal positions of M_{R_3} both contain 1's. Such positions may both have 0's, however, and the diagonal elements are unrestricted. The relation R_3 is not asymmetric because M_{R_3} has 1's on the main diagonal.

Relation R_4 has none of the three properties. M_{R_4} is not symmetric. The presence of the 1's in positions 4, 1 and 1, 4 of M_{R_4} violates both asymmetry and antisymmetry.

Finally, R_5 is antisymmetric but not asymmetric, and R_6 is both asymmetric and antisymmetric.

$$\begin{matrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{matrix} = M_{R_1} \qquad \begin{matrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{matrix} = M_{R_2}$$

(a) (b)

$$\begin{matrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{matrix} = M_{R_3} \qquad \begin{matrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{matrix} = M_{R_4}$$

(c) (d)

$$\begin{matrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{matrix} = M_{R_5} \qquad \begin{matrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{matrix} = M_{R_6}$$

(e) (f)

Figure 19

We now consider the digraphs of these three types of relations. If R is an asymmetric relation, then the digraph of R cannot simultaneously have an edge from vertex i to vertex j and an edge from vertex j to vertex i. This is true for any i and j, and in particular if i equals j. Thus there can be no cycles of length 1, and all edges are "one-way streets."

If R is an antisymmetric relation, then for different vertices i and j there cannot be an edge from vertex i to vertex j and an edge from vertex j to vertex i.

When $i = j$, no condition is imposed. Thus there may be cycles of length 1, but again all edges are "one way."

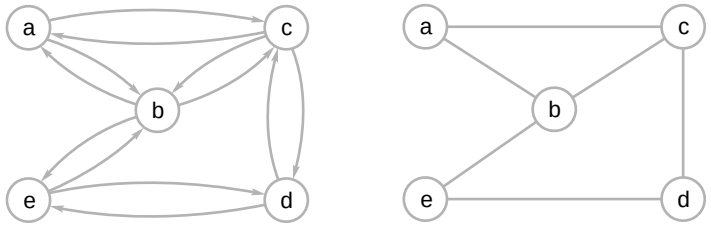
We consider the digraphs of symmetric relations in more detail.

The digraph of a symmetric relation R has the property that if there is an edge from vertex i to vertex j, then there is an edge from vertex j to vertex i. Thus, if two vertices are connected by an edge, they must always be connected in both directions. Because of this, it is possible and quite useful to give a different representation of a symmetric relation. We keep the vertices as they appear in the digraph, but if two vertices a and b are connected by edges in each direction, we replace these two edges with one undirected edge, or a "two-way street." This undirected edge is just a single line without arrows and connects a and b. The resulting diagram will be called the *graph* of the symmetric relation.

Example 7 Let $A = \{a, b, c, d, e\}$ and let R be the symmetric relation given by

$$R = \{(a, b), (b, a), (a, c), (c, a), (b, c), (c, b),$$
$$(b, e), (e, b), (e, d), (d, e), (c, d), (d, c)\}.$$

The usual digraph of R is shown in Figure 20(a), while Figure 20(b) shows the graph of R. Note that each undirected edge corresponds to two ordered pairs in the relation R.



Digraph of R (a) Graph of R (b)

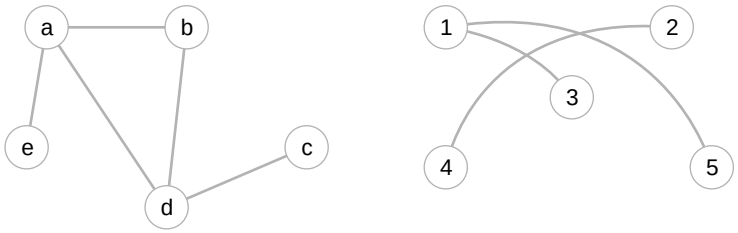
Figure 20

An undirected edge between a and b, in the graph of a symmetric relation R, corresponds to a set $\{a, b\}$ such that $(a, b) \in R$ and $(b, a) \in R$. Sometimes we will also refer to such a set $\{a, b\}$ as an *undirected edge* of the relation R and call a and b adjacent vertices

A symmetric relation R on a set A is called *connected* if there is a path from any element of A to any other element of A. This simply means that the graph of R is all in one piece. In Figure 21 we show the graphs of two symmetric relations. The graph in Figure 21(a) is connected, whereas that in Figure 21(b) is not connected.

Transitive Relations

We say that a relation R on a set A is transitive if whenever $a \, R \, b$ and $b \, R \, c$, then $a \, R \, c$. It is often convenient to say what it means for a relation to be not transitive. A relation R on A is not transitive if there exist a, b, and c in A so that $a \, R \, b$ and $b \, R \, c$, but $a \not R \, c$. If such a, b, and c do not exist, then R is transitive.



(a) (b)

Figure 21

Example 8 Let $A = Z$, the set of integers, and let R be the relation less than. To see whether R is transitive, we assume that $a \mathrel{R} b$ and $b \mathrel{R} c$. Thus $a < b$ and $b < c$. It then follows that $a < c$, so $a \mathrel{R} c$. Hence R is transitive.

Example 9 Let $A = Z^+$ and let R be the relation considered in Example 5. Is R transitive?

Solution

Suppose that $a \mathrel{R} b$ and $b \mathrel{R} c$, so that $a \mid b$ and $b \mid c$. It then does follow that $a \mid c$. Thus R is transitive.

Example 10 Let $A = \{1, 2, 3, 4\}$ and let

$$R = \{(1, 2), (1, 3), (4, 2)\}.$$

Is R transitive?

Solution

Since there are no elements a, b, and c in A such that $a \mathrel{R} b$ and $b \mathrel{R} c$, but $a \not\mathrel{R} c$, we conclude that R is transitive.

A relation R is transitive if and only if its matrix $M_R = [m_{ij}]$ has the property

$$\text{if } m_{ij} = 1 \text{ and } m_{jk} = 1, \text{ then } m_{ik} = 1.$$

The left-hand side of this statement simply means that $(M_R)^2$ has a 1 in position i, k. Thus the transitivity of R means that if $(M_R)^2$ has a 1 in any position, then M_R must have a 1 in the same position. Thus, in particular, if $(M_R)^2 = M_R$, then R is transitive. The converse is not true.

Example 11 Let $A = \{1, 2, 3\}$ and let R be the relation on A whose matrix is

$$M_R = \begin{matrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{matrix}.$$

Show that R is transitive.

Solution

By direct computation, $(M_R)^2 = M_R$; therefore, R is transitive.

To see what transitivity means for the digraph of a relation, we translate the definition of transitivity into geometric terms.

If we consider particular vertices a and c, the conditions $a \mathrel{R} b$ and $b \mathrel{R} c$ mean that there is a path of length 2 from a to c. In other words, $a \mathrel{R^2} c$.

Therefore, we may rephrase the definition of transitivity as follows: R is transitive if whenever $a\ R^2\ c$, then $a\ R\ c$; that is, $R^2 \subseteq R$ (as subsets of $A \times A$). In other words, if a and c are connected by a path of length 2 in R, then they must be connected by a path of length 1.

We can slightly generalize the foregoing geometric characterization of transitivity as follows.

THEOREM 1 A relation R is transitive if and only if it satisfies the following property: If there is a path of length greater than 1 from vertex a to vertex b, there is a path of length 1 from a to b (that is, a is related to b). Algebraically stated, R is transitive if and only if $R^n \subseteq R$ for all $n \geq 1$.

Proof
The proof is left to the reader.

It will be convenient to have a restatement of some of these relational properties in terms of R-relative sets. We list these statements without proof.

THEOREM 2 Let R be a relation on a set A. Then

(a) Reflexivity of R means that $a \in R(a)$ for all a in A.
(b) Symmetry of R means that $a \in R(b)$ if and only if $b \in R(a)$.
(c) Transitivity of R means that if $b \in R(a)$ and $c \in R(b)$, then $c \in R(a)$.

4 Exercises

In Exercises 1 through 8, let $A = \{1, 2, 3, 4\}$. Determine whether the relation is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, or transitive.

1. $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (4, 3), (4, 4)\}$
2. $R = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$
3. $R = \{(1, 3), (1, 1), (3, 1), (1, 2), (3, 3), (4, 4)\}$
4. $R = \{(1, 1), (2, 2), (3, 3)\}$
5. $R =$
6. $R = A \times A$
7. $R = \{(1, 2), (1, 3), (3, 1), (1, 1), (3, 3), (3, 2), (1, 4), (4, 2), (3, 4)\}$
8. $R = \{(1, 3), (4, 2), (2, 4), (3, 1), (2, 2)\}$

In Exercises 9 and 10 (Figures 22 and 23), let $A = \{1, 2, 3, 4, 5\}$. Determine whether the relation R whose digraph is given is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, or transitive.

9.

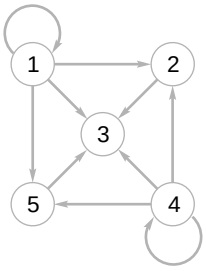


Figure 22

10.

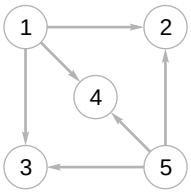


Figure 23

In Exercises 11 and 12, let $A = \{1, 2, 3, 4\}$. Determine whether the relation R whose matrix M_R is given is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, or transitive.

11. $\begin{matrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{matrix}$ 12. $\begin{matrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{matrix}$

In Exercises 13 through 24, determine whether the relation R on the set A is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, or transitive.

13. $A = Z$; $a\ R\ b$ if and only if $a \leq b + 1$.
14. $A = Z^+$; $a\ R\ b$ if and only if $|a - b| \geq 2$.

15. $A = Z^+$; $a \mathrel{R} b$ if and only if $a = b^k$ for some $k \in Z^+$.
16. $A = Z$; $a \mathrel{R} b$ if and only if $a + b$ is even.
17. $A = Z$; $a \mathrel{R} b$ if and only if $|a - b| = 2$.
18. $A =$ the set of real numbers; $a \mathrel{R} b$ if and only if $a^2 + b^2 = 4$.
19. $A = Z^+$; $a \mathrel{R} b$ if and only if $\text{GCD}(a, b) = 1$. In this case, we say that a and b are *relatively prime*. (See Section 4 for GCD.)
20. A is the set of all ordered pairs of real numbers; $(a, b) \mathrel{R} (c, d)$ if and only if $a = c$.
21. $S = \{1, 2, 3, 4\}$, $A = S \times S$; $(a, b) \mathrel{R} (c, d)$ if and only if $ad = bc$.
22. A is the set of all lines in the plane; $l_1 \mathrel{R} l_2$ if and only if l_1 is parallel to l_2.
23. A is the set of all triangles in the plane; $t_1 \mathrel{R} t_2$ if and only if the three angles of t_1 have the same measures as the three angles of t_2.
24. A is the set of all people in the world; $a \mathrel{R} b$ if and only if a is the sister of b.
25. Let R be the following symmetric relation on the set $A = \{1, 2, 3, 4, 5\}$:

$$R = \{(1, 2), (2, 1), (3, 4), (4, 3), (3, 5), (5, 3), (4, 5), (5, 4), (5, 5)\}.$$

Draw the graph of R.

26. Let $A = \{a, b, c, d\}$ and let R be the symmetric relation

$$R = \{(a, b), (b, a), (a, c), (c, a), (a, d), (d, a)\}.$$

Draw the graph of R.

27. Consider the graph of a symmetric relation R on $A = \{1, 2, 3, 4, 5, 6, 7\}$ shown in Figure 24. Determine R (list all pairs).

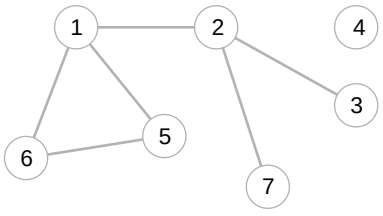


Figure 24

28. Consider the graph of a symmetric relation R on $A = \{a, b, c, d, e\}$ shown in Figure 25. Determine R (list all pairs).

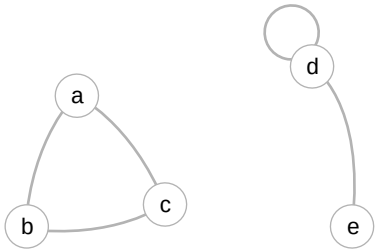


Figure 25

29. Let R be a symmetric relation given by its matrix M_R. Describe a procedure for using M_R to determine if the graph of R is connected.
30. Let R be a relation on A and $B \subseteq A$. Which relational properties of R would be inherited by the restriction of R to B?
31. Prove or disprove that if a relation on a set A is transitive and irreflexive, then it is asymmetric.
32. Prove or disprove that if a relation R on A is transitive, then R^2 is also transitive.
33. Let R be a nonempty relation on a set A. Suppose that R is symmetric and transitive. Show that R is not irreflexive.
34. Prove that if a relation R on a set A is symmetric, then the relation R^2 is also symmetric.
35. Prove by induction that if a relation R on a set A is symmetric, then R^n is symmetric for $n \geq 1$.
36. Define a relation on Z^+ that is reflexive, symmetric, and transitive and has not been defined previously.
37. Define a relation on the set $\{a, b, c, d\}$ that is
    (a) reflexive and symmetric, but not transitive.
    (b) reflexive and transitive, but not symmetric.
38. Define a relation on the set $\{a, b, c, d\}$ that is
    (a) irreflexive and transitive, but not symmetric.
    (b) antisymmetric and reflexive, but not transitive.
39. Define a relation on the set $\{a, b, c, d\}$ that is
    (a) transitive, reflexive, and symmetric.
    (b) asymmetric and transitive.
40. Give a direct proof of Theorem 1 of this section.

5 Equivalence Relations

A relation R on a set A is called an **equivalence relation** if it is reflexive, symmetric, and transitive.

Example 1 Let A be the set of all triangles in the plane and R be the relation on A defined as follows:

$$R = \{(a, b) \in A \times A \mid a \text{ is congruent to } b\}.$$

It is easy to see that R is an equivalence relation.

Example 2 Let $A = \{1, 2, 3, 4\}$ and let

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 3), (3, 3), (4, 4)\}.$$

It is easy to verify that R is an equivalence relation.

Example 3 Let $A = Z$, the set of integers, and let R be defined by $a \; R \; b$ if and only if $a \geq b$. Is R an equivalence relation?

Solution

Since $a \geq a$, R is reflexive. If $a \geq b$, it need not follow that $b \geq a$, so R is not symmetric. Incidentally, R is transitive, since $a \geq b$ and $b \geq c$ imply that $a \geq c$. We see that R is not an equivalence relation.

Example 4 Let $A = Z$ and let

$$R = \{(a, b) \in A \times A \mid a \text{ and } b \text{ yield the same remainder when divided by } 2\}$$

In this case, we call 2 the **modulus** and write $a \equiv b \pmod 2$, read "a is congruent to b mod 2."

Show that congruence mod 2 is an equivalence relation.

Solution

First, clearly $a \equiv a \pmod 2$. Thus R is reflexive.

Second, if $a \equiv b \pmod 2$, then a and b yield the same remainder when divided by 2, so $b \equiv a \pmod 2$. R is symmetric.

Finally, suppose that $a \equiv b \pmod 2$ and $b \equiv c \pmod 2$. Then a, b, and c yield the same remainder when divided by 2. Thus, $a \equiv c \pmod 2$. Hence congruence mod 2 is an equivalence relation.

Example 5 Let $A = Z$ and let $n \in Z^+$. We generalize the relation defined in Example 4 as follows. Let

$$R = \{(a, b) \in A \times A \mid a \equiv b \pmod n\}.$$

That is, $a \equiv b \pmod n$ if and only if a and b yield the same remainder when divided by n. Proceeding exactly as in Example 4, we can show that congruence mod n is an equivalence relation.

We note that if $a \equiv b \pmod n$, then $a = qn + r$ and $b = tn + r$ and $a - b$ is a multiple of n. Thus, $a \equiv b \pmod n$ if and only if $n \mid (a - b)$.

Equivalence Relations and Partitions

The following result shows that if P is a partition of a set A (see Section 1), then P can be used to construct an equivalence relation on A.

THEOREM 1 Let P be a partition of a set A. Recall that the sets in P are called the blocks of P. Define the relation R on A as follows:

$$a \; R \; b \quad \text{if and only if} \quad a \text{ and } b \text{ are members of the same block}$$

Then R is an equivalence relation on A.

Proof

(a) If $a \in A$, then clearly a is in the same block as itself; so $a \; R \; a$.
(b) If $a \; R \; b$, then a and b are in the same block; so $b \; R \; a$.
(c) If $a \; R \; b$ and $b \; R \; c$, then a, b, and c must all lie in the same block of P. Thus $a \; R \; c$.

Since R is reflexive, symmetric, and transitive, it is an equivalence relation. R will be called the equivalence relation determined by P.

Example 6 Let $A = \{1, 2, 3, 4\}$ and consider the partition $P = \{\{1, 2, 3\}, \{4\}\}$ of A. Find the equivalence relation R on A determined by P.

Solution

The blocks of P are $\{1, 2, 3\}$ and $\{4\}$. Each element in a block is related to every other element in the same block and only to those elements. Thus, in this case,

$$R = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (4, 4)\}.$$

If P is a partition of A and R is the equivalence relation determined by P, then the blocks of P can easily be described in terms of R. If A_1 is a block of P and $a \in A_1$, we see by definition that A_1 consists of all elements x of A with $a \; R \; x$. That is, $A_1 = R(a)$. Thus the partition P is $\{R(a) \mid a \in A\}$. In words, P consists of all distinct R-relative sets that arise from elements of A. For instance, in Example 6 the blocks $\{1, 2, 3\}$ and $\{4\}$ can be described, respectively, as $R(1)$ and $R(4)$. Of course, $\{1, 2, 3\}$ could also be described as $R(2)$ or $R(3)$, so this way of representing the blocks is not unique.

The foregoing construction of equivalence relations from partitions is very simple. We might be tempted to believe that few equivalence relations could be produced in this way. The fact is, as we will now show, that all equivalence relations on A can be produced from partitions.

We begin with the following result. Since its proof uses Theorem 2 of Section 4, the reader might first want to review that theorem.

Lemma 1 Let R be an equivalence relation on a set A, and let $a \in A$ and $b \in A$. Then

$$a \; R \; b \quad \text{if and only if} \quad R(a) = R(b).$$

Proof

First suppose that $R(a) = R(b)$. Since R is reflexive, $b \in R(b)$; therefore, $b \in R(a)$, so $a \; R \; b$.

A lemma is a theorem whose main purpose is to aid in proving some other theorem.

Conversely, suppose that $a R b$. Then note that

1. $b \in R(a)$ by definition. Therefore, since R is symmetric,
2. $a \in R(b)$, by Theorem 2(b) of Section 4.

We must show that $R(a) = R(b)$. First, choose an element x at $R(b)$. Since R is transitive, the fact that $x \in R(b)$, together with (1), implies by Theorem 2(c) of Section 4 that $x \in R(a)$. Thus $R(b) \subseteq R(a)$. Now choose $y \in R(a)$. This fact and (2) imply, as before, that $y \in R(b)$. Thus $R(a) \subseteq R(b)$, so we must have $R(a) = R(b)$.

Note the two-part structure of the lemma's proof. Because we want to prove a biconditional, $p \leftrightarrow q$, we must show $q \rightarrow p$ as well as $p \rightarrow q$.

We now prove our main result.

THEOREM 2 Let R be an equivalence relation on A, and let P be the collection of all distinct relative sets $R(a)$ for a in A. Then P is a partition of A, and R is the equivalence relation determined by P.

Proof

According to the definition of a partition, we must show the following two properties:

(a) Every element of A belongs to some relative set.

(b) If $R(a)$ and $R(b)$ are not identical, then $R(a) \cap R(b) = \emptyset$.

Now property (a) is true, since $a \in R(a)$ by reflexivity of R. To show property (b) we prove the following equivalent statement:

$$\text{If } R(a) \cap R(b) = \emptyset, \text{ then } R(a) = R(b).$$

To prove this, we assume that $R(a) \cap R(b) \neq \emptyset$. Then $a R c$ and $b R c$.

Since R is symmetric, we have $c R b$. Then $a R c$ and $c R b$, so, by transitivity of R, $a R b$. Lemma 1 then tells us that $R(a) = R(b)$. We have now proved that R is a partition. By Lemma 1 we see that $a R b$ if and only if a and b belong to the same block of P. Thus P determines R, and the theorem is proved.

Note the use of the contrapositive in this proof.

If R is an equivalence relation on A, then the sets $R(a)$ are traditionally called equivalence classes of R. Some authors denote the class $R(a)$ by $[a]$. The partition P constructed in Theorem 2 therefore consists of all equivalence classes $R(a)$, and this partition will be denoted by A/R. Recall that partitions of A are also called quotient sets of A, and the notation A/R reminds us that P is the quotient set of A that is constructed from and determines R.

Example 7 Let R be the relation defined in Example 2. Determine A/R.

Solution

From Example 2 we have $R(1) = \{1, 2\} = R(2)$. Also, $R(3) = \{3, 4\} = R(4)$. Hence $A/R = \{\{1, 2\}, \{3, 4\}\}$.

Example 8 Let R be the equivalence relation defined in Example 4. Determine A/R.

Solution

First, $R(0) = \{\ldots, -6, -4, -2, 0, 2, 4, 6, 8, \ldots\}$, the set of even integers, since the remainder is zero when each of these numbers is divided by 2. $R(1) =$

{. . . , −5, −3, −1, 1, 3, 5, 7, . . . }, the set of odd integers, since each gives a remainder of 1 when divided by 2. Hence A/R consists of the set of even integers and the set of odd integers.

From Examples 7 and 8 we can extract a general procedure for determining partitions A/R for A finite or countable. The procedure is as follows:

Step 1 Choose any element a of A and compute the equivalence class $R(a)$.

Step 2 If $R(a) = A$, choose an element b not included in $R(a)$, and compute the equivalence class $R(b)$.

Step 3 If A is not the union of previously computed equivalence classes, then choose an element x of A that is not in any of those equivalence classes and compute $R(x)$.

Step 4 Repeat step 3 until all elements of A are included in the computed equivalence classes. If A is countable, this process could continue indefinitely. In that case, continue until a pattern emerges that allows you to describe or give a formula for all equivalence classes.

5 Exercises

In Exercises 1 and 2, let $A = \{a, b, c\}$. Determine whether the relation R whose matrix M_R is given is an equivalence relation.

1. $M_R = \begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{matrix}$

2. $M_R = \begin{matrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$

In Exercises 3 and 4 (Figures 26 and 27), determine whether the relation R whose digraph is given is an equivalence relation.

3.

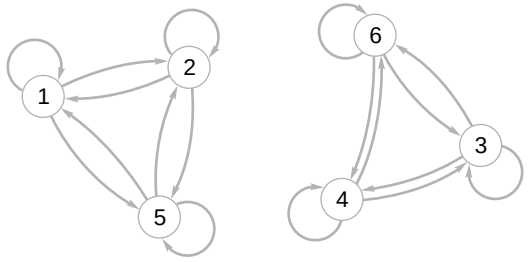


Figure 26

4.

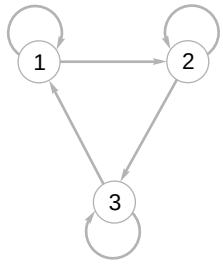


Figure 27

In Exercises 5 through 12, determine whether the relation R on the set A is an equivalence relation.

5. $A = \{a, b, c, d\}$,
$R = \{(a, a), (b, a), (b, b), (c, c), (d, d), (d, c)\}$

6. $A = \{1, 2, 3, 4, 5\}$, $R = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1), (2, 3), (3, 3), (4, 4), (3, 2), (5, 5)\}$

7. $A = \{1, 2, 3, 4\}$, $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 3), (1, 3), (4, 1), (4, 4)\}$

8. $A =$ the set of all members of the Software-of-the-Month Club; a R b if and only if a and b buy the same number of programs.

9. $A =$ the set of all members of the Software-of-the-Month Club; a R b if and only if a and b buy the same programs.

10. $A =$ the set of all people in the Social Security database; a R b if and only if a and b have the same last name.

11. $A =$ the set of all triangles in the plane; a R b if and only if a is similar to b.

12. $A = Z^+ \times Z^+$; (a, b) R (c, d) if and only if $b = d$.

13. If $\{\{a, c, e\}, \{b, d, f\}\}$ is a partition of the set $A = \{a, b, c, d, e, f\}$, determine the corresponding equivalence relation R.

14. If $\{\{1, 3, 5\}, \{2, 4\}\}$ is a partition of the set $A = \{1, 2, 3, 4, 5\}$, determine the corresponding equivalence relation R.

15. If $\{\{1, 3, 5, 7, 9\}, \{2, 4, 6, 8, 10\}\}$ is a partition of the set $A = \{1, 2, 3, \ldots, 10\}$, determine the corresponding equivalence relation R.

16. If $\{\{a, i\}, \{e, o\}, \{u\}\}$ is a partition of the set $A = \{a, e, i, o, u\}$, determine the corresponding equivalence relation R.

Solution

(a) We first find

$$A \times B = \{(1,a),(1,b),(1,c),(2,a),(2,b),(2,c),(3,a),(3,b),(3,c),(4,a),(4,b),(4,c)\}.$$

Then the complement of R in $A \times B$ is

$$\overline{R} = \{(1,c),(2,a),(3,a),(3,c),(4,b),(4,c)\}.$$

(b) We have $R \cap S = \{(1,b),(3,b),(2,c)\}$.

(c) We have

$$R \cup S = \{(1,a),(1,b),(2,b),(2,c),(3,b),(4,a),(4,b)\}.$$

(d) Since $(x,y) \in R^{-1}$ if and only if $(y,x) \in R$, we have

$$R^{-1} = \{(a,1),(b,1),(b,2),(c,2),(b,3),(a,4)\}.$$

Example 2 Let $A = R$. Let R be the relation $<$ on A and let S be $>$. Then the complement of R is the relation $\geq$, since $a \not< b$ means that $a > b$. Similarly, the complement of S is $\leq$. On the other hand, $R^{-1} = S$, since for any numbers a and b,

$$a \, R^{-1} \, b \quad \text{if and only if} \quad b \, R \, a \quad \text{if and only if} \quad b < a \quad \text{if and only if} \quad a > b.$$

Similarly, we have $S^{-1} = R$. Also, we note that $R \cup S$ is the relation of equality, since $a \,(R \cup S)\, b$ if and only if $a < b$ and $a > b$ if and only if $a = b$. Since, for any a and b, $a < b$ or $a > b$ must hold, we see that $R \cup S = A \times A$; that is, $R \cup S$ is the universal relation in which any a is related to any b.

Example 3 Let $A = \{a,b,c,d,e\}$ and let R and S be two relations on A whose corresponding digraphs are shown in Figure 38. Then the reader can verify the following facts:

$$\overline{R} = \{(a,a),(b,b),(a,c),(b,a),(c,b),(c,d),(c,e),(c,a),(d,b),(d,a),(d,e),(e,b),(e,a),(e,d),(e,c)\}$$

$$R^{-1} = \{(b,a),(e,b),(c,c),(c,d),(d,d),(d,b),(c,b),(d,a),(e,e),(e,a)\}$$

$$R \cap S = \{(a,b),(b,e),(c,c)\}.$$

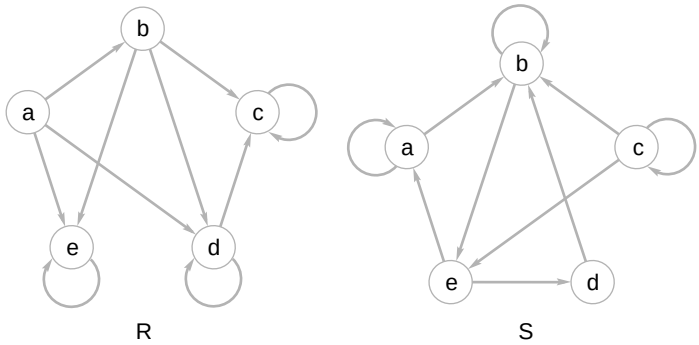


Figure 38

Example 4 Let $A = \{1, 2, 3\}$ and let R and S be relations on A. Suppose that the matrices of R and S are

$$M_R = \begin{matrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{matrix} \quad \text{and} \quad M_S = \begin{matrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{matrix}.$$

Then we can verify that

$$M_{\overline{R}} = \begin{matrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{matrix}, \quad M_{R^{-1}} = \begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{matrix},$$

$$M_{R \cup S} = \begin{matrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{matrix}, \quad M_{R \cap S} = \begin{matrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{matrix}.$$

Example 4 illustrates some general facts. We can show (Exercise 31) that if R and S are relations on set A, then

$$M_{R \cup S} = M_R \vee M_S$$
$$M_{R \cap S} = M_R \wedge M_S$$
$$M_{R^{-1}} = (M_R)^T.$$

Moreover, if M is a Boolean matrix, we define the *complement* $\overline{M}$ of M as the matrix obtained from M by replacing every 1 in M by a 0 and every 0 by a 1. Thus, if

$$M = \begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{matrix},$$

then

$$\overline{M} = \begin{matrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{matrix}.$$

We can also show (Exercise 31) that if R is a relation on a set A, then

$$M_{\overline{R}} = \overline{M}_R.$$

We know that a symmetric relation is a relation R such that $M_R = (M_R)^T$, and since $(M_R)^T = M_{R^{-1}}$, we see that R is symmetric if and only if $R = R^{-1}$.

We now prove a few useful properties about combinations of relations.

THEOREM 1 Suppose that R and S are relations from A to B.

(a) If $R \subseteq S$, then $R^{-1} \subseteq S^{-1}$.
(b) If $R \subseteq S$, then $\overline{S} \subseteq \overline{R}$.
(c) $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$ and $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$.
(d) $\overline{R \cup S} = \overline{R} \cap \overline{S}$ and $\overline{R \cap S} = \overline{R} \cup \overline{S}$.

Proof
Parts (b) and (d) are special cases of general set properties.

We now prove part (a). Suppose that $R \subseteq S$ and let $(a, b) \in R^{-1}$. Then $(b, a) \in R$, so $(b, a) \in S$. This, in turn, implies that $(a, b) \in S^{-1}$. Since each element of R^{-1} is in S^{-1}, we are done.

We next prove part (c). For the first part, suppose that $(a, b) \in (R \cap S)^{-1}$. Then $(b, a) \in R \cap S$, so $(b, a) \in R$ and $(b, a) \in S$. This means that $(a, b) \in R^{-1}$ and $(a, b) \in S^{-1}$, so $(a, b) \in R^{-1} \cap S^{-1}$. The converse containment can be proved by reversing the steps. A similar argument works to show that $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$.

The relations $\overline{R}$ and R^{-1} can be used to check if R has the properties of relations that we presented in Section 4. For instance, we saw earlier that R is symmetric if and only if $R = R^{-1}$. Here are some other connections between operations on relations and properties of relations.

THEOREM 2 Let R and S be relations on a set A.

(a) If R is reflexive, so is R^{-1}.
(b) If R and S are reflexive, then so are $R \cup S$ and $R \cap S$.
(c) R is reflexive if and only if $\overline{R}$ is irreflexive.

Proof
Let Δ be the equality relation on A. We know that R is reflexive if and only if $\Delta \subseteq R$. Clearly, $\Delta = \Delta^{-1}$, so if $\Delta \subseteq R$, then $\Delta = \Delta^{-1} \subseteq R^{-1}$ by Theorem 1, so R^{-1} is also reflexive. This proves part (a). To prove part (b), we note that if $\Delta \subseteq R$ and $\Delta \subseteq S$, then $\Delta \subseteq R \cup S$ and $\Delta \subseteq R \cap S$. To show part (c), we note that a relation S is irreflexive if and only if $S \cap \Delta = \emptyset$. Then R is reflexive if and only if $\Delta \subseteq R$ if and only if $\overline{R} \cap \Delta = \emptyset$ if and only if $\overline{R}$ is irreflexive.

Example 5 Let $A = \{1, 2, 3\}$ and consider the two reflexive relations

$$R = \{(1, 1), (1, 2), (1, 3), (2, 2), (3, 3)\}$$

and

$$S = \{(1, 1), (1, 2), (2, 2), (3, 2), (3, 3)\}.$$

Then

(a) $R^{-1} = \{(1, 1), (2, 1), (3, 1), (2, 2), (3, 3)\}$; R and R^{-1} are both reflexive.
(b) $\overline{R} = \{(2, 1), (2, 3), (3, 1), (3, 2)\}$ is irreflexive while R is reflexive.
(c) $R \cap S = \{(1, 1), (1, 2), (2, 2), (3, 3)\}$ and $R \cup S = \{(1, 1), (1, 2), (1, 3), (2, 2), (3, 2), (3, 3)\}$ are both reflexive.

THEOREM 3 Let R be a relation on a set A. Then

(a) R is symmetric if and only if $R = R^{-1}$.
(b) R is antisymmetric if and only if $R \cap R^{-1} \subseteq \Delta$.
(c) R is asymmetric if and only if $R \cap R^{-1} = \emptyset$.

Proof
The proof is straightforward and is left as an exercise.

THEOREM 4 Let R and S be relations on A.

(a) If R is symmetric, so are R^{-1} and $\overline{R}$.

(b) If R and S are symmetric, so are $R \cup S$ and $R \cap S$.

Proof
If R is symmetric, $R = R^{-1}$ and thus $(R^{-1})^{-1} = R = R^{-1}$, which means that R^{-1} is also symmetric. Also, $(a, b) \in (\overline{R})^{-1}$ if and only if $(b, a) \in \overline{R}$ if and only if $(b, a) \notin R$ if and only if $(a, b) \notin R^{-1} = R$ if and only if $(a, b) \in \overline{R}$, so $\overline{R}$ is symmetric and part (a) is proved. The proof of part (b) follows immediately from Theorem 1(c).

Example 6 Let $A = \{1, 2, 3\}$ and consider the symmetric relations

$$R = \{(1, 1), (1, 2), (2, 1), (1, 3), (3, 1)\}$$

and

$$S = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}.$$

Then

(a) $R^{-1} = \{(1, 1), (2, 1), (1, 2), (3, 1), (1, 3)\}$ and $\overline{R} = \{(2, 2), (2, 3), (3, 2), (3, 3)\}$; R^{-1} and $\overline{R}$ are symmetric.

(b) $R \cup S = \{(1, 1), (1, 2), (2, 1)\}$ and $R \cap S = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1), (3, 3)\}$, which are both symmetric.

THEOREM 5 Let R and S be relations on A.

(a) $(R \cup S)^2 \supseteq R^2 \cup S^2$.

(b) If R and S are transitive, so is $R \cap S$.

(c) If R and S are equivalence relations, so is $R \circ S$.

Proof
We prove part (a) geometrically. We have $a (R \cup S)^2 b$ if and only if there is a path of length 2 from a to b in $R \cup S$. Both edges of this path lie in R and in S, so $a R^2 b$ and $a S^2 b$, which implies that $a (R^2 \cup S^2) b$. To show part (b), recall from Section 4 that a relation T is transitive if and only if $T^2 \subseteq T$. If R and S are transitive, then $R^2 \subseteq R$, $S^2 \subseteq S$, so $(R \cap S)^2 \subseteq R^2 \cap S^2$ [by part (a)] $\subseteq R \cap S$, so $R \cap S$ is transitive. We next prove part (c). Relations R and S are each reflexive, symmetric, and transitive. The same properties hold for S^{-1} from Theorems 2(b), 4(b), and 5(b), respectively. Hence $R \circ S$ is an equivalence relation.

Example 7 Let R and S be equivalence relations on a finite set A, and let A/R and A/S be the corresponding partitions (see Section 5). Since $R \circ S$ is an equivalence relation, it corresponds to a partition $A/(R \circ S)$. We now describe $A/(R \circ S)$ in terms of A/R and A/S. Let W be a block of $A/(R \circ S)$ and suppose that a and b belong to W. Then $a (R \circ S) b$, so $a R b$ and $a S b$. Thus a and b belong to the same block, say X, of A/R and to the same block, say Y, of A/S. This means that $W \subseteq X \cap Y$. The steps in this argument are reversible; therefore, $W = X \cap Y$. Thus we can directly compute the partition $A/(R \circ S)$ by forming all possible intersections of blocks in A/R with blocks in A/S.

Closures

If R is a relation on a set A, it may well happen that R lacks some of the important relational properties discussed in Section 4, especially reflexivity, symmetry, and transitivity. If R does not possess a particular property, we may wish to add pairs to R until we get a relation that *does* have the required property. Naturally, we want to add as few new pairs as possible, so what we need to find is the smallest relation R_1 on A that contains R and possesses the property we desire. Sometimes R_1 does not exist. If a relation such as R_1 does exist, we call it the *closure* of R with respect to the property in question.

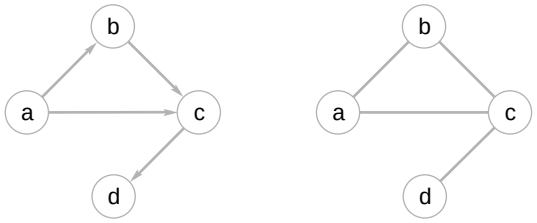
Example 8 Suppose that R is a relation on a set A, and R is not reflexive. This can only occur because some pairs of the diagonal relation Δ are not in R. Thus $R_1 = R \cup \Delta$ is the smallest reflexive relation on A containing R; that is, the *reflexive closure* of R is $R \cup \Delta$.

Example 9 Suppose now that R is a relation on A that is not symmetric. Then there must exist pairs (x, y) in R such that (y, x) is not in R. Of course $(y, x) \in R^{-1}$, so if R is to be symmetric we must add all pairs from R^{-1}; that is, we must enlarge R to $R \cup R^{-1}$. Clearly, $(R \cup R^{-1})^{-1} = R \cup R^{-1}$, so $R \cup R^{-1}$ is the smallest symmetric relation containing R; that is, $R \cup R^{-1}$ is the *symmetric closure* of R.

If $A = \{ a, b, c, d\}$ and $R = \{ (a, b), (b, c), (a, c), (c, d) \}$, then $R^{-1} = \{ (b, a), (c, b), (c, a), (d, c) \}$, so the symmetric closure of R is

$$R \cup R^{-1} = \{ (a, b), (b, a), (b, c), (c, b), (a, c), (c, a), (c, d), (d, c) \}.$$

The symmetric closure of a relation R is very easy to visualize geometrically. All edges in the digraph of R become "two-way streets" in $R \cup R^{-1}$. Thus the graph of the symmetric closure of R is simply the digraph of R with all edges made bidirectional. We show in Figure 39(a) the digraph of the relation R of Example 9. Figure 39(b) shows the graph of the symmetric closure $R \cup R^{-1}$.



(a) R (b) $R \cup R^{-1}$

Figure 39

The *transitive closure* of a relation R is the smallest transitive relation containing R. We will discuss the transitive closure in the next section.

Composition

Now suppose that A, B, and C are sets, R is a relation from A to B, and S is a relation from B to C. We can then define a new relation, the *composition* of R and S, written $S \circ R$. The relation $S \circ R$ is a relation from A to C and is defined as follows. If a is in A and c is in C, then $a\ (S \circ R)\ c$ if and only if for some b in B, we have $a\ R\ b$ and $b\ S\ c$. In other words, a is related to c by $S \circ R$ if we can get from a to c in two stages: first to an intermediate vertex b by relation R and then

from b to c by relation S. The relation $S \circ R$ might be thought of as "S following R" since it represents the combined effect of two relations R, then S.

Example 10

Let $A = \{1, 2, 3, 4\}$, $R = \{(1, 2), (1, 1), (1, 3), (2, 4), (3, 2)\}$, and $S = \{(1, 4), (1, 3), (2, 3), (3, 1), (4, 1)\}$. Since $(1, 2) \in R$ and $(2, 3) \in S$, we must have $(1, 3) \in S \circ R$. Similarly, since $(1, 1) \in R$ and $(1, 4) \in S$, we see that $(1, 4) \in S \circ R$. Proceeding in this way, we find that $S \circ R = \{(1, 4), (1, 3), (1, 1), (2, 1), (3, 3)\}$.

The following result shows how to compute relative sets for the composition of two relations.

THEOREM 6

Let R be a relation from A to B and let S be a relation from B to C. Then, if A_1 is any subset of A, we have

$$(S \circ R)(A_1) = S(R(A_1)). \tag{1}$$

Proof

If an element $z \in C$ is in $(S \circ R)(A_1)$, then $x (S \circ R) z$ for some x in A_1. By the definition of composition, this means that $x R y$ and $y S z$ for some y in B. Thus $y \in R(x)$, so $z \in S(R(x))$. Since $\{x\} \subseteq A_1$, Theorem 1(a) of Section 2 tells us that $S(R(x)) \subseteq S(R(A_1))$. Hence $z \in S(R(A_1))$, so $(S \circ R)(A_1) \subseteq S(R(A_1))$.

Conversely, suppose that $z \in S(R(A_1))$. Then $z \in S(y)$ for some y in $R(A_1)$ and, similarly, $y \in R(x)$ for some x in A_1. This means that $x R y$ and $y S z$, so $x (S \circ R) z$. Thus $z \in (S \circ R)(A_1)$, so $S(R(A_1)) \subseteq (S \circ R)(A_1)$. This proves the theorem.

Example 11

Let $A = \{a, b, c\}$ and let R and S be relations on A whose matrices are

$$M_R = \begin{matrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{matrix}, \qquad M_S = \begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{matrix}.$$

We see from the matrices that

$$\begin{array}{lllllll} (a, a) \in R & \text{and} & (a, a) \in S, & \text{so} & (a, a) \in S \circ R \\ (a, c) \in R & \text{and} & (c, a) \in S, & \text{so} & (a, a) \in S \circ R \\ (a, c) \in R & \text{and} & (c, c) \in S, & \text{so} & (a, c) \in S \circ R. \end{array}$$

It is easily seen that $(a, b) \notin S \circ R$ since, if we had $(a, x) \in R$ and $(x, b) \in S$, then matrix M_R tells us that x would have to be a or c; but matrix M_S tells us that neither (a, b) nor (c, b) is an element of S.

We see that the first row of $M_{S \circ R}$ is $1 \quad 0 \quad 1$. The reader may show by similar analysis that

$$M_{S \circ R} = \begin{matrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{matrix}.$$

We note that $M_{S \circ R} = M_R \odot M_S$ (verify this).

Example 11 illustrates a general and useful fact. Let A, B, and C be finite sets with n, p, and m elements, respectively, let R be a relation from A to B, and let S be a relation from B to C. Then R and S have Boolean matrices M_R and M_S with respective sizes $n \times p$ and $p \times m$. Thus $M_R \odot M_S$ can be computed, and it equals $M_{S \circ R}$.

To see this let $A = \{a_1, \ldots, a_n\}$, $B = \{b_1, \ldots, b_p\}$, and $C = \{c_1, \ldots, c_m\}$. Also, suppose that $M_R = [r_{ij}]$, $M_S = [s_{ij}]$, and $M_{S \circ R} = [t_{ij}]$. Then $t_{ij} = 1$ if and only if $(a_i, c_j) \in S \circ R$, which means that for some k, $(a_i, b_k) \in R$ and $(b_k, c_j) \in S$. In other words $r_{ik} = 1$ and $s_{kj} = 1$ for some k between 1 and p. This condition is identical to the condition needed for $M_R \odot M_S$ to have a 1 in position i, j, and thus $M_{S \circ R}$ and $M_R \odot M_S$ are equal.

In the special case where R and S are equal, we have $S \circ R = R^2$ and $M_{S \circ R} = M_{R^2} = M_R \odot M_R$, as was shown in Section 3.

Example 12 Let us redo Example 10 using matrices. We see that

$$M_R = \begin{matrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{matrix} \quad \text{and} \quad M_S = \begin{matrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{matrix}.$$

Then

$$M_R \odot M_S = \begin{matrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{matrix},$$

so

$$S \circ R = \{(1, 1), (1, 3), (1, 4), (2, 1), (3, 3)\}$$

as we found before. In cases where the number of pairs in R and S is large, the matrix method is much more reliable.

THEOREM 7 Let A, B, C, and D be sets, R a relation from A to B, S a relation from B to C, and T a relation from C to D. Then

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

Proof
The relations R, S, and T are determined by their Boolean matrices M_R, M_S, and M_T, respectively. As we showed after Example 11, the matrix of the composition is the Boolean matrix product; that is, $M_{S \circ R} = M_R \odot M_S$. Thus

$$M_{T \circ (S \circ R)} = M_{S \circ R} \odot M_T = (M_R \odot M_S) \odot M_T.$$

Similarly,

$$M_{(T \circ S) \circ R} = M_R \odot (M_S \odot M_T).$$

Since Boolean matrix multiplication is associative, we must have

$$(M_R \odot M_S) \odot M_T = M_R \odot (M_S \odot M_T),$$

and therefore

$$M_{T \circ (S \circ R)} = M_{(T \circ S) \circ R}.$$

Then

$$T \circ (S \circ R) = (T \circ S) \circ R$$

since these relations have the same matrices.

The proof illustrates the advantage of having several ways to represent a relation. Here using the matrix of the relation produces a simple proof.

In general, $R \circ S = S \circ R$, as shown in the following example.

Example 13 Let $A = \{a, b\}$, $R = \{(a, a), (b, a), (b, b)\}$, and $S = \{(a, b), (b, a), (b, b)\}$. Then $S \circ R = \{(a, b), (b, a), (b, b)\}$, while $R \circ S = \{(a, a), (a, b), (b, a), (b, b)\}$.

THEOREM 8 Let A, B, and C be sets, R a relation from A to B, and S a relation from B to C. Then $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$.

Proof
Let $c \in C$ and $a \in A$. Then $(c, a) \in (S \circ R)^{-1}$ if and only if $(a, c) \in S \circ R$, that is, if and only if there is a $b \in B$ with $(a, b) \in R$ and $(b, c) \in S$. Finally, this is equivalent to the statement that $(c, b) \in S^{-1}$ and $(b, a) \in R^{-1}$; that is, $(c, a) \in R^{-1} \circ S^{-1}$.

7 Exercises

In Exercises 1 and 2, let R and S be the given relations from A to B. Compute (a) $\overline{R}$; (b) $R \cup S$; (c) $R \cap S$; (d) S^{-1}.

1. $A = B = \{1, 2, 3\}$
   $R = \{(1, 1), (1, 2), (2, 3), (3, 1)\}$
   $S = \{(2, 1), (3, 1), (3, 2), (3, 3)\}$

2. $A = \{a, b, c\}$; $B = \{1, 2, 3\}$
   $R = \{(a, 1), (b, 1), (c, 2), (c, 3)\}$
   $S = \{(a, 1), (a, 2), (b, 1), (b, 2)\}$

3. Let A = a set of people. Let $a\ R\ b$ if and only if a and b are brothers; let $a\ S\ b$ if and only if a and b are sisters. Describe $R \cup S$.

4. Let A = a set of people. Let $a\ R\ b$ if and only if a is older than b; let $a\ S\ b$ if and only if a is a brother of b. Describe $R \circ S$.

5. Let A = a set of people. Let $a\ R\ b$ if and only if a is the father of b; let $a\ S\ b$ if and only if a is the mother of b. Describe $R \cup S$.

6. Let $A = \{2, 3, 6, 12\}$ and let R and S be the following relations on A: $x\ R\ y$ if and only if $2 \mid (x - y)$; $x\ S\ y$ if and only if $3 \mid (x - y)$. Compute
   (a) $\overline{R}$ (b) $R \cup S$ (c) $R \cap S$ (d) S^{-1}.

In Exercises 7 and 8, let R and S be two relations whose corresponding digraphs are shown in Figures 40 and 41. Compute (a) $\overline{R}$; (b) $R \cup S$; (c) $R \cap S$; (d) S^{-1}.

7.

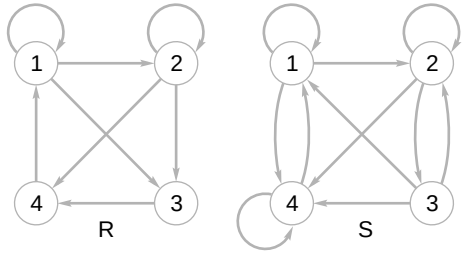


Figure 40

8.

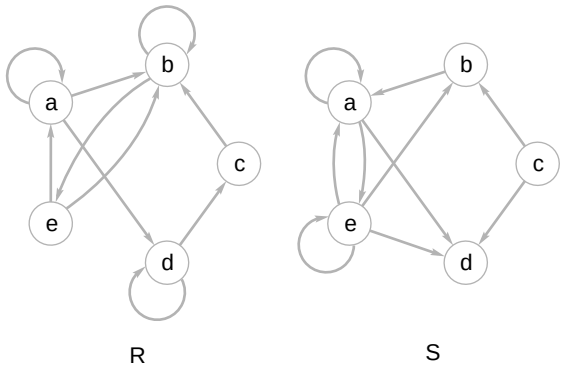


Figure 41

In Exercises 9 and 10, let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$. Let R and S be the relations from A to B whose matrices are given. Compute (a) $\overline{S}$; (b) $R \cup S$; (c) $R \cap S$; (d) R^{-1}.

9. $M_R = \begin{matrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{matrix}$, $M_S = \begin{matrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{matrix}$

10. $M_R = \begin{matrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{matrix}$, $M_S = \begin{matrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{matrix}$

In Exercises 11 and 12, let $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3\}$. Given the matrices M_R and M_S of the relations R and S from A to B, compute (a) $M_{R \cup S}$; (b) $M_{R \cap S}$; (c) $M_{R^{-1}}$; (d) $M_{\overline{S}}$.

11. $M_R = \begin{matrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{matrix}$, $M_S = \begin{matrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{matrix}$

12. $M_R = \begin{matrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{matrix}$, $M_S = \begin{matrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{matrix}$

13. Let $A = B = \{1, 2, 3, 4\}$, $R = \{(1, 1), (1, 3), (2, 3), (3, 1), (4, 2), (4, 4)\}$, and $S = \{(1, 2), (2, 3), (3, 1), (3, 2), (4, 3)\}$. Compute (a) $M_{R \circ S}$; (b) $M_{R \circ R}$; (c) $M_{R^{-1}}$; (d) $M_{\overline{S}}$.

14. Let

$$A = \{1, 2, 3, 4, 5, 6\},$$
$$R = \{(1, 2), (1, 1), (2, 1), (2, 2), (3, 3), (4, 4), (5, 5), (5, 6), (6, 5), (6, 6)\}, \text{ and}$$
$$S = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (4, 6), (4, 4), (6, 4), (6, 6), (5, 5)\}$$

be equivalence relations on A. Compute the partition corresponding to $R \cap S$.

15. Let $A = \{a, b, c, d, e\}$ and let the equivalence relations R and S on A be given by

$$M_R = \begin{matrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{matrix}$$

$$M_S = \begin{matrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{matrix}.$$

Compute the partition of A corresponding to $R \cap S$.

16. Let $A = \{1, 2, 3, 4\}$ and $R = \{(2, 1), (2, 3), (3, 2), (3, 3), (2, 2), (4, 2)\}$.
    (a) Find the reflexive closure of R.
    (b) Find the symmetric closure of R.

17. Let R be the relation whose matrix is

$$\begin{matrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{matrix}.$$

    (a) Find the reflexive closure of R.
    (b) Find the symmetric closure of R.

18. (a) Let R be a relation on a set A. Explain how to use the digraph of R to create the digraph of the reflexive closure of R.
    (b) Let R be a relation on a set A. Explain how to use the digraph of R to create the digraph of the symmetric closure of R.

19. Explain why the concept of closure is not applicable for irreflexivity, asymmetry, or antisymmetry.

20. Let $A = B = C =$ the set of real numbers. Let R and S be the following relations from A to B and from B to C, respectively:

$$R = \{(a, b) \mid a \quad 2b\} \quad \text{and} \quad S = \{(b, c) \mid b \quad 3c\}.$$

    (a) Is $(2, 3) \quad R \quad S$?
    (b) Is $(8, 1) \quad R \quad S$?

21. Let $A = B = C =$ the set of real numbers. Let R and S be the following relations from A to B and from B to C, respectively:

$$R = \{(a, b) \mid a \quad 2b\}$$
$$S = \{(b, c) \mid b \quad 3c\}.$$

    (a) Is $(1, 5) \quad S \quad R$?
    (b) Is $(2, 3) \quad S \quad R$?
    (c) Describe $S \quad R$.

22. Let $A = \{1, 2, 3, 4\}$. Let

$$R = \{(1, 1), (1, 2), (2, 3), (2, 4), (3, 4), (4, 1), (4, 2)\}$$
$$S = \{(3, 1), (4, 4), (2, 3), (2, 4), (1, 1), (1, 4)\}.$$

    (a) Is $(1, 3) \quad R \quad R$?
    (b) Is $(4, 3) \quad S \quad R$?
    (c) Is $(1, 1) \quad R \quad S$?
    (d) Compute $R \quad R$.
    (e) Compute $S \quad R$.
    (f) Compute $R \quad S$.
    (g) Compute $S \quad S$.

23. (a) Which properties of relations on a set A are preserved by composition? Prove your conclusion.
    (b) If R and S are equivalence relations on a set A, is $S \quad R$ an equivalence relation on A? Prove your conclusion.

In Exercises 24 and 25, let $A = \{1, 2, 3, 4, 5\}$ and let M_R and M_S be the matrices of the relations R and S on A. Compute (a) $M_{R \circ R}$; (b) $M_{S \circ R}$; (c) $M_{R \circ S}$; (d) $M_{S \circ S}$.

24.

$$M_R = \begin{matrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 \end{matrix}$$

$$M_S = \begin{matrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{matrix}$$

25.

$$M_R = \begin{matrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{matrix}$$

$$M_S = \begin{matrix} 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \end{matrix}$$

26. Let R and S be relations on a set A. If R and S are asymmetric, prove or disprove that $R \cap S$ and $R - S$ are asymmetric.

27. Let R and S be relations on a set A. If R and S are antisymmetric, prove or disprove that $R \cap S$ and $R - S$ are antisymmetric.

In Exercises 28 and 29, let R be a relation from A to B and let S and T be relations from B to C. Prove or disprove.

28. $(S \cup T) \circ R = (S \circ R) \cup (T \circ R)$

29. $(S \cap T) \circ R = (S \circ R) \cap (T \circ R)$

30. Let R and S be relations from A to B and let T be a relation from B to C. Show that if $R \subseteq S$, then $T \circ R \subseteq T \circ S$.

31. Show that if R and S are relations on a set A, then
    (a) $M_{R \cup S} = M_R \vee M_S$ (b) $M_{R \cap S} = M_R \wedge M_S$
    (c) $M_{R^{-1}} = (M_R)^T$ (d) $M_{\overline{R}} = \overline{M}_R$

32. Let R and S be relations on a set A. Prove that $(R \circ S)^n \subseteq R^n \circ S^n$, for $n \geq 1$.

In Exercises 33 through 35, let R and S be relations on a finite set A. Describe how to form the digraph of the specified relation directly from the digraphs of R and S.

33. R^{-1} 34. $R \cup S$ 35. $R \cap S$

36. Let R and S be symmetric relations on a set A. Prove or disprove that $R \circ S$ is also a symmetric relation on A.

37. Prove Theorem 3.

8 Transitive Closure and Warshall's Algorithm

Transitive Closure

In this section we consider a construction that has several interpretations and many important applications. Suppose that R is a relation on a set A and that R is not transitive. We will show that the transitive closure of R (see Section 7) is just the connectivity relation R^*, defined in Section 3.

THEOREM 1 Let R be a relation on a set A. Then R^* is the transitive closure of R.

Proof
We recall that if a and b are in the set A, then $a \, R^* \, b$ if and only if there is a path in R from a to b. Now R^* is certainly transitive since, if $a \, R^* \, b$ and $b \, R^* \, c$, the composition of the paths from a to b and from b to c forms a path from a to c in R, and so $a \, R^* \, c$. To show that R^* is the smallest transitive relation containing R, we must show that if S is any transitive relation on A and $R \subseteq S$, then $R^* \subseteq S$. Theorem 1 of Section 4 tells us that if S is transitive, then $S^n \subseteq S$ for all n; that is, if a and b are connected by a path of length n, then $a \, S \, b$. It follows that $S^* = \bigcup_{n=1}^{\infty} S^n \subseteq S$. It is also true that if $R \subseteq S$, then $R^* \subseteq S^*$, since any path in R is also a path in S. Putting these facts together, we see that if S and S^* is transitive on A, then $R^* \subseteq S^* \subseteq S$. This means that R^* is the smallest of all transitive relations on A that contain R.

We see that R^* has several interpretations. From a geometric point of view, it is called the connectivity relation, since it specifies which vertices are connected (by paths) to other vertices. If we include the relation I (see Section 4), then $R^* \cup I$ is the reachability relation R^* (see Section 3), which is frequently more useful. On the other hand, from the algebraic point of view, R^* is the transitive closure of R, as we have shown in Theorem 1. In this form, it plays important roles in the theory of equivalence relations and in the theory of certain languages.

Example 1 Let $A = \{1, 2, 3, 4\}$, and let $R = \{(1, 2), (2, 3), (3, 4), (2, 1)\}$. Find the transitive closure of R.

Solution

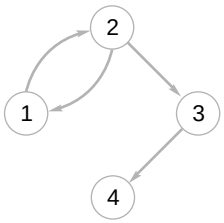


Figure 42

Method 1: The digraph of R is shown in Figure 42. Since R^* is the transitive closure, we can proceed geometrically by computing all paths. We see that from vertex 1 we have paths to vertices 2, 3, 4, and 1. Note that the path from 1 to 1 proceeds from 1 to 2 to 1. Thus we see that the ordered pairs $(1, 1)$, $(1, 2)$, $(1, 3)$, and $(1, 4)$ are in R^*. Starting from vertex 2, we have paths to vertices 2, 1, 3, and 4, so the ordered pairs $(2, 1)$, $(2, 2)$, $(2, 3)$, and $(2, 4)$ are in R^*. The only other path is from vertex 3 to vertex 4, so we have

$$R^* = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4)\}.$$

Method 2: The matrix of R is

$$M_R = \begin{matrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{matrix}.$$

We may proceed algebraically and compute the powers of M_R. Thus

$$(M_R)^2 = \begin{matrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{matrix}, \qquad (M_R)^3 = \begin{matrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{matrix},$$

$$(M_R)^4 = \begin{matrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{matrix}.$$

Continuing in this way, we can see that $(M_R)^n$ equals $(M_R)^2$ if n is even and equals $(M_R)^3$ if n is odd and greater than 1. Thus

$$M_{R^*} = M_R \vee (M_R)^2 \vee (M_R)^3 = \begin{matrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{matrix}$$

and this gives the same relation as Method 1.

In Example 1 we did not need to consider all powers R^n to obtain R^*. This observation is true whenever the set A is finite, as we will now prove.

THEOREM 2 Let A be a set with $|A| = n$, and let R be a relation on A. Then

$$R^* = R \cup R^2 \cup \cdots \cup R^n.$$

In other words, powers of R greater than n are not needed to compute R^*.

Proof

Let a and b be in A, and suppose that $a, x_1, x_2, \ldots, x_m, b$ is a path from a to b in R; that is, $(a, x_1), (x_1, x_2), \ldots, (x_m, b)$ are all in R. If x_i and x_j are the same vertex, say $i < j$, then the path can be divided into three sections. First, a path from a to x_i, then a path from x_i to x_j, and finally a path from x_j to b. The middle path is a cycle, since $x_i = x_j$, so we simply leave it out and put the remaining two paths together. This gives us a shorter path from a to b (see Figure 43).

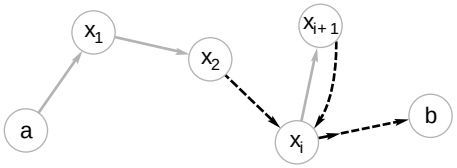


Figure 43

Now let $a, x_1, x_2, \ldots, x_k, b$ be the shortest path from a to b. If $a = b$, then all vertices $a, x_1, x_2, \ldots, x_k, b$ are distinct. Otherwise, the preceding discussion shows that we could find a shorter path. Thus the length of the path is at most $n-1$ (since $|A| = n$). If $a = b$, then for similar reasons, the vertices $a, x_1, x_2, \ldots, x_k$ are distinct, so the length of the path is at most n. In other words, if $a\,R\,b$, then $a\,R^k\,b$, for some k, $1 \le k \le n$. Thus $R^* = R \cup R^2 \cup \cdots \cup R^n$.

The methods used to solve Example 1 each have certain difficulties. The graphical method is impractical for large sets and relations and is not systematic. The matrix method can be used in general and is systematic enough to be programmed for a computer, but it is inefficient and, for large matrices, can be prohibitively costly. Fortunately, a more efficient algorithm for computing transitive closure is available. It is known as Warshall's algorithm, after its creator, and we describe it next.

Warshall's Algorithm

Let R be a relation on a set $A = \{a_1, a_2, \ldots, a_n\}$. If $x_1, x_2, \ldots, x_m$ is a path in R, then any vertices other than x_1 and x_m are called *interior vertices* of the path. Now, for $1 \le k \le n$, we define a Boolean matrix W_k as follows. W_k has a 1 in position i, j if and only if there is a path from a_i to a_j in R whose interior vertices, if any, come from the set $\{a_1, a_2, \ldots, a_k\}$.

Since any vertex must come from the set $\{a_1, a_2, \ldots, a_n\}$, it follows that the matrix W_n has a 1 in position i, j if and only if some path in R connects a_i with a_j. In other words, $W_n = M_R$. If we define W_0 to be M_R, then we will have a sequence $W_0, W_1, \ldots, W_n$ whose first term is M_R and whose last term is M_R^*. We will show how to compute each matrix W_k from the previous matrix W_{k-1}. Then we can begin with the matrix of R and proceed one step at a time until, in n steps, we reach the matrix of R^*. This procedure is called *Warshall's algorithm*. The matrices W_k are different from the powers of the matrix M_R, and this difference results in a considerable savings of steps in the computation of the transitive closure of R.

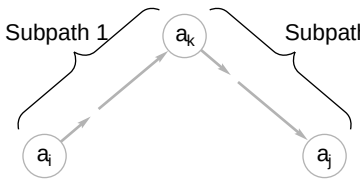


Figure 44

Suppose that $W_k = t_{ij}$ and $W_{k-1} = s_{ij}$. If $t_{ij} = 1$, then there must be a path from a_i to a_j whose interior vertices come from the set $\{a_1, a_2, \ldots, a_k\}$. If the vertex a_k is not an interior vertex of this path, then all interior vertices must actually come from the set $\{a_1, a_2, \ldots, a_{k-1}\}$, so $s_{ij} = 1$. If a_k is an interior vertex of the path, then the situation is as shown in Figure 44. As in the proof of Theorem 2, we may assume that all interior vertices are distinct. Thus a_k appears only once in the path, so all interior vertices of subpaths 1 and 2 must come from the set $\{a_1, a_2, \ldots, a_{k-1}\}$. This means that $s_{ik} = 1$ and $s_{kj} = 1$.

Thus $t_{ij} = 1$ if and only if either

(1) $s_{ij} = 1$ or
(2) $s_{ik} = 1$ and $s_{kj} = 1$.

This is the basis for Warshall's algorithm. If W_{k-1} has a 1 in position i, j then, by (1), so will W_k. By (2), a new 1 can be added in position i, j of W_k if and only if column k of W_{k-1} has a 1 in position i and row k of W_{k-1} has a 1 in position j. Thus we have the following procedure for computing W_k from W_{k-1}.

Step 1 First transfer to W_k all 1's in W_{k-1}.

Step 2 List the locations $p_1, p_2, \ldots,$ in column k of W_{k-1}, where the entry is 1, and the locations $q_1, q_2, \ldots,$ in row k of W_{k-1}, where the entry is 1.

Step 3 Put 1's in all the positions p_i, q_j of W_k (if they are not already there).

Example 2 Consider the relation R defined in Example 1. Then

$$W_0 = M_R = \begin{matrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{matrix}$$

and $n = 4$.

First we find W_1 so that $k = 1$. W_0 has 1's in location 2 of column 1 and location 2 of row 1. Thus W_1 is just W_0 with a new 1 in position 2, 2.

$$W_1 = \begin{matrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{matrix}$$

Now we compute W_2 so that $k = 2$. We must consult column 2 and row 2 of W_1. Matrix W_1 has 1's in locations 1 and 2 of column 2 and locations 1, 2, and 3 of row 2.

Thus, to obtain W_2, we must put 1's in positions 1, 1; 1, 2; 1, 3; 2, 1; 2, 2; and 2, 3 of matrix W_1 (if 1's are not already there). We see that

$$W_2 = \begin{matrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{matrix} .$$

Proceeding, we see that column 3 of W_2 has 1's in locations 1 and 2, and row 3 of W_2 has a 1 in location 4. To obtain W_3, we must put 1's in positions 1, 4 and 2, 4 of W_2, so

$$W_3 = \begin{matrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{matrix} .$$

Finally, W_3 has 1's in locations 1, 2, 3 of column 4 and no 1's in row 4, so no new 1's are added and $M_R = W_4 = W_3$. Thus we have obtained the same result as in Example 1.

The procedure illustrated in Example 2 yields the following algorithm for computing the matrix, CLOSURE, of the transitive closure of a relation R represented by the $N \times N$ matrix MAT.

Algorithm WARSHALL

```
1. CLOSURE    MAT
2. FOR K = 1 THRU N
    a. FOR I = 1 THRU N
        1. FOR J = 1 THRU N
            a. CLOSURE[I,J]    CLOSURE[I,J]
                  (CLOSURE[I,K]    CLOSURE[K,J])
```

This algorithm was set up to proceed exactly as we have outlined previously. With some slight rearrangement of the steps, it can be made a little more efficient. If we think of the testing and assignment line as one step, then algorithm WARSHALL requires n^3 steps in all. The Boolean product of two $n \times n$ Boolean matrices A and B also requires n^3 steps, since we must compute n^2 entries, and each of these requires n comparisons. To compute all products M_R, $(M_R)^2$, $(M_R)^3$, . . . , $(M_R)^n$, we require $n^3(n - 1)$ steps, since we will need $n - 1$ matrix multiplications. The formula

$$M_R = M_R \vee (M_R)^2 \vee \cdots \vee (M_R)^n, \tag{1}$$

if implemented directly, would require about n^4 steps without the final joins. Thus Warshall's algorithm is a significant improvement over direct computation of M_R using formula (1).

An interesting application of the transitive closure is to equivalence relations. We showed in Section 7 that if R and S are equivalence relations on a set A, then $R \cap S$ is also an equivalence relation on A. The relation $R \cap S$ is the largest equivalence relation contained in both R and S, since it is the largest subset of $A \times A$ contained in both R and S. We would like to know the smallest equivalence relation that contains both R and S. The natural candidate is $R \cup S$, but this relation is not necessarily transitive. The solution is given in the next theorem.

THEOREM 3 If R and S are equivalence relations on a set A, then the smallest equivalence relation containing both R and S is $(R \cup S)^\infty$.

Proof

Recall that Δ is the relation of equality on A and that a relation is reflexive if and only if it contains Δ. Then $R \subseteq R \cup S$, $S \subseteq R \cup S$ since both are reflexive, so $R \cup S \subseteq (R \cup S)^\infty$, and $(R \cup S)^\infty$ is also reflexive.

Since R and S are symmetric, $R = R^{-1}$ and $S = S^{-1}$, so $(R \cup S)^{-1} = R^{-1} \cup S^{-1} = R \cup S$, and $R \cup S$ is also symmetric. Because of this, all paths in $R \cup S$ are "two-way streets," and it follows from the definitions that $(R \cup S)^\infty$ must also be symmetric. Since we already know that $(R \cup S)^\infty$ is transitive, it is an equivalence relation containing $R \cup S$. It is the smallest one, because no smaller set containing $R \cup S$ can be transitive, by definition of the transitive closure.

Example 3 Let $A = \{1, 2, 3, 4, 5\}$, $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (4, 3), (4, 4), (5, 5)\}$, and $S = \{(1, 1), (2, 2), (3, 3), (4, 4), (4, 5), (5, 4), (5, 5)\}$. The reader may verify that both R and S are equivalence relations. The partition A/R of A corresponding to R is $\{\{1, 2\}, \{3, 4\}, \{5\}\}$, and the partition A/S of A corresponding to S is $\{\{1\}, \{2\}, \{3\}, \{4, 5\}\}$. Find the smallest equivalence relation containing R and S, and compute the partition of A that it produces.

Solution

We have

$$M_R = \begin{matrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{matrix} \quad \text{and} \quad M_S = \begin{matrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{matrix},$$

so

$$M_{R \cup S} = M_R \vee M_S = \begin{matrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{matrix}.$$

We now compute $M_{(R \cup S)}$ by Warshall's algorithm. First $W_0 = M_{R \cup S}$. We next compute W_1, so $k = 1$. Since W_0 has 1's in locations 1 and 2 of column 1 and in locations 1 and 2 of row 1, we find that no new 1's must be adjoined to W_1. Thus

$$W_1 = W_0.$$

We now compute W_2, so $k = 2$. Since W_1 has 1's in locations 1 and 2 of column 2 and in locations 1 and 2 of row 2, we find that no new 1's must be added to W_1. Thus

$$W_2 = W_1.$$

We next compute W_3, so $k = 3$. Since W_2 has 1's in locations 3 and 4 of column 3 and in locations 3 and 4 of row 3, we find that no new 1's must be added to W_2. Thus

$$W_3 = W_2.$$

Things change when we now compute W_4. Since W_3 has 1's in locations 3, 4, and 5 of column 4 and in locations 3, 4, and 5 of row 4, we must add new 1's to W_3 in positions 3, 5 and 5, 3. Thus

$$W_4 = \begin{matrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{matrix}.$$

The reader may verify that $W_5 = W_4$ and thus

$$(R \cup S) = \{(1,1), (1,2), (2,1), (2,2), (3,3), (3,4), (3,5), (4,3), (4,4), (4,5), (5,3), (5,4), (5,5)\}.$$

The corresponding partition of A is then (verify) $\{\{1, 2\}, \{3, 4, 5\}\}$.

8 Exercises

1. (a) Let $A = \{1, 2, 3\}$ and let $R = \{(1,1), (1,2), (2,3), (1,3), (3,1), (3,2)\}$. Compute the matrix M_R of the transitive closure R^∞ by using the formula

$$M_R = M_R \vee (M_R)^2 \vee (M_R)^3.$$

   (b) List the relation R whose matrix was computed in part (a).

2. For the relation R of Exercise 1, compute the transitive closure R^∞ by using Warshall's algorithm.

3. Let $A = \{a_1, a_2, a_3, a_4, a_5\}$ and let R be a relation on A whose matrix is

$$M_R = \begin{matrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{matrix} = W_0.$$

Compute W_1, W_2, and W_3 as in Warshall's algorithm.

4. Find R^* for the relation in Exercise 3.

5. Let $A = Z^+$ and R be the relation on A defined by $a\ R\ b$ if and only if $b = a + 1$. Give the transitive closure of R.

6. Let A be the set of all people and R be the relation on A defined by $a\ R\ b$ if and only if b is the mother of a. Give the transitive closure of R.

7. Prove that if R is reflexive and transitive, then $R^n = R$ for all n.

8. Let R be a relation on a set A, and let $S = R^2$. Prove that if $a, b \in A$, then $a\ S\ b$ if and only if there is a path in R from a to b having an even number of edges.

In Exercises 9 through 12, let $A = \{1, 2, 3, 4\}$. For the relation R whose matrix is given, find the matrix of the transitive closure by using Warshall's algorithm.

9. $M_R = \begin{matrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{matrix}$

10. $M_R = \begin{matrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{matrix}$

11. $M_R = \begin{matrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{matrix}$

12. $M_R = \begin{matrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{matrix}$

13. Let $A = \{1, 2, 3, 4\}$ and $R = \{(2, 1), (2, 3), (3, 2), (3, 3), (2, 2), (4, 2)\}$. Define R_r to be the reflexive closure of R and R_s to be the symmetric closure of R. Prove or disprove that the symmetric closure of R_r is the same relation as the reflexive closure of R_s.

14. Let $A = \{1, 2, 3, 4\}$ and $R = \{(2, 1), (2, 3), (3, 2), (3, 3), (2, 2), (4, 2)\}$. Define R_t to be the transitive closure of R and R_s to be the symmetric closure of R. Prove or disprove that the symmetric closure of R_t is the same relation as the transitive closure of R_s.

In Exercises 15 and 16, let $A = \{1, 2, 3, 4, 5\}$ and let R and S be the equivalence relations on A whose matrices are given. Compute the matrix of the smallest equivalence relation containing R and S, and list the elements of this relation.

15. $M_R = \begin{matrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{matrix}$ $M_S = \begin{matrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{matrix}$

16. $M_R = \begin{matrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{matrix}$ $M_S = \begin{matrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{matrix}$

17. Compute A/R, A/S, and the partition of A that corresponds to the equivalence relation found in Exercise 15.

18. Compute A/R, A/S, and the partition of A that corresponds to the equivalence relation found in Exercise 16.

19. Examine the results of Example 3 and Exercises 17 and 18. Based on these, give a procedure for producing $A/(R \cup S)$ from A/R and A/S. Explain why the procedure works.

20. Why is the procedure developed in Exercise 19 not a replacement for Warshall's algorithm?

21. Let $A = \{1, 2, 3, 4\}$ and let R and S be the relations on A described by

$$M_R = \begin{matrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{matrix}$$

and

$$M_S = \begin{matrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{matrix}.$$

Use Warshall's algorithm to compute the transitive closure of $R \cup S$.

22. Let $A = \{a, b, c, d, e\}$ and let R and S be the relations on A described by

$$M_R = \begin{matrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{matrix}$$

and

$$M_S = \begin{matrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{matrix}.$$

Use WarshallÕs algorithm to compute the transitive closure of $R \cup S$.

23. Outline the strategy of the proof of Theorem 1. What type of proof is it?

24. Outline the strategy of the proof of Theorem 2. What type of proof is it?

25. Let $A = \mathbb{R}$ and R be the relation defined by $a \, R \, b$ if and only if $|a| < |b|$. Compute the smallest equivalence relation containing R.

Tips for Proofs

Before beginning a proof, you should be able to restate the statement in your own words. Consider carefully what the statement says. For example, Theorem 1, Section 1, tells how to count the elements of $|A \times B|$. Thus to prove it you should try to apply one of the counting methods.

Many statements about relations are statements about them as sets. As an example, consider Section 2, Theorem 1. Remember that a very common way to show that two sets are equal is to show that each is a subset of the other. We do have other representations for relations, too, and in some cases a proof based on matrix or digraph ideas may be clearer. Also, this chapter contains many facts about relational properties, operations on relations, and their interactions. These facts can form the basis of a proof at the name level rather than at the element level; see, for example, Section 7, Theorem 2.

Many definitions in this chapter are biconditional statements. A biconditional, *p if and only if q*, is generally proved in two parts: *if p then q* and *if q then p*. This is done for Lemma 1, Section 5. A frequently used grammatical structure is to introduce the second part of the proof with the word *conversely*. See Theorem 6, Section 7. Occasionally the proof of a biconditional is of the form $p \Leftrightarrow q \Leftrightarrow r \Leftrightarrow \cdots \Leftrightarrow t$ [Theorem 2(c), Section 7].

Checking whether a relation is an equivalence relation or has a certain relational property is the same as proving or disproving the statement that the property $P(R)$. For this reason, you must work with generic elements of R. If R is small, check all cases.

Some exercises in this chapter ask you to analyze a proof and outline its strategy or identify its key points. These exercises should help you develop the habit of looking at all proofs for these features. Understanding how proofs you read are carried out will help you create proofs yourself. At this point you should be able to read a simple proof with understanding and recognize its structure.

Key Ideas for Review

- $A \times B$ (product set or Cartesian product): $\{(a, b) \mid a \in A \text{ and } b \in B\}$
- $|A \times B| = |A| \, |B|$
- Database: collection of data
- Relational database
- Attribute: characteristic of data
- Select: database operation that retrieves a set of records
- Project: database operation that reports partial records
- Partition or quotient set
- Relation from A to B: subset of $A \times B$
- Domain and range of a relation
- Relative sets $R(a)$, a in A, and $R(B)$, B a subset of A
- Matrix of a relation
- Digraph of a relation: pictorial representation of a relation
- Path of length n from a to b in a relation R: finite sequence $a, x_1, x_2, \ldots, x_{n-1}, b$ such that $a \, R \, x_1$, $x_1 \, R \, x_2$, $\ldots$, $x_{n-1} \, R \, b$
- $x \, R^n \, y$ (R a relation on A): There is a path of length n from x to y in R.
- $x \, R^* \, y$ (connectivity relation for R): Some path exists in R from x to y.

- Theorem: $M_{R^n} = M_R \odot M_R \odot \cdots \odot M_R$ (n factors)
- Properties of relations on a set A

Reflexive	$(a, a) \in R$ for all $a \in A$
Irreflexive	$(a, a) \notin R$ for all $a \in A$
Symmetric	$(a, b) \in R$ implies that $(b, a) \in R$
Asymmetric	$(a, b) \in R$ implies that $(b, a) \notin R$
Antisymmetric	$(a, b) \in R$ and $(b, a) \in R$ imply that $a = b$
Transitive	$(a, b) \in R$ and $(b, c) \in R$ imply that $(a, c) \in R$

- Graph of a symmetric relation
- Adjacent vertices
- Equivalence relation: reflexive, symmetric, and transitive relation
- Equivalence relation determined by a partition
- Linked-list computer representation of a relation
- Data structure: conceptual way to organize data
- $a \overline{R} b$ (complement of R): $a \overline{R} b$ if and only if $a \not R b$
- R^{-1}: $(x, y) \in R^{-1}$ if and only if $(y, x) \in R$
- $M_{R \cup S} = M_R \vee M_S$
- $M_{R \cap S} = M_R \wedge M_S$
- $M_{R^{-1}} = (M_R)^T$
- $M_{\overline{R}} = \overline{M_R}$
- If R and S are equivalence relations, so is $R \cap S$
- $S \circ R$
- $M_{S \circ R} = M_R \odot M_S$
- Theorem: R^* is the smallest transitive relation on A that contains R
- Theorem: If $|A| = n$, $R^* = R \cup R^2 \cup \cdots \cup R^n$
- Warshall's algorithm: computes M_{R^*} efficiently
- Theorem: If R and S are equivalence relations on A, $(R \cup S)^*$ is the smallest equivalence relation on A containing both R and S.

Chapter Self-Test

1. What kind of mathematical object is $A \times B$?
2. What kind of mathematical object is a partition of A?
3. What are the ways to represent a relation on a set?
4. Which representations of a relation R on a finite set A would you use to test the transitivity of R? Why?
5. What are two interpretations of the transitive closure of a relation R?
6. Let $A = \{2, 5, 7\}$ and $B = \{x \mid x \in Z^+ \text{ and } x^3 < 100\}$.
   (a) What is $|A \times B|$? (b) List $A \times B$.
7. Let A and B be subsets of the universal set U. Then $A \times B \subseteq U \times U$. Is $\overline{A \times B} = \overline{A} \times \overline{B}$? Justify your answer.
8. Give all two-element partitions of $\{a, b, c, d, e\}$.
9. Let $C = \{2, 8, 14, 18\}$. Define a relation on C by $x \; R \; y$ if and only if $x - y > 5$.
   (a) Draw the digraph of R.
   (b) Give M_R.
10. Let $B = \{a, b, c, d\}$ and $R = \{(a, a), (a, b), (b, c), (c, d), (d, b)\}$.
    (a) Draw the digraphs of R and R^2.
    (b) Give M_R and M_{R^2}.
    (c) Give M_{R^*}.
11. Determine whether the relation R on the set A is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, or transitive, if $A = Z^+$; $x \; R \; y$ if and only if $x \mid 3y$.
12. Let $D = \{1, 2, 3, 4, 5, 6\}$ and R be the relation on D whose matrix is

$$M_R = \begin{matrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \end{matrix}.$$

    Determine whether R is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, or transitive.
13. Suppose R is a relation on a set A and that R is asymmetric. Can R also be antisymmetric? Must R be antisymmetric? Explain your answers.
14. Let $B = \{1, 2, 3, 4, 5\}$, $A = B \times B$, and define R on A as follows: $(u, v) \; R \; (x, y)$ if and only if $u - v = x - y$.
    (a) Prove that R is an equivalence relation.
    (b) Find $[(2, 3)]$.
    (c) Compute A/R.
15. The following arrays describe a relation R on the set $A = \{1, 2, 3, 4\}$. Give the matrix of R.

VERT = [5, 3, 1, 8]
TAIL = [3, 3, 2, 2, 1, 1, 4, 4]
HEAD = [1, 4, 1, 3, 2, 3, 4, 2]
NEXT = [2, 0, 4, 0, 6, 0, 0, 7]

16. Let R and S be relations on $\{a, b, c, d, e\}$ where $R = \{(a, b), (a, c), (b, c), (c, e), (e, a), (a, a), (d, c)\}$ and $S = \{(a, a), (a, b), (b, a), (c, c), (c, d), (d, e), (b, e), (e, d)\}$.

(a) Give R^{-1}. (b) Compute $R \circ S$.

17. Let $R = \{(1, 4), (2, 1), (2, 5), (2, 4), (4, 3), (5, 3), (3, 2)\}$. Use Warshall's algorithm to find the matrix of the connectivity relation based on R.

Experiment 4

Equivalence relations and partial orders are defined as relations with certain properties. In this experiment, you will investigate compatibility relations that are also defined by the relation properties they have in common. A compatibility relation is a relation that is reflexive and symmetric. Every equivalence relation is a compatibility relation, but here you will focus on compatibility relations that are not equivalence relations.

Part I.

1. Verify that the relation R on A is a compatibility relation.
   (a) A is the set of students at your college; $x R y$ if and only if x and y have taken the same course.
   (b) A is the set of all triangles; $x R y$ if and only if x and y have an angle with the same measure.
   (c) $A = \{1, 2, 3, 4, 5\}$; $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (2, 3), (3, 2), (4, 1), (1, 4), (2, 4), (4, 2), (1, 2), (2, 1), (4, 5), (5, 4), (1, 3), (3, 1)\}$.
2. In Part I.1(c), the relation is given as a set of ordered pairs. A relation can also be represented by a matrix or a digraph. Describe how to determine if R is a compatibility relation using its
   (a) Matrix.
   (b) Digraph.
3. Give another example of a compatibility relation that is not an equivalence relation.

Part II. Every relation has several associated relations that may or may not share its properties.

1. If R is a compatibility relation, is R^{-1}, the inverse of R, also a compatibility relation? If so, prove this. If not, give a counterexample.
2. If R is a compatibility relation, is $\overline{R}$, the complement of R, also a compatibility relation? If so, prove this. If not, give a counterexample.
3. If R and S are compatibility relations, is $R \circ S$ also a compatibility relation? If so, prove this. If not, give a counterexample.

Part III. In Section 5, we showed that each equivalence relation R on a set A gives a partition of A. A compatibility relation R on a set A gives instead a covering of A. A covering of A is a set of subsets of A, $\{A_1, A_2, \ldots, A_k\}$, such that $\bigcup_{i=1}^{k} A_i = A$. We define a *maximal compatibility block* to be a subset B of A with each element of B related by R to every other element of B, and no element of $A - B$ is R-related to every element of B. For example, in Part I.1(c), the sets $\{1, 2, 3\}$ and $\{1, 2, 4\}$ are maximal compatibility blocks. The set of all maximal compatibility blocks relative to a compatibility relation R forms a covering of A.

1. Give all maximal compatibility blocks for the relation in Part I.1(c). Verify that they form a covering of A.

2. Describe the maximal compatibility blocks for the relation in Part I.1(b). The set of all maximal compatibility blocks form a covering of A. Is this covering also a partition for this example? Explain.
3. The digraph of a compatibility relation R can be simpliÞed by omitting the loop at each vertex and using a single edge with no arrow between related vertices.
   (a) Draw the simpliÞed graph for the relation in Part I.1(c).
   (b) Describe how to Þnd the maximal compatibility blocks of a compatibility relation, given its simpliÞed graph.
4. Find the covering of A associated with the relation whose simpliÞed graph is given in
   (a) Figure 1. (b) Figure 2. (c) Figure 3.

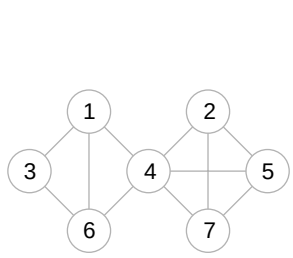


Figure 1

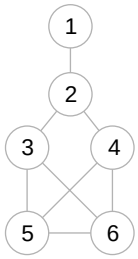


Figure 2

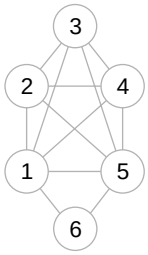


Figure 3

5. Given the following covering of A, produce an associated compatibility relation R; that is, one whose maximal compatibility blocks are the elements of the covering.

$$\{\{1, 2\}, \{1, 3, 6, 7\}, \{4, 5, 11\},$$
$$\{5, 10\}, \{8, 5\}, \{2, 8, 9\}, \{3, 9\}, \{9, 10\}\}$$

   Is there another compatibility relation that would produce the same covering of A?

Coding Exercises

1. Write a program CROSS with input positive integers m and n and output the set $A \times B$ where $A = \{1, 2, 3, \ldots, m\}$ and $B = \{1, 2, 3, \ldots, n\}$.
2. (a) Write a subroutine that has as input the matrix of a relation and determines whether the relation is reßexive.
   (b) Write a subroutine that has as input the matrix of a relation and determines whether the relation is symmetric.
3. Write a program that has as input the matrix of a relation and determines whether the relation is an equivalence relation.
4. Let R and S be relations represented by matrices M_R and M_S, respectively. Write a subroutine to produce the matrix of
   (a) R S (b) R S (c) R S
5. Let R be a relation represented by the matrix M_R. Write a subroutine to produce the matrix of
   (a) R^{-1} (b) $\overline{R}$

17. (a) {(a, a), (a, d), (a, e), (b, b), (b, c), (b, e), (c, a) , (c, b), (c, c), (d, b), (d, c), (d, d), (e, c), (e, e) }.

    (b) {(a, a), (a, d), (d, a), (a, e), (e, a), (b, c), (c, b) , (b, e), (e, b), (c, a), (a, c), (c, c), (d, b), (b, d), (d, c) , (c, d), (e, c), (c, e), (e, e)}.

19. The definitions of irreflexive, asymmetric, and antisymmetric each require that a certain pair does not belong to R. We cannot "fix" this by including more pairs in R.

21. (a) Yes. (b) Yes.

    (c) x (S ∘ R) y if and only if x ≠ y.

23. (a) Reflexive. a R a ⇒ a S a ⇒ a S ∘ R a.
    Irreflexive. No. 1 R 2 ⇒ 2 S 1 ⇒ 1 S ∘ R 1.
    Symmetric. No. 1 R 3, 3 R 1, 3 S 2, 2 S 3 ⇒ 1 S ∘ R 2, but 2 S ∘ R 1.
    Asymmetric. No. R = {(1, 2), (3, 4)} and S = {(2, 3), (4, 1)} provide a counterexample.
    Antisymmetric. No. R = {(a, b), (c, d)} and S = {(b, c), (d, a)} provide a counterexample.
    Transitive. No. R = {(a, d), (b, e)} and S = {(d, b), (e, c)} provide a counterexample.

    (b) No, symmetric and transitive properties are not preserved.

25. (a) $\begin{matrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{matrix}$.

    (b) $\begin{matrix} 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{matrix}$.

    (c) $\begin{matrix} 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{matrix}$.

    (d) $\begin{matrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{matrix}$.

27. R ∘ S is antisymmetric. If a(R ∘ S)b and b(R ∘ S)a, then a R b and b R a. Hence a = b because R is antisymmetric. R ∘ S may not be antisymmetric. Let R = {(1, 2)}, S = {(2, 1)}.

29. Let R = {(x, z), (x, m)}, S = {(z, y)}, and T = {(m, y)}. Then (x, y) ∈ (S ∘ R) ∩ (T ∘ R), but (S ∩ T) ∘ R = { }.

31. (a) Let $M_{R \cap S} = [m_{ij}]$, $M_R = [r_{ij}]$, $M_S = [s_{ij}]$. $m_{ij} = 1$ if and only if (i, j) ∈ R ∩ S. (i, j) ∈ R if and only if $r_{ij} = 1$ and (i, j) ∈ S if and only if $s_{ij} = 1$. But this happens if and only if the i, j th entry of $M_R \wedge M_S$ is 1.

    (b) Let $M_{R \cap S} = [m_{ij}]$, $M_R = [r_{ij}]$, $M_S = [s_{ij}]$. $m_{ij} = 1$ if and only if (i, j) ∈ R ∩ S. (i, j) ∈ R if and only if $r_{ij} = 1$ or (i, j) ∈ S if and only if $s_{ij} = 1$. But this happens if and only if the i, j th entry of $M_R \vee M_S$ is 1.

    (c) The i, j th entry of $M_{R^{-1}}$ is 1 if and only if (i, j) ∈ R^{-1} if and only if (j, i) ∈ R if and only if the j, i th entry of M_R is 1 if and only if the i, j th entry of M_R^T is 1.

    (d) The i, j th entry of $M_{\overline{R}}$ is 1 if and only if (i, j) ∈ $\overline{R}$ if and only if (i, j) ∉ R if and only if the i, j th entry of M_R is 0 if and only if the i, j th entry of $\overline{M_R}$ is 1.

33. To form the digraph of R^{-1}, reverse the arrows in the digraph of R.

35. The edges of the digraph of R ∩ S are the edges that appear on either the digraph of R or the digraph of S.

37. (a) R is symmetric if and only if x R y ⇔ y R x if and only if R = R^{-1} ∘ R.

    (b) Suppose R is antisymmetric. Let (x, y) ∈ R ∩ R^{-1}, then x = y and (x, y) ∈ . Suppose R ∩ R^{-1} = . If x R y and y R x, then x R y and x R^{-1} y. Thus (x, y) ∈ , so x = y.

    (c) Suppose R is asymmetric. Let (x, y) ∈ R ∩ R^{-1}. This contradicts the fact that R is asymmetric. Hence R ∩ R^{-1} = . Let x R y and y R x. Then (x, y) ∈ R ∩ R^{-1} = . Hence R is asymmetric.

Exercise Set 8

1. (a) $\begin{matrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{matrix}$.

    (b) {(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)}.

3. $W_1 = \begin{matrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{matrix}$, $W_2 = W_1 = W_3$.

5. a R b if and only if b > a.

7. Let R be reflexive and transitive. Suppose that x R^n y. Then $x, a_1, a_2, \ldots, a_{n-1}, y$ is a path of length n from x to y. x R a_1 ⇒ a_1 R a_2 ⇒ x R a_2. Similarly, we have x R a_k ⇒ a_k R a_{k+1} ⇒ x R a_{k+1} and finally x R a_{n-1} ⇒ a_{n-1} R y ⇒ x R y. Hence R^n ⊆ R. If x R y, then since R is reflexive we can build a path of length n: x, x, x, . . . , x, y from x to y and x R^n y.

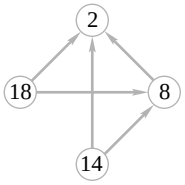
9. $\begin{matrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{matrix}$.

11. $\begin{matrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{matrix}$.

13. The sets are the same. This can be shown by computing each set.
15. $A \times A$.
17. $A/R = \{\{1, 2, 3\}, \{4, 5\}\}$, $A/S = \{\{1\}, \{2, 3, 4\}, \{5\}\}$, $A/(R \cap S) = \{A\}$.
19. The collection of elements in A/R and A/S can be separated into subcollections of nondisjoint sets. Each element of $A/(R \cap S)$ is the union of the sets in one of these subcollections.
21. $\{(1, 1), (1, 2), (1, 4), (2, 2), (3, 2), (3, 3), (4, 2), (4, 3), (4, 4), (1, 3)\}$.
23. We first show R is transitive. Then we show it is the smallest relation that contains R. It is a direct proof.
25. $R \times R$.

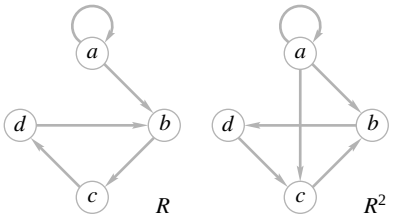
Answers to Chapter Self-Tests

1. $A \times B$ is a set of ordered pairs.
2. A partition of A is a set of nonempty subsets of A.
3. A relation may be given by a verbal description, a set of ordered pairs, a digraph (in some cases) and as a matrix (in some cases).
4. Unless $|R|$ is very small, using the matrix representation will be the most efficient way to determine if R is transitive.
5. The transitive closure of R is the connectivity relation R^*; it is also the smallest transitive relation containing R.
6. (a) 12.
   (b) $\{(2, 1), (2, 2), (2, 3), (2, 4), (5, 1), (5, 2), (5, 3), (5, 4), (7, 1), (7, 2), (7, 3), (7, 4)\}$.
7. Let $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, $B = \{2, 3\}$. Then $(2, 5) \in \overline{A \times B}$, but $(2, 5) \notin \overline{A} \times \overline{B}$.
8. $\{\{a, b, c\}, \{d, e\}\}$, $\{\{a, b, d\}, \{c, e\}\}$, $\{\{a, b, e\}, \{c, d\}\}$, $\{\{b, c, d\}, \{a, e\}\}$, $\{\{b, c, e\}, \{a, d\}\}$, $\{\{b, d, e\}, \{a, c\}\}$, $\{\{c, d, e\}, \{a, b\}\}$, $\{\{a, c, d\}, \{b, e\}\}$, $\{\{a, c, e\}, \{b, d\}\}$, $\{\{a, d, e\}, \{b, c\}\}$, $\{\{a, b, c, d\}, \{e\}\}$, $\{\{a, b, d, e\}, \{c\}\}$, $\{\{b, c, d, e\}, \{a\}\}$, $\{\{a, b, c, e\}, \{d\}\}$, $\{\{a, c, d, e\}, \{b\}\}$.
9. (a)



   (b) $\begin{matrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{matrix}$.

10. (a)



    (b) $M_R = \begin{matrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{matrix}$.

    $M_{R^2} = \begin{matrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{matrix}$.

    (c) $M_R = \begin{matrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{matrix}$.

11. Reflexive, not irreflexive, not symmetric, not asymmetric, not antisymmetric, not transitive.
12. Not reflexive, not irreflexive, not symmetric, not asymmetric, antisymmetric, not transitive.
13. Since $a \mathrel{R} b$ implies $b \not\mathrel{R} a$, $a \mathrel{R} b \rightarrow b \mathrel{R} a$ is always false. Hence $a \mathrel{R} b \rightarrow b \mathrel{R} a \rightarrow a = b$ is always true. R must be antisymmetric.
14. (a) $(u, v) \mathrel{R} (u, v)$ since $u \le v = u \le v$ is true. Thus R is reflexive. If $(u, v) \mathrel{R} (x, y)$, then $u \le v = x \le y$ and $(x, y) \mathrel{R} (u, v)$. Thus R is symmetric.
    If $((u, v) \mathrel{R} (x, y)) \wedge ((x, y) \mathrel{R} (w, z))$, then $u \le v = x \le y = w \le z$ and $(u, v) \mathrel{R} (w, z)$. Hence R is transitive.
    (b) $[(2, 3)] = \{(2, 3), (1, 2), (3, 4), (4, 5)\}$.
    (c) $A/R = \{[(2, 3)], [(2, 4)], [(2, 5)], [(2, 2)], [(2, 1)], [(3, 1)], [(4, 1)], [(5, 1)], [(1, 5)]\}$.
15. $M_R = \begin{matrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{matrix}$.
16. (a) $R^{-1} = \{(a, a), (a, e), (b, a), (c, a), (c, b), (c, d), (e, c)\}$.
    (b) $R \circ S = \{(a, a), (a, b), (a, c), (b, b), (b, a), (b, c), (c, e), (c, c), (d, a), (e, c)\}$.
17. $M_R = \begin{matrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{matrix}$.

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Functions

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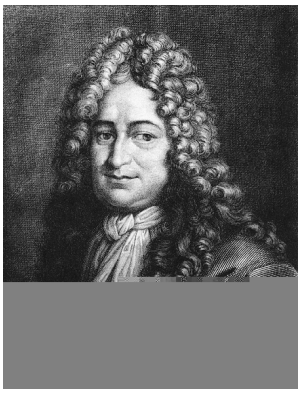
Functions

In this chapter we focus our attention on a special type of relation, a function, that plays an important role in mathematics, computer science, and many applications. We also define some functions used in computer science and examine the growth of functions.

Looking Back

The origins of the notion of a function can be traced back to the great Italian philosopher, astronomer, and mathematician Galileo Galilei (1564–1642), who in the 1630s observed the relationship between two variables. The early work on functions in the second half of the seventeenth century concentrated on the study of special functions as curves. These included the power, exponential, logarithmic, and trigonometric functions. Gottfried Wilhelm Leibniz (1646–1716) was the first person to use the word *function* for a quantity whose value varies as a point moves on a curve. Leibniz was an extraordinary person who made brilliant contributions in a number of diverse areas, including logic, philosophy, law, metaphysics, religion, mathematics, diplomacy, and literature. He has often been called a "universal genius." Leibniz was born in Leipzig, and died in Hanover, both in Germany. Early in his career, he developed the foundations for what would later be called symbolic logic. Leibniz began his study of advanced contemporary mathematics in 1672 at the age of 26. Three years later he discovered the Fundamental Theorem of Calculus independently of Newton, who had also discovered the same result. Indeed, a heated battle raged over a number of years between the supporters of Leibniz and Newton as to who had discovered calculus first. Today, both Newton and Leibniz are considered the fathers of calculus. It is quite surprising to learn what a visionary Leibniz was. In the 1670s he invented a mechanical calculator, known as the *Leibniz wheel*, capable of adding, subtracting, multiplying, and dividing. He almost envisioned the modern age of computing!



Gottfried Wilhelm Leibniz
Library of Congress

Leonhard Euler
Bavarian State Library, Munich

The commonly used notation for a function value, $f(x)$, is due to Leonhard Euler (1707–1783), who was born in Basel, Switzerland, and died in St. Petersburg, Russia. Euler is one of the greatest and most prolific mathematicians in history. After his death, it took nearly 50 years to publish all his papers and his collected works comprise more than 75 volumes. He was also able to carry out complex calculations in his head. During the last 17 years of his life, Euler was totally blind, but his mathematical output remained undiminished. Euler made significant contributions to many areas of mathematics and used mathematics to solve a wide variety of problems in the sciences.

1 Functions

In this section we define the notion of a function, a special type of relation. We study its basic properties and then discuss several special types of functions. A number of important applications of functions will occur in later sections so it is essential to get a good grasp of the material in this section.

Let A and B be nonempty sets. A **function** f from A to B, which is denoted $f : A \to B$, is a relation from A to B such that for all $a \in \mathrm{Dom}(f)$, $f(a)$, the f-relative set of a, contains just one element of B. Naturally, if a is not in $\mathrm{Dom}(f)$, then $f(a) = \emptyset$. If $f(a) = \{b\}$, it is traditional to identify the set $\{b\}$ with the element b and write $f(a) = b$. We will follow this custom, since no confusion results. The relation f can then be described as the set of pairs $\{(a, f(a)) \mid a \in \mathrm{Dom}(f)\}$. Functions are also called **mappings** or **transformations**, since they can be geometrically viewed as rules that assign to each element $a \in A$ the unique element $f(a) \in B$ (see Figure 1). The element a is called an **argument** of the function f, and $f(a)$ is called the **value** of the function for the argument a and is also referred to as the **image** of a under f. Figure 1 is a schematic or pictorial display of our definition of a function, and we will use several other similar diagrams. They should not be confused with the digraph of the relation f, which we will not generally display.

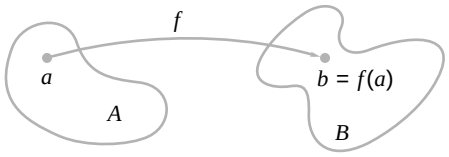


Figure 1

Example 1 Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c, d\}$, and let

$$f = \{(1, a), (2, a), (3, d), (4, c)\}.$$

Here we have

$$f(1) = a$$
$$f(2) = a$$
$$f(3) = d$$
$$f(4) = c.$$

Since each set $f(n)$ is a single value, f is a function.

Note that the element $a \in B$ appears as the second element of two different ordered pairs in f. This does not conflict with the definition of a function. Thus a function may take the same value at two different elements of A.

Example 2 Let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$. Consider the relations

$$R = \{(1, x), (2, x)\} \quad \text{and} \quad S = \{(1, x), (1, y), (2, z), (3, y)\}.$$

The relation S is not a function since $S(1) = \{x, y\}$. The relation R is a function with $\mathrm{Dom}(R) = \{1, 2\}$ and $\mathrm{Ran}(R) = \{x\}$.

Example 3 Let P be a computer program that accepts an integer as input and produces an integer as output. Let $A = B = Z$. Then P determines a relation f_P defined as

follows: $(m, n) \in f_P$ means that n is the output produced by program P when the input is m.

It is clear that f_P is a function, since any particular input corresponds to a unique output. (We assume that computer results are reproducible; that is, they are the same each time the program is run.)

Example 3 can be generalized to a program with any set A of possible inputs and set B of corresponding outputs. In general, therefore, we may think of functions as **input-output** relations.

Example 4 Let $A = R$ be the set of real numbers, and let $p(x) = a_0 + a_1x + \cdots + a_nx^n$ be a real polynomial. Then p may be viewed as a relation on R. For each r in R we determine the relative set $p(r)$ by substituting r into the polynomial. Then, since all relative sets $p(r)$ are known, the relation p is determined. Since a unique value is produced by this substitution, the relation p is actually a function.

If the formula defining the function does not make sense for all elements of A, then the domain of the function is taken to be the set of elements for A for which the formula does make sense.

In elementary mathematics, the *formula* (in the case of Example 4, the polynomial) is sometimes confused with the *function* it produces. This is not harmful, unless the student comes to expect a formula for every type of function.

Suppose that, in the preceding construction, we used a formula that produced more than one element in $p(x)$, for example, $p(x) = \pm \sqrt{x}$. Then the resulting relation would not be a function. For this reason, in older texts, relations were sometimes called multiple-valued functions.

Example 5 A **labeled digraph** is a digraph in which the vertices or the edges (or both) are labeled with information from a set. If V is the set of vertices and L is the set of labels of a labeled digraph, then the labeling of V can be specified to be a function $f : V \to L$, where, for each $v \in V$, $f(v)$ is the label we wish to attach to v. Similarly, we can define a labeling of the edges E as a function $g : E \to L$, where, for each $e \in E$, $g(e)$ is the label we wish to attach to e. An example of a labeled digraph is a map on which the vertices are labeled with the names of cities and the edges are labeled with the distances or travel times between the cities. Figure 2 shows an example of a labeled digraph. Another example is a flow chart of a program in which the vertices are labeled with the steps that are to be performed at that point in the program; the edges indicate the flow from one part of the program to another part.

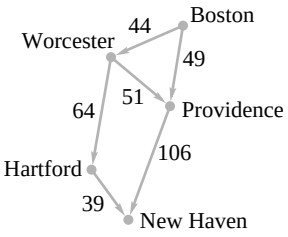


Figure 2

Example 6 Let $A = B = Z$ and let $f : A \to B$ be defined by

$$f(a) = a + 1, \quad \text{for } a \in A.$$

Here, as in Example 4, f is defined by giving a formula for the values $f(a)$.

Example 7 Let $A = Z$ and let $B = \{0, 1\}$. Let $f : A \to B$ be found by

$$f(a) = \begin{cases} 0 & \text{if } a \text{ is even} \\ 1 & \text{if } a \text{ is odd.} \end{cases}$$

Then f is a function, since each set $f(a)$ consists of a single element. Unlike the situation in Examples 4 and 6, the elements $f(a)$ are not specified through an algebraic formula. Instead, a verbal description is given.

Example 8 Let A be an arbitrary nonempty set. The **identity function on *A***, denoted by 1_A, is defined by $1_A(a) = a$.

The reader may notice that 1_A is the relation we previously called Δ which stands for the diagonal subset of $A \times A$. In the context of functions, the notation 1_A is preferred, since it emphasizes the input-output or functional nature of the relation. Clearly, if $A_1 \subseteq A$, then $1_A(A_1) = A_1$.

Suppose that $f : A \to B$ and $g : B \to C$ are functions. Then the composition of f and g, $g \circ f$ is a relation. Let $a \in \mathrm{Dom}(g \circ f)$. Then $(g \circ f)(a) = g(f(a))$. Since f and g are functions, $f(a)$ consists of a single element $b \in B$, so $g(f(a)) = g(b)$. Since g is also a function, $g(b)$ contains just one element of C. Thus each set $(g \circ f)(a)$, for a in $\mathrm{Dom}(g \circ f)$, contains just one element of C, so $g \circ f$ is a function. This is illustrated in Figure 3.

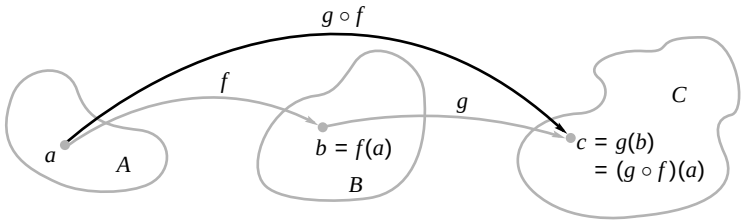


Figure 3

Example 9 Let $A = B = Z$, and C be the set of even integers. Let $f : A \to B$ and $g : B \to C$ be defined by

$$f(a) = a + 1$$
$$g(b) = 2b.$$

Find $g \circ f$.

Solution

We have

$$(g \circ f)(a) = g(f(a)) = g(a + 1) = 2(a + 1).$$

Thus, if f and g are functions specified by giving formulas, then so is $g \circ f$ and the formula for $g \circ f$ is produced by substituting the formula for f into the formula for g.

Special Types of Functions

Let f be a function from A to B. Then we say that f is **everywhere defined** if $\mathrm{Dom}(f) = A$. We say that f is **onto** if $\mathrm{Ran}(f) = B$. Finally, we say that f is **one to one** if we cannot have $f(a) = f(a')$ for two distinct elements a and a' of A. The definition of one to one may be restated in the following equivalent form:

$$\text{If } f(a) = f(a'), \quad \text{then } a = a'.$$

The latter form is often easier to verify in particular examples.

Example 10 Consider the function f defined in Example 1. Since $\mathrm{Dom}(f) = A$, f is everywhere defined. On the other hand, $\mathrm{Ran}(f) = \{a, c, d\} \subseteq B$; therefore, f is not onto. Since

$$f(1) = f(2) = a,$$

we can conclude that f is not one to one.

Example 11 Consider the function f defined in Example 6. Which of the special properties, if any, does f possess?

Solution

Since the formula defining f makes sense for all integers, $\text{Dom}(f) = Z = A$, and so f is everywhere defined.

Suppose that

$$f(a) = f(a')$$

for a and a' in A. Then

$$a + 1 = a' + 1$$

so

$$a = a'.$$

Hence f is one to one.

To see if f is onto, let b be an arbitrary element of B. Can we find an element $a \in A$ such that $f(a) = b$? Since

$$f(a) = a + 1,$$

we need an element a in A such that

$$a + 1 = b.$$

Of course,

$$a = b - 1$$

will satisfy the desired equation since $b - 1$ is in A. Hence $\text{Ran}(f) = B$; therefore, f is onto.

Example 12 Let $A = \{a_1, a_2, a_3\}$, $B = \{b_1, b_2, b_3\}$, $C = \{c_1, c_2\}$, and $D = \{d_1, d_2, d_3, d_4\}$. Consider the following four functions, from A to B, A to D, B to C, and D to B, respectively.

(a) $f_1 = \{(a_1, b_2), (a_2, b_3), (a_3, b_1)\}$
(b) $f_2 = \{(a_1, d_2), (a_2, d_1), (a_3, d_4)\}$
(c) $f_3 = \{(b_1, c_2), (b_2, c_2), (b_3, c_1)\}$
(d) $f_4 = \{(d_1, b_1), (d_2, b_2), (d_3, b_1)\}$

Determine whether each function is one to one, whether each function is onto, and whether each function is everywhere defined.

Solution

(a) f_1 is everywhere defined, one to one, and onto.
(b) f_2 is everywhere defined and one to one, but not onto.
(c) f_3 is everywhere defined and onto, but is not one to one.
(d) f_4 is not everywhere defined, not one to one, and not onto.

If $f : A \to B$ is a one-to-one function, then f assigns to each element a of $\text{Dom}(f)$ an element $b = f(a)$ of $\text{Ran}(f)$. Every b in $\text{Ran}(f)$ is matched, in this way, with one and only one element of $\text{Dom}(f)$. For this reason, such an f is often called a **bijection** between $\text{Dom}(f)$ and $\text{Ran}(f)$. If f is also everywhere defined and onto, then f is called a **one-to-one correspondence between *A* and *B***.

Example 13 Let R be the set of all equivalence relations on a given set A, and let $\square$ be the set of all partitions on A. Then we can define a function $f : R \to \square$ as follows. For each equivalence relation R on A, let $f(R) = A/R$, the partition of A that corresponds to R; f is a one-to-one correspondence between R and $\square$.

Invertible Functions

A function $f : A \to B$ is said to be **invertible** if its inverse relation, f^{-1}, is also a function. The next example shows that a function is not necessarily invertible.

Example 14 Let f be the function of Example 1. Then

$$f^{-1} = \{(a, 1), (a, 2), (d, 3), (c, 4)\}.$$

We see that f^{-1} is not a function, since $f^{-1}(a) = \{1, 2\}$.

The following theorem is frequently used.

THEOREM 1 Let $f : A \to B$ be a function.

(a) Then f^{-1} is a function from B to A if and only if f is one to one.

If f^{-1} is a function, then

(b) the function f^{-1} is also one to one.

(c) f^{-1} is everywhere defined if and only if f is onto.

(d) f^{-1} is onto if and only if f is everywhere defined.

Proof

(a) We prove the following equivalent statement.

f^{-1} is not a function if and only if f is not one to one.

Suppose first that f^{-1} is not a function. Then, for some b in B, $f^{-1}(b)$ must contain at least two distinct elements, a_1 and a_2. Then $f(a_1) = b = f(a_2)$, so f is not one to one.

Conversely, suppose that f is not one to one. Then $f(a_1) = f(a_2) = b$ for two distinct elements a_1 and a_2 of A. Thus $f^{-1}(b)$ contains both a_1 and a_2, so f^{-1} cannot be a function.

(b) Since $(f^{-1})^{-1}$ is the function f, part (a) shows that f^{-1} is one to one.

(c) Recall that $\text{Dom}(f^{-1}) = \text{Ran}(f)$. Thus $B = \text{Dom}(f^{-1})$ if and only if $B = \text{Ran}(f)$. In other words, f^{-1} is everywhere defined if and only if f is onto.

(d) Since $\text{Ran}(f^{-1}) = \text{Dom}(f)$, $A = \text{Dom}(f)$ if and only if $A = \text{Ran}(f^{-1})$. That is, f is everywhere defined if and only if f^{-1} is onto.

As an immediate consequence of Theorem 1, we see that if f is a one-to-one correspondence between A and B, then f^{-1} is a one-to-one correspondence between B and A. Note also that if $f : A \to B$ is a one-to-one function, then the equation $b = f(a)$ is equivalent to $a = f^{-1}(b)$.

Example 15 Consider the function f defined in Example 6. Since it is everywhere defined, one to one, and onto, f is a one-to-one correspondence between A and B. Thus f is invertible, and f^{-1} is a one-to-one correspondence between B and A.

Example 16 Let R be the set of real numbers, and let $f : R \to R$ be defined by $f(x) = x^2$. Is f invertible?

Solution

We must determine whether f is one to one. Since

$$f(2) = f(-2) = 4,$$

we conclude that f is not one to one. Hence f is not invertible.

There are some useful results concerning the composition of functions. We summarize these in the following theorem.

THEOREM 2 Let $f : A \to B$ be any function. Then

(a) $1_B \circ f = f$.

(b) $f \circ 1_A = f$.

If f is a one-to-one correspondence between A and B, then

(c) $f^{-1} \circ f = 1_A$.

(d) $f \circ f^{-1} = 1_B$.

Proof

(a) $(1_B \circ f)(a) = 1_B(f(a)) = f(a)$, for all a in $\mathrm{Dom}(f)$. Thus, by Theorem 2 of Section 4.2, $1_B \circ f = f$.

(b) $(f \circ 1_A)(a) = f(1_A(a)) = f(a)$, for all a in $\mathrm{Dom}(f)$, so $f \circ 1_A = f$.

Suppose now that f is a one-to-one correspondence between A and B. As we pointed out, the equation $b = f(a)$ is equivalent to the equation $a = f^{-1}(b)$. Since f and f^{-1} are both everywhere defined and onto, this means that, for all a in A and b in B, $f(f^{-1}(b)) = b$ and $f^{-1}(f(a)) = a$.

(c) For all a in A, $1_A(a) = a = f^{-1}(f(a)) = (f^{-1} \circ f)(a)$. Thus $1_A = f^{-1} \circ f$.

(d) For all b in B, $1_B(b) = b = f(f^{-1}(b)) = (f \circ f^{-1})(b)$. Thus $1_B = f \circ f^{-1}$.

THEOREM 3

(a) Let $f : A \to B$ and $g : B \to A$ be functions such that $g \circ f = 1_A$ and $f \circ g = 1_B$. Then f is a one-to-one correspondence between A and B, g is a one-to-one correspondence between B and A, and each is the inverse of the other.

(b) Let $f : A \to B$ and $g : B \to C$ be invertible. Then $g \circ f$ is invertible, and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof

(a) The assumptions mean that

$$g(f(a)) = a \quad \text{and} \quad f(g(b)) = b, \quad \text{for all } a \text{ in } A \text{ and } b \text{ in } B.$$

This shows in particular that $\mathrm{Ran}(f) = B$ and $\mathrm{Ran}(g) = A$, so each function is onto. If $f(a_1) = f(a_2)$, then $a_1 = g(f(a_1)) = g(f(a_2)) = a_2$. Thus f is one to one. In a similar way, we see that g is one to one, so both f and g are invertible. Note that f^{-1} is everywhere defined since $\mathrm{Dom}(f^{-1}) = \mathrm{Ran}(f) = B$. Now, if b is any element in B,

$$f^{-1}(b) = f^{-1}(f(g(b)) = (f^{-1} \circ f)g(b)) = 1_A(g(b)) = g(b).$$

Thus $g = f^{-1}$, so also $f = (f^{-1})^{-1} = g^{-1}$. Then, since g and f are onto, f^{-1} and g^{-1} are onto, so f and g must be everywhere defined. This proves all parts of part (a).

(b) We know that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$, since this is true for any two relations. Since g^{-1} and f^{-1} are functions by assumption, so is their composition, and then $(g \circ f)^{-1}$ is a function. Thus $g \circ f$ is invertible.

Example 17 Let $A = B = R$, the set of real numbers. Let $f : A \to B$ be given by the formula $f(x) = 2x^3 - 1$ and let $g : B \to A$ be given by

$$g(y) = \sqrt[3]{\tfrac{1}{2}y + \tfrac{1}{2}}.$$

Show that f is a bijection between A and B and g is a bijection between B and A.

Solution

Let $x \in A$ and $y = f(x) = 2x^3 - 1$. Then $\frac{1}{2}(y + 1) = x^3$; therefore,

$$x = \sqrt[3]{\tfrac{1}{2}y + \tfrac{1}{2}} = g(y) = g(f(x)) = (g \circ f)(x).$$

Thus $g \circ f = 1_A$. Similarly, $f \circ g = 1_B$, so by Theorem 3(a) both f and g are bijections.

As Example 17 shows, it is often easier to show that a function, such as f, is one to one and onto by constructing an inverse instead of proceeding directly.

Finally, we discuss briefly some special results that hold when A and B are finite sets. Let $A = \{a_1, \ldots, a_n\}$ and $B = \{b_1, \ldots, b_n\}$, and let f be a function from A to B that is everywhere defined. If f is one to one, then $f(a_1), f(a_2), \ldots, f(a_n)$ are n distinct elements of B. Thus we must have all of B, so f is also onto. On the other hand, if f is onto, then $f(a_1), \ldots, f(a_n)$ form the entire set B, so they must all be different. Hence f is also one to one. We have therefore shown the following:

THEOREM 4 Let A and B be two finite sets with the same number of elements, and let $f : A \to B$ be an everywhere defined function.

(a) If f is one to one, then f is onto.

(b) If f is onto, then f is one to one.

Thus for finite sets A and B with the same number of elements, and particularly if $A = B$, we need only prove that a function is one to one *or* onto to show that it is a bijection. This is an application of the pigeonhole principle.

One-to-one functions are a fundamental tool in cryptology, because of the need to both encode and decode. Many secret codes are simple **substitution codes** created as follows. Let $A = \{a, b, \ldots, z\}$ be the English alphabet, and let $f : A \to A$ be a function agreed on in advance by each party to a correspondence. A message is encoded by replacing each letter with its f image. In order for the message to be decoded, the function f must have an inverse. The recipient decodes the message by applying f^{-1} to each letter. We know by Theorem 3(a) that f must therefore be one to one.

Example 18 Suppose that f is defined by the following table:

A	B	C	D	E	F	G	H	I	J	K	L	M
D	E	S	T	I	N	Y	A	B	C	F	G	H

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
J	K	L	M	O	P	Q	R	U	V	W	X	Z

Thus $f(D) = T$, $f(R) = O$, and so on. The rearrangement of the alphabet that defines this function is an example of using a keyword to begin, then listing all remaining letters in order.

The phrase THE TRUCK ARRIVES TONIGHT is encoded as

QAIQORSFDOOBUIPQKJBYAQ.

In this case, the inverse function is easily found by using the table from bottom to top. Thus, we can decode the phrase CKAJADPDGKJYEIDOT as

JOHNHASALONGBEARD;

that is, JOHN HAS A LONG BEARD.

1 Exercises

1. Let $A = \{a, b, c, d\}$ and $B = \{1, 2, 3\}$. Determine whether the relation R from A to B is a function. If it is a function, give its range.

(a) $R = \{(a, 1), (b, 2), (c, 1), (d, 2)\}$

(b) $R = \{(a, 1), (b, 2), (a, 2), (c, 1), (d, 2)\}$

2. Let $A = \{a, b, c, d\}$ and $B = \{1, 2, 3\}$. Determine whether the relation R from A to B is a function. If it is a function, give its range.

(a) $R = \{(a, 3), (b, 2), (c, 1)\}$

(b) $R = \{(a, 1), (b, 1), (c, 1), (d, 1)\}$

3. Determine whether the relation R from A to B is a function.

A = the set of all recipients of Medicare in the United States,
$B = \{x \mid x$ is a nine-digit number$\}$,
$a\ R\ b$ if b is a's Social Security number.

4. Determine whether the relation R from A to B is a function.

A = a set of people in the United States,
$B = \{x \mid x$ is a nine-digit number$\}$,
$a\ R\ b$ if b is a's passport number.

In Exercises 5 through 8, verify that the formula yields a function from A to B.

5. $A = B = Z$; $f(a) = a^2$

6. $A = B = R$; $f(a) = e^a$

7. $A = R$, $B = \{0, 1\}$; let Z be the set of integers and note that $Z \subseteq R$. Then for any real number a, let

$$f(a) = \begin{cases} 0 & \text{if } a \notin Z \\ 1 & \text{if } a \in Z. \end{cases}$$

8. $A = R$, $B = Z$; $f(a)$ = the greatest integer less than or equal to a.

9. Let $A = B = C = R$, and let $f : A \to B$, $g : B \to C$ be defined by $f(a) = a - 1$ and $g(b) = b^2$. Find

(a) $(f \circ g)(2)$ **(b)** $(g \circ f)(2)$

(c) $(g \circ f)(x)$ **(d)** $(f \circ g)(x)$

(e) $(f \circ f)(y)$ **(f)** $(g \circ g)(y)$

10. Let $A = B = C = R$, and let $f : A \to B$, $g : B \to C$ be defined by $f(a) = a + 1$ and $g(b) = b^2 + 2$. Find

(a) $(g \circ f)(-2)$ **(b)** $(f \circ g)(-2)$

(c) $(g \circ f)(x)$ **(d)** $(f \circ g)(x)$

(e) $(f \circ f)(y)$ **(f)** $(g \circ g)(y)$

11. In each part, sets A and B and a function from A to B are given. Determine whether the function is one to one or onto (or both or neither).

(a) $A = \{1, 2, 3, 4\} = B$;
$f = \{(1, 1), (2, 3), (3, 4), (4, 2)\}$

(b) $A = \{1, 2, 3\}$; $B = \{a, b, c, d\}$;
$f = \{(1, a), (2, a), (3, c)\}$

12. In each part, sets A and B and a function from A to B are given. Determine whether the function is one to one or onto (or both or neither).

(a) $A = \left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}\right\}$; $B = \{x, y, z, w\}$;
$f = \left\{\left(\frac{1}{2}, x\right), \left(\frac{1}{4}, y\right), \left(\frac{1}{3}, w\right)\right\}$

(b) $A = \{1.1, 7, 0.06\}$; $B = \{p, q\}$; $f = \{(1.1, p), (7, q), (0.06, p)\}$

13. In each part, sets A and B and a function from A to B are given. Determine whether the function is one to one or onto (or both or neither).

(a) $A = B = Z$; $f(a) = a - 1$

(b) $A = R$, $B = \{x \mid x \text{ is real and } x \geq 0\}$; $f(a) = |a|$

14. In each part, sets A and B and a function from A to B are given. Determine whether the function is one to one or onto (or both or neither).

(a) $A = R \times R$, $B = R$; $f((a, b)) = a$

(b) Let $S = \{1, 2, 3\}$, $T = \{a, b\}$. Let $A = B = S \times T$ and let f be defined by $f(n, a) = (n, b)$, $n = 1, 2, 3$, and $f(n, b) = (1, a)$, $n = 1, 2, 3$.

15. In each part, sets A and B and a function from A to B are given. Determine whether the function is one to one or onto (or both or neither).

(a) $A = B = R \times R$; $f((a, b)) = (a + b, a - b)$

(b) $A = R$, $B = \{x \mid x \text{ is real and } x \geq 0\}$; $f(a) = a^2$

16. Let $f(n)$ be the number of divisors of n, $n \in Z^+$. Determine whether f is one to one or onto (or both or neither).

17. Let $f(n)$ be the maximum of n and 50, $n \in Z^+$. Determine whether f is one to one or onto (or both or neither).

18. Explain why Theorem 1(a) is equivalent to "f^{-1} is not a function if and only if f is not one to one."

19. Let $f: A \to B$ and $g: B \to A$. Verify that $g = f^{-1}$.

(a) $A = B = R$; $f(a) = \frac{a+1}{2}$, $g(b) = 2b - 1$

(b) $A = \{x \mid x \text{ is real and } x \geq 0\}$; $B = \{y \mid y \text{ is real and } y \geq -1\}$; $f(a) = a^2 - 1$, $g(b) = \sqrt{b + 1}$

20. Let $f: A \to B$ and $g: B \to A$. Verify that $g = f^{-1}$.

(a) $A = B = P(S)$, where S is a set. If $X \in P(S)$, let $f(X) = \overline{X} = g(X)$.

(b) $A = B = \{1, 2, 3, 4\}$; $f = \{(1, 4), (2, 1), (3, 2), (4, 3)\}$; $g = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$

21. Let f be a function from A to B. Find f^{-1}.

(a) $A = \{x \mid x \text{ is real and } x \geq -1\}$; $B = \{x \mid x \text{ is real and } x \geq 0\}$; $f(a) = \sqrt{a + 1}$

(b) $A = B = R$; $f(a) = a^3 + 1$

22. Let f be a function from A to B. Find f^{-1}.

(a) $A = B = R$; $f(a) = \frac{2a-1}{3}$

(b) $A = B = \{1, 2, 3, 4, 5\}$; $f = \{(1, 3), (2, 2), (3, 4), (4, 5), (5, 1)\}$

23. Let $f(x, y) = (2x - y, x - 2y)$, $(x, y) \in R \times R$.

(a) Show that f is one to one.

(b) Find f^{-1}.

In Exercises 24 and 25, let f be a function from $A = \{1, 2, 3, 4\}$ to $B = \{a, b, c, d\}$. Determine whether f^{-1} is a function.

24. $f = \{(1, a), (2, a), (3, c), (4, d)\}$

25. $f = \{(1, a), (2, c), (3, b), (4, d)\}$

26. Let $A = B = C = R$ and consider the functions $f: A \to B$ and $g: B \to C$ defined by $f(a) = 2a + 1$, $g(b) = b/3$. Verify Theorem 3(b): $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

27. If a set A has n elements, how many functions are there from A to A?

28. If a set A has n elements, how many bijections are there from A to A?

29. If A has m elements and B has n elements, how many functions are there from A to B?

30. Complete the following proof.

If $f: A \to B$ and $g: B \to C$ are one-to-one functions, then $g \circ f$ is one to one.

Proof: Let $a_1, a_2 \in A$. Suppose $(g \circ f)(a_1) = (g \circ f)(a_2)$. Then $g(f(a_1)) = g(f(a_2))$ and $f(a_1) = f(a_2)$, because _____. Thus $a_1 = a_2$, because _____. Hence $g \circ f$ is one to one.

31. Complete the following proof.

If $f: A \to B$ and $g: B \to C$ are onto functions, then $g \circ f$ is onto.

Proof: Choose $x \in$ _____. Then there exists $y \in$ _____ such that $g(y) = x$. (Why?) Then there exists $z \in$ _____ such that $f(z) = y$ (why?) and $(g \circ f)(z) = x$. Hence, $g \circ f$ is onto.

32. Let $f: A \to B$ and $g: B \to C$ be functions. Show that if $g \circ f$ is one to one, then f is one to one.

33. Let $f: A \to B$ and $g: B \to C$ be functions. Show that if $g \circ f$ is onto, then g is onto.

34. Let A be a set, and let $f: A \to A$ be a bijection. For any integer $k \geq 1$, let $f^k = f \circ f \circ \cdots \circ f$ (k factors), and let $f^{-k} = f^{-1} \circ f^{-1} \circ \cdots \circ f^{-1}$ (k factors). Define f^0 to be 1_A. Then f^n is defined for all $n \in Z$. For any $a \in A$, let $O(a, f) = \{f^n(a) \mid n \in Z\}$. Prove that if $a_1, a_2 \in A$, and $O(a_1, f) \cap O(a_2, f) \neq \emptyset$, then $O(a_1, f) = O(a_2, f)$.

35. Let $f: A \to B$ be a function with finite domain and range. Suppose that $|\text{Dom}(f)| = n$ and $|\text{Ran}(f)| = m$. Prove that

(a) if f is one to one, then $m = n$.

(b) if f is not one to one, then $m < n$.

36. Let $|A| = |B| = n$ and let $f: A \to B$ be an everywhere defined function. Prove that the following three statements are equivalent.

(a) f is one to one. **(b)** f is onto.

(c) f is a one-to-one correspondence (that is, f is one to one and onto).

37. Give a one-to-one correspondence between Z^+, the set of positive integers, and $A = \{x \mid x \text{ is a positive even integer}\}$.

38. Give a one-to-one correspondence between Z^+, the set of positive integers, and $A = \{x \mid x \text{ is a positive odd integer}\}$.

39. Based on Exercises 37 and 38, does $|Z^+| = |A| = |B|$? Justify your conclusion.

40. **(a)** Let $A = R$ and $f : A \to R$ be defined by $f(a) = a^2$. Prove or disprove that $f(a_1 + a_2) = f(a_1) + f(a_2)$.

(b) Let $A = \{a, b\}$ and $f : A^* \to Z$ be defined by $f(s)$ is the length of the string s. Prove or disprove that $f(s_1 \cdot s_2) = f(s_1) + f(s_2)$.

41. Let $A = \{0, 1\}$ and define $a \oplus b = (a + b) \bmod 2$. Let $B = \{true, false\}$. Define $f : A \to B$ by $f(0) = true$ and $f(1) = false$.

(a) Prove or disprove that $f(a \oplus b) = f(a) \oplus f(b)$.

(b) Prove or disprove that $f(a \oplus b) = f(a) \odot f(b)$.

42. **(a)** Use the function in Example 18 to encode the message COME BACK AT ONCE.

(b) Decode the following message that was encoded using the function of Example 18.

QODLLITSDJJKQOIQROJ

43. Use the method of Example 18 and the keyword JOURNALISM to encode the message ALL PROJECTS ARE ON TRACK.

44. Substitution codes like the one in Example 18 are not very secure. Describe a commonsense method to break such a code.

2 Functions for Computer Science

In this section we review some functions commonly used in computer science applications and define some others.

Example 1 Let A be a subset of the universal set $U = \{u_1, u_2, u_3, \ldots, u_n\}$. The **characteristic function of *A*** is defined as a function from U to $\{0, 1\}$ by the following:

$$f_A(u_i) = \begin{cases} 1 & \text{if } u_i \in A \\ 0 & \text{if } u_i \notin A. \end{cases}$$

If $A = \{4, 7, 9\}$ and $U = \{1, 2, 3, \ldots, 10\}$, then $f_A(2) = 0$, $f_A(4) = 1$, $f_A(7) = 1$, and $f_A(12)$ is undefined. It is easy to check that f_A is everywhere defined and onto, but is not one to one.

Example 2 We define a family of mod-n functions. As one for each positive integer n. We call these functions f_n; that is, $f_n(m) = m \pmod n$. Each f_n is a function from the nonnegative integers to the set $\{0, 1, 2, 3, \ldots, n-1\}$. For a fixed n, any nonnegative integer z can be written as $z = kn + r$ with $0 \le r < n$. Then $f_n(z) = r$. We can also express this relation as $z \equiv r \pmod n$. Each member of the mod function family is everywhere defined and onto, but not one to one.

Example 3 Let A be the set of nonnegative integers, $B = Z^+$, and let $f : A \to B$ be defined by $f(n) = n!$.

Example 4 The general version of the pigeonhole principle requires the **floor function**, which is defined for rational numbers as $f(q)$ is the largest integer less than or equal to q. Here again is an example of a function that is not defined by a formula. The notation $\lfloor q \rfloor$ is often used for $f(q)$. Thus

$$f(1.5) = \lfloor 1.5 \rfloor = 1, \quad f(-3) = \lfloor -3 \rfloor = -3, \quad f(-2.7) = \lfloor -2.7 \rfloor = -3.$$

Example 5 A function similar to that in Example 4 is the **ceiling function**, which is defined for rational numbers as $c(q)$ is the smallest integer greater than or equal to q. The notation $\lceil q \rceil$ is often used for $c(q)$. Thus

$$c(1.5) = \lceil 1.5 \rceil = 2, \quad c(-3) = \lceil -3 \rceil = -3, \quad c(-2.7) = \lceil -2.7 \rceil = -2.$$

Many common algebraic functions are used in computer science, often with domains restricted to subsets of the integers.

Example 6

(a) Any polynomial with integer coefficients, p, can be used to define a function on Z as follows: If $p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ and $z \in Z$, then $f(z)$ is the value of p evaluated at z.

(b) Let $A = B = Z^+$ and let $f : A \to B$ be defined by $f(z) = 2^z$. We call f the **base 2 exponential function**. Other bases may be used to define similar functions.

(c) Let $A = B = R$ and let $f_n : A \to B$ be defined for each positive integer $n > 1$ as $f_n(x) = \log_n(x)$, the logarithm to the base n of x. In computer science applications, the bases 2 and 10 are particularly useful.

In general, the unary operations discussed in previous sections can be used to create functions similar to the function in Example 3. The sets A and B in the definition of a function need not be sets of numbers, as seen in the following examples.

Example 7

(a) Let A be a finite set and define $l : A^* \to Z$ as $l(w)$ is the length of the string w.

(b) Let B be a finite subset of the universal set U and define $pow(B)$ to be the power set of B. Then pow is a function from V, the power set of U, to the power set of V.

(c) Let $A = B =$ the set of all 2×2 matrices with real number entries and let $t(\mathbf{M}) = \mathbf{M}^T$, the transpose of $\mathbf{M}$. Then t is everywhere defined, onto, and one to one.

Example 8

(a) For elements of $Z^+ \times Z^+$, define $g(z_1, z_2)$ to be $GCD(z_1, z_2)$. Then g is a function from $Z^+ \times Z^+$ to Z^+.

(b) In a similar fashion we can define $m(z_1, z_2)$ to be $LCM(z_1, z_2)$.

Another type of function, a Boolean function, plays a key role in nearly all computer programs. Let $B = \{\text{true}, \text{false}\}$. Then a function from a set A to B is called a **Boolean function**.

Example 9

Let $P(x)$: x is even and $Q(y)$: y is odd. Then P and Q are functions from Z to B. We see that $P(4)$ is true and $Q(4)$ is false. The predicate $R(x, y)$: x is even or y is odd is a Boolean function of two variables from $Z \times Z$ to B. Here $R(3, 4)$ is false and $R(6, 4)$ is true.

Hashing Functions

Here we consider a more general problem of storing data. Suppose that we must store and later examine a large number of data records, customer accounts for example. In general we do not know how many records we may have to store at any given time. This suggests that linked-list storage is appropriate, because storage space is only used when we assign a record to it and we are not holding idle storage space. In order to examine a record we will have to be able to find it, so storing the data in a single linked list may not be practical because looking for an item may take a very long time (relatively speaking). One technique for handling such storage problems is to create a number of linked lists and to provide a method for deciding onto which list a new item should be linked. This method will also determine which list to search for a desired item. A key

point is to attempt to assign an item to one of the lists at random. This will have the effect of making the lists roughly the same size and thus keep the searching time about the same for any item.

Suppose we must maintain the customer records for a large company and will store the information as computer records. We begin by assigning each customer a unique seven-digit account number. A unique identifier for a record is called its **key**. For now we will not consider exactly how and what information will be stored for each customer account, but will describe only the storage of a location in the computer's memory where this information will be found. In order to determine to which list a particular record should be assigned, we create a **hashing function** from the set of keys to the set of list numbers. Hashing functions frequently use a mod-n function, as shown in the next example.

Example 10 Suppose that (approximately) 10,000 customer account records must be stored and processed. The company's computer is capable of searching a list of 100 items in an acceptable amount of time. We decide to create 101 linked lists for storage, because if the hashing function works well in "randomly" assigning records to lists, we would expect to see roughly 100 records per list. We define a hashing function from the set of seven-digit account numbers to the set $\{0, 1, 2, 3, \ldots, 100\}$ as follows:

$$h(n) = n \pmod{101}.$$

That is, h is the mod-101 function. Thus,

$$h(2473871) = 2473871 \pmod{101} = 78.$$

This means that the record with account number 2473871 will be assigned to list 78. Note that the range of h is the set $\{0, 1, 2, \ldots, 100\}$.

Because the function h in Example 10 is not one to one, different account numbers may be assigned to the same list by the hashing function. If the first position on list 78 is already occupied when the record with key 2473871 is to be stored, we say a collision has occurred. There are many methods for resolving collisions. One very simple method that will be sufficient for our work is to insert the new record at the end of the existing list. Using this method, when we wish to find a record, its key will be hashed and the list h(key) will be searched sequentially.

Many other hashing functions are suitable for this situation. For example, we may break the seven-digit account number into a three-digit number and a four-digit number, add these, and then apply the mod-101 function. Chopping the key into pieces to create the function is the origin of the name hashing function. Many factors are considered in addition to the number of records to be stored; the speed with which an average-length list can be searched and the time needed to compute the list number for an account are two possible factors to be taken into account. For reasons that will not be discussed here, the modulus used in the mod function should be a prime. Thus, in Example 10, we chose 101 as the modulus, rather than 100 or 102. Determining a "good" hashing function for a particular application is a challenging task.

Hashing functions are also employed in other applications such as cryptology where they are used to produce digital fingerprints and other electronic means to verify the authenticity of messages.

Fuzzy Sets

Sets are precisely defined objects. If A is a subset of a universal set U, then given any element x of U, it must be possible, in principle at least, to say that x is a member of A or x is not a member of A. However, the descriptions of many collections used in daily life and in practical technology are not so precise, but "fuzzy" in nature. An example is the "collection of tall humans" in the universal set U of human beings. The adjective "tall" is imprecise; its meaning is likely to depend on the people using it and the situations in which it is used.

Given a fixed range of height, the difficulty centers around the boundary of the collection. We may regard some people as definitely tall, and some as definitely not tall, but in between there will be people that we feel are "somewhat tall" or "a little tall." Membership in the collection of tall people is not a yes-or-no question, but has degrees. One can be a member, somewhat of a member, or definitely not a member. This is the idea of a fuzzy set. Other examples are

- the collection of rich people,
- the collection of automobile speeds that are too fast,
- the collection of dangerously close distances between moving objects, and
- the collection of useful laws.

What is needed is a precise way to deal with these useful "fuzzy" collections that are not sets. A set is completely described by its characteristic function. By definition, the set A determines the characteristic function f_A. On the other hand, $A = \{x \mid f_A(x) = 1\}$, so the function, in turn, determines the set. Which representation of a set we choose to use depends on the task at hand. Any mathematical operation that we can perform on sets has a corresponding operation on the characteristic functions. One way to define a fuzzy set precisely is to use a function representation that is a generalization of a characteristic function.

A **fuzzy set** in a universal set U is a function f defined on U and having values in the interval $[0, 1]$. This function is a generalization of the characteristic function of a set. If $f(x) = 0$, then we say that x is not in the fuzzy set. If $f(x) = 1$, then we say that x is in the fuzzy set. But if $0 < f(x) < 1$, then $f(x)$ can be thought of as the degree to which x is in the fuzzy set. If the range of f is only the numbers 0 and 1, then f is the characteristic function of an ordinary set. Thus, ordinary sets are special cases of fuzzy sets. Here, a function such as f is the fuzzy set, but some other authors call a function such as f the **membership function for the fuzzy set**.

Example 11

We define a universal set U to consist of all yearly incomes between \$0 and \$1,000,000, in thousands. Thus, U is the interval $[0, 1000]$. Suppose we wish to define the (fuzzy) set of rich people, based on yearly income. We may consider that a person whose yearly income is greater than \$250,000 is definitely rich, a person with an income less than \$30,000 is definitely not, and a person with an income in between has some degree of "richness." With these initial conditions, a possible fuzzy set of rich people is the function f in Figure 4. (The horizontal line continues to 1000.)

An equation of the slanted line segment is $f(x) = (x - 30)/220, 30 \le x \le 250$. (Why?) Based on this, a person with a yearly salary of \$100,000 (or $x = 100$)

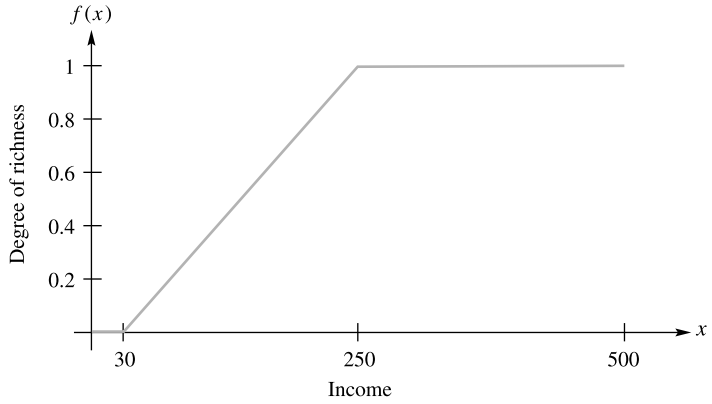


Figure 4

has degree of membership $(100 - 30)/220 = 7/22$ or about 0.32 in the fuzzy set f, a person making less than \$30,000 a year is not in this fuzzy set at all, and a person making \$300,000 a year is definitely rich. Many other fuzzy sets with the same initial conditions could define the collection of rich people. The choice of a defining function is highly subjective and may depend on the situation under discussion.

Figure 5 shows another appropriate fuzzy set, g, where $0 \le x \le 1000$, $g(x) = 0$ for $0 \le x \le 30$, and $g(x) = 1$ for $x \ge 250$.

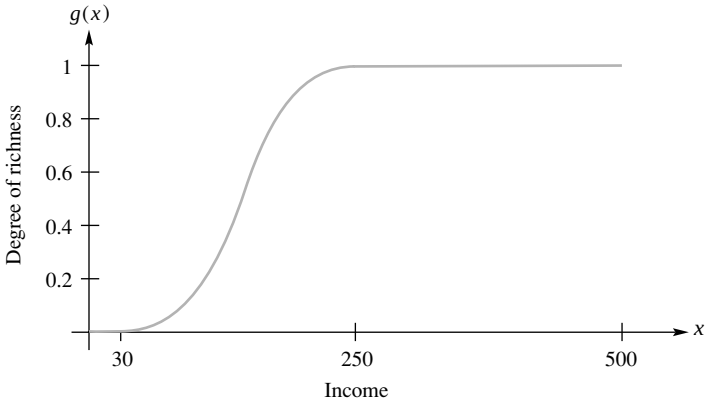


Figure 5

Example 12 Determine in which fuzzy set f, or g, shown in Figures 4 and 5, the given income has a higher degree of richness.

(a) \$30,000 **(b)** \$75,000 **(c)** \$200,000 **(d)** \$260,000

Solution

Although no explicit definition of g is provided, a comparison of the graphs shows that $f(30) = g(30) = 0$ and $f(260) = g(260) = 1$, so the answer for parts (a) and (d) is neither. It is geometrically clear that $g(75) < f(75)$ and $g(200) > f(200)$. Thus, (b) and (c) have greater degrees of richness in f and g, respectively.

36. Sketch the graph of $\bar{g}$.

37. For the graph shown in Figure 8(b), give the coordinates of the lowest point between $(3, 1)$ and $(5, 1)$.

38. **(a)** Define the complement of fuzzy set g with respect to the fuzzy set f by generalizing the definition of $A - B$ for sets.

(b) Define the symmetric difference of fuzzy sets f and g by generalizing the definition of symmetric difference for sets.

In Exercises 39 and 40, use the fuzzy sets f and g shown in Figure 9.

39. Compute the degree of membership in the complement of g with respect to f, $f - g$, of

(a) 2 **(b)** 9.

40. Compute the degree of membership in the symmetric difference of f and g, $f \square g$, of

(a) 2 **(b)** 4.5.

41. Prove Theorem 1.

Exercises 42 through 46 use ideas from this section. Pairs of parentheses are often used in mathematical expressions to indicate the order in which operations are to be done. A compiler (or interpreter) for a programming language must check that pairs of parentheses are properly placed. This may involve a number of things, but one simple check is that the number of left and right parentheses are equal and that in reading from left to right the number of left parentheses is always greater than or equal to the number of right parentheses read. An expression that passes this check is called well formed. The task here is to count the number of well-formed strings of n left and n right parentheses. This number is C_n, the nth Catalan number.

42. How many strings of n left and n right parentheses can be made (not just well-formed ones)?

43. List all well-formed strings of n left and n right parentheses for $n = 1, 2, 3$. What are the values of C_1, C_2, and C_3?

44. We will count the strings that are not well formed by making a one-to-one correspondence between them and a set of easier to count strings. Suppose $p_1 p_2 p_3 \cdots p_{2n}$ is not well formed; then there is a first p_i that is a right parenthesis and there are fewer left parentheses than right parentheses in $p_1 p_2 \cdots p_i$. How many fewer are there? So to the right of p_i the number of left parentheses is _____ than the number of right parentheses. Make a new string $q_1 q_2 \cdots q_{2n}$ as follows:

$$q_j = p_j, \quad j = 1, 2, \ldots, i$$

and

$$q_j = \square \begin{cases} (& \text{if } p_j =) \\ ) & \text{if } p_j = (\end{cases} \quad \text{for } j = i+1, i+2, \ldots, 2n.$$

This new string $q_1 q_2 \cdots q_{2n}$ has _____ left and _____ right parentheses. Explain your reasoning.

45. To complete the one-to-one correspondence between the p and the q strings of Exercise 44, we must show that any string with $n - 1$ left and $n + 1$ right parentheses can be paired with exactly one string with n left and n right parentheses that is not well formed. Let $r_1 r_2 r_3 \cdots r_{2n}$ consist of $n - 1$ left and $n + 1$ right parentheses. There must be a first position j where the number of right parentheses is greater than the number of left parentheses. Why? So in $r_1 r_2 r_3 \cdots r_j$ there is one more right than left parenthesis. Hence in $r_{j+1} \cdots r_{2n}$, the number of left parentheses is _____ than the number of right parentheses. Make a new string $s_1 s_2 \cdots s_{2n}$ as follows:

$$s_k = r_k, \quad k = 1, 2, \ldots, j$$

and

$$s_k = \square \begin{cases} (& \text{if } r_k =) \\ ) & \text{if } r_k = (\end{cases} \quad \text{for } k = j+1, j+2, \ldots, 2n.$$

This new string $s_1 s_2 \cdots s_{2n}$ has _____ left and _____ right parentheses. Explain how you know $s_1 s_2 \cdots s_{2n}$ is not well formed.

46. Using the results of Exercises 44 and 45, the number of strings with n left and n right parentheses that are not well formed is equal to the number of strings with $n - 1$ left and $n + 1$ right parentheses.

47. Use the results of Exercises 42 and 46 to give a formula for C_n. Confirm this result by comparing its values with those found in Exercise 43.

48. Express C_n using the notation for combinations and without this notation.

*Another application of mod functions occurs in assigning an ISBN (International Standard Book Number) to each title published. The 10-digit ISBN encodes information about the language of publication, the publisher, and the book itself. This is an example of coding for error checking rather than for security purposes. For example, if the ISBN for the fourth edition of a book is 0-13-083143-3; the 0 indicates the book was published in an English-speaking country and the 13 identifies the publisher. The last digit is a **check digit** chosen to help prevent transcription errors. If $d_1 d_2 d_3 \cdots d_9 c$ is an ISBN, then c is chosen so that*

$$(d_1 + 2d_2 + 3d_3 + \cdots + 9d_9 + 10c) \equiv 0 \pmod{11}.$$

If c is 10, then the Roman numeral X is used.

49. **(a)** Verify that 3 is the correct check digit for the book's ISBN.

(b) Compute the check digit c for the following ISBNs.

(i) 0-471-80075-c **(ii)** 0-80504826-c

(iii) 88-8117-275-c **(iv)** 5-05-001801-c

50. **(a)** Make one change in the first nine digits of the ISBN 0-183-47381-7 so that the check digit will indicate an error.

(b) Make two changes in the first nine digits of the ISBN 0-183-47381-7 so that the check digit will not indicate an error.

3 Growth of Functions

The idea of one function growing more rapidly than another arises naturally when working with functions. In this section we formalize this notion.

Example 1 Let R be a relation on a set A with $|A| = n$ and $|R| = \frac{1}{2}n^2$. If R is stored as a matrix, then $t(n) = \frac{1}{2}n^3 + \frac{1}{2}n^2$ is a function that describes (roughly) the average number of steps needed to determine if R is transitive using the algorithm TRANS. Storing R with a linked list and using NEWTRANS, the average number of steps needed is (roughly) given by $s(n) = \frac{1}{8}n^4$. Table 1 shows that s grows faster than t.

TABLE 1

n	$t(n)$	$s(n)$
10	550	1250
50	63,750	781,250
100	505,000	12,500,000

Let f and g be functions whose domains are subsets of Z^+, the positive integers. We say that f is $O(g)$, read f **is big-Oh of** g, if there exist constants c and k such that $|f(n)| \le c \cdot |g(n)|$ for all $n \ge k$. If f is $O(g)$, then f grows no faster than g does.

Example 2 The function $f(n) = \frac{1}{2}n^3 + \frac{1}{2}n^2$ is $O(g)$ for $g(n) = n^3$. To see this, consider

$$\frac{1}{2}n^3 + \frac{1}{2}n^2 \le \frac{1}{2}n^3 + \frac{1}{2}n^3, \qquad \text{if } n \ge 1.$$

Thus,

$$\frac{1}{2}n^3 + \frac{1}{2}n^2 \le 1 \cdot n^3, \qquad \text{if } n \ge 1.$$

Choosing 1 for c and 1 for k, we have shown that $|f(n)| \le c \cdot |g(n)|$ for all $n \ge 1$ and f is $O(g)$.

The reader can see from Example 2 that other choices of c, k, and even g are possible. If $|f(n)| \le c|g(n)|$ for all $n \ge k$, then we have $|f(n)| \le C \cdot |g(n)|$ for all $n \ge k$ for any $C \ge c$, and $|f(n)| \le c \cdot |g(n)|$ for all $n \ge K$ for any

$K \geq k$. For the function t in Example 2, t is $O(h)$ for $h(n) = dn^3$, if $d \geq 1$, since $|t(n)| \leq 1 \cdot |g(n)| \leq |h(n)|$. Observe also that t is $O(r(n))$ for $r(n) = n^4$, because $\frac{1}{2}n^3 + \frac{1}{2}n^2 \leq n^3 \leq n^4$ for all $n \geq 1$. When analyzing algorithms, we want to know the "slowest growing" simple function g for which f is $O(g)$.

It is common to replace g in $O(g)$ with the formula that defines g. Thus we write that t is $O(n^3)$. This is called big-O notation.

We say that f and g have the **same order** if f is $O(g)$ and g is $O(f)$.

Example 3 Let $f(n) = 3n^4 - 5n^2$ and $g(n) = n^4$ be defined for positive integers n. Then f and g have the same order. First,

$$\begin{aligned} 3n^4 - 5n^2 &\leq 3n^4 + 5n^2 \\ &\leq 3n^4 + 5n^4, \qquad \text{if } n \geq 1 \\ &= 8n^4. \end{aligned}$$

Let $c = 8$ and $k = 1$, then $|f(n)| \leq c \cdot |g(n)|$ for all $n \geq k$. Thus f is $O(g)$. Conversely, $n^4 = 3n^4 - 2n^4 \leq 3n^4 - 5n^2$ if $n \geq 2$. This is because if $n \geq 2$, then $n^2 > \frac{5}{2}$, $2n^2 > 5$, and $2n^4 > 5n^2$. Using 1 for c and 2 for k, we conclude that g is $O(f)$.

If f is $O(g)$ but g is not $O(f)$, we say that f is **lower order** than g or that f grows more slowly than g.

Example 4 The function $f(n) = n^5$ is lower order than $g(n) = n^7$. Clearly, if $n \geq 1$, then $n^5 \leq n^7$. Suppose that there exist c and k such that $n^7 \leq cn^5$ for all $n \geq k$. Choose N so that $N > k$ and $N^2 > c$. Then $N^7 \leq cN^5 < N^2 \cdot N^5$, but this is a contradiction. Hence f is $O(g)$, but g is not $O(f)$, and f is lower order than g. This agrees with our experience that n^5 grows more slowly than n^7.

We define a relation Θ, **big-theta**, on functions whose domains are subsets of Z^+ as $f \Theta g$ if and only if f and g have the same order.

THEOREM 1 The relation Θ, big-theta, is an equivalence relation.

Proof

Clearly, Θ is reflexive since every function has the same order as itself. Because the definition of same order treats f and g in the same way, this definition is symmetric and the relation Θ is symmetric.

To see that Θ is transitive, suppose f and g have the same order. Then there exist c_1 and k_1 with $|f(n)| \leq c_1 \cdot |g(n)|$ for all $n \geq k_1$, and there exist c_2 and k_2 with $|g(n)| \leq c_2 \cdot |f(n)|$ for all $n \geq k_2$. Suppose that g and h have the same order; then there exist c_3, k_3 with $|g(n)| \leq c_3 \cdot |h(n)|$ for all $n \geq k_3$, and there exist c_4, k_4 with $|h(n)| \leq c_4 \cdot |g(n)|$ for all $n \geq k_4$.

Then $|f(n)| \leq c_1 \cdot |g(n)| \leq c_1(c_3 \cdot |h(n)|)$ if $n \geq k_1$ and $n \geq k_3$. Thus $|f(n)| \leq c_1c_3 \cdot |h(n)|$ for all $n \geq$ maximum of k_1 and k_3.

Similarly, $|h(n)| \leq c_2c_4 \cdot |f(n)|$ for all $n \geq$ maximum of k_2 and k_4. Thus f and h have the same order and Θ is transitive.

The equivalence classes of Θ consist of functions that have the same order. We use any simple function in the equivalence class to represent the order of all functions in that class. One Θ-class is said to be **lower** than another Θ-class if a representative function from the first is of lower order than one from the second class. This means functions in the first class grow more slowly than those in the

second. It is the Θ-class of a function that gives the information we need for algorithm analysis.

Example 5 All functions that have the same order as $g(n) = n^3$ are said to have order $\Theta(n^3)$. The most common orders in computer science applications are $\Theta(1)$, $\Theta(n)$, $\Theta(n^2)$, $\Theta(n^3)$, $\Theta(lg(n))$, $\Theta(nlg(n))$, and $\Theta(2^n)$. Here $\Theta(1)$ represents the class of constant functions and *lg* is the base 2 log function. The continuous versions of some of these functions are shown in Figure 10.

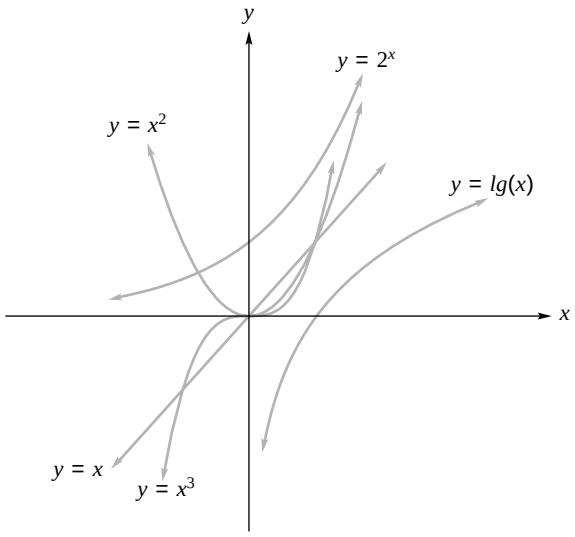


Figure 10

Example 6 Every logarithmic function $f(n) = \log_b(n)$ has the same order as $g(n) = lg(n)$. There is a logarithmic change-of-base identity

$$\log_b(x) = \frac{\log_a(x)}{\log_a(b)}$$

in which $\log_a(b)$ is a constant. Thus

$$\left|\log_b(n)\right| \le \frac{1}{lg(b)}|lg(n)|$$

and, conversely,

$$|lg(n)| \le lg(b) \cdot |\log_b(n)|.$$

Hence g is $O(f)$ and f is $O(g)$.

It is sometimes necessary to combine functions that give the number of steps required for pieces of an algorithm as is done in the analysis of TRANS where functions are added, and in the analysis of NEWTRANS, where functions are multiplied. There are some general rules regarding the ordering of the Θ-equivalence classes that can be used to determine the class of many functions and the class of the sum or product of previously classified functions.

Rules for Determining the Θ-Class of a Function

1. $\Theta(1)$ functions are constant and have zero growth, the slowest growth possible.
2. $\Theta(lg(n))$ is lower than $\Theta(n^k)$ if $k > 0$. This means that any logarithmic function grows more slowly than any power function with positive exponent.
3. $\Theta(n^a)$ is lower than $\Theta(n^b)$ if and only if $0 < a < b$.
4. $\Theta(a^n)$ is lower than $\Theta(b^n)$ if and only if $0 < a < b$.
5. $\Theta(n^k)$ is lower than $\Theta(a^n)$ for any power n^k and any $a > 1$. This means that any exponential function with base greater than 1 grows more rapidly than any power function.
6. If r is not zero, then $\Theta(rf) = \Theta(f)$ for any function f.
7. If h is a nonzero function and $\Theta(f)$ is lower than (or the same as) $\Theta(g)$, then $\Theta(fh)$ is lower than (or the same as) $\Theta(gh)$.
8. If $\Theta(f)$ is lower than $\Theta(g)$, then $\Theta(f + g) = \Theta(g)$.

Example 7 Determine the Θ-class of each of the following.

(a) $f(n) = 4n^4 - 6n^7 + 25n^3$
(b) $g(n) = lg(n) - 3n$
(c) $h(n) = 1.1^n + n^{15}$

Solution

(a) By Rules 3, 6, and 8, the degree of the polynomial determines the Θ-class of a polynomial function. $\Theta(f) = \Theta(n^7)$.
(b) Using Rules 2, 6, and 8, we have that $\Theta(g) = \Theta(n)$.
(c) By Rules 5 and 8, $\Theta(h) = \Theta(1.1^n)$.

Example 8 Using the rules for ordering Θ-classes, arrange the following in order from lowest to highest.

$\Theta(nlg(n))$ $\Theta(1000n^2 - n)$ $\Theta(n^{0.2})$ $\Theta(1{,}000{,}000)$ $\Theta(1.3^n)$ $\Theta(n + 10^7)$

Solution

$\Theta(1{,}000{,}000)$ is the class of constant functions, so it is the first on the list. By Rules 5 and 8, $\Theta(n + 10^7)$ is lower than $\Theta(1000n^2 - n)$, but higher than $\Theta(n^{0.2})$. To determine the position of $\Theta(nlg(n))$ on the list, we apply Rules 2 and 7. These give that $\Theta(nlg(n))$ is lower than $\Theta(n^2)$ and higher than $\Theta(n)$. Rule 5 says that $\Theta(1.3^n)$ is the highest class on this list. In order, the classes are

$$\Theta(1{,}000{,}000) \quad \Theta(n^{0.2}) \quad \Theta(n + 10^7)$$
$$\Theta(nlg(n)) \quad \Theta(1000n^2 - n) \quad \Theta(1.3^n).$$

The Θ-class of a function that describes the number of steps performed by an algorithm is frequently referred to as the **running time** of the algorithm. For example, the algorithm TRANS has an average running time of n^3. In general, algorithms with exponential running times are impractical for all but very small values of n. In many cases the running time of an algorithm is estimated by examining best, worst, or average cases.

3 Exercises

In Exercises 1 and 2, let f *be a function that describes the number of steps required to carry out a certain algorithm. The number of items to be processed is represented by* n*. For each function, describe what happens to the number of steps if the number of items is doubled.*

1. **(a)** $f(n) = 1001$ **(b)** $f(n) = 3n$
(c) $f(n) = 5n^2$ **(d)** $f(n) = 2.5n^3$

2. **(a)** $f(n) = 1.4lg(n)$ **(b)** $f(n) = 2^n$
(c) $f(n) = nlg(n)$ **(d)** $f(n) = 100n^4$

3. Show that $g(n) = n!$ is $O(n^n)$.

4. Show that $h(n) = 1 + 2 + 3 + \cdots + n$ is $O(n^2)$.

5. Show that $f(n) = 8n + lg(n)$ is $O(n)$.

6. Show that $g(n) = n^2(7n - 2)$ is $O(n^3)$.

7. Show that $f(n) = nlg(n)$ is $O(g)$ for $g(n) = n^2$, but that g is not $O(f)$.

8. Show that $f(n) = n^{100}$ is $O(g)$ for $g(n) = 2^n$, but that g is not $O(f)$.

9. Show that f and g have the same order for $f(n) = 5n^2 + 4n + 3$ and $g(n) = n^2 + 100n$.

10. Show that f and g have the same order for $f(n) = lg(n^3)$ and $g(n) = \log_5(6n)$.

11. Determine which of the following are in the same Θ-class. A function may be in a class by itself.

$$f_1(n) = 5nlg(n), \quad f_2(n) = 6n^2 - 3n + 7,$$
$$f_3(n) = 1.5^n, \quad f_4(n) = lg(n^4),$$
$$f_5(n) = 13{,}463, \quad f_6(n) = -15n,$$
$$f_7(n) = lg(lg(n)), \quad f_8(n) = 9n^{0.7},$$
$$f_9(n) = n!, \quad f_{10}(n) = n + lg(n),$$
$$f_{11}(n) = \sqrt{n} + 12n, \quad f_{12}(n) = lg(n!)$$

12. Order the Θ-classes in Exercise 11 from lowest to highest.

13. Consider the functions $f_1, f_2, f_4, f_5, f_6, f_{10}, f_{11}$ in Exercise 11. Match each of the functions with its Θ-class from the following list: $\Theta(1)$, $\Theta(n)$, $\Theta(nlg(n))$, $\Theta(lg(n))$, $\Theta(n^2)$, $\Theta(\sqrt{n})$, $\Theta(2^n)$.

In Exercises 14 through 21, analyze the operation performed by the given piece of pseudocode and write a function that describes the number of steps required. Give the Θ*-class of the function.*

14.
```
1. A ← 1
2. B ← 1
3. UNTIL (B > 100)
        a. B ← 2A − 2
        b. A ← A + 3
```

15.
```
1. X ← 1
2. Y ← 100
3. WHILE (X < Y)
        a. X ← X + 2
        b. Y ← ½Y
```

16.
```
1. I ← 1
2. X ← 0
3. WHILE (I ≤ N)
        a. X ← X + 1
        b. I ← I + 1
```

17.
```
1. SUM ← 0
2. FOR I = 0 THRU 2(N − 1) BY 2
        a. SUM ← SUM + I
```

18. Assume that N is a power of 2.
```
1. X ← 1
2. K ← N
3. WHILE (K ≥ 1)
        a. X ← 3X
        b. K ← ⌊K/2⌋
```

19.
```
1. I ← 1
2. SUM ← 0
3. WHILE (I ≤ N)
        a. FOR K = 1 THRU I
             1. SUM ← SUM + K
        b. I ← I + 1
```

20.
```
1. K ← 0
2. FOR I = 0 THRU N
        a. WHILE K ≤ I
             1. K ← K + 1
```

21.
```
SUBROUTINE MATMUL(A,B,N,M,P,Q;C)
1. IF (M = P) THEN
        a. FOR I = 1 THRU N
             1. FOR J = 1 THRU Q
                  a. C[I,J] ← 0
                  b. FOR K = 1 THRU M
                       1. C[I,J] ←
                        C[I,J] +
                        (A[I,K] × B[K,J])
2. ELSE
        a. CALL PRINT ('INCOMPATIBLE')
3. RETURN
END OF SUBROUTINE MATMUL
```

22. **(a)** Write a recurrence relation to count the number of ways a 3×3 square can be placed on an $n \times n$ square with the edges of the squares parallel.

(b) What is the running time of an algorithm that uses the recurrence relation in (a) to count the number of placements?

23. Prove Rule 3.

24. Prove Rule 4.

25. Prove Rule 6.

26. Prove Rule 7.

27. Prove that if $\Theta(f) = \Theta(g) = \Theta(h)$, then $f + g$ is $O(h)$.

28. Prove that if $\Theta(f) = \Theta(g)$ and $c \neq 0$, then $\Theta(cf) = \Theta(g)$.

4 Permutation Functions

In this section we discuss bijections from a set A to itself. Of special importance is the case when A is finite. Bijections on a finite set occur in a wide variety of applications in mathematics, computer science, and physics.

A bijection from a set A to itself is called a **permutation** of A.

Example 1 Let $A = R$ and let $f : A \to A$ be defined by $f(a) = 2a + 1$. Since f is one to one and onto (verify), it follows that f is a permutation of A.

If $A = \{a_1, a_2, \ldots, a_n\}$ is a finite set and p is a bijection on A, we list the elements of A and the corresponding function values $p(a_1), p(a_2), \ldots, p(a_n)$ in the following form:

$$\begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ p(a_1) & p(a_2) & \cdots & p(a_n) \end{pmatrix}. \tag{1}$$

Observe that (1) completely describes p since it gives the value of p for every element of A. We often write

$$p = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ p(a_1) & p(a_2) & \cdots & p(a_n) \end{pmatrix}.$$

Thus, if p is a permutation of a finite set $A = \{a_1, a_2, \ldots, a_n\}$, then the sequence $p(a_1), p(a_2), \ldots, p(a_n)$ is just a rearrangement of the elements of A and so corresponds exactly to a permutation of A.

Example 2 Let $A = \{1, 2, 3\}$. Then all the permutations of A are

$$1_A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \quad p_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \quad p_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix},$$

$$p_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \quad p_4 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \quad p_5 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

Example 3 Using the permutations of Example 2, compute **(a)** p_4^{-1}; **(b)** $p_3 \circ p_2$.

Solution

(a) Viewing p_4 as a function, we have

$$p_4 = \{(1, 3), (2, 1), (3, 2)\}.$$

Then

$$p_4^{-1} = \{(3, 1), (1, 2), (2, 3)\}$$

or, when written in increasing order of the first component of each ordered pair, we have

$$p_4^{-1} = \{(1, 2), (2, 3), (3, 1)\}.$$

Thus

$$p_4^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = p_3.$$

(b) The function p_2 takes 1 to 2 and p_3 takes 2 to 3, so $p_3 \circ p_2$ takes 1 to 3. Also, p_2 takes 2 to 1 and p_3 takes 1 to 2, so $p_3 \circ p_2$ takes 2 to 2. Finally, p_2 takes 3 to 3 and p_3 takes 3 to 1, so $p_3 \circ p_2$ takes 3 to 1. Thus

$$p_3 \circ p_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

We may view the process of forming $p_3 \circ p_2$ as shown in Figure 11. Observe that $p_3 \circ p_2 = p_5$.

$$p_3 \circ p_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ ③ & 2 & 1 \end{pmatrix}$$

Figure 11

The composition of two permutations is another permutation, usually referred to as the **product** of these permutations. In the remainder of this chapter, we will follow this convention.

THEOREM 1 If $A = \{a_1, a_2, \ldots, a_n\}$ is a set containing n elements, then there are

$$n! = n \cdot (n-1) \cdots 2 \cdot 1 \quad \text{permutations of } A. \tag{2}$$

Proof
This result follows by letting $r = n$.

Let $b_1, b_2, \ldots, b_r$ be r distinct elements of the set $A = \{a_1, a_2, \ldots, a_n\}$. The permutation $p: A \to A$ defined by

$$\begin{aligned} p(b_1) &= b_2 \\ p(b_2) &= b_3 \\ &\vdots \\ p(b_{r-1}) &= b_r \\ p(b_r) &= b_1 \\ p(x) &= x, \quad \text{if } x \in A, x \notin \{b_1, b_2, \ldots, b_r\}, \end{aligned}$$

is called a **cyclic permutation** of length r, or simply a **cycle** of length r, and will be denoted by $(b_1, b_2, \ldots, b_r)$. Do not confuse this terminology with that used for cycles in a digraph. The two concepts are different and we use slightly different notations. If the elements $b_1, b_2, \ldots, b_r$ are arranged uniformly on a circle, as shown in Figure 12, then a cycle p of length r moves these elements in a clockwise direction so that b_1 is sent to b_2, b_2 to b_3, $\ldots$, b_{r-1} to b_r, and b_r to b_1. All the other elements of A are left fixed by p.

Example 4 Let $A = \{1, 2, 3, 4, 5\}$. The cycle $(1, 3, 5)$ denotes the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 4 & 1 \end{pmatrix}.$$

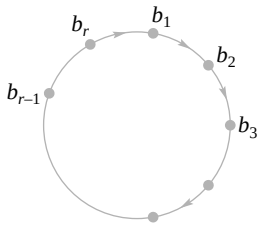


Figure 12

Observe that if $p = (b_1, b_2, \ldots, b_r)$ is a cycle of length r, then we can also write p by starting with any b_i, $1 \le i \le r$, and moving in a clockwise direction, as shown in Figure 12. Thus, as cycles,

$$(3, 5, 8, 2) = (5, 8, 2, 3) = (8, 2, 3, 5) = (2, 3, 5, 8).$$

Note also that the notation for a cycle does not include the number of elements in the set A. Thus the cycle $(3, 2, 1, 4)$ could be a permutation of the set $\{1, 2, 3, 4\}$ or of $\{1, 2, 3, 4, 5, 6, 7, 8\}$. We need to be told explicitly the set on which a cycle is defined. It follows from the definition that a cycle on a set A is of length 1 if and only if it is the identity permutation, 1_A.

Since cycles are permutations, we can form their product. However, as we show in the following example, the product of two cycles need not be a cycle.

Example 5 Let $A = \{1, 2, 3, 4, 5, 6\}$. Compute $(4, 1, 3, 5) \circ (5, 6, 3)$ and $(5, 6, 3) \circ (4, 1, 3, 5)$.

Solution

We have

$$(4, 1, 3, 5) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 5 & 1 & 4 & 6 \end{pmatrix}$$

and

$$(5, 6, 3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 5 & 4 & 6 & 3 \end{pmatrix}.$$

Then

$$(4, 1, 3, 5) \circ (5, 6, 3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 5 & 1 & 4 & 6 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 5 & 4 & 6 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 4 & 1 & 6 & 5 \end{pmatrix}$$

and

$$(5, 6, 3) \circ (4, 1, 3, 5) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 5 & 4 & 6 & 3 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 5 & 1 & 4 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 2 & 6 & 1 & 4 & 3 \end{pmatrix}.$$

Observe that

$$(4, 1, 3, 5) \circ (5, 6, 3) \neq (5, 6, 3) \circ (4, 1, 3, 5)$$

and that neither product is a cycle.

Two cycles of a set A are said to be **disjoint** if no element of A appears in both cycles.

Example 6 Let $A = \{1, 2, 3, 4, 5, 6\}$. Then the cycles $(1, 2, 5)$ and $(3, 4, 6)$ are disjoint, whereas the cycles $(1, 2, 5)$ and $(2, 4, 6)$ are not.

It is not difficult to show that if $p_1 = (a_1, a_2, \ldots, a_r)$ and $p_2 = (b_1, b_2, \ldots, b_s)$ are disjoint cycles of A, then $p_1 \circ p_2 = p_2 \circ p_1$. This can be seen by observing that p_1 affects only the a's, while p_2 affects only the b's.

We shall now present a fundamental theorem and, instead of giving its proof, we shall give an example that imitates the proof.

THEOREM 2 A permutation of a finite set that is not the identity or a cycle can be written as a product of disjoint cycles of length ≥ 2.

Example 7 Write the permutation

$$p = \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 6 & 5 & 2 & 1 & 8 & 7 \end{matrix}$$

of the set $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ as a product of disjoint cycles.

Solution

We start with 1 and find that $p(1) = 3$, $p(3) = 6$, and $p(6) = 1$, so we have the cycle $(1, 3, 6)$. Next we choose the first element of A that has not appeared in a previous cycle. We choose 2, and we have $p(2) = 4$, $p(4) = 5$, and $p(5) = 2$, so we obtain the cycle $(2, 4, 5)$. We now choose 7, the first element of A that has not appeared in a previous cycle. Since $p(7) = 8$ and $p(8) = 7$, we obtain the cycle $(7, 8)$. We can then write p as a product of disjoint cycles as

$$p = (7, 8) \circ (2, 4, 5) \circ (1, 3, 6).$$

It is not difficult to show that in Theorem 2, when a permutation is written as a product of disjoint cycles, the product is unique except for the order of the cycles.

We saw in Section 1 how a permutation of the alphabet produces a substitution code. Permutations are also used to produce **transposition codes**. Unlike a substitution code in which each letter is replaced by a substitute, the letters in transposition coded messages are not changed, but are rearranged. Thus if a message TEST THE WATERS is subjected to the permutation

$$(1, 2, 3) \circ (4, 7) \circ (5, 10, 11) \circ (6, 8, 12, 13, 9),$$

where the numbers refer to the positions of the letters in the message, the message becomes STEEEATHSTTWR. If the permutation is known to both parties, then the receiver of the message has only to apply the inverse permutation to decode.

Example 8 One commonly used transposition code is the **keyword columnar transposition**. For this it is only necessary to remember a keyword, say JONES. The message to be encoded is written under the keyword in successive rows, padding at the end if necessary. For example, the message THE FIFTH GOBLET CONTAINS THE GOLD would be arranged as shown:

J	**O**	**N**	**E**	**S**
T	H	E	F	I
F	T	H	G	O
B	L	E	T	C
O	N	T	A	I
N	S	T	H	E
G	O	L	D	X

Note that the message has been padded with an X to fill out the row. Then the coded message is constructed by writing the columns in succession, beginning with column 4 (since E is the first keyword letter to appear in the alphabet) and following with the letters in columns 1, 3, 2, 5. The encoded message is thus

FGTAHDTFBOGCEHETTLHTLNSOIOCIEX.

The recipient of the message divides the number of letters by 5 to find that there are 6 rows. She writes the coded message, six letters at a time, in columns 4, 1, 3, 2, 5, then reads the original message from the rows.

Notice that although the encoded message is a permutation of the original message string, this permutation depends on the length of the message. In Example 8, the permutation of the 30 positions begins

$$\begin{pmatrix} 1 & 2 & 3 & \cdots \\ 7 & 19 & 13 & \cdots \end{pmatrix}.$$

But using the keyword JONES to encode MAKE ME AN OFFER produces a permutation of 15 positions that begin

$$\begin{pmatrix} 1 & 2 & 3 & \cdots \\ 4 & 10 & 7 & \cdots \end{pmatrix}.$$

A common variation of this idea, used in the U.S. Civil War, is to transpose words rather than letters and add some superfluous known words for confusion. These extra words were called arbitraries during the Civil War.

Even and Odd Permutations

A cycle of length 2 is called a **transposition**. That is, a transposition is a cycle $p = (a_i, a_j)$, where $p(a_i) = a_j$ and $p(a_j) = a_i$.

Observe that if $p = (a_i, a_j)$ is a transposition of A, then $p \circ p = 1_A$, the identity permutation of A.

Every cycle can be written as a product of transpositions. In fact,

$$(b_1, b_2, \ldots, b_r) = (b_1, b_r) \circ (b_1, b_{r-1}) \circ \cdots \circ (b_1, b_3) \circ (b_1, b_2).$$

This case can be verified by induction on r, as follows:

Basis Step

If $r = 2$, then the cycle is just (b_1, b_2), which already has the proper form.

Induction Step

We use $P(k)$ to show $P(k + 1)$. Let $(b_1, b_2, \ldots, b_k, b_{k+1})$ be a cycle of length $k + 1$. Then $(b_1, b_2, \ldots, b_k, b_{k+1}) = (b_1, b_{k+1}) \circ (b_1, b_2, \ldots, b_k)$, as may be verified by computing the composition. Using $P(k)$, $(b_1, b_2, \ldots, b_k) = (b_1, b_k) \circ (b_1, b_{k-1}) \circ \cdots \circ (b_1, b_2)$. Thus, by substitution,

$$(b_1, b_2, \ldots, b_{k+1}) = (b_1, b_{k+1}) \circ (b_1, b_k) \circ \cdots \circ (b_1, b_3)(b_1, b_2).$$

This completes the induction step. Thus, by the principle of mathematical induction, the result holds for every cycle. For example,

$$(1, 2, 3, 4, 5) = (1, 5) \circ (1, 4) \circ (1, 3) \circ (1, 2).$$

We now obtain the following corollary of Theorem 2.

Corollary 1 Every permutation of a finite set with at least two elements can be written as a product of transpositions.

Observe that the transpositions in Corollary 1 need not be disjoint.

Example 9 Write the permutation p of Example 7 as a product of transpositions.

Solution

We have $p = (7, 8) \circ (2, 4, 5) \circ (1, 3, 6)$. Since we can write

$$(1, 3, 6) = (1, 6) \circ (1, 3)$$
$$(2, 4, 5) = (2, 5) \circ (2, 4),$$

we have $p = (7, 8) \circ (2, 5) \circ (2, 4) \circ (1, 6) \circ (1, 3)$.

We have observed that every cycle can be written as a product of transpositions. However, this can be done in many different ways. For example,

$$\begin{aligned}(1, 2, 3) &= (1, 3) \circ (1, 2)\\ &= (2, 1) \circ (2, 3)\\ &= (1, 3) \circ (3, 1) \circ (1, 3) \circ (1, 2) \circ (3, 2) \circ (2, 3).\end{aligned}$$

It then follows that every permutation on a set of two or more elements can be written as a product of transpositions in many ways. However, the following theorem, whose proof we omit, brings some order to the situation.

THEOREM 3 If a permutation of a finite set can be written as a product of an even number of transpositions, then it can never be written as a product of an odd number of transpositions, and conversely.

A permutation of a finite set is called **even** if it can be written as a product of an even number of transpositions, and it is called **odd** if it can be written as a product of an odd number of transpositions.

Example 10 Is the permutation

$$p = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 4 & 5 & 7 & 6 & 3 & 1 \end{pmatrix}$$

even or odd?

Solution

We first write p as a product of disjoint cycles, obtaining

$$p = (3, 5, 6) \circ (1, 2, 4, 7). \qquad \text{(Verify this.)}$$

Next we write each of the cycles as a product of transpositions:

$$(1, 2, 4, 7) = (1, 7) \circ (1, 4) \circ (1, 2)$$
$$(3, 5, 6) = (3, 6) \circ (3, 5).$$

Then $p = (3, 6) \circ (3, 5) \circ (1, 7) \circ (1, 4) \circ (1, 2)$. Since p is a product of an odd number of transpositions, it is an odd permutation.

From the definition of even and odd permutations, it follows (see Exercises 22 through 24) that

(a) the product of two even permutations is even.
(b) the product of two odd permutations is even.
(c) the product of an even and an odd permutation is odd.

THEOREM 4 Let $A = \{a_1, a_2, \ldots, a_n\}$ be a finite set with n elements, $n \geq 2$. There are $n!/2$ even permutations and $n!/2$ odd permutations.

Proof
Let A_n be the set of all even permutations of A, and let B_n be the set of all odd permutations. We shall define a function $f : A_n \to B_n$, which we show is one to one and onto, and this will show that A_n and B_n have the same number of elements.

Since $n \geq 2$, we can choose a particular transposition q_0 of A. Say that $q_0 = (a_{n-1}, a_n)$. We now define the function $f : A_n \to B_n$ by

$$f(p) = q_0 \circ p, \qquad p \in A_n.$$

Observe that if $p \in A_n$, then p is an even permutation, so $q_0 \circ p$ is an odd permutation and thus $f(p) \in B_n$. Suppose now that p_1 and p_2 are in A_n and $f(p_1) = f(p_2)$. Then

$$q_0 \circ p_1 = q_0 \circ p_2. \tag{3}$$

We now compose each side of equation (3) with q_0:

$$q_0 \circ (q_0 \circ p_1) = q_0 \circ (q_0 \circ p_2);$$

so, by the associative property, $(q_0 \circ q_0) \circ p_1 = (q_0 \circ q_0) \circ p_2$ or, since $q_0 \circ q_0 = 1_A$,

$$1_A \circ p_1 = 1_A \circ p_2$$
$$p_1 = p_2.$$

Thus f is one to one.

Now let $q \in B_n$. Then $q_0 \circ q \in A_n$, and

$$f(q_0 \circ q) = q_0 \circ (q_0 \circ q) = (q_0 \circ q_0) \circ q = 1_A \circ q = q,$$

which means that f is an onto function. Since $f : A_n \to B_n$ is one to one and onto, we conclude that A_n and B_n have the same number of elements. Note that $A_n \cap B_n =$ since no permutation can be both even and odd. Also, by Theorem 1, $|A_n \cup B_n| = n!$. Thus,

$$n! = |A_n \cup B_n| = |A_n| + |B_n| - |A_n \cap B_n| = 2|A_n|.$$

We then have

$$|A_n| = |B_n| = \frac{n!}{2}.$$

4 Exercises

1. Which of the following functions $f : R \to R$ are permutations of R?

(a) f is defined by $f(a) = a - 1$.

(b) f is defined by $f(a) = a^2$.

2. Which of the following functions $f : R \to R$ are permutations of R?

(a) f is defined by $f(a) = a^3$.

(b) f is defined by $f(a) = e^a$.

3. Which of the following functions $f : Z \to Z$ are permutations of Z?

(a) f is defined by $f(a) = a + 1$.

(b) f is defined by $f(a) = (a - 1)^2$.

4. Which of the following functions $f : Z \to Z$ are permutations of Z?

(a) f is defined by $f(a) = a^2 + 1$.

(b) f is defined by $f(a) = a^3 - 3$.

In Exercises 5 through 8, let $A = \{1, 2, 3, 4, 5, 6\}$ *and*

$$p_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 1 & 2 & 6 & 5 \end{pmatrix},$$

$$p_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 1 & 5 & 4 & 6 \end{pmatrix},$$

$$p_3 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 3 & 2 & 5 & 4 & 1 \end{pmatrix}.$$

5. Compute

(a) ρ_1^{-1} **(b)** $\rho_3 \circ \rho_1$

6. Compute

(a) $(\rho_2 \circ \rho_1) \circ \rho_2$ **(b)** $\rho_1 \circ (\rho_3 \circ \rho_2^{-1})$

7. Compute

(a) ρ_3^{-1} **(b)** $\rho_1^{-1} \circ \rho_2^{-1}$

8. Compute

(a) $(\rho_3 \circ \rho_2) \circ \rho_1$ **(b)** $\rho_3 \circ (\rho_2 \circ \rho_1)^{-1}$

In Exercises 9 and 10, let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. *Compute the products.*

9. **(a)** $(3, 5, 7, 8) \circ (1, 3, 2)$

(b) $(2, 6) \circ (3, 5, 7, 8) \circ (2, 5, 3, 4)$

10. **(a)** $(1, 4) \circ (2, 4, 5, 6) \circ (1, 4, 6, 7)$

(b) $(5, 8) \circ (1, 2, 3, 4) \circ (3, 5, 6, 7)$

11. Let $A = \{a, b, c, d, e, f, g\}$. Compute the products.

(a) $(a, f, g) \circ (b, c, d, e)$

(b) $(f, g) \circ (b, c, f) \circ (a, b, c)$

In Exercises 12 and 13, let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. *Write each permutation as the product of disjoint cycles.*

12. **(a)**
$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 3 & 2 & 5 & 1 & 8 & 7 & 6 \end{pmatrix}$$

(b)
$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 4 & 1 & 7 & 5 & 8 & 6 \end{pmatrix}$$

13. **(a)**
$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 6 & 5 & 7 & 8 & 4 & 3 & 2 & 1 \end{pmatrix}$$

(b)
$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 1 & 4 & 6 & 7 & 8 & 5 \end{pmatrix}$$

14. Let $A = \{a, b, c, d, e, f, g\}$. Write each permutation as the product of disjoint cycles.

(a)
$$\begin{pmatrix} a & b & c & d & e & f & g \\ g & d & b & a & c & f & e \end{pmatrix}$$

(b)
$$\begin{pmatrix} a & b & c & d & e & f & g \\ d & e & a & b & g & f & c \end{pmatrix}$$

15. Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. Write each permutation as a product of transpositions.

(a) $(2, 1, 4, 5, 8, 6)$ **(b)** $(3, 1, 6) \circ (4, 8, 2, 5)$

16. Code the message WHERE ARE YOU by applying the permutation $(1, 7, 3, 5, 11) \circ (2, 6, 9) \circ (4, 8, 10)$.

17. Decode the message ATEHAOMOMNTI, which was encoded using the permutation

$$(3, 7, 1, 12) \circ (2, 5, 8) \circ (4, 10, 6, 11, 9).$$

18. **(a)** Give the complete permutation of the positions for the message in Example 8.

(b) Write the permutation found in part (a) as the product of disjoint cycles.

19. **(a)** Encode the message MAKE ME AN OFFER using the keyword JONES and the method of Example 8.

(b) Write the permutation of the positions for the message in part (a).

In Exercises 20 and 21, let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. *Determine whether the permutation is even or odd.*

20. **(a)**
$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 2 & 1 & 6 & 5 & 8 & 7 & 3 \end{pmatrix}$$

(b)
$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 7 & 3 & 4 & 2 & 1 & 8 & 6 & 5 \end{pmatrix}$$

21. **(a)** $(6, 4, 2, 1, 5)$

(b) $(4, 8) \circ (3, 5, 2, 1) \circ (2, 4, 7, 1)$

22. Prove that the product of two even permutations is even.

23. Prove that the product of two odd permutations is even.

24. Prove that the product of an even and an odd permutation is odd.

25. Let $A = \{1, 2, 3, 4, 5\}$. Let $f = (5, 2, 3)$ and $g = (3, 4, 1)$ be permutations of A. Compute each of the following and write the result as the product of disjoint cycles.

(a) $f \circ g$ **(b)** $f^{-1} \circ g^{-1}$

26. Show that if ρ is a permutation of a finite set A, then $\rho^2 = \rho \circ \rho$ is a permutation of A.

27. Let $A = \{1, 2, 3, 4, 5, 6\}$ and

$$\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 3 & 1 & 5 & 6 \end{pmatrix}$$

be a permutation of A.

(a) Write ρ as a product of disjoint cycles.

(b) Compute ρ^{-1}. **(c)** Compute ρ^2.

(d) Find the period of ρ, that is, the smallest positive integer k such that $\rho^k = 1_A$.

28. Let $A = \{1, 2, 3, 4, 5, 6\}$ and

$$\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 3 & 5 & 1 & 2 & 6 \end{pmatrix}$$

be a permutation of A.

(a) Write ρ as a product of disjoint cycles.

(b) Compute ρ^{-1}. **(c)** Compute ρ^2.

(d) Find the period of ρ, that is, the smallest positive integer k such that $\rho^k = 1_A$.

29. **(a)** Use mathematical induction to show that if ρ is a permutation of a finite set A, then $\rho^n = \rho \circ \rho \circ \cdots \circ \rho$ (n factors) is a permutation of A for $n \in Z^+$.

(b) If A is a finite set and ρ is a permutation of A, show that $\rho^m = 1_A$ for some $m \in Z^+$.

30. Let ρ be a permutation of a set A. Define the following relation R on A: $a \mathrel{R} b$ if and only if $\rho^n(a) = b$ for some $n \in Z$. [ρ^0 is defined as the identity permutation and ρ^{-n} is defined as $(\rho^{-1})^n$.] Show that R is an equivalence relation and describe the equivalence classes.

31. Build a composition table for the permutations of $A = \{1, 2, 3\}$ given in Example 2.

32. Describe how to use the composition table in Exercise 31 to identify p^{-1} for any permutation p of A.

33. Find all subsets of $\{1_A, p_1, p_2, p_3, p_4, p_5\}$, the permutations in Example 2, that satisfy the closure property for composition.

34. For each permutation, p, of A in Example 2, determine its period. How does this relate to the subset in Exercise 33 to which p belongs?

35. Let $A = \{1, 2, 3, \ldots, n\}$. How many permutations of A, $p = (a_1, a_2, \ldots, a_n)$, are there for which $a_i < a_{i+1}$, $1 \leq i \leq n - 1$? How many permutations of A, $p = (a_1, a_2, \ldots, a_n)$, are there for which $a_i > a_{i+1}$, $1 \leq i \leq n - 1$?

36. Let $A = \{1, 2, 3, 4, 5\}$. How many different sequences of length 3 can be formed using the elements of A and such that $a_1 < a_2 < a_3$?

37. We call a permutation $p = (a_1, a_2, \ldots, a_n)$ **up-down** if the elements in the odd positions form an increasing sequence and the elements in the even positions form a decreasing sequence.

(a) Let $A = \{1, 2, 3\}$. How many up-down permutations of A are there?

(b) Let $A = \{1, 2, 3, 4\}$. How many up-down permutations of A are there?

38. Let $A = \{1, 2, 3, 4, 5\}$. How many up-down permutations of A are there?

39. Prove that the number of up-down permutations for $A = \{1, 2, 3, \ldots, n\}$ is the same as the number of increasing sequences of length $\lceil \frac{n}{2} \rceil$ that can be formed from elements of A.

Tips for Proofs

Before beginning a proof, you might find it helpful to consider what the statement does *not* say. This can help clarify your thinking about what facts and tools are available for the proof. Consider Theorem 4, Section 1. It does not say that if f is one to one, then f is onto. The additional facts that $|A| = |B|$ and that f is everywhere defined will need to be used in the proof.

To show that a function is one to one or onto, we need to use generic elements. See Example 11, Section 1. Either the definition of one-to-oneness or its contrapositive may be used to prove this property. We also have the fact that if $f : A \to B$ is everywhere defined and $|A| = |B| = n$, then f is one to one if and only if f is onto. In addition, if we wish to show f is one to one and onto, we may do this by constructing the inverse function f^{-1}. Establishing a one-to-one correspondence is a powerful counting strategy, because it allows us to count a different set than the original one. For example, see Theorem 4, Section 4, and Exercises 32 through 34, Section 2.

To prove that f and g have the same order or one is of lower order than the other, the principal tools are the rules for Θ-classes or manipulation of inequalities (Section 3, Examples 2 and 3).

Key Ideas for Review

- Function
- Identity function, 1_A: $1_A(a) = a$
- One-to-one function f from A to B: $a \neq a'$ implies $f(a) \neq f(a')$
- Onto function f from A to B: $\text{Ran}(f) = B$
- Bijection: one-to-one and onto function
- One-to-one correspondence: onto, one-to-one, everywhere defined function
- If f is a function from A to B, $1_B \circ f = f$; $f \circ 1_A = f$
- If f is an invertible function from A to B, $f^{-1} \circ f = 1_A$; $f \circ f^{-1} = 1_B$
- $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$
- Substitution code
- Boolean function f: $\text{Ran}(f) \subseteq \{\text{true}, \text{false}\}$
- Hashing function
- Fuzzy set
- Fuzzy predicate
- $O(g)$ (big Oh of g)
- f and g of the same order: f is $O(g)$ and g is $O(f)$
- Theorem: The relation Θ, $f \Theta g$ if and only if f and g have the same order, is an equivalence relation.
- Lower Θ-class
- Rules for determining Θ-class of a function

- Running time of an algorithm: Θ-class of a function that describes the number of steps performed by the algorithm
- Permutation function: a bijection from a set A to itself
- Theorem: If A is a set that has n elements, then there are $n!$ permutations of A.
- Cycle of length r: $(b_1, b_2, \ldots, b_r)$
- Theorem: A permutation of a finite set that is not the identity or a cycle can be written as a product of disjoint cycles.
- Transposition: a cycle of length 2
- Corollary: Every permutation of a finite set with at least two elements can be written as a product of transpositions.
- Transposition code
- Keyword columnar transposition code
- Even (odd) permutation: one that can be written as a product of an even (odd) number of transpositions
- Theorem: If a permutation of a finite set can be written as a product of an even number of transpositions, then it can never be written as a product of an odd number of transpositions, and conversely.
- The product of
  - **(a)** Two even permutations is even.
  - **(b)** Two odd permutations is even.
  - **(c)** An even and an odd permutation is odd.
- Theorem: If A is a set that has n elements, then there are $n!/2$ even permutations and $n!/2$ odd permutations of A.

Chapter Self-Test

1. How does a function differ from a general relation?

2. What is a common strategy to prove that a function is one to one?

3. What is a common strategy to prove that a function is onto?

4. Why are mod functions often used in constructing hashing functions?

5. What does the Θ-class of a function represent?

6. Let $A = \{a, b, c, d\}$, $B = \{1, 2, 3\}$, and $R = \{(a, 2), (b, 1), (c, 2), (d, 1)\}$. Is R a function? Is R^{-1} a function? Explain your answers.

7. Let $A = B = R$. Let $f : A \to B$ be the function defined by $f(x) = -5x + 8$. Show that f is one to one and onto.

8. Compute

(a) $\lfloor 16.29 \rfloor$ **(b)** $\lceil -1.6 \rceil$

9. Compute

(a) $\lfloor 16.29 \rfloor$ **(b)** $\lceil -1.6 \rceil$

10. Compute

(a) $lg(1)$ **(b)** $lg(64)$

11. Let Q be the propositional function defined by

$$Q(x) : \lfloor y \; xy = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix} \rfloor .$$

Evaluate $Q \begin{matrix} 2 & 1 \\ 0 & 3 \end{matrix}$ and $Q \begin{matrix} 2 & 3 \\ 4 & 6 \end{matrix}$.

12. Assume that 9500 account records need to be stored using the hashing function h, which takes the first two digits of the account number as one number and the last four digits as another number, adds them, and then applies the mod-89 function.

(a) How many linked lists will be needed?

(b) If an approximately even distribution of records is achieved, roughly how many records will be stored in each linked list?

(c) Compute $h(473810)$, $h(125332)$, and $h(308691)$.

13. Show that $f(n) = 2n^2 + 9n + 5$ is $O(n^2)$.

14. Determine the Θ-class of $f(n) = lg(n) + n^2 + 2^n$.

15. Consider the following pseudocode.

```
1. X ← 10
2. I ← 0
3. UNTIL (I > N)
        a. X ← X + I
        b. I ← I + 2
```

Write a function of N that describes the number of steps required and give the Θ-class of the function.

16. Let $A = \{1, 2, 3, 4, 5, 6\}$ and let $p_1 = (3, 6, 2)$ and $p_2 = (5, 1, 4)$ be permutations of A.

(a) Compute $p_1 \circ p_2$ and write the result as a product of cycles and as the product of transpositions.

(b) Compute $p_1^{-1} \circ p_2^{-1}$.

17. Let $p_1 = \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 3 & 2 & 1 & 4 & 5 & 6 \end{matrix}$ and $p_2 = \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 3 & 2 & 1 & 5 & 4 & 7 \end{matrix}$.

(a) Compute $p_1 \circ p_2$.

(b) Compute p_1^{-1}.

(c) Is p_1 an even or odd permutation? Explain.

Experiment 5

The □-class of a function that describes the number of steps performed by an algorithm is referred to as the **running time** of the algorithm. In this experiment you will analyze several algorithms, presented in pseudocode, to determine their running times.

Part I. The first algorithm is one method for computing the product of two $n \times n$ matrices. Assume that the matrices are each stored in an array of dimension 2 and that **A**[i, j] holds the element of **A** in row i and column j.

Algorithm MATMUL(**A**, **B**; **C**)

```
1. FOR I = 1 THRU N
    a. FOR J = 1 THRU N
        1. C[I,J] ← 0
        2. FOR K = 1 THRU N
            a. C[I,J] ← C[I,J] + A[I,K] × B[K,J]
```

Assume that each assignment of a value, each addition, and each element multiplication take the same fixed amount of time.

1. How many assignments will be done in the second **FOR** loop?
2. How many element multiplications are done in the third **FOR** loop?
3. What is the running time of MATMUL? Justify your answer.

Part II. The following recursive algorithm will compute n! for any positive integer n.

Algorithm FAC(N)

```
1. IF (N = 1) THEN
    a. A ← 1
2. ELSE
    a. A ← N × FAC(N − 1)
3. RETURN (A)
```

1. Let S_n be the number of steps needed to calculate n! using FAC. Write a recurrence relation for S_n in terms of S_{n-1}.
2. Solve the recurrence relation in question 1 and use the result to determine the running time of FAC.

Part III. The function SEEK will give the cell in which a specified value is stored in cells i through $i + n - 1$ (inclusive) of an array A. Assume that $i \geq 1$.

```
FUNCTION SEEK(ITEM, I, I + N − 1)
1. CELL ← 0
2. FOR J = I THRU I + N − 1
    a. IF (A[J] = ITEM) THEN
    b. CELL ← J
3. RETURN (CELL)
END OF FUNCTION SEEK
```

1. How many cells are there from A[i] to A[i + n − 1] (inclusive)?
2. Give a verbal description of how SEEK operates.

3. What is the running time of SEEK? Justify your answer.

Part IV. The algorithm HUNT will give the cell in which a specified value is stored in cells i through $i + n - 1$ (inclusive) of an array A. Assume that $i \geq 1$. To simplify the analysis of this algorithm, assume that n, the number of cells to be inspected, is a power of 2.

Algorithm HUNT (*ITEM*, I, I + N − 1)

```
1. CELL ← 0
2. IF (N = 1 AND A[I] = ITEM) THEN
    a. CELL ← I
3. ELSE
    a. CELL1 ← HUNT(ITEM, I, I + N/2 − 1)
    b. CELL2 ← HUNT(ITEM, I + N/2, I + N − 1)
4. IF (CELL1 ≠ 0) THEN
    a. CELL ← CELL1
5. ELSE
    a. CELL ← CELL2
6. RETURN (CELL)
```

1. Give a verbal description of how HUNT operates.
2. What is the running time of HUNT? Justify your answer.
3. Under what circumstances would it be better to use SEEK (Part III) rather than HUNT? When would it be better to use HUNT rather than SEEK?

Coding Exercises

1. Let $U = \{u_1, u_2, \ldots, u_n\}$ be the universal set for possible input sets. Write a function CHARFCN that given a set as input returns the characteristic function of the set as a sequence.

2. Write a function TRANSPOSE that, given an $n \times n$ matrix, returns its transpose.

3. Write a program that writes a given permutation as a product of disjoint cycles.

4. Write a program that writes a given permutation as a product of transpositions.

5. Use the program in Exercise 4 as a subroutine in a program that determines whether a given permutation is even or odd.

Answers to Odd-Numbered Exercises

Exercise Set 1

1. (a) Yes. $\text{Ran}(R) = \{1, 2\}$. (b) No.

3. Yes.

5. Each integer has a unique square that is also an integer.

7. Each $r \in R$ is either an integer or it is not.

9. **(a)** 3. **(b)** 1. **(c)** $(x - 1)^2$. **(d)** $x^2 - 1$. **(e)** $y - 2$. **(f)** y^4.

11. **(a)** Both. **(b)** Neither.

13. **(a)** Both. **(b)** Onto.

15. **(a)** Both. **(b)** Onto.

17. Neither.

19. **(a)** $(g \circ f)(a) = g\left(\frac{a+1}{2}\right) = 2\left(\frac{a+1}{2}\right) - 1 = a.$

(b) $(g \circ f)(a) = g(a^2 - 1) = \sqrt{a^2 + 1 - 1} = |a| = a$, since $a \geq 0$.

Observe that in a partially ordered set every pair of elements need not be comparable. For example, consider the poset in Example 3. The elements 2 and 7 are not comparable, since $2 \not| 7$ and $7 \not| 2$. Thus the word partial in partially ordered set means that some elements may not be comparable. If every pair of elements in a poset A is comparable, we say that A is a **linearly ordered** set, and the partial order is called a **linear order**. We also say that A is a **chain**.

Example 7 The poset of Example 2 is linearly ordered.

The following theorem is sometimes useful since it shows how to construct a new poset from given posets.

THEOREM 1 If $(A, \leq)$ and $(B, \leq)$ are posets, then $(A \times B, \leq)$ is a poset, with partial order $\leq$ defined by

$$(a, b) \leq (a', b') \quad \text{if } a \leq a' \text{ in } A \text{ and } b \leq b' \text{ in } B.$$

Note that the symbol $\leq$ is being used to denote three distinct partial orders. The reader should find it easy to determine which of the three is meant at any time.

Proof

If $(a, b) \in A \times B$, then $(a, b) \leq (a, b)$ since $a \leq a$ in A and $b \leq b$ in B, so $\leq$ satisfies the reflexive property in $A \times B$. Now suppose that $(a, b) \leq (a', b')$ and $(a', b') \leq (a, b)$, where a and $a' \in A$ and b and $b' \in B$. Then

$$a \leq a' \quad \text{and} \quad a' \leq a \quad \text{in } A$$

and

$$b \leq b' \quad \text{and} \quad b' \leq b \quad \text{in } B.$$

Since A and B are posets, the antisymmetry of the partial orders on A and B implies that

$$a = a' \quad \text{and} \quad b = b'.$$

Hence $\leq$ satisfies the antisymmetry property in $A \times B$.

Finally, suppose that

$$(a, b) \leq (a', b') \quad \text{and} \quad (a', b') \leq (a'', b''),$$

where $a, a', a'' \in A$, and $b, b', b'' \in B$. Then

$$a \leq a' \quad \text{and} \quad a' \leq a'',$$

so $a \leq a''$, by the transitive property of the partial order on A. Similarly,

$$b \leq b' \quad \text{and} \quad b' \leq b'',$$

so $b \leq b''$, by the transitive property of the partial order on B. Hence

$$(a, b) \leq (a'', b'').$$

Consequently, the transitive property holds for the partial order on $A \times B$, and we conclude that $A \times B$ is a poset.

The partial order $\leq$ defined on the Cartesian product $A \times B$ is called the **product partial order**.

If $(A, \leq)$ is a poset, we say that $a < b$ if $a \leq b$ but $a \neq b$. Suppose now that $(A, \leq)$ and $(B, \leq)$ are posets. In Theorem 1 we have defined the product partial

order on $A \times B$. Another useful partial order on $A \times B$, denoted by $\sqsubset$, is de ned as follows:

$$(a, b) \sqsubset (a', b') \quad \text{if } a < a' \text{ or if } a = a' \text{ and } b \leq b'.$$

This ordering is called **lexicographic**, or dictionary order. The ordering of the elements in the rst coordinate dominates, except in case of ties, when attention passes to the second coordinate. If $(A, \leq)$ and $(B, \leq)$ are linearly ordered sets, then the lexicographic order $\sqsubset$ on $A \times B$ is also a linear order.

Example 8 Let $A = R$, with the usual ordering $\leq$. Then the plane $R^2 = R \times R$ may be given lexicographic order. This is illustrated in Figure 1. We see that the plane is linearly ordered by lexicographic order. Each vertical line has the usual order, and points on one line are less than points on a line farther to the right. Thus, in Figure 1, $p_1 \sqsubset p_2$, $p_1 \sqsubset p_3$, and $p_2 \sqsubset p_3$.

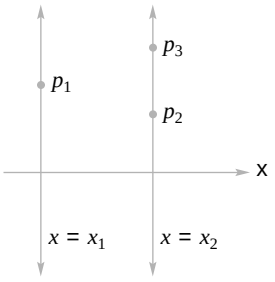


Figure 1

Lexicographic ordering is easily extended to Cartesian products $A_1 \times A_2 \times \cdots \times A_n$ as follows:

$$\begin{aligned}(a_1, a_2, \ldots, a_n) \sqsubset (a_1', a_2', \ldots, a_n') \quad &\text{if and only if} \\ a_1 < a_1' \quad &\text{or} \\ a_1 = a_1' \quad &\text{and} \quad a_2 < a_2' \quad \text{or} \\ a_1 = a_1', \quad a_2 = a_2', \quad &\text{and} \quad a_3 < a_3' \quad \text{or} \ldots \\ a_1 = a_1', \quad a_2 = a_2', \quad &\ldots, \quad a_{n-1} = a_{n-1}' \quad \text{and} \quad a_n \leq a_n'.\end{aligned}$$

Thus the rst coordinate dominates except for equality, in which case we consider the second coordinate. If equality holds again, we pass to the next coordinate, and so on.

Example 9 Let $S = \{a, b, \ldots, z\}$ be the ordinary alphabet, linearly ordered in the usual way $(a \leq b, b \leq c, \ldots, y \leq z)$. Then $S^n = S \times S \times \cdots \times S$ (n factors) can be identi ed with the set of all words having length n. Lexicographic order on S^n has the property that if $w_1 \sqsubset w_2$ ($w_1, w_2 \sqsubset S^n$), then w_1 would precede w_2 in a dictionary listing. This fact accounts for the name of the ordering.

Thus *park* $\sqsubset$ *part*, *help* $\sqsubset$ *hind*, *jump* $\sqsubset$ *mump*. The third is true since $j < m$; the second, since $h = h$, $e < i$; and the rst is true since $p = p$, $a = a$, $r = r$, $k < t$.

If S is a poset, we can extend lexicographic order to $S^\sqsubset$ in the following way.

If $x = a_1 a_2 \cdots a_n$ and $y = b_1 b_2 \cdots b_k$ are in $S^\sqsubset$ with $n \leq k$, we say that $x \sqsubset y$ if $(a_1, \ldots, a_n) \sqsubset (b_1, \ldots, b_n)$ in S^n under lexicographic ordering of S^n.

In the previous paragraph, we use the fact that the n-tuple $(a_1, a_2, \ldots, a_n) \sqsubset S^n$, and the string $a_1 a_2 \cdots a_n \sqsubset S^\sqsubset$ are really the same sequence of length n, written in two different notations. The notations differ for historical reasons, and we will use them interchangeably depending on context.

Example 10 Let S be $\{a, b, \ldots, z\}$, ordered as usual. Then S^* is the set of all possible words of any length, whether such words are meaningful or not.

Thus we have

$$help \prec helping$$

in S^* since

$$help \prec help$$

in S^4. Similarly, we have

$$helper \prec helping$$

since

$$helper \prec helpin$$

in S^6. As the example

$$help \prec helping$$

shows, this order includes *prefix order*; that is, any word is greater than all of its prefixes (beginning parts). This is also the way that words occur in the dictionary. Thus we have dictionary ordering again, but this time for words of any finite length.

Since a partial order is a relation, we can look at the digraph of any partial order on a finite set. We shall find that the digraphs of partial orders can be represented in a simpler manner than those of general relations. The following theorem provides the first result in this direction.

THEOREM 2 The digraph of a partial order has no cycle of length greater than 1.

Proof

Suppose that the digraph of the partial order $\leq$ on the set A contains a cycle of length $n \geq 2$. Then there exist distinct elements $a_1, a_2, \ldots, a_n \in A$ such that

$$a_1 \leq a_2, \quad a_2 \leq a_3, \quad \ldots, \quad a_{n-1} \leq a_n, \quad a_n \leq a_1.$$

By the transitivity of the partial order, used $n - 1$ times, $a_1 \leq a_n$. By antisymmetry, $a_n \leq a_1$ and $a_1 \leq a_n$ imply that $a_n = a_1$, a contradiction to the assumption that $a_1, a_2, \ldots, a_n$ are distinct.

Hasse Diagrams

Let A be a finite set. Theorem 2 has shown that the digraph of a partial order on A has only cycles of length 1. Indeed, since a partial order is reflexive, every vertex in the digraph of the partial order is contained in a cycle of length 1. To simplify matters, we shall delete all such cycles from the digraph. Thus the digraph shown in Figure 2(a) would be drawn as shown in Figure 2(b).

We shall also eliminate all edges that are implied by the transitive property. Thus, if $a \leq b$ and $b \leq c$, it follows that $a \leq c$. In this case, we omit the edge from a to c; however, we do draw the edges from a to b and from b to c. For example, the digraph shown in Figure 3(a) would be drawn as shown in Figure 3(b). We also agree to draw the digraph of a partial order with all edges pointing upward, so that arrows may be omitted from the edges. Finally, we replace the circles representing the vertices by dots. Thus the diagram shown in Figure 4 gives the final form of the digraph shown in Figure 2(a). The resulting diagram of a partial order, much simpler than its digraph, is called the **Hasse diagram** of the partial order of the poset. Since the Hasse diagram completely describes the associated partial order, we shall find it to be a very useful tool.

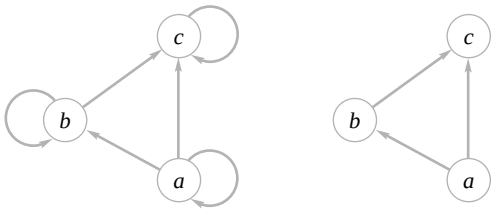


Figure 2

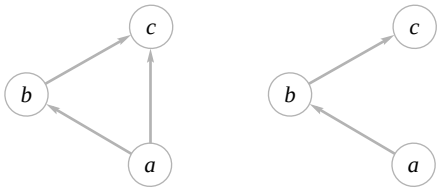


Figure 3

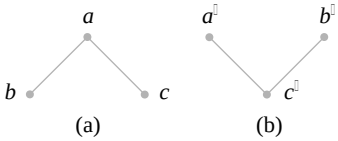


Figure 4

Example 11 Let $A = \{1, 2, 3, 4, 12\}$. Consider the partial order of divisibility on A. That is, if a and $b \in A$, $a \leq b$ if and only if $a \mid b$. Draw the Hasse diagram of the poset $(A, \leq)$.

Solution

The Hasse diagram is shown in Figure 5. To emphasize the simplicity of the Hasse diagram, we show in Figure 6 the digraph of the poset in Figure 5.

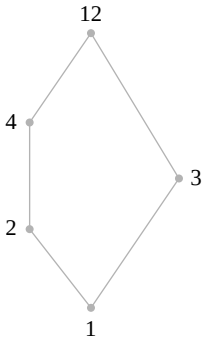


Figure 5

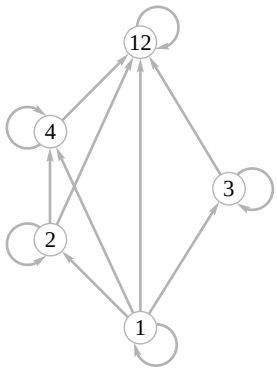


Figure 6

Example 12 Let $S = \{a, b, c\}$ and $A = P(S)$. Draw the Hasse diagram of the poset A with the partial order $\subseteq$ (set inclusion).

Solution

We first determine A, obtaining

$$A = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

The Hasse diagram can then be drawn as shown in Figure 7.

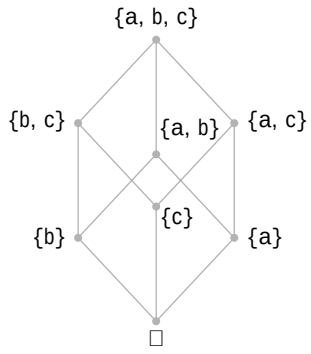


Figure 7

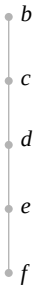


Figure 8

Observe that the Hasse diagram of a finite linearly ordered set is always of the form shown in Figure 8.

It is easily seen that if $(A, \leq)$ is a poset and $(A, \geq)$ is the dual poset, then the Hasse diagram of $(A, \geq)$ is just the Hasse diagram of $(A, \leq)$ turned upside down.

Example 13 Figure 9(a) shows the Hasse diagram of a poset $(A, \leq)$, where

$$A = \{a, b, c, d, e, f\}.$$

Figure 9(b) shows the Hasse diagram of the dual poset $(A, \geq)$. Notice that, as stated, each of these diagrams can be constructed by turning the other upside down.

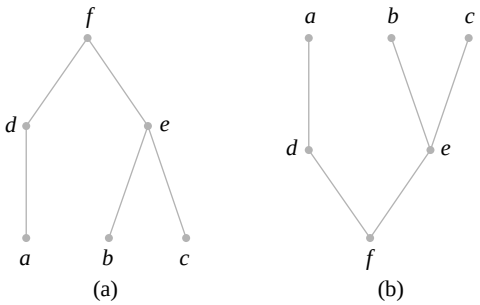


Figure 9

Topological Sorting

If A is a poset with partial order $\leq$, we sometimes need to find a linear order $\sqsubseteq$ for the set A that will merely be an extension of the given partial order in the sense that if $a \leq b$, then $a \sqsubseteq b$. The process of constructing a linear order such as $\sqsubseteq$ is called **topological sorting**. This problem might arise when we have to enter a finite poset A into a computer. The elements of A must be entered in some order, and we might want them entered so that the partial order is preserved. That is, if $a \leq b$, then a is entered before b. A topological sorting $\sqsubseteq$ will give an order of entry of the elements that meets this condition.

Example 14 Give a topological sorting for the poset whose Hasse diagram is shown in Figure 10.

Solution

The partial order $\prec$ whose Hasse diagram is shown in Figure 11(a) is clearly a linear order. It is easy to see that every pair in $\leq$ is also in the order $\prec$, so $\prec$ is a topological sorting of the partial order $\leq$. Figures 11(b) and (c) show two other solutions to this problem.

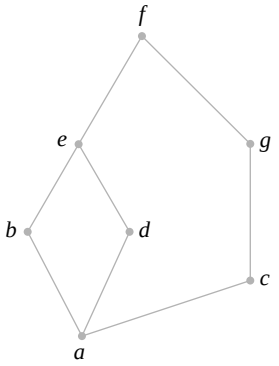


Figure 10

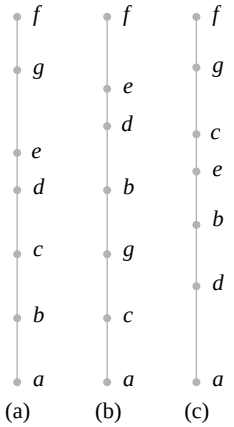


Figure 11

Isomorphism

Let $(A, \leq)$ and $(A', \leq')$ be posets and let $f : A \to A'$ be a one-to-one correspondence between A and A'. The function f is called an **isomorphism** from $(A, \leq)$ to $(A', \leq')$ if, for any a and b in A,

$$a \leq b \quad \text{if and only if} \quad f(a) \leq' f(b).$$

If $f : A \to A'$ is an isomorphism, we say that $(A, \leq)$ and $(A', \leq')$ are **isomorphic** posets.

Example 15 Let A be the set Z^+ of positive integers, and let $\leq$ be the usual partial order on A (see Example 2). Let A' be the set of positive even integers, and let $\leq'$ be the usual partial order on A'. The function $f : A \to A'$ given by

$$f(a) = 2a$$

is an isomorphism from $(A, \leq)$ to $(A', \leq')$.

First, f is one to one since, if $f(a) = f(b)$, then $2a = 2b$, so $a = b$. Next, $\text{Dom}(f) = A$, so f is everywhere de ned. Finally, if $c \in A'$, then $c = 2a$ for some $a \in Z^+$; therefore, $c = f(a)$. This shows that f is onto, so we see that f is a one-to-one correspondence. Finally, if a and b are elements of A, then it is clear that $a \leq b$ if and only if $2a \leq 2b$. Thus f is an isomorphism.

Suppose that $f : A \to A'$ is an isomorphism from a poset $(A, \leq)$ to a poset $(A', \leq')$. Suppose also that B is a subset of A, and $B' = f(B)$ is the corresponding subset of A'. Then we see from the de nition of isomorphism that the following principle must hold.

THEOREM 3
Principle of Correspondence

If the elements of B have any property relating to one another or to other elements of A, and if this property can be de ned entirely in terms of the relation $\leq$, then the elements of B' must possess exactly the same property, de ned in terms of $\leq'$.

Example 16

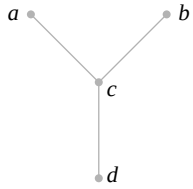


Figure 12

Let $(A, \leq)$ be the poset whose Hasse diagram is shown in Figure 12, and suppose that f is an isomorphism from $(A, \leq)$ to some other poset $(A', \leq')$. Note first that $d \leq x$ for any x in A (later we will call an element such as d a least element of A). Then the corresponding element $f(d)$ in A' must satisfy the property $f(d) \leq' y$ for all y in A'. As another example, note that $a \not\leq b$ and $b \not\leq a$. Such a pair is called **incomparable** in A. It then follows from the principle of correspondence that $f(a)$ and $f(b)$ must be incomparable in A'.

For a finite poset, one of the objects that is defined entirely in terms of the partial order is its Hasse diagram. It follows from the principle of correspondence that two finite isomorphic posets must have the same Hasse diagrams.

To be precise, let $(A, \leq)$ and $(A', \leq')$ be finite posets, let $f : A \to A'$ be a one-to-one correspondence, and let H be any Hasse diagram of $(A, \leq)$. Then

1. If f is an isomorphism and each label a of H is replaced by $f(a)$, then H will become a Hasse diagram for $(A', \leq')$.

Conversely,

2. If H becomes a Hasse diagram for $(A', \leq')$, whenever each label a is replaced by $f(a)$, then f is an isomorphism.

This justifies the name isomorphism, since isomorphic posets have the same (*iso*-) shape (*morph*) as described by their Hasse diagrams.

Example 17

Let $A = \{1, 2, 3, 6\}$ and let $\leq$ be the relation $|$ (divides). Figure 13(a) shows the Hasse diagram for $(A, \leq)$. Let

$$A' = P(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\},$$

and let $\leq'$ be set containment, $\subseteq$. If $f : A \to A'$ is defined by

$$f(1) = \emptyset, \quad f(2) = \{a\}, \quad f(3) = \{b\}, \quad f(6) = \{a, b\},$$

then it is easily seen that f is a one-to-one correspondence. If each label $a \in A$ of the Hasse diagram is replaced by $f(a)$, the result is as shown in Figure 13(b). Since this is clearly a Hasse diagram for $(A', \leq')$, the function f is an isomorphism.

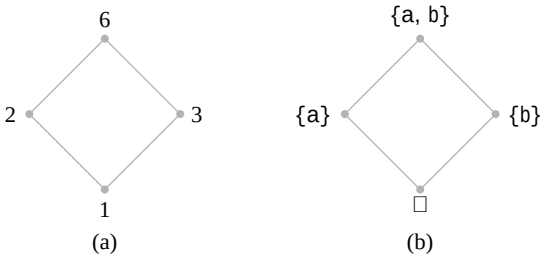


Figure 13

1 Exercises

1. Determine whether the relation R is a partial order on the set A.

(a) $A = Z$, and $a \mathrel{R} b$ if and only if $a = 2b$.

(b) $A = Z$, and $a \mathrel{R} b$ if and only if $b^2 \mid a$.

2. Determine whether the relation R is a partial order on the set A.

(a) $A = Z$, and $a \mathrel{R} b$ if and only if $a = b^k$ for some $k \in Z^+$. Note that k depends on a and b.

(b) $A = R$, and $a \mathrel{R} b$ if and only if $a \le b$.

3. Determine whether the relation R is a linear order on the set A.

(a) $A = R$, and $a \mathrel{R} b$ if and only if $a \le b$.

(b) $A = R$, and $a \mathrel{R} b$ if and only if $a \ge b$.

4. Determine whether the relation R is a linear order on the set A.

(a) $A = P(S)$, where S is a set. The relation R is set inclusion.

(b) $A = R \times R$, and $(a, b) \mathrel{R} (a', b')$ if and only if $a \le a'$ and $b \le b'$, where $\le$ is the usual partial order on R.

5. On the set $A = \{a, b, c\}$, nd all partial orders $\le$ in which $a \le b$.

6. What can you say about the relation R on a set A if R is a partial order and an equivalence relation?

7. Outline the structure of the proof given for Theorem 1.

8. Outline the structure of the proof given for Theorem 2.

In Exercises 9 and 10, determine the Hasse diagram of the relation R.

9. $A = \{1, 2, 3, 4\}$, $R = \{(1, 1), (1, 2), (2, 2), (2, 4), (1, 3), (3, 3), (3, 4), (1, 4), (4, 4)\}$.

10. $A = \{a, b, c, d, e\}$, $R = \{(a, a), (b, b), (c, c), (a, c), (c, d), (c, e), (a, d), (d, d), (a, e), (b, c), (b, d), (b, e), (e, e)\}$.

In Exercises 11 and 12, describe the ordered pairs in the relation determined by the Hasse diagram on the set $A = \{1, 2, 3, 4\}$ *(Figures 14 and 15).*

11.

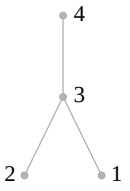


Figure 14

12.

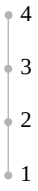


Figure 15

In Exercises 13 and 14, determine the Hasse diagram of the partial order having the given digraph (Figures 16 and 17).

13.

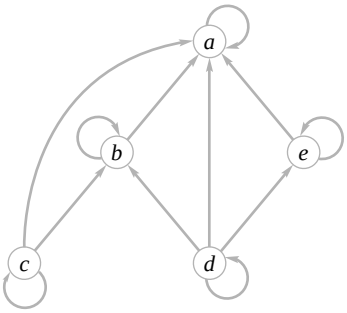


Figure 16

14.

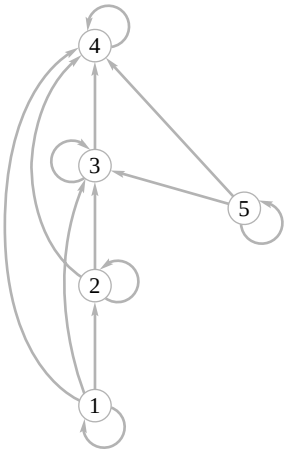


Figure 17

15. Determine the Hasse diagram of the relation on $A = \{1, 2, 3, 4, 5\}$ whose matrix is shown.

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

16. Determine the Hasse diagram of the relation on $A = \{1, 2, 3, 4, 5\}$ whose matrix is shown.

$$\begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

In Exercises 17 and 18, determine the matrix of the partial order whose Hasse diagram is given (Figures 18 and 19).

17.

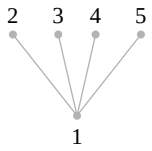


Figure 18

18.

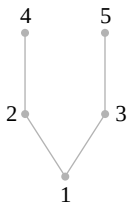


Figure 19

19. Let $A = \{\square, A, B, C, E, O, M, P, S\}$ have the usual alphabetical order, where $\square$ represents a blank character and $\square \le x$ for all $x \in A$. Arrange the following in lexicographic order (as elements of $A \times A \times A \times A$).

(a) MOP$\square$ **(b)** MOPE **(c)** CAP$\square$
(d) MAP$\square$ **(e)** BASE **(f)** ACE$\square$
(g) MACE **(h)** CAPE

20. Let $A = Z^+ \times Z^+$ have lexicographic order. Mark each of the following as true or false.

(a) $(2, 12) \in (5, 3)$ **(b)** $(3, 6) \in (3, 24)$
(c) $(4, 8) \in (4, 6)$ **(d)** $(15, 92) \in (12, 3)$

In Exercises 21 through 24, consider the partial order of divisibility on the set A. Draw the Hasse diagram of the poset and determine which posets are linearly ordered.

21. $A = \{1, 2, 3, 5, 6, 10, 15, 30\}$

22. $A = \{2, 4, 8, 16, 32\}$

23. $A = \{3, 6, 12, 36, 72\}$

24. $A = \{1, 2, 3, 4, 5, 6, 10, 12, 15, 30, 60\}$

25. Describe how to use $\mathbf{M}_R$ to determine if R is a partial order.

26. A partial order may or may not be a linear order, but any poset can be partitioned into subsets that are each a linear order. Give a partition into linearly ordered subsets with as few subsets as possible for each of the following. The poset whose Hasse diagram is given in

(a) Figure 5 **(b)** Figure 9(a)
(c) Figure 15 **(d)** Figure 20
(e) Figure 21

27. For each of the posets whose Hasse diagram is indicated below, give as large a set of elements as possible that are incomparable to one another.

(a) Figure 5 **(b)** Figure 9(a)
(c) Figure 15 **(d)** Figure 20
(e) Figure 21

28. What is the relationship between the smallest number of subsets needed to partition a partial order into linearly ordered subsets and the cardinality of the largest set of incomparable elements in the partial order? Justify your response.

In Exercises 29 and 30, draw the Hasse diagram of a topological sorting of the given poset (Figures 20 and 21).

29.

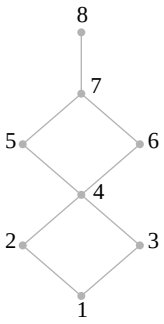


Figure 20

30.

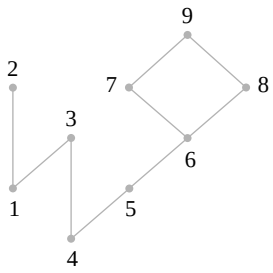


Figure 21

31. If $(A, \le)$ is a poset and A' is a subset of A, show that $(A', \le')$ is also a poset, where $\le'$ is the restriction of $\le$ to A'.

32. Show that if R is a linear order on the set A, then R^{-1} is also a linear order on A.

33. A relation R on a set A is called a **quasiorder** if it is transitive and irre exive. Let $A = P(S)$ be the power set of a set S, and consider the following relation R on A: $U \; R \; T$ if and only if $U \subset T$ (proper containment). Show that R is a quasiorder.

34. Let $A = \{x \mid x \text{ is a real number and } -5 \le x \le 20\}$. Show that the usual relation $<$ is a quasiorder (see Exercise 33) on A.

35. If R is a quasiorder on A (see Exercise 33), prove that R^{-1} is also a quasiorder on A.

36. Modify the relation in Example 3 to produce a quasiorder on Z^+.

37. Let $B = \{2, 3, 6, 9, 12, 18, 24\}$ and let $A = B \times B$. De ne the following relation on A: $(a, b) \preceq (a', b')$ if and only if $a \mid a'$ and $b \le b'$, where $\le$ is the usual partial order. Show that $\preceq$ is a partial order.

38. Let A be the set of 2×2 Boolean matrices with $\mathbf{M} \; R \; \mathbf{N}$ if and only if $m_{ij} \le n_{ij}$ $1 \le i \le 2, 1 \le j \le 2$. Prove that R is a partial order on A.

39. Let $A = \{1, 2, 3, 5, 6, 10, 15, 30\}$ and consider the partial order $\le$ of divisibility on A. That is, de ne $a \le b$ to mean that $a \mid b$. Let $A' = P(S)$, where $S = \{e, f, g\}$, be the poset with partial order $\subseteq$. Show that $(A, \le)$ and $(A', \subseteq)$ are isomorphic.

40. Let $A = \{1, 2, 4, 8\}$ and let $\le$ be the partial order of divisibility on A. Let $A' = \{0, 1, 2, 3\}$ and let $\le'$ be the usual relation less than or equal to on integers. Show that $(A, \le)$ and $(A', \le')$ are isomorphic posets.

41. Show that the partial order (A, R) of Exercise 38 is isomorphic to $(P(\{a, b, c, d\}), \subseteq)$.

2 Extremal Elements of Partially Ordered Sets

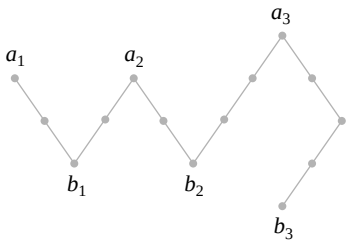


Figure 22

Certain elements in a poset are of special importance for many of the properties and applications of posets. In this section we discuss these elements, and in later sections we shall see the important role played by them. In this section we consider a poset $(A, \leq)$.

An element $a \in A$ is called a **maximal element** of A if there is no element c in A such that $a < c$ (see Section 1). An element $b \in A$ is called a **minimal element** of A if there is no element c in A such that $c < b$.

It follows immediately that, if $(A, \leq)$ is a poset and $(A, \geq)$ is its dual poset, an element $a \in A$ is a maximal element of $(A, \geq)$ if and only if a is a minimal element of $(A, \leq)$. Also, a is a minimal element of $(A, \geq)$ if and only if it is a maximal element of $(A, \leq)$.

Example 1 Consider the poset A whose Hasse diagram is shown in Figure 22. The elements a_1, a_2, and a_3 are maximal elements of A, and the elements b_1, b_2, and b_3 are minimal elements. Observe that, since there is no line between b_2 and b_3, we can conclude neither that $b_3 \leq b_2$ nor that $b_2 \leq b_3$.

Example 2 Let A be the poset of nonnegative real numbers with the usual partial order $\leq$. Then 0 is a minimal element of A. There are no maximal elements of A.

Example 3 The poset Z with the usual partial order $\leq$ has no maximal elements and has no minimal elements.

THEOREM 1 Let A be a finite nonempty poset with partial order $\leq$. Then A has at least one maximal element and at least one minimal element.

Proof

Let a be any element of A. If a is not maximal, we can find an element $a_1 \in A$ such that $a < a_1$. If a_1 is not maximal, we can find an element $a_2 \in A$ such that $a_1 < a_2$. This argument cannot be continued indefinitely, since A is a finite set. Thus we eventually obtain the finite chain

$$a < a_1 < a_2 < \cdots < a_{k-1} < a_k,$$

which cannot be extended. Hence we cannot have $a_k < b$ for any $b \in A$, so a_k is a maximal element of $(A, \leq)$.

This same argument says that the dual poset $(A, \geq)$ has a maximal element, so $(A, \leq)$ has a minimal element.

By using the concept of a minimal element, we can give an algorithm for finding a topological sorting of a given finite poset $(A, \leq)$. We remark first that if $a \in A$ and $B = A - \{a\}$, then B is also a poset under the restriction of $\leq$ to $B \times B$. We then have the following algorithm, which produces a linear array named SORT. We assume that SORT is ordered by increasing index, that is, SORT[1] $\preceq$ SORT[2] $\preceq \cdots$. The relation $\preceq$ on A defined in this way is a topological sorting of $(A, \leq)$.

Algorithm

For finding a topological sorting of a finite poset $(A, \leq)$.

Step 1 Choose a minimal element a of A.

Step 2 Make a the next entry of SORT and replace A with $A - \{a\}$.

Step 3 Repeat steps 1 and 2 until $A = \{\ \}$.

Example 4 Let $A = \{a, b, c, d, e\}$, and let the Hasse diagram of a partial order $\leq$ on A be as shown in Figure 23(a). A minimal element of this poset is the vertex labeled d (we could also have chosen e). We put d in SORT[1] and in Figure 23(b) we show the Hasse diagram of $A - \{d\}$. A minimal element of the new A is e, so e becomes SORT[2], and $A - \{e\}$ is shown in Figure 23(c). This process continues until we have exhausted A and filled SORT. Figure 23(f) shows the completed array SORT and the Hasse diagram of the poset corresponding to SORT. This is a topological sorting of $(A, \leq)$.

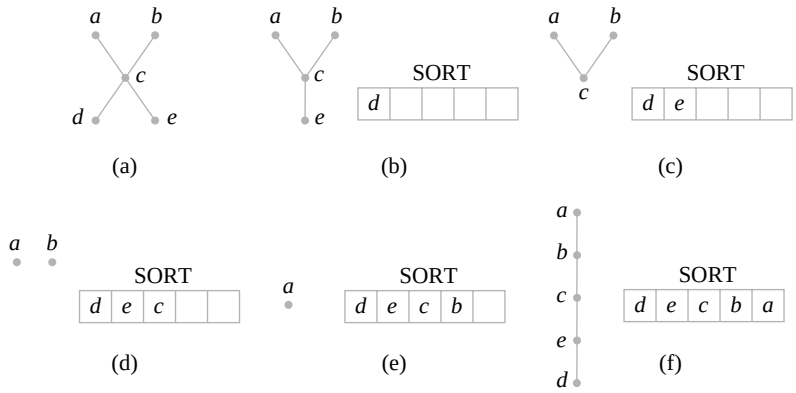


Figure 23

An element $a \in A$ is called a **greatest element** of A if $x \leq a$ for all $x \in A$. An element $a \in A$ is called a **least element** of A if $a \leq x$ for all $x \in A$.

As before, an element a of $(A, \leq)$ is a greatest (or least) element if and only if it is a least (or greatest) element of $(A, \geq)$.

Example 5 Consider the poset defined in Example 2. Then 0 is a least element; there is no greatest element.

Example 6 Let $S = \{a, b, c\}$ and consider the poset $A = P(S)$ defined in Example 12 of Section 1. The empty set is a least element of A, and the set S is a greatest element of A.

Example 7 The poset Z with the usual partial order has neither a least nor a greatest element.

THEOREM 2 A poset has at most one greatest element and at most one least element.

Proof

Suppose that a and b are greatest elements of a poset A. Then, since b is a greatest element, we have $a \leq b$. Similarly, since a is a greatest element, we have $b \leq a$. Hence $a = b$ by the antisymmetry property. Thus, if the poset has a greatest element, it only has one such element. Since this fact is true for all posets, the dual poset $(A, \geq)$ has at most one greatest element, so $(A, \leq)$ also has at most one least element.

The greatest element of a poset, if it exists, is denoted by I and is often called the **unit element**. Similarly, the least element of a poset, if it exists, is denoted by *0* and is often called the **zero element**.

Consider a poset A and a subset B of A. An element $a \in A$ is called an **upper bound** of B if $b \leq a$ for all $b \in B$. An element $a \in A$ is called a **lower bound** of B if $a \leq b$ for all $b \in B$.

Example 8 Consider the poset $A = \{a, b, c, d, e, f, g, h\}$, whose Hasse diagram is shown in Figure 24. Find all upper and lower bounds of the following subsets of A: (a) $B_1 = \{a, b\}$; (b) $B_2 = \{c, d, e\}$.

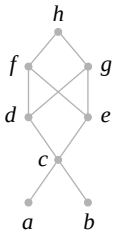


Figure 24

Solution

(a) B_1 has no lower bounds; its upper bounds are c, d, e, f, g, and h.

(b) The upper bounds of B_2 are f, g, and h; its lower bounds are c, a, and b.

As Example 8 shows, a subset B of a poset may or may not have upper or lower bounds (in A). Moreover, an upper or lower bound of B may or may not belong to B itself.

Let A be a poset and B a subset of A. An element $a \in A$ is called a **least upper bound** of B, (LUB(B)), if a is an upper bound of B and $a \leq a'$, whenever a' is an upper bound of B. Thus $a = \text{LUB}(B)$ if $b \leq a$ for all $b \in B$, and if whenever $a' \in A$ is also an upper bound of B, then $a \leq a'$.

Similarly, an element $a \in A$ is called a **greatest lower bound** of B, (GLB(B)), if a is a lower bound of B and $a' \leq a$, whenever a' is a lower bound of B. Thus $a = \text{GLB}(B)$ if $a \leq b$ for all $b \in B$, and if whenever $a' \in A$ is also a lower bound of B, then $a' \leq a$.

As usual, upper bounds in $(A, \leq)$ correspond to lower bounds in $(A, \geq)$ (for the same set of elements), and lower bounds in $(A, \leq)$ correspond to upper bounds in $(A, \geq)$. Similar statements hold for greatest lower bounds and least upper bounds.

Example 9 Let A be the poset considered in Example 8 with subsets B_1 and B_2 as defined in that example. Find all least upper bounds and all greatest lower bounds of **(a)** B_1; **(b)** B_2.

Solution

(a) Since B_1 has no lower bounds, it has no greatest lower bounds. However,

$$\text{LUB}(B_1) = c.$$

(b) Since the lower bounds of B_2 are c, a, and b, we find that

$$\text{GLB}(B_2) = c.$$

The upper bounds of B_2 are f, g, and h. Since f and g are not comparable, we conclude that B_2 has no least upper bound.

THEOREM 3 Let $(A, \leq)$ be a poset. Then a subset B of A has at most one LUB and at most one GLB.

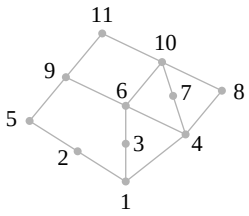


Figure 25

Proof

The proof is similar to the proof of Theorem 2.

We conclude this section with some remarks about LUB and GLB in a finite poset A, as viewed from the Hasse diagram of A. Let $B = \{b_1, b_2, \ldots, b_r\}$. If $a = \text{LUB}(B)$, then a is the first vertex that can be reached from $b_1, b_2, \ldots, b_r$ by upward paths. Similarly, if $a = \text{GLB}(B)$, then a is the first vertex that can be reached from $b_1, b_2, \ldots, b_r$ by downward paths.

Example 10 Let $A = \{1, 2, 3, 4, 5, \ldots, 11\}$ be the poset whose Hasse diagram is shown in Figure 25. Find the LUB and GLB of $B = \{6, 7, 10\}$, if they exist.

Solution

Exploring all upward paths from vertices 6, 7, and 10, we find that $\text{LUB}(B) = 10$. Similarly, by examining all downward paths from 6, 7, and 10, we find that $\text{GLB}(B) = 4$.

The next result follows immediately from the principle of correspondence (see Section 1).

THEOREM 4 Suppose that $(A, \leq)$ and $(A', \leq')$ are isomorphic posets under the isomorphism $f : A \to A'$.

- **(a)** If a is a maximal (minimal) element of $(A, \leq)$, then $f(a)$ is a maximal (minimal) element of $(A', \leq')$.
- **(b)** If a is the greatest (least) element of $(A, \leq)$, then $f(a)$ is the greatest (least) element of $(A', \leq')$.
- **(c)** If a is an upper bound (lower bound, least upper bound, greatest lower bound) of a subset B of A, then $f(a)$ is an upper bound (lower bound, least upper bound, greatest lower bound) for the subset $f(B)$ of A'.
- **(d)** If every subset of $(A, \leq)$ has a LUB (GLB), then every subset of $(A', \leq')$ has a LUB (GLB).

Example 11 Show that the posets $(A, \leq)$ and $(A', \leq')$, whose Hasse diagrams are shown in Figures 26(a) and (b), respectively, are not isomorphic.

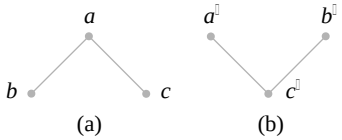


Figure 26

Solution

The two posets are not isomorphic because $(A, \leq)$ has a greatest element a, while $(A', \leq')$ does not have a greatest element. We could also argue that they are not isomorphic because $(A, \leq)$ does not have a least element, while $(A', \leq')$ does have a least element.

2 Exercises

In Exercises 1 through 8, determine all maximal and minimal elements of the poset.

1.

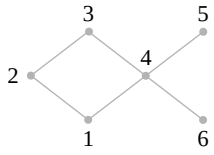


Figure 27

2.

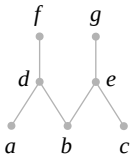


Figure 28

3.

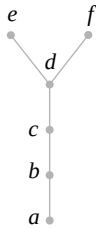


Figure 29

4.

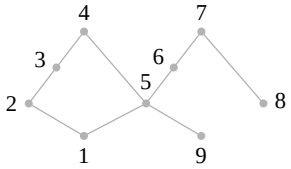


Figure 30

5. $A = R$ with the usual partial order $\leq$.

6. $A = \{x \mid x \text{ is a real number and } 0 \leq x < 1\}$ with the usual partial order $\leq$.

7. $A = \{x \mid x \text{ is a real number and } 0 < x \leq 1\}$ with the usual partial order $\leq$.

8. $A = \{2, 3, 4, 6, 8, 24, 48\}$ with the partial order of divisibility.

In Exercises 9 through 16, determine the greatest and least elements, if they exist, of the poset.

9.

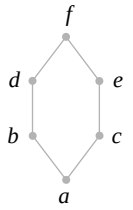


Figure 31

10.

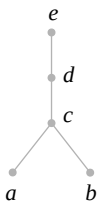


Figure 32

11.

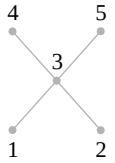


Figure 33

12.

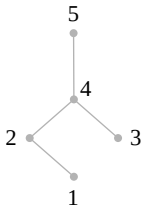


Figure 34

13. $A = \{x \mid x \text{ is a real number and } 0 < x < 1\}$ with the usual partial order $\leq$.

14. $A = \{x \mid x \text{ is a real number and } 0 \leq x \leq 1\}$ with the usual partial order $\leq$.

15. $A = \{2, 4, 6, 8, 12, 18, 24, 36, 72\}$ with the partial order of divisibility.

16. $A = \{2, 3, 4, 6, 12, 18, 24, 36\}$ with the partial order of divisibility.

In Exercises 17 and 18, determine if the statements are equivalent. Justify your conclusion.

17. **(a)** If $a \in A$ is a maximal element, then there is no $c \in A$ such that $a < c$.

(b) If $a \in A$ is a maximal element, then for all $b \in A$, $b \leq a$.

18. **(a)** If $a \in A$ is a minimal element, then there is no $c \in A$ such that $c < a$.

(b) If $a \in A$ is a minimal element, then for all $b \in A$, $a \leq b$.

19. Determine if the given statement is true or false. Explain your reasoning.

(a) A nonempty nite poset has a maximal element.

(b) A nonempty nite poset has a greatest element.

(c) A nonempty nite poset has a minimal element.

(d) A nonempty nite poset has a least element.

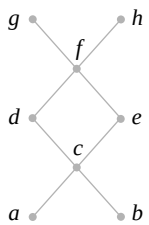
20. Prove that if $(A, \leq)$ has a greatest element, then $(A, \leq)$ has a unique greatest element.

21. Prove that if $(A, \leq)$ has a least element, then $(A, \leq)$ has a unique least element.

22. Prove Theorem 3.

In Exercises 23 through 32 find, if they exist, ***(a)*** *all upper bounds of* B*;* ***(b)*** *all lower bounds of* B*;* ***(c)*** *the least upper bound of* B*;* ***(d)*** *the greatest lower bound of* B.

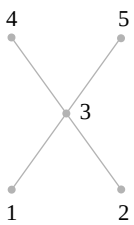
23.



$B = \{c, d, e\}$

Figure 35

24.



$B = \{1, 2, 3, 4, 5\}$

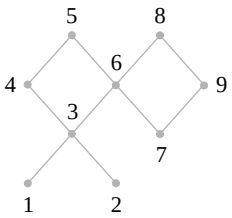
Figure 36

25.

$B = \{b, c, d\}$

Figure 37

26.



$B = \{3, 4, 6\}$

Figure 38

27. $(A, \leq)$ is the poset in Exercise 23; $B = \{b, g, h\}$.

28. **(a)** $(A, \leq)$ is the poset in Exercise 26; $B = \{4, 6, 9\}$.

(b) $(A, \leq)$ is the poset in Exercise 26; $B = \{3, 4, 8\}$.

29. $A = R$ and $\leq$ denotes the usual partial order; $B = \{x \mid x \text{ is a real number and } 1 < x < 2\}$.

30. $A = R$ and $\leq$ denotes the usual partial order; $B = \{x \mid x \text{ is a real number and } 1 \leq x < 2\}$.

31. A is the set of 2×2 Boolean matrices and $\leq$ denotes the relation R with $\mathbf{M}\ R\ \mathbf{N}$ if and only if $m_{ij} \leq n_{ij}$, $1 \leq i \leq 2$, $1 \leq j \leq 2$; B is the set of matrices in A with exactly two ones.

32. A is the set of 2×2 Boolean matrices and $\leq$ denotes the relation R with $\mathbf{M}\ R\ \mathbf{N}$ if and only if $m_{ij} \leq n_{ij}$, $1 \leq i \leq 2$, $1 \leq j \leq 2$; $B = \left\{ \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$.

33. Construct the Hasse diagram of a topological sorting of the poset whose Hasse diagram is shown in Figure 35. Use the algorithm SORT.

34. Construct the Hasse diagram of a topological sorting of the poset whose Hasse diagram is shown in Figure 36. Use the algorithm SORT.

35. Let R be a partial order on a finite set A. Describe how to use $\mathbf{M}_R$ to find the least and greatest elements of A if they exist.

36. Give an example of a partial order on $A = \{a, b, c, d, e\}$ that has two maximal elements and no least element.

37. Let $A = \{2, 3, 4, \ldots, 100\}$ with the partial order of divisibility.

(a) How many maximal elements does $(A, \leq)$ have?

(b) Give a subset of A that is a linear order under divisibility and is as large as possible.

38. Let $(A, \leq)$ be as in Exercise 37. How many minimal elements does $(A, \leq)$ have?

3 Lattices

A **lattice** is a poset $(L, \leq)$ in which every subset $\{a, b\}$ consisting of two elements has a least upper bound and a greatest lower bound. We denote $\text{LUB}(\{a, b\})$ by $a \sqcup b$ and call it the **join** of a and b. Similarly, we denote $\text{GLB}(\{a, b\})$ by $a \sqcap b$ and call it the **meet** of a and b. Lattice structures often appear in computing and mathematical applications. Observe that a lattice is a mathematical structure with two binary operations, join and meet.

Example 1 Let S be a set and let $L = P(S)$. As we have seen, $\subseteq$, containment, is a partial order on L. Let A and B belong to the poset $(L, \subseteq)$. Then $A \sqcup B$ is the set $A \cup B$. To see this, note that $A \subseteq A \cup B$, $B \subseteq A \cup B$, and, if $A \subseteq C$ and $B \subseteq C$, then it follows that $A \cup B \subseteq C$. Similarly, we can show that the element $A \sqcap B$ in $(L, \subseteq)$ is the set $A \cap B$. Thus, L is a lattice.

Example 2 Consider the poset $(Z^+, \leq)$, where for a and b in Z^+, $a \leq b$ if and only if $a \mid b$. Then L is a lattice in which the join and meet of a and b are their least common multiple and greatest common divisor, respectively. That is,

$$a \sqcup b = \text{LCM}(a, b) \quad \text{and} \quad a \sqcap b = \text{GCD}(a, b).$$

Example 3 Let n be a positive integer and let D_n be the set of all positive divisors of n. Then D_n is a lattice under the relation of divisibility as considered in Example 2. Thus, if $n = 20$, we have $D_{20} = \{1, 2, 4, 5, 10, 20\}$. The Hasse diagram of D_{20} is shown in Figure 39(a). If $n = 30$, we have $D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$. The Hasse diagram of D_{30} is shown in Figure 39(b).

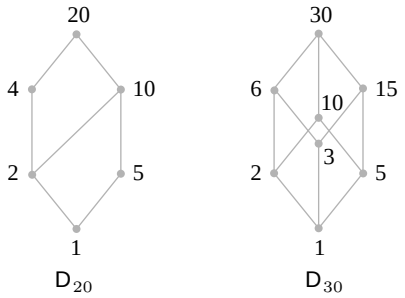


Figure 39

Example 4 Which of the Hasse diagrams in Figure 40 represent lattices?

Solution

Hasse diagrams (a), (b), (d), and (e) represent lattices. Diagram (c) does not represent a lattice because $f \sqcup g$ does not exist. Diagram (f) does not represent a lattice because neither $d \sqcup e$ nor $b \sqcup c$ exist. Diagram (g) does not represent a lattice because $c \sqcup d$ does not exist.

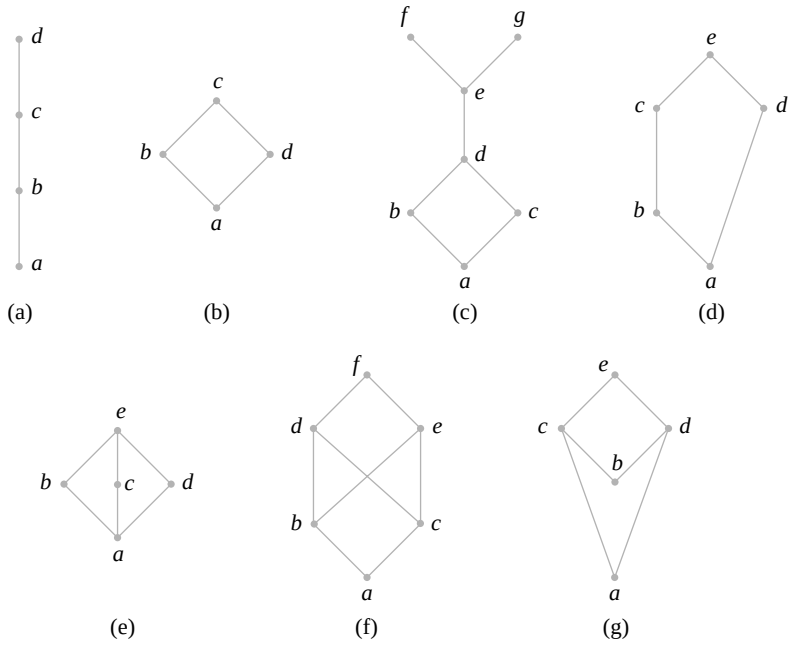


Figure 40

Example 5 We have already observed in Example 4 of Section 1 that the set R of all equivalence relations on a set A is a poset under the partial order of set containment. We can now conclude that R is a lattice where the meet of the equivalence relations R and S is their intersection $R \cap S$ and their join is $(R \sqcup S)^{\infty}$, the transitive closure of their union.

Let $(L, \leq)$ be a poset and let $(L, \geq)$ be the dual poset. If $(L, \leq)$ is a lattice, we can show that $(L, \geq)$ is also a lattice. In fact, for any a and b in L, the least

upper bound of a and b in $(L, \leq)$ is equal to the greatest lower bound of a and b in $(L, \geq)$. Similarly, the greatest lower bound of a and b in $(L, \leq)$ is equal to the least upper bound of a and b in $(L, \geq)$. If L is a finite set, this property can easily be seen by examining the Hasse diagrams of the poset and its dual.

Example 6 Let S be a set and $L = P(S)$. Then $(L, \subseteq)$ is a lattice, and its dual lattice is $(L, \supseteq)$, where $\subseteq$ is contained in and $\supseteq$ is contains. The discussion preceding this example then shows that in the poset $(L, \subseteq)$ the join $A \vee B$ is the set $A \cup B$, and the meet $A \wedge B$ is the set $A \cap B$.

THEOREM 1 If $(L_1, \leq)$ and $(L_2, \leq)$ are lattices, then $(L, \leq)$ is a lattice, where $L = L_1 \times L_2$, and the partial order $\leq$ of L is the product partial order.

Proof

We denote the join and meet in L_1 by $\vee_1$ and $\wedge_1$, respectively, and the join and meet in L_2 by $\vee_2$ and $\wedge_2$, respectively. We already know from Theorem 1 of Section 1 that L is a poset. We now need to show that if (a_1, b_1) and $(a_2, b_2) \in L$, then $(a_1, b_1) \vee (a_2, b_2)$ and $(a_1, b_1) \wedge (a_2, b_2)$ exist in L. We leave it as an exercise to verify that

$$(a_1, b_1) \vee (a_2, b_2) = (a_1 \vee_1 a_2, b_1 \vee_2 b_2)$$
$$(a_1, b_1) \wedge (a_2, b_2) = (a_1 \wedge_1 a_2, b_1 \wedge_2 b_2).$$

Thus L is a lattice.

Example 7 Let L_1 and L_2 be the lattices shown in Figures 41(a) and (b), respectively. Then $L = L_1 \times L_2$ is the lattice shown in Figure 41(c).

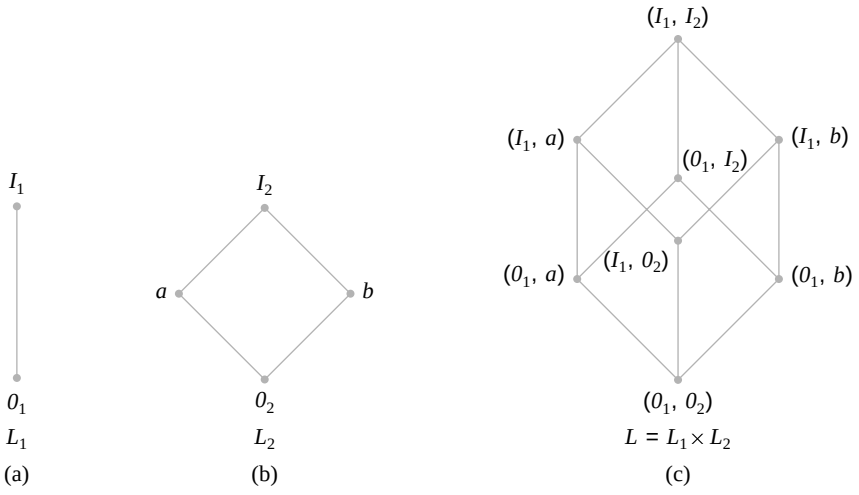


Figure 41

Let $(L, \leq)$ be a lattice. A nonempty subset S of L is called a **sublattice** of L if $a \vee b \in S$ and $a \wedge b \in S$ whenever $a \in S$ and $b \in S$.

Example 8 The lattice D_n of all positive divisors of n (see Example 3) is a sublattice of the lattice Z^+ under the relation of divisibility (see Example 2).

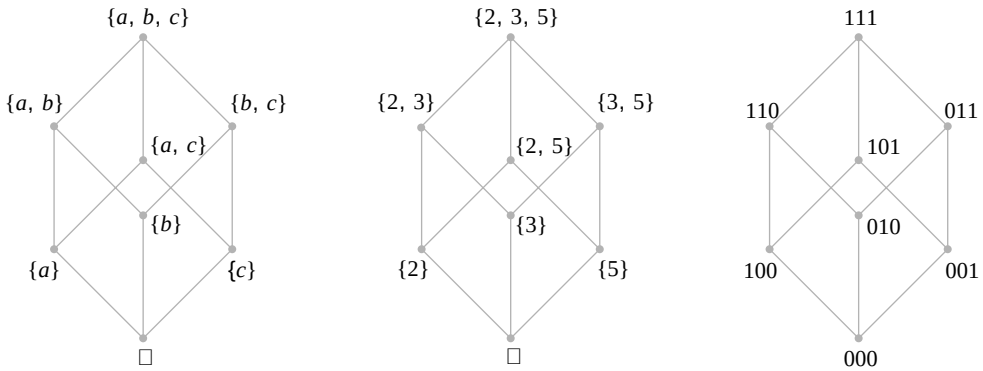


Figure 60

1 s of length n. We can therefore label the Hasse diagram of a lattice $(P(S), \subseteq)$ by such sequences. In doing so, we free the diagram from dependence on a particular set S and emphasize the fact that it depends only on n.

Example 2 Figure 60(c) shows how the diagrams that appear in Figures 60(a) and (b) can be labeled by sequences of 0 s and 1 s. This labeling serves equally well to describe the lattice of Figure 60(a) or (b), or for that matter the lattice $(P(S), \subseteq)$ that arises from any set S having three elements.

If the Hasse diagram of the lattice corresponding to a set with n elements is labeled by sequences of 0 s and 1 s of length n, as described previously, then the resulting lattice is named B_n. The properties of the partial order on B_n can be described directly as follows. If $x = a_1a_2 \cdots a_n$ and $y = b_1b_2 \cdots b_n$ are two elements of B_n, then

1. $x \leq y$ if and only if $a_k \leq b_k$ (as numbers 0 or 1) for $k = 1, 2, \ldots, n$.
2. $x \sqcup y = c_1c_2 \cdots c_n$, where $c_k = \min\{a_k, b_k\}$.
3. $x \sqcap y = d_1d_2 \cdots d_n$, where $d_k = \max\{a_k, b_k\}$.
4. x has a complement $x' = z_1z_2 \cdots z_n$, where $z_k = 1$ if $x_k = 0$, and $z_k = 0$ if $x_k = 1$.

The truth of these statements can be seen by noting that $(B_n, \leq)$ is isomorphic with $(P(S), \subseteq)$, so each x and y in B_n correspond to subsets A and B of S. Then $x \leq y$, $x \sqcup y$, $x \sqcap y$, and x' correspond to $A \subseteq B$, $A \cap B$, $A \cup B$, and $\overline{A}$ (set complement), respectively (verify). Figure 61 shows the Hasse diagrams of the lattices B_n for $n = 0, 1, 2, 3$.

We have seen that each lattice $(P(S), \subseteq)$ is isomorphic with B_n, where $n = |S|$. Other lattices may also be isomorphic with one of the B_n and thus possess all the special properties that the B_n possess.

Example 3 In Example 17 of Section 1, we considered the lattice D_6 consisting of all positive integer divisors of 6 under the partial order of divisibility. The Hasse diagram of D_6 is shown in that example, and we now see that D_6 is isomorphic with B_2. In fact, $f : D_6 \to B_2$ is an isomorphism, where

$$f(1) = 00, \quad f(2) = 10, \quad f(3) = 01, \quad f(6) = 11.$$

We are therefore led to make the following definition. A finite lattice is called a **Boolean algebra** if it is isomorphic with B_n for some nonnegative integer n.

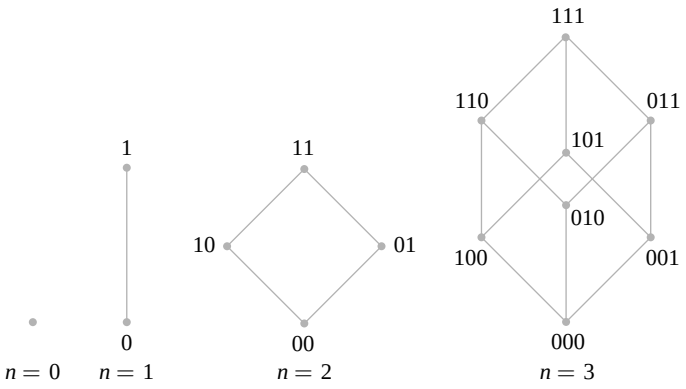


Figure 61

Thus each B_n is a Boolean algebra and so is each lattice $(P(S), \subseteq)$, where S is a finite set. Example 3 shows that D_6 is also a Boolean algebra.

We will work only with finite posets in this section. For the curious, however, we note that there are infinite posets that share all the relevant properties of the lattices $(P(S), \subseteq)$ (for infinite sets S, of course), but that are not isomorphic with one these lattices. This necessitates the restriction of our definition of Boolean algebra to the finite case, which is sufficient for the applications that we present.

Example 4 Consider the lattices D_{20} and D_{30} of all positive integer divisors of 20 and 30, respectively, under the partial order of divisibility. These posets were introduced in Example 3 of Section 3, and their Hasse diagrams were shown in Figure 39. Since D_{20} has six elements and $6 \neq 2^n$ for any integer $n \geq 0$, we conclude that D_{20} is not a Boolean algebra. The poset D_{30} has eight elements, and since $8 = 2^3$, it could be a Boolean algebra. By comparing Figure 39(b) and Figure 61, we see that D_{30} is isomorphic with B_3. In fact, we see that the one-to-one correspondence $f : D_{30} \to B_3$ defined by

$$\begin{aligned} f(1) &= 000, \quad f(2) = 100, \quad f(3) = 010, \\ f(5) &= 001, \quad f(6) = 110, \quad f(10) = 101, \\ f(15) &= 011, \quad f(30) = 111 \end{aligned}$$

is an isomorphism. Thus D_{30} is a Boolean algebra.

If a finite lattice L does not contain 2^n elements for some nonnegative integer n, we know that L cannot be a Boolean algebra. If $|L| = 2^n$, then L may or may not be a Boolean algebra. If L is relatively small, we may be able to compare its Hasse diagram with the Hasse diagram of B_n. In this way we saw in Example 4 that D_{30} is a Boolean algebra. However, this technique may not be practical if L is large. In that case, we may be able to show that L is a Boolean algebra by directly constructing an isomorphism with some B_n or, equivalently, with $(P(S), \subseteq)$ for some finite set S. Suppose, for example, that we want to know whether a lattice D_n is a Boolean algebra, and we want a method that works no matter how large n is. The following theorem gives a partial answer.

THEOREM 2 Let

$$n = p_1 p_2 \cdots p_k,$$

where the p_i are distinct primes. Then D_n is a Boolean algebra.

Proof
Let $S = \{p_1, p_2, \ldots, p_k\}$. If $T \subseteq S$ and a_T is the product of the primes in T, then $a_T \mid n$. Any divisor of n must be of the form a_T for some subset T of S (where we let $a_\emptyset = 1$). The reader may verify that if V and T are subsets of S, $V \subseteq T$ if and only if $a_V \mid a_T$. Also, it follows that $a_{V \cap T} = a_V \wedge a_T = \text{GCD}(a_V, a_T)$ and $a_{V \cup T} = a_V \vee a_T = \text{LCM}(a_V, a_T)$. Thus the function $f : P(S) \to D_n$ given by $f(T) = a_T$ is an isomorphism from $P(S)$ to D_n. Since $P(S)$ is a Boolean algebra, so is D_n.

Example 5 Since $210 = 2 \cdot 3 \cdot 5 \cdot 7$, $66 = 2 \cdot 3 \cdot 11$, and $646 = 2 \cdot 17 \cdot 19$, we see from Theorem 2 that D_{210}, D_{66}, and D_{646} are all Boolean algebras.

In other cases of large lattices L, we may be able to show that L is not a Boolean algebra by showing that the partial order of L does not have the necessary properties. A Boolean algebra is isomorphic with some B_n and therefore with some lattice $(P(S), \subseteq)$. Thus a Boolean algebra L must be a bounded lattice and a complemented lattice (see Section 3). In other words, it will have a greatest element I corresponding to the set S and a least element 0 corresponding to the subset $\emptyset$. Also, every element x of L will have a complement x'. According to Example 16, Section 3, L must also be distributive. The principle of correspondence (see Section 1) then tells us that the following rule holds.

THEOREM 3
Substitution Rule for Boolean Algebras

Any formula involving $\subseteq$ or $\cap$ that holds for arbitrary subsets of a set S will continue to hold for arbitrary elements of a Boolean algebra L if $\leq$ is substituted for $\cap$ and $\geq$ for $\subseteq$.

Example 6 If L is any Boolean algebra and x, y, and z are in L, then the following three properties hold.

1. $(x')' = x$ *Involution Property*
2. $(x \vee y)' = x' \wedge y'$
3. $(x \wedge y)' = x' \vee y'$ *De Morgan's Laws*

This is true by the substitution rule for Boolean algebras, since we know that the corresponding formulas

1′. $\overline{(\overline{A})} = A$
2′. $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$
3′. $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$

hold for arbitrary subsets A and B of a set S.

In a similar way, we can list other properties that must hold in any Boolean algebra by the substitution rule. Next we summarize all the basic properties of a Boolean algebra $(L, \leq)$ and, next to each one, we list the corresponding property for subsets of a set S. We suppose that x, y, and z are arbitrary elements in L, and A, B, and C are arbitrary subsets of S. Also, we denote the greatest and least elements of L by I and 0, respectively.

1. $x \le y$ if and only if $x \vee y = y$.	**1′.** $A \subseteq B$ if and only if $A \cup B = B$.
2. $x \le y$ if and only if $x \wedge y = x$.	**2′.** $A \subseteq B$ if and only if $A \cap B = A$.
3. (a) $x \vee x = x$.	**3′. (a)** $A \cup A = A$.
(b) $x \wedge x = x$.	**(b)** $A \cap A = A$.
4. (a) $x \vee y = y \vee x$.	**4′. (a)** $A \cup B = B \cup A$.
(b) $x \wedge y = y \wedge x$.	**(b)** $A \cap B = B \cap A$.
5. (a) $x \vee (y \vee z) = (x \vee y) \vee z$.	**5′. (a)** $A \cup (B \cup C) = (A \cup B) \cup C$.
(b) $x \wedge (y \wedge z) = (x \wedge y) \wedge z$.	**(b)** $A \cap (B \cap C) = (A \cap B) \cap C$.
6. (a) $x \vee (x \wedge y) = x$.	**6′. (a)** $A \cup (A \cap B) = A$.
(b) $x \wedge (x \vee y) = x$.	**(b)** $A \cap (A \cup B) = A$.
7. $0 \le x \le I$ for all x in L.	**7′.** $\emptyset \subseteq A \subseteq S$ for all A in $P(S)$.
8. (a) $x \vee 0 = x$.	**8′. (a)** $A \cup \emptyset = A$.
(b) $x \wedge 0 = 0$.	**(b)** $A \cap \emptyset = \emptyset$.
9. (a) $x \vee I = I$.	**9′. (a)** $A \cup S = S$.
(b) $x \wedge I = x$.	**(b)** $A \cap S = A$.
10. (a) $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$.	**10′. (a)** $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
(b) $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$.	**(b)** $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
11. Every element x has a *unique* complement x' satisfying	**11′.** Every element A has a *unique* complement $\overline{A}$ satisfying
(a) $x \vee x' = I$.	**(a)** $A \cup \overline{A} = S$.
(b) $x \wedge x' = 0$.	**(b)** $A \cap \overline{A} = \emptyset$.
12. (a) $0' = I$.	**12′. (a)** $\overline{\emptyset} = S$.
(b) $I' = 0$.	**(b)** $\overline{S} = \emptyset$.
13. $(x')' = x$.	**13′.** $\overline{(\overline{A})} = A$.
14. (a) $(x \vee y)' = x' \wedge y'$.	**14′. (a)** $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$.
(b) $(x \wedge y)' = x' \vee y'$.	**(b)** $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$.

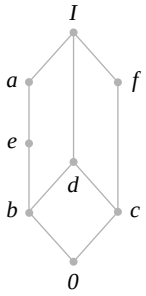


Figure 62

Thus we may be able to show that a lattice L is not a Boolean algebra by showing that it does not possess one or more of these properties.

Example 7 Show that the lattice whose Hasse diagram is shown in Figure 62 is not a Boolean algebra.

Solution

Elements a and e are both complements of c; that is, they both satisfy properties 11(a) and 11(b) with respect to the element c. But property 11 says that such an element is unique in any Boolean algebra. Thus the given lattice cannot be a Boolean algebra.

Example 8 Show that if n is a positive integer and $p^2 \mid n$, where p is a prime number, then D_n is not a Boolean algebra.

Solution

Suppose that $p^2 \mid n$ so that $n = p^2q$ for some positive integer q. Since p is also a divisor of n, p is an element of D_n. Thus, by the remarks given previously, if D_n is a Boolean algebra, then p must have a complement p'. Then $\text{GCD}(p, p') = 1$

and LCM($p, p^{\square}$) = n. Similarly, $pp^{\square} = n$, so $p^{\square} = n/p = pq$. This shows that GCD(p, pq) = 1, which is impossible, since p and pq have p as a common divisor. Hence D_n cannot be a Boolean algebra.

If we combine Example 8 and Theorem 2, we see that D_n is a Boolean algebra if and only if n is the product of distinct primes, that is, if and only if no prime divides n more than once.

Example 9 If $n = 40$, then $n = 2^3 \cdot 5$, so 2 divides n three times. If $n = 75$, then $n = 3 \cdot 5^2$, so 5 divides n twice. Thus neither D_{40} nor D_{75} are Boolean algebras.

Let us summarize what we have shown about Boolean algebras. We may attempt to show that a lattice L is a Boolean algebra by examining its Hasse diagram or constructing directly an isomorphism between L and B_n or $(P(S), \square)$. We may attempt to show that L is not a Boolean algebra by checking the number of elements in L or the properties of its partial order. If L is a Boolean algebra, then we may use any of the properties 1 through 14 to manipulate or rewrite expressions involving elements of L. Simply proceed as if the elements were subsets and the manipulations were those that arise in set theory. We call such a lattice an algebra, because we use properties 1 through 14 just as the properties of real numbers are used in high school algebra.

From now on we will denote the Boolean algebra B_1 simply as B. Thus B contains only the two elements 0 and 1. It is a fact that any of the Boolean algebras B_n can be described in terms of B. The following theorem gives this description.

THEOREM 4 For any $n \geq 1$, B_n is the product $B \times B \times \cdots \times B$ of B, n factors, where $B \times B \times \cdots \times B$ is given the product partial order.

Proof

By de nition, B_n consists of all n-tuples of 0 s and 1 s, that is, all n-tuples of elements from B. Thus, as a set, B_n is equal to $B \times B \times \cdots \times B$ (n factors). Moreover, if $x = x_1x_2 \cdots x_n$ and $y = y_1y_2 \cdots y_n$ are two elements of B_n, then we know that

$$x \leq y \quad \text{if and only if} \quad x_k \leq y_k \qquad \text{for all } k.$$

Thus B_n, identi ed with $B \times B \times \cdots \times B$ (n factors), has the product partial order.

4 Exercises

In Exercises 1 through 10, determine whether the poset is a Boolean algebra. Explain.

1.

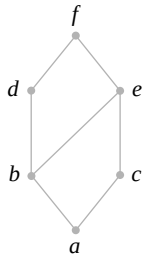


Figure 63

2.

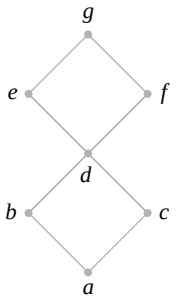


Figure 64

3.

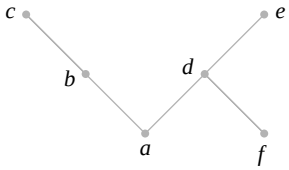


Figure 65

4.

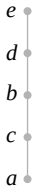


Figure 66

5.

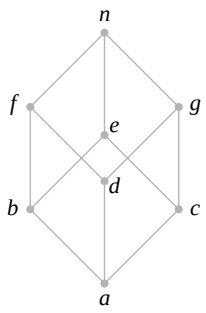


Figure 67

6.

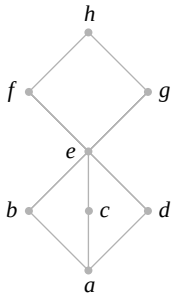


Figure 68

7.

Figure 69

8.

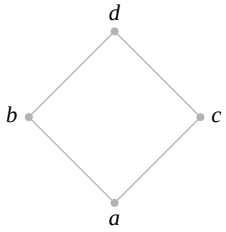


Figure 70

9. D_{385} **10.** D_{60}

11. Are there any Boolean algebras having three elements? Why or why not?

12. Show that in a Boolean algebra, for any a and b, $a \le b$ if and only if $b' \le a'$.

13. Show that in a Boolean algebra, for any a and b, $a = b$ if and only if $(a \wedge b') \vee (a' \wedge b) = 0$.

14. Show that in a Boolean algebra, for any a, b, and c, if $a \le b$, then $a \wedge c \le b \wedge c$.

15. Show that in a Boolean algebra, for any a, b, and c, if $a \le b$, then $a \vee c \le b \vee c$.

16. Show that in a Boolean algebra the following statements are equivalent for any a and b.

(a) $a \wedge b = b$ **(b)** $a \vee b = a$

(c) $a' \vee b = 1$ **(d)** $a \wedge b' = 0$

(e) $a \le b$

17. Show that in a Boolean algebra, for any a and b,

$$(a \wedge b) \vee (a \wedge b') = a.$$

18. Show that in a Boolean algebra, for any a and b,

$$b \vee (a \wedge (a' \vee (b \wedge b'))) = b.$$

19. Show that in a Boolean algebra, for any a, b, and c,

$$(a \wedge b \wedge c) \vee (b \wedge c) = b \wedge c.$$

20. Show that in a Boolean algebra, for any a, b, and c,

$$((a \vee c) \wedge (b' \vee c))' = (a' \wedge b) \vee c'.$$

21. Show that in a Boolean algebra, for any a, b, and c, if $a \le b$, then

$$a \vee (b \wedge c) = b \wedge (a \vee c).$$

22. Explain the connection between Examples 7 and 8.

For Exercises 23 through 26, let $A = \{a, b, c, d, e, f, g, h\}$ *and* R *be the relation defined by*

$$\mathbf{M}_R = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

23. Show that (A, R) is a poset.

24. Does the poset (A, R) have a least element? a greatest element? If so, identify them.

25. Show that the poset (A, R) is complemented and give all pairs of complements.

26. Prove or disprove that (A, R) is a Boolean algebra.

27. Let $A = \{a, b, c, d, e, f, g, h\}$ and R be the relation de ned by

$$\mathbf{M}_R = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Prove or disprove that (A, R) is a Boolean algebra.

28. Let A be the set of 2×2 Boolean matrices with $\mathbf{M}\ R\ \mathbf{N}$ if and only if $m_{ij} \le n_{ij}$, $1 \le i \le 2$, $1 \le j \le 2$. Prove that (A, R) is a Boolean algebra.

29. An **atom** of a Boolean algebra, B, is a nonzero element a such that there is no x in B with $0 < x < a$.

(a) Identify the atoms of the Boolean algebra in Figure 60(a).

(b) Identify the atoms of D_{30}.

30. How many atoms does B_n have?

31. Give the atoms of (A, R) as de ned in Exercise 28.

32. Like physical atoms, the atoms of a Boolean algebra serve as building blocks for the elements of the Boolean algebra. Write each nonzero element of the given Boolean algebra as the join of distinct atoms.

(a) B_3 **(b)** D_{42}

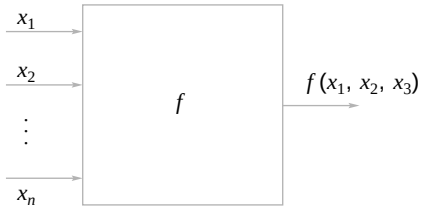
(c) (A, R) in Exercise 28

5 Functions on Boolean Algebras

Tables listing the values of a function f for all elements of B_n, such as shown in Figure 71(a), are often called truth tables for f. This is because they are analogous to tables that arise in logic. Suppose that the x_k represent propositions, and $f(x_1, x_2, \ldots, x_n)$ represents a compound sentence constructed from the x_k s. If we think of the value 0 for a sentence as meaning that the sentence is false, and 1 as meaning that the sentence is true, then tables such as Figure 71(a) show us how truth or falsity of $f(x_1, x_2, \ldots, x_n)$ depends on the truth or falsity of its component sentences x_k. Thus such tables are called **truth tables**, even when they arise in areas other than logic, such as in Boolean algebras.

x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

(a)



(b)

Figure 71

The reason that such functions are important is that, as shown schematically in Figure 71(b), they may be used to represent the output requirements of a circuit for all possible input values. Thus each x_i represents an input circuit capable of carrying two indicator voltages (one voltage for 0 and a different voltage for 1). The function f represents the desired output response in all cases. Such requirements occur at the design stage of all combinational and sequential computer circuitry.

Note carefully that the speci cation of a function $f : B_n \to B$ simply lists circuit requirements. It gives no indication of how these requirements can be met. One important way of producing functions from B_n to B is by using Boolean polynomials, which we now consider.

Boolean Polynomials

Let $x_1, x_2, \ldots, x_n$ be a set of n symbols or variables. A **Boolean polynomial** $p(x_1, x_2, \ldots, x_n)$ in the variables x_k is de ned recursively as follows:

1. $x_1, x_2, \ldots, x_n$ are all Boolean polynomials.
2. The symbols 0 and 1 are Boolean polynomials.
3. If $p(x_1, x_2, \ldots, x_n)$ and $q(x_1, x_2, \ldots, x_n)$ are two Boolean polynomials, then so are
$$p(x_1, x_2, \ldots, x_n) \cdot q(x_1, x_2, \ldots, x_n)$$
and
$$p(x_1, x_2, \ldots, x_n) + q(x_1, x_2, \ldots, x_n).$$
4. If $p(x_1, x_2, \ldots, x_n)$ is a Boolean polynomial, then so is
$$(p(x_1, x_2, \ldots, x_n))'.$$

By tradition, $(0)'$ is denoted $0'$, $(1)'$ is denoted $1'$, and $(x_k)'$ is denoted x_k'.

5. There are no Boolean polynomials in the variables x_k other than those that can be obtained by repeated use of rules 1, 2, 3, and 4.

Boolean polynomials are also called **Boolean expressions**.

Example 1 The following are Boolean polynomials in the variables x, y, and z.

$$p_1(x, y, z) = (x \sqcup y) \sqcap z$$
$$p_2(x, y, z) = (x \sqcup y') \sqcap (y \sqcup 1)$$
$$p_3(x, y, z) = (x \sqcup (y' \sqcap z)) \sqcap (x \sqcup (y \sqcup 1))$$
$$p_4(x, y, z) = (x \sqcup (y \sqcup z')) \sqcap ((x' \sqcap z)' \sqcap (y' \sqcap 0))$$

Ordinary polynomials in several variables, such as $x^2y + z^4$, $xy + yz + x^2y^2$, $x^3y^3 + xz^4$, and so on, are generally interpreted as expressions representing algebraic computations with unspeci ed numbers. As such, they are subject to the usual rules of arithmetic. Thus the polynomials $x^2 + 2x + 1$ and $(x + 1)(x + 1)$ are considered equivalent, and so are $x(xy + yz)(x + z)$ and $x^3y + 2x^2yz + xyz^2$, since in each case we can turn one into the other with algebraic manipulation.

Similarly, Boolean polynomials may be interpreted as representing Boolean computations with unspeci ed elements of B, that is, with 0 s and 1 s. As such, these polynomials are subject to the rules of Boolean arithmetic, that is, to the rules obeyed by $\sqcup$, $\sqcap$, and $'$ in Boolean algebras. As with ordinary polynomials, two Boolean polynomials are considered equivalent if we can turn one into the other with Boolean manipulations.

Ordinary polynomials can produce functions by substitution. This process works whether the polynomials involve one or several variables. Thus the polynomial $xy + yz^3$ produces a function $f : R^3 \to R$ by letting $f(x, y, z) = xy + yz^3$. For example, $f(3, 4, 2) = (3)(4) + (4)(2^3)$ or 44. In a similar way, Boolean polynomials involving n variables produce functions from B_n to B.

Example 2 Consider the Boolean polynomial

$$p(x_1, x_2, x_3) = (x_1 \sqcup x_2) \sqcap (x_1 \sqcup (x_2' \sqcap x_3)).$$

Construct the truth table for the Boolean function $f : B_3 \to B$ determined by this Boolean polynomial.

Solution

The Boolean function $f : B_3 \to B$ is described by substituting all the 2^3 ordered triples of values from B for x_1, x_2, and x_3. The truth table for the resulting function is shown in Figure 72.

Boolean polynomials can also be written in a graphical or schematic way. If x and y are variables, then the basic polynomials $x \sqcup y$, $x \sqcap y$, and x' are shown schematically in Figure 73. Each symbol has lines for the variables on the left and a line on the right representing the polynomial as a whole. The symbol for $x \sqcup y$ is called an **or gate**, that for $x \sqcap y$ is called an **and gate**, and the symbol for x' is called an **inverter**. The logical names arise because the truth tables showing the functions represented by $x \sqcup y$ and $x \sqcap y$ are exact analogs of the truth table for the connectives or and and, respectively.

Recall that functions from B_n to B can be used to describe the desired behavior of circuits with n 0-or-1 inputs and one 0-or-1 output. In the case of the

x_1	x_2	x_3	$f(x_1, x_2, x_3) = (x_1 \sqcap x_2) \sqcup (x_1 \sqcap (x'_2 \sqcup x_3))$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Figure 72

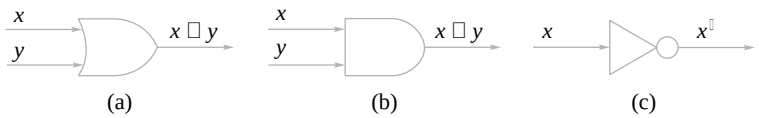


Figure 73

functions corresponding to the Boolean polynomials $x \sqcap y$, $x \sqcup y$, and x', the desired circuits can be implemented, and the schematic forms of Figure 73 are also used to represent these circuits. By repeatedly using these schematic forms for $\sqcap$, $\sqcup$, and $'$, we can make a schematic form to represent any Boolean polynomial. For the reasons given previously, such diagrams are called **logic diagrams** for the polynomial.

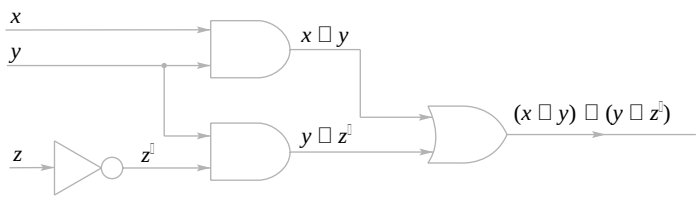
Example 3 Let

$$p(x, y, z) = (x \sqcap y) \sqcup (y \sqcap z').$$

Figure 74(a) shows the truth table for the corresponding function $f : B_3 \rightarrow B$. Figure 74(b) shows the logic diagram for p.

x	y	z	$f(x, y, z) = (x \sqcap y) \sqcup (y \sqcap z')$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

(a)



(b)

Figure 74

Suppose that p is a Boolean polynomial in n variables, and f is the corresponding function from B_n to B. We know that f may be viewed as a description of the behavior of a circuit having n inputs and one output. In the same way, the logic diagram of p can be viewed as a description of the construction of such a circuit, at least in terms of and gates, or gates, and inverters. Thus if the function f, describing the desired behavior of a circuit, can be produced by a Boolean polynomial p, then the logic diagram for p will give one way to construct a circuit having that behavior. In general, many different polynomials will produce the same function. The logic diagrams of these polynomials will represent alternative methods for constructing the desired circuit. It is almost impossible to overestimate the importance of these facts for the study of computer circuitry. Nearly all circuit design is done by software packages, but it is important to understand the basic principles of circuit design outlined in this section and the next.

5 Exercises

1. Consider the Boolean polynomial

$$p(x, y, z) = x \sqcap (y \sqcap z').$$

If $B = \{0, 1\}$, compute the truth table of the function $f : B_3 \to B$ de ned by p.

2. Consider the Boolean polynomial

$$p(x, y, z) = (x \sqcap y) \sqcup (z \sqcap x').$$

If $B = \{0, 1\}$, compute the truth table of the function $f : B_3 \to B$ de ned by p.

3. Consider the Boolean polynomial

$$p(x, y, z) = (x \sqcap y') \sqcup (y \sqcap (x' \sqcup y)).$$

If $B = \{0, 1\}$, compute the truth table of the function $f : B_3 \to B$ de ned by p.

4. Consider the Boolean polynomial

$$p(x, y, z) = (x \sqcap y) \sqcup (x' \sqcup (y \sqcap z')).$$

If $B = \{0, 1\}$, compute the truth table of the function $f : B_3 \to B$ de ned by p.

In Exercises 5 through 10, apply the rules of Boolean arithmetic to show that the given Boolean polynomials are equivalent.

5. $(x \sqcup y) \sqcap (x' \sqcup y)$; y

6. $x \sqcup (y \sqcap (y' \sqcup (y \sqcup y')))$; x

7. $(z' \sqcap x) \sqcup ((x \sqcup y) \sqcap z) \sqcup (z' \sqcap y)$; $x \sqcup y$

8. $[(x \sqcup z) \sqcap (y' \sqcup z)'] \sqcup [(y \sqcup z) \sqcap (x \sqcup z')]$; $x \sqcup y$

9. $(x' \sqcap y)' \sqcap z \sqcup (x \sqcap ((y \sqcup z) \sqcap (y' \sqcap z')))$; $(x \sqcup y') \sqcap z$

10. $(x \sqcap z) \sqcup (y' \sqcap (y' \sqcap z)) \sqcap ((x \sqcup y') \sqcap z')$; $(x \sqcap z) \sqcup y'$

In Exercises 11 through 14, rewrite the given Boolean polynomial to obtain the requested format.

11. $(x \sqcap y' \sqcap z) \sqcup (x \sqcap y \sqcap z)$; two variables and one operation

12. $(z \sqcap (y \sqcap (x \sqcap x'))) \sqcap (y \sqcap z')'$; one variable

13. $(y \sqcap z) \sqcap x' \sqcap (w \sqcap w')' \sqcap (y \sqcap z')$; two variables and two operations

14. $(x' \sqcap y' \sqcap z' \sqcap w) \sqcup (x' \sqcap z' \sqcap w' \sqcap y') \sqcup (w' \sqcap x' \sqcap y \sqcap z') \sqcup (w \sqcap x' \sqcap y \sqcap z')$; two variables and three operations

15. Construct a logic diagram implementing the function f of

(a) Exercise 1 **(b)** Exercise 2.

16. Construct a logic diagram implementing the function f of

(a) Exercise 3 **(b)** Exercise 4.

17. Construct a logic diagram implementing the function f for

(a) $f(x, y, z) = (x \sqcap (y' \sqcap z)) \sqcup (x \sqcap (y \sqcup 1))$.

(b) $f(x, y, z) = (x \sqcap (y \sqcup z')) \sqcup ((x' \sqcap z)' \sqcup (y' \sqcap 0))$.

18. Give the Boolean function described by the logic diagram in Figure 75.

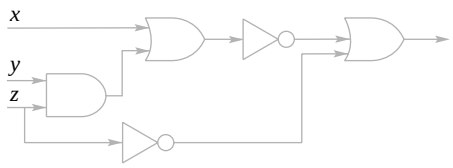


Figure 75

19. Give the Boolean function described by the logic diagram in Figure 76.

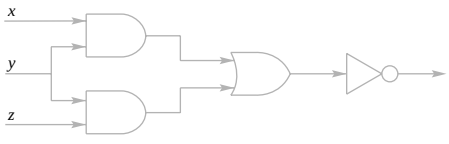


Figure 76

20. Give the Boolean function described by the logic diagram in Figure 77.

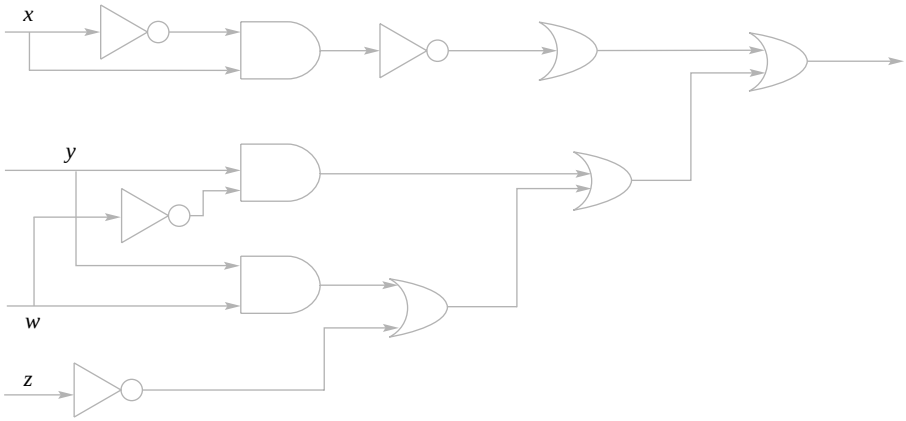


Figure 77

21. Use the properties of a Boolean algebra to re ne the function in Exercise 18 to use the minimal number of variables and operations. Draw the logic diagram for the new function.

22. Use the properties of a Boolean algebra to re ne the function in Exercise 19 to use the minimal number of variables and operations. Draw the logic diagram for the new function.

23. Use the properties of a Boolean algebra to re ne the function in Exercise 20 to use the minimal number of variables and operations. Draw the logic diagram for the new function.

6 Circuit Design

In Section 5 we considered functions from B_n to B, where B is the Boolean algebra $\{0, 1\}$. We noted that such functions can represent input-output requirements for models of many practical computer circuits. We also pointed out that if the function is given by some Boolean expression, then we can construct a logic diagram for it and thus model the implementation of the function. In this section we show that all functions from B_n to B are given by Boolean expressions, and thus logic diagrams can be constructed for any such function. Our discussion illustrates a method for nding a Boolean expression that produces a given function.

If $f : B_n \to B$, we will let $S(f) = \{b \in B_n \mid f(b) = 1\}$. We then have the following result.

THEOREM 1 Let f, f_1, and f_2 be three functions from B_n to B.

(a) If $S(f) = S(f_1) \cup S(f_2)$, then $f(b) = f_1(b) + f_2(b)$ for all b in B.

(b) If $S(f) = S(f_1) \cap S(f_2)$, then $f(b) = f_1(b) \cdot f_2(b)$ for all b in B.

($+$ and $\cdot$ are LUB and GLB, respectively, in B.)

Proof

(a) Let $b \in B_n$. If $b \in S(f)$, then, by the de nition of $S(f)$, $f(b) = 1$. Since $S(f) = S(f_1) \cup S(f_2)$, either $b \in S(f_1)$ or $b \in S(f_2)$, or both. In any case, $f_1(b) + f_2(b) = 1$. Now, if $b \notin S(f)$, then $f(b) = 0$. This means that $f_1(b) + f_2(b) = 0$. Thus, for all $b \in B_n$, $f(b) = f_1(b) + f_2(b)$.

(b) This part is proved in a manner completely analogous to that used in part (a).

Recall that a function $f : B_n \to B$ can be viewed as a function $f(x_1, x_2, \ldots, x_n)$ of n variables, each of which may assume the values 0 or 1. If $E(x_1, x_2, \ldots, x_n)$ is a Boolean expression, then the function that it produces is generated by substituting all combinations of 0 s and 1 s for the x_i s in the expression.

Example 1 Let $f_1 : B_2 \to B$ be produced by the expression $E(x, y) = x\bar{}$, and let $f_2 : B_2 \to B$ be produced by the expression $E(x, y) = y\bar{}$. Then the truth tables of f_1 and f_2 are shown in Figures 78(a) and (b), respectively. Let $f : B_2 \to B$ be the function whose truth table is shown in Figure 78(c). Clearly, $S(f) = S(f_1) \cup S(f_2)$, since f_1 is 1 at the elements $(0, 0)$ and $(0, 1)$ of B_2, f_2 is 1 at the elements $(0, 0)$ and $(1, 0)$ of B_2, and f is 1 at the elements $(0, 0)$, $(0, 1)$, and $(1, 0)$ of B_2. By Theorem 1, $f = f_1 \cup f_2$, so a Boolean expression that produces f is $x\bar{} \cup y\bar{}$. This is easily veri ed.

x	y	$f_1(x, y)$
0	0	1
0	1	1
1	0	0
1	1	0

(a)

x	y	$f_2(x, y)$
0	0	1
0	1	0
1	0	1
1	1	0

(b)

x	y	$f(x, y)$
0	0	1
0	1	1
1	0	1
1	1	0

(c)

Figure 78

It is not hard to show that any function $f : B_n \to B$ for which $S(f)$ has exactly one element is produced by a Boolean expression. Table 1 shows the correspondence between functions of two variables that are 1 at just one element and the Boolean expression that produces these functions.

TABLE 1

$S(f)$	**Expression Producing** f
$\{(0, 0)\}$	$x\bar{} \cdot y\bar{}$
$\{(0, 1)\}$	$x\bar{} \cdot y$
$\{(1, 0)\}$	$x \cdot y\bar{}$
$\{(1, 1)\}$	$x \cdot y$

Example 2 Let $f : B_2 \to B$ be the function whose truth table is shown in Figure 79(a). This function is equal to 1 only at the element $(0, 1)$ of B_2; that is, $S(f) = \{(0, 1)\}$. Thus $f(x, y) = 1$ only when $x = 0$ and $y = 1$. This is also true for the expression $E(x, y) = x\bar{} \cdot y$, so f is produced by this expression.

The function $f : B_3 \to B$ whose truth table is shown in Figure 79(b) has $S(f) = \{(0, 1, 1)\}$; that is, f equals 1 only when $x = 0$, $y = 1$, and $z = 1$. This is also true for the Boolean expression $x\bar{} \cdot y \cdot z$, which must therefore produce f.

If $b \in B_n$, then b is a sequence $(c_1, c_2, \ldots, c_n)$ of length n, where each c_k is 0 or 1. Let E_b be the Boolean expression $\bar{x}_1 \cdot \bar{x}_2 \cdot \cdots \cdot \bar{x}_n$, where $\bar{x}_k = x_k$ when $c_k = 1$ and $\bar{x}_k = x_k\bar{}$ when $c_k = 0$. Such an expression is called a **minterm**. Example 2 illustrates the fact that any function $f : B_n \to B$ for which $S(f)$ is a single element of B_n is produced by a minterm expression. In fact, if $S(f) = \{b\}$, it is easily seen that the minterm expression E_b produces f. We then have the following result.

x	y	$f(x, y)$
0	0	0
0	1	1
1	0	0
1	1	0

(a)

x	y	z	$f(x, y, z)$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

(b)

Figure 79

THEOREM 2 Any function $f : B_n \to B$ is produced by a Boolean expression.

Proof
Let $S(f) = \{b_1, b_2, \ldots, b_k\}$, and for each i, let $f_i : B_n \to B$ be the function de ned by

$$f_i(b_i) = 1$$
$$f_i(b) = 0, \qquad \text{if } b \neq b_i.$$

Then $S(f_i) = \{b_i\}$, so $S(f) = S(f_1) \cup \cdots \cup S(f_n)$ and by Theorem 1,

$$f = f_1 \vee f_2 \vee \cdots \vee f_n.$$

By the preceding discussion, each f_i is produced by the minterm E_{b_i}. Thus f is produced by the Boolean expression

$$E_{b_1} \vee E_{b_2} \vee \cdots \vee E_{b_n}$$

and this completes the proof.

Example 3 Consider the function $f : B_3 \to B$ whose truth table is shown in Figure 80. Since $S(f) = \{(0, 1, 1), (1, 1, 1)\}$, Theorem 2 shows that f is produced by the Boolean expression $E(x, y, z) = E_{(0,1,1)} \vee E_{(1,1,1)} = (x' \wedge y \wedge z) \vee (x \wedge y \wedge z)$. This expression, however, is not the simplest Boolean expression that produces f. Using properties of Boolean algebras, we have

$$(x' \wedge y \wedge z) \vee (x \wedge y \wedge z) = (x' \vee x) \wedge (y \wedge z) = 1 \wedge (y \wedge z) = y \wedge z.$$

Thus f is also produced by the simple expression $y \wedge z$.

x	y	z	$f(x, y, z)$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Figure 80

The process of writing a function as an or combination of minterms and simplifying the resulting expression can be systematized in various ways. We will demonstrate a graphical procedure utilizing what is known as a Karnaugh map. This procedure is easy for human beings to use with functions $f : B_n \to B$, if n is not too large. We will illustrate the method for $n = 2$, 3, and 4. If n is large or if a programmable algorithm is desired, other techniques may be preferable. The Exercises explore a nongraphical method.

We consider rst the case where $n = 2$ so that f is a function of two variables, say x and y. In Figure 81(a), we show a 2×2 matrix of squares with each square containing one possible input b from B_2. In Figure 81(b), we have replaced each input b with the corresponding minterm E_b. The labeling of the squares in Figure

81 is for reference only. In the future we will not exhibit these labels, but we will assume that the reader remembers their locations. In Figure 81(b), we note that x' appears everywhere in the first row and x appears everywhere in the second row. We label these rows accordingly, and we perform a similar labeling of the columns.

00	01
10	11

(a)

	y'	y
x'	$x' \square y'$	$x' \square y$
x	$x \square y'$	$x \square y$

(b)

Figure 81

Example 4 Let $f : B_2 \rightarrow B$ be the function whose truth table is shown in Figure 82(a). In Figure 82(b), we have arranged the values of f in the appropriate squares, and we have kept the row and column labels. The resulting 2×2 array of 0 s and 1 s is called the **Karnaugh map of** f. Since $S(f) = \{(0, 0), (0, 1)\}$, the corresponding expression for f is $(x' \square y') \square (x' \square y) = x' \square (y' \square y) = x'$.

x	y	$f(x, y)$
0	0	1
0	1	1
1	0	0
1	1	0

Truth table of f

(a)

	y'	y
x'	1	1
x	0	0

Karnaugh map of f

(b)

Figure 82

The outcome of Example 4 is typical. When the 1-values of a function $f : B_2 \rightarrow B$ exactly fill one row or one column, the label of that row or column gives the Boolean expression for f. Of course, we already know that if the 1-values of f fill just one square, then f is produced by the corresponding minterm. It can be shown that the larger the rectangle of 1-values of f, the smaller the expression for f will be. Finally, if the 1-values of f do not lie in a rectangle, we can decompose these values into the union of (possibly overlapping) rectangles. Then, by Theorem 1, the Boolean expression for f can be found by computing the expressions corresponding to each rectangle and combining them with $\square$.

Example 5 Consider the function $f : B_2 \rightarrow B$ whose truth table is shown in Figure 83(a). In Figure 83(b), we show the Karnaugh map of f and decompose the 1-values

x	y	$f(x, y)$
0	0	1
0	1	1
1	0	1
1	1	0

(a)

	y'	y
x'	1	1
x	1	0

(b)

	y'	y
x'	1	1
x	1	0

(c)

Figure 83

into the two indicated rectangles. The expression for the function having 1 s in the horizontal rectangle is x' (verify). The function having all its 1 s in the vertical rectangle corresponds to the expression y' (verify). Thus f corresponds to the expression $x' + y'$. In Figure 83(c), we show a different decomposition of the 1-values of f into rectangles. This decomposition is also correct, but it leads to the more complex expression $y' + (x' \cdot y)$. We see that the decomposition into rectangles is not unique and that we should try to use the largest possible rectangles.

We now turn to the case of a function $f : B_3 \to B$, which we consider to be a function of x, y, and z. We could proceed as in the case of two variables and construct a cube of side 2 to contain the values of f. This would work, but three-dimensional figures are awkward to draw and use, and the idea would not generalize. Instead, we use a rectangle of size 2×4. In Figures 84(a) and (b), respectively, we show the inputs (from B_3) and corresponding minterms for each square of such a rectangle.

	0 0	0 1	1 1	1 0
0	0 0 0	0 0 1	0 1 1	0 1 0
1	1 0 0	1 0 1	1 1 1	1 1 0

(a)

	y'		y	
x'	$x' \cdot y' \cdot z'$	$x' \cdot y' \cdot z$	$x' \cdot y \cdot z$	$x' \cdot y \cdot z'$
x	$x \cdot y' \cdot z'$	$x \cdot y' \cdot z$	$x \cdot y \cdot z$	$x \cdot y \cdot z'$
		z		
	z'			

(b)

Figure 84

Consider the rectangular areas shown in Figure 85. If the 1-values for a function $f : B_3 \to B$ exactly fill one of the rectangles shown, then the Boolean expression for this function is one of the six expressions x, y, z, x', y', or z', as indicated in Figure 85.

Consider the situation shown in Figure 85(a). Theorem 1(a) shows that f can be computed by joining all the minterms corresponding to squares of the region with the symbol $+$. Thus f is produced by

$$\begin{aligned}(x' \cdot y' \cdot z') + (x' \cdot y' \cdot z) + (x \cdot y' \cdot z') + (x \cdot y' \cdot z) &= ((x' \cdot x) + (y' \cdot z')) + ((x' \cdot x) + (y' \cdot z)) \\ &= (1 + (y' \cdot z')) + (1 + (y' \cdot z)) \\ &= (y' \cdot z') + (y' \cdot z) \\ &= y' \cdot (z' + z) = y' \cdot 1 = y'.\end{aligned}$$

A similar computation shows that the other five regions are correctly labeled.

If we think of the left and right edges of our basic rectangle as glued together to make a cylinder, as we show in Figure 86, we can say that the six large regions

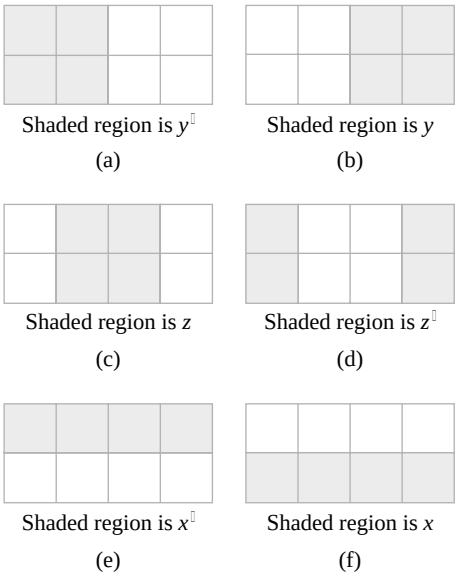


Figure 85

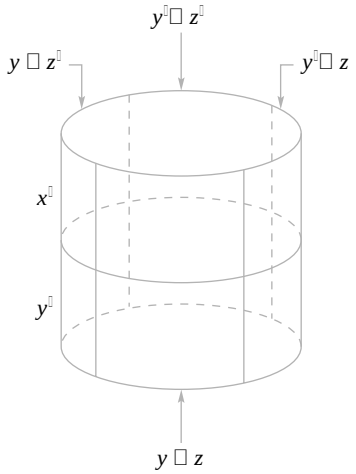


Figure 86

shown in Figure 85 consist of any two adjacent columns of the cylinder, or of the top or bottom half-cylinder.

The six basic regions shown in Figure 85 are the only ones whose corresponding Boolean expressions need be considered. That is why we used them to label Figure 84(b), and we keep them as labels for all Karnaugh maps of functions from B_3 to B. Theorem 1(b) tells us that, if the 1-values of a function $f : B_3 \to B$ form exactly the intersection of two or three of the basic six regions, then a Boolean expression for f can be computed by combining the expressions for these basic regions with $\cdot$ symbols.

Thus, if the 1-values of the function f are as shown in Figure 87(a), then we get them by intersecting the regions shown in Figures 85(a) and (d). The Boolean expression for f is therefore $y' \cdot z'$. Similar derivations can be given for the other three columns. If the 1-values of f are as shown in Figure 87(b), we get them by intersecting the regions of Figures 85(c) and (e), so a Boolean expression for f is $z \cdot x'$. In a similar fashion, we can compute the expression for any function whose 1-values ll two horizontally adjacent squares. There are eight such functions if we again consider the rectangle to be formed into a cylinder. Thus we include the case where the 1-values of f are as shown in Figure 87(c). The resulting Boolean expression is $z' \cdot x'$.

If we intersect three of the basic regions and the intersection is not empty, the intersection must be a single square, and the resulting Boolean expression is a minterm. In Figure 87(d), the 1-values of f form the intersection of the three regions shown in Figures 85(a), (c), and (f). The corresponding minterm is $y' \cdot z \cdot x$. Thus we need not remember the placement of minterms in Figure 84(b), but instead may reconstruct it.

We have seen how to compute a Boolean expression for any function $f : B_3 \to B$ whose 1-values form a rectangle of adjacent squares (in the cylinder) of size $2^n \times 2^m$, $n = 0, 1$; $m = 0, 1, 2$. In general, if the set of 1-values of f do not form such a rectangle, we may write this set as the union of such rectangles. Then a Boolean expression for f is computed by combining the expression associated with each rectangle with $\cdot$ symbols. This is true by Theorem 1(a).

The preceding discussion shows that the larger the rectangles that are chosen, the simpler will be the resulting Boolean expression.

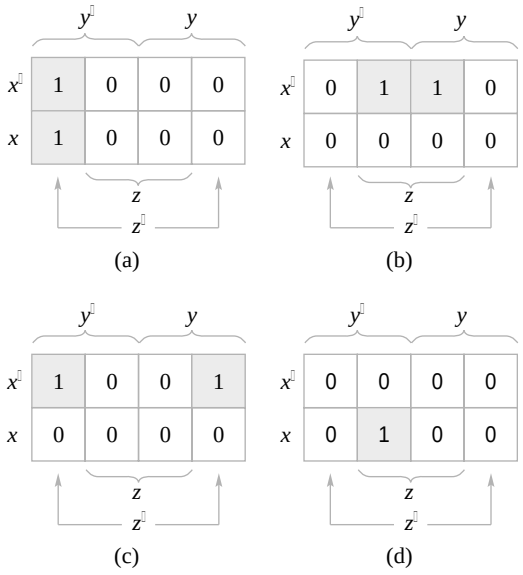


Figure 87

Example 6 Consider the function f whose truth table and corresponding Karnaugh map are shown in Figure 88. The placement of the 1 s can be derived by locating the corresponding inputs in Figure 84(a). One decomposition of the 1-values of f is shown in Figure 88(b). From this we see that a Boolean expression for f is $(y' \cdot z') + (x' \cdot y') + (y \cdot z)$.

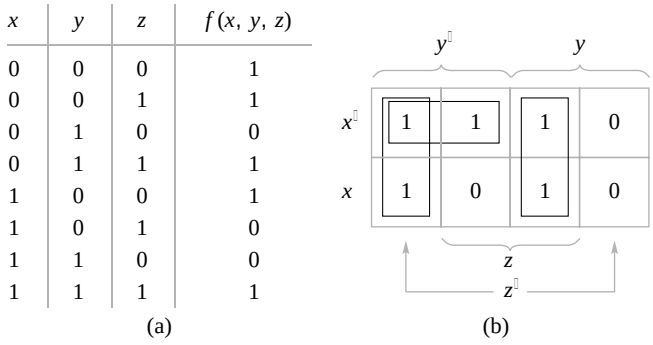
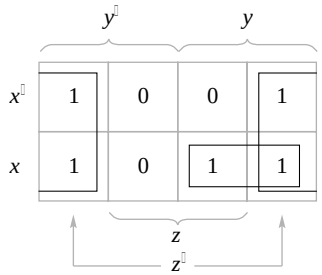


Figure 88

Example 7 Figure 89 shows the truth table and corresponding Karnaugh map for a function f. The decomposition into rectangles shown in Figure 89(b) uses the idea that the first and last columns are considered adjacent (by wrapping around the cylinder). Thus the symbols are left open ended to signify that they join in one 2×2 rectangle corresponding to z'. The resulting Boolean expression is $z' \cdot (x + y)$ (verify).

x	y	z	$f(x, y, z)$
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

(a)



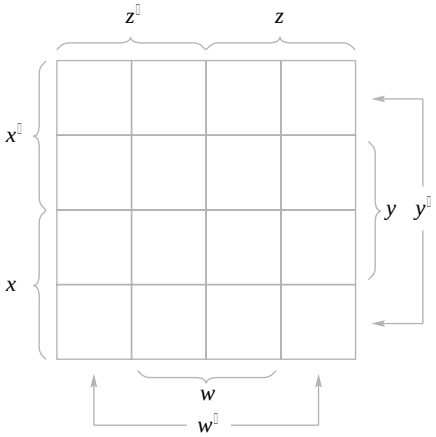
(b)

Figure 89

Finally, without additional comment, we present in Figure 90 the distribution of inputs and corresponding labeling of rectangles for the case of a function $f : B_4 \to B$, considered as a function of x, y, z, and w. Here again, we consider the first and last columns to be adjacent, and the first and last rows to be adjacent, both by wraparound, and we look for rectangles with sides of length some power of 2, so the length is 1, 2, or 4. The expression corresponding to such rectangles is given by intersecting the large labeled rectangles of Figure 91.

	00	01	11	10
00	0000	0001	0011	0010
01	0100	0101	0111	0110
11	1100	1101	1111	1110
10	1000	1001	1011	1010

(a)



(b)

Figure 90

Example 8 Figure 92 shows the Karnaugh map of a function $f : B_4 \to B$. The 1-values are placed by considering the location of inputs in Figure 90(a). Thus $f(0101) = 1$, $f(0001) = 0$, and so on.

The center 2×2 square represents the Boolean expression $w \cdot y$ (verify).

The four corners also form a 2×2 square, since the right and left edges and the top and bottom edges are considered adjacent. From a geometric point of view, we can see that if we wrap the rectangle around horizontally, getting a cylinder, then when we further wrap around vertically, we will get a torus or inner tube. On this inner tube, the four corners form a 2×2 square, which represents the Boolean expression $w' \cdot y'$ (verify).

It then follows that the decomposition leads to the Boolean expression

$$(w \cdot y) + (w' \cdot y')$$

for f.

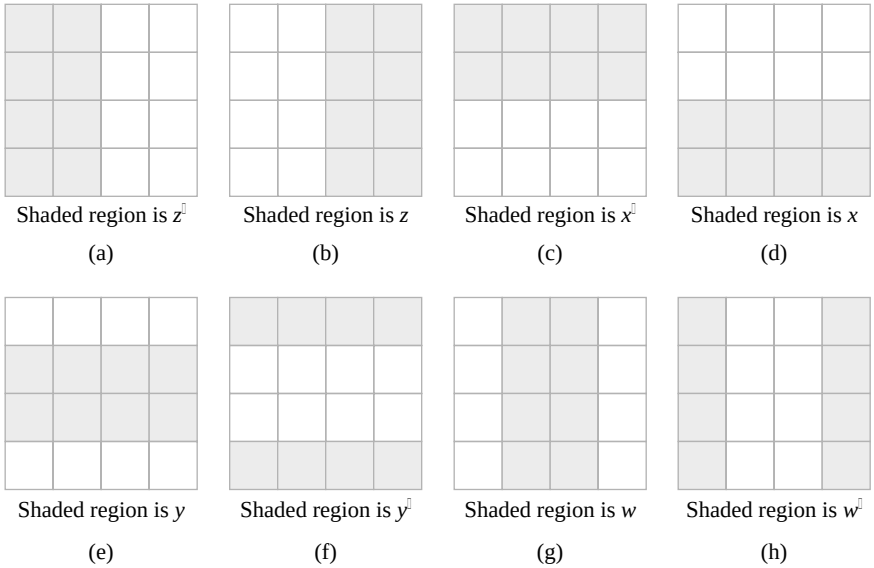


Figure 91

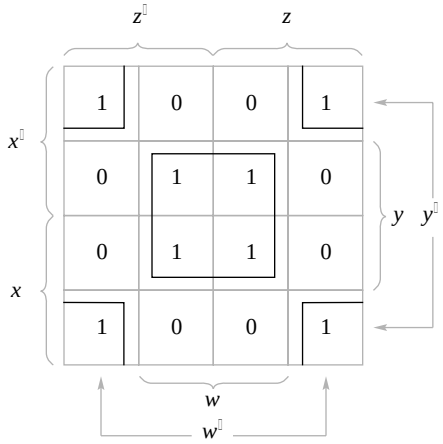


Figure 92

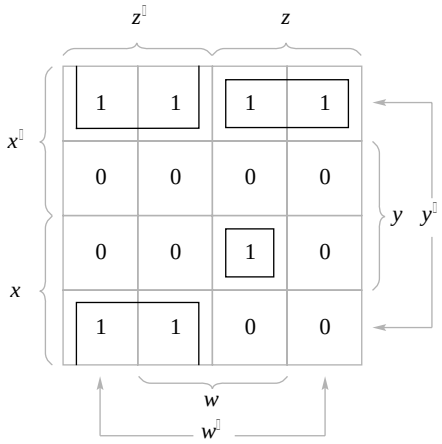


Figure 93

Example 9 In Figure 93, we show the Karnaugh map of a function $f : B_4 \to B$. The decomposition of 1-values into rectangles of sides 2^n, shown in this figure, again uses the wraparound property of top and bottom rows. The resulting expression for f is (verify)

$$(z' \cdot y') + (x' \cdot y' \cdot z) + (x \cdot y \cdot z \cdot w).$$

The first term comes from the 2×2 square formed by joining the 1×2 rectangle in the upper-left corner and the 1×2 rectangle in the lower-left corner. The second comes from the rectangle of size 1×2 in the upper-right corner, and the last is a minterm corresponding to the isolated square.

We see here that each transition in T can be enabled and fired repeatedly. A marking of a Petri net is **live** if every transition can be fired by some sequence of transition firings. Not every marking is live. In the net shown in Figure 6, once t_1 or t_2 is fired, either t_3 or t_4 is enabled (but not both). After that transition is fired, no further action is possible.

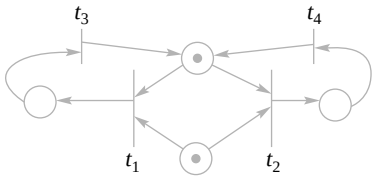


Figure 6

If no transitions can be fired, we say there is a **deadlock**.

1. Is it possible to mark Figure 5 using some tokens so that deadlock occurs? If so, draw this marked Petri net. If not, explain why.
2. Is it possible to mark Figure 6 using some tokens so a deadlock never occurs? If so, draw this marked Petri net. If not, explain why.

Part IV. Figure 7 gives a model of a simple computer network where two workers share a high-speed Internet connection and a database. Only one user of a resource is allowed at a time, but both resources are needed for some tasks the workers perform.

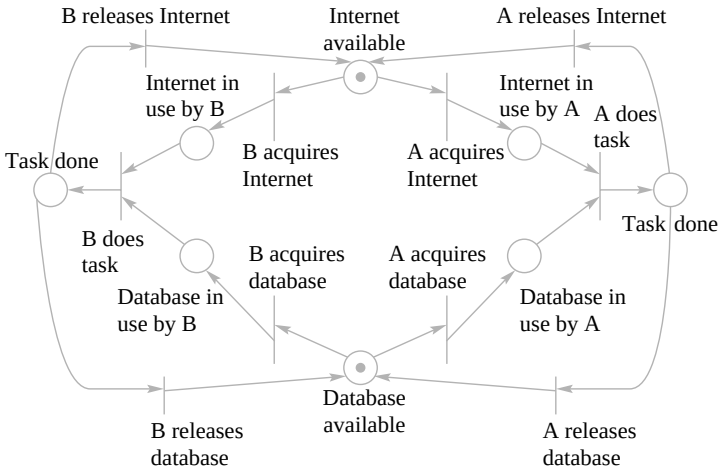


Figure 7

1. Give a sequence of transition firings that returns the net to its original marking. Explain in words what the workers have done.
2. Give a sequence of transition firings that produces deadlock in this net. Explain in words what the workers have done.
3. If they were in the same room, the workers might choose to communicate to avoid deadlock, but this is not always possible. Redesign the Petri net in Figure 7 so that deadlock cannot occur. Consider using conditions such as p: The system is available; q: A is in control of the system; and r: B is in control of the system. Clearly identify all conditions and actions.

Coding Exercises

1. Write a subroutine that determines if a relation R represented by its matrix is a partial order.
For Exercises 2 through 4, let

$$B_n = \{(x_1, x_2, x_3, \ldots, x_n) \mid x_i \in \{0, 1\}\} \quad \textit{and } x, y \in B_n.$$

2. Write a subroutine that determines if $x \leq y$.

3. **(a)** Write a function that computes $x \vee y$.
(b) Write a function that computes $x \wedge y$.
(c) Write a function that computes x'.

4. Write a subroutine that given x produces the corresponding minterm.

5. Let $B = \{0, 1\}$. Write a program that prints a truth table for the function $f : B_3 \to B$ de ned by $p(x, y, z) = (x \vee y') \wedge (y \wedge (x' \vee y))$.

Answers to Odd-Numbered Exercises

Exercise Set 1

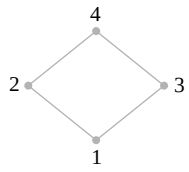
1. **(a)** No. **(b)** No.

3. **(a)** Yes. **(b)** Yes.

5. {(a, a), (b, b), (c, c), (a, b)},
{(a, a), (b, b), (c, c), (a, b), (a, c)},
{(a, a), (b, b), (c, c), (a, b), (c, b)},
{(a, a), (b, b), (c, c), (a, b), (b, c), (a, c)},
{(a, a), (b, b), (c, c), (a, b), (c, b), (c, a)},
{(a, a), (b, b), (c, c), (a, c), (c, b), (a, b)}.

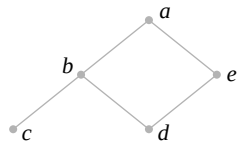
7. The structure of the proof is to check directly each of the three properties required for a partial order.

9.



11. {(1, 1), (2, 2), (3, 3), (4, 4), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)}.

13.

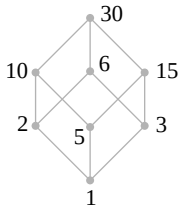


15.

17.
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

19. ACE, BASE, CAP, CAPE, MACE, MAP, MOP, MOPE.

21.



23. Linear.

25. If the main diagonal of $\mathbf{M}_R$ is all 1 s, then R is re exive. If $\mathbf{M}_R \odot \mathbf{M}_R = \mathbf{M}_R$, then R is transitive. If $a_{ij} = 1, i \neq j$, in $\mathbf{M}_R$, then a_{ji} must be 0 in order for R to be antisymmetric.

27. **(a)** {2, 3}. **(b)** {b, c, d}. **(c)** {3}. **(d)** {2, 3}. **(e)** {2, 3, 7, 8}.

29.

31. $a \leq a$ gives $a \leq' a$ for all $a \in A$. $\leq'$ is re exive.
Suppose $a \leq' b$ and $b \leq' a$. Then $a \leq b$, $b \leq a$, and $a = b$. Hence $\leq'$ is antisymmetric.

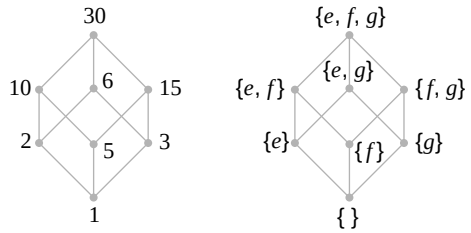
Suppose $a \preceq b$ and $b \preceq c$. Then $a \le b$, $b \le c$, and $a \le c$. Hence $a \preceq c$ and $\preceq$ is transitive.

33. Suppose $U \subset T$ and $T \subset V$. Then $U \subset V$ and R is transitive. No set is a proper subset of itself so R is irre exive. Hence R is a quasiorder.

35. Suppose $a\,R^{-1}\,b$ and $b\,R^{-1}\,c$. Then $c\,R\,b$, $b\,R\,a$, and $c\,R\,a$. Hence $a\,R^{-1}\,c$ and R^{-1} is transitive. Suppose that $x\,R^{-1}\,x$. Then $x\,R\,x$, but this is a contradiction. Hence R^{-1} is irre exive and a quasiorder.

37. $(a, b) \subset (a, b)$ since $a \mid a$ and $b \le b$. Thus $\subset$ is re exive. Suppose $(a, b) \subset (c, d)$ and $(c, d) \subset (a, b)$. Then $a \mid c$ and $c \mid a$. This means $c = ka = k(mc)$ and for a and c in B, $km = 1$ implies $k = m = 1$. Hence $a = c$. Also, $b \le d$ and $d \le b$ so $b = d$. Thus $\subset$ is antisymmetric.

39.



De ne F as follows: $F(1) = \{\ \}$; $F(2) = \{e\}$; $F(5) = \{f\}$; $F(3) = \{g\}$; $F(10) = \{e, f\}$; $F(6) = \{e, g\}$; $F(15) = \{f, g\}$; $F(30) = \{e, f, g\}$.

41. Let $U = \{a, b, c, d\}$, S a subset of U, and f_S be the characteristic function of S (relative to U). De ne

$$g(S) = \begin{bmatrix} f_S(a) & f_S(b) \\ f_S(c) & f_S(d) \end{bmatrix}.$$

Then g is a one-to-one correspondence between $P(U)$ and A. If $S \le T$ in $(U, \subset)$, then $f_S(x) \le f_T(x)$ for all x in U. Hence $g(S) \le g(T)$.

Exercise Set 2

1. Maximal: 3, 5; minimal: 1, 6.

3. Maximal: e, f; minimal: a.

5. Maximal: none; minimal: none.

7. Maximal: 1; minimal: none.

9. Greatest: f; least: a.

11. No greatest or least.

13. Greatest: none; least: none.

15. Greatest: 72; least: 2.

17. No, a may be maximal and there exists an element of A, b, such that a and b are incomparable.

19. **(a)** True. There cannot be $a_1 < a_2 < \cdots$ since A is nite.

(b) False. Not all elements have to be comparable.

(c) True. There cannot be $\cdots < a_2 < a_1$ since A is nite.

(d) False. Not all elements have to be comparable.

21. Suppose a and b are least elements of $(A, \le)$. Then $a \le b$ and $b \le a$. Since $\le$ is antisymmetric, $a = b$. Note: This is a restatement of Theorem 2.

23. **(a)** f, g, h. **(b)** a, b, c. **(c)** f. **(d)** c.

25. **(a)** d, e, f. **(b)** b, a **(c)** d. **(d)** b.

27. **(a)** None. **(b)** b. **(c)** None. **(d)** b.

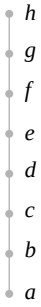
29. **(a)** $x \subset [2, \infty)$. **(b)** $x \subset (-\infty, 1]$.

(c) 2. **(d)** 1.

31. **(a)** $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. **(b)** $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

(c) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. **(d)** $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

33.



35. The least element of A is the label on the row that is all ones. The greatest element of A is the label on the column that is all ones.

37. **(a)** 49. **(b)** {2, 4, 8, 16, 32, 64}.

Exercise Set 3

1. Yes, all the properties are satis ed.

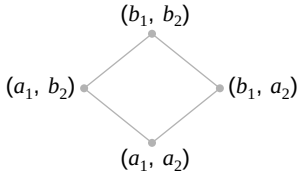
3. No, GLB({e, b}) does not exist.

5. Yes, all the properties are satis ed.

7. No.

9. Yes, $\text{LUB}(\mathbf{M}, \mathbf{N}) = [a_{ij} = \max\{m_{ij}, n_{ij}\}]$ and $\text{GLB}(\mathbf{M}, \mathbf{N}) = [b_{ij} = \min\{m_{ij}, n_{ij}\}]$

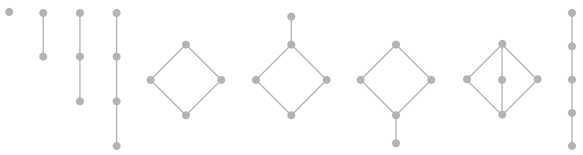
11.



13. For each $T_1, T_2 \subset T$, $T_1 \cap T_2$, and $T_1 \subset T_2$ are subsets of T so $P(T)$ is a sublattice of $P(S)$.

15. For any elements x, y of a linearly ordered poset, $x \le y$ or $y \le x$. Say $x \le y$. Then $x = x \wedge y$ and $y = x \vee y$. Hence any subset of a linearly ordered poset is a sublattice.

17.



19. Suppose $a \vee b = a$. $a \le a \vee b = (a \vee b) \vee b = (a \vee b) \vee b \le b$. Thus $a \le b$.

Suppose $a \le b$. $a \wedge b \le a$ and $a \le a$, $a \le b$ gives $a \le a \wedge b$. Hence $a \wedge b = a$.

21. **(a)** 12.

(b) Figure 44(a): 3; Figure 44(b): 3.

23. $(a_1, a_2) \vee ((b_1, b_2) \vee (c_1, c_2))$
$= (a_1, a_2) \vee (b_1 \vee c_1, b_2 \vee c_2)$
$= (a_1 \vee (b_1 \vee c_1), a_2 \vee (b_2 \vee c_2))$
$= ((a_1 \vee b_1) \vee (a_1 \vee c_1), (a_2 \vee b_2) \vee (a_2 \vee c_2))$
$= ((a_1, a_2) \vee (b_1, b_2)) \vee ((a_1, a_2) \vee (c_1, c_2))$.

A similar argument establishes the other distributive property.

25. Suppose $a \wedge x = a \wedge y$ and $a \vee x = a \vee y$. Then

$$\begin{aligned} y \le y \vee (y \wedge a) &= (y \vee y) \wedge (y \vee a) \\ &= y \wedge (y \vee a) \\ &= y \wedge (a \vee x) \\ &= (y \wedge a) \vee (y \wedge x) \\ &= (a \wedge x) \vee (y \wedge x) \\ &= x \wedge (a \vee y) \le x. \end{aligned}$$

Hence $y \le x$. A similar argument shows $x \le y$. Thus $x = y$.

27. $1' = 42$, $42' = 1$, $2' = 21$, $21' = 2$, $3' = 14$, $14' = 3$, $7' = 6$, $6' = 7$.

29. Neither.

31. Distributive, but not complemented.

33. If $x = x'$, then $x = x \wedge x = I$ and $x = x \vee x = 0$. But by Exercise 18, $0 \ne I$. Hence, $x \ne x'$.

35. Suppose $P_1 \le P_2$. Then $R_1 \subseteq R_2$. Let $x \in A_i$. Then $A_i = \{y \mid a\ R_1\ y\}$ and $A_i \subseteq \{y \mid x\ R_2\ y\} = B_j$, where $x \in B_j$. Suppose each $A_i \subseteq B_j$. Then $x\ R_1\ y$ implies $x\ R_2\ y$ and $R_1 \subseteq R_2$. Thus $P_1 \le P_2$.

37. The sublattice $\{a, b, d\}$ of Figure 57 is not complemented.

39. For any a, b, c in the sublattice with $a \le c$, $a \vee (b \wedge c) = (a \vee b) \wedge c$, because this is true in the full lattice.

Exercise Set 4

1. No, it has 6 elements, not 2^n elements.

3. No, it has 6 elements, not 2^n elements.

5. Yes, it is B_3.

7. Yes, it is B_1.

9. Yes; $385 = 5 \cdot 7 \cdot 11$.

11. No, each Boolean algebra must have 2^n elements.

13. Suppose $a = b$. $(a \wedge b') \vee (a' \wedge b) = (b \wedge b') \vee (a' \wedge a) = 0 \vee 0 = 0$.

Suppose $(a \wedge b') \vee (a' \wedge b) = 0$. Then $a \wedge b' = 0$ and $a' \wedge b = 0$. We have $I = 0' = (a \wedge b')' = a' \vee b$. So a' is the complement of b; $b' = a'$.

15. Suppose $a \le b$. Then $a \wedge c \le a \le b$ and $a \wedge c \le c$ so $a \wedge c \le b \wedge c$.

17. $(a \vee b) \wedge (a \vee b') = a \vee (b \wedge b') = a \vee I = a$.

19. $(a \vee b \vee c) \wedge (b \vee c) = (a \vee I) \vee (b \vee c) = I \vee (b \vee c) = b \vee c$.

21. Suppose $a \le b$. Then $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c) = b \wedge (a \vee c)$.

23. R is reflexive because $m_{ii} = 1$, $i = 1, 2, \ldots, 8$. R is antisymmetric since if $m_{ij} = 1$ and $i \ne j$, then $m_{ji} = 0$. R is transitive, because $\mathbf{M}_R \odot \mathbf{M}_R$ shows that $R^2 \subseteq R$.

25. Complement pairs are a, h; b, g; c, f; d, e. Since each element has a unique complement, (A, R) is complemented.

27. (A, R) is not a Boolean algebra; complements are not unique.

29. **(a)** {a}, {b}, {c}. **(b)** 2, 3, 5.

31. Matrices with exactly one 1.

Exercise Set 5

1.

x	y	z	x ∨ (y ∧ z′)
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1
			↑

3.

x	y	z	(x ∨ y′)	∧	(y ∧	(x′ ∨ y)
0	0	0	0	0	0	1
0	0	1	0	0	0	1
0	1	0	0	1	1	1
0	1	1	0	1	1	1
1	0	0	1	1	0	0
1	0	1	1	1	0	0
1	1	0	0	1	1	1
1	1	1	0	1	1	1
			(1)	(4)	(3)	(2)

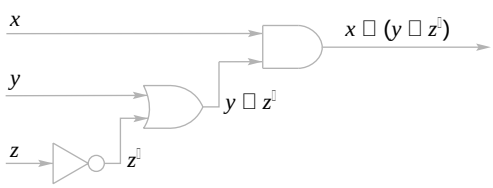
5. $(x \wedge y) \vee (x' \wedge y) = (x \vee x') \wedge y = 0 \wedge y = y$.

7. $(z' \wedge x) \vee ((x \vee y) \wedge z) \vee (z' \wedge y)) =$
$(z' \wedge (x \vee y)) \vee ((x \vee y) \wedge z) = (x \vee y) \wedge (z' \vee z) =$
$(x \vee y) \wedge 0 = x \vee y$.

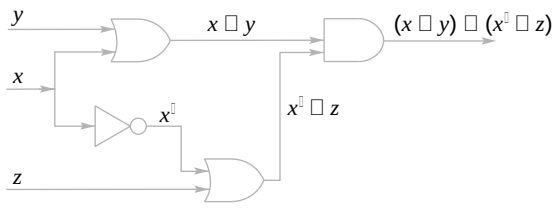
9. $(x' \vee y) \wedge z \wedge x \vee ((y \vee z) \wedge (y' \vee z')) = (x \vee y') \wedge z \vee (x \vee y \vee z) \wedge (x \vee y' \vee z') = ((x \vee y') \wedge (1 \vee z')) \wedge z \vee (1 \wedge (x \vee y)) = (x \vee y') \wedge z$.

11. $x \wedge z$. **13.** $y \wedge x'$.

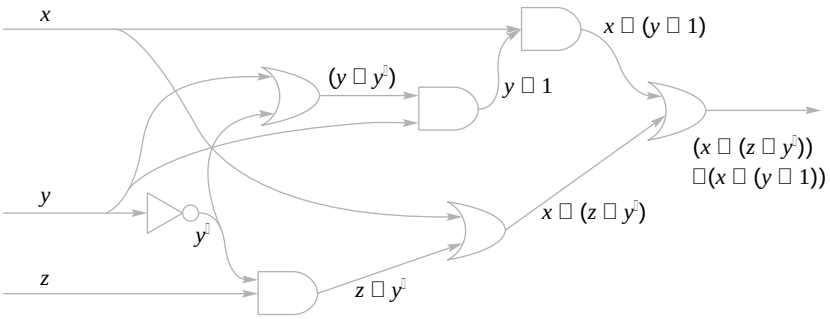
15. (a)



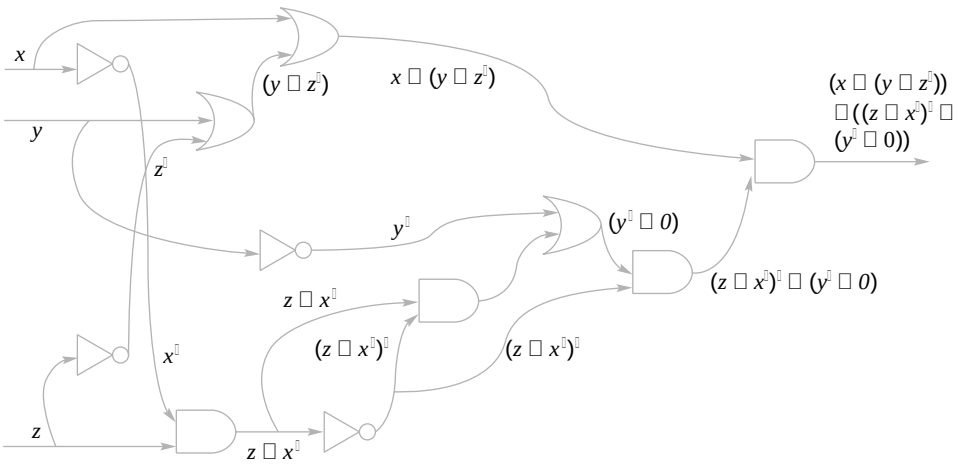
(b)



17. (a)

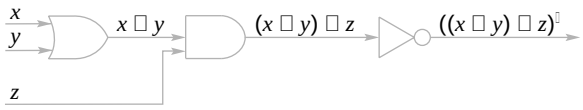


(b)

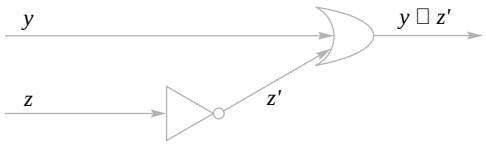


19. $((x \vee y) \wedge (y \vee z))'$.

21. $((x \vee y) \wedge z)'$.



23. $y \vee z'$.

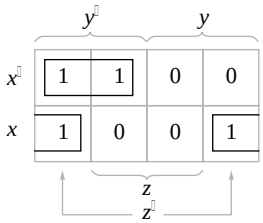


Exercise Set 6

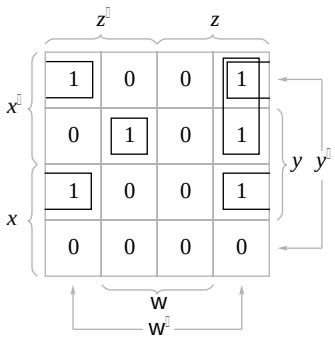
1.

	y'	y
x'	[1]	0
x	0	[1]

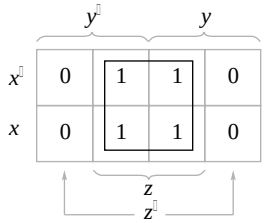
3.



5.



7.



9. $(x' \cdot y') + (x \cdot y)$.

11. $z' + (x' \cdot z)$.

13. $(z' \cdot y) + (x \cdot y') + (y' \cdot z)$.

15. $(z \cdot x') + (w' \cdot x \cdot y) + (w \cdot x \cdot y')$.

17. $(x' \cdot y') + (x \cdot y)$.

19. $(x' \cdot y') + (x \cdot z')$.

21. z.

23. $(x' \cdot y' \cdot w') + (y \cdot z \cdot w') + (x' \cdot z' \cdot y \cdot w)$.

25. **(a)** $x' \cdot y'$, $x' \cdot y$, $x \cdot y'$

(b) Since $\cdot$ is commutative and associative, we need only consider the case $(w_1 \cdot w_2 \cdot \cdots \cdot w_n \cdot y) + (w_1 \cdot w_2 \cdot \cdots \cdot w_n \cdot y')$. But this is equivalent to $w_1 \cdot w_2 \cdot \cdots \cdot w_n$.

27. $x' \cdot z'$, $y' \cdot z'$, $y \cdot z'$, $x \cdot z'$, $x \cdot y$.

29. **(a)** $(x \cdot y) + z'$.

(b) A simple check of the values of $f(x, y, z)$ will verify this.

Answers to Chapter Self-Tests

1. Partial order is a generalization of less than or equal for the real numbers.

2. In a partial order not every pair of elements must have a least upper bound and a greatest lower bound.

3. A model for any Boolean algebra is the power set of a set and the subset relation.

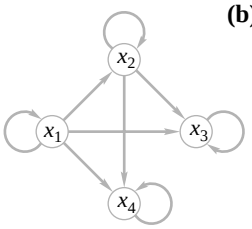
4. The Karnaugh map of a function is an $n \times n$ array of 0 s and 1 s used to create a Boolean expression that produces the function.

5. Every Boolean function can be produced by a Boolean expression; every Boolean expression produces a Boolean function.

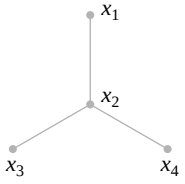
6. **(a)** R is re exive, antisymmetric, and transitive. Hence, R is a partial order on A.

(b) R is re exive and transitive, but not antisymmetric. R is not a partial order.

7. **(a)**



(b)



8. **(a)** Minimal: d, e; maximal: a.

(b) Least: none; greatest: a.

9. **(a)** Upper bounds: 12, 24, 48.

(b) Lower bounds: 2.

(c) LUB(B) = 12. **(d)** GLB(B) = 2.

10. **(a)** $a + (b \cdot c) = \begin{cases} a & \text{if } a \le b + c \text{ or } a \le b, c \quad (1) \\ b \cdot c & \text{if } b + c \le a \text{ or } b, c \le a \quad (2) \end{cases}$

Thus, $(a + b) \cdot (a + c) = \begin{cases} a + a \text{ or } a & (1) \\ b \cdot c & (2) \end{cases}$

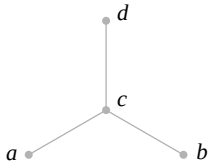
(b) $a + (b \cdot c) = \begin{cases} a & \text{if } (b + c) \le a \text{ or } b, c \le a \quad (3) \\ b \cdot c & \text{if } a \le (b + c) \text{ or } a \le b, c \quad (4) \end{cases}$

Thus, $(a + b) \cdot (a + c) = \begin{cases} a + a \text{ or } a & (3) \\ b \cdot c & (4) \end{cases}$

11. $1' = 105$; $3' = 35$; $5' = 21$; $7' = 15$; $15' = 7$; $21' = 5$; $35' = 3$; $105' = 1$.

12. **(a)** R is re exive, because the main diagonal of $\mathbf{M}_R$ is all ones. R is antisymmetric, because if $m_{ij} = 1$, then $m_{ji} = 0$. $\mathbf{M}_{R^2} = \mathbf{M}_R$, so R is transitive.

(b)

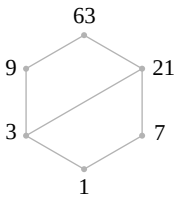


13. Since $a \le b \le b + d$ and $c \le d \le b + d$, we have $a + c \le b + d$. $(a + c$ is the LUB of a and c.) Also, $a + c \le a \le b$ and $a + c \le c \le d$ since $a + c$ is the GLB of a and c. Thus, $a + c \le b + d$.

14. **(a)** (1) is not a lattice; $b + c$ does not exist.
(2), (3), and (4) are lattices.

(b) (1), (2), and (3) are not Boolean algebras; the number of vertices is not a power of 2. (4) is B_3.

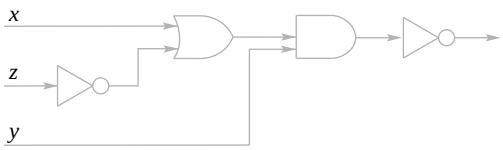
15. (a)



(b) D_{63} is not a Boolean algebra; there are 6 elements.

16. (a) $((x \sqcap y) \sqcup (y \sqcap z^{\sim}))^{\sim}$. **(b)** $(y \sqcup (x \sqcap z^{\sim}))^{\sim}$.

(c)



17. $(x \sqcap y^{\sim}) \sqcup (y \sqcap z^{\sim})$.

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Trees

From *Discrete Mathematical Structures*, Sixth Edition, Bernard Kolman, Robert C. Busby, Sharon Cutler Ross.
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Trees

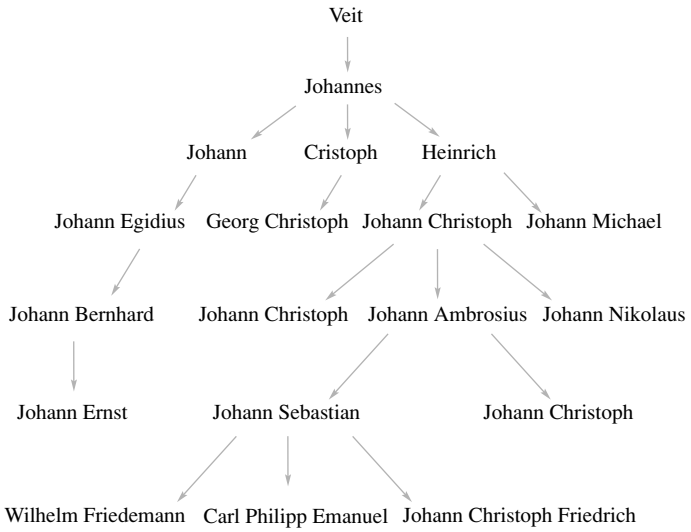
In this chapter we study a special type of relation that is exceptionally useful in a variety of biology and computer science applications and is usually represented by its digraph. These relations are essential for the construction of databases and language compilers, to name just two important computer science areas. They are called trees or sometimes rooted trees, because of the appearance of their digraphs.

Looking Back

Trees were first used in 1847 by the German mathematician Karl Georg Christian von Staudt in his work on projective geometry, and in the same year by the German physicist Gustav Robert Kirchhoff in a paper on electrical networks. However, it is Arthur Cayley who first used the word *tree* in a mathematics paper on differential transformations.

Trees are used in a wide variety of applications, such as family trees, organizational charts, sports competitions, biology, and computer file structures.

The great German composer Johann Sebastian Bach was very proud of his long family musical heritage, dating back to the late 1500s. In 1735, he prepared a genealogy of his family entitled, *Ursprung der musicalisch-Bachischen Familie* ("Origin of the Musical Bach Family"). A small part of this family is shown in the tree below. During their annual family reunions they performed together, and on some occasions there were more than 100 family members present.



Family tree of Johann Sebastian Bach
Bernard Kolman

1 Trees

Let A be a set, and let T be a relation on A. We say that T is a **tree** if there is a vertex v_0 in A with the property that there exists a unique path in T from v_0 to every other vertex in A, but no path from v_0 to v_0.

We show below that the vertex v_0, described in the definition of a tree, is unique. It is often called the **root** of the tree T, and T is then referred to as a **rooted tree**. We write (T, v_0) to denote a rooted tree T with root v_0.

If (T, v_0) is a rooted tree on the set A, an element v of A will often be referred to as a **vertex in *T***. This terminology simplifies the discussion, since it often happens that the underlying set A of T is of no importance.

To help us see the nature of trees, we will prove some simple properties satisfied by trees.

THEOREM 1 Let (T, v_0) be a rooted tree. Then

(a) There are no cycles in T.

(b) v_0 is the only root of T.

(c) Each vertex in T, other than v_0, has in-degree one, and v_0 has in-degree zero.

Proof

(a) Suppose that there is a cycle q in T, beginning and ending at vertex v. By definition of a tree, we know that $v \neq v_0$, and there must be a path p from v_0 to v. Then $q \circ p$ is a path from v_0 to v that is different from p, and this contradicts the definition of a tree.

(b) If v_0' is another root of T, there is a path p from v_0 to v_0' and a path q from v_0' to v_0 (since v_0' is a root). Then $q \circ p$ is a cycle from v_0 to v_0, and this is impossible by definition. Hence the vertex v_0 is the unique root.

(c) Let w_1 be a vertex in T other than v_0. Then there is a unique path $v_0, \ldots, v_k, w_1$ from v_0 to w_1 in T. This means that $(v_k, w_1) \in T$, so w_1 has in-degree at least one. If the in-degree of w_1 is more than one, there must be distinct vertices w_2 and w_3 such that (w_2, w_1) and (w_3, w_1) are both in T. If $w_2 \neq v_0$ and $w_3 \neq v_0$, there are paths p_2 from v_0 to w_2 and p_3 from v_0 to w_3, by definition. Then $(w_2, w_1) \circ p_2$ and $(w_3, w_1) \circ p_3$ are two different paths from v_0 to w_1, and this contradicts the definition of a tree with root v_0. Hence, the in-degree of w_1 is one. We leave it as an exercise to complete the proof if $w_2 = v_0$ or $w_3 = v_0$ and to show that v_0 has in-degree zero. ■

Theorem 1 summarizes the geometric properties of a tree. With these properties in mind, we can see how the digraph of a typical tree must look.

Let us first draw the root v_0. No edges enter v_0, but several may leave, and we draw these edges downward. The terminal vertices of the edges beginning at v_0 will be called the **level 1** vertices, while v_0 will be said to be at **level 0**. Also, v_0 is sometimes called the **parent** of these level 1 vertices, and the level 1 vertices are called the **offspring** of v_0. This is shown in Figure 1(a). Each vertex at level 1 has no other edges entering it, by part (c) of Theorem 1, but each of these vertices may have edges leaving the vertex. The edges leaving a vertex of level 1 are drawn downward and terminate at various vertices, which are said to be at **level 2**. Figure 1(b) shows the situation at this point. A parent-offspring relationship holds also for these levels (and at every consecutive pair of levels). For example, v_3 would

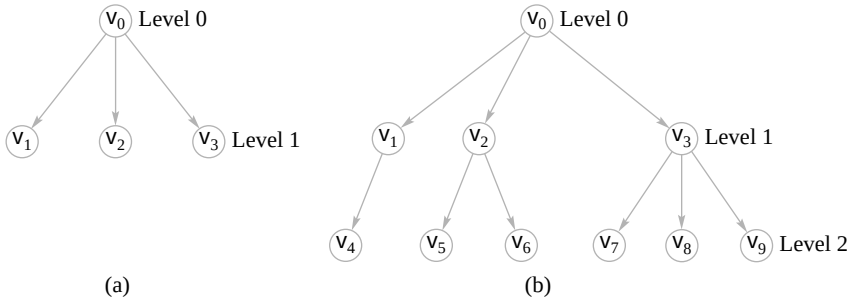


Figure 1

be called the parent of the three offspring v_7, v_8, and v_9. The offspring of any one vertex are sometimes called **siblings**.

The preceding process continues for as many levels as are required to complete the digraph. If we view the digraph upside down, we will see why these relations are called trees. The largest level number of a tree is called the **height** of the tree.

We should note that a tree may have infinitely many levels and that any level other than level 0 may contain an infinite number of vertices. In fact, any vertex could have infinitely many offspring. However, in all our future discussions, trees will be assumed to have a finite number of vertices. Thus the trees will always have a bottom (highest-numbered) level consisting of vertices with no offspring. The vertices of the tree that have no offspring are called the **leaves** of the tree.

The vertices of a tree that lie at any one level simply form a subset of A. Often, however, it is useful to suppose that the offspring of each vertex of the tree are linearly ordered. Thus, if a vertex v has four offspring, we may assume that they are ordered, so we may refer to them as the first, second, third, or fourth offspring of v. Whenever we draw the digraph of a tree, we automatically assume some ordering of the offspring of each vertex by arranging them from left to right. Such a tree will be called an **ordered tree**. Generally, ordering of offspring in a tree is not explicitly mentioned. If ordering is needed, it is usually introduced at the time when the need arises, and it often is specified by the way the digraph of the tree is drawn. The following relational properties of trees are easily verified.

THEOREM 2 Let (T, v_0) be a rooted tree on a set A. Then

(a) T is irreflexive.

(b) T is asymmetric.

(c) If $(a, b) \in T$ and $(b, c) \in T$, then $(a, c) \notin T$, for all a, b, and c in A.

Proof
The proof is left as an exercise. ■

Example 1 Let A be the set consisting of a given woman v_0 and all of her female descendants. We now define the following relation T on A: If v_1 and v_2 are elements of A, then $v_1 \, T \, v_2$ if and only if v_1 is the mother of v_2. The relation T on A is a rooted tree with root v_0. ◆

Example 2 Let $A = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8, v_9, v_{10}\}$ and let $T = \{(v_2, v_3), (v_2, v_1), (v_4, v_5), (v_4, v_6), (v_5, v_8), (v_6, v_7), (v_4, v_2), (v_7, v_9), (v_7, v_{10})\}$. Show that T is a rooted tree and identify the root.

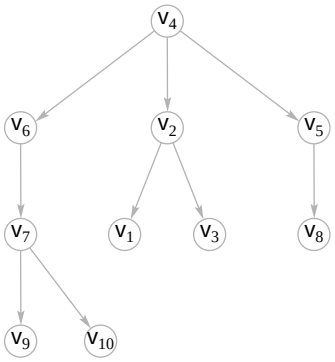


Figure 2

Solution

Since no paths begin at vertices v_1, v_3, v_8, v_9, and v_{10}, these vertices cannot be roots of a tree. There are no paths from vertices v_6, v_7, v_2, and v_5 to vertex v_4, so we must eliminate these vertices as possible roots. Thus, if T is a rooted tree, its root must be vertex v_4. It is easy to show that there is a path from v_4 to every other vertex. For example, the path v_4, v_6, v_7, v_9 leads from v_4 to v_9, since (v_4, v_6), (v_6, v_7), and (v_7, v_9) are all in T. We draw the digraph of T, beginning with vertex v_4, and with edges shown downward. The result is shown in Figure 2. A quick inspection of this digraph shows that paths from vertex v_4 to every other vertex are unique, and there are no paths from v_4 to v_4. Thus T is a tree with root v_4. ◆

If n is a positive integer, we say that a tree T is an ***n*-tree** if every vertex has at most n offspring. If all vertices of T, other than the leaves, have exactly n offspring, we say that T is a **complete *n*-tree**. In particular, a 2-tree is often called a **binary tree**, and a complete 2-tree is often called a **complete binary tree**.

Binary trees are extremely important, since there are efficient methods of implementing them and searching through them on computers. We will see some of these methods in Section 3, and we will also see that any tree can be represented as a binary tree.

Let (T, v_0) be a rooted tree on the set A, and let v be a vertex of T. Let B be the set consisting of v and all its **descendants**, that is, all vertices of T that can be reached by a path beginning at v. Observe that $B \subseteq A$. Let $T(v)$ be the restriction of the relation T to B, that is, $T \cap (B \times B)$. In other words, $T(v)$ is the relation that results from T in the following way. Delete all vertices that are not descendants of v and all edges that do not begin and end at any such vertex. Then we have the following result.

THEOREM 3 If (T, v_0) is a rooted tree and $v \in T$, then $T(v)$ is also a rooted tree with root v. We will say that $T(v)$ is the ***subtree*** of T beginning at v.

Proof

By the definition of $T(v)$, we see that there is a path from v to every other vertex in $T(v)$. If there is a vertex w in $T(v)$ such that there are two distinct paths q and q' from v to w, and if p is the path in T from v_0 to v, then $q \circ p$ and $q' \circ p$ would be two distinct paths in T from v_0 to w. This is impossible, since T is a tree with root v_0. Thus each path from v to another vertex w in $T(v)$ must be unique. Also, if q is a cycle at v in $T(v)$, then q is also a cycle in T. This contradicts Theorem 1(a); therefore, q cannot exist. It follows that $T(v)$ is a tree with root v. ■

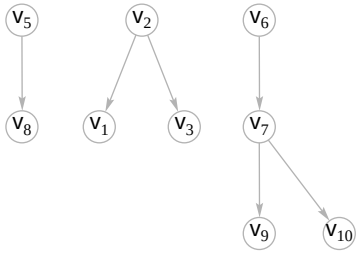


Figure 3

Subtrees and sublattices are examples of substructures of a mathematical structure. In general, if a set A and a collection of operations and their properties form a mathematical structure, then a substructure of the same type is a subset of A with the same operations that satisfies all the properties that define this type of structure.

Example 3 Consider the tree T of Example 2. This tree has root v_4 and is shown in Figure 2. In Figure 3 we have drawn the subtrees $T(v_5)$, $T(v_2)$, and $T(v_6)$ of T. ◆

1 Exercises

In Exercises 1 through 8, each relation R *is defined on the set* A. *In each case determine if* R *is a tree and, if it is, find the root.*

1. A = {a, b, c, d, e}
R = {(a, d), (b, c), (c, a), (d, e)}

2. A = {a, b, c, d, e}
R = {(a, b), (b, e), (c, d), (d, b), (c, a)}

3. A = {a, b, c, d, e, f }
R = {(a, b), (c, e), (f, a), (f, c), (f, d)}

4. A = {1, 2, 3, 4, 5, 6}
R = {(2, 1), (3, 4), (5, 2), (6, 5), (6, 3)}

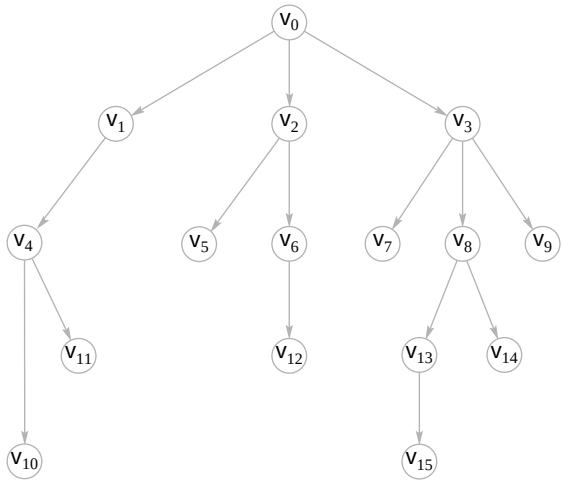
5. A = {1, 2, 3, 4, 5, 6}
R = {(1, 1), (2, 1), (2, 3), (3, 4), (4, 5), (4, 6)}

6. A = {1, 2, 3, 4, 5, 6}
R = {(1, 2), (1, 3), (4, 5), (4, 6)}

7. A = {t, u, v, w, x, y, z}
R = {(t, u), (u, w), (u, x), (u, v), (v, z), (v, y)}

8. A = {u, v, w, x, y, z}
R = {(u, x), (u, v), (w, v), (x, z), (x, y)}

In Exercises 9 through 13, consider the rooted tree (T, v_0) *shown in Figure 4.*



(T, v_0)

Figure 4

9. **(a)** List all level-3 vertices.
(b) List all leaves.

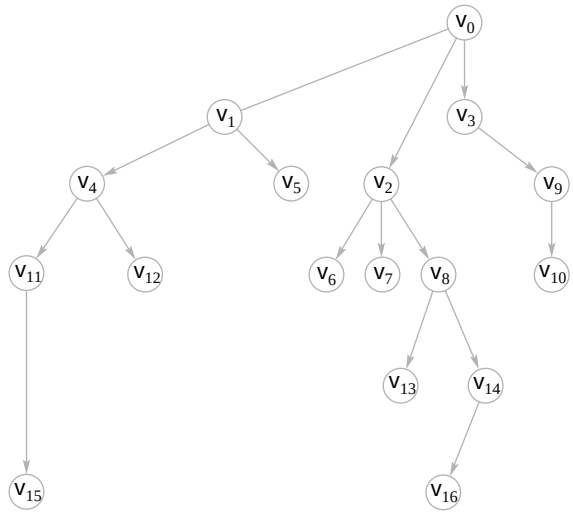
10. **(a)** What are the siblings of v_8?
(b) What are the descendants of v_8?

11. **(a)** Compute the tree $T(v_2)$.
(b) Compute the tree $T(v_3)$.

12. **(a)** What is the height of (T, v_0)?
(b) What is the height of $T(v_3)$?

13. Is (T, v_0) an n-tree? If so, for what integer n? Is (T, v_0) a complete n-tree? If so, for what integer n?

In Exercises 14 through 18, consider the rooted tree (T, v_0) *shown in Figure 5.*



(T, v_0)

Figure 5

14. **(a)** List all level-4 vertices.
(b) List all leaves.

15. **(a)** What are the siblings of v_2?
(b) What are the descendants of v_2?

16. **(a)** Compute the tree $T(v_4)$.
(b) Compute the tree $T(v_2)$.

17. **(a)** What is the height of (T, v_0)?
(b) What is the height of $T(v_4)$?

18. What is the minimal number of vertices that would need to be added to make (T, v_0) a complete 3-tree? Draw the new tree.

19. Give all the subtrees of the rooted tree (T, v_0) shown in Figure 2.

20. Show that the maximum number of vertices in a binary tree of height n is $2^{n+1} - 1$.

21. Prove that the largest possible number of leaves in an n-tree of height k is n^k.

22. If T is a complete n-tree with exactly three levels, prove that the number of vertices of T must be $1 + kn$, where $2 \le k \le n + 1$.

23. Let T be a complete n-tree with m vertices of which k are nonleaves and l are leaves. (That is, $m = k + l$.) Prove that $m = nk + 1$ and $l = (n - 1)k + 1$.

24. Prove Theorem 2(a).

25. Prove Theorem 2(b).

26. Prove Theorem 2(c).

27. Let T be a tree. Suppose that T has r vertices and s edges. Find a formula relating r to s.

28. Draw all possible unordered trees on the set $S = \{a, b, c\}$.

29. What is the maximum height for a tree on $S = \{a, b, c, d, e\}$? Explain.

30. What is the maximum height for a complete binary tree on $S = \{a, b, c, d, e\}$?

31. Show that if (T, v_0) is a rooted tree, then v_0 has in-degree zero.

32. An n-tree of height k is **balanced** if all the leaves are at level k or $k - 1$. Which of the following are balanced trees? The tree shown in

(a) Figure 1(a) **(b)** Figure 1(b)

(c) Figure 4 **(d)** Figure 5.

33. Let T be a balanced n-tree with 125 leaves.

(a) What are the possible values of n?

(b) What are the possible values for the height of T?

34. Let T be a balanced n-tree of height 4 with 125 leaves. What are the possible values for n?

2 Labeled Trees

It is sometimes useful to label the vertices or edges of a digraph to indicate that the digraph is being used for a particular purpose. This is especially true for many uses of trees in computer science and biology. We will now give a series of examples in which the sets of vertices of the trees are not important, but rather the utility of the tree is best emphasized by the labels on these vertices. Thus we will represent the vertices simply as dots and show the label of each vertex next to the dot representing that vertex.

Consider the fully parenthesized, algebraic expression

$$(3 - (2 \times x)) + ((x - 2) - (3 + x)).$$

We assume, in such an expression, that no operation such as $-$, $+$, $\times$, or $\div$ can be performed until both of its arguments have been evaluated, that is, until all computations inside both the left and right arguments have been performed. Therefore, we cannot perform the central addition until we have evaluated $(3 - (2 \times x))$ and $((x - 2) - (3 + x))$. We cannot perform the central subtraction in $((x - 2) - (3 + x))$ until we evaluate $(x - 2)$ and $(3 + x)$, and so on. It is easy to see that each such expression has a **central operator**, corresponding to the last computation that can be performed. Thus $+$ is central to the main expression, $-$ is central to $(3 - (2 \times x))$, and so on. An important graphical representation of such an expression is as a labeled binary tree. In this tree the root is labeled with the central operator of the main expression. The two offspring of the root are labeled with the central operator of the expressions for the left and right arguments, respectively. If either argument is a constant or variable, instead of an expression, this constant or variable is used to label the corresponding offspring vertex. This process continues until the expression is exhausted. Figure 6 shows the tree for the original expression of this example. To illustrate the technique further, we have shown in Figure 7 the tree corresponding to the full parenthesized expression

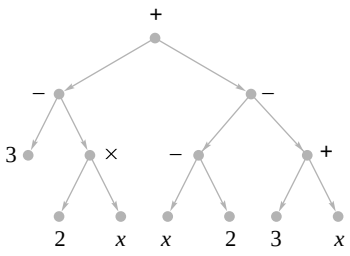


Figure 6

$$((3 \times (1 - x)) \div ((4 + (7 - (y + 2))) \times (7 + (x \div y)))).$$

Our next example of a labeled tree is important for the computer implementation of a tree data structure. We start with an n-tree (T, v_0). Each vertex in T has at most n offspring. We imagine that each vertex *potentially* has exactly n offspring, which would be ordered from 1 to n, but that some of the offspring in the sequence may be missing. The remaining offspring are labeled with the position that they occupy in the hypothetical sequence. Thus the offspring of any vertex are labeled with distinct numbers from the set $\{1, 2, \ldots, n\}$.

and then searches subtree 3. Subtree 3 has just one vertex, and so, as previously mentioned, a search of this tree yields just the symbol D. Up to this point, the search has yielded the string *ABCD*. Note that we have had to interrupt the search of each tree (except subtree 3, which is a leaf of T) in order to apply the search procedure to a subtree. Thus we cannot finish the search of T by searching subtree 7 until we apply the search procedure to subtrees 2 and 4. We could not complete the search of subtree 2 until we search subtree 3, and so on. The bookkeeping brought about by these interruptions produces the labels in the desired order, and recursion is a simple way to specify this bookkeeping.

Returning to the search, we have completed searching subtree 2, and we now must search subtree 4, since this is the right subtree of tree 1. Thus we print E and search 5 and 6 in order. These searches produce F and G. The search of subtree 1 is now complete, and we go to subtree 7. Applying the same procedure, we can see that the search of subtree 7 will ultimately produce the string *HIJKL*. The result, then, of the complete search of T is to print the string *ABCDEFGHIJKL*. ◆

Example 2 Consider the completely parenthesized expression $(a-b)\times(c+(d\div e))$. Figure 15(a) shows the digraph of the labeled, positional binary tree representation of this expression. We apply the search procedure PREORDER to this tree, as we did to the tree in Example 1. Figure 15(b) shows the various subtrees encountered in the search. Proceeding as in Example 1 and supposing again that visiting v simply prints out the label of v, we see that the string × − a b + c ÷ d e is the result of the search. This is the **prefix** or **Polish form** of the given algebraic expression. Once again, the numbering of the boxes in Figure 15(b) shows the order in which the algorithm PREORDER is applied to subtrees. ◆

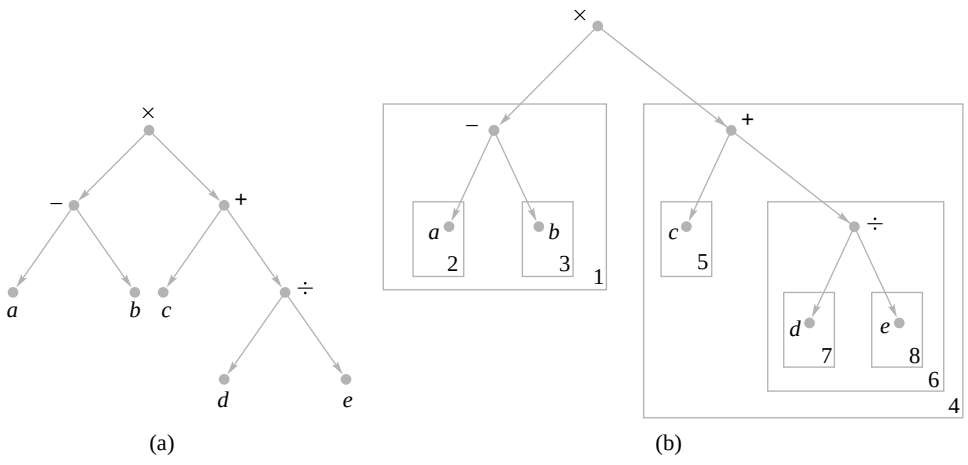


Figure 15

The Polish form of an algebraic expression is interesting because it represents the expression unambiguously, without the need for parentheses. To evaluate an expression in Polish form, proceed as follows. Move from left to right until we find a string of the form *Fxy*, where F is the symbol for a binary operation (+, −, ×, and so on) and x and y are numbers. Evaluate x F y and substitute the answer for the string *Fxy*. Continue this procedure until only one number remains.

For example, in the preceding expression, suppose that a = 6, b = 4, c = 5, d = 2, and e = 2. Then we are to evaluate × − 6 4 + 5 ÷ 2 2. This is done in the following sequence of steps.

1. × − 6 4 + 5 ÷ 2 2
2. × 2 + 5 ÷ 2 2 since the first string of the correct type is − 6 4 and 6 − 4 = 2
3. × 2 + 5 1 replacing ÷ 2 2 by 2 ÷ 2 or 1
4. × 2 6 replacing + 5 1 by 5 + 1 or 6
5. 12 replacing × 2 6 by 2 × 6

This example is one of the primary reasons for calling this type of search the preorder search, because here the operation symbol precedes the arguments.

Consider now the following informal descriptions of two other procedures for searching a positional binary tree T with root v.

Algorithm INORDER

Step 1 Search the left subtree $(T(v_L), v_L)$, if it exists.
Step 2 Visit the root, v.
Step 3 Search the right subtree $(T(v_R), v_R)$, if it exists. ●

Algorithm POSTORDER

Step 1 Search the left subtree $(T(v_L), v_L)$, if it exists.
Step 2 Search the right subtree $(T(v_R), v_R)$, if it exists.
Step 3 Visit the root, v. ●

As indicated by the naming of the algorithms, these searches are called, respectively, the **inorder** and **postorder** searches. The names indicate when the root of the (sub)tree is visited relative to when the left and right subtrees are searched. Informally, in a preorder search, the order is root, left, right; for an inorder search, it is left, root, right; and for a postorder search, it is left, right, root.

Example 3 Consider the tree of Figure 14(b) and apply the algorithm INORDER to search it. First we must search subtree 1. This requires us to first search subtree 2, and this in turn requires us to search subtree 3. As before, a search of a tree with only one vertex simply prints the label of the vertex. Thus D is the first symbol printed. The search of subtree 2 continues by printing C and then stops, since there is no right subtree at C. We then visit the root of subtree 1 and print B, and then proceed to the search of subtree 4, which yields F, E, and G, in that order. We then visit the root of T and print A and proceed to search subtree 7. The reader may complete the analysis of the search of subtree 7 to show that the subtree yields the string *IJHKL*. Thus the complete search yields the string *DCBFEGAIJHKL*.

Suppose now that we apply algorithm POSTORDER to search the same tree. Again, the search of a tree with just one vertex will yield the label of that vertex. In general, we must search both the left and the right subtrees of a tree with root v before we print out the label at v.

Referring again to Figure 14(b), we see that both subtree 1 and subtree 7 must be searched before A is printed. Subtrees 2 and 4 must be searched before B is printed, and so on.

The search of subtree 2 requires us to search subtree 3, and D is the first symbol printed. The search of subtree 2 continues by printing C. We now search subtree 4, yielding F, G, and E. We next visit the root of subtree 1 and print B. Then we proceed with the search of subtree 7 and print the symbols J, I, L, K, and H. Finally, we visit the root of T and print A. Thus we print out the string *DCFGEBJILKHA*. ◆

Example 4 Let us now apply the inorder and postorder searches to the algebraic expression tree of Example 2 [see Figure 15(a)]. The use of INORDER produces the string $a - b \times c + d \div e$. Notice that this is exactly the expression that we began with in Example 2, with all parentheses removed. Since the algebraic symbols lie between their arguments, this is often called the **infix notation**, and this explains the name INORDER. The preceding expression is ambiguous without parentheses. It could have come from the expression $a - (b \times ((c + d) \div e))$, which would have produced a different tree. Thus the tree cannot be recovered from the output of search procedure INORDER, while it can be shown that the tree is recoverable from the Polish form produced by PREORDER. For this reason, Polish notation is often better for computer applications, although infix form is more familiar to human beings.

The use of search procedure POSTORDER on this tree produces the string $ab - cde \div + \times$. This is the **postfix** or **reverse Polish** form of the expression. It is evaluated in a manner similar to that used for Polish form, except that the operator symbol is *after* its arguments rather than *before* them. If $a = 2$, $b = 1$, $c = 3$, $d = 4$, and $e = 2$, the preceding expression is evaluated in the following sequence of steps.

1. $2\ 1 - 3\ 4\ 2 \div + \times$
2. $1\ 3\ 4\ 2 \div + \times$ replacing $2\ 1 -$ by $2 - 1$ or 1
3. $1\ 3\ 2 + \times$ replacing $4\ 2 \div$ by $4 \div 2$ or 2
4. $1\ 5 \times$ replacing $3\ 2 +$ by $3 + 2$ or 5
5. 5 replacing $1\ 5 \times$ by 1×5 or 5 ◆

Reverse Polish form is also parentheses free, and from it we can recover the tree of the expression. It is used even more frequently than the Polish form and is the method of evaluating expressions in some calculators.

Searching General Trees

Until now, we have only shown how to search binary positional trees. We now show that any ordered tree T (see Section 1) may be represented as a binary positional tree that, although different from T, captures all the structure of T and can be used to re-create T. With the binary positional description of the tree, we may apply the computer implementation and search methods previously developed. Since any tree may be ordered, we can use this technique on any (finite) tree.

Let T be any ordered tree and let A be the set of vertices of T. Define a binary positional tree $B(T)$ on the set of vertices A, as follows. If $v \in A$, then the left offspring v_L of v in $B(T)$ is the first offspring of v in T (in the given order of siblings in T), if it exists. The right offspring v_R of v in $B(T)$ is the next sibling of v in T (in the given order of siblings in T), if it exists.

Example 5 Figure 16(a) shows the digraph of a labeled tree T. We assume that each set of siblings is ordered from left to right, as they are drawn. Thus the offspring of vertex 1, that is, vertices 2, 3, and 4, are ordered with vertex 2 first, 3 second, and 4 third. Similarly, the first offspring of vertex 5 is vertex 11, the second is vertex 12, and the third is vertex 13.

In Figure 16(b), we show the digraph of the corresponding binary positional tree, $B(T)$. To obtain Figure 16(b), we simply draw a left edge from each vertex v to its first offspring (if v has offspring). Then we draw a right edge from each vertex v to its next sibling (in the order given), if v has a next sibling. Thus the

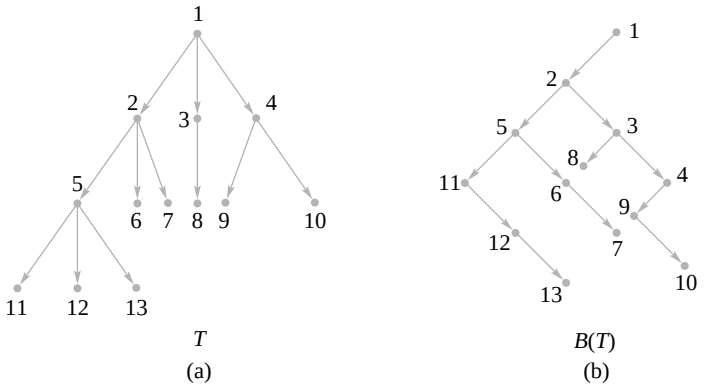
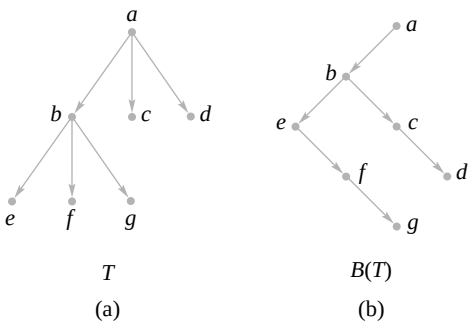


Figure 16

left edge from vertex 2, in Figure 16(b), goes to vertex 5, because vertex 5 is the first offspring of vertex 2 in the tree T. Also, the right edge from vertex 2, in Figure 16(b), goes to vertex 3, since vertex 3 is the next sibling in line (among all offspring of vertex 1). A doubly-linked-list representation of B(T) is sometimes simply referred to as a **linked-list representation of *T***. ◆

Example 6 Figure 17(a) shows the digraph of another labeled tree, with siblings ordered from left to right, as indicated. Figure 17(b) shows the digraph of the corresponding tree B(T), and Figure 17(c) gives an array representation of B(T). As mentioned, the data in Figure 17(c) would be called a linked-list representation of T. ◆



INDEX	LEFT	DATA	RIGHT
1	2		0
2	3	*a*	0
3	6	*b*	4
4	0	*c*	5
5	0	*d*	0
6	0	*e*	7
7	0	*f*	8
8	0	*g*	0

(c)

Figure 17

Pseudocode Versions

The three search algorithms in this section have straightforward pseudocode versions, which we present here. In each, we assume that the subroutine VISIT has been previously defined.

SUBROUTINE `PREORDER(T,v)`

```
1. CALL VISIT (v)
2. IF (v_L exists) THEN
     a. CALL PREORDER(T(v_L),v_L)
3. IF (v_R exists) THEN
     a. CALL PREORDER(T(v_R),v_R)
4. RETURN
END OF SUBROUTINE PREORDER
```

```
SUBROUTINE INORDER(T,v)
1. IF (v_L exists) THEN
    a. CALL INORDER(T(v_L),v_L)
2. CALL VISIT(v)
3. IF (v_R exists) THEN
    a. CALL INORDER(T(v_R),v_R)
4. RETURN
END OF SUBROUTINE INORDER
```

```
SUBROUTINE POSTORDER(T,v)
1. IF (v_L exists) THEN
    a. CALL POSTORDER(T(v_L),v_L)
2. IF (v_R exists) THEN
    a. CALL POSTORDER(T(v_R),v_R)
3. CALL VISIT(v)
4. RETURN
END OF SUBROUTINE POSTORDER
```

3 Exercises

In Exercises 1 through 5 (Figures 18 through 22), the digraphs of labeled, positional binary trees are shown. In each case we suppose that visiting a node results in printing out the label of that node. For each exercise, show the result of performing a preorder search of the tree whose digraph is shown.

1.

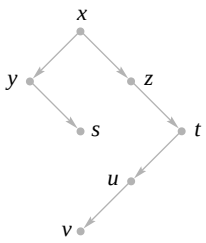


Figure 18

2.

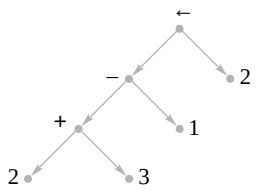


Figure 19

3.

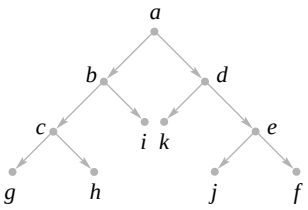


Figure 20

4.

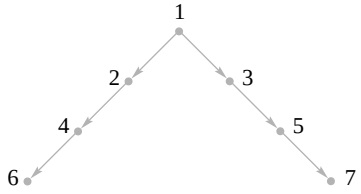


Figure 21

5.

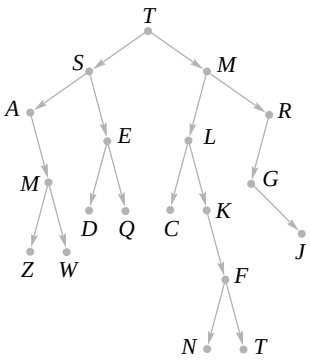


Figure 22

In Exercises 6 through 15, visiting a node means printing out the label of the node.

6. Show the result of performing an inorder search of the tree shown in Figure 18.

7. Show the result of performing an inorder search of the tree shown in Figure 19.

8. Show the result of performing an inorder search of the tree shown in Figure 20.

9. Show the result of performing an inorder search of the tree shown in Figure 21.

10. Show the result of performing a postorder search of the tree shown in Figure 22.

11. Show the result of performing a postorder search of the tree shown in Figure 18.

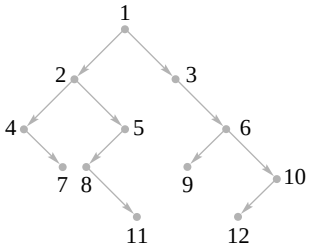
12. Show the result of performing a postorder search of the tree shown in Figure 19.

13. Show the result of performing a postorder search of the tree shown in Figure 20.

14. Show the result of performing a postorder search of the tree shown in Figure 21.

15. Show the result of performing an inorder search of the tree shown in Figure 22.

For Exercises 16 through 18, consider the tree whose digraph is shown in Figure 23 and the accompanying list of words. Suppose that visiting a node means printing out the word corresponding to the number that labels the node.



1. ONE	7. I
2. COW	8. A
3. SEE	9. I
4. NEVER	10. I
5. PURPLE	11. SAW
6. NEVER	12. HOPE

Figure 23

16. Give the sentence that results from doing a preorder search of the tree.

17. Give the sentence that results from doing an inorder search of the tree.

18. Give the sentence that results from doing a postorder search of the tree.

In Exercises 19 and 20, evaluate the expression, which is given in Polish, or prefix, notation.

19. × − + 3 4 − 7 2 ÷ 12 × 3 − 6 4

20. ÷ − × 3 x × 4 y + 15 × 2 − 6 y, where x is 2 and y is 3.

In Exercises 21 through 24, evaluate the expression, which is given in reverse Polish, or postfix, notation.

21. 4 3 2 ÷ − 5 × 4 2 × 5 × 3 ÷ ÷

22. 3 7 × 4 − 9 × 6 5 × 2 + ÷

23. x 2 − 3 + 2 3 y + − w 3 − × ÷, where x is 7, y is 2, and w is 1.

24. 7 x × y − 8 x × w + × where x is 7, y is 2, and w is 1.

25. Draw a binary tree whose preorder search produces the string *JBACDIHEGF*.

26. Draw a binary tree whose preorder search produces the string *CATSANDDOGS*.

27. Draw a binary tree whose postorder search produces the string *SEARCHING*.

28. Draw a binary tree whose postorder search produces the string *TREEHOUSE*.

29. **(a)** Every binary tree whose preorder search produces the string *JBACDIHEGF* must have 10 vertices. What else do the trees have in common?

(b) Every binary tree whose postorder search produces the string *SEARCHING* must have 9 vertices. What else do the trees have in common?

30. Show that any element of the string *ABCDEF* may be the root of a binary tree whose inorder search produces this string.

In Exercises 31 and 32 (Figures 24 and 25), draw the digraph of the binary positional tree B(T) *that corresponds to the tree shown. Label the vertices of* B(T) *to show their correspondence to the vertices of* T.

31.

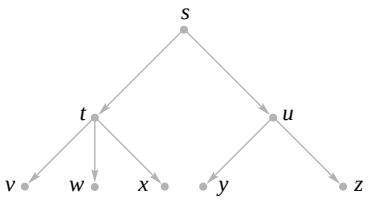


Figure 24

32.

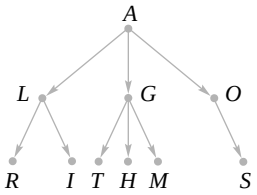


Figure 25

In Exercises 33 and 34, we give, in array form, the doubly-linked-list representation of a labeled tree T *(not binary). Draw the digraph of both the labeled binary tree* B(T) *actually stored in the arrays and the labeled tree* T *of which* B(T) *is the binary representation.*

33.

INDEX	LEFT	DATA	RIGHT
1	2	▫	0
2	3	a	0
3	4	b	5
4	6	c	7
5	8	d	0
6	0	e	10
7	0	f	0
8	0	g	11
9	0	h	0
10	0	i	9
11	0	j	12
12	0	k	0

34.

INDEX	LEFT	DATA	RIGHT
1	12	□	0
2	0	T	0
3	0	W	0
4	2	O	3
5	0	B	0
6	0	R	0
7	0	A	0
8	6	N	7
9	5	C	8
10	4	H	9
11	0	E	10
12	11	S	0

In Exercises 35 and 36 (Figures 26 and 27), consider the digraph of the labeled binary positional tree shown. If this tree is the binary form $B(T)$ *of some tree* T*, draw the digraph of the labeled tree* T*.*

35.

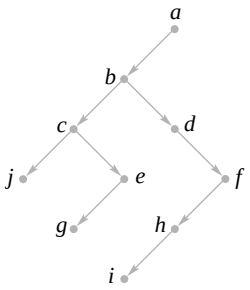


Figure 26

36.

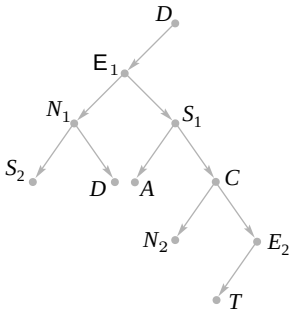


Figure 27

37. Finding information stored in a binary tree is a common task. We can improve the efficiency of such a search by having a "balanced" tree. An **AVL tree**, (T, v_0), is a binary tree where for each $v \in T$, the height of $T(v_L)$ and the height of $T(v_R)$ differ by at most one. For the given height, draw an AVL tree using the smallest possible number of vertices.

(a) $ht = 0$ **(b)** $ht = 1$

(c) $ht = 2$ **(d)** $ht = 3$

38. Is an AVL-tree a balanced 2-tree as defined in Section 1, Exercise 32?

39. Write a recurrence relation for AVL_n, the minimum number of vertices needed for an AVL tree of height n. Justify your recurrence relation by giving a procedure for forming an AVL tree of height k with a minimum number of vertices from smaller AVL trees with minimal numbers of vertices.

4 Undirected Trees

An **undirected tree** is simply the symmetric closure of a tree; that is, it is the relation that results from a tree when all the edges are made bidirectional. As is the custom with symmetric relations, we represent an undirected tree by its graph, rather than by its digraph. The graph of an undirected tree T will have a single line without arrows connecting vertices a and b whenever (a, b) and (b, a) belong to T. The set $\{a, b\}$, where (a, b) and (b, a) are in T, is called an **undirected edge** of T. In this case, the vertices a and b are called **adjacent vertices**. Thus each undirected edge $\{a, b\}$ corresponds to two ordinary edges, (a, b) and (b, a). The lines in the graph of an undirected tree T correspond to the undirected edges in T.

Example 1 Figure 28(a) shows the graph of an undirected tree T. In Figures 28(b) and (c), we show digraphs of ordinary trees T_1 and T_2, respectively, which have T as symmetric closure. This shows that an undirected tree will, in general, correspond to many directed trees. Labels are included to show the correspondence of underlying vertices in the three relations. Note that the graph of T in Figure 28(a) has six lines (undirected edges), although the relation T contains 12 pairs. ◆

We want to present some useful alternative definitions of an undirected tree, and to do so we must make a few remarks about symmetric relations.

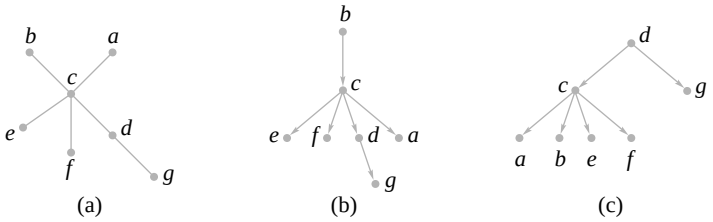


Figure 28

Let R be a symmetric relation, and let $p: v_1, v_2, \ldots, v_n$ be a path in R. We will say that p is **simple** if no two edges of p correspond to the same undirected edge. If, in addition, v_1 equals v_n (so that p is a cycle), we will call p a **simple cycle**.

Example 2

Figure 29 shows the graph of a symmetric relation R. The path a, b, c, e, d is simple, but the path f, e, d, c, d, a is not simple, since d, c and c, d correspond to the same undirected edge. Also, f, e, a, d, b, a, f and d, a, b, d are simple cycles, but f, e, d, c, e, f is not a simple cycle, since f, e and e, f correspond to the same undirected edge. ◆

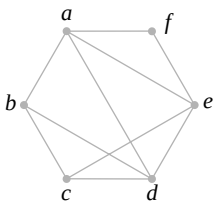


Figure 29

We will say that a symmetric relation R is **acyclic** if it contains no simple cycles. A symmetric relation R is connected if there is a path in R from any vertex to any other vertex.

The following theorem provides a useful statement equivalent to the previous definition of an undirected tree.

THEOREM 1

Let R be a symmetric relation on a set A. Then the following statements are equivalent.

(a) R is an undirected tree.

(b) R is connected and acyclic.

Proof

We will prove that part (a) implies part (b), and we will omit the proof that part (b) implies part (a). We suppose that R is an undirected tree, which means that R is the symmetric closure of some tree T on A. Note first that if $(a, b) \in R$, we must have either $(a, b) \in T$ or $(b, a) \in T$. In geometric terms, this means that every undirected edge in the graph of R appears in the digraph of T, directed one way or the other.

We will show by contradiction that R has no simple cycles. Suppose that R has a simple cycle $p: v_1, v_2, \ldots, v_n, v_1$. For each edge (v_i, v_j) in p, choose whichever pair (v_i, v_j) or (v_j, v_i) is in T. The result is a closed figure with edges in T, where each edge may be pointing in either direction. Now there are three possibilities. Either all arrows point clockwise, as in Figure 30(a), all point counterclockwise, or some pair must be as in Figure 30(b). Figure 30(b) is impossible, since in a tree T every vertex (except the root) has in-degree 1 (see Theorem 1 of Section 1). But either of the other two cases would mean that T contains a cycle, which is also impossible. Thus the existence of the cycle p in R leads to a contradiction and so is impossible.

We must also show that R is connected. Let v_0 be the root of the tree T. Then, if a and b are any vertices in A, there must be paths p from v_0 to a and q from v_0 to b, as shown in Figure 30(c). Now all paths in T are reversible in R, so the path

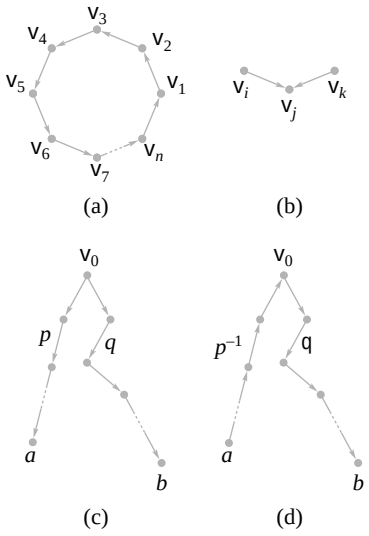


Figure 30

$q \circ p^{-1}$, shown in Figure 30(d), connects a with b in R, where p^{-1} is the reverse path of p. Since a and b are arbitrary, R is connected, and part (b) is proved. ■

There are other useful characterizations of undirected trees. We state two of these without proof in the following theorem.

THEOREM 2 Let R be a symmetric relation on a set A. Then R is an undirected tree if and only if either of the following statements is true.

(a) R is acyclic, and if any undirected edge is added to R, the new relation will not be acyclic.

(b) R is connected, and if any undirected edge is removed from R, the new relation will not be connected. ■

Note that Theorems 1 and 2 tell us that an undirected tree must have exactly the "right" number of edges; one too many and a cycle will be created; one too few and the tree will become disconnected.

The following theorem will be useful in finding certain types of trees.

THEOREM 3 A tree with n vertices has $n - 1$ edges.

Proof

Because a tree is connected, there must be at least $n - 1$ edges to connect the n vertices. Suppose that there are more than $n - 1$ edges. Then either the root has in-degree 1 or some other vertex has in-degree at least 2. But by Theorem 1, Section 1, this is impossible. Thus there are exactly $n - 1$ edges. ■

Spanning Trees of Connected Relations

If R is a symmetric, connected relation on a set A, we say that a tree T on A is a **spanning tree** for R if T is a tree with exactly the same vertices as R and which can be obtained from R by deleting some edges of R.

Example 3 The symmetric relation R whose graph is shown in Figure 31(a) has the tree T_1, whose digraph is shown in Figure 31(b), as a spanning tree. Also, the tree T_2, whose digraph is shown in Figure 31(c), is a spanning tree for R. Since R, T_1, and T_2 are all relations on the same set A, we have labeled the vertices to show the correspondence of elements. As this example illustrates, spanning trees are not unique. ◆

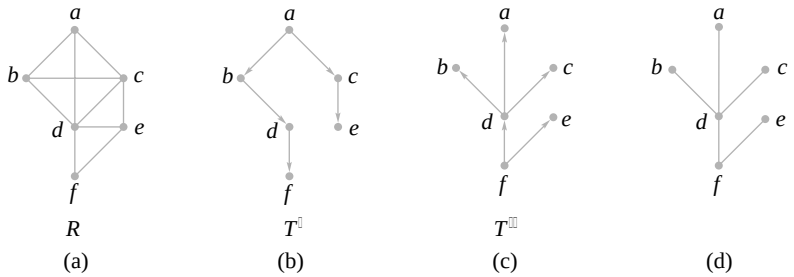


Figure 31

An **undirected spanning tree** for a symmetric connected relation R is useful in some applications. This is just the symmetric closure of a spanning tree. Figure 31(d) shows an undirected spanning tree for R that is derived from the

spanning tree of Figure 31(c). If R is a complicated relation that is symmetric and connected, it might be difficult to devise a scheme for searching R, that is, for visiting each of its vertices once in some systematic manner. If R is reduced to a spanning tree, the searching algorithms discussed in Section 3 can be used.

Theorem 2(b) suggests an algorithm for finding an undirected spanning tree for a relation R. Simply remove undirected edges from R until we reach a point where removal of one more undirected edge will result in a relation that is not connected. The result will be an undirected spanning tree.

Example 4 In Figure 32(a), we repeat the graph of Figure 31(a). We then show the result of successive removal of undirected edges, culminating in Figure 32(f), the undirected spanning tree, which agrees with Figure 31(d). ◆

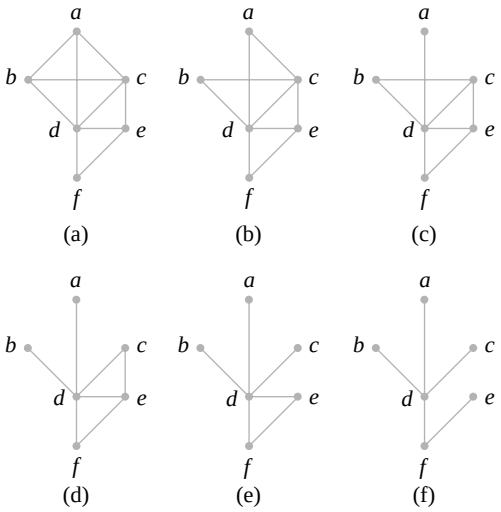


Figure 32

This algorithm is fine for small relations whose graphs are easily drawn. For large relations, perhaps stored in a computer, it is inefficient because at each stage we must check for connectedness, and this in itself requires a complicated algorithm. We now introduce a more efficient method, which also yields a spanning tree, rather than an undirected spanning tree.

Let R be a relation on a set A, and let $a, b \in A$. Let $A_0 = A - \{a, b\}$, and $A' = A_0 \cup \{a'\}$, where a' is some new element not in A. Define a relation R' on A' as follows. Suppose $u, v \in A'$, $u \neq a'$, $v \neq a'$. Let $(a', u) \in R'$ if and only if $(a, u) \in R$ or $(b, u) \in R$. Let $(u, a') \in R'$ if and only if $(u, a) \in R$ or $(u, b) \in R$. Finally, let $(u, v) \in R'$ if and only if $(u, v) \in R$. We say that R' is a result of merging the vertices a and b.

Imagine, in the digraph of R, that the vertices are pins, and the edges are elastic bands that can be shrunk to zero length. Now physically move pins a and b together, shrinking the edge between them, if there is one, to zero length. The resulting digraph is the digraph of R'. If R is symmetric, we may perform this operation on the graph of R. The result is the graph of the symmetric relation R'.

Example 5 Figure 33(a) shows the graph of a symmetric relation R. In Figure 33(b), we show the result of merging vertices v_0 and v_1 into a new vertex v_0'. In Figure 33(c), we show the result of merging vertices v_0' and v_2 of the relation whose graph is shown in Figure 33(b) into a new vertex v_0''. Notice in Figure 33(c) that the undirected edges that were previously present between v_0' and v_5 and between v_2 and v_5 have been combined into one undirected edge. ◆

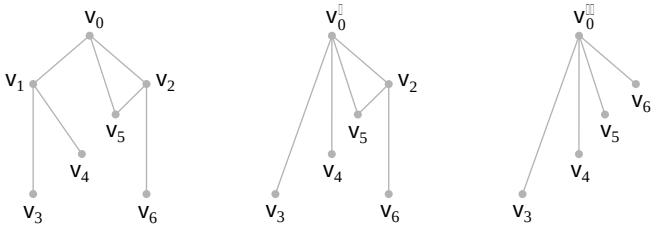


Figure 33

The algebraic form of this merging process is also very important. Let us restrict our attention to symmetric relations and their graphs.

If R is a relation on A, we will temporarily refer to elements of A as vertices of R. This will facilitate the discussion.

Suppose now that vertices a and b of a relation R are merged into a new vertex a' that replaces a and b to obtain the relation R'. To determine the matrix of R', we proceed as follows.

Step 1 Let row i represent vertex a and row j represent vertex b. Replace row i by the join of rows i and j. The join of two n-tuples of 0's and 1's has a 1 in some position exactly when either of those two n-tuples has a 1 in that position.

Step 2 Replace column i by the join of columns i and j.

Step 3 Restore the main diagonal to its original values in R.

Step 4 Delete row j and column j.

We make the following observation regarding Step 3. If $e = (a, b) \in R$ and we merge a and b, then e would become a cycle of length 1 at a'. We do not want to create this situation, since it does not correspond to "shrinking (a, b) to zero." Step 3 corrects for this occurrence.

Example 6 Figure 34 gives the matrices for the corresponding symmetric relations whose graphs are given in Figure 33. In Figure 34(b), we have merged vertices v_0 and v_1 into v_0'. Note that this is done by taking the join of the first two rows and entering the result in row 1, doing the same for the columns, then restoring the diagonal, and removing row 2 and column 2. If vertices v_0' and v_2 in the graph whose matrix is given by Figure 34(b) are merged, the resulting graph has the matrix given by Figure 34(c). ◆

	v_0	v_1	v_2	v_3	v_4	v_5	v_6
v_0	0	1	1	0	0	1	0
v_1	1	0	0	1	1	0	0
v_2	1	0	0	0	0	1	1
v_3	0	1	0	0	0	0	0
v_4	0	1	0	0	0	0	0
v_5	1	0	1	0	0	0	0
v_6	0	0	1	0	0	0	0

(a)

	v_0'	v_2	v_3	v_4	v_5	v_6
v_0'	0	1	1	1	1	0
v_2	1	0	0	0	1	1
v_3	1	0	0	0	0	0
v_4	1	0	0	0	0	0
v_5	1	1	0	0	0	0
v_6	0	1	0	0	0	0

(b)

	v_0''	v_3	v_4	v_5	v_6
v_0''	0	1	1	1	1
v_3	1	0	0	0	0
v_4	1	0	0	0	0
v_5	1	0	0	0	0
v_6	1	0	0	0	0

(c)

Figure 34

We can now give an algorithm for finding a spanning tree for a symmetric, connected relation R on the set $A = \{v_1, v_2, \ldots, v_n\}$. The method is equivalent to a special case of an algorithm called **Prim's algorithm**. The steps are as follows:

Step 1 Choose a vertex v_1 of R, and arrange the matrix of R so that the first row corresponds to v_1.

Step 2 Choose a vertex v_2 of R such that $(v_1, v_2) \in R$, merge v_1 and v_2 into a new vertex v_1', representing $\{v_1, v_2\}$, and replace v_1 by v_1'. Compute the matrix of the resulting relation R'. Call the vertex v_1' a merged vertex.

Step 3 Repeat Steps 1 and 2 on R' and on all subsequent relations until a relation with a single vertex is obtained. At each stage, keep a record of the set of original vertices that is represented by each merged vertex.

Step 4 Construct the spanning tree as follows. At each stage, when merging vertices a and b, select an edge in R from one of the original vertices represented by a to one of the original vertices represented by b.

Example 7 We apply Prim's algorithm to the symmetric relation whose graph is shown in Figure 35. In Table 1, we show the matrices that are obtained when the original set of vertices is reduced by merging until a single vertex is obtained, and at each stage we keep track of the set of original vertices represented by each merged vertex, as well as of the new vertex that is about to be merged.

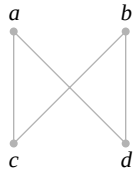


Figure 35

TABLE 1

Matrix	Original Vertices Represented by Merged Vertices	New Vertex to Be Merged (with First Row)
$\begin{array}{c\|cccc} & a & b & c & d \\ a & 0 & 0 & 1 & 1 \\ b & 0 & 0 & 1 & 1 \\ c & 1 & 1 & 0 & 0 \\ d & 1 & 1 & 0 & 0 \end{array}$	—	c
$\begin{array}{c\|ccc} & a' & b & d \\ a' & 0 & 1 & 1 \\ b & 1 & 0 & 1 \\ d & 1 & 1 & 0 \end{array}$	$a' \leftrightarrow \{a, c\}$	b
$\begin{array}{c\|cc} & a'' & d \\ a'' & 0 & 1 \\ d & 1 & 0 \end{array}$	$a'' \leftrightarrow \{a, c, b\}$	d
$\begin{array}{c\|c} & a''' \\ a''' & 0 \end{array}$	$a''' \leftrightarrow \{a, c, d, b\}$	—

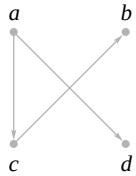


Figure 36

The first vertex chosen is a, and we choose c as the vertex to be merged with a, since there is a 1 at vertex c in row 1. We also select the edge (a, c) from the original graph. At the second stage, there is a 1 at vertex b in row 1, so we merge b with vertex a'. We select an edge in the original relation R from a vertex of $\{a, c\}$ to b, say (c, b). At the third stage, we have to merge d with vertex a''. Again, we need an edge in R from a vertex of $\{a, c, b\}$ to d, say (a, d). The selected edges (a, c), (c, b), and (a, d) form the spanning tree for R, which is shown in Figure 36. Note that the first vertex selected becomes the root of the spanning tree that is constructed. ◆

4 Exercises

In Exercises 1 through 6, construct an undirected spanning tree for the connected graph G *by removing edges in succession. Show the graph of the resulting undirected tree.*

1. Let G be the graph shown in Figure 37.

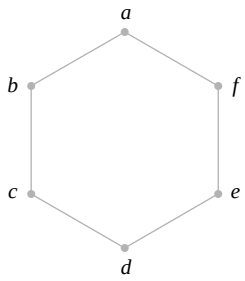


Figure 37

2. Let G be the graph shown in Figure 38.

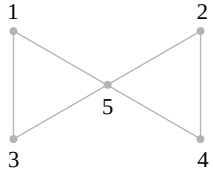


Figure 38

3. Let G be the graph shown in Figure 39.

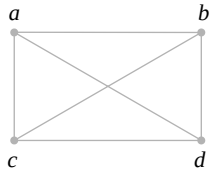


Figure 39

4. Let G be the graph shown in Figure 40.

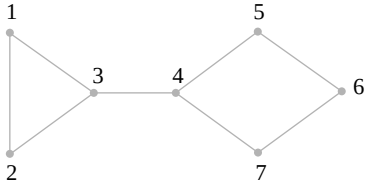


Figure 40

5. Let G be the graph shown in Figure 41.

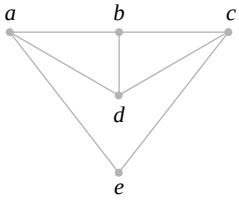


Figure 41

6. Let G be the graph shown in Figure 42.

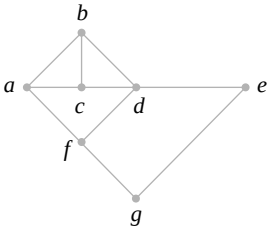


Figure 42

In Exercises 7 through 12 (Figures 37 through 42), use Prim's algorithm to construct a spanning tree for the connected graph shown. Use the indicated vertex as the root of the tree and draw the digraph of the spanning tree produced.

7. Figure 37; use e as the root.

8. Figure 38; use 5 as the root.

9. Figure 39; use c as the root.

10. Figure 40; use 4 as the root.

11. Figure 41; use e as the root.

12. Figure 42; use d as the root.

13. Consider the connected graph shown in Figure 43. Show the graphs of three different undirected spanning trees.

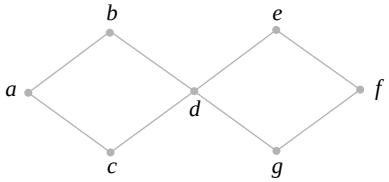


Figure 43

14. For the connected graph shown in Figure 44, show the graphs of all undirected spanning trees.

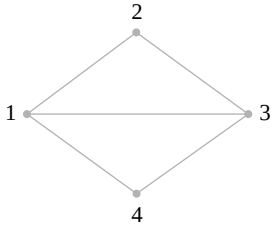


Figure 44

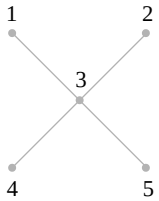


Figure 45

15. For the undirected tree shown in Figure 45, show the digraphs of all spanning trees. How many are there?

Two trees, T_1 *and* T_2*, are* ***isomorphic*** *if they are isomorphic posets; that is, there is a one-to-one correspondence,* f*, between the vertices of* T_1 *and* T_2 *and* (a, b) *is an edge in* T_1 *if and only if* (f(a), f(b)) *is an edge in* T_2.

16. For the graph shown in Figure 44, show the graphs of all nonisomorphic undirected spanning trees.

17. For the graph shown in Figure 45, show the graphs of all nonisomorphic spanning trees.

18. For the graph in Figure 46, give all spanning trees.

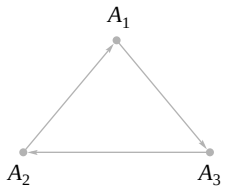


Figure 46

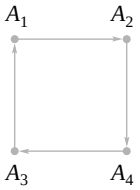


Figure 47

19. For the graph in Figure 47, give all spanning trees.

20. For the graph in Figure 47, show the graphs of all nonisomorphic spanning trees.

21. For the graph in Figure 48, how many different spanning trees are there?

22. For the graph shown in Figure 48, show the graphs of all nonisomorphic spanning trees.

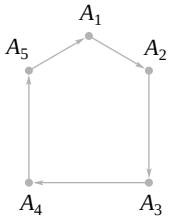


Figure 48

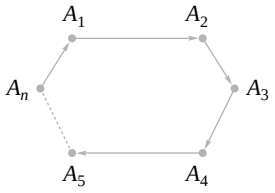


Figure 49

23. For the graph in Figure 49, how many different spanning trees are there?

24. State your conclusion for Figure 49 as a theorem and prove it.

25. For the graph shown in Figure 49, how many nonisomorphic spanning trees are there? Justify your answer.

26. Prove that a symmetric connected relation has an undirected spanning tree.

5 Minimal Spanning Trees

In many applications of symmetric connected relations, the (undirected) graph of the relation models a situation in which the edges as well as the vertices carry information. A **weighted graph** is a graph for which each edge is labeled with a numerical value called its **weight**.

Example 1 The small town of Social Circle maintains a system of walking trails between the recreational areas in town. The system is modeled by the weighted graph in Figure 50, where the weights represent the distances in kilometers between sites. ◆

Example 2 A communications company is investigating the costs of upgrading links between the relay stations it owns. The weighted graph in Figure 51 shows the stations and the cost in millions of dollars for upgrading each link. ◆

The weight of an edge (v_i, v_j) is sometimes referred to as the **distance between vertices** v_i and v_j. A vertex u is a **nearest neighbor of vertex** v if u and v are

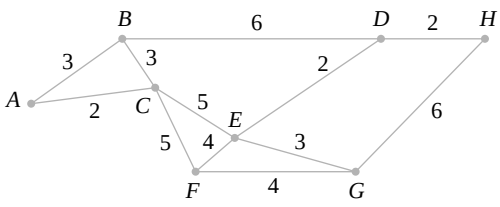


Figure 50

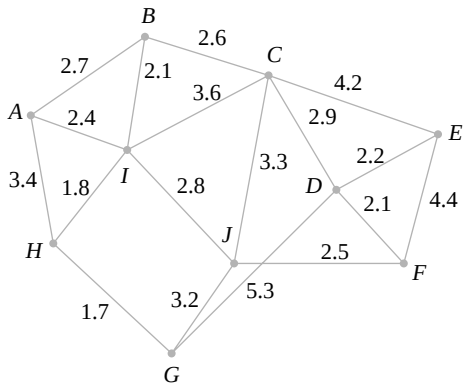


Figure 51

adjacent and no other vertex is joined to v by an edge of lesser weight than (u, v). Notice that, ungrammatically, v may have more than one nearest neighbor.

Example 3 In the graph shown in Figure 50, vertex C is a nearest neighbor of vertex A. Vertices E and G are both nearest neighbors of vertex F. ◆

A vertex v is a **nearest neighbor of a set of vertices** $V = \{v_1, v_2, \ldots, v_k\}$ in a graph if v is adjacent to some member v_i of V and no other vertex adjacent to a member of V is joined by an edge of lesser weight than (v, v_i). This vertex v may belong to V.

Example 4 Referring to the graph given in Figure 51, let $V = \{C, E, J\}$. Then vertex D is a nearest neighbor of V, because (D, E) has weight 2.2 and no other vertex adjacent to C, E, or J is linked by an edge of lesser weight to one of these vertices. ◆

With applications of weighted graphs, it is often necessary to find an undirected spanning tree for which the total weight of the edges in the tree is as small as possible. Such a spanning tree is called a **minimal spanning tree**. Prim's algorithm (Section 4) can easily be adapted to produce a minimal spanning tree for a weighted graph. We restate Prim's algorithm as it would be applied to a symmetric, connected relation given by its undirected weighted graph.

Algorithm PRIM'S ALGORITHM

Let R be a symmetric, connected relation with n vertices.

Step 1 Choose a vertex v_1 of R. Let $V = \{v_1\}$ and $E = \{\ \}$.

Step 2 Choose a nearest neighbor v_i of V that is adjacent to v_j, $v_j \in V$, and for which the edge (v_i, v_j) does not form a cycle with members of E. Add v_i to V and add (v_i, v_j) to E.

Step 3 Repeat Step 2 until $|E| = n - 1$. Then V contains all n vertices of R, and E contains the edges of a minimal spanning tree for R. ●

In this version of Prim's algorithm, we begin at any vertex of R and construct a minimal spanning tree by adding an edge to a nearest neighbor of the set of vertices already linked, as long as adding this edge does not complete a cycle. This is an example of a **greedy algorithm**. At each stage we chose what is "best" based on local conditions, rather than looking at the global situation. Greedy algorithms do not always produce optimal solutions, but we can show that for Prim's algorithm the solution is optimal.

THEOREM 1 Prim's algorithm, as given, produces a minimal spanning tree for the relation.

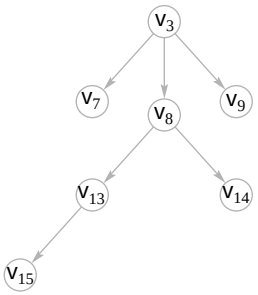
Proof

Let R have n vertices. Let T be the spanning tree for R produced by Prim's algorithm. Suppose that the edges of T, in the order in which they were selected, are $t_1, t_2, \ldots, t_{n-1}$. For each i from 1 to $n - 1$, we define T_i to be the tree with edges $t_1, t_2, \ldots, t_i$ and $T_0 = \{\ \}$. Then $T_0 \subseteq T_1 \subseteq \cdots \subseteq T_{n-1} = T$. We now prove, by mathematical induction, that each T_i is contained in a minimal spanning tree for R.

Basis Step

Clearly $P(0)$: $T_0 = \{\ \}$ is contained in every minimal spanning tree for R.

(b)



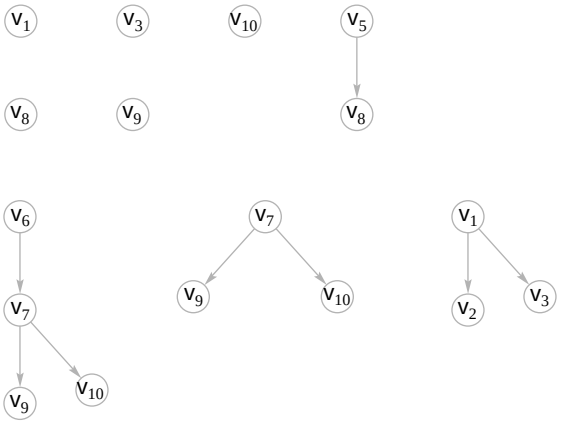
13. (T, v_0) may be an n-tree for $n \geq 3$. It is not a complete 3-tree.

15. **(a)** v_1, v_3.

(b) v_6, v_7, v_8, v_{13}, v_{14}, v_{16}, v_{10}.

17. **(a)** 4. **(b)** 2.

19.



21. Basis step: $n = 1$. An n-tree of height 1 can have at most n leaves by definition.
Induction step: Use $P(i)$: An n-tree of height i has at most n^i leaves to show $P(i + 1)$: An n-tree of height $i + 1$ has at most n^{i+1} leaves. The leaves of a tree T of height $i + 1$ belong to the subtrees of T whose roots are at level 1. Each of these subtrees has height at most i, and there are at most n of them. Hence the maximum number of leaves of T is $n \cdot n^i$ or n^{i+1}.

23. The total number of vertices is $1 + kn$, where k is the number of nonleaves and 1 counts the root, because every vertex except the root is an offspring. Since $l = m - k$, $l = 1 + kn - k$ or $1 + k(n - 1)$.

25. If both $v \; T \; u$ and $u \; T \; v$, then v, u, v is a cycle in T. Thus $v \; T \; u$ implies $u \not T \; v$. T is asymmetric.

27. Each vertex except the root has in-degree 1. Thus $s = r - 1$.

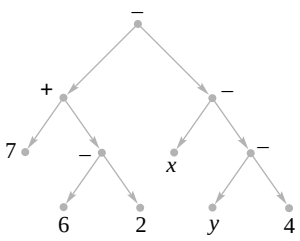
29. 4. The tree of maximum height has one vertex on each level.

31. Assume that the in-degree of $v_0 \neq 0$. Then there is a cycle that begins and ends at v_0. This is impossible. Hence the in-degree of v_0 must be 0.

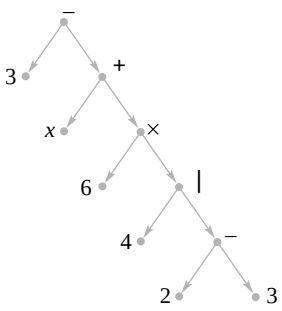
33. **(a)** $2 \leq n$. **(b)** $1 \leq k \leq 7$.

Exercise Set 2

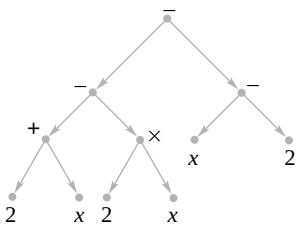
1.



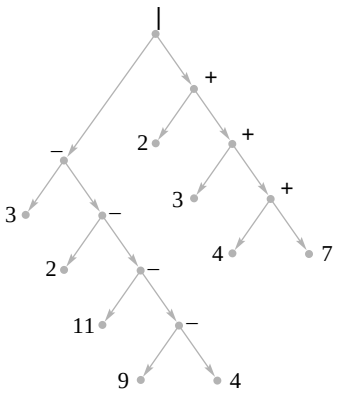
3.



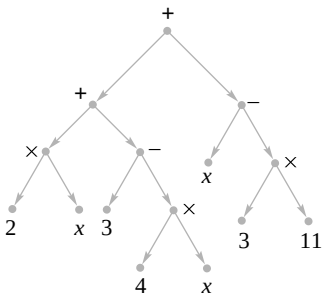
5.



7.

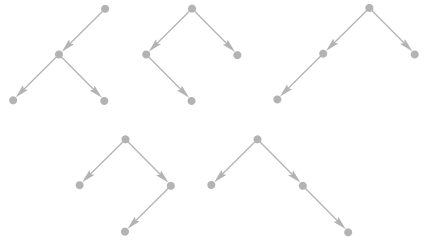


9.



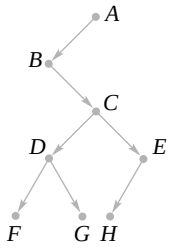
11. 6.

13.

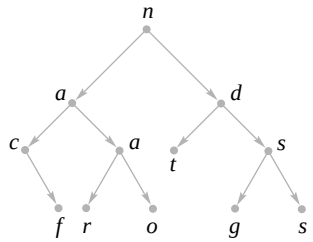


15. 721.

17.



19.



21.

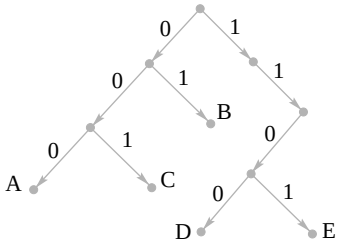
LEFT	DATA	RIGHT
2	⊠	0
3	−	4
5	+	6
9	−	10
0	7	0
7	−	8
0	6	0
0	2	0
0	*x*	0
11	−	12
0	*y*	0
0	4	0

23.

LEFT	DATA	RIGHT
2	⊠	0
3	−	4
5	−	6
9	−	10
7	+	8
11	×	12
0	2	0
0	*x*	0
0	*x*	0
0	2	0
0	2	0
0	*x*	0

25. **(a)** CAR. **(b)** SEAR. **(c)** RACE. **(d)** SCAR.

27.



Exercise Set 3

1. *x y s z t u v.*

3. *a b c g h i d k e j f.*

5. TSAMZWEDQMLCKFNTRGJ.

7. 2 + 3 − 1 × 2.

9. 6 4 2 1 3 5 7.

11. *s y v u t z x.*

13. *g h c i b k j f e d a.*

15. ZWMADQESCNTFKLJGRMT.

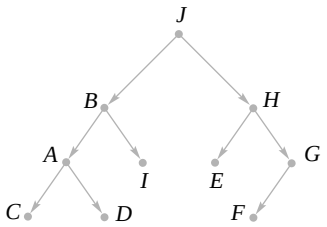
17. NEVER I COW A SAW PURPLE ONE SEE I NEVER HOPE I.

19. 4.

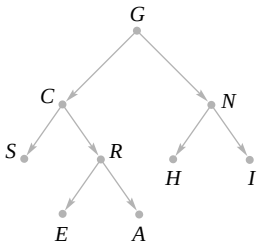
21. $\frac{15}{16}$.

23. $\frac{8}{6}$.

25.



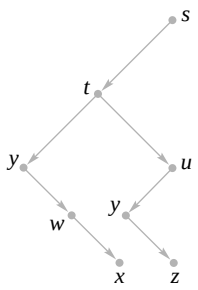
27.



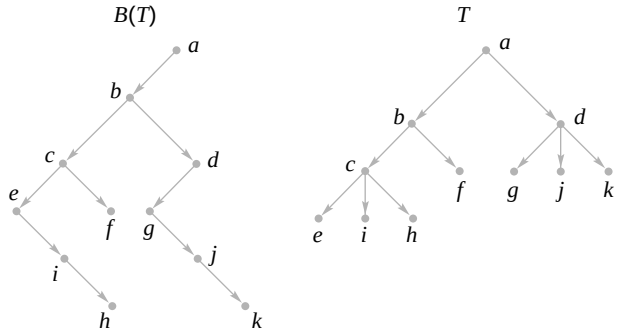
29. **(a)** The root must be labeled J; if J has a left offspring, it must be labeled B, otherwise the right offspring is labeled B.

(b) The root must be labeled G; if G has a left offspring, it must be labeled N, otherwise the right offspring is labeled N.

31.



33.



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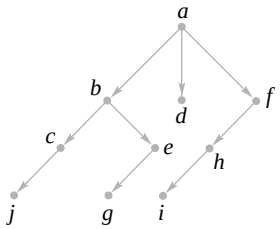
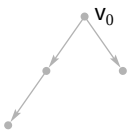


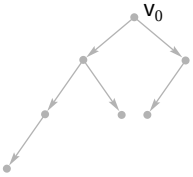
37. **(a)**

(b)

(c)



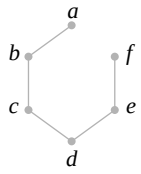
(d)



39. $AVL_n = AVL_{n-1} + AVL_{n-2} + 1$. Let v be a new root. Let $T(v_L)$ be an AVL tree of height $n - 1$ using a minimal number of vertices. Let $T(v_R)$ be an AVL tree of height $n - 2$ using a minimal number of vertices. Then $T(v)$, where the left offspring of v is v_L and its right offspring is v_R, is an AVL tree of height n using a minimal number of vertices.

Exercise Set 4

1.



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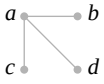

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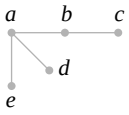

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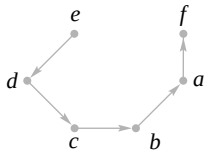

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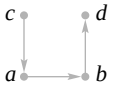

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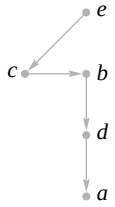

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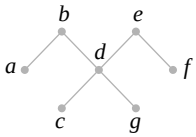
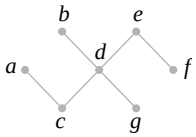
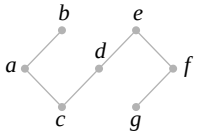
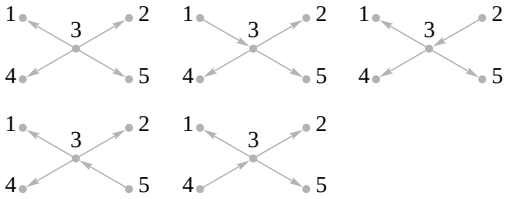


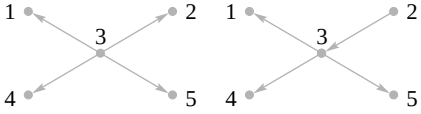




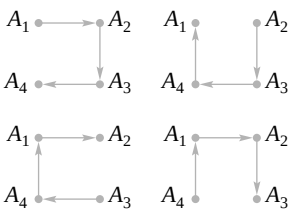
15. There are 5 spanning trees.



17.



19.



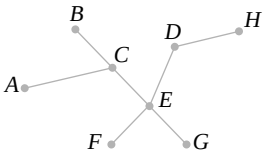
21. Five.

23. n.

25. There is only one. Any spanning tree is formed by omitting one edge from the graph. A clockwise shift of the labels gives an isomorphism between any spanning tree and the tree formed by omitting the edge (v_n, v_1).

Exercise Set 5

1.



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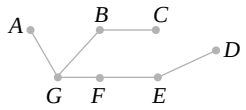
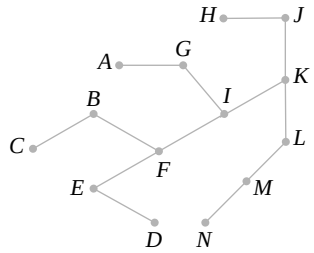


5.



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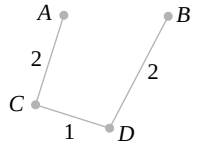
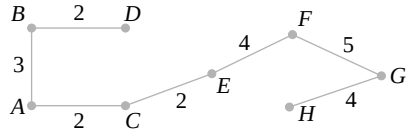
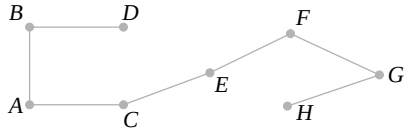


9.



11.



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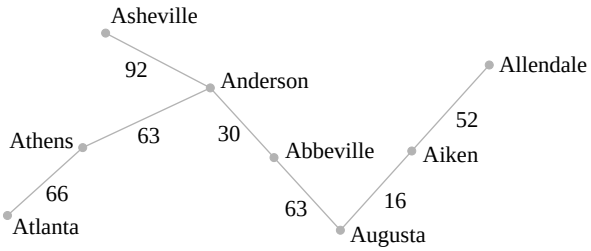
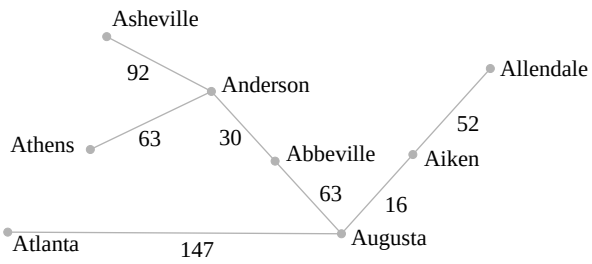
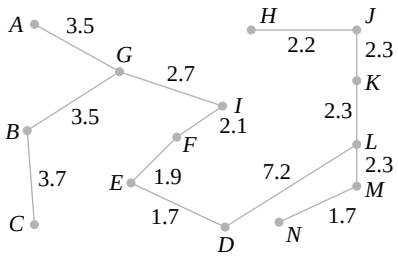


15.

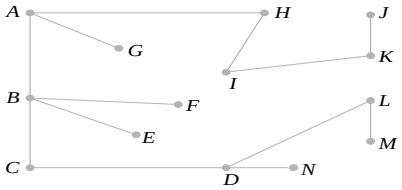


It would be 81 miles longer.

17.



19.



21. The maximal spanning tree is the same as that in Exercise 19.

23. If each edge has a distinct weight, there will be a unique maximal spanning tree since only one choice can be made at each step.

25. One example is the sum 50. The greedy algorithm would select four 11-xebec coins and then six 1-xebec coins for a total of 10 coins. But the amount can be made from seven 7-xebec coins and one 1-xebec coin, a total of eight coins.

Answers to Chapter Self-Tests

1. A tree is a relation with certain conditions.

2. The only change in performing a preorder, inorder, or postorder search is when the root is visited.

3. The other three sequences are (i) right, root, left, (ii) right, left, root, and (iii) root, right, left.

4. A complete n-tree usually has smaller height than a general n-tree with the same number of vertices. This would shorten a search of the tree that begins at the root.

5. An estimate of how many vertices and edges the graph has would help in the decision process.

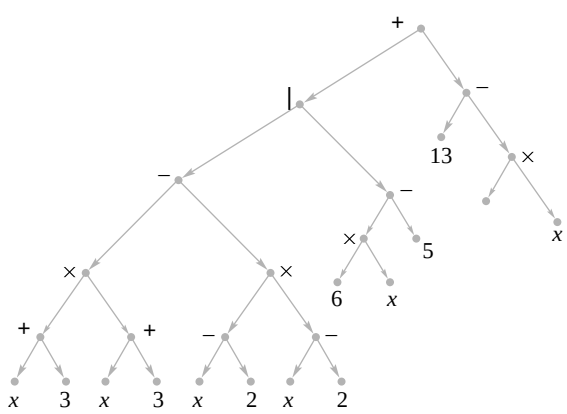
6. It is not a tree; deleting either $(2, 3)$ or $(5, 3)$ will give a tree with root 4.

7. **(a)** 4. **(b)** $v_4, v_{10}, v_6, v_9, v_8$.

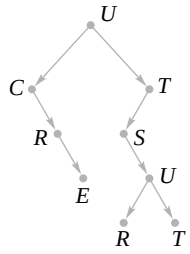
(c) 4. **(d)** v_6, v_8.

8. Every edge (v_i, v_j) in (T, v_0) belongs to a unique path from v_0 to v_j. Hence removing (v_i, v_j) would mean there is no path from v_0 to v_j.

9.

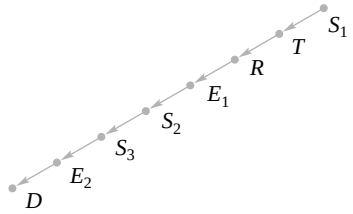


10.

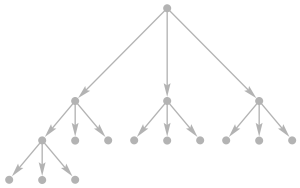


11. **(a)** UCRETSURT. **(b)** ERCRTUSTU.

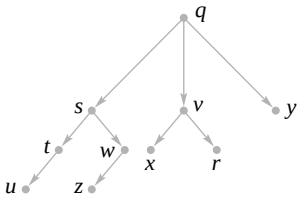
12.



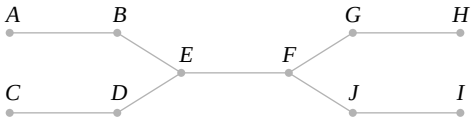
13.



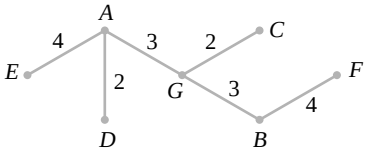
14.



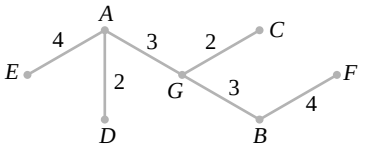
15. One solution is



16. *EA*, *AD*, *AG*, *GC*, *GB*, *BF*.



17. *GC*, *AD*, *GB*, *GA*, *BF*, *AE*.



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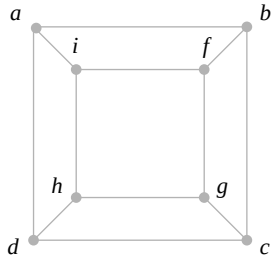


Figure 10

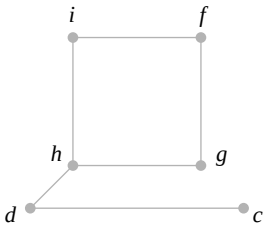


Figure 11

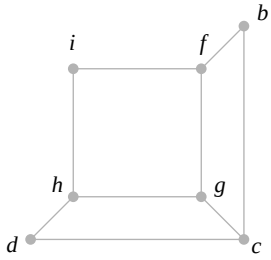


Figure 12

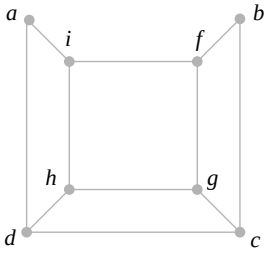


Figure 13

Our second important construction is defined for graphs without multiple edges between the same vertices. Suppose that $G = (V, E, \gamma)$ is such a graph and that R is an equivalence relation on the set V. Then we construct the **quotient graph** G^R in the following way. The vertices of G^R are the equivalence classes of V produced by R. If $[v]$ and $[w]$ are the equivalence classes of vertices v and w of G, then there is an edge in G^R from $[v]$ to $[w]$ if and only if some vertex in $[v]$ is connected to some vertex in $[w]$ in the graph G. Informally, this just says that we get G^R by merging all the vertices in each equivalence class into a single vertex and combining any edges that are superimposed by such a process.

Example 9 Let G be the graph of Figure 14 (which has no multiple edges), and let R be the equivalence relation on V defined by the partition

$$\{\{a, m, i\}, \{b, f, j\}, \{c, g, k\}, \{d, h, l\}\}.$$

Then G^R is shown in Figure 15.

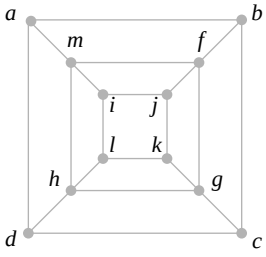


Figure 14

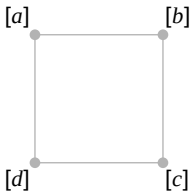


Figure 15

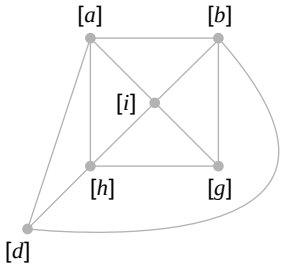


Figure 16

If S is also an equivalence relation on V defined by the partition

$$\{\{i, j, k, l\}, \{a, m\}, \{f, b, c\}, \{d\}, \{g\}, \{h\}\},$$

then the quotient graph G^S is shown in Figure 16.

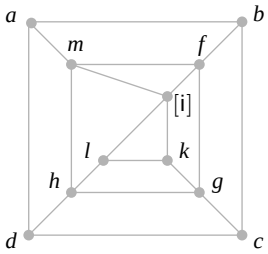


Figure 17

Again, one of the most important cases arises from using just one edge. If e is an edge between vertex v and vertex w in a graph $G = \{V, E, \gamma\}$, then we consider the equivalence relation whose partition consists of $\{v, w\}$ and $\{v_i\}$, for each $v_i \neq v$, $v_i \neq w$. That is, we merge v and w and leave everything else alone. The resulting quotient graph is denoted G^e. If G is the graph of Figure 14, and $e = \{i, j\}$, then G^e is the graph shown in Figure 17.

1 Exercises

In Exercises 1 through 4 (Figures 18 through 21), give V*, the set of vertices, and* E*, the set of edges, for the graph* $G = (V, E, \gamma)$.

1.

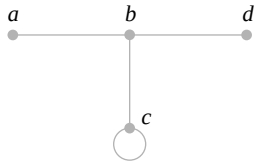


Figure 18

2.

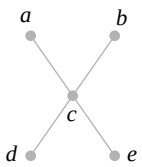


Figure 19

3.

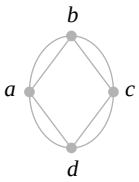


Figure 20

4.

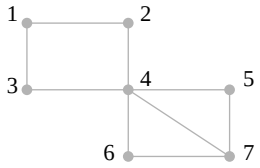


Figure 21

5. Give two subgraphs with three vertices for the graph shown in Figure 20.

6. Give three subgraphs with four vertices and four edges for the graph shown in Figure 21.

7. Draw a picture of the graph $G = (V, E, \gamma)$, where $V = \{a, b, c, d, e\}$, $E = \{e_1, e_2, e_3, e_4, e_5, e_6\}$, and $\gamma(e_1) = \gamma(e_5) = \{a, c\}$, $\gamma(e_2) = \{a, d\}$, $\gamma(e_3) = \{e, c\}$, $\gamma(e_4) = \{b, c\}$, and $\gamma(e_6) = \{e, d\}$.

8. Draw a picture of the graph $G = \{V, E, \gamma)$, where $V = \{a, b, c, d, e, f, g, h\}$, $E = \{e_1, e_2, \ldots, e_9\}$, and $\gamma(e_1) = \{a, c\}$, $\gamma(e_2) = \{a, b\}$, $\gamma(e_3) = \{d, c\}$, $\gamma(e_4) = \{b, d\}$, $\gamma(e_5) = \{e, a\}$, $\gamma(e_6) = \{e, d\}$, $\gamma(e_7) = \{f, e\}$, $\gamma(e_8) = \{e, g\}$, and $\gamma(e_9) = \{f, g\}$.

9. Give the degree of each vertex in Figure 18.

10. Give the degree of each vertex in Figure 20.

11. List all paths that begin at a in Figure 19.

12. List three circuits that begin at 5 in Figure 21.

13. Draw the complete graph on seven vertices.

14. Consider K_n, the complete graph on n vertices. What is the degree of each vertex?

15. Which of the graphs in Exercises 1 through 4 are regular?

16. Give an example of a regular, connected graph on six vertices that is not complete.

17. Give an example of a graph on five vertices with exactly two components.

18. Give an example of a graph that is regular, but not complete, with each vertex having degree three.

For Exercises 19 through 22, use the graph G *in Figure 22.*

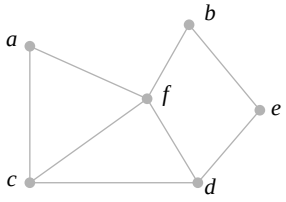


Figure 22

19. If R is the equivalence relation defined by the partition $\{\{a, f\}, \{e, b, d\}, \{c\}\}$, find the quotient graph, G^R.

20. If R is the equivalence relation defined by the partition $\{\{a, b\}, \{e\}, \{d\}, \{f, c\}\}$, find the quotient graph, G^R.

21. **(a)** Give the largest subgraph of G that does not contain f.

(b) Give the largest subgraph of G that does not contain a.

22. Let e_1 be the edge between c and f. Draw the graph of

(a) G_{e_1} **(b)** G^{e_1}

For Exercises 23 and 24, use the graph G *in Figure 23.*

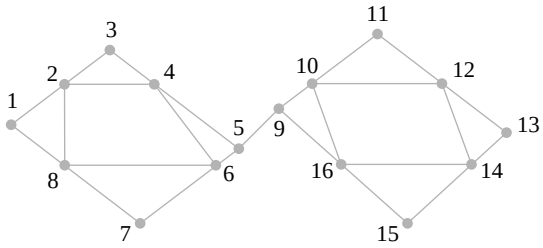


Figure 23

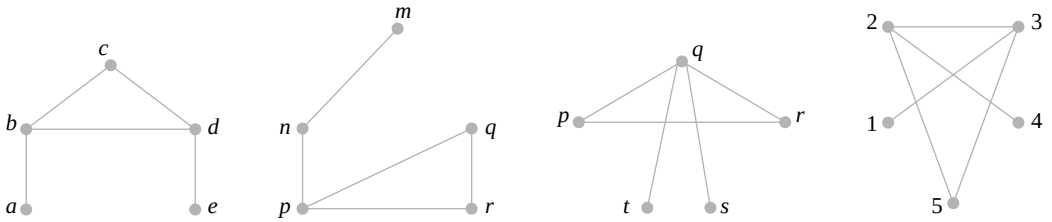
23. Let $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7), (8, 8), (9, 9), (10, 10), (11, 11), (12, 12), (13, 13), (14, 14), (15, 15), (16, 16), (1, 10), (10, 1), (3, 12), (12, 3), (5, 14), (14, 5), (2, 11), (11, 2), (4, 13), (13, 4), (6, 15), (15, 6), (7, 16), (16, 7), (8, 9), (9, 8)\}$. Draw the quotient graph G^R.

24. Let $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7), (8, 8), (9, 9), (10, 10), (11, 11), (12, 12), (13, 13), (14, 14), (15, 15), (16, 16), (1, 2), (2, 1), (3, 4), (4, 3), (5, 6), (6, 5), (7, 8), (8, 7), (9, 16), (16, 9), (10, 11), (11, 10), (12, 13), (13, 12), (14, 15), (15, 14)\}$. Draw the quotient graph G^R.

25. Complete the following statement. Every linear graph on n vertices must have _____ edges. Explain your answer.

26. What is the total number of edges in K_n, the complete graph on n vertices? Justify your answer.

27. Restate the definition of isomorphic trees to define isomorphic graphs.



(a) (b) (c) (d)

Figure 24

28. Prove or disprove that the graphs shown in Figures 24(a) and (b) are isomorphic.

29. Prove or disprove that the graphs shown in Figures 24(a) and (c) are isomorphic.

30. Prove or disprove that the graphs shown in Figures 24(a) and (d) are isomorphic.

31. Prove that if a graph G has no loops or multiple edges, then twice the number of edges is equal to the sum of the degrees of all vertices.

32. Use the result of Exercise 31 to prove that if a graph G has no loops or multiple edges, then the number of vertices of odd degree is an even number.

2 Euler Paths and Circuits

In this section and the next, we consider broad categories of problems for which graph theory is used. In the first type of problem, the task is to travel a path using each edge of the graph exactly once. It may or may not be necessary to begin and end at the same vertex. A simple example of this is the common puzzle problem that asks the solver to trace a geometric figure without lifting pencil from paper or tracing an edge more than once.

A path in a graph G is called an **Euler path** if it includes every edge exactly once. An **Euler circuit** is an Euler path that is a circuit. Leonhard Euler (1707–1783) worked in many areas of mathematics. The names "Euler path" and "Euler circuit" recognize his work with the Königsberg Bridge problem.

Example 1 Figure 25 shows the street map of a small neighborhood. A recycling ordinance has been passed, and those responsible for picking up the recyclables must start and end each trip by having the truck in the recycling terminal. They would like

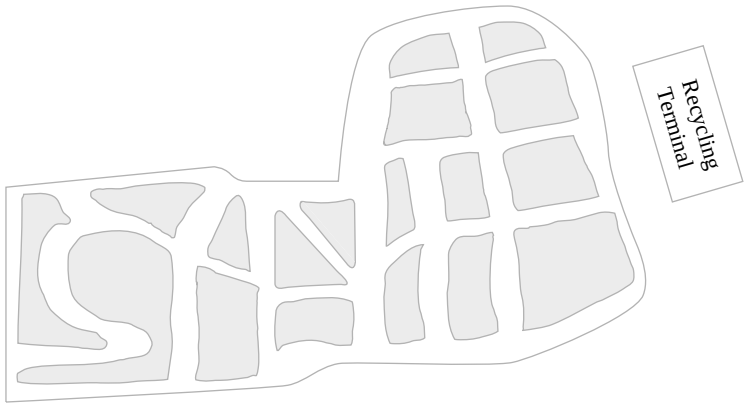


Figure 25

to plan the truck route so that the entire neighborhood can be covered and each street need be traveled only once. A graph can be constructed having one vertex for each intersection and an edge for each street between any two intersections. The problem then is to find an Euler circuit for this graph.

Example 2 **(a)** An Euler path in Figure 26 is π: E, D, B, A, C, D.

(b) One Euler circuit in the graph of Figure 27 is π: 5, 3, 2, 1, 3, 4, 5.

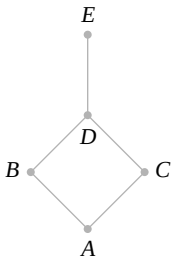


Figure 26

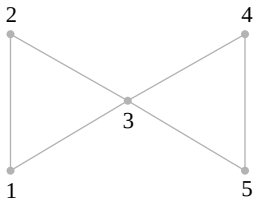


Figure 27

A little experimentation will show that no Euler circuit is possible for the graph in Figure 26. We also see that an Euler path is not possible for the graph in Figure 6 in Section 1. (Why?)

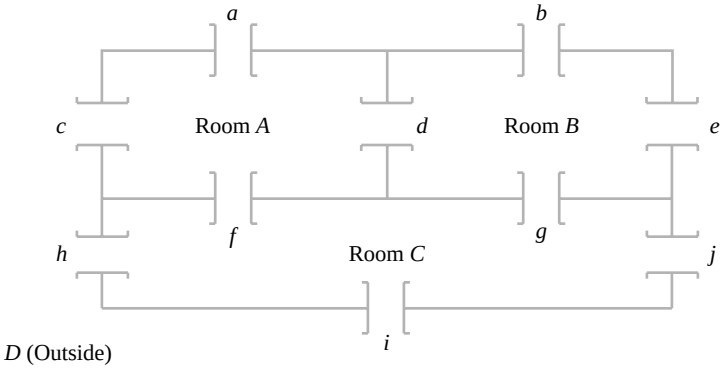


Figure 28

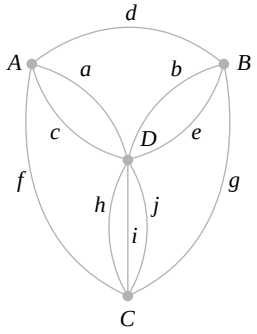


Figure 29

Example 3 Consider the floor plan of a three-room structure that is shown in Figure 28. Each room is connected to every room that it shares a wall with and to the outside along each wall. The problem is this: Is it possible to begin in a room or outside and take a walk that goes through each door exactly once? This diagram can also be formulated as a graph where each room and the outside constitute a vertex and an edge corresponds to each door. A possible graph for this structure is shown in Figure 29. The translation of the problem is whether or not there exists an Euler path for this graph. We will solve this problem later.

Two questions arise naturally at this point. Is it possible to determine whether an Euler path or Euler circuit exists without actually finding it? If there must be an Euler path or circuit, is there an efficient way to find one?

Consider again the graphs in Example 2. In Figure 26 the edge {D, E} must be either the first or the last traveled, because there is no other way to travel to or from vertex E. This means that if G has a vertex of degree 1, there cannot be an

Euler circuit, and if there is an Euler path, it must begin or end at this vertex. A similar argument applies to any vertex v of odd degree, say $2n + 1$. We may travel in on one of these edges and out on another one n times, leaving one edge from v untraveled. This last edge may be used for leaving v or arriving at v, but not both, so a circuit cannot be completed. We have just shown the first of the following results.

THEOREM 1

(a) If a graph G has a vertex of odd degree, there can be no Euler circuit in G.

(b) If G is a connected graph and every vertex has even degree, then there is an Euler circuit in G.

Proof

(b) Suppose that there are connected graphs where every vertex has even degree, but there is no Euler circuit. Choose such a G with the smallest number of edges. G must have more than one vertex since, if there were only one vertex of even degree, there is clearly an Euler circuit.

We show first that G must have at least one circuit. If v is a fixed vertex of G, then since G is connected and has more than one vertex, there must be an edge between v and some other vertex v_1. This is a simple path (of length 1) and so simple paths exist. Let π_0 be a simple path in G having the longest possible length, and let its vertex sequence be $v_1, v_2, \ldots, v_s$. Since v_s has even degree and π_0 uses only one edge that has v_s as a vertex, there must be an edge e not in π_0 that also has v_s as a vertex. If the other vertex of e is not one of the v_i, then we could construct a simple path longer than π_0, which is a contradiction. Thus e has some v_i as its other vertex, and therefore we can construct a circuit $v_i, v_{i+1}, \ldots, v_s, v_i$ in G.

Since we now know that G has circuits, we may choose a circuit π in G that has the longest possible length. Since we assumed that G has no Euler circuits, π cannot contain all the edges of G. Let G_1 be the graph formed from G by deleting all edges in π (but no vertices). Since π is a circuit, deleting its edges will reduce the degree of every vertex by 0 or 2, so G_1 is also a graph with all vertices of even degree. The graph G_1 may not be connected, but we can choose a largest connected component (piece) and call this graph G_2 (G_2 may be G_1). Now G_2 has fewer edges than G, and so (because of the way G was chosen), G_2 must have an Euler path π'.

If π' passes through all the vertices on G, then π and π' clearly have vertices in common. If not, then there must be an edge e in G between some vertex v' in π', and some vertex v not in π'. Otherwise we could not get from vertices in π' to the other vertices in G, and G would not be connected. Since e is not in π', it must have been deleted when G_1 was created from G, and so must be an edge in π. Then v' is also in the vertex sequence of π, and so in any case π and π' have at least one vertex v' in common. We can then construct a circuit in G that is longer than π by combining π and π' at v'. This is a contradiction, since π was chosen to be the longest possible circuit in G. Hence the existence of the graph G always produces a contradiction, and so no such graph is possible.

The strategy of this proof is one we have used before: Suppose there is a largest (smallest) object and construct a larger (smaller) object of the same type thereby creating a contradiction. Here we have π, the longest possible circuit that

begins and ends at v, in G, and we construct a longer circuit that begins and ends at v.

We have proved that if G has vertices of odd degree, it is not possible to construct an Euler circuit for G, but an Euler path may be possible. Our earlier discussion noted that a vertex of odd degree must be either the beginning or the end of any possible Euler path. We have the following theorem.

THEOREM 2

(a) If a graph G has more than two vertices of odd degree, then there can be no Euler path in G.

(b) If G is connected and has exactly two vertices of odd degree, there is an Euler path in G. Any Euler path in G must begin at one vertex of odd degree and end at the other.

Proof

(a) Let v_1, v_2, v_3 be vertices of odd degree. Any possible Euler path must leave (or arrive at) each of v_1, v_2, v_3 with no way to return (or leave) since each of these vertices has odd degree. One vertex of these three vertices may be the beginning of the Euler path and another the end, but this leaves the third vertex at one end of an untraveled edge. Thus there is no Euler path.

(b) Let u and v be the two vertices of odd degree. Adding the edge $\{u, v\}$ to G produces a connected graph G' all of whose vertices have even degree. By Theorem 1(b), there is an Euler circuit π' in G'. Omitting $\{u, v\}$ from π' produces an Euler path that begins at u (or v) and ends at v (or u).

Example 4 Which of the graphs in Figures 30, 31, and 32 have an Euler circuit, an Euler path but not an Euler circuit, or neither?

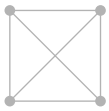


Figure 30

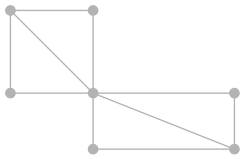


Figure 31

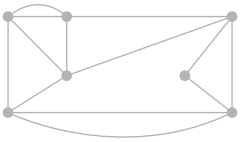


Figure 32

Solution

(a) In Figure 30, each of the four vertices has degree 3; thus, by Theorems 1 and 2, there is neither an Euler circuit nor an Euler path.

(b) The graph in Figure 31 has exactly two vertices of odd degree. There is no Euler circuit, but there must be an Euler path.

(c) In Figure 32, every vertex has even degree; thus the graph must have an Euler circuit.

Example 5 Let us return to Example 3. We see that the four vertices have degrees 4, 4, 5, and 7, respectively. Thus the problem can be solved by Theorem 2; that is, there is an Euler path. One is shown in Figure 33. Using the labels of Figure 29, this path π is specified by V_π: C, D, C, A, D, A, B, D, B, C and E_π: i, h, f, c, a, d, b, e, g, j.

Theorems 1 and 2 are examples of existence theorems. They guarantee the existence of an object of a certain type, but they give no information on how to

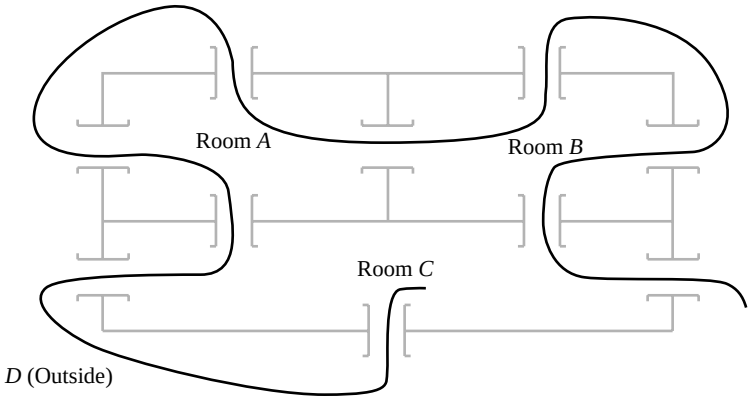


Figure 33

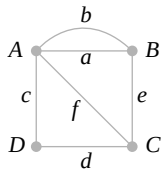


Figure 34

produce the object. There is one hint in Theorem 2(b) about how to proceed. In Figure 34, an Euler path must begin (or end) at B and end (or begin) at C. One possible Euler path for this graph is specified by V_π: B, A, D, C, A, B, C and E_π: a, c, d, f, b, e.

We next give an algorithm that produces an Euler circuit for a connected graph with no vertices of odd degree. We require an additional definition before stating the algorithm. An edge is a **bridge** in a connected graph G if deleting it would create a disconnected graph. For example, in the graph of Figure 4, Section 1, {B, E} is a bridge.

Algorithm FLEURY'S ALGORITHM

Let $G = (V, E, \gamma)$ be a connected graph with each vertex of even degree.

Step 1 Select an edge e_1 that is not a bridge in G. Let its vertices be v_1, v_2. Let π be specified by V_π: v_1, v_2 and E_π: e_1. Remove e_1 from E and let G_1 be the resulting subgraph of G.

Step 2 Suppose that V_π: $v_1, v_2, \ldots, v_k$ and E_π: $e_1, e_2, \ldots, e_{k-1}$ have been constructed so far, and that all of these edges and any resulting isolated vertices have been removed from V and E to form G_{k-1}. Since v_k has even degree, and e_{k-1} ends there, there must be an edge e_k in G_{k-1} that also has v_k as a vertex. If there is more than one such edge, select one that is not a bridge for G_{k-1}. Denote the vertex of e_k other than v_k by v_{k+1}, and extend V_π and E_π to V_π: $v_1, v_2, \ldots, v_k, v_{k+1}$ and E_π: $e_1, e_2, \ldots, e_{k-1}, e_k$. Then delete e_k and any isolated vertices from G_{k-1} to form G_k.

Step 3 Repeat Step 2 until no edges remain in E.

Example 6 Use Fleury's algorithm to construct an Euler circuit for the graph in Figure 35.

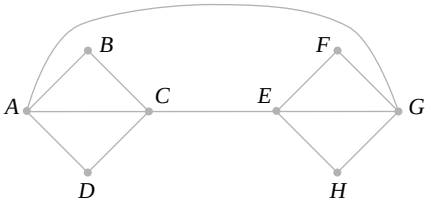


Figure 35

Solution

According to Step 1, we may begin anywhere. Arbitrarily choose vertex A. We summarize the results of applying Step 2 repeatedly in Table 1.

TABLE 1

Current Path	Next Edge	Reasoning
π: A	{A, B}	No edge from A is a bridge. Choose any one.
π: A, B	{B, C}	Only one edge from B remains.
π: A, B, C	{C, A}	No edge from C is a bridge. Choose any one.
π: A, B, C, A	{A, D}	No edge from A is a bridge. Choose any one.
π: A, B, C, A, D	{D, C}	Only one edge from D remains.
π: A, B, C, A, D, C	{C, E}	Only one edge from C remains.
π: A, B, C, A, D, C, E	{E, G}	No edge from E is a bridge. Choose any one.
π: A, B, C, A, D, C, E, G	{G, F}	{A, G} is a bridge. Choose {G, F} or {G, H}.
π: A, B, C, A, D, C, E, G, F	{F, E}	Only one edge from F remains.
π: A, B, C, A, D, C, E, G, F, E	{E, H}	Only one edge from E remains.
π: A, B, C, A, D, C, E, G, F, E, H	{H, G}	Only one edge from H remains.
π: A, B, C, A, D, C, E, G, F, E, H, G	{G, A}	Only one edge from G remains.
π: A, B, C, A, D, C, E, G, F, E, H, G, A		

The edges in Figure 36 have been numbered in the order of their choice in applying Step 2. In several places, other choices could have been made. In general, if a graph has an Euler circuit, it is likely to have several different Euler circuits.

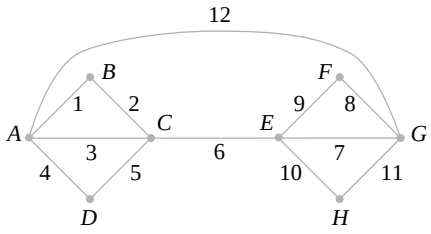


Figure 36

2 Exercises

In Exercises 1 through 8, (Figures 37 through 44), tell whether the graph has an Euler circuit, an Euler path but no Euler circuit, or neither. Give reasons for your choice.

1.

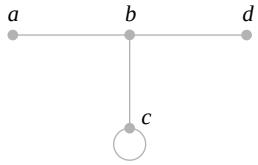


Figure 37

2.

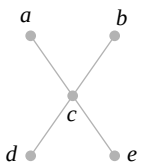


Figure 38

3.

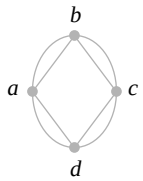


Figure 39

4.

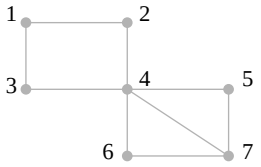


Figure 40

5.

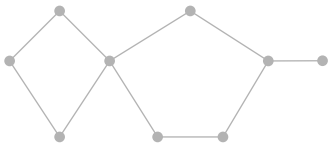


Figure 41

6.

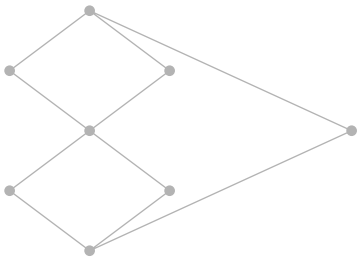


Figure 42

7.

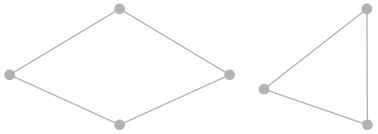


Figure 43

8.

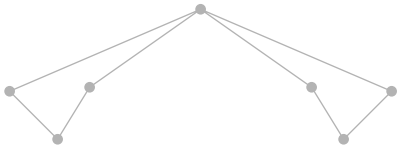


Figure 44

In Exercises 9 and 10, (Figures 45 and 46), tell if it is possible to trace the figure without lifting the pencil. Explain your reasoning.

9.

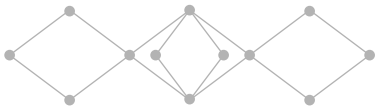


Figure 45

10.

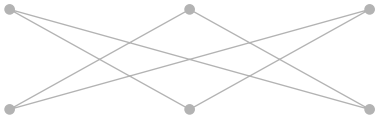


Figure 46

11. Use Fleury's algorithm to produce an Euler circuit for the graph in Figure 47.

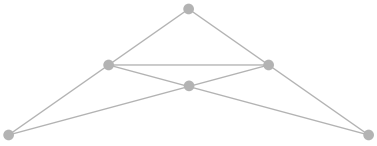


Figure 47

12. Use Fleury's algorithm to produce an Euler circuit for the graph in Figure 45.

13. An art museum arranged its current exhibit in the five rooms shown in Figure 48. Is there a way to tour the exhibit so that you pass through each door exactly once? If so, give a sketch of your tour.

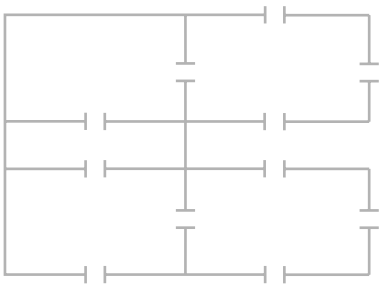


Figure 48

14. At the door of an historical mansion, you receive a copy of the floor plan for the house (Figure 49). Is it possible to visit every room in the house by passing through each door exactly once? Explain your reasoning.

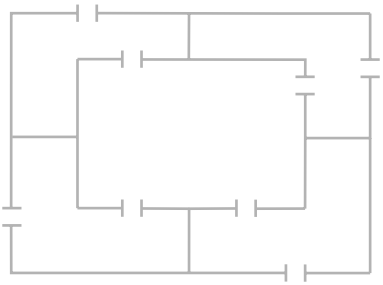


Figure 49

In Exercises 15 through 18 (Figures 50 through 53), no Euler circuit is possible for the graph given. For each graph, show the minimum number of edges that would need to be traveled twice in order to travel every edge and return to the starting vertex.

15.

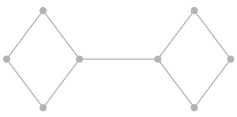


Figure 50

16.

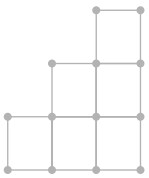


Figure 51

17.

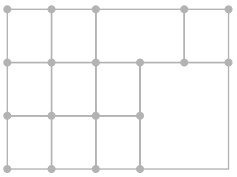


Figure 52

18.

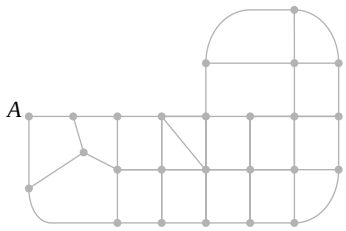


Figure 53

19. Modify the graph in Figure 52 by adding the minimum number of duplicate edges needed to make all vertices have even degree. Use Fleury's algorithm to find an Euler circuit for the modified version of the graph. Begin at the upper-left corner.

20. Modify the graph in Figure 53 by adding the minimum number of duplicate edges needed to make all vertices have even degree. Use Fleury's algorithm to find an Euler circuit for the modified version of the graph. Begin at A.

21. For which values of n does the complete graph on n vertices, K_n, have an Euler circuit?

22. Let G and H be a pair of isomorphic graphs. Prove or disprove that if there is an Euler circuit in G, then there is one in H also.

23. Consider the Hasse diagram of the Boolean algebra B_n as an undirected graph with the vertices labeled as usual by strings of 0's and 1's of length n. Prove that there is an edge between two vertices in this graph if and only if the strings labeling the end points differ in exactly one position.

24. With B_n as described in Exercise 23, which of the following graphs have an Euler circuit?

(a) B_2 **(b)** B_3 **(c)** B_4 **(d)** B_5

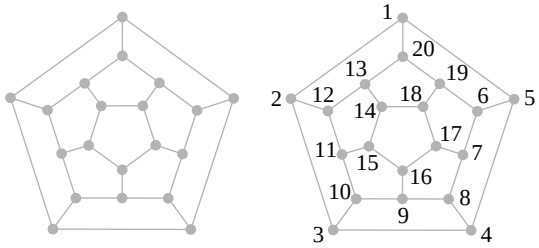
25. In general, the Hasse diagrams of which B_n have an Euler circuit? Justify your answer.

3 Hamiltonian Paths and Circuits

We turn now to the second category of graph problems in which the task is to visit each vertex exactly once, with the exception of the beginning vertex if it must also be the last vertex. For example, such a path would be useful to someone who must service a set of vending machines on a regular basis. Each vending machine could be represented by a vertex.

A **Hamiltonian path** is a path that contains each vertex exactly once. A **Hamiltonian circuit** is a circuit that contains each vertex exactly once except for the first vertex, which is also the last. This sort of path is named for the mathematician Sir William Hamilton, who developed and marketed a game consisting of a wooden graph in the shape of a regular dodecahedron and instructions to find what we have called a Hamiltonian circuit. A planar version of this solid is shown in Figure 54(a), with a Hamiltonian circuit (one of many) shown in Figure 54(b) by the consecutively numbered vertices.

It is clear that loops and multiple edges are of no use in finding Hamiltonian circuits, since loops could not be used, and only one edge can be used between any two vertices. Thus we will suppose that any graph we mention has no loops or multiple edges.



(a) (b)

Figure 54

Example 1 Consider the graph in Figure 55. The path a, b, c, d, e is a Hamiltonian path because it contains each vertex exactly once. It is not hard to see, however, that there is no Hamiltonian circuit for this graph. For the graph shown in Figure 56, the path A, D, C, B, A is a Hamiltonian circuit. In Figures 57 and 58, no Hamiltonian path is possible. (Verify this.)

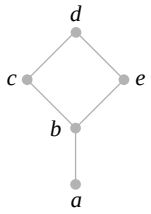


Figure 55

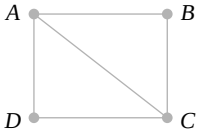


Figure 56

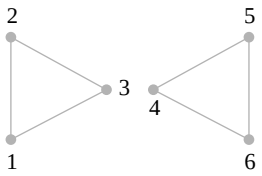


Figure 57

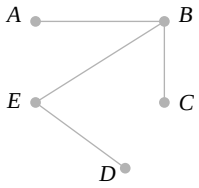


Figure 58

Example 2 Any complete graph K_n has Hamiltonian circuits. In fact, starting at any vertex, you can visit the other vertices sequentially in any desired order.

Questions analogous to those about Euler paths and circuits can be asked about Hamiltonian paths and circuits. Is it possible to determine whether a Hamiltonian path or circuit exists? If there must be a Hamiltonian path or circuit, is there an efficient way to find it? Surprisingly, considering Theorems 1 and 2 of Section 2, the first question about Hamiltonian paths and circuits has not been completely answered and the second is still unanswered as well. However, we can make several observations based on the examples.

If a graph G on n vertices has a Hamiltonian circuit, then G must have at least n edges.

We now state some partial answers that say if a graph G has "enough" edges, a Hamiltonian circuit can be found. These are again existence statements; no method for constructing a Hamiltonian circuit is given.

THEOREM 1 Let G be a connected graph with n vertices, $n > 2$, and no loops or multiple edges. G has a Hamiltonian circuit if for any two vertices u and v of G that are not adjacent, the degree of u plus the degree of v is greater than or equal to n.

We omit the proof of this result, but from it we can prove the following:

Corollary 1 G has a Hamiltonian circuit if each vertex has degree greater than or equal to $n/2$.

Proof
The sum of the degrees of any two vertices is at least $\frac{n}{2} + \frac{n}{2} = n$, so the hypotheses of Theorem 1 hold.

THEOREM 2 Let the number of edges of G be m. Then G has a Hamiltonian circuit if $m \geq \frac{1}{2}(n^2 - 3n + 6)$ (recall that n is the number of vertices).

Proof
Suppose that u and v are any two vertices of G that are not adjacent. We write deg(u) for the degree of u. Let H be the graph produced by eliminating u and v from G along with any edges that have u or v as end points. Then H has $n - 2$

vertices and $m - \deg(u) - \deg(v)$ edges (one fewer edge would have been removed if u and v had been adjacent). The maximum number of edges that H could possibly have is ${}_{n-2}C_2$. This happens when there is an edge connecting every distinct pair of vertices. Thus the number of edges of H is at most

$$ {}_{n-2}C_2 = \frac{(n-2)(n-3)}{2} \quad \text{or} \quad \frac{1}{2}(n^2 - 5n + 6). $$

We then have $m - \deg(u) - \deg(v) \le \frac{1}{2}(n^2 - 5n + 6)$. Therefore, $\deg(u) + \deg(v) \ge m - \frac{1}{2}(n^2 - 5n + 6)$. By the hypothesis of the theorem,

$$ \deg(u) + \deg(v) \ge \frac{1}{2}(n^2 - 3n + 6) - \frac{1}{2}(n^2 - 5n + 6) = n. $$

Thus the result follows from Theorem 1.

Example 3 The converses of Theorems 1 and 2 given previously are not true; that is, the conditions given are sufficient, but not necessary, for the conclusion. Consider the graph represented by Figure 59. Here n, the number of vertices, is 8, each vertex has degree 2, and $\deg(u) + \deg(v) = 4$ for every pair of nonadjacent vertices u and v. The total number of edges is also 8. Thus the premises of Theorems 1 and 2 fail to be satisfied, but there are certainly Hamiltonian circuits for this graph.

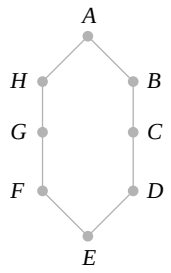


Figure 59

The problem we have been considering has a number of important variations. In one case, the edges may have *weights* representing distance, cost, and the like. The problem is then to find a Hamiltonian circuit (or path) for which the total sum of weights in the path is a minimum. For example, the vertices might represent cities; the edges, lines of transportation; and the weight of an edge, the cost of traveling along that edge. This version of the problem is often called the traveling salesperson problem.

3 Exercises

In Exercises 1 through 6 (Figures 60 through 65), determine whether the graph shown has a Hamiltonian circuit, a Hamiltonian path but no Hamiltonian circuit, or neither. If the graph has a Hamiltonian circuit, give the circuit.

1.

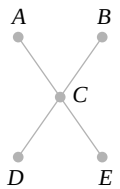


Figure 60

2.

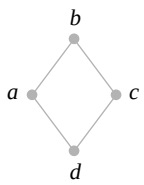


Figure 61

3.

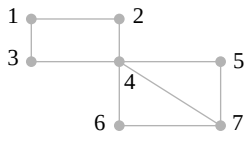


Figure 62

4.

Figure 63

5.

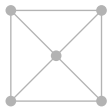


Figure 64

6.

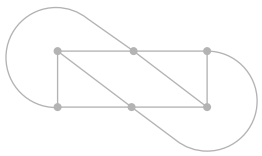


Figure 65

7. Give two Hamiltonian circuits in K_5 that have no edges in common.

In Exercises 8 through 11 (Figures 66 through 69), find a Hamiltonian circuit for the graph given.

8.

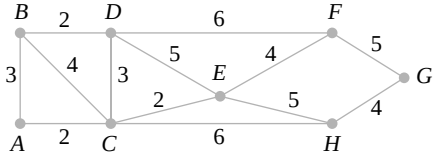


Figure 66

9.

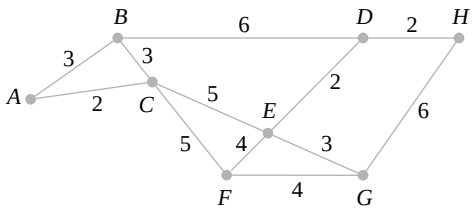


Figure 67

10.

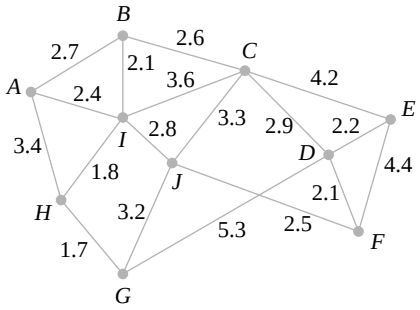


Figure 68

11.

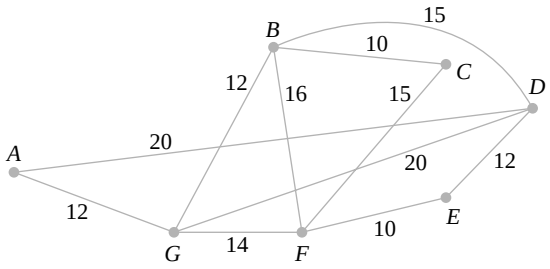


Figure 69

In Exercises 12 through 15, find a Hamiltonian circuit of minimal weight for the graph represented by the given figure.

12. Figure 66 **13.** Figure 67

14. Figure 68 **15.** Figure 69

16. Find a Hamiltonian circuit of minimal weight for the graph represented by Figure 66 if you must begin and end at D.

17. Find a Hamiltonian circuit of minimal weight for the graph represented by Figure 67 if you must begin and end at F.

18. For the complete graph on n vertices, K_n, $n \geq 3$,

(a) how many edges must a Hamiltonian circuit have?

(b) how many different Hamiltonian circuits, beginning at a fixed vertex, are there?

19. Prove that K_n, the complete graph on n vertices with $n \geq 3$, has $(n-1)!$ Hamiltonian circuits.

20. Give an example of a graph with at least four vertices with a circuit that is both an Euler and a Hamiltonian circuit.

21. Give an example of a graph that has an Euler circuit and a Hamiltonian circuit that are not the same.

22. Let $G = (V, E, \gamma)$ be a graph with $|V| = n$ that has no multiple edges. The relation R on V defined by G can be represented by a matrix, M_R. Explain how to use M_{R^∞} to determine if G is connected.

23. Using the usual vertex labels of strings of 0's and 1's, give a Hamiltonian circuit for the Hasse diagram of B_3.

24. Using the usual vertex labels of strings of 0's and 1's, give a Hamiltonian circuit for the Hasse diagram of B_4.

25. The problem of finding a Hamiltonian circuit for the Hasse diagram of B_n is equivalent to a problem about strings of 0's and 1's of length n. State this problem.

4 Transport Networks

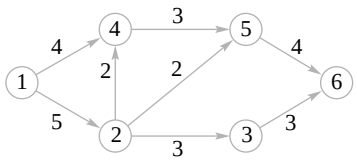


Figure 70

We have previously examined several uses of labeled graphs. In this section we return to the idea of a directed graph (digraph). An important use of labeled digraphs is to model what are commonly called transport networks. Consider the labeled digraph shown in Figure 70. This might represent a pipeline that carries water from vertex 1 to vertex 6 as part of a municipal water system. The label on an edge represents the maximum flow that can be passed through that edge and is called the **capacity** of the edge. Many situations can be modeled in this way. For instance, Figure 70 might as easily represent an oil pipeline, a highway system, a communications network, or an electric power grid. The vertices of a network are usually called nodes and may denote pumping stations, shipping depots, relay stations, or highway interchanges.

More formally, a **transport network**, or a **network**, is a connected digraph N with the following properties:

(a) There is a unique node, the **source**, that has in-degree 0. We generally label the source node 1.

In Exercises 5 through 11, find a maximum flow in the given network by using the labeling algorithm.

5. The network shown in Figure 85

6. The network shown in Figure 86

7. The network shown in Figure 87

8. The network shown in Figure 88

9. The network shown in Figure 89

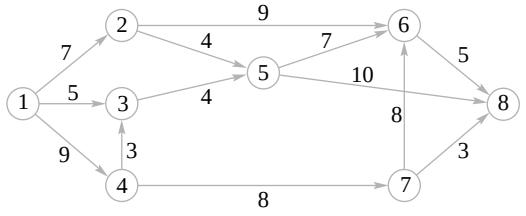


Figure 89

10. The network shown in Figure 90

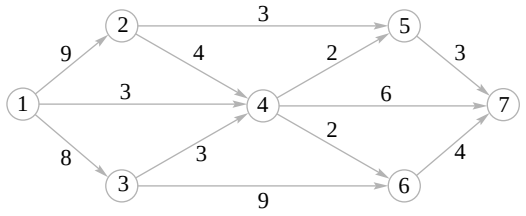


Figure 90

11. The network shown in Figure 91

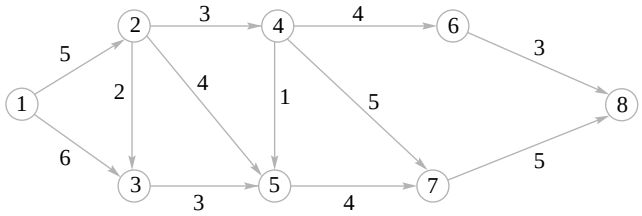


Figure 91

12. Give two cuts and their capacities for the network in Figure 87.

13. Give two cuts and their capacities for the network in Figure 89.

14. Give three cuts and their capacities for the network in Figure 91.

In Exercises 15 through 21, find the minimum cut that corresponds to the maximum flow for the given network.

15. The network of Exercise 5

16. The network of Exercise 6

17. The network of Exercise 7

18. The network of Exercise 8

19. The network of Exercise 9

20. The network of Exercise 10

21. The network of Exercise 11

5 Matching Problems

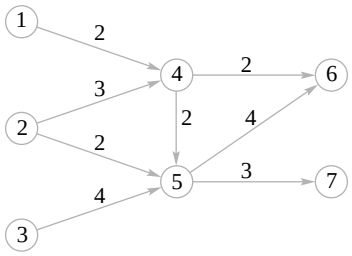


Figure 92

The definition of a transport network can be extended, and the concept of a maximal flow in a network can be used to model situations that, at first glance, do not seem to be network problems. We consider two examples in this section.

The first example is to allow a network to have multiple sources or multiple sinks as many real networks do. In the network N shown in Figure 92, nodes 1, 2, and 3 are all sources, and nodes 6 and 7 are sinks. For example, the sources could carry water from different pumping stations on a lake to two tanks (nodes 6 and 7) that supply two towns' drinking water.

In a case like this, we want to maximize the flow from all sources taken together to all the sinks taken together. As before, a flow F consists of the quantities F_{ij} assigned to each edge (i, j). We require that flow be conserved at each node that is not a source or a sink and that the flow in each edge not exceed the capacity of that edge. The value of the flow F, *value*(F), is the sum of the flows in all edges that come from any source. It is not hard to show that this value must equal the sum of flows in all edges that lead to any sink. To find the maximal flow in a general network N, we change the problem into a single-source, single-sink network problem by enlarging N to N' as follows. We add two nodes, which we call a and b. Node a is the source for N' and is connected to all nodes that are sources in N. Similarly, node b is the sink for N', and all nodes that were sinks in

N are connected to it. Nodes a and b are called, respectively, a **supersource** and a **supersink**. To complete the new network we set the capacities of all new edges to some number C_0 that is so large that no real flow through N′ can pass more than C_0 units through any of the new edges. Some authors choose C_0 to be infinity.

Example 1 An enlarged network N′ for the network of Figure 92 is shown in Figure 93. We have set C_0 to be 11, the sum of all capacities leaving sources of N.

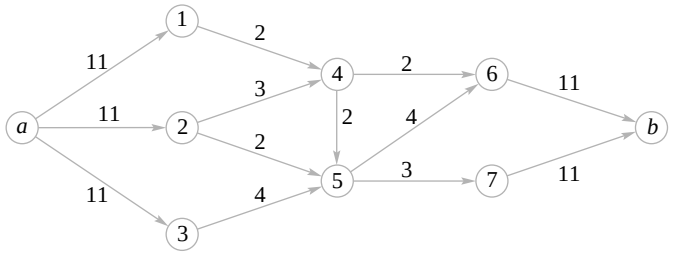


Figure 93

By adding a supersource and a supersink (if necessary) to a network, we can apply the labeling algorithm to find a maximal flow for the enlarged network. This flow will also be maximal for the original network.

Example 2 Find the maximal flow for the network N given in Figure 92.

Solution

Applying the labeling algorithm to the enlarged network N′ given in Figure 93 produces the result shown in Figure 94. (Verify.) The value of this flow is 9. This is the maximal flow from nodes 1, 2, and 3 to nodes 6 and 7 in N.

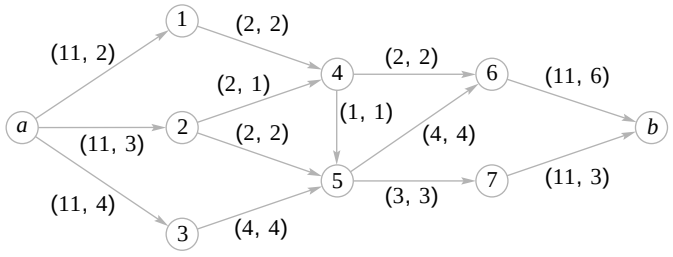


Figure 94

The Matching Problem

We consider now an important class of problems, matching problems, that can also be modeled by network flows. We begin with two finite sets A and B and a relation R from A to B. A **matching function** M is a one-to-one function from a subset of A to a subset of B. We say a is matched with b if M(a) = b. A matching function M is **compatible with *R*** if M ⊆ R; that is, if M(a) = b, then a R b.

Example 3 Let A be a set of girls and B a set of boys attending a school dance. Define R by a R b if and only if a knows b. A matching function M is defined from A to B by M(a) = b if a and b dance the third dance together. M is compatible with R if each girl knows her partner for the third dance.

Example 4 Let $A = \{s_1, s_2, s_3, s_4, s_5\}$ be a set of students working on a research project and $B = \{b_1, b_2, b_3, b_4, b_5\}$ be a set of reference books on reserve in the library for the project. Define R by $s_i \, R \, b_k$ if and only if student s_i wants to sign out book b_k. A matching of students to books would be compatible with R if each student is matched with a book that he or she wants to sign out.

Given any relation R from A to B, a matching M that is compatible with R is called **maximal** if its domain is as large as possible and is **complete** if its domain is A. In general, a matching problem is, given A, B, and R, find a maximal matching from A to B that is compatible with R. Somewhat surprisingly, matching problems can be solved using networks. We create a network to model the situation by using the elements of A as sources and the elements of B as sinks. There is a directed edge (a, b) if and only if $a \, R \, b$. To complete the network, each edge is assigned capacity 1.

Example 5 Let A, B, and R be as in Example 4. Suppose student s_1 wants books b_2 and b_3; s_2 wants b_1, b_2, b_3, b_4; s_3 wants b_2, b_3; s_4 wants b_2, b_3, b_4; s_5 wants b_2, b_3. Then the network N that represents this situation is given in Figure 95.

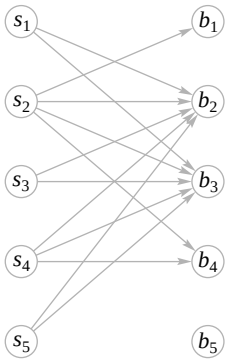


Figure 95

For the network in Figure 95, one maximal matching M can be found by inspection: $M(s_1) = b_2$, $M(s_2) = b_1$, $M(s_3) = b_3$, $M(s_4) = b_4$. It is easy to see that no complete matching is possible for this case. Usually it can be very difficult to find a maximal matching, so let us examine a network solution to the matching problem. In Figure 96, a supersource x and a super sink y have been provided and new edges have been assigned capacity 1 to create N'.

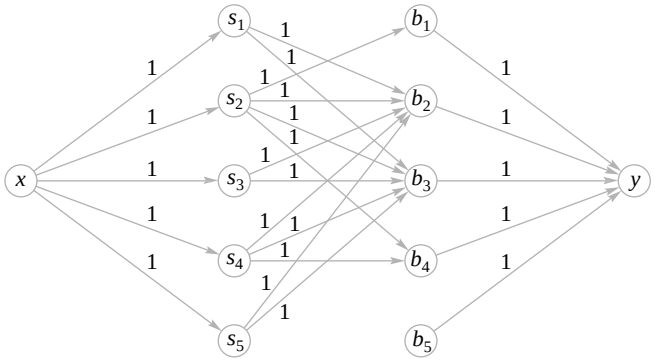


Figure 96

Every flow in N provides a matching of students to books that is compatible with R, and every compatible matching arises from a flow in N. To see this, suppose that F is a flow in N. If the flow into node s_m is 1, then the flow out must be 1, so the flow from s_m (if any) can go to only one node, say b_n. Similarly, flow can enter a node b_n from at most one s_m since the flow out of b_n is 1. We match s_m to b_n if and only if there is flow between these two nodes. The matching function M that we construct is clearly compatible with R. Conversely, if we have an R-compatible matching M, we can define a flow F by letting the flow be 1 between any two matched nodes, 1 from x to each student matched with a book, and from each matched book to y, and 0 on all other edges. This flow yields the matching M again. Hence, there is a one-to-one correspondence between flows in N and matching functions compatible with R. This means that we can use the labeling algorithm to solve the matching problem by constructing N and N' as in this example.

We saw that no complete matching is possible in the student-book example, but whether or not there is a complete matching for a particular problem is generally not obvious. A condition that will guarantee the existence of a complete matching was first found for the matching problem for a set of men and a set of women with the relation "suitable for marriage." Because the question posed was "Is it possible for each man to marry a suitable woman?" this is often referred to as the marriage problem.

The notion of an R-relative set defined in Section 2 simplifies the statement of the next theorem.

THEOREM 1
Hall's Marriage Theorem

Let R be a relation from A to B. Then there exists a complete matching M if and only if for each $X \subseteq A$, $|X| \leq |R(X)|$.

Proof

If a complete matching M exists, then $M(X) \subseteq R(X)$ for every subset X of A. But M is one to one, so $|X| = |M(X)| \leq |R(X)|$.

Conversely, suppose that for any $X \subseteq A$, $|X| \leq |R(X)|$. Construct the network N that corresponds to R. Suppose $|A| = n$. We want to show that there is a flow in N with value n, which will correspond to a complete matching. We know by the Max Flow Min Cut Theorem that it is sufficient to show that the minimal cut in N has value n. A typical situation is shown in Figure 97, with the wavy line representing a cut in N.

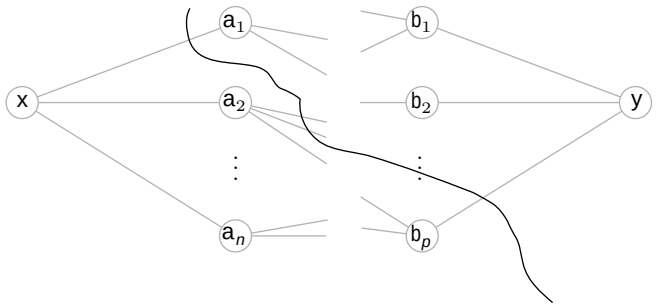


Figure 97

Remember that all edges are directed to the right and have capacity 1. We say that two vertices of N are **connected** if there is a path from one to the other; if not, the two vertices are **disconnected**. With this language, a cut in N is a set of edges whose removal will make the supersource x and the supersink y disconnected.

Let $n = |A|$ and $p = |B|$. Suppose K is a minimal cut of N. Divide the edges of K into three sets: S_1 contains the edges that begin at x, S_2 contains the edges that correspond to pairs in R, S_3 contains the edges that end with y. Consider removing the edges of K one set at a time. Suppose that $|S_1| = k$ so x is connected to k elements of A. Let A_1 be this subset of A. When the edges in S_1 are removed, no path from x to y can pass through an element of A_1. Since K is minimal, we can suppose that no edges in K begin with elements in A_1. Let $A_2 = A - A_1$ so $|A_2| = n - k$. Let $B_2 = R(A_2)$. Thus each element of B_2 labels the terminal node of an edge that begins in A_2. Since the supersource x is connected to each element in A_2, x is also connected to each element in B_2. We know that $|A_2| \leq |R(A_2)| = |B_2|$ so there are at least $n - k$ elements in B_2.

Let $|S_2| = r$. Each of these edges connects some element $a \in A_2$ to some element $b \in B_2$. Removing an edge in S_2 may or may not disconnect x from any element of B_2, but it certainly cannot disconnect x from more than one element

of B_2. Thus, when the edges in S_2 are removed, x is still connected to at least $(n - k) - r$ elements of B_2. These elements must then be disconnected from y to ensure there is no path from x to y, and so S_3 must contain at least $(n - k) - r$ edges. Hence the capacity of K = the number of edges in $K = |S_1| + |S_2| + |S_3| \geq k + r + ((n - k) - r) = n$.

Since the cut that removes all edges beginning at x has capacity n, we see that the minimal cut, and therefore the maximum flow, is exactly n.

Note that this is an example of an existence theorem; no method is given for finding a complete matching, if one exists.

Example 6 Let $\mathbf{M}_R$ be the matrix of a marriage suitability relation between five men and five women. Can each man marry a suitable woman?

$$\mathbf{M}_R = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Solution

We could construct an enlarged network N' to model this problem and apply the labeling algorithm in order to answer the question. Instead, we use Hall's marriage theorem. Note that each man considers exactly two women suitable and each woman is considered suitable by exactly two men. Consider the network that represents this problem. Let S be any subset of the men and E be the set of edges that begin in S. Clearly $|E| = 2|S|$. Each edge in E must terminate in a node of $R(S)$. But we know the number of edges terminating at elements of $R(S)$ is exactly $2|R(S)|$. Thus, $2|S| \leq 2|R(S)|$, and so $|S| \leq |R(S)|$. By Hall's marriage theorem, a complete match is possible.

5 Exercises

1. For Example 5, find a different maximal matching. Is it possible to find a maximal matching where s_5 is matched with a book?

2. Verify that the labeling algorithm used on the network of Figure 96 gives the maximal matching $M(s_1) = b_2$, $M(s_2) = b_1$, $M(s_3) = b_3$, $M(s_4) = b_4$.

In Exercises 3 through 5 (Figures 98 through 100), find a maximum flow through the network.

3.

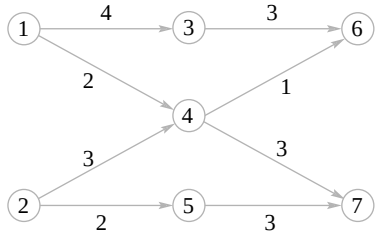


Figure 98

4.

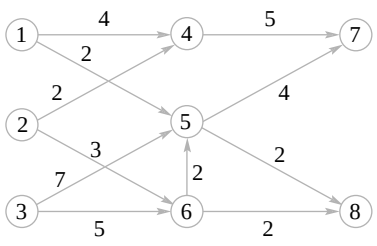


Figure 99

5.

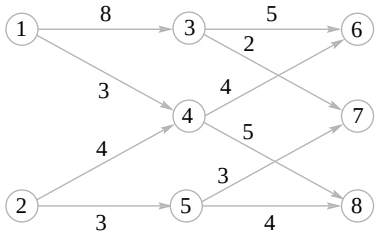


Figure 100

In Exercises 6 through 10, the matrix $\mathbf{M}_R$ *for a relation from* A *to* B *is given. Find a maximal matching for* A, B, *and* R.

6.
$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

7.
$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

8.
$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

9.
$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

10.
$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

11. Which of the matchings found in Exercises 6 through 10 are complete matchings?

12. Ann, Bing, Carmen, D'Andrea, Emilio, and Henri have inherited six paintings (of roughly equal value). Each heir has written on a slip of paper the letters that identify the paintings he or she would like to have. Ann: E; Bing: B, D; Carmen: A, E; D'Andrea: B, D; Emilio: C; Henri: A, D, F. As the executor of the estate, can you give each heir a painting he or she wants using the information on the slips of paper? If so, give the matching of heir to painting.

13. Eight friends are taking a dog-sledding trip with four sleds. Of course, not everyone is a good partner for everyone else. Here is a list of the friends each with the people with whom they can be paired.

Sam: Jane, Gail
Jane: Sam, Kip, Rufus
Kip: Kirk, Jane, Gail
Gail: Sam, Kip, Rufus
Homer: Kirk, Rufus, Stacey
Stacey: Homer, Kirk
Kirk: Kip, Homer, Stacey
Rufus: Sam, Homer, Jane, Gail

(a) Give a set of four good partnerships for the trip.

(b) A mistake was made and you've been given the list of people with whom people cannot be paired. Is a matching of good partners still possible? If so, give one.

14. Let R be a relation from A to B with $|A| = |B| = n$. Prove that if the number of ones in each column of $\mathbf{M}_R$ is k and the number of ones in each row of $\mathbf{M}_R$ is k, then there is a complete matching for A, B, and R.

15. Let R be a relation from A to B with $|A| = |B| = n$. Let j be the maximum number of ones in any column of $\mathbf{M}_R$ and k be the minimum number of ones in any row of $\mathbf{M}_R$. Prove that if $j \le k$, then there is a complete matching for A, B, and R.

16. Let R be a relation from A to B with $|A| = |B| = n$. Prove that a complete matching exists for A, B, and R if each node in the network corresponding to A, B, and R has degree at least $\frac{n}{2}$. (*Hint*: Use a result from Section 3.)

Figure 95 is an example of a ***bipartite graph****; there is a two-set partition of the set of vertices such that all edges in the graph connect members from different sets in the partition.*

17. Which of the following graphs are bipartite? If the graph is bipartite, give a partition to show this. If not, explain why this is not possible.

(a)

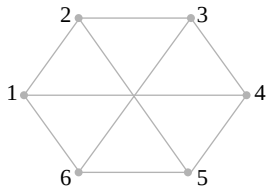


Figure 101

(b)

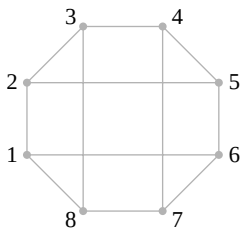


Figure 102

18. Prove that the Hasse diagram of the Boolean algebra B_n with n vertices, $n \ge 2$, is bipartite.

19. Prove that if an undirected graph contains a triangle, then the graph is not bipartite.

6 Coloring Graphs

Suppose that $G = (V, E, \gamma)$ is a graph with no multiple edges, and $C = \{c_1, c_2, \ldots, c_n\}$ is any set of n "colors." Any function $f : V \to C$ is called a **coloring of the graph *G* using *n* colors** (or using the colors of C). For each vertex v, $f(v)$ is the color of v. As we usually present a graph pictorially, so we also think of a coloring in the intuitive way of simply painting each vertex with a color from C.

However, graph-coloring problems have a wide variety of practical applications in which "color" may have almost any meaning. For example, if the graph represents a connected grid of cities, each city can be marked with the name of the airline having the most flights to and from that city. In this case, the vertices are cities and the colors are airline names. Other examples are given later.

A coloring is **proper** if any two adjacent vertices v and w have different colors.

Example 1

Let $C = \{r, w, b, y\}$ so that $n = 4$. Figure 103 shows a graph G properly colored with the colors from C in two different ways, one using three colors from C and one using all four. We show the colors as labels, which helps to explain why we avoid giving names to vertices. There are many ways to color this graph properly with three or four colors, but it is not hard to see that this cannot be done with two or fewer colors. (Experiment to convince yourself that this is true.)

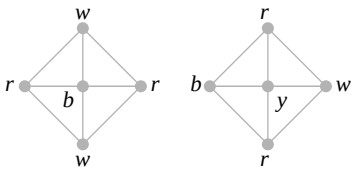


Figure 103

The smallest number of colors needed to produce a proper coloring of a graph G is called the **chromatic number of *G***, denoted by $\chi(G)$. For the graph G of Figure 103, our discussion leads us to believe that $\chi(G) = 3$.

Of the many problems that can be viewed as graph-coloring problems, one of the oldest is the map-coloring problem. Consider the map shown in Figure 104.

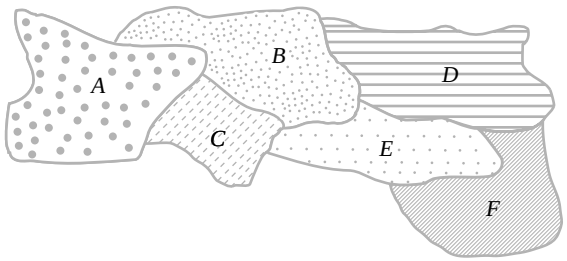


Figure 104

A coloring of a map is a way to color each region (country, state, county, province, etc.) so that no two distinct regions sharing a common border have the same color. The map-coloring problem is to find the smallest number of colors that can be used. We can view this problem as a proper graph-coloring problem as follows. Given a map M, construct a graph G_M with one vertex for each region and an edge connecting any two vertices whose corresponding regions share a common boundary. Then the proper colorings of G_M correspond exactly to the colorings of M.

Example 2

Consider the map M shown in Figure 104. Then G_M is represented by Figure 105.

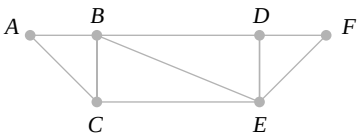


Figure 105

The map-coloring problem dates back to the mid-nineteenth century and has been an active subject of research at various times since then. A conjecture was that four colors are always enough to color any map drawn on a plane. This conjecture was proved to be true in 1976 with the aid of computer computations performed on almost 2000 configurations of graphs. There is still no proof known that does not depend on computer checking.

The graph corresponding to a map is an example of a **planar graph**, meaning that it can be drawn in a plane so that no edges cross except at vertices. Figure 105 illustrates the planarity of the graph corresponding to the map of Figure 104. The complete graph K_5 is not planar, so graph-coloring problems are more

general than map-coloring problems. Later, we will see that five colors are required to color K_5.

Graph-coloring problems also arise from counting problems.

Example 3 Fifteen different foods are to be held in refrigerated compartments within the same refrigerator. Some of them can be kept together, but other foods must be kept apart. For example, spicy meats and cheeses should be kept separate from bland meats and vegetables. Apples, eggs, and onions should be isolated or they will contaminate many other foods. Butter, margarine, and cream cheese can be kept together, but must be separated from foods with strong odors. We can construct a graph G as follows. Construct one vertex for each food and connect two with an edge if they must be kept in separate compartments in the refrigerator. Then $\chi(G)$ is the smallest number of separate containers needed to store the 15 foods properly.

A similar method could be used to calculate the minimum number of laboratory drawers needed to store chemicals if we need to separate chemicals that will react with one another if stored close to each other.

Chromatic Polynomials

Closely related to the problem of computing $\chi(G)$ is the problem of computing the total number of different proper colorings of a graph G using a set $C = \{c_1, c_2, \ldots, c_n\}$ of colors.

If G is a graph and $n \geq 0$ is an integer, let $P_G(n)$ be the number of ways to color G properly using n or fewer colors. Since $P_G(n)$ is a definite number for each n, we see that P_G is a function. What may not be obvious is that P_G is a polynomial in n. This can be shown in general and is clearly seen in the examples of this section. We call P_G the **chromatic polynomial** of G.

Example 4 Consider the linear graph L_4 defined in Section 1 and shown in Figure 9. Suppose that we have x colors. The first vertex can be colored with any color. No matter how this is done, the second can be colored with any color that was not chosen for vertex 1. Thus there are $x - 1$ choices for vertex 2. Vertex 3 can then be colored with any of the $x - 1$ colors not used for vertex 2. A similar result holds for vertex 4. By the multiplication principle of counting, the total number of proper colorings is $x(x-1)^3$. Thus $P_{L_4} = x(x-1)^3$.

We can see from Example 4 that $P_{L_4}(0) = 0$, $P_{L_4}(1) = 0$, and $P_{L_4}(2) = 2$. Thus there are no proper colorings of L_4 using zero colors (obviously) or one color, and there are two using two colors. From this we see that $\chi(L_4) = 2$. This connection holds in general, and we have the following principle: If G is a graph with no multiple edges, and P_G is the chromatic polynomial of G, then $\chi(G)$ is the smallest positive integer x for which $P_G(x) \neq 0$.

An argument similar to the one given in Example 4 shows that for L_n, $n \geq 1$, $P_{L_n}(x) = x(x-1)^{n-1}$. Thus, $\chi(L_n) = 2$ for every n.

Example 5 For any $n \geq 1$, consider the complete graph K_n defined in Section 1. Suppose that we again have x colors to use in coloring K_n. If $x < n$, no proper coloring is possible. So let $x \geq n$. Vertex v_1 can be colored with any of the x colors. For vertex v_2, only $x - 1$ remain since v_2 is connected to v_1. We can only color v_3 with $x - 2$ colors, since v_3 is connected to v_1 and v_2 and so the colors of v_1 and v_2 cannot be used again. Similarly, only $x - 3$ colors remain for v_4 and so on. Again using the multiplication principle of counting, we find that $P_{K_n}(x) =$

$x(x-1)(x-2)\cdots(x-n+1)$. This shows that $\chi(K_n) = n$. Note that if there are at least n colors, then $P_{K_n}(x)$ is the number of permutations of x objects taken n at a time (see Section 3.1).

Suppose that a graph G is not connected and that G_1 and G_2 are two components of G. This means that no vertex in G_1 is connected to any vertex in G_2. Thus any coloring of G_1 can be paired with any coloring of G_2. This can be extended to any number of components, so the multiplication principle of counting gives the following result.

THEOREM 1 If G is a disconnected graph with components $G_1, G_2, \ldots, G_m$, then $P_G(x) = P_{G_1}(x)P_{G_2}(x)\cdots P_{G_m}(x)$, the product of the chromatic polynomials for each component.

Example 6 Let G be the graph shown in Figure 6. Then G has two components, each of which is K_3. The chromatic polynomial of K_3 is $x(x-1)(x-2)$, $x \geq 3$. Thus, by Theorem 1, $P_G(x) = x^2(x-1)^2(x-2)^2$. We see that $\chi(G) = 3$ and that the number of distinct ways to color G using three colors is $P_G(3) = 36$. If x is 4, then the total number of proper colorings of G is $4^2 \cdot 3^2 \cdot 2^2$ or 576.

Example 7 Consider the discrete graph U_n of Section 1, having n vertices and no edges. All n components are single points. The chromatic polynomial of a single point is x, so, by Theorem 1, $P_{U_n}(x) = x^n$. Thus $\chi(D_n) = 1$ as can also be seen directly.

There is a useful theorem for computing chromatic polynomials using the subgraph and quotient graph constructions of Section 1. Let $G = \{V, E, \gamma\}$ be a graph with no multiple edges, and let $e \in E$, say $e = \{a, b\}$. As in Section 1, let G_e be the subgraph of G obtained by deleting e, and let G^e be the quotient graph of G obtained by merging the end points of e. Then we have the following result.

THEOREM 2 With the preceding notation and using x colors,

$$P_G(x) = P_{G_e}(x) - P_{G^e}(x). \tag{1}$$

Proof

Consider all the proper colorings of G_e. They are of two types, those for which a and b have different colors and those for which a and b have the same color. Now a coloring of the first type is also a proper coloring for G, since a and b are connected in G, and this coloring gives them different colors. On the other hand, a coloring of G_e of the second type corresponds to a proper coloring of G^e. In fact, since a and b are combined in G^e, they must have the same color there. All other vertices of G_e have the same connections as in G. Thus we have proved that $P_{G_e}(x) = P_G(x) + P_{G^e}(x)$ or $P_G(x) = P_{G_e}(x) - P_{G^e}(x)$.

Example 8 Let us compute $P_G(x)$ for the graph G shown in Figure 106, using the edge e. Then G^e is K_3 and G_e has two components, one being a single point and the other being K_3. By Theorem 1,

$$P_{G_e}(x) = x(x(x-1)(x-2)) = x^2(x-1)(x-2),$$

if $x \geq 2$. Also,

$$P_{G^e}(x) = x(x-1)(x-2).$$

Thus, by Theorem 2, we see that

$$P_G(x) = x^2(x-1)(x-2) - x(x-1)(x-2)$$

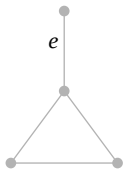


Figure 106

or

$$x(x-1)^2(x-2).$$

Clearly, $P_G(1) = P_G(2) = 0$, and $P_G(3) = 12$. This shows that $\chi(G) = 3$.

Using this model allows us to use chromatic polynomials as tools for counting the number of solutions and to answer the question of whether there is a unique solution.

6 Exercises

In Exercises 1 through 4 (Figures 107 through 110), construct a graph for the map given as done in Example 2.

1.

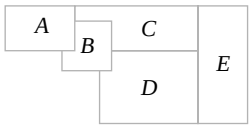


Figure 107

2.

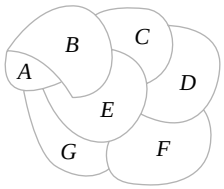


Figure 108

3.

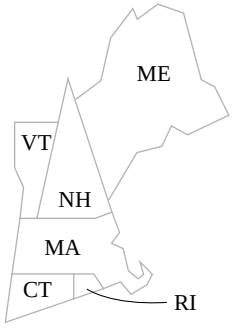


Figure 109

4.

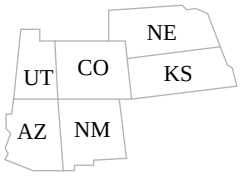


Figure 110

In Exercises 5 through 8 (Figures 111 through 114), determine the chromatic number of the graph by inspection.

5.

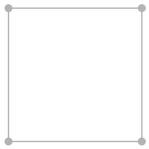


Figure 111

6.

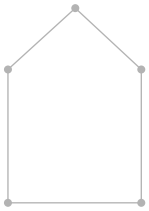


Figure 112

7.

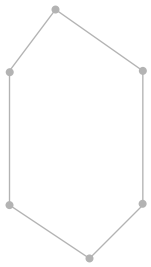


Figure 113

8.

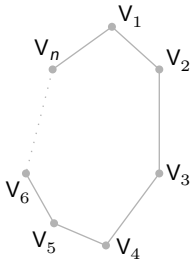


Figure 114

In Exercises 9 through 12, find the chromatic polynomial for the graph represented by the given figure. Confirm each of these by using the results of Exercises 5 through 8.

9. Figure 111 **10.** Figure 112

11. Figure 113 **12.** Figure 114

In Exercises 13 through 16 (Figures 115 through 118), find the chromatic polynomial P_G for the given graph and use P_G to find $\chi(G)$.

13.

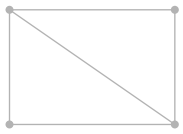


Figure 115

14.

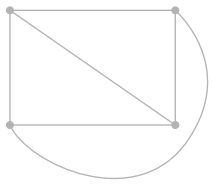


Figure 116

15.

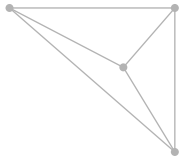


Figure 117

16.

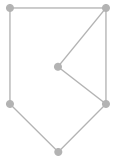


Figure 118

17. Find P_G and $\chi(G)$ for the graph G of the map in Exercise 1.

18. Find P_G and $\chi(G)$ for the graph G of the map in Exercise 3.

19. Find P_G and $\chi(G)$ for the graph G of the map in Exercise 4. Consider using Theorem 2 to do this.

20. Let G be the graph represented by Figure 64, in Section 3, Exercise 5. Find P_G and compute $\chi(G)$.

21. Let G be a bipartite graph. What is $\chi(G)$? Explain your answer.

22. Prove that if G is a graph where the longest cycle has odd length n and n is greater than 2, then $\chi(G) \leq n + 1$.

23. Prove by mathematical induction that

$$P_{L_n}(x) = x(x-1)^{n-1}, \quad n \geq 1.$$

24. **(a)** Give an example of a connected graph with five vertices that is planar.

(b) Give an example of a connected graph with five vertices that is not planar.

25. Show that the graph given is planar.

(a)

Figure 119

(b)

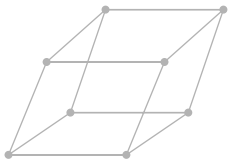


Figure 120

26. The problem of constructing a final exam schedule with a minimum number of time slots so that students do not have time conflicts can be modeled as a graph-coloring problem. If the vertices represent the courses,

(a) What do the edges represent?

(b) What should the colors represent?

(c) What question does $\chi(G)$ answer for the graph constructed?

27. The Pet Fishe Store has 20 species of fish in stock. Since some species eat others, the store owners would like to know the minimum number of tanks needed to prevent any fish tragedies. Model this problem for them as a graph-coloring problem.

28. To model an $n^2 \times n^2$ Sudoku grid as a graph-coloring problem, choose the vertices to represent the cells of the grid. To color a vertex means to fill the corresponding cell with a digit.

(a) How many vertices does the graph have?

(b) In order to represent the requirements that each digit appears once in each row, column, and subgrid, what should the edges represent?

(c) How many colors are needed for a solution to the puzzle?

29. **(a)** For the model described in Exercise 28, explain why the graph produced is regular.

(b) For $n = 2$ what is the degree of each vertex in the model described in Exercise 28?

(c) For $n = 3$ what is the degree of each vertex in the model described in Exercise 28?

Tips for Proofs

For statements about graphs, a typical structure for a contradiction proof is to examine the largest (or smallest) object with certain properties and show that if the statement were not true, then a larger (or smaller) object with the same properties could be constructed. This is the structure of the proof of Theorem 1(b), Section 2. Often indirect proofs are also existence proofs since they tell us only that an object with certain properties must exist, but not how to construct one. The proofs of Theorems 1 and 2 in Section 2 are of this type. Induction proofs are common in graph theory, because many theorems are stated in terms of the number of vertices or the number of edges. Analysis of all possible cases is also common; the proof of the four-color theorem that was mentioned earlier is an extreme case of this technique.

Key Ideas for Review

- Graph: $G = (V, E, \gamma)$, where V is a finite set of objects, called vertices, E is a set of objects, called edges, and γ is a function that assigns to each edge a two-element subset of V.
- Degree of a vertex: number of edges at the vertex
- Path
- Circuit: path that begins and ends at the same vertex
- Simple path or circuit
- Connected graph: There is a path from any vertex to any other vertex.
- Subgraph

- Euler path (circuit): path (circuit) that contains every edge of the graph exactly once
- Theorem: (a) If a graph G has a vertex of odd degree, there can be no Euler circuit in G. (b) If G is a connected graph and every vertex has even degree, then there is an Euler circuit in G.
- Theorem: (a) If a graph G has more than two vertices of odd degree, then there can be no Euler path in G. (b) If G is connected and has exactly two vertices of odd degree, there is an Euler path in G.
- Bridge: edge whose deletion would cause the graph to become disconnected
- Fleury's algorithm
- Hamiltonian path: path that includes each vertex of the graph exactly once
- Hamiltonian circuit: circuit that includes each vertex exactly once except for the first vertex, which is also the last
- Theorem: Let G be a graph on n vertices with no loops or multiple edges, $n > 2$. If for any two vertices u and v of G, the degree of u plus the degree of v is at least n, then G has a Hamiltonian circuit.
- Theorem: Let G be a graph on n vertices that has no loops or multiple edges, $n > 2$. If the number of edges in G is at least $\frac{1}{2}(n^2 - 3n + 6)$, then G has a Hamiltonian circuit.
- Transport network
- Capacity: maximum flow that can be passed through an edge
- Flow in a network: function that assigns a flow to each edge of a network
- Value of a flow: the sum of flows entering the sink
- Labeling algorithm
- Cut in a network: set of edges in a network such that every path from the source to the sink contains at least one edge in the set.
- Capacity of a cut
- Max Flow Min Cut Theorem: A maximum flow in a network has value equal to the capacity of a minimum cut of the network
- Matching (function) M compatible with a relation R: one-to-one function such that $M \subseteq R$.
- Hall's Marriage Theorem: Let R be a relation from A to B. There exists a complete matching M if and only if for each $X \subseteq A$, $|X| \leq |R(X)|$.
- Coloring of a graph using n colors
- Proper coloring of a graph: adjacent edges have different colors.
- Chromatic number of a graph G, $\chi(G)$: smallest number of colors needed for a proper coloring of G
- Planar graph: graph that can be drawn in a plane with no crossing edges
- Chromatic polynomial of a graph G, P_G: number of proper colorings of G in terms of the number of colors available

Chapter Self-Test

1. What is the relationship between a quotient graph and a quotient set?

2. What is the difference between an Euler circuit and a Hamiltonian circuit?

3. What is needed for a subset of vertices and a subset of edges to form a subgraph?

4. How are maximum flow problems related to matching problems?

5. How does the chromatic polynomial of a graph aid us in finding the chromatic number of the graph?

6. Draw a picture of the graph $G = (V, E, \gamma)$, where $V = \{j, k, l, m, n\}$, $E = \{e_1, e_2, e_3, e_4, e_5\}$, and

$$\gamma(e_1) = \{m, k\}, \quad \gamma(e_2) = \{j, l\}, \quad \gamma(e_3) = \{n, k\},$$
$$\gamma(e_4) = \{m, j\}, \quad \text{and} \quad \gamma(e_5) = \{n, l\}.$$

7. Give an example of a graph with seven vertices and exactly three components.

8. If R is the equivalence relation defined by the partition $\{\{v_2, v_4, v_5\}, \{v_1, v_3\}, \{v_7, v_8\}, \{v_6, v_9\}\}$, find the quotient graph G^R of the graph G represented by Figure 121.

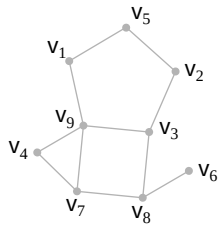


Figure 121

Use Figures 122 and 123 for Problems 9 and 10.

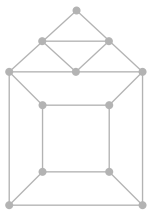


Figure 122

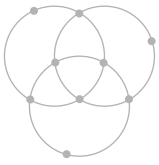
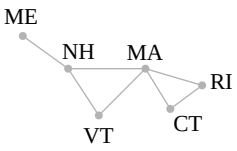


Figure 123

9. Tell whether the graph in the specified figure has an Euler circuit, an Euler path but no Euler circuit, or neither.

(a) Figure 122 **(b)** Figure 123

3. 

5. 2.

7. 2.

9. $P_G(x) = x(x-1)(x^2-3x+3)$; $P_G(0) = P_G(1) = 0$, $P_G(2) = 2$.

11. $P_G(x) = x(x-1)(x^4-5x^3+10x^2-10x+5)$; $P_G(0) = P_G(1) = 0$, $P_G(2) = 2$.

13. $P_G(x) = x(x-1)(x-2)^2$; $\chi(G) = 3$.

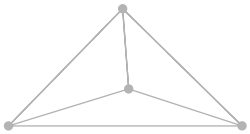
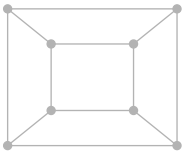
15. $P_G(x) = x(x-1)(x-2)(x-3)$; $\chi(G) = 4$.

17. $P_G(x) = x(x-1)(x-2)^3$; $\chi(G) = 3$.

19. $P_G(x) = x(x-1)^2(x-2)(x^2-3x+3)$; $\chi(G) = 3$.

21. $\chi(G) = 2$. Let A and B be the subsets that partition the vertices. Color vertices in A one color and those in B the other. It is easy to see that this will give a proper coloring of G.

23. (Outline) Basis step: $n = 1$ $P(1)$: $P_{L_1}(x) = x$ is true, because L_1 consists of a single vertex.
Induction step: We use $P(k)$ to show $P(k+1)$. Let $G = L_{k+1}$ and e be an edge $\{u, v\}$ with $\deg(v) = 1$. Then G_e has two components, L_k and v. Using Theorem 1 and $P(k)$, we have $P_{G_e}(x) = x \cdot x(x-1)^{k-1}$. Merging v with u gives $G^e = L_k$. Thus $P_{G^e}(x) = x(x-1)^{k-1}$. By Theorem 2, $P_{L_{k+1}}(x) = x^2(x-1)^{k-1} - x(x-1)^{k-1} =$
$x(x-1)^{k-1}(x-1)$ or $x(x-1)^k$.

25. These are possible answers.
(a)  **(b)** 

27. Label the vertices with the fish species. An edge connects two vertices if one species eats the other. The colors represent the tanks, so $\chi(G)$ will be the minimum number of tanks needed.

29. **(a)** Each row, column, and subgrid has the same number of elements.
(b) 7. **(c)** 20.

Answers to Chapter Self-Tests

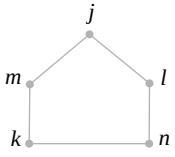
1. A quotient graph is a quotient set of the vertices with a graph structure of inherited edges.

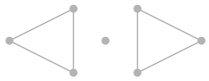
2. An Euler circuit must use all edges exactly once; a Hamiltonian circuit must use all vertices exactly once except for the starting vertex, which is also the ending vertex.

3. The subset of edges cannot contain edges whose end points do not belong to the subsets of vertices.

4. Any matching problem can be converted to a maximum flow problem by adding to the graph a supersource and a supersink, and giving each edge capacity 1.

5. Since the chromatic polynomial counts the number of ways the graph can be colored with x colors, the smallest value of x for which $P(x) \neq 0$ is the minimum number of colors needed; that is, the chromatic number.

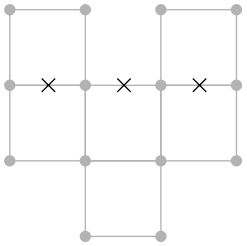
6. 

7. One solution is 

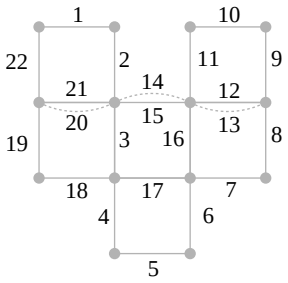
8.

9. **(a)** Neither an Euler circuit nor path. There are more than 2 vertices of odd degree.
(b) An Euler circuit. All vertices have even degree.

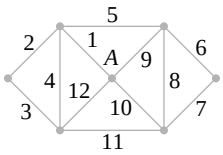
10. **(a)** A Hamiltonian circuit.
(b) A Hamiltonian path, but not a circuit.

11. **(a)** 

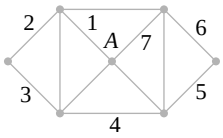
(b) One solution is



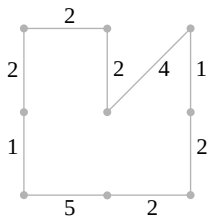
12. One solution is



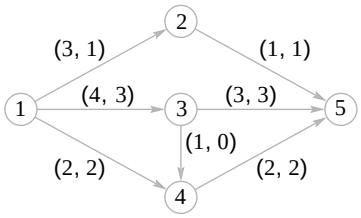
13. One solution is



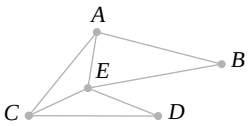
14.



15. *value*(F) = 6.



16. (a)



(b) 3.

17. $P_G(x) = x(x-1)(x-2)^3$

$P_G(0) = P_G(1) = P_G(2) = 0, P_G(3) = 6.$

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Semigroups and Groups

From *Discrete Mathematical Structures*, Sixth Edition, Bernard Kolman, Robert C. Busby, Sharon Cutler Ross.
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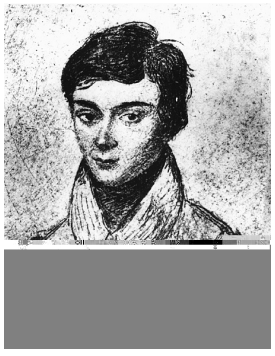
Semigroups and Groups

In this chapter, we identify more types of mathematical structures, namely, semigroups, groups, rings, and fields.

Looking Back

The term *group* was applied by Evariste Galois (1811–1832) in 1830 to a set of permutations of a finite set that satisfies certain properties. Galois was born and died in Paris. He was educated by his mother at home until the age of 12. He attended a prestigious lycée in Paris and by the age of 16 became fully absorbed in the study of mathematics, even neglecting other subjects. He failed in his two tries at admission to the highly regarded École Polytechnique and enrolled in the École Normale, a lesser institution of higher learning. In his first year there he published four papers. Three additional papers that he wrote shortly thereafter were lost by the distinguished mathematicians that he had asked to present his papers to the Academy of Sciences. In 1831, Galois wrote another paper carefully presenting the results of his research. This paper was rejected as being "incomprehensible." During the 1830 revolution, Galois criticized the director of his school and was expelled. He also spent some time in jail because of his political activities. On May 30, 1832 he was mortally wounded in a duel and died the next day at the age of 20. Before his duel, Galois left a letter to a friend detailing the results of his research. His results were so advanced for his time that a full exposition of this work did not appear until 1870.

We are all familiar with the quadratic formula for the roots of a quadratic polynomial. It uses arithmetic operations and radicals, and so a quadratic is said to be solvable by radicals. Similar formulas for the roots of a cubic and fourth-degree polynomial in terms of their coefficients were discovered in the 1500s. For the next 300 years, mathematicians tried, unsuccessfully, to solve the general fifth-degree polynomial by radicals. The Norwegian mathematician Niels Henrik Abel (1802–1829) showed at the age of 19 that the general polynomial of degree 5 or higher cannot be solved by radicals. Since many special polynomials of degree 5 or higher can be solved by radicals, it became important to determine which polynomials have this property. Galois characterized polynomials that are solvable by radicals by studying the properties of a group (now called a *Galois group*) that is associated with the polynomial.



Evariste Galois
CORBIS BETTMANN

1 Binary Operations Revisited

A binary operation may be used to define a function. Here we turn the process around and define a binary operation as a function with certain properties.

A **binary operation on a set *A*** is an everywhere defined function $f : A \times A \to A$. Observe the following properties that a binary operation must satisfy:

1. Since $\mathrm{Dom}(f) = A \times A$, f assigns an element $f(a, b)$ of A to each ordered pair (a, b) in $A \times A$. That is, the binary operation must be defined for each ordered pair of elements of A.
2. Since a binary operation is a function, only one element of A is assigned to each ordered pair.

Thus we can say that a binary operation is a rule that assigns to each ordered pair of elements of A a unique element of A. The reader should note that this definition is restrictive, but we have made the change to simplify the discussion in this chapter. We shall now turn to a number of examples.

It is customary to denote binary operations by a symbol such as $\square$, instead of f, and to denote the element assigned to (a, b) by $a \square b$ [instead of $\square(a, b)$]. It should be emphasized that if a and b are elements in A, then $a \square b \in A$, and this property is often described by saying that A is **closed** under the operation $\square$.

Example 1 Let $A = Z$. Define $a \square b$ as $a + b$. Then $\square$ is a binary operation on Z. ◆

Example 2 Let $A = R$. Define $a \square b$ as a/b. Then $\square$ is not a binary operation, since it is not defined for every ordered pair of elements of A. For example, $3 \square 0$ is not defined, since we cannot divide by zero. ◆

Example 3 Let $A = Z^+$. Define $a \square b$ as $a - b$. Then $\square$ is not a binary operation since it does not assign an element of A to every ordered pair of elements of A; for example, $2 \square 5 \notin A$. ◆

Example 4 Let $A = Z$. Define $a \square b$ as a number less than both a and b. Then $\square$ is not a binary operation, since it does not assign a *unique* element of A to each ordered pair of elements of A; for example, $8 \square 6$ could be 5, 4, 3, 1, and so on. Thus, in this case, $\square$ would be a relation from $A \times A$ to A, but not a function. ◆

Example 5 Let $A = Z$. Define $a \square b$ as $\max\{a, b\}$. Then $\square$ is a binary operation; for example, $2 \square 4 = 4$, $-3 \square (-5) = -3$. ◆

Example 6 Let $A = P(S)$, for some set S. If V and W are subsets of S, define $V \square W$ as $V \cup W$. Then $\square$ is a binary operation on A. Moreover, if we define $V \square' W$ as $V \cap W$, then $\square'$ is another binary operation on A. ◆

As Example 6 shows, it is possible to define many binary operations on the same set.

Example 7 Let M be the set of all $n \times n$ Boolean matrices for a fixed n. Define $\mathbf{A} \square \mathbf{B}$ as $\mathbf{A} \square \mathbf{B}$. Then $\square$ is a binary operation. This is also true of $\mathbf{A} \square \mathbf{B}$. ◆

Example 8 Let L be a lattice. Define $a \square b$ as $a \square b$ (the greatest lower bound of a and b). Then $\square$ is a binary operation on L. This is also true of $a \square b$ (the least upper bound of a and b). ◆

Tables

If $A = \{a_1, a_2, \ldots, a_n\}$ is a finite set, we can define a binary operation on A by means of a table as shown in Figure 1. The entry in position i, j denotes the element $a_i \square a_j$.

$*$	a_1	a_2	$\cdots$	a_j	$\cdots$	a_n
a_1						
a_2						
$\vdots$						
a_i				$a_i * a_j$		
$\vdots$						
a_n						

Figure 1

Example 9 Let $A = \{0, 1\}$. We define binary operations $\square$ and $\square$ by the following tables:

$\square$	0	1
0	0	1
1	1	1

$\square$	0	1
0	0	0
1	0	1

◆

For $A = \{a, b\}$, we shall determine the number of binary operations that can be defined on A. Every binary operation $\square$ on A can be described by a table

$\square$	a	b
a		
b		

Since every blank can be filled in with the element a or b, we conclude that there are $2 \cdot 2 \cdot 2 \cdot 2 = 2^4$ or 16 ways to complete the table. Thus, there are 16 binary operations on A.

Properties of Binary Operations

A binary operation on a set A is said to be **commutative** if

$$a \square b = b \square a$$

for all elements a and b in A.

Example 10 The binary operation of addition on Z (as discussed in Example 1) is commutative. ◆

Example 11 The binary operation of subtraction on Z is not commutative, since

$$2 - 3 \neq 3 - 2.$$ ◆

A binary operation that is described by a table is commutative if and only if the entries in the table are symmetric with respect to the main diagonal.

Example 12 Which of the following binary operations on $A = \{a, b, c, d\}$ are commutative?

$\square$	a	b	c	d
a	a	c	b	d
b	b	c	b	a
c	c	d	b	c
d	a	a	b	b

(a)

$\square$	a	b	c	d
a	a	c	b	d
b	c	d	b	a
c	b	b	a	c
d	d	a	c	d

(b)

Solution

The operation in (a) is not commutative, since $a \square b$ is c while $b \square a$ is b. The operation in (b) is commutative, since the entries in the table are symmetric with respect to the main diagonal. ◆

A binary operation $\square$ on a set A is said to be **associative** if

$$a \square (b \square c) = (a \square b) \square c$$

for all elements a, b, and c in A.

Example 13 The binary operation of addition on Z is associative. ◆

Example 14 The binary operation of subtraction on Z is not associative, since

$$2 - (3 - 5) \neq (2 - 3) - 5.$$ ◆

Example 15 Let L be a lattice. The binary operation defined by $a \square b = a \sqcap b$ (see Example 8) is commutative and associative. It also satisfies the **idempotent** property $a \square a = a$. A partial converse of this example is also true, as shown in Example 16. ◆

Example 16 Let $\square$ be a binary operation on a set A, and suppose that $\square$ satisfies the following properties for any a, b, and c in A.

1.	$a = a \square a$	Idempotent property
2.	$a \square b = b \square a$	Commutative property
3.	$a \square (b \square c) = (a \square b) \square c$	Associative property

Define a relation $\leq$ on A by

$$a \leq b \quad \text{if and only if} \quad a = a \square b.$$

Show that $(A, \leq)$ is a poset, and for all a, b in A, $\text{GLB}(a, b) = a \square b$.

Solution

We must show that $\leq$ is reflexive, antisymmetric, and transitive. Since $a = a \square a$, $a \leq a$ for all a in A, and $\leq$ is reflexive.

Now suppose that $a \leq b$ and $b \leq a$. Then, by definition and property 2, $a = a \square b = b \square a = b$, so $a = b$. Thus $\leq$ is antisymmetric.

If $a \leq b$ and $b \leq c$, then $a = a \square b = a \square (b \square c) = (a \square b) \square c = a \square c$, so $a \leq c$ and $\leq$ is transitive.

Finally, we must show that, for all a and b in A, $a \square b = a \sqcap b$ (the greatest lower bound of a and b with respect to $\leq$). We have $a \square b = a \square (b \square b) = (a \square b) \square b$, so $a \square b \leq b$. In a similar way, we can show that $a \square b \leq a$, so $a \square b$ is a lower bound for a and b. Now, if $c \leq a$ and $c \leq b$, then $c = c \square a$ and $c = c \square b$ by definition. Thus $c = (c \square a) \square b = c \square (a \square b)$, so $c \leq a \square b$. This shows that $a \square b$ is the greatest lower bound of a and b. ◆

1 Exercises

In Exercises 1 through 8, determine whether the description of $\square$ is a valid definition of a binary operation on the set.

1. On R, where $a \square b$ is ab (ordinary multiplication).

2. On Z^+, where $a \square b$ is a/b.

3. On Z, where $a \square b$ is a^b.

4. On Z^+, where $a \square b$ is a^b.

5. On Z^+, where $a \square b$ is $\sqrt{a} - b$.

6. On R, where $a \square b$ is $a \sqrt{b}$.

7. On R, where $a \square b$ is the largest rational number that is less than ab.

8. On Z, where $a \square b$ is $2a + b$.

In Exercises 9 through 19, determine whether the binary operation $\square$ is commutative and whether it is associative on the set.

9. On Z^+, where $a \square b$ is $a + b + 2$.

10. On Z, where $a \square b$ is ab.

11. On R, where $a \square b$ is $a \times |b|$.

12. On the set of nonzero real numbers, where $a \square b$ is a/b.

13. On R, where $a \square b$ is the minimum of a and b.

14. On the set of $n \times n$ Boolean matrices, where $\mathbf{A} \square \mathbf{B}$ is $\mathbf{A} \square \mathbf{B}$.

15. On R, where $a \square b$ is $ab/3$.

16. On R, where $a \square b$ is $ab + 2b$.

17. On a lattice A, where $a \square b$ is $a \square b$.

18. On the set of 2×1 matrices, where

$$\begin{bmatrix} a \\ b \end{bmatrix} \square \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a + c \\ b + d + 1 \end{bmatrix}.$$

19. On the set of rational numbers, where $a \square b = \dfrac{a + b}{2}$.

20. Prove or disprove that the binary operation on Z^+ of $a \square b = \text{GCD}(a, b)$ has the idempotent property.

21. Prove or disprove that the binary operation in Exercise 19 has the idempotent property.

22. Fill in the following table so that the binary operation $\square$ is commutative.

$\square$	a	b	c
a	b		
b	c	b	a
c	a		c

23. Fill in the following table so that the binary operation $\square$ is commutative and has the idempotent property.

$\square$	a	b	c
a		c	
b			
c	c	a	

24. Consider the binary operation $\square$ defined on the set $A = \{a, b, c\}$ by the following table.

$\square$	a	b	c
a	b	c	b
b	a	b	c
c	c	a	b

(a) Is $\square$ a commutative operation?

(b) Compute $a \square (b \square c)$ and $(a \square b) \square c$.

(c) Is $\square$ an associative operation?

25. Consider the binary operation $\square$ defined on the set $A = \{a, b, c, d\}$ by the following table.

$\square$	a	b	c	d
a	a	c	b	d
b	d	a	b	c
c	c	d	a	a
d	d	b	a	c

Compute

(a) $c \square d$ and $d \square c$. **(b)** $b \square d$ and $d \square b$.

(c) $a \square (b \square c)$ and $(a \square b) \square c$.

(d) Is $\square$ commutative? associative?

In Exercises 26 and 27, complete the given table so that the binary operation $\square$ is associative.

26.

$\square$	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	a	b
d				

27.

$\square$	a	b	c	d
a	b	a	c	d
b	b	a	c	d
c				
d	d	c	c	d

28. Let A be a set with n elements. How many binary operations can be defined on A?

29. Let A be a set with n elements. How many commutative binary operations can be defined on A?

30. Let $A = \{a, b\}$.

(a) Make a table for each of the 16 binary operations that can be defined on A.

(b) Using part (a), identify the binary operations on A that are commutative.

31. Let $A = \{a, b\}$.

(a) Using Exercise 30, identify the binary operations on A that are associative.

(b) Using Exercise 30, identify the binary operations on A that satisfy the idempotent property.

32. Let $\square$ be a binary operation on a set A, and suppose that $\square$ satisfies the idempotent, commutative, and associative properties, as discussed in Example 16. Define a relation $\leq$ on A by $a \leq b$ if and only if $b = a \square b$. Show

that $(A, \leq)$ is a poset and, for all a and b, LUB$(a, b) = a \sqcup b$.

33. Define a binary operation on a set S by $a \sqcup b = b$. Is $\sqcup$ associative? commutative? idempotent?

2 Semigroups

In this section we define a simple mathematical structure, consisting of a set together with a binary operation, that has many important applications.

A **semigroup** is a nonempty set S together with an associative binary operation $\sqcup$ defined on S. We shall denote the semigroup by $(S, \sqcup)$ or, when it is clear what the operation $\sqcup$ is, simply by S. We also refer to $a \sqcup b$ as the **product** of a and b. The semigroup $(S, \sqcup)$ is said to be commutative if $\sqcup$ is a commutative operation.

Example 1 It follows from Section 1 that $(Z, +)$ is a commutative semigroup. ◆

Example 2 The set $P(S)$, where S is a set, together with the operation of union is a commutative semigroup. ◆

Example 3 The set Z with the binary operation of subtraction is not a semigroup, since subtraction is not associative. ◆

Example 4 Let S be a fixed nonempty set, and let S^S be the set of all functions $f : S \to S$. If f and g are elements of S^S, we define $f \sqcup g$ as $f \circ g$, the composite function. Then $\sqcup$ is a binary operation on S^S, and it follows that $\sqcup$ is associative. Hence $(S^S, \sqcup)$ is a semigroup. The semigroup S^S is not commutative. ◆

Example 5 Let $(L, \leq)$ be a lattice. Define a binary operation on L by $a \sqcup b = a \sqcup b$. Then L is a semigroup. ◆

Example 6 Let $A = \{a_1, a_2, \ldots, a_n\}$ be a nonempty set. $A^\sqcup$ is the set of all finite sequences of elements of A. That is, $A^\sqcup$ consists of all words that can be formed from the alphabet A. Let α and β be elements of $A^\sqcup$. Observe that catenation is a binary operation $\cdot$ on $A^\sqcup$. Recall that if $\alpha = a_1 a_2 \cdots a_n$ and $\beta = b_1 b_2 \cdots b_k$, then $\alpha \cdot \beta = a_1 a_2 \cdots a_n b_1 b_2 \cdots b_k$. It is easy to see that if α, β, and γ are any elements of $A^\sqcup$, then

$$\alpha \cdot (\beta \cdot \gamma) = (\alpha \cdot \beta) \cdot \gamma$$

so that $\cdot$ is an associative binary operation, and $(A^\sqcup, \cdot)$ is a semigroup. The semigroup $(A^\sqcup, \cdot)$ is called the **free semigroup generated by *A***. ◆

In a semigroup $(S, \sqcup)$ we can establish the following generalization of the associative property; we omit the proof.

THEOREM 1 If $a_1, a_2, \ldots, a_n$, $n \geq 3$, are arbitrary elements of a semigroup, then all products of the elements $a_1, a_2, \ldots, a_n$ that can be formed by inserting meaningful parentheses arbitrarily are equal. ■

Example 7 Theorem 1 shows that the products

$$((a_1 \sqcup a_2) \sqcup a_3) \sqcup a_4, \quad a_1 \sqcup (a_2 \sqcup (a_3 \sqcup a_4)), \quad (a_1 \sqcup (a_2 \sqcup a_3)) \sqcup a_4$$

are all equal. ◆

If $a_1, a_2, \ldots, a_n$ are elements in a semigroup $(S, \square)$, we shall write their product as

$$a_1 \square a_2 \square \cdots \square a_n,$$

omitting the parentheses.

An element e in a semigroup $(S, \square)$ is called an **identity** element if

$$e \square a = a \square e = a$$

for all $a \in S$. An identity element must be unique.

Example 8 The number 0 is an identity in the semigroup $(Z, +)$. ◆

Example 9 The semigroup $(Z^+, +)$ has no identity element. ◆

A **monoid** is a semigroup $(S, \square)$ that has an identity.

Example 10 The semigroup $P(S)$ defined in Example 2 has the identity $\square$, since

$$\square \square A = \square \square A = A = A \square \square = A \square \square$$

for any element $A \in P(S)$. Hence $P(S)$ is a monoid. ◆

Example 11 The semigroup S^S defined in Example 4 has the identity 1_S, since

$$1_S \square f = 1_S \circ f = f \circ 1_S = f \square 1_S$$

for any element $f \in S^S$. Thus, we see that S^S is a monoid. ◆

Example 12 The semigroup $A^\square$ defined in Example 6 is actually a monoid with identity $\square$, the empty sequence, since $\alpha \cdot \square = \square \cdot \alpha = \alpha$ for all $\alpha \in A^\square$. ◆

Example 13 The set of all relations on a set A is a monoid under the operation of composition. The identity element is the equality relation $\square$. ◆

Let $(S, \square)$ be a semigroup and let T be a subset of S. If T is closed under the operation $\square$ (that is, $a \square b \in T$ whenever a and b are elements of T), then $(T, \square)$ is called a **subsemigroup** of $(S, \square)$. Similarly, let $(S, \square)$ be a monoid with identity e, and let T be a nonempty subset of S. If T is closed under the operation $\square$ and $e \in T$, then $(T, \square)$ is called a **submonoid** of $(S, \square)$.

Observe that the associative property holds in any subset of a semigroup so that a subsemigroup $(T, \square)$ of a semigroup $(S, \square)$ is itself a semigroup. Similarly, a submonoid of a monoid is itself a monoid.

Example 14 If T is the set of all even integers, then $(T, \times)$ is a subsemigroup of the monoid $(Z, \times)$, where $\times$ is ordinary multiplication, but it is not a submonoid since the identity of Z, the number 1, does not belong to T. ◆

Example 15 If $(S, \square)$ is a semigroup, then $(S, \square)$ is a subsemigroup of $(S, \square)$. Similarly, let $(S, \square)$ be a monoid. Then $(S, \square)$ is a submonoid of $(S, \square)$, and if $T = \{e\}$, then $(T, \square)$ is also a submonoid of $(S, \square)$. ◆

Suppose that $(S, \square)$ is a semigroup, and let $a \in S$. For $n \in Z^+$, we define the powers of a^n recursively as follows:

$$a^1 = a, \quad a^n = a^{n-1} \square a, \qquad n \geq 2.$$

Moreover, if $(S, \Box)$ is a monoid, we also define

$$a^0 = e.$$

It can be shown that if m and n are nonnegative integers, then

$$a^m \Box a^n = a^{m+n}.$$

Example 16 **(a)** If $(S, \Box)$ is a semigroup, $a \in S$, and

$$T = \{a^i \mid i \in Z^+\},$$

then $(T, \Box)$ is a subsemigroup of $(S, \Box)$.

(b) If $(S, \Box)$ is a monoid, $a \in S$, and

$$T = \{a^i \mid i \in Z^+ \text{ or } i = 0\},$$

then $(T, \Box)$ is a submonoid of $(S, \Box)$. ◆

Isomorphism and Homomorphism

An isomorphism between two posets is a one-to-one correspondence that preserved order relations, the distinguishing feature of posets. We now define an isomorphism between two semigroups as a one-to-one correspondence that preserves the binary operations. In general, an isomorphism between two mathematical structures of the same type should preserve the distinguishing features of the structures.

Let $(S, \Box)$ and $(T, \Box')$ be two semigroups. A function $f : S \to T$ is called an **isomorphism** from $(S, \Box)$ to $(T, \Box')$ if it is a one-to-one correspondence from S to T, and if

$$f(a \Box b) = f(a) \Box' f(b)$$

for all a and b in S.

If f is an isomorphism from $(S, \Box)$ to $(T, \Box')$, then, since f is a one-to-one correspondence, it follows that f^{-1} exists and is a one-to-one correspondence from T to S. We now show that f^{-1} is an isomorphism from $(T, \Box')$ to $(S, \Box)$. Let a' and b' be any elements of T. Since f is onto, we can find elements a and b in S such that $f(a) = a'$ and $f(b) = b'$. Then $a = f^{-1}(a')$ and $b = f^{-1}(b')$. Now

$$\begin{aligned} f^{-1}(a' \Box' b') &= f^{-1}(f(a) \Box' f(b)) \\ &= f^{-1}(f(a \Box b)) \\ &= (f^{-1} \circ f)(a \Box b) \\ &= a \Box b = f^{-1}(a') \Box f^{-1}(b'). \end{aligned}$$

Hence f^{-1} is an isomorphism.

We now say that the semigroups $(S, \Box)$ and $(T, \Box')$ are **isomorphic** and we write $S \cong T$.

To show that two semigroups $(S, \Box)$ and $(T, \Box')$ are isomorphic, we use the following procedure:

Step 1 Define a function $f : S \to T$ with $\text{Dom}(f) = S$.

Step 2 Show that f is one-to-one.

Step 3 Show that f is onto.

Step 4 Show that $f(a \Box b) = f(a) \Box' f(b)$.

Example 17 Let T be the set of all even integers. Show that the semigroups $(\mathbb{Z}, +)$ and $(T, +)$ are isomorphic.

Solution

Step 1 We define the function $f : \mathbb{Z} \to T$ by $f(a) = 2a$.

Step 2 We now show that f is one to one as follows. Suppose that $f(a_1) = f(a_2)$. Then $2a_1 = 2a_2$, so $a_1 = a_2$. Hence f is one to one.

Step 3 We next show that f is onto. Suppose that b is any even integer. Then $a = b/2 \in \mathbb{Z}$ and

$$f(a) = f(b/2) = 2(b/2) = b,$$

so f is onto.

Step 4 We have

$$f(a + b) = 2(a + b) = 2a + 2b = f(a) + f(b).$$

Hence $(\mathbb{Z}, +)$ and $(T, +)$ are isomorphic semigroups. ◆

In general, it is rather straightforward to verify that a given function $f : S \to T$ is or is not an isomorphism. However, it is generally more difficult to show that two semigroups are isomorphic, because one has to create the isomorphism f.

As in the case of poset or lattice isomorphisms, when two semigroups $(S, \circ)$ and $(T, \circ')$ are isomorphic, they can differ only in the nature of their elements; their semigroup structures are identical. If S and T are finite semigroups, their respective binary operations can be given by tables. Then S and T are isomorphic if we can rearrange and relabel the elements of S so that its table is identical with that of T.

Example 18 Let $S = \{a, b, c\}$ and $T = \{x, y, z\}$. It is easy to verify that the following operation tables give semigroup structures for S and T, respectively.

$\circ$	a	b	c
a	a	b	c
b	b	c	a
c	c	a	b

$\circ'$	x	y	z
x	z	x	y
y	x	y	z
z	y	z	x

Let

$$f(a) = y$$
$$f(b) = x$$
$$f(c) = z.$$

Replacing the elements in S by their images and rearranging the table, we obtain exactly the table for T. Thus S and T are isomorphic. ◆

THEOREM 2 Let $(S, \circ)$ and $(T, \circ')$ be monoids with identities e and e', respectively. Let $f : S \to T$ be an isomorphism. Then $f(e) = e'$.

Proof
Let b be any element of T. Since f is onto, there is an element a in S such that $f(a) = b$. Then

$$\begin{aligned} a &= a \circ e \\ b = f(a) &= f(a \circ e) = f(a) \circ' f(e) \\ &= b \circ' f(e). \end{aligned}$$

Similarly, since $a = e \circ a$, $b = f(e) \circ' b$. Thus for any $b \in T$,

$$b = b \circ' f(e) = f(e) \circ' b,$$

which means that $f(e)$ is an identity for T. Thus since the identity is unique, it follows that $f(e) = e'$. ■

If $(S, \circ)$ and $(T, \circ')$ are semigroups such that S has an identity and T does not, it then follows from Theorem 2 that $(S, \circ)$ and $(T, \circ')$ cannot be isomorphic.

Example 19 Let T be the set of all even integers and let $\times$ be ordinary multiplication. Then the semigroups $(Z, \times)$ and $(T, \times)$ are not isomorphic, since Z has an identity and T does not. ◆

By dropping the conditions of one to one and onto in the definition of an isomorphism of two semigroups, we get another important method for comparing the algebraic structures of the two semigroups.

Let $(S, \circ)$ and $(T, \circ')$ be two semigroups. An everywhere-defined function $f : S \to T$ is called a **homomorphism** from $(S, \circ)$ to $(T, \circ')$ if

$$f(a \circ b) = f(a) \circ' f(b)$$

for all a and b in S. If f is also onto, we say that T is a **homomorphic image** of S.

Example 20 Let $A = \{0, 1\}$ and consider the semigroups $(A^*, \cdot)$ and $(A, +)$, where $\cdot$ is the catenation operation and $+$ is defined by the table

+	0	1
0	0	1
1	1	0

Define the function $f : A^* \to A$ by

$$f(\alpha) = \begin{cases} 1 & \text{if } \alpha \text{ has an odd number of 1's} \\ 0 & \text{if } \alpha \text{ has an even number of 1's.} \end{cases}$$

It is easy to verify that if α and β are any elements of A^*, then

$$f(\alpha \cdot \beta) = f(\alpha) + f(\beta).$$

Thus f is a homomorphism. The function f is onto since

$$\begin{aligned} f(0) &= 0 \\ f(1) &= 1 \end{aligned}$$

but f is not an isomorphism, since it is not one to one. ◆

The difference between an isomorphism and a homomorphism is that an isomorphism must be one to one and onto. For both an isomorphism and a homomorphism, the image of a product is the product of the images.

The proof of the following theorem, which is left as an exercise for the reader, is completely analogous to the proof of Theorem 2.

THEOREM 3 Let $(S, \circ)$ and $(T, \circ')$ be monoids with identities e and e', respectively. Let $f : S \to T$ be a homomorphism from $(S, \circ)$ onto $(T, \circ')$. Then $f(e) = e'$. ■

Theorem 3 is a stronger, or more general, statement than Theorem 2, because it requires fewer (weaker) conditions for the conclusion.

Theorem 3, together with the following two theorems, shows that, if a semigroup $(T, \circ')$ is a homomorphic image of the semigroup $(S, \circ)$, then $(T, \circ')$ has a strong algebraic resemblance to $(S, \circ)$.

THEOREM 4 Let f be a homomorphism from a semigroup $(S, \circ)$ to a semigroup $(T, \circ')$. If S' is a subsemigroup of $(S, \circ)$, then

$$f(S') = \{t \in T \mid t = f(s) \text{ for some } s \in S'\},$$

the image of S' under f, is a subsemigroup of $(T, \circ')$.

Proof

If t_1 and t_2 are any elements of $f(S')$, then there exist s_1 and s_2 in S' with

$$t_1 = f(s_1) \quad \text{and} \quad t_2 = f(s_2).$$

Then

$$t_1 \circ' t_2 = f(s_1) \circ' f(s_2) = f(s_1 \circ s_2) = f(s_3),$$

where $s_3 = s_1 \circ s_2 \in S'$. Hence $t_1 \circ' t_2 \in f(S')$.

Thus $f(S')$ is closed under the operation $\circ'$. Since the associative property holds in T, it holds in $f(S')$, so $f(S')$ is a subsemigroup of $(T, \circ')$. ■

THEOREM 5 If f is a homomorphism from a commutative semigroup $(S, \circ)$ onto a semigroup $(T, \circ')$, then $(T, \circ')$ is also commutative.

Proof

Let t_1 and t_2 be any elements of T. Then there exist s_1 and s_2 in S with

$$t_1 = f(s_1) \quad \text{and} \quad t_2 = f(s_2).$$

Therefore,

$$t_1 \circ' t_2 = f(s_1) \circ' f(s_2) = f(s_1 \circ s_2) = f(s_2 \circ s_1) = f(s_2) \circ' f(s_1) = t_2 \circ' t_1.$$

Hence $(T, \circ')$ is also commutative. ■

2 Exercises

1. Let $A = \{a, b\}$. Which of the following tables define a semigroup on A? Which define a monoid on A?

(a)

$\circ$	a	b
a	a	b
b	a	a

(b)

$\circ$	a	b
a	a	b
b	b	b

2. Let $A = \{a, b\}$. Which of the following tables define a semigroup on A? Which define a monoid on A?

(a)

$\circ$	a	b
a	b	a
b	a	b

(b)

$\circ$	a	b
a	a	b
b	b	a

3. Let $A = \{a, b\}$. Which of the following tables define a semigroup on A? Which define a monoid on A?

(a)

$\square$	a	b
a	a	a
b	b	b

(b)

$\square$	a	b
a	b	b
b	a	a

In Exercises 4 through 16, determine whether the set together with the binary operation is a semigroup, a monoid, or neither. If it is a monoid, specify the identity. If it is a semigroup or a monoid, determine if it is commutative.

4. Z^+, where $\square$ is defined as ordinary multiplication.

5. Z^+, where $a \square b$ is defined as $\max\{a, b\}$.

6. Z^+, where $a \square b$ is defined as $\text{GCD}\{a, b\}$.

7. Z^+, where $a \square b$ is defined as a.

8. The nonzero real numbers, where $\square$ is ordinary multiplication.

9. $P(S)$, with S a set, where $\square$ is defined as intersection.

10. A Boolean algebra B, where $a \square b$ is defined as $a \square b$.

11. $S = \{1, 2, 3, 6, 12\}$, where $a \square b$ is defined as $\text{GCD}(a, b)$.

12. $S = \{1, 2, 3, 6, 9, 18\}$, where $a \square b$ is defined as $\text{LCM}(a, b)$.

13. Z, where $a \square b = a + b - ab$.

14. The even integers, where $a \square b$ is defined as $\dfrac{ab}{2}$.

15. The set of 2×1 matrices, where

$$\begin{bmatrix} a \\ b \end{bmatrix} \square \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a + c \\ b + d + 1 \end{bmatrix}.$$

16. The set of integers of the form $3k + 1$, $k \in Z^+$, where $\square$ is ordinary multiplication.

17. Does the following table define a semigroup or a monoid?

$\square$	a	b	c
a	c	b	a
b	b	c	b
c	a	b	c

18. Does the following table define a semigroup or a monoid?

$\square$	a	b	c
a	a	c	b
b	c	b	a
c	b	a	c

19. Complete the following table to obtain a semigroup.

$\square$	a	b	c
a	c	a	b
b	a	b	c
c			a

20. Complete the following table so that it defines a monoid.

$\square$	a	b	c	d
a	c	d	a	b
b		a	b	
c			c	
d	b		d	a

21. Let $S = \{a, b\}$. Write the operation table for the semigroup S^S. Is the semigroup commutative?

22. Let $S = \{a, b\}$. Write the operation table for the semigroup $(P(S), \square)$.

23. Let $A = \{a, b, c\}$ and consider the semigroup $(A^\square, \cdot)$, where $\cdot$ is the operation of catenation. If $\alpha = abac$, $\beta = cba$, and $\gamma = babc$, compute

(a) $(\alpha \cdot \beta) \cdot \gamma$ **(b)** $\gamma \cdot (\alpha \cdot \alpha)$ **(c)** $(\gamma \cdot \beta) \cdot \alpha$

24. What is required for a subset of the elements of a semigroup to be a subsemigroup?

25. What is required for a subset of the elements of a monoid to be a submonoid?

26. Prove or disprove that the intersection of two subsemigroups of a semigroup $(S, \square)$ is a subsemigroup of $(S, \square)$.

27. Prove or disprove that the intersection of two submonoids of a monoid $(S, \square)$ is a submonoid of $(S, \square)$.

28. Let $A = \{0, 1\}$, and consider the semigroup $(A^\square, \cdot)$, where $\cdot$ is the operation of catenation. Let T be the subset of $A^\square$ consisting of all sequences having an odd number of 1's. Is $(T, \cdot)$ a subsemigroup of $(A, \cdot)$?

29. Let $A = \{a, b\}$. Are there two semigroups $(A, \square)$ and $(A, \square')$ that are not isomorphic?

30. An element x in a monoid is called an **idempotent** if $x^2 = x \square x = x$. Show that the set of all idempotents in a commutative monoid S is a submonoid of S.

31. Let $(S_1, \square_1)$, $(S_2, \square_2)$, and $(S_3, \square_3)$ be semigroups and $f : S_1 \to S_2$ and $g : S_2 \to S_3$ be homomorphisms. Prove that $g \circ f$ is a homomorphism from S_1 to S_3.

32. Let $(S_1, \square)$, $(S_2, \square')$, and $(S_3, \square'')$ be semigroups, and let $f : S_1 \to S_2$ and $g : S_2 \to S_3$ be isomorphisms. Show that $g \circ f : S_1 \to S_3$ is an isomorphism.

33. Which properties of f are used in the proof of Theorem 2?

34. Explain why the proof of Theorem 1 can be used as a proof of Theorem 3.

35. Let R^+ be the set of all positive real numbers. Show that the function $f : R^+ \to R$ defined by $f(x) = \ln x$ is an isomorphism of the semigroup $(R^+, \times)$ to the semigroup $(R, +)$, where $\times$ and $+$ are ordinary multiplication and addition, respectively.

36. Let $(S, \square)$ be a semigroup and A, a finite subet of S. Define $\hat{A}$ to be the set of all finite products of elements in A.

(a) Prove that $\hat{A}$ is a subsemigroup of $(S, \square)$.

(b) Prove that $\hat{A}$ is the smallest subsemigroup of $(S, \square)$ that contains A.

3 Products and Quotients of Semigroups

In this section we shall obtain new semigroups from existing semigroups.

THEOREM 1 If $(S, \circ)$ and $(T, \circ')$ are semigroups, then $(S \times T, \circ'')$ is a semigroup, where $\circ''$ is defined by $(s_1, t_1) \circ'' (s_2, t_2) = (s_1 \circ s_2, t_1 \circ' t_2)$.

Proof
The proof is left as an exercise. ■

It follows at once from Theorem 1 that if S and T are monoids with identities e_S and e_T, respectively, then $S \times T$ is a monoid with identity (e_S, e_T).

We now turn to a discussion of equivalence relations on a semigroup $(S, \circ)$. Since a semigroup is not merely a set, we shall find that certain equivalence relations on a semigroup give additional information about the structure of the semigroup.

An equivalence relation R on the semigroup $(S, \circ)$ is called a **congruence relation** if

$$a \, R \, a' \quad \text{and} \quad b \, R \, b' \quad \text{imply} \quad (a \circ b) \, R \, (a' \circ b').$$

Example 1 Consider the semigroup $(Z, +)$ and the equivalence relation R on Z defined by

$$a \, R \, b \quad \text{if and only if} \quad a \equiv b \pmod 2.$$

If $a \equiv b \pmod 2$, then $2 \mid (a - b)$. We now show that this relation is a congruence relation as follows.

If

$$a \equiv b \pmod 2 \quad \text{and} \quad c \equiv d \pmod 2,$$

then 2 divides $a - b$ and 2 divides $c - d$, so

$$a - b = 2m \quad \text{and} \quad c - d = 2n,$$

where m and n are in Z. Adding, we have

$$(a - b) + (c - d) = 2m + 2n$$

or

$$(a + c) - (b + d) = 2(m + n),$$

so

$$a + c \equiv b + d \pmod 2.$$

Hence the relation is a congruence relation. ◆

Example 2 Let $A = \{0, 1\}$ and consider the free semigroup $(A^+, \cdot)$ generated by A. Define the following relation on A:

$$\alpha \, R \, \beta \quad \text{if and only if} \quad \alpha \text{ and } \beta \text{ have the same number of 1's.}$$

Show that R is a congruence relation on $(A^+, \cdot)$.

Solution

We first show that R is an equivalence relation. We have

1. $\alpha \, R \, \alpha$ for any $\alpha \in A^+$.

2. If $\alpha \, R \, \beta$, then α and β have the same number of 1's, so $\beta \, R \, \alpha$.

3. If $\alpha \, R \, \beta$ and $\beta \, R \, \gamma$, then α and β have the same number of 1's and β and γ have the same number of 1's, so α and γ have the same number of 1's. Hence $\alpha \, R \, \gamma$.

We next show that R is a congruence relation. Suppose that $\alpha \, R \, \alpha'$ and $\beta \, R \, \beta'$. Then α and α' have the same number of 1's and β and β' have the same number of 1's. Since the number of 1's in $\alpha \cdot \beta$ is the sum of the number of 1's in α and the number of 1's in β, we conclude that the number of 1's in $\alpha \cdot \beta$ is the same as the number of 1's in $\alpha' \cdot \beta'$. Hence

$$(\alpha \cdot \beta) \, R \, (\alpha' \cdot \beta')$$

and thus R is a congruence relation. ◆

Example 3 Consider the semigroup $(Z, +)$, where $+$ is ordinary addition. Let $f(x) = x^2 - x - 2$. We now define the following relation on Z:

$$a \, R \, b \quad \text{if and only if} \quad f(a) = f(b).$$

It is straightforward to verify that R is an equivalence relation on Z. However, R is not a congruence relation since we have

$$-1 \, R \, 2 \qquad \text{since } f(-1) = f(2) = 0$$

and

$$-2 \, R \, 3 \qquad \text{since } f(-2) = f(3) = 4$$

but

$$(-1 + (-2)) \not R \, (2 + 3)$$

since $f(-3) = 10$ and $f(5) = 18$. ◆

An equivalence relation R on the semigroup $(S, \square)$ determines a partition of S. We let $[a] = R(a)$ be the equivalence class containing a and S/R denote the set of all equivalence classes. The notation $[a]$ is more traditional in this setting and produces less confusing computations.

THEOREM 2 Let R be a congruence relation on the semigroup $(S, \square)$. Consider the relation $\square$ from $S/R \times S/R$ to S/R in which the ordered pair $([a], [b])$ is, for a and b in S, related to $[a \square b]$.

(a) $\square$ is a function from $S/R \times S/R$ to S/R, and as usual we denote $\square([a], [b])$ by $[a] \square [b]$. Thus $[a] \square [b] = [a \square b]$.

(b) $(S/R, \square)$ is a semigroup.

Proof

Suppose that $([a], [b]) = ([a'], [b'])$. Then $a \, R \, a'$ and $b \, R \, b'$, so we must have $a \square b \, R \, a' \square b'$, since R is a congruence relation. Thus $[a \square b] = [a' \square b']$; that is, $\square$ is a function. This means that $\square$ is a binary operation on S/R.

Next, we must verify that $\square$ is an associative operation. We have

$$\begin{aligned}
[a] \square ([b] \square [c]) &= [a] \square [b \square c] \\
&= [a \square (b \square c)] \\
&= [(a \square b) \square c] \quad \text{by the associative property of } \square \text{ in } S \\
&= [a \square b] \square [c] \\
&= ([a] \square [b]) \square [c].
\end{aligned}$$

Hence S/R is a semigroup. We call S/R the **quotient semigroup** or **factor semigroup**. Observe that $\square$ is a type of "quotient binary relation" on S/R that is constructed from the original binary relation $\square$ on S by the congruence relation R. ■

Corollary 1 Let R be a congruence relation on the monoid $(S, \square)$. If we define the operation $\square$ in S/R by $[a] \square [b] = [a \square b]$, then $(S/R, \square)$ is a monoid.

Proof
If e is the identity in $(S, \square)$, then it is easy to verify that $[e]$ is the identity in $(S/R, \square)$. ■

Example 4 Consider the situation in Example 2. Since R is a congruence relation on the monoid $S = (A^{\square}, \cdot)$, we conclude that $(S/R, \square)$ is a monoid, where

$$[\alpha] \square [\beta] = [\alpha \cdot \beta].$$ ◆

Example 5 We define the following relation on the semigroup $(Z, +)$:

$$a \; R \; b \quad \text{if and only if} \quad a \equiv b \pmod{n}.$$

We show that R is an equivalence relation and, as in the case of $n = 2$, $a \equiv b \pmod{n}$ implies $n \mid (a - b)$. Thus, if n is 4, then

$$2 \equiv 6 \pmod{4}$$

and 4 divides $(2 - 6)$. We also leave it for the reader to show that $\equiv \pmod{n}$ is a congruence relation on Z.

We now let $n = 4$ and we compute the equivalence classes determined by the congruence relation $\equiv \pmod{4}$ on Z. We obtain

$$\begin{aligned}
[0] &= \{\ldots, -8, -4, 0, 4, 8, 12, \ldots\} = [4] = [8] = \cdots \\
[1] &= \{\ldots, -7, -3, 1, 5, 9, 13, \ldots\} = [5] = [9] = \cdots \\
[2] &= \{\ldots, -6, -2, 2, 6, 10, 14, \ldots\} = [6] = [10] = \cdots \\
[3] &= \{\ldots, -5, -1, 3, 7, 11, 15, \ldots\} = [7] = [11] = \cdots .
\end{aligned}$$

These are all the distinct equivalence classes that form the quotient set $Z/\equiv \pmod{4}$. It is customary to denote the quotient set $Z/\equiv \pmod{n}$ by Z_n; Z_n is a monoid with operation $\square$ and identity $[0]$. We now determine the addition table for the semigroup Z_4 with operation $\square$.

$\square$	$[0]$	$[1]$	$[2]$	$[3]$
$[0]$	$[0]$	$[1]$	$[2]$	$[3]$
$[1]$	$[1]$	$[2]$	$[3]$	$[0]$
$[2]$	$[2]$	$[3]$	$[0]$	$[1]$
$[3]$	$[3]$	$[0]$	$[1]$	$[2]$

The entries in this table are obtained from

$$[a] \square [b] = [a + b].$$

Thus

$$[1] \square [2] = [1 + 2] = [3]$$
$$[1] \square [3] = [1 + 3] = [4] = [0]$$
$$[2] \square [3] = [2 + 3] = [5] = [1]$$
$$[3] \square [3] = [3 + 3] = [6] = [2].$$

It can be shown that, in general, Z_n has the n equivalence classes

$$[0], [1], [2], \ldots, [n - 1]$$

and that

$$[a] \square [b] = [r],$$

where r is the remainder when $a + b$ is divided by n. Thus, if n is 6,

$$[2] \square [3] = [5]$$
$$[3] \square [5] = [2]$$
$$[3] \square [3] = [0].$$ ◆

We shall now examine the connection between the structure of a semigroup $(S, \square)$ and the quotient semigroup $(S/R, \square)$, where R is a congruence relation on $(S, \square)$.

THEOREM 3

Let R be a congruence relation on a semigroup $(S, \square)$, and let $(S/R, \square)$ be the corresponding quotient semigroup. Then the function $f_R : S \to S/R$ defined by

$$f_R(a) = [a]$$

is an onto homomorphism, called the **natural homomorphism**.

Proof

If $[a] \in S/R$, then $f_R(a) = [a]$, so f_R is an onto function. Moreover, if a and b are elements of S, then

$$f_R(a \square b) = [a \square b] = [a] \square [b] = f_R(a) \square f_R(b),$$

so f_R is a homomorphism. ■

THEOREM 4

Fundamental Homomorphism Theorem

Let $f : S \to T$ be a homomorphism of the semigroup $(S, \square)$ onto the semigroup $(T, \square')$. Let R be the relation on S defined by $a \, R \, b$ if and only if $f(a) = f(b)$, for a and b in S. Then

(a) R is a congruence relation.

(b) $(T, \square')$ and the quotient semigroup $(S/R, \square)$ are isomorphic.

Proof

(a) We show that R is an equivalence relation. First, $a \, R \, a$ for every $a \in S$, since $f(a) = f(a)$. Next, if $a \, R \, b$, then $f(a) = f(b)$, so $b \, R \, a$. Finally, if $a \, R \, b$ and $b \, R \, c$, then $f(a) = f(b)$ and $f(b) = f(c)$, so $f(a) = f(c)$ and $a \, R \, c$. Hence R is an equivalence relation. Now suppose that $a \, R \, a_1$ and $b \, R \, b_1$. Then

$$f(a) = f(a_1) \quad \text{and} \quad f(b) = f(b_1).$$

Multiplying in T, we obtain

$$f(a) \square' f(b) = f(a_1) \square' f(b_1).$$

Since f is a homomorphism, this last equation can be rewritten as

$$f(a \square b) = f(a_1 \square b_1).$$

Hence

$$(a \square b) \; R \; (a_1 \square b_1)$$

and R is a congruence relation.

(b) We now consider the relation $\overline{f}$ from S/R to T defined as follows:

$$\overline{f} = \{([a], f(a)) \mid [a] \square S/R\}.$$

We first show that $\overline{f}$ is a function. Suppose that $[a] = [a']$. Then $a \; R \; a'$, so $f(a) = f(a')$, which implies that $\overline{f}$ is a function. We may now write $\overline{f} : S/R \to T$, where $\overline{f}([a]) = f(a)$ for $[a] \square S/R$.

We next show that $\overline{f}$ is one to one. Suppose that $\overline{f}([a]) = \overline{f}([a'])$. Then

$$f(a) = f(a').$$

So $a \; R \; a'$, which implies that $[a] = [a']$. Hence $\overline{f}$ is one to one.

Now we show that $\overline{f}$ is onto. Suppose that $b \square T$. Since f is onto, $f(a) = b$ for some element a in S. Then

$$\overline{f}([a]) = f(a) = b.$$

So $\overline{f}$ is onto.

Finally,

$$\begin{aligned} \overline{f}([a] \square [b]) &= \overline{f}([a \square b]) \\ &= f(a \square b) = f(a) \square f(b) \\ &= \overline{f}([a]) \square \overline{f}([b]). \end{aligned}$$

Hence $\overline{f}$ is an isomorphism. ■

Example 6 Let $A = \{0, 1\}$, and consider the free semigroup $A^\square$ generated by A under the operation of catenation. Note that $A^\square$ is a monoid with the empty string $\square$ as its identity. Let N be the set of all nonnegative integers. Then N is a semigroup under the operation of ordinary addition, denoted by $(N, +)$. The function $f : A^\square \to N$ defined by

$$f(\alpha) = \text{the number of 1's in } \alpha$$

is readily checked to be a homomorphism. Let R be the following relation on $A^\square$:

$$\alpha \; R \; \beta \quad \text{if and only if} \quad f(\alpha) = f(\beta).$$

That is, $\alpha \; R \; \beta$ if and only if α and β have the same number of 1's. Theorem 4 implies that $A^\square/R \square N$ under the isomorphism $\overline{f} : A^\square/R \to N$ defined by

$$\overline{f}([\alpha]) = f(\alpha) = \text{the number of 1's in } \alpha.$$ ◆

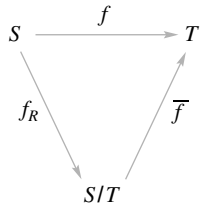


Figure 2

Theorem 4(b) can be described by the diagram shown in Figure 2. Here f_R is the natural homomorphism. It follows from the definitions of f_R and $\overline{f}$ that

$$\overline{f} \circ f_R = f$$

since

$$(\overline{f} \circ f_R)(a) = \overline{f}(f_R(a)) = \overline{f}([a]) = f(a).$$

3 Exercises

1. Let $(S, \square)$ and $(T, \square')$ be commutative semigroups. Show that $S \times T$ (see Theorem 1) is also a commutative semigroup.

2. Let $(S, \square)$ and $(T, \square')$ be monoids. Show that $S \times T$ is also a monoid. Show that the identity of $S \times T$ is (e_S, e_T).

3. Let $(S, \square)$ and $(T, \square')$ be semigroups. Show that the function $f : S \times T \to S$ defined by $f(s, t) = s$ is a homomorphism of the semigroup $S \times T$ onto the semigroup S.

4. Let $(S, \square)$ and $(T, \square')$ be semigroups. Show that $S \times T$ and $T \times S$ are isomorphic semigroups.

5. Prove Theorem 1.

In Exercises 6 through 16, determine whether the relation R *on the semigroup* S *is a congruence relation.*

6. $S = Z$ under the operation of ordinary addition; $a \mathrel{R} b$ if and only if 2 does not divide $a - b$.

7. $S = Z$ under the operation of ordinary addition; $a \mathrel{R} b$ if and only if $a + b$ is even.

8. S is any semigroup; $a \mathrel{R} b$ if and only if $a = b$.

9. S is the set of all rational numbers under the operation of addition; $a/b \mathrel{R} c/d$ if and only if $ad = bc$.

10. S is the set of all rational numbers under the operation of multiplication; $a/b \mathrel{R} c/d$ if and only if $ad = bc$.

11. $S = Z$ under the operation of ordinary addition; $a \mathrel{R} b$ if and only if $a \equiv b \pmod 3$.

12. $S = Z$ under the operation of ordinary addition; $a \mathrel{R} b$ if and only if a and b are both even or a and b are both odd.

13. $S = Z^+$ under the operation of ordinary multiplication; $a \mathrel{R} b$ if and only if $|a - b| \leq 2$.

14. $A = \{0, 1\}$ and $S = A^{\square}$, the free semigroup generated by A under the operation of catenation; $\alpha \mathrel{R} \beta$ if and only if α and β both have an even number of 1's or both have an odd number of 1's.

15. $S = \{0, 1\}$ under the operation $\square$ defined by the table

$\square$	0	1
0	0	1
1	1	0

$a \mathrel{R} b$ if and only if $a \square a = b \square b$. (*Hint*: Observe that if x is any element in S, then $x \square x = 0$.)

16. $S = \{3k + 1, k \in Z^+\}$ under the operation of ordinary multiplication; $a \mathrel{R} b$ if and only if $a \equiv b \pmod 5$.

17. Describe the quotient semigroup for S and R given in Exercise 16.

18. Show that the intersection of two congruence relations on a semigroup is a congruence relation.

19. Show that the composition of two congruence relations on a semigroup need not be a congruence relation.

20. Describe the quotient semigroup for S and R given in Exercise 10.

21. Describe the quotient semigroup for S and R given in Exercise 11.

22. Describe the quotient semigroup for S and R given in Exercise 12.

23. Describe the quotient semigroup for $S = Z$ with ordinary addition and R defined by $a \mathrel{R} b$ if and only if $a \equiv b \pmod 5$.

24. Consider the semigroup $S = \{a, b, c, d\}$ with the following operation table.

$\square$	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	a	b
d	d	c	b	a

Consider the congruence relation $R = \{(a, a), (a, b), (b, a), (b, b), (c, c), (c, d), (d, c), (d, d)\}$ on S.

(a) Write the operation table of the quotient semigroup S/R.

(b) Describe the natural homomorphism $f_R : S \to S/R$.

25. Consider the monoid $S = \{e, a, b, c\}$ with the following operation table.

$\square$	e	a	b	c
e	e	a	b	c
a	a	e	b	c
b	b	c	b	c
c	c	b	b	c

Consider the congruence relation $R = \{(e, e), (e, a), (a, e), (a, a), (b, b), (b, c), (c, b), (c, c)\}$ on S.

(a) Write the operation table of the quotient monoid S/R.

(b) Describe the natural homomorphism $f_R : S \to S/R$.

26. Let $A = \{0, 1\}$ and consider the free semigroup $A^{\square}$ generated by A under the operation of catenation. Let N be the semigroup of all nonnegative integers under the operation of ordinary addition.

(a) Verify that the function $f : A^{\square} \to N$, defined by $f(\alpha) =$ the number of digits in α, is a homomorphism.

(b) Let R be the following relation on $A^{\square}$: $\alpha \mathrel{R} \beta$ if and only if $f(\alpha) = f(\beta)$. Show that R is a congruence relation on $A^{\square}$.

(c) Show that $A^{\square}/R$ and N are isomorphic.

27. Prove or disprove that Z_2 is isomorphic to the semigroup in Exercise 22.

28. Prove or disprove that Z_4 is isomorphic to the semigroup in Exercise 24.

29. Describe the strategy of the proof of Theorem 4. Outline the proof.

30. Let S be a nonempty set with $a \square b = b$. Prove that any equivalence relation on S is a congruence relation.

4 Groups

In this section we examine a special type of monoid, called a group, that has applications in every area where symmetry occurs. Applications of groups can be found in mathematics, physics, and chemistry, as well as in less obvious areas such as sociology. Recent and exciting applications of group theory have arisen in fields such as particle physics and in the solutions of puzzles such as Rubik's cube.

A **group** $(G, \square)$ is a monoid, with identity e, that has the additional property that for every element $a \square G$ there exists an element $a' \square G$ such that $a \square a' = a' \square a = e$. Thus a group is a set together with a binary operation $\square$ on G such that

1. $(a \square b) \square c = a \square (b \square c)$ for any elements a, b, and c in G.

2. There is a unique element e in G such that

$$a \square e = e \square a \qquad \text{for any } a \square G.$$

3. For every $a \square G$, there is an element $a' \square G$, called an **inverse** of a, such that

$$a \square a' = a' \square a = e.$$

Observe that if $(G, \square)$ is a group, then $\square$ is a binary operation, so G must be closed under $\square$; that is,

$$a \square b \square G \qquad \text{for any elements } a \text{ and } b \text{ in } G.$$

To simplify our notation, from now on when only one group $(G, \square)$ is under consideration and there is no possibility of confusion, we shall write the product $a \square b$ of the elements a and b in the group $(G, \square)$ simply as ab, and we shall also refer to $(G, \square)$ simply as G.

A group G is said to be **Abelian** if $ab = ba$ for all elements a and b in G.

Example 1 The set of all integers Z with the operation of ordinary addition is an Abelian group. If $a \square Z$, then an inverse of a is its opposite $-a$. ◆

Example 2 The set Z^+ under the operation of ordinary multiplication is not a group since, for example, the element 2 in Z^+ has no inverse. However, this set together with the given operation is a monoid. ◆

Example 3 The set of all nonzero real numbers under the operation of ordinary multiplication is a group. An inverse of $a \boxminus 0$ is $1/a$. ◆

Example 4 Let G be the set of all nonzero real numbers and let

$$a \square b = \frac{ab}{2}.$$

Show that $(G, \square)$ is an Abelian group.

Solution

We first verify that $\ast$ is a binary operation. If a and b are elements of G, then $ab/2$ is a nonzero real number and hence is in G. We next verify associativity. Since

$$(a \ast b) \ast c = \left(\frac{ab}{2}\right) \ast c = \frac{(ab)c}{4}$$

and since

$$a \ast (b \ast c) = a \ast \left(\frac{bc}{2}\right) = \frac{a(bc)}{4} = \frac{(ab)c}{4},$$

the operation $\ast$ is associative.

The number 2 is the identity in G, for if $a \in G$, then

$$a \ast 2 = \frac{(a)(2)}{2} = a = \frac{(2)(a)}{2} = 2 \ast a.$$

Finally, if $a \in G$, then $a' = 4/a$ is an inverse of a, since

$$a \ast a' = a \ast \frac{4}{a} = \frac{a(4/a)}{2} = 2 = \frac{(4/a)(a)}{2} = \frac{4}{a} \ast a = a' \ast a.$$

Since $a \ast b = b \ast a$ for all a and b in G, we conclude that G is an Abelian group. ◆

Before proceeding with additional examples of groups, we develop several important properties that are satisfied in any group G.

THEOREM 1 Let G be a group. Each element a in G has only one inverse in G.

Proof

Let a' and a'' be inverses of a. Then

$$a'(aa'') = a'e = a'$$

and

$$(a'a)a'' = ea'' = a''.$$

Hence, by associativity,

$$a' = a''.$$ ■

From now on we shall denote the inverse of a by a^{-1}. Thus in a group G we have

$$aa^{-1} = a^{-1}a = e.$$

THEOREM 2 Let G be a group and let a, b, and c be elements of G. Then

(a) $ab = ac$ implies that $b = c$ (**left cancellation property**).

(b) $ba = ca$ implies that $b = c$ (**right cancellation property**).

Proof

(a) Suppose that

$$ab = ac.$$

Multiplying both sides of this equation by a^{-1} on the left, we obtain

$$\begin{aligned} a^{-1}(ab) &= a^{-1}(ac) \\ (a^{-1}a)b &= (a^{-1}a)c && \text{by associativity} \\ eb &= ec && \text{by the definition of an inverse} \\ b &= c && \text{by definition of an identity.} \end{aligned}$$

(b) The proof is similar to that of part (a). ■

Corollary 1 Let G be a group and $a \in G$. Define a function $M_a: G \to G$ by the formula $M_a(g) = ag$. Then M_a is one to one.

Proof
This is a direct consequence of Theorem 2. ■

THEOREM 3 Let G be a group and let a and b be elements of G. Then

(a) $(a^{-1})^{-1} = a$.
(b) $(ab)^{-1} = b^{-1}a^{-1}$.

Proof

(a) We show that a acts as an inverse for a^{-1}:

$$a^{-1}a = aa^{-1} = e.$$

Since the inverse of an element is unique, we conclude that $(a^{-1})^{-1} = a$.

(b) We easily verify that

$$(ab)(b^{-1}a^{-1}) = a(b(b^{-1}a^{-1})) = a((bb^{-1})a^{-1}) = a(ea^{-1}) = aa^{-1} = e$$

and, similarly,

$$(b^{-1}a^{-1})(ab) = e,$$

so

$$(ab)^{-1} = b^{-1}a^{-1}.$$ ■

THEOREM 4 Let G be a group, and let a and b be elements of G. Then

(a) The equation $ax = b$ has a unique solution in G.
(b) The equation $ya = b$ has a unique solution in G.

Proof

(a) The element $x = a^{-1}b$ is a solution of the equation $ax = b$, since

$$a(a^{-1}b) = (aa^{-1})b = eb = b.$$

Suppose now that x_1 and x_2 are two solutions of the equation $ax = b$. Then

$$ax_1 = b \quad \text{and} \quad ax_2 = b.$$

Hence

$$ax_1 = ax_2.$$

Theorem 2 implies that $x_1 = x_2$.

(b) The proof is similar to that of part (a). ■

From our discussion of monoids, we know that if a group G has a finite number of elements, then its binary operation can be given by a table, which is generally called a **multiplication table**. The multiplication table of a group $G = \{a_1, a_2, \ldots, a_n\}$ under the binary operation $\square$ must satisfy the following properties:

1. The row labeled by e must be

$$a_1, a_2, \ldots, a_n$$

and the column labeled by e must be

$$\begin{matrix} a_1 \\ a_2 \\ \vdots \\ a_n. \end{matrix}$$

2. From Theorem 4, it follows that each element b of the group must appear exactly once in each row and column of the table. Thus each row and column is a permutation of the elements $a_1, a_2, \ldots, a_n$ of G, and each row (and each column) determines a different permutation.

TABLE 1

	e	a
e	e	a
a	a	

TABLE 2

	e	a
e	e	a
a	a	e

TABLE 3

	e	a	b
e	e	a	b
a	a		
b	b		

TABLE 4

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

If G is a group that has a finite number of elements, we say that G is a **finite group**, and the **order** of G is the number of elements $|G|$ in G. We shall now determine the multiplication tables of all nonisomorphic groups of orders 1, 2, 3, and 4.

If G is a group of order 1, then $G = \{e\}$, and we have $ee = e$. Now let $G = \{e, a\}$ be a group of order 2. Then we obtain a multiplication table (Table 1) where we need to fill in the blank. The blank can be filled in by e or by a. Since there can be no repeats in any row or column, we must write e in the blank. The multiplication table shown in Table 2 satisfies the associativity property and the other properties of a group, so it is the multiplication table of a group of order 2.

Next, let $G = \{e, a, b\}$ be a group of order 3. We have a multiplication table (Table 3) where we must fill in four blanks. A little experimentation shows that we can only complete the table as shown in Table 4. It can be shown (a tedious task) that Table 4 satisfies the associative property and the other properties of a group. Thus it is the multiplication table of a group of order 3. Observe that the groups of orders 1, 2, and 3 are also Abelian and that there is just one group of each order for a fixed labeling of the elements.

We next come to a group $G = \{e, a, b, c\}$ of order 4. It is not difficult to show that the possible multiplication table for G can be completed as shown in Tables 5 through 8. It can be shown that each of these tables satisfies the associative property and the other properties of a group. Thus there are four possible multiplication tables for a group of order 4. Again, observe that a group of order 4 is Abelian. We shall return to groups of order 4 toward the end of this section, where

TABLE 5

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

TABLE 6

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

TABLE 7

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

TABLE 8

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

we shall see that there are only two and not four different nonisomorphic groups of order 4.

Example 5 Let $B = \{0, 1\}$, and let $+$ be the operation defined on B as follows:

$+$	0	1
0	0	1
1	1	0

Then B is a group. In this group, every element is its own inverse. ◆

We next turn to an important example of a group.

Example 6 Consider the equilateral triangle shown in Figure 3 with vertices 1, 2, and 3. A **symmetry** of the triangle (or of any geometrical figure) is a one-to-one correspondence from the set of points forming the triangle (the geometrical figure) to itself that preserves the distance between adjacent points. Since the triangle is determined by its vertices, a symmetry of the triangle is merely a permutation of the vertices that preserves the distance between adjacent points. Let l_1, l_2, and l_3 be the angle bisectors of the corresponding angles as shown in Figure 3, and let O be their point of intersection.

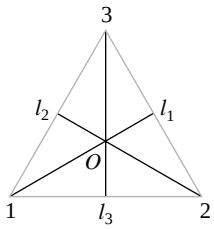


Figure 3

We now describe the symmetries of this triangle. First, there is a counterclockwise rotation f_2 of the triangle about O through 120°. Then f_2 can be written as the permutation

$$f_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}.$$

We next obtain a counterclockwise rotation f_3 about O through 240°, which can be written as the permutation

$$f_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

Finally, there is a counterclockwise rotation f_1 about O through 360°, which can be written as the permutation

$$f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

Of course, f_1 can also be viewed as the result of rotating the triangle about O through 0°.

We may also obtain three additional symmetries of the triangle, g_1, g_2, and g_3, by reflecting about the lines l_1, l_2, and l_3, respectively. We may denote these reflections as the following permutations:

$$g_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \quad g_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \quad g_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}.$$

Observe that the set of all symmetries of the triangle is described by the set of permutations of the set $\{1, 2, 3\}$, which is denoted by S_3. Thus

$$S_3 = \{f_1, f_2, f_3, g_1, g_2, g_3\}.$$

We now introduce the operation $\square$, *followed by*, on the set S_3, and we obtain the multiplication table shown in Table 9. Each of the entries in this table can be obtained in one of two ways: algebraically or geometrically. For example,

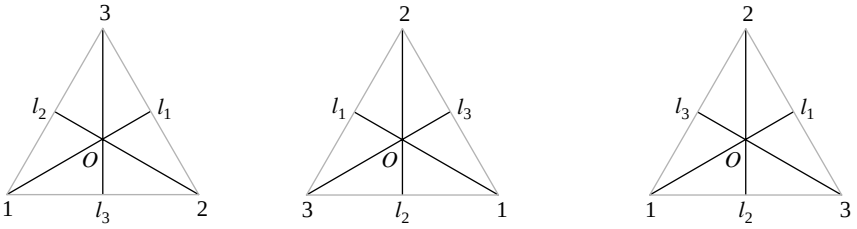
TABLE 9

$\square$	f_1	f_2	f_3	g_1	g_2	g_3
f_1	f_1	f_2	f_3	g_1	g_2	g_3
f_2	f_2	f_3	f_1	g_3	g_1	g_2
f_3	f_3	f_1	f_2	g_2	g_3	g_1
g_1	g_1	g_2	g_3	f_1	f_2	f_3
g_2	g_2	g_3	g_1	f_3	f_1	f_2
g_3	g_3	g_1	g_2	f_2	f_3	f_1

suppose that we want to compute $f_2 \square g_2$. Geometrically, we proceed as in Figure 4. Note that "followed by" here refers to the geometric order. To compute $f_2 \square g_2$ algebraically, we compute $f_2 \circ g_2$,

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = g_1$$

and find that $f_2 \square g_2 = g_1$.



Given triangle

Triangle resulting after applying f_2

Triangle resulting after applying g_2 to the triangle at the left

Figure 4

Since composition of functions is always associative, we see that $\square$ is an associative operation on S_3. Observe that f_1 is the identity in S_3 and that every element of S_3 has a unique inverse in S_3. For example, $f_2^{-1} = f_3$. Hence S_3 is a group called the **group of symmetries of the triangle**. Observe that S_3 is the first example that we have given of a group that is not Abelian. ◆

Example 7 The set of all permutations of n elements is a group of order $n!$ under the operation of composition. This group is called the **symmetric group on *n* letters** and is denoted by S_n. We have seen that S_3 also represents the group of symmetries of the equilateral triangle. ◆

As in Example 6, we can also consider the group of symmetries of a square. However, it turns out that this group is of order 8, so it does not agree with the group S_4, whose order is $4! = 24$.

Example 8 In Section 3 we discussed the monoid Z_n. We now show that Z_n is a group as follows. Let $[a] \square Z_n$. Then we may assume that $0 \le a < n$. Moreover, $[n - a] \square Z_n$ and since

$$[a] \square [n - a] = [a + n - a] = [n] = [0],$$

we conclude that $[n - a]$ is the inverse of $[a]$. Thus, if n is 6, then $[2]$ is the inverse of $[4]$. Observe that Z_n is an Abelian group. ◆

(b) Determine all left cosets of the subgroup $\{1, 9\}$.

37. Let G be a finite group and H a subgroup of G. Prove that the set of distinct left cosets of H is a partition of G.

38. Use the results of Exercises 27 and 37 to describe the relationship between the order of H and the order of G.

6 Other Mathematical Structures

Rings

In earlier sections, we have seen many cases where a set S has two binary operations defined on it. Here we study such structures in more detail. In particular, let S be a nonempty set with two binary operations $+$ and $\square$ such that $(S, +)$ is an Abelian group and $\square$ is distributive over $+$. (The operation symbols are the same as those for the most well-known such structure, the real numbers.) The structure $(S, +, \square)$ is called a **ring** if $\square$ is associative. If $\square$ is associative and commutative, we call $(S, +, \square)$ a **commutative ring**. If $(S, \square)$ is a monoid, then $(S, +, \square)$ is a **ring with identity**. The identity for $\square$ is usually denoted by 1; the identity for $+$ is usually denoted by 0.

Example 1 Let $S = Z$, the integers, and let $+$ and $\square$ be the usual addition and multiplication of integers. Then $(S, +, \square)$ is a commutative ring with identity. ◆

Example 2 Let S be the set of all 2×2 matrices, and let $+$ and $\square$ be the operations of addition and multiplication of matrices. Then it follows that S is a noncommutative ring. Let $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then I is an identity for matrix multiplication, that is, $AI = IA = A$ for all A in S. This means that $(S, +, \square)$ is a ring with identity that is not commutative. ◆

Recall that if a, b, and n are integers, with $n > 1$, then we say that a is congruent to b mod n, written $a \equiv b \pmod{n}$, if $a - b$ is a multiple of n, or, alternatively, if a and b have the same remainder when divided by n. We showed in Section 4 that congruence mod n is an equivalence relation on the integers and that the set Z_n consisting of all equivalence classes is an Abelian group with respect to addition mod n. If we denote the equivalence class of an integer a by the expression $\bar{a}$, then $Z_n = \{\bar{0}, \bar{1}, \bar{2}, \ldots, \overline{n-1}\}$, and $\bar{a} + \bar{b} = \overline{a + b}$.

We now define a multiplication in Z_n. Suppose that a, b, x, and y are integers and that $a \equiv x \pmod{n}$ and $b \equiv y \pmod{n}$. These assumptions imply that for some integers s and t, we have $a = x + sn$ and $b = y + tn$. Then $ab = xy + xtn + ysn + stn^2$, which means that $ab - xy = n(xt + ys + stn)$, so $ab \equiv xy \pmod{n}$. Thus we can define $\bar{a} \square \bar{b}$ to be $\overline{ab}$ and the definition does not depend on the integers picked to represent each equivalence class.

Example 3 The set Z_n with addition mod n and the multiplication defined previously is a commutative ring with identity. The computations

$$(\bar{a} \square \bar{b}) \square \bar{c} = \overline{ab} \square \bar{c} = \overline{(ab)c} = \overline{a(bc)} = \bar{a} \square \overline{bc} = \bar{a} \square (\bar{b} \square \bar{c})$$

and

$$\begin{aligned} \bar{a} \square (\bar{b} + \bar{c}) &= \bar{a} \square \overline{(b + c)} \\ &= \overline{a(b + c)} = \overline{ab + ac} = \overline{ab} + \overline{ac} = (\bar{a} \square \bar{b}) + (\bar{a} \square \bar{c}) \end{aligned}$$

show that multiplication is associative and distributive over addition. In a similar way we can prove that multiplication is associative and that $\bar{1}$ is the identity for multiplication. ◆

Generally, we will refer to $+$ and $\cdot$ as addition and multiplication even when they are not the usual operations with these names.

Many properties of the ring of integers are true for any commutative ring with identity. Two examples are given in the next theorem.

THEOREM 1 Let R be a commutative ring with additive identity 0 and multiplicative identity 1. Then

(a) For any x in R, $0 \cdot x = 0$.

(b) For any x in R, $-x = (-1) \cdot x$.

Proof

(a) Let y denote the element $0 \cdot x$. Since R is a ring, we have

$$y + y = 0 \cdot x + 0 \cdot x = (0 + 0) \cdot x = 0 \cdot x = y.$$

But $(R, +)$ is an Abelian group, so

$$0 = (-y) + y = (-y) + (y + y) = [(-y) + y] + y = 0 + y = y,$$

which shows part (a).

(b) Since $x + ((-1) \cdot x) = (1 \cdot x) + ((-1) \cdot x) = (1 + (-1)) \cdot x = 0 \cdot x = 0$, part (b) follows. ■

In the proof of Theorem 1(b), we use the fact that an inverse in an Abelian group is unique, so that if an element behaves as an inverse, then it must be an inverse.

A nonzero element x of a commutative ring R with identity 1 is said to have a multiplicative inverse y if $x \cdot y = y \cdot x = 1$. If such a y exists, it is unique. We therefore speak of *the* multiplicative inverse of x and denote it by x^{-1}, or sometimes by $1/x$.

The only integers with inverses in Z are 1 and -1, but the situation in the rings Z_n is different. We can show that if a is relatively prime to n, that is, if $\text{GCD}(a, n) = 1$, then $\bar{a}$ has a multiplicative inverse in Z_n. In fact, it follows that there are integers k and s satisfying the equation $ak + ns = 1$, or $1 - ak = ns$. This means that $\bar{1} = \overline{ak} = \bar{a} \cdot \bar{k}$, and we see that $\bar{a}$ has the multiplicative inverse $\bar{k}$.

Example 4 The integer 25 is relatively prime to 384, so $\overline{25}$ has a multiplicative inverse in Z_{384}. To find it, we use a Euclidean algorithm.

$$\begin{aligned} 384 &= 15 \times 25 + 9 \\ 25 &= 2 \times 9 + 7 \\ 9 &= 1 \times 7 + 2 \\ 7 &= 3 \times 2 + 1 \end{aligned}$$

By successive substitutions, we get

$$\begin{aligned} 1 &= 7 - 3 \cdot 2 = 7 - 3(9 - 7) = (4 \cdot 7) - (3 \cdot 9) \\ &= 4(25 - 2 \cdot 9) - (3 \cdot 9) = (4 \cdot 25) - (11 \cdot 9) \\ &= (4 \cdot 25) - 11(384 - 15 \cdot 25) = (169 \cdot 25) - 11 \cdot (384). \end{aligned}$$

This shows that $169 \cdot 25 \equiv 1 \pmod{384}$, so the multiplicative inverse of $\overline{25}$ in Z_{384} is $\overline{169}$. ◆

Fields

Suppose that F is a commutative ring with identity. We say that F is a **field** if every nonzero element x in F has a multiplicative inverse. In the following table, we summarize the properties of a field F.

Field Properties

F has two binary operations: an addition $+$ and a multiplication $\cdot$, and two special elements denoted 0 and 1, so that for all x, y, and z in F,

(1) $x + y = y + x$ **(2)** $x \cdot y = y \cdot x$

(3) $(x + y) + z = x + (y + z)$ **(4)** $(x \cdot y) \cdot z = x \cdot (y \cdot z)$

(5) $x + 0 = x$ **(6)** $x \cdot 1 = x$

(7) $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ **(8)** $(y + z) \cdot x = (y \cdot x) + (z \cdot x)$

(9) For each x in F there is a unique element in F denoted by $-x$ so that $x + (-x) = 0$.

(10) For each $x \neq 0$ in F there is a unique element in F denoted by x^{-1} so that $x \cdot x^{-1} = 1$.

Example 5 The collection R of all real numbers, with ordinary addition and multiplication, is a field. Here $x^{-1} = 1/x$. The field properties shown in the preceding table are the standard rules of arithmetic. ◆

Example 6 The collection Q of all rational numbers, with ordinary addition and multiplication, is a field. ◆

The preceding examples are typical of fields. Fields obey virtually all the familiar rules of arithmetic and algebra, and most algebraic techniques can be used in any field. Remarkably, there are fields with only a finite number of elements. The following theorem introduces the finite fields most important to our future discussions.

THEOREM 2 The ring Z_n is a field when n is a prime.

Proof

Recall that n is a prime if it has no divisors other than itself and 1. If $\bar{a}$ is any nonzero element of Z_n, then a is not divisible by n, so $\text{GCD}(a, n) = 1$. It follows from the discussion preceding Example 4 that $\bar{a}$ has a multiplicative inverse, so Z_n is a field. ■

Example 7 By Theorem 2, $Z_5 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}\}$ is a field. Since $2 + 3 = 5$, we have $\bar{2} + \bar{3} = \bar{0}$, so $-\bar{2} = \bar{3}$ and $-\bar{3} = \bar{2}$. Similarly, $-\bar{4} = \bar{1}$ and $-\bar{1} = \bar{4}$. For notational convenience, we denote the multiplicative inverse of a nonzero element $\bar{a}$ in this field by $\frac{1}{\bar{a}}$, and the product of elements $\bar{a}$ and $\bar{b}$ by $\bar{a} \cdot \bar{b}$. Then, since $2 \cdot 3 = 6 = 1 \cdot 5 + 1$, we see that $\bar{2} \cdot \bar{3} = \bar{1}$. Thus $\frac{1}{\bar{2}} = \bar{3}$ and $\frac{1}{\bar{3}} = \bar{2}$. Similarly, since $4 \cdot 4 = 16 = 3 \cdot 5 + 1$, we

have $\frac{1}{\bar{4}} = \bar{4}$ and, as in the real number field, $\bar{1}$ is also its own multiplicative inverse. We can use these facts in the same way we would for real numbers. For example, suppose we want to solve the following system of equations simultaneously:

$$\begin{cases} \bar{3}x + \bar{2}y = \bar{4} \\ \bar{2}x + \bar{4}y = \bar{2}. \end{cases}$$

We could begin by multiplying the first equation by $\frac{1}{\bar{3}} = \bar{2}$, to obtain $x + \bar{4}y = \bar{3}$ (since $\bar{2} \cdot \bar{4} = \bar{3}$), or $x = \bar{3} - \bar{4}y = \bar{3} + (-\bar{4})y = \bar{3} + y$, using Theorem 1(b). We then substitute for x in the second equation and obtain

$$\bar{2} \cdot (\bar{3} + y) + \bar{4}y = \bar{1} + y = \bar{2},$$

where we have used the facts that $\bar{2} \cdot \bar{3} = \bar{1}$ and $\bar{2} + \bar{4} = \bar{1}$. We see that $y = \bar{1}$, so $x = \bar{4}$. ◆

The reader is invited to check this result by substituting into the system of equations.

Fermat's Little Theorem

An important property of any field F is that the set F^* of nonzero elements of F is an Abelian group under multiplication. We need to show that F^* is closed under multiplication, that is, that the product of nonzero elements of F is nonzero. Then the result will follow from properties (2), (4), (6), and (10) of fields. Suppose that $a \cdot b = 0$ in F. If a is not 0, then we can multiply both sides of the equation $a \cdot b = 0$ by a^{-1} and obtain

$$b = a^{-1} \cdot 0 = 0$$

by Theorem 1(a). Thus either a or b must be 0. It follows that the product of nonzero elements in F is nonzero, and thus F^* is closed under multiplication and is therefore an Abelian group.

The following result has many mathematical uses.

THEOREM 3

(a) If $G = \{g_1, g_2, \ldots, g_n\}$ is a finite Abelian group with identity denoted by e, and a is any element of G, then $a^n = e$.

(b) Fermat's Little Theorem: If p is a prime number, and $\text{GCD}(a, p) = 1$, then $a^{p-1} \equiv 1 \pmod{p}$.

(c) If p is a prime number, and a is any integer, then $a^p \equiv a \pmod{p}$.

Proof

(a) Corollary 1 in Section 4 shows that multiplication by an element in a group is a one-to-one function. Therefore, the products $ag_1, ag_2, \ldots, ag_n$ are all distinct, and are simply the elements $g_1, g_2, \ldots, g_n$ possibly arranged in a different order. It follows from this and the commutativity of multiplication in G that

$$g_1 g_2 \cdots g_n = (ag_1)(qg_2) \cdots (ag_n) = g_1 g_2 \cdots g_n (a^n).$$

Part (a) results from multiplying each side of this equation on the left by $(g_1 g_2 \cdots g_n)^{-1}$.

(b) If p is a prime, then Z_p is a field by Theorem 2, so the nonzero elements form an Abelian group under multiplication. The identity of this group is $\bar{1}$. Since this group has $p - 1$ elements, part (a) implies that if $\bar{a} \neq \bar{0}$, then $[\bar{a}]^{p-1} = \bar{1}$. This is equivalent to part (b).

(c) If a is not divisible by p, then we can apply Fermat's Little Theorem, and the result follows by multiplying both sides of the congruence by a. If a is divisible by p, then $a^p \equiv 0 \pmod{p}$ and $a \equiv 0 \pmod{p}$, so a^p and a are congruent to one another. ■

Example 8 By Fermat's Little Theorem, $12^{30} \equiv 1 \pmod{31}$ and $74^{83} \equiv 74 \pmod{83}$. ◆

Example 9 What is the remainder when 4^{900} is divided by 53?

Solution

We know by Fermat's Little Theorem that $4^{52} \equiv 1 \pmod{53}$. Since

$$900 = (17 \times 52) + 16,$$

we have

$$4^{900} = 4^{(17\times 52)+16} = (4^{52})^{17}4^{16} \equiv 4^{16} \pmod{53}.$$

Now

$$4^3 = 64 \equiv 11 \pmod{53}$$
$$4^6 \equiv 11^2 \equiv 15 \pmod{53}$$
$$4^{12} \equiv 15^2 \equiv 13 \pmod{53}$$
$$4^{16} \equiv 4^{12} \cdot 128 \equiv 13 \cdot 22 \equiv 21 \pmod{53}.$$

Thus the remainder after dividing 4^{900} by 53 is 21. ◆

6 Exercises

In Exercises 1 through 6, determine if the mathematical structure given is a ring, a commutative ring, or a ring with identity.

1. (2×2 matrices, $+$, $\square$)

2. ($n \times n$ diagonal matrices, $+$, $\square$)

3. $n \times n$ Boolean matrices, where $+$ is $\square$ and $\square$ is $\square$.

4. $S = \{0, 1\}$ where $+$ and $\square$ are defined by the following tables:

$+$	0	1
0	0	1
1	1	0

$\square$	0	1
0	0	0
1	0	1

5. $S = \{a + b\sqrt{2}, a, b \square Z\}$, where $+$ and $\square$ are ordinary addition and multiplication.

6. $S = \{a + b\sqrt{5}, a, b \square Z\}$, where $+$ and $\square$ are ordinary addition and multiplication.

A ring R *has* ***zero divisors*** *if there exist elements a and b in R such that* $a \neq 0, b \neq 0$, *and* $a \square b = 0$.

7. Show that (2×2 matrices, $+$, $\square$) is a ring with zero divisors.

8. Show that Z_{10} is a ring with zero divisors.

An element of a ring R is called a ***unit*** *of* R *if* r *has a multiplicative inverse,* r^{-1}, *in* R. *In Exercises 9 through 12, give all units of the given ring.*

9. Z_4 **10.** Z_7 **11.** Z_{10} **12.** Z_{11}

T *is a* ***subring*** *of a ring* R *if* (T, +) *is a subgroup of* (R, +) *and* (T, $\square$) *is a subsemigroup of* (R, $\square$).

13. Show that the set of 2×2 matrices of the form $\begin{matrix} 0 & a \\ 0 & b \end{matrix}$ is a subring of the ring in Exercise 1.

14. Show that the integers form a subring of the ring given in Exercise 5.

15. For each of the structures in Exercises 1 through 6, determine if the structure is a field. Explain your decisions.

16. In the field Z_7, find each of the following.

(a) $-\bar{3}$ **(b)** $-\bar{2}$ **(c)** $-\bar{6}$

(d) $\bar{\frac{1}{2}}$ **(e)** $\bar{\frac{2}{3}}$

17. Find the multiplicative inverse of $\overline{55}$ in Z_{196}.

18. Find the multiplicative inverse of $\overline{29}$ in Z_{196}.

19. Solve the following system of equations in Z_5.

$$\begin{aligned} \bar{4}x - \bar{3}y &= \bar{1} \\ \bar{2}x + \ \ y &= \bar{3} \end{aligned}$$

20. Solve the following system of equations in Z_7.

$$\begin{aligned} \bar{4}x - \bar{3}y &= \bar{1} \\ \bar{2}x + \bar{4}y &= \bar{2} \end{aligned}$$

21. Find all solutions of each equation in Z_7.

(a) $x^2 + \bar{2}x + \bar{3} = \bar{4}$ **(b)** $x^2 + \bar{4}x + \bar{1} = \bar{3}$

22. Find all solutions of each equation in Z_5.

(a) $x^2 + \bar{2}x + \bar{3} = \bar{4}$ **(b)** $x^2 + \bar{4}x + \bar{1} = \bar{3}$

23. What is the remainder when 3^{850} is divided by 17?

24. What is the remainder when 5^{219} is divided by 17?

25. Prove that a field cannot have any zero divisors.

26. What condition on the set of units of a ring R will guarantee that R is a field?

27. Prove that if n is not a prime, then Z_n is not a field.

28. Prove that Z_n is a field if and only if n is a prime.

Tips for Proofs

The proofs in this chapter are mostly simple direct proofs, in part because we have introduced several new mathematical structures (semigroup, monoids, groups, Abelian groups, rings, and fields). With a new structure we first explore the simple consequences of the definitions; for example, Theorem 1, Section 2. However, proofs of uniqueness are frequently indirect as in Theorems 1 and 4 in Section 4.

The idea of a substructure appears several times in this chapter. In general, to prove that a subset forms a substructure of a mathematical structure, we show that the subset together with the operation(s) satisfy the definition of this type of structure. But any global property such as associativity is inherited by the subset so we need only check closure properties and properties involving special elements. Thus, to show that a subset is a subgroup, we check closure for the multiplication, that the identity belongs to the subset, and that the inverse of each element in the subset belongs to the subset.

Isomorphism is a powerful tool for proving statements, since, roughly speaking, establishing an isomorphism between two structures allows us to transfer knowledge about one structure to the other. This can be seen in Theorem 4, Section 2.

Key Ideas for Review

- Binary operation on A: everywhere defined function $f : A \times A \to A$
- Commutative binary operation: $a \square b = b \square a$
- Associative binary operation: $a \square (b \square c) = (a \square b) \square c$
- Semigroup: nonempty set S together with an associative binary operation $\square$ defined on S
- Monoid: semigroup that has an identity
- Subsemigroup $(T, \square)$ of semigroup $(S, \square)$: T is a nonempty subset of S and $a \square b \in T$ whenever a and b are in T.
- Submonoid $(T, \square)$ of monoid $(S, \square)$: T is a nonempty subset of S, $e \in T$, and $a \square b \in T$ whenever a and b are in T.
- Isomorphism.
- Homomorphism.
- Theorem: Let $(S, \square)$ and $(T, \square')$ be monoids with identities e and e', respectively, and suppose that $f : S \to T$ is an isomorphism. Then $f(e) = e'$.
- Theorem: If $(S, \square)$ and $(T, \square')$ are semigroups, then $(S \times T, \square'')$ is a semigroup, where $\square''$ is defined by

$$(s_1, t_1) \square'' (s_2, t_2) = (s_1 \square s_2, t_1 \square' t_2).$$

- Congruence relation R on semigroup $(S, \square)$: equivalence relation R such that $a \, R \, a'$ and $b \, R \, b'$ imply that $(a \square b) \, R \, (a' \square b')$
- Theorem: Let R be a congruence relation on the semigroup $(S, \square)$. Define the operation $\square$ in S/R as follows:

$$[a] \square [b] = [a \square b].$$

Then $(S/R, \square)$ is a semigroup.

- Quotient semigroup or factor semigroup S/R.

- Z_n.
- Theorem (Fundamental Homomorphism Theorem): Let $f : S \to T$ be a homomorphism of the semigroup $(S, \Box)$ onto the semigroup $(T, \Box')$. Let R be the relation on S defined by $a \; R \; b$ if and only if $f(a) = f(b)$, for a and b in S. Then
  (a) R is a congruence relation.
  (b) T is isomorphic to S/R.
- Group $(G, \Box)$: monoid with identity e such that for every $a \in G$ there exists $a' \in G$ with the property that $a \Box a' = a' \Box a = e$.
- Theorem: Let G be a group, and let a, b, and c be elements of G. Then
  (a) $ab = ac$ implies that $b = c$ (left cancellation property).
  (b) $ba = ca$ implies that $b = c$ (right cancellation property).
- Theorem: Let G be a group, and let a and b be elements of G. Then
  (a) $(a^{-1})^{-1} = a$.
  (b) $(ab)^{-1} = b^{-1}a^{-1}$.
- Order of a group G: $|G|$, the number of elements in G
- S_n: the symmetric group on n letters
- Subgroup.
- Theorem: Let R be a congruence relation on the group $(G, \Box)$. Then the semigroup $(G/R, \Box)$ is a group, where the operation $\Box$ is defined in G/R by

$$[a] \Box [b] = [a \Box b].$$

- Left coset aH of H in G determined by a: $\{ah \mid h \in H\}$
- Normal subgroup: subgroup H such that $aH = Ha$ for all a in G
- Theorem: Let R be a congruence relation on a group G, and let $H = [e]$, the equivalence class containing the identity. Then H is a normal subgroup of G and, for each $a \in G$, $[a] = aH = Ha$.
- Theorem: Let N be a normal subgroup of a group G, and let R be the following relation on G:

$$a \; R \; b \quad \text{if and only if} \quad a^{-1}b \in N.$$

  Then
  (a) R is a congruence relation on G.
  (b) N is the equivalence class $[e]$ relative to R, where e is the identity of G.
- Ring $(S, +, \Box)$: nonempty set S such that $(S, +)$ is an Abelian group, $\Box$ is associative, and $\Box$ distributes over $+$.
- Commutative ring: ring in which the operation $\Box$ is commutative.
- Theorem: Let R be a commutative ring with additive identity 0 and multiplicative identity 1. Then
  (a) For any x in R, $0 \Box x = 0$.
  (b) For any x in R, $-x = (-1) \Box x$.
- Field: commutative ring with identity in which every nonzero element has a multiplicative inverse
- Theorem: The ring Z_n is a field when n is a prime.
- Theorem:
  (a) If $G = \{g_1, g_2, \ldots, g_n\}$ is a finite Abelian group with identity denoted by e, and a is any element of G, then $a^n = e$.
  (b) (Fermat's Little Theorem) If p is a prime number, and $\text{GCD}(a, p) = 1$, then $a^{p-1} \equiv 1 \pmod{p}$.
  (c) If p is a prime number and a is any integer, then $a^p \equiv a \pmod{p}$.

Chapter Self-Test

1. What does it mean to say a set is closed with respect to a binary operation?

2. How does an isomorphism of semigroups differ from an isomorphism of posets? How are an isomorphism of groups and an isomorphism of posets alike?

3. What are the properties that define a congruence relation?

4. Why are groups said to have more structure than semigroups?

5. How does a field differ from a ring?

6. For each of the following, determine whether the description of $\Box$ is a valid definition of a binary operation on the given set.
  (a) On the set of 2×2 Boolean matrices, where $\mathbf{A} \Box \mathbf{B} = (a_{ij} + b_{ij}) \pmod 2$
  (b) On the set of even integers, where $a \Box b = a + b$
  (c) On Z^+, where $a \Box b = 2^{ab}$

7. Complete the table so that $\Box$ is a commutative and idempotent binary operation.

$\Box$	a	b	c
a		c	
b			
c		b	

8. Let Q be the set of rational numbers and define $a \Box b = a + b - ab$.
  (a) Is $(Q, \Box)$ a monoid? Justify your answer.
  (b) If $(Q, \Box)$ is a monoid, which elements of Q have an inverse?

9. Determine whether the set together with the operation is a semigroup, a monoid, or neither for each of the pairs given in Exercise 1.

10. Let $A = \{0, 1\}$, and consider the semigroup $(A^+, \cdot)$, where $\cdot$ is the operation of catenation. Define a relation R on this semigroup by $\alpha \; R \; \beta$ if and only if α and β have the same length. Prove that R is a congruence relation.

11. Let G be a group and define $f : G \to G$ by $f(a) = a^{-1}$. Is f a homomorphism? Justify your answer.

12. Let G be the group whose multiplication table is given below and let H be the subgroup $\{c, d, e\}$.

$\square$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e	f	a	c
c	c	f	a	d	e	b
d	d	b	f	e	c	a
f	f	c	d	a	b	e

Find the right cosets of H in G.

13. Let $f : G_1 \to G_2$ be a homomorphism from the group $(G_1, \square_1)$ onto the group $(G_2, \square_2)$. If N is a normal subgroup of G_1, show that its image $f(N)$ is a normal subgroup of G_2.

14. Let G be a group with identity e. Show that if $x^2 = x$ for some x in G, then $x = e$.

15. Let G be the group of integers under the operation of addition and G' be the group of all even integers under the operation of addition. Show that the function $f : G \to G'$ defined by $f(a) = 2a$ is an isomorphism.

16. Let $H_1, H_2, \ldots, H_k$ be subgroups of a group G. Prove that $\bigcap_{i=1}^{k} H_i$ is also a subgroup of G.

17. Prove that if $\sqrt{n}$ is an irrational number, then the set of numbers of the form $a + b\sqrt{n}$, a, b integers, together with ordinary addition and multiplication, is a field.

Experiment 9

The purpose of this experiment is to investigate relationships among groups, subgroups, and elements. Five groups are given as examples to use in the investigation. You may decide to look at other groups as well to test your conjectures.

S_3 is the group of permutations of $\{1, 2, 3\}$ with the operation of composition. It is also the group of symmetries of a triangle. (See Section 4.)

D is the group of symmetries of a square. (This group is presented in Exercise 19, Section 4.)

S_4 is the group of permutations of $\{1, 2, 3, 4\}$ with the operation of composition.

G_1 is the group whose multiplication table is given in Table 1.

G_2 is the group whose multiplication table is given in Table 2.

You may find it helpful to write out the multiplication tables for S_3, D, and S_4.

TABLE 1

	1	2	3	4	5	6	7	8
1	1	2	3	4	5	6	7	8
2	2	5	4	7	6	1	8	3
3	3	8	5	2	7	4	1	6
4	4	3	6	5	8	7	2	1
5	5	6	7	8	1	2	3	4
6	6	1	8	3	2	5	4	7
7	7	4	1	6	3	8	5	2
8	8	7	2	1	4	3	6	5

TABLE 2

	1	2	3	4	5
1	1	2	3	4	5
2	2	3	4	5	1
3	3	4	5	1	2
4	4	5	1	2	3
5	5	1	2	3	4

1. Identify the identity element e for each of the five groups.

2. For each of the five groups, do the following. For each element g in the group, find the smallest k for which $g^k = e$, the identity. This number k is called the **order of *g***.

3. What is the relationship between the order of an element of a group and the order of the group? (The order of a group is the number of elements.)
4. For each of the five groups, find all subgroups of the group.
5. A group is called **cyclic** if its elements are the powers of one of the elements. Identify any cyclic groups among the subgroups of each group.
6. What is the relationship between the order of a subgroup and the order of the group?
7. The groups G_1 and D are both of order 8. Are they isomorphic? Explain your reasoning.

Coding Exercise

Let Z_n be as defined in Section 3.

1. Write a function SUM that takes two elements of Z_n, [x] and [y] and returns their sum [x] □ [y]. The user should be able to input a choice for n.

2. Let $H = \{[0], [2]\}$. Write a subroutine that computes the left cosets of H in Z_6.

3. Let $H = \{[0], [2], [4], [6]\}$. Write a subroutine that computes the right cosets of H in Z_8.

4. Write a program that given a finite operation table will determine if the operation satisfies the associative property.

5. Write a program that given a finite group G and a subgroup H determines if H is a normal subgroup of G.

Answers to Odd-Numbered Exercises

Exercise Set 1

1. Yes. **3.** No. **5.** No. **7.** No.

9. Commutative, associative.

11. Not commutative, associative.

13. Commutative, associative.

15. Commutative, associative.

17. Commutative, associative.

19. Commutative; not associative

21. The operation has the idempotent property, because $a \square a = \frac{a+a}{2} = a$.

23. One solution is

□	a	b	c
a	a	c	c
b	c	b	a
c	c	a	c

25. **(a)** a, a. **(b)** c, b. **(c)** c, a. **(d)** Neither.

27.

□	a	b	c	d
a	a	b	c	d
b	b	a	c	d
c	c	d	c	d
d	d	c	c	d

29. $n^{\frac{n(n+1)}{2}}$ commutative operations.

31. **(a)** Associative: (1), (5), (8), (9), (10), (11), (15), (16).

(b) Idempotent: (5), (10), (11), (15).

33. A binary operation on a set S must be defined for every a, b in S. According to the earlier definition, $a \square b$ may be undefined for some a, b in S.

Exercise Set 2

1. Semigroup: (b). monoid: (b).

3. Semigroup: (a). monoid: neither.

5. Monoid: identity is 1; commutative.

7. Semigroup.

9. Monoid: identity is S; commutative.

11. Monoid: identity is 12; commutative.

13. Monoid: identity is 0; commutative.

15. monoid; identity $\begin{bmatrix} 0 & \\ & -1 \end{bmatrix}$; commutative.

17. Neither.

19.

□	a	b	c
a	c	a	b
b	a	b	c
c	b	c	a

21. Let $f_1(a) = a$, $f_1(b) = a$; $f_2(a) = a$, $f_2(b) = b$; $f_3(a) = b$, $f_3(b) = a$; $f_4(a) = b$, $f_4(b) = b$. These are the only functions on S. It is not commutative.

$\circ$	f_1	f_2	f_3	f_4
f_1	f_1	f_1	f_4	f_4
f_2	f_1	f_2	f_3	f_4
f_3	f_1	f_3	f_2	f_1
f_4	f_1	f_4	f_4	f_4

23. **(a)** *abaccbababc*.

(b) *babcabacabac*.

(c) *babccbaabac*.

25. The subset must form a subsemigroup and the identity element must belong to the subset.

27. By Exercise 26, we need only check that $e \in S_1 \cap S_2$. But $e \in S_1$ and $e \in S_2$, because each is a submonoid of $(S, \square)$.

29. Yes. Refer to Exercise 1.

31. Let $x, y \in S_1$.

$$\begin{aligned}(g \circ f)(x \square_1 y) &= g(f(x \square_1 y)) \\ &= g(f(x) \square_2 f(y)) \\ &= g(f(x)) \square_3 g(f(y)) \\ &= (g \circ f)(x) \square_3 (g \circ f)(y).\end{aligned}$$

Hence $g \circ f$ is a homomorphism from $(S_1, \square_1)$ to $(S_3, \square_3)$.

33. Let $x, y \in R^+$. $\ln(x \square y) = \ln(x) + \ln(y)$ so ln is a homomorphism. Suppose $x \in R$. Then $e^x \in R^+$ and $\ln(e^x) = x$ so ln is onto R^+. Suppose $\ln(x) = \ln(y)$; then $e^{\ln(x)} = e^{\ln(y)}$ and $x = y$. Hence ln is one to one and an isomorphism between $(R^+, \times)$ and $(R, +)$.

Exercise Set 3

1. Let $(s_1, t_1), (s_2, t_2) \in S \times T$. $(s_1, t_1) \square\!\square (s_2, t_2) = (s_1 \square s_2, t_1 \square' t_2)$, so $\square\!\square$ is a binary operation. Consider $(s_1, t_1) \square\!\square ((s_2, t_2) \square\!\square (s_3, t_3)) = (s_1, t_1) \square\!\square (s_2 \square s_3, t_2 \square' t_3) = (s_1 \square (s_2 \square s_3), t_1 \square' (t_2 \square' t_3)) = ((s_1 \square s_2) \square s_3, (t_1 \square' t_2) \square' t_3) = ((s_1, t_1) \square\!\square (s_2, t_2)) \square\!\square (s_3, t_3)$. Thus $(S \times T, \square\!\square)$ is a semigroup. $(s_1, t_1) \square\!\square (s_2, t_2) = (s_1 \square s_2, t_1 \square' t_2) = (s_2 \square s_1, t_2 \square' t_1) = (s_2, t_2) \square\!\square (s_1, t_1)$. Hence $\square\!\square$ is commutative.

3. Let $(s_1, t_1), (s_2, t_2) \in S \times T$. Then $f((s_1, t_1) \square\!\square (s_2, t_2)) = f(s_1 \square s_2, t_1 \square' t_2) = s_1 \square s_2 = f(s_1, t_1) \square f(s_2, t_2)$. f is a homomorphism.

5. $\square\!\square$ is a binary operation, because both $\square$ and $\square'$ are. Consider $(s_1, t_1) \square\!\square ((s_2, t_2) \square\!\square (s_3, t_3))$.

$$\begin{aligned}(s_1, t_1) \square\!\square ((s_2, t_2) \square\!\square (s_3, t_3)) &= (s_1, t_1) \square\!\square (s_2 \square s_3, t_2 \square' t_3) \\ &= (s_1 \square (s_2 \square s_3), t_1 \square' (t_2 \square' t_3)) \\ &= ((s_1 \square s_2) \square s_3, (t_1 \square' t_2) \square' t_3) \\ &= ((s_1, t_1) \square\!\square (s_2, t_2)) \square\!\square (s_3, t_3)\end{aligned}$$

Thus, $\square\!\square$ is associative.

7. Yes. **9.** Yes. **11.** Yes. **13.** No. **15.** Yes.

17. $S/R = \{[4], [7], [10], [13], [16]\}$.

19. The composition of two equivalence relations need not be an equivalence relation.

21. $S/R = \{[0], [1], [2]\}$. $[0] = \{0, \pm 3, \pm 6, \ldots\}$, $[1] = \{\pm 1, \pm 4, \pm 7, \ldots\}$, $[2] = \{\pm 2, \pm 5, \pm 8, \ldots\}$.

$\square$	[0]	[1]	[2]
[0]	[0]	[1]	[2]
[1]	[1]	[2]	[0]
[2]	[2]	[0]	[1]

23. $S/R = \{[0], [1], [2], [3], [4]\}$, $[a] = \{z \mid z = 5k + a, k \in Z\}$, $a = 0, 1, 2, 3, 4$.

$\square$	[0]	[1]	[2]	[3]	[4]
[0]	[0]	[1]	[2]	[3]	[4]
[1]	[1]	[2]	[3]	[4]	[0]
[2]	[2]	[3]	[4]	[0]	[1]
[3]	[3]	[4]	[0]	[1]	[2]
[4]	[4]	[0]	[1]	[2]	[3]

25. **(a)**

$\square$	[a]	[b]
[a]	[a]	[b]
[b]	[b]	[b]

(b) $f_R(e) = [a] = f_R(a)$, $f_R(b) = [b] = f_R(c)$.

27. An examination of the two multiplication tables shows that they are identical and so Z_2 is isomorphic to S/R.

29. This is a direct proof. For part (a), we check the three properties for an equivalence relation and then the property for a congruence relation. In part (b), we first check that $\bar{f}$ is a function and then the properties of an isomorphism are confirmed.

Exercise Set 4

1. No.

3. Yes; Abelian; identity is 0; a^{-1} is $-a$.

5. No. **7.** No. **9.** No.

11. Yes; Abelian; identity is $\{\ \}$; a^{-1} is a.

13. Since g_1, g_2, g_3 in S_3 each have order 2, they must be paired somehow with f_2, f_3, f_4 of Example 12 if the groups are isomorphic. But no rearrangement of the columns and rows labeled f_2, f_3, f_4 in Example 12 will give the "block" pattern shown by g_1, g_2, g_3 in the table for S_3. Hence the groups are not isomorphic.

15. **(a)** $\frac{8}{3}$. **(b)** $-\frac{4}{5}$.

17. $H_1 = \{1\}$, $H_2 = \{1, -1\}$, $H_3 = \{1, -1, i, -i\}$.

19.

$\circ$	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8
f_1	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8
f_2	f_2	f_3	f_4	f_1	f_8	f_7	f_5	f_6
f_3	f_3	f_4	f_1	f_2	f_6	f_5	f_8	f_7
f_4	f_4	f_1	f_2	f_3	f_7	f_8	f_6	f_5
f_5	f_5	f_7	f_6	f_8	f_1	f_3	f_2	f_4
f_6	f_6	f_8	f_5	f_7	f_3	f_1	f_4	f_2
f_7	f_7	f_6	f_8	f_5	f_4	f_2	f_1	f_3
f_8	f_8	f_5	f_7	f_6	f_2	f_4	f_3	f_1

21. Consider the sequence $e, a, a^2, a^3, \ldots$. Since G is finite, not all terms of this sequence can be distinct; that is, for some $i \le j$, $a^i = a^j$. Then $(a^{-1})^i a^i = (a^{-1})^i a^j$ and $e = a^{j-i}$. Note that $j - i \ge 0$.

23. Yes.

25. Clearly, $e \in H$. Let $a, b \in H$. Consider $(ab)y = a(by) = a(yb) = (ay)b = (ya)b = y(ab)$ $\forall y \in G$. Hence H is closed under multiplication and is a subgroup of G.

27. The identity permutation is an even permutation. If p_1 and p_2 are even permutations, then each can be written as the product of an even number of transpositions. Then $p_1 \circ p_2$ can be written as the product of these representations of p_1 and p_2. But this gives $p_1 \circ p_2$ as the product of an even number of transpositions. Thus $p_1 \circ p_2 \in A_n$ and A_n is a subgroup of S_n.

29. $\{f_1\}$, $\{f_1, f_2, f_3, f_4\}$, $\{f_1, f_3, f_5, f_6\}$, $\{f_1, f_3, f_7, f_8\}$, $\{f_1, f_5\}$, $\{f_1, f_6\}$, $\{f_1, f_3\}$, $\{f_1, f_7\}$, $\{f_1, f_8\}$, D_4.

31. $|xy| = |x| \cdot |y|$. Thus $f(xy) = f(x)f(y)$.

33. Suppose $f : G \to G$ defined by $f(a) = a^2$ is a homomorphism. Then $f(ab) = f(a)f(b)$ or $(ab)^2 = a^2b^2$. Hence $a^{-1}(abab)b^{-1} = a^{-1}(a^2b^2)b^{-1}$ and $ba = ab$. Suppose G is Abelian. By Exercise 37, $f(ab) = f(a)f(b)$.

35. Let $x, y \in G$. $f_a(xy) = axya^{-1} = axa^{-1}aya^{-1} = f_a(x)f_a(y)$. f_a is a homomorphism. Suppose $x \in G$. Then $f_a(a^{-1}xa) = aa^{-1}xaa^{-1} = x$ so f_a is onto. Suppose $f_a(x) = f_a(y)$, then $axa^{-1} = aya^{-1}$. Now $a^{-1}(axa^{-1})a = a^{-1}(aya^{-1})a$ and $x = y$. Thus f_a is one to one and an isomorphism.

37. (Outline) Basis step: $n = 1$ P(1): $(ab)^1 = a^1b^1$ is true.
Induction step: LHS of P(k + 1): $(ab)^{k+1} = (ab)^kab = a^kb^kab = a^kab^kb = a^{k+1}b^{k+1}$
RHS of P(k + 1).

39. One table is

$\circ$	a	b	c
a	b	a	c
b	c	b	a
c	a	c	b

$\circ$ has no identity element.

Exercise Set 5

1.

	$(\bar{0},\bar{0})$	$(\bar{0},\bar{1})$	$(\bar{0},\bar{2})$	$(\bar{1},\bar{0})$	$(\bar{1},\bar{1})$	$(\bar{1},\bar{2})$
$(\bar{0},\bar{0})$	$(\bar{0},\bar{0})$	$(\bar{0},\bar{1})$	$(\bar{0},\bar{2})$	$(\bar{1},\bar{0})$	$(\bar{1},\bar{1})$	$(\bar{1},\bar{2})$
$(\bar{0},\bar{1})$	$(\bar{0},\bar{1})$	$(\bar{0},\bar{2})$	$(\bar{0},\bar{0})$	$(\bar{1},\bar{1})$	$(\bar{1},\bar{2})$	$(\bar{1},\bar{0})$
$(\bar{0},\bar{2})$	$(\bar{0},\bar{2})$	$(\bar{0},\bar{0})$	$(\bar{0},\bar{1})$	$(\bar{1},\bar{2})$	$(\bar{1},\bar{0})$	$(\bar{1},\bar{1})$
$(\bar{1},\bar{0})$	$(\bar{1},\bar{0})$	$(\bar{1},\bar{1})$	$(\bar{1},\bar{2})$	$(\bar{0},\bar{0})$	$(\bar{0},\bar{1})$	$(\bar{0},\bar{2})$
$(\bar{1},\bar{1})$	$(\bar{1},\bar{1})$	$(\bar{1},\bar{2})$	$(\bar{1},\bar{0})$	$(\bar{0},\bar{1})$	$(\bar{0},\bar{2})$	$(\bar{0},\bar{0})$
$(\bar{1},\bar{2})$	$(\bar{1},\bar{2})$	$(\bar{1},\bar{0})$	$(\bar{1},\bar{1})$	$(\bar{0},\bar{2})$	$(\bar{0},\bar{0})$	$(\bar{0},\bar{1})$

3. Define $f : G_1 \to G_2$ by $f((g_1, g_2)) = (g_2, g_1)$. By Exercise 4, Section 3, f is an isomorphism.

5.

	[0]	[1]	[2]
[0]	[0]	[1]	[2]
[1]	[1]	[2]	[0]
[2]	[2]	[0]	[1]

7. $\ker(f) = \{(e_1, g_2), e_1$, identity of $G_1, g_2 \in G_2\}$.

9. $\{[0]\}, \{[1]\}, \{[2]\}, \{[3]\}$.

11. $\{[0], [1], [2], [3]\}$.

13. The groups are isomorphic. Define $f : G \to Z_4$ by $f(1) = [0]$, $f(i) = [1]$, $f(-1) = [2]$, $f(-i) = [3]$. A comparison of the multiplication tables shows that f preserves the operation.

15. $\{f_1, g_3\}, \{f_2, g_2\}, \{f_3, g_1\}$.

17. $\{f_1\}, \{f_2\}, \{f_3\}, \{g_1\}, \{g_2\}, \{g_3\}$.

19. $\{[0], [4]\}, \{[1], [5]\}, \{[2], [6]\}, \{[3], [7]\}$.

21. $\{(m + x, n + x) \mid x \in Z\}$ for $(m, n) \in Z \times Z$.

23. If N is a normal subgroup of G, Exercise 22 shows that $a^{-1}Na \subseteq N$ for all $a \in G$.

Suppose $a^{-1}Na \subseteq N$ for all $a \in G$. Again the proof in Exercise 22 shows that N is a normal subgroup of G.

25. $\{f_1\}, \{f_1, f_3\}, \{f_1, f_3, f_5, f_6\}, \{f_1, f_2, f_3, f_4\}, \{f_1, f_3, f_7, f_8\}, D$.

27. Suppose $f_a(h_1) = f_a(h_2)$. Then $ah_1 = ah_2$ and $a^{-1}(ah_1) = a^{-1}(ah_2)$. Hence $h_1 = h_2$ and f_a is one to one. Let $x \in aH$. Then $x = ah$, $h \in H$ and $f_a(h) = x$. Thus f_a is onto and since it is everywhere defined as well, f_a is a one-to-one correspondence between H and aH. Hence $|H| = |aH|$.

29. Suppose $f(aH) = f(bH)$. Then $Ha^{-1} = Hb^{-1}$ and $a^{-1} = hb^{-1}$, $h \in H$. Hence $a = bh^{-1} \in bH$ so $aH \subseteq bH$. Similarly, $bH \subseteq aH$ so $aH = bH$. This means f is one to one. If Hc is a right coset of H, then $f(c^{-1}H) = Hc$ so f is also onto.

31. Consider $f(aba^{-1}b^{-1}) = f(a)f(b)f(a^{-1})f(b^{-1}) = f(a)f(a^{-1})f(b)f(b^{-1}) = f(a)(f(a))^{-1}f(b)(f(b))^{-1}$ (by Theorem 5, Section 9.4) $= ee = e$. Hence $\{aba^{-1}b^{-1} \mid a, b$ in $G_1\} \subseteq \ker(f)$.

33. Let $a \notin H$. The left cosets of H are H and aH. The right cosets are H and Ha. $H \cap aH = H \cap Ha = \{\}$ and $H \cup aH = H \cup Ha$. Thus $aH = Ha$. Since $a \in H \cup aH = H$, we have $xH = Hx$ $\forall x \in G$. H is a normal subgroup of G.

35. Suppose $f : G \to G'$ is one to one. Let $x \in \ker(f)$. Then $f(x) = e' = f(e)$. Thus $x = e$ and $\ker(f) = \{e\}$. Conversely, suppose $\ker(f) = \{e\}$. If $f(g_1) = f(g_2)$, then $f(g_1g_2^{-1}) = f(g_1)f(g_2^{-1}) = f(g_1)(f(g_2))^{-1} = f(g_1)(f(g_1))^{-1} = e$. Hence $g_1g_2^{-1} \in \ker(f)$. Thus $g_1g_2^{-1} = e$ and $g_1 = g_2$. Hence f is one to one.

37. Since H is a subgroup of G, the identity element e belongs to H. For any $g \in G$, $g = g \cdot e \in gH$, so every element of G belongs to some left coset of H. If aH and bH are distinct left cosets of H, this means that $aH \cap bH = \{\}$. Hence the set of distinct left cosets of H forms a partition of G.

Exercise Set 6

1. Noncommutative ring with identity.

3. Not a ring.

5. Commutative ring with identity.

7. This is a ring from Exercise 1. An example of zero divisors are the matrices

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -2 & -2 \\ 1 & 1 \end{bmatrix}.$$

9. $\bar{1}, \bar{3}$.

11. $\bar{1}, \bar{3}, \bar{7}, \bar{9}$.

13. This subset is a subgroup with respect to $+$ since it contains the zero element and $\mathbf{A} - \mathbf{B}$ belongs to the subset if $\mathbf{A}$ and $\mathbf{B}$ do. This subset is a subsemigroup with respect to $\cdot$ since if $\mathbf{A}$ and $\mathbf{B}$ are in the subset so is $\mathbf{A} \cdot \mathbf{B}$.

15. The structures in Exercises 1 and 2 are not fields, because they lack multiplicative inverses. The structure in Exercise 3 is not a ring so it is not a field. The structures in Exercises 4, 5, and 6 are fields.

17. $\overline{55} \cdot \overline{57} = \bar{1}$ in Z_{196}.

19. $(\bar{4}, \bar{0})$.

21. **(a)** $\bar{2}, \bar{3}$. **(b)** There are no solutions.

23. 9.

26. The set of units must contain all nonzero elements of R.

28. The statement Z_n is a field implies n is prime is proven in Exercise 27. Any $a \in Z_n$ is relatively prime to n so we have $1 = sa + tn$ for some integers s, t and $\bar{s}$ is the multiplicative inverse of a.

Answers to Chapter Self-Tests

1. A set is closed with respect to a binary operation if using the operation with any two elements of the set yields an element that belongs to the set.

2. An isomorphism between two semigroups must preserve the operations, but an isomorphism between posets must preserve the orders. In each case the mapping must be a one-to-one correspondence and preserve the defining structure, multiplications for groups and orders for posets.

3. The relation must be reflexive, symmetric, and transitive and preserve the binary operations.

4. Groups are said to have more structure than semigroups, because they must satisfy more conditions.

5. A field must contain a multiplicative inverse for each nonzero element; this is not required in a ring.

6. **(a)** Yes, $\mathbf{A} \cdot \mathbf{B}$ is well defined for all 2×2 Boolean matrices.

(b) Yes, this is ordinary addition for even numbers.

(c) Yes, 2^{ab} is defined uniquely for all a, b in Z^+.

7.

$\cdot$	a	b	c
a	a	c	
b	c	b	b
c		b	c

where $\cdot$ represents a, b, or c.

8. **(a)** If $a, b \in Q$, then $a \cdot b$ is also a rational number.

$$\begin{aligned} a \cdot (b \cdot c) &= a \cdot (b + c - bc) \\ &= a + (b + c - bc) - a(b + c - bc) \\ &= a + b + c - bc - ab - ac + abc. \\ (a \cdot b) \cdot c &= (a + b - ab) \cdot c \\ &= a + b - ab + c - (a + b - ab)c \\ &= a + b + c - ab - ac - bc + abc. \end{aligned}$$

Hence, $\cdot$ is associative. Zero is the identity for $(Q, \cdot)$, which is a monoid.

(b) If $a \neq 1$, then

$$\begin{aligned} a \cdot \frac{a}{a-1} &= a + \frac{a}{a-1} - a\left(\frac{a}{a-1}\right) \\ &= \frac{a^2 - a + a - a^2}{a-1} \\ &= 0. \end{aligned}$$

Thus all rational numbers except 1 have a $\cdot$-inverse.

9. (a) and (b) are monoids. (c) is neither.

10. R has previously been shown to be an equivalence relation, because it is equality for the string lengths. Suppose $a \, R \, b$ and $\alpha \, R \, \beta$, then $\text{length}(a \cdot \alpha) = \text{length}(a) + \text{length}(\alpha) = \text{length}(b) + \text{length}(\beta) = \text{length}(b \cdot \beta)$. Thus $a \cdot \alpha \, R \, b \cdot \beta$ and R is a congruence relation.

11. No, $f(ab) = (ab)^{-1} = b^{-1}a^{-1} \neq f(a) \cdot f(b)$.

12. $\{c, d, e\} = H = He = Hc = Hd$; $Ha = \{a, b, f\} = Hb = Hf$.

13. Let $g \in G_2$ and $n \in f(N)$. Since f is onto, there is a $g' \in G_1$ such that $f(g') = g$. Since $n \in f(N)$, there is an $n' \in N$ such that $f(n') = n$. Then $gn = f(g')f(n') = f(g'n')$. N is normal in G_1 so $g'n' = n''g'$ for some $n'' \in N$. Then $f(g'n') = f(n''g') = f(n'')f(g') = f(n'')g \in f(N)g$. Thus $g \cdot f(N) \subseteq f(N) \cdot g$. Similarly, we can show $f(N) \cdot g \subseteq g \cdot f(N)$ and hence $g \cdot f(N) = f(N) \cdot g$ for all $g \in G_2$.

14. Suppose $x^2 = x$. Then $x^{-1}(xx) = x^{-1}x$ and $(x^{-1}x)x = e$. So $x = e$.

15. $f(a + b) = 2(a + b) = 2a + 2b = f(a) + f(b)$ so f is a homomorphism. For any even integer n, $n = 2k$, $k \in Z$, and $f(k) = n$ so f is onto. Suppose $f(a) = f(b)$. Then $2a = 2b$ and $a = b$. Hence f is one to one.

16. Since the identity e belongs to every subgroup, $e \in \bigcap_{i=1}^{k} H_i$. Suppose h and h' belong to $\bigcap_{i=1}^{k} H_i$. Then h, h', and hh' belong to each H_i and $hh' \in \bigcap_{i=1}^{k} H_i$. Let $h \in \bigcap_{i=1}^{k} H_i$. Then h and h^{-1} belong to each H_i, since each H_i is a subgroup of G and so $h^{-1} \in \bigcap_{i=1}^{k} H_i$.

17. The inverse of $a + b\sqrt{n}$ must be of the form

$$\frac{a - b\sqrt{n}}{a^2 - nb^2}.$$

But this exists in the set if and only if $a^2 - nb^2 \neq 0$ if and only if $n \neq \frac{a}{b}$ if and only if $\sqrt{n} \neq \frac{a}{b}$.

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Groups and Coding

In today s modern world of communication, data items are constantly being transmitted from point to point. This transmission may result from the simple task of a computer terminal interacting with the mainframe computer 200 feet away via a satellite that is parked in an orbit 20,000 miles from the earth, or from a telephone call or letter to another part of the country. The basic problem in transmission of data is that of receiving the data as sent and not receiving a distorted piece of data. Distortion can be caused by a number of factors.

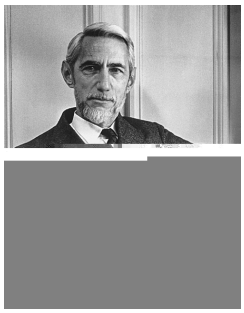
Coding theory has developed techniques for introducing redundant information in transmitted data that help in detecting, and sometimes in correcting, errors. Some of these techniques make use of group theory.

Another entirely different problem that arises frequently in the transmission of data is that of modifying the data being sent so that only the intended recipient is able to reconstitute the original data. This problem dates back to the days of the early Greeks. Cryptology is the discipline that studies techniques, called cryptosystems, for the secure transmission of data. With the widespread use of e-commerce and ATMs, cryptology has become of vital importance in today s society. In this chapter, we present a brief introduction to the important topic of public key cryptology.

Looking Back

The American mathematician Claude E. Shannon (1916 2001), who worked at the Bell Laboratories, published a paper in 1948 that described a mathematical theory of communication and thereby founded the eld of information theory. Shortly thereafter, Richard Hamming and his colleagues at Bell Laboratories laid the foundations for error-correcting codes.

During the rst half of the twentieth century, most of the work in cryptology was carried out by and for the military. In 1949, Shannon published the paper The Communication Theory of Secrecy Systems, which broke new ground in the eld of cryptology. The eld was dormant until 1975, when two researchers at Stanford University discovered public key cryptology, which resulted in a burst of activity in this area. In 1976, three researchers at M.I.T. discovered a public key cryptosystem, known as the RSA (Rivest, Shamir, and Adelman) system, which was widely used. Today, the most widely used system is known as the DES system, a private key system.



Claude E. Shannon
M.I.T. Museum and Historical Collections

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1 Coding of Binary Information and Error Detection

The basic unit of information, called a **message**, is a nite sequence of characters from a nite alphabet. We shall choose as our alphabet the set $B = \{0, 1\}$. Every character or symbol that we want to transmit is now represented as a sequence of m elements from B. That is, every character or symbol is represented in binary form. Our basic unit of information, called a **word**, is a sequence of m 0 s and 1 s.

TABLE 1

+	0	1
0	0	1
1	1	0

The set B is a group under the binary operation + whose table is shown in Table 1. If we think of B as the group Z_2, then + is merely mod 2 addition. It follows that $B^m = B \times B \times \cdots \times B$ (m factors) is a group under the operation $\square$ de ned by

$$(x_1, x_2, \ldots, x_m) \square (y_1, y_2, \ldots, y_m) = (x_1 + y_1, x_2 + y_2, \ldots, x_m + y_m).$$

This group s identity is $\overline{0} = (0, 0, \ldots, 0)$ and every element is its own inverse. An element in B^m will be written as $(b_1, b_2, \ldots, b_m)$ or more simply as $b_1 b_2 \cdots b_m$. Observe that B^m has 2^m elements. That is, the order of the group B^m is 2^m.

Figure 1 shows the basic process of sending a word from one point to another point over a transmission channel. An element $x \square B^m$ is sent through the transmission channel and is received as an element $x_t \square B^m$. In actual practice, the transmission channel may suffer disturbances, which are generally called **noise**, due to weather interference, electrical problems, and so on, that may cause a 0 to be received as a 1, or vice versa. This erroneous transmission of digits in a word being sent may give rise to the situation where the word received is different from the word that was sent; that is, $x \boxminus x_t$. If an error does occur, then x_t could be any element of B^m.

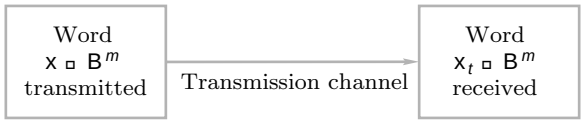


Figure 1

The basic task in the transmission of information is to reduce the likelihood of receiving a word that differs from the word that was sent. This is done as follows. We rst choose an integer $n > m$ and a one-to-one function $e: B^m \to B^n$. The function e is called an (m, n) **encoding function**, and we view it as a means of representing every word in B^m as a word in B^n. If $b \square B^m$, then $e(b)$ is called the **code word** representing b. The additional 0 s and 1 s can provide the means to detect or correct errors produced in the transmission channel.

We now transmit the code words by means of a transmission channel. Then each code word $x = e(b)$ is received as the word x_t in B^n. This situation is illustrated in Figure 2.

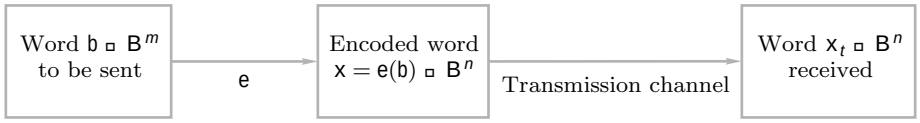


Figure 2

Observe that we want an encoding function e to be one to one so that different words in B^m will be assigned different code words.

If the transmission channel is noiseless, then $x_t = x$ for all x in B^n. In this case $x = e(b)$ is received for each $b \in B^m$, and since e is a known function, b may be identified.

In general, errors in transmission do occur. We will say that the code word $x = e(b)$ has been transmitted with ***k* or fewer errors** if x and x_t differ in at least 1 but no more than k positions.

Let $e: B^m \to B^n$ be an (m, n) encoding function. We say that e **detects *k* or fewer errors** if whenever $x = e(b)$ is transmitted with k or fewer errors, then x_t is not a code word (thus x_t could not be x and therefore could not have been correctly transmitted). For $x \in B^n$, the number of 1 s in x is called the **weight** of x and is denoted by $|x|$.

Example 1 Find the weight of each of the following words in B^5:

(a) $x = 01000$ **(b)** $x = 11100$ **(c)** $x = 00000$ **(d)** $x = 11111$

Solution

(a) $|x| = 1$ **(b)** $|x| = 3$ **(c)** $|x| = 0$ **(d)** $|x| = 5$ ◆

Example 2 Parity Check Code

The following encoding function $e: B^m \to B^{m+1}$ is called the **parity** $(m, m + 1)$ **check code**: If $b = b_1 b_2 \cdots b_m \in B^m$, define

$$e(b) = b_1 b_2 \cdots b_m b_{m+1},$$

where

$$b_{m+1} = \begin{cases} 0 & \text{if } |b| \text{ is even} \\ 1 & \text{if } |b| \text{ is odd.} \end{cases}$$

Observe that b_{m+1} is zero if and only if the number of 1 s in b is an even number. It then follows that every code word $e(b)$ has even weight. A single error in the transmission of a code word will change the received word to a word of odd weight and therefore can be detected. In the same way we see that any odd number of errors can be detected.

For a concrete illustration of this encoding function, let $m = 3$. Then

$$\left.\begin{array}{l} e(000) = 0000 \\ e(001) = 0011 \\ e(010) = 0101 \\ e(011) = 0110 \end{array}\right\} \text{code words.}$$

$$\begin{array}{l} e(100) = 1001 \\ e(101) = 1010 \\ e(110) = 1100 \\ e(111) = 1111 \end{array}$$

Suppose now that $b = 111$. Then $x = e(b) = 1111$. If the transmission channel transmits x as $x_t = 1101$, then $|x_t| = 3$, and we know that an odd number of errors (at least one) has occurred. ◆

It should be noted that if the received word has even weight, then we cannot conclude that the code word was transmitted correctly, since this encoding function does not detect an even number of errors. Despite this limitation, the parity check code is widely used.

Example 3 Consider the following $(m, 3m)$ encoding function $e: B^m \to B^{3m}$. If

$$b = b_1 b_2 \cdots b_m \in B^m,$$

de ne

$$e(b) = e(b_1b_2 \cdots b_m) = b_1b_2 \cdots b_mb_1b_2 \cdots b_mb_1b_2 \cdots b_m.$$

That is, the encoding function e repeats each word of B^m three times. For a concrete example, let $m = 3$. Then

$$\left.\begin{array}{l} e(000) = 000000000 \\ e(001) = 001001001 \\ e(010) = 010010010 \\ e(011) = 011011011 \\ e(100) = 100100100 \\ e(101) = 101101101 \\ e(110) = 110110110 \\ e(111) = 111111111 \end{array}\right\} \text{code words.}$$

Suppose now that $b = 011$. Then $e(011) = 011011011$. Assume now that the transmission channel makes an error in the underlined digit and that we receive the word 011111011. This is not a code word, so we have detected the error. It is not hard to see that any single error and any two errors can be detected. ◆

Let x and y be words in B^m. The **Hamming distance** $\delta(x, y)$ between x and y is the weight, $|x \oplus y|$, of $x \oplus y$. Thus the distance between $x = x_1x_2 \cdots x_m$ and $y = y_1y_2 \cdots y_m$ is the number of values of i such that $x_i \neq y_i$, that is, the number of positions in which x and y differ. Using the weight of $x \oplus y$ is a convenient way to count the number of different positions.

Example 4 Find the distance between x and y:

(a) $x = 110110$, $y = 000101$

(b) $x = 001100$, $y = 010110$

Solution

(a) $x \oplus y = 110011$, so $|x \oplus y| = 4$

(b) $x \oplus y = 011010$, so $|x \oplus y| = 3$ ◆

THEOREM 1
Properties of the Distance Function

Let x, y, and z be elements of B^m. Then

(a) $\delta(x, y) = \delta(y, x)$

(b) $\delta(x, y) \geq 0$

(c) $\delta(x, y) = 0$ if and only if $x = y$

(d) $\delta(x, y) \leq \delta(x, z) + \delta(z, y)$

Proof

Properties (a), (b), and (c) are simple to prove and are left as exercises.

(d) For a and b in B^m,

$$|a \oplus b| \leq |a| + |b|,$$

since at any position where a and b differ one of them must contain a 1. Also, if $a \in B^m$, then $a \oplus a = \bar{0}$, the identity element in B^m. Then

$$\begin{aligned} \delta(x, y) &= |x \oplus y| = |x \oplus \bar{0} \oplus y| = |x \oplus z \oplus z \oplus y| \\ &\leq |x \oplus z| + |z \oplus y| \\ &= \delta(x, z) + \delta(z, y). \end{aligned}$$

■

The **minimum distance** of an encoding function $e: B^m \to B^n$ is the minimum of the distances between all distinct pairs of code words; that is,

$$\min\{\delta(e(x), e(y)) \mid x, y \in B^m\}.$$

Example 5 Consider the following $(2, 5)$ encoding function e:

$$\left.\begin{array}{l} e(00) = 00000 \\ e(10) = 00111 \\ e(01) = 01110 \\ e(11) = 11111 \end{array}\right\} \text{ code words.}$$

The minimum distance is 2, as can be checked by computing the minimum of the distances between all six distinct pairs of code words. ◆

THEOREM 2 An (m, n) encoding function $e: B^m \to B^n$ can detect k or fewer errors if and only if its minimum distance is at least $k + 1$.

Proof

Suppose that the minimum distance between any two code words is at least $k + 1$. Let $b \in B^m$, and let $x = e(b) \in B^n$ be the code word representing b. Then x is transmitted and is received as x_t. If x_t were a code word different from x, then $\delta(x, x_t) \geq k + 1$, so x would be transmitted with $k + 1$ or more errors. Thus, if x is transmitted with k or fewer errors, then x_t cannot be a code word. This means that e can detect k or fewer errors.

Conversely, suppose that the minimum distance between code words is $r \leq k$, and let x and y be code words with $\delta(x, y) = r$. If $x_t = y$, that is, if x is transmitted and is mistakenly received as y, then $r \leq k$ errors have been committed and have not been detected. Thus it is not true that e can detect k or fewer errors. ■

Example 6 Consider the $(3, 8)$ encoding function $e: B^3 \to B^8$ de ned by

$$\left.\begin{array}{l} e(000) = 00000000 \\ e(001) = 10111000 \\ e(010) = 00101101 \\ e(011) = 10010101 \\ e(100) = 10100100 \\ e(101) = 10001001 \\ e(110) = 00011100 \\ e(111) = 00110001 \end{array}\right\} \text{ code words.}$$

How many errors will e detect?

Solution

The minimum distance of e is 3, as can be checked by computing the minimum of the distances between all 28 distinct pairs of code words. By Theorem 2, the code will detect k or fewer errors if and only if its minimum distance is at least $k + 1$. Since the minimum distance is 3, we have $3 \geq k + 1$ or $k \leq 2$. Thus the code will detect two or fewer errors. ◆

Group Codes

So far, we have not made use of the fact that $(B^n, \oplus)$ is a group. We shall now consider an encoding function that makes use of this property of B^n.

An (m, n) encoding function $e: B^m \to B^n$ is called a **group code** if

$$e(B^m) = \{e(b) \mid b \in B^m\} = \text{Ran}(e)$$

is a subgroup of B^n.

N is a subgroup of B^n if (a) the identity of B^n is in N, (b) if x and y belong to N, then $x \oplus y \in N$, and (c) if x is in N, then its inverse is in N. Property (c) need not be checked here, since every element in B^n is its own inverse. Moreover, since B^n is Abelian, every subgroup of B^n is a normal subgroup.

Example 7 Consider the $(3, 6)$ encoding function $e: B^3 \to B^6$ defined by

$$\left.\begin{array}{l} e(000) = 000000 \\ e(001) = 001100 \\ e(010) = 010011 \\ e(011) = 011111 \\ e(100) = 100101 \\ e(101) = 101001 \\ e(110) = 110110 \\ e(111) = 111010 \end{array}\right\} \text{code words.}$$

Show that this encoding function is a group code.

Solution

We must show that the set of all code words

$$N = \{000000, 001100, 010011, 011111, 100101, 101001, 110110, 111010\}$$

is a subgroup of B^6. This is done by first noting that the identity of B^6 belongs to N. Next we verify, by trying all possibilities, that if x and y are elements in N, then $x \oplus y$ is in N. Hence N is a subgroup of B^6, and the given encoding function is a group code. ◆

The strategy of the next proof is similar to the way we often show two sets A and B are the same by showing that $A \subseteq B$ and $B \subseteq A$. Here we show that $\delta = \eta$ by proving $\delta \leq \eta$ and $\eta \leq \delta$.

THEOREM 3 Let $e: B^m \to B^n$ be a group code. The minimum distance of e is the minimum weight of a nonzero code word.

Proof

Let δ be the minimum distance of the group code, and suppose that $\delta = \delta(x, y)$, where x and y are distinct code words. Also, let η be the minimum weight of a nonzero code word and suppose that $\eta = |z|$ for a code word z. Since e is a group code, $x \oplus y$ is a nonzero code word. Thus

$$\delta = \delta(x, y) = |x \oplus y| \geq \eta.$$

On the other hand, since 0 and z are distinct code words,

$$\eta = |z| = |z \oplus 0| = \delta(z, 0) \geq \delta.$$

Hence $\eta = \delta$. ■

One advantage of a group code is given in the following example.

Example 8 The minimum distance of the group code in Example 7 is 2, since by Theorem 3 this distance is equal to the smallest number of 1 s in any of the seven nonzero code words. To check this directly would require 28 different calculations. ◆

We shall now take a brief look at a procedure for generating group codes. First, we need several additional results on Boolean matrices. Consider the set B with operation $+$ de ned in Table 1. Now let $\mathbf{D} = [d_{ij}]$ and $\mathbf{E} = [e_{ij}]$ be $m \times n$ Boolean matrices. We de ne the **mod-2 sum** $\mathbf{D} \oplus \mathbf{E}$ as the $m \times n$ Boolean matrix $\mathbf{F} = [f_{ij}]$, where

$$f_{ij} = d_{ij} + e_{ij}, \qquad 1 \le i \le m, \quad 1 \le j \le n. \qquad \text{(Here } + \text{ is addition in } B.)$$

Example 9 We have

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \oplus \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+1 & 0+1 & 1+0 & 1+1 \\ 0+1 & 1+1 & 1+0 & 0+1 \\ 1+0 & 0+1 & 0+1 & 1+1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}.$$

Observe that if $\mathbf{F} = \mathbf{D} \oplus \mathbf{E}$, then f_{ij} is zero when *both* d_{ij} and e_{ij} are zero or both are one. ◆

TABLE 2

$\cdot$	0	1
0	0	0
1	0	1

Next, consider the set $B = \{0, 1\}$ with the binary operation given in Table 2. This operation has been seen earlier in a different setting and with a different symbol. B is the unique Boolean algebra with two elements. In particular, B is a lattice with partial order $\le$ de ned by $0 \le 0$, $0 \le 1$, $1 \le 1$. Then the reader may easily check that if a and b are any two elements of B,

$$a \cdot b = a \wedge b \qquad \text{(the greatest lower bound of } a \text{ and } b\text{)}.$$

Thus Table 2 for $\cdot$ is just a copy of the table for $\wedge$.

Let $\mathbf{D} = [d_{ij}]$ be an $m \times p$ Boolean matrix, and let $\mathbf{E} = [e_{ij}]$ be a $p \times n$ Boolean matrix. We de ne the **mod-2 Boolean product** $\mathbf{D} \odot \mathbf{E}$ as the $m \times n$ matrix $\mathbf{F} = [f_{ij}]$, where

$$f_{ij} = d_{i1} \cdot e_{1j} + d_{i2} \cdot e_{2j} + \cdots + d_{ip} \cdot e_{pj}, \qquad 1 \le i \le m, \quad 1 \le j \le n.$$

This type of multiplication is illustrated in Figure 3.

Example 10 We have

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

◆

The proof of the following theorem is left as an exercise.

THEOREM 4 Let $\mathbf{D}$ and $\mathbf{E}$ be $m \times p$ Boolean matrices, and let $\mathbf{F}$ be a $p \times n$ Boolean matrix. Then

$$(\mathbf{D} \oplus \mathbf{E}) \odot \mathbf{F} = (\mathbf{D} \odot \mathbf{F}) \oplus (\mathbf{E} \odot \mathbf{F}).$$

That is, a distributive property holds for $\odot$ and $\oplus$. ■

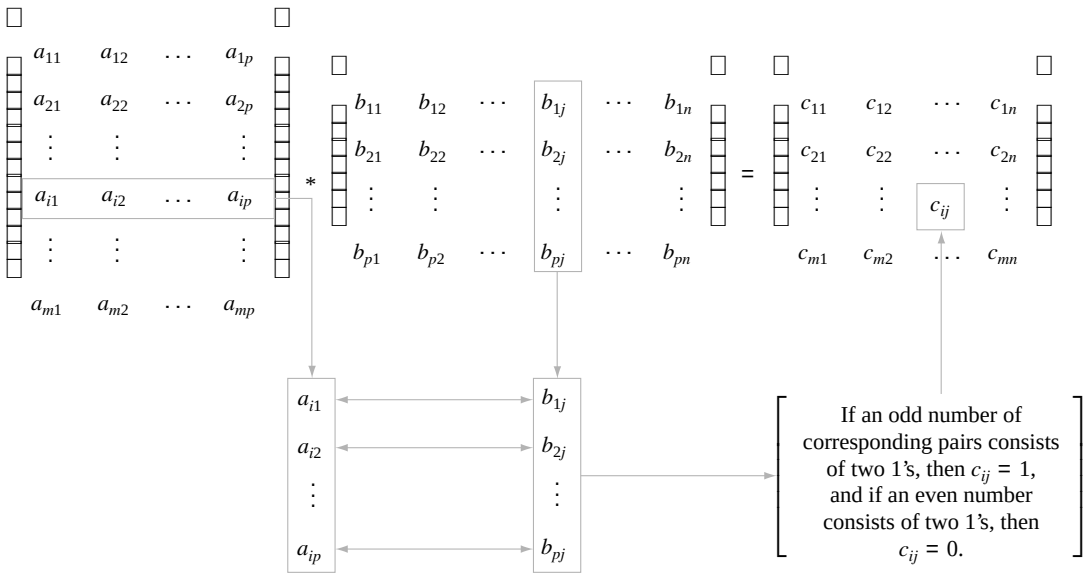


Figure 3

We shall now consider the element $x = x_1x_2 \cdots x_n \in B^n$ as the $1 \times n$ matrix $\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}$.

THEOREM 5 Let m and n be nonnegative integers with $m < n$, $r = n - m$, and let $\mathbf{H}$ be an $n \times r$ Boolean matrix. Then the function $f_H : B^n \to B^r$ defined by

$$f_H(x) = x \cdot \mathbf{H}, \qquad x \in B^n$$

is a homomorphism from the group B^n to the group B^r.

Proof
If x and y are elements in B^n, then

$$\begin{aligned} f_H(x \cdot y) &= (x \cdot y) \cdot \mathbf{H} \\ &= (x \cdot \mathbf{H}) \cdot (y \cdot \mathbf{H}) \qquad \text{by Theorem 4} \\ &= f_H(x) \cdot f_H(y). \end{aligned}$$

Hence f_H is a homomorphism from B^n to B^r. ■

Corollary 1 Let m, n, r, $\mathbf{H}$, and f_H be as in Theorem 5. Then

$$N = \{x \in B^n \mid x \cdot \mathbf{H} = \bar{0}\}$$

is a normal subgroup of B^n.

Proof
It follows that N is the kernel of the homomorphism f_H, so it is a normal subgroup of B^n. ■

Let $m < n$ and $r = n - m$. An $n \times r$ Boolean matrix

$$\mathbf{H} = \begin{matrix} \\ \\ \\ \\ \\ \\ \\ \\ \end{matrix}\!\!\begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}\!\!\begin{matrix} \\ \\ \\ \\ \\ \\ \\ \\ \end{matrix}, \qquad n - m = r \text{ rows}$$

whose last r rows form the $r \times r$ identity matrix, is called a **parity check matrix**. We use $\mathbf{H}$ to de ne an encoding function $e_H : B^m \to B^n$. If $b = b_1 b_2 \cdots b_m$, let $x = e_H(b) = b_1 b_2 \cdots b_m x_1 x_2 \cdots x_r$, where

$$\begin{aligned} x_1 &= b_1 \cdot h_{11} + b_2 \cdot h_{21} + \cdots + b_m \cdot h_{m1} \\ x_2 &= b_1 \cdot h_{12} + b_2 \cdot h_{22} + \cdots + b_m \cdot h_{m2} \\ &\vdots \\ x_r &= b_1 \cdot h_{1r} + b_2 \cdot h_{2r} + \cdots + b_m \cdot h_{mr}. \end{aligned} \tag{1}$$

THEOREM 6 Let $x = y_1 y_2 \cdots y_m x_1 \cdots x_r \in B^n$. Then $x \cdot \mathbf{H} = \bar{0}$ if and only if $x = e_H(b)$ for some $b \in B^m$.

Proof
Suppose that $x \cdot \mathbf{H} = \bar{0}$. Then

$$\begin{aligned} y_1 \cdot h_{11} + y_2 \cdot h_{21} + \cdots + y_m \cdot h_{m1} + x_1 &= 0 \\ y_1 \cdot h_{12} + y_2 \cdot h_{22} + \cdots + y_m \cdot h_{m2} + x_2 &= 0 \\ &\vdots \\ y_1 \cdot h_{1r} + y_2 \cdot h_{2r} + \cdots + y_m \cdot h_{mr} + x_r &= 0. \end{aligned}$$

The first equation is of the form

$$a + x_1 = 0, \quad \text{where } a = y_1 \cdot h_{11} + y_2 \cdot h_{21} + \cdots + y_m \cdot h_{m1}.$$

Adding a to both sides, we obtain

$$\begin{aligned} a + (a + x_1) &= a + 0 = a \\ (a + a) + x_1 &= a \\ 0 + x_1 &= a \quad \text{since } a + a = 0 \\ x_1 &= a. \end{aligned}$$

This can be done for each row; therefore,

$$x_i = y_1 \cdot h_{1i} + y_2 \cdot h_{2i} + \cdots + y_m \cdot h_{mi}, \qquad 1 \le i \le r.$$

Letting $b_1 = y_1$, $b_2 = y_2$, ..., $b_m = y_m$, we see that $x_1, x_2, \ldots, x_r$ satisfy the equations in (1). Thus $b = b_1 b_2 \cdots b_m \in B^m$ and $x = e_H(b)$.

Conversely, if $x = e_H(b)$, the equations in (1) can be rewritten by adding x_i to both sides of the ith equation, $i = 1, 2, \ldots, n$, as

$$\begin{aligned} b_1 \cdot h_{11} + b_2 \cdot h_{21} + \cdots + b_m \cdot h_{m1} + x_1 &= 0 \\ b_1 \cdot h_{12} + b_2 \cdot h_{22} + \cdots + b_m \cdot h_{m2} + x_2 &= 0 \\ &\vdots \\ b_1 \cdot h_{1r} + b_2 \cdot h_{2r} + \cdots + b_m \cdot h_{mr} + x_r &= 0, \end{aligned}$$

which shows $x \in \mathbf{H} = \overline{0}$. ■

Corollary 2 $e_H(B^m) = \{e_H(b) \mid b \in B^m\}$ is a subgroup of B^n.

Proof
The result follows from the observation that

$$e_H(B^m) = \ker(f_H)$$

and from Corollary 1. Thus e_H is a group code. ■

Example 11 Let $m = 2$, $n = 5$, and

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Determine the group code $e_H : B^2 \to B^5$.

Solution
We have $B^2 = \{00, 10, 01, 11\}$. Then

$$e(00) = 00x_1x_2x_3,$$

where x_1, x_2, and x_3 are determined by the equations in (1). Thus

$$x_1 = x_2 = x_3 = 0$$

and

$$e(00) = 00000.$$

Next

$$e(10) = 10x_1x_2x_3.$$

Using the equations in (1) with $b_1 = 1$ and $b_2 = 0$, we obtain

$$\begin{aligned} x_1 &= 1 \cdot 1 + 0 \cdot 0 = 1 \\ x_2 &= 1 \cdot 1 + 0 \cdot 1 = 1 \\ x_3 &= 1 \cdot 0 + 0 \cdot 1 = 0. \end{aligned}$$

Thus $x_1 = 1$, $x_2 = 1$, and $x_3 = 0$, so

$$e(10) = 10110.$$

Similarly (verify),

$$\begin{aligned} e(01) &= 01011 \\ e(11) &= 11101. \end{aligned}$$

◆

1 Exercises

In Exercises 1 and 2, find the weight of the given words.

1. **(a)** 1011 **(b)** 0110 **(c)** 1110

2. **(a)** 011101 **(b)** 11111 **(c)** 010101

3. Consider the $(3, 4)$ parity check code. For each of the received words, determine whether an error will be detected.

(a) 0100 **(b)** 1100

4. Consider the $(3, 4)$ parity check code. For each of the received words, determine whether an error will be detected.

(a) 0010 **(b)** 1001

5. Consider the $(6, 7)$ parity check code. For each of the received words, determine whether an error will be detected.

(a) 1101010 **(b)** 1010011

(c) 0011111 **(d)** 1001101

In Exercises 6 through 8, use the $(m, 3m)$ *encoding function of Example 3 for the given value of* m*. For each of the received words, determine whether an error will be detected.*

6. $m = 3$

(a) 110111110 **(b)** 110011011

7. $m = 4$

(a) 011010011111 **(b)** 110110010110

8. $m = 4$

(a) 010010110010 **(b)** 001001111001

9. The ISBN system is also an error-detecting code. One error in transmission will be detected, but two may not be. Show, however, that any transposition of adjacent digits before the check digit will be detected.

10. Twelve-digit bar codes use the twelfth digit as a check digit by choosing it so that the sum of the digits in even-numbered positions and three times the sum of the digits in odd-numbered positions is congruent to 0 mod 10. Show that this code will detect a single error in an even-numbered position but may not detect two errors in even-numbered positions.

11. Explain how $|x \square y|$ counts the number of positions in which x and y differ.

12. Find the distance between x and y.

(a) $x = 1100010, y = 1010001$

(b) $x = 0100110, y = 0110010$

13. Find the distance between x and y.

(a) $x = 00111001, y = 10101001$

(b) $x = 11010010, y = 00100111$

14. **(a)** Prove Theorem 1(a).

(b) Prove Theorem 1(b).

15. Prove Theorem 1(c).

16. Find the minimum distance of the $(2, 4)$ encoding function e.

$$e(00) = 0000 \qquad e(10) = 0110$$
$$e(01) = 1011 \qquad e(11) = 1100$$

17. Find the minimum distance of the $(3, 8)$ encoding function e.

$$e(000) = 00000000 \qquad e(100) = 01100101$$
$$e(001) = 01110010 \qquad e(101) = 10110000$$
$$e(010) = 10011100 \qquad e(110) = 11110000$$
$$e(011) = 01110001 \qquad e(111) = 00001111$$

18. Consider the $(2, 6)$ encoding function e.

$$e(00) = 000000 \qquad e(10) = 101010$$
$$e(01) = 011110 \qquad e(11) = 111000$$

(a) Find the minimum distance of e.

(b) How many errors will e detect?

19. Consider the $(3, 9)$ encoding function e.

$$e(000) = 000000000 \qquad e(100) = 010011010$$
$$e(001) = 011100101 \qquad e(101) = 111101011$$
$$e(010) = 010101000 \qquad e(110) = 001011000$$
$$e(011) = 110010001 \qquad e(111) = 110000111$$

(a) Find the minimum distance of e.

(b) How many errors will e detect?

20. Show that the $(2, 5)$ encoding function $e: B^2 \to B^5$ de ned by

$$e(00) = 00000 \qquad e(10) = 10101$$
$$e(01) = 01110 \qquad e(11) = 11011$$

is a group code.

21. Show that the $(3, 7)$ encoding function $e: B^3 \to B^7$ de ned by

$$e(000) = 0000000 \qquad e(100) = 1000101$$
$$e(001) = 0010110 \qquad e(101) = 1010011$$
$$e(010) = 0101000 \qquad e(110) = 1101101$$
$$e(011) = 0111110 \qquad e(111) = 1111011$$

is a group code.

22. Find the minimum distance of the group code de ned in Exercise 20.

23. Find the minimum distance of the group code de ned in Exercise 21.

24. Compute

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

25. Compute

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

26. Compute

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

27. Compute

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

28. Which of the following properties does the structure ($m \times n$ Boolean matrices, $\square$) have?

(a) Associative **(b)** Commutative

(c) Identity **(d)** Inverses

29. Which of the following properties does the structure ($n \times n$ Boolean matrices, $\square$) have?

(a) Associative **(b)** Commutative

(c) Identity **(d)** Inverses

30. Let

$$\mathbf{H} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Determine the $(2, 5)$ group code function $e_H : B^2 \to B^5$.

31. Let

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Determine the $(3, 6)$ group code $e_H : B^3 \to B^6$.

32. Prove Theorem 4.

33. Outline the structure of the proof given for Theorem 2.

2 Decoding and Error Correction

Consider an (m, n) encoding function $e : B^m \to B^n$. Once the encoded word $x = e(b) \in B^n$, for $b \in B^m$, is received as the word x_t, we are faced with the problem of identifying the word b that was the original message.

An onto function $d : B^n \to B^m$ is called an (n, m) **decoding function associated with *e*** if $d(x_t) = b' \in B^m$ is such that when the transmission channel has no noise, then $b' = b$, that is,

$$d \circ e = 1_{B^m},$$

where 1_{B^m} is the identity function on B^m. The decoding function d is required to be onto so that every received word can be decoded to give a word in B^m. It decodes properly received words correctly, but the decoding of improperly received words may or may not be correct.

Example 1 Consider the parity check code that is de ned in Example 2 of Section 1. We now de ne the decoding function $d : B^{m+1} \to B^m$. If $y = y_1 y_2 \cdots y_m y_{m+1} \in B^{m+1}$, then

$$d(y) = y_1 y_2 \cdots y_m.$$

Observe that if $b = b_1 b_2 \cdots b_m \in B^m$, then

$$(d \circ e)(b) = d(e(b)) = b,$$

so $d \circ e = 1_{B^m}$.

For a concrete example, let $m = 4$. Then we obtain $d(10010) = 1001$ and $d(11001) = 1100$. ◆

Let e be an (m, n) encoding function and let d be an (n, m) decoding function associated with e. We say that the pair (e, d) **corrects *k* or fewer errors** if whenever $x = e(b)$ is transmitted correctly or with k or fewer errors and x_t is received, then $d(x_t) = b$. Thus x_t is decoded as the correct message b.

Example 2 Consider the $(m, 3m)$ encoding function defined in Example 3 of Section 1. We now define the decoding function $d: B^{3m} \to B^m$. Let

$$y = y_1 y_2 \cdots y_m y_{m+1} \cdots y_{2m} y_{2m+1} \cdots y_{3m}.$$

Then

$$d(y) = z_1 z_2 \cdots z_m,$$

where

$$z_i = \begin{cases} 1 & \text{if } \{y_i, y_{i+m}, y_{i+2m}\} \text{ has at least two 1 s} \\ 0 & \text{if } \{y_i, y_{i+m}, y_{i+2m}\} \text{ has less than two 1 s.} \end{cases}$$

That is, the decoding function d examines the ith digit in each of the three blocks transmitted. The digit that occurs at least twice in these three blocks is chosen as the decoded ith digit. For a concrete example, let $m = 3$. Then

$$e(100) = 100100100$$
$$e(011) = 011011011$$
$$e(001) = 001001001.$$

Suppose now that $b = 011$. Then $e(011) = 011\underline{0}11011$. Assume now that the transmission channel makes an error in the underlined digit and that we receive the word $x_t = 011111011$. Then, since the first digits in two out of the three blocks are 0, the first digit is decoded as 0. Similarly, the second digit is decoded as 1, since all three second digits in the three blocks are 1. Finally, the third digit is also decoded as 1, for the analogous reason. Hence $d(x_t) = 011$; that is, the decoded word is 011, which is the word that was sent. Therefore, the single error has been corrected. A similar analysis shows that if e is this $(m, 3m)$ code for any value of m and d is as defined, then (e, d) corrects any single error. ◆

Given an (m, n) encoding function $e: B^m \to B^n$, we often need to determine an (n, m) decoding function $d: B^n \to B^m$ associated with e. We now discuss a method, called the **maximum likelihood technique**, for determining a decoding function d for a given e.

Since B^m has 2^m elements, there are 2^m code words in B^n. We first list the code words in a fixed order:

$$x^{(1)}, x^{(2)}, \ldots, x^{(2^m)}.$$

If the received word is x_t, we compute $\delta(x^{(i)}, x_t)$ for $1 \le i \le 2^m$ and choose the first code word, say it is $x^{(s)}$, such that

$$\min_{1 \le i \le 2^m} \{\delta(x^{(i)}, x_t)\} = \delta(x^{(s)}, x_t).$$

That is, $x^{(s)}$ is a code word that is closest to x_t and the first in the list. If $x^{(s)} = e(b)$, we define the **maximum likelihood decoding function** d associated with e by

$$d(x_t) = b.$$

Observe that d depends on the particular order in which the code words in $e(B^n)$ are listed. If the code words are listed in a different order, we may obtain a different maximum likelihood decoding function d associated with e.

THEOREM 1 Suppose that e is an (m, n) encoding function and d is a maximum likelihood decoding function associated with e. Then (e, d) can correct k or fewer errors if and only if the minimum distance of e is at least $2k + 1$.

Proof
Assume that the minimum distance of e is at least $2k + 1$. Let $b \in B^m$ and $x = e(b) \in B^n$. Suppose that x is transmitted with k or fewer errors, and x_t is received. This means that $\delta(x, x_t) \leq k$. If z is any other code word, then

$$2k + 1 \leq \delta(x, z) \leq \delta(x, x_t) + \delta(x_t, z) \leq k + d(x_t, z).$$

Thus $\delta(x_t, z) \geq 2k + 1 - k = k + 1$. This means that x is the unique code word that is closest to x_t, so $d(x_t) = b$. Hence (e, d) corrects k or fewer errors.

Conversely, assume that the minimum distance between code words is $r \leq 2k$, and let $x = e(b)$ and $x' = e(b')$ be code words with $\delta(x, x') = r$. Suppose that x' precedes x in the list of code words used to de ne d. Write $x = b_1 b_2 \cdots b_n$, $x' = b'_1 b'_2 \cdots b'_n$. Then $b_i \neq b'_i$ for exactly r integers i between 1 and n. Assume, for simplicity, that $b_1 \neq b'_1$, $b_2 \neq b'_2$, $\ldots$, $b_r \neq b'_r$, but $b_i = b'_i$ when $i > r$. Any other case is handled in the same way.

(a) Suppose that $r \leq k$. If x is transmitted as $x_t = x'$, then $r \leq k$ errors have been committed, but $d(x_t) = b'$; so (e, d) has not corrected the r errors.

(b) Suppose that $k + 1 \leq r \leq 2k$, and let

$$y = b'_1 b'_2 \cdots b'_k b_{k+1} \cdots b_n.$$

If x is transmitted as $x_t = y$, then $\delta(x_t, x') = r - k \leq k$ and $\delta(x_t, x) \geq k$. Thus x' is at least as close to x_t as x is, and x' precedes x in the list of code words; so $d(x_t) \neq b$. Then we have committed k errors, which (e, d) has not corrected. ■

Example 3 Let e be the $(3, 8)$ encoding function de ned in Example 6 of Section 1, and let d be an $(8, 3)$ maximum likelihood decoding function associated with e. How many errors can (e, d) correct?

Solution

Since the minimum distance of e is 3, we have $3 \geq 2k + 1$, so $k \leq 1$. Thus (e, d) can correct one error. ◆

We now discuss a simple and effective technique for determining a maximum likelihood decoding function associated with a given group code. First, we prove the following result.

THEOREM 2 If K is a nite subgroup of a group G, then every left coset of K in G has exactly as many elements as K.

Proof
Let aK be a left coset of K in G, where $a \in G$. Consider the function $f : K \to aK$ de ned by

$$f(k) = ak, \qquad \text{for } k \in K.$$

We show that f is one to one and onto.

To show that f is one to one, we assume that

$$f(k_1) = f(k_2), \qquad k_1, k_2 \in K.$$

Then

$$ak_1 = ak_2.$$

$k_1 = k_2$. Hence f is one to one.

To show that f is onto, let b be an arbitrary element in aK. Then $b = ak$ for some $k \in K$. We now have

$$f(k) = ak = b,$$

so f is onto. Since f is one to one and onto, K and aK have the same number of elements. ■

Let $e: B^m \to B^n$ be an (m, n) encoding function that is a group code. Thus the set N of code words in B^n is a subgroup of B^n whose order is 2^m, say $N = \{x^{(1)}, x^{(2)}, \ldots, x^{(2^m)}\}$.

Suppose that the code word $x = e(b)$ is transmitted and that the word x_t is received. The left coset of x_t is

$$x_t + N = \{\alpha_1, \alpha_2, \ldots, \alpha_{2^m}\},$$

where $\alpha_i = x_t + x^{(i)}$. The distance from x_t to code word $x^{(i)}$ is just $|\alpha_i|$, the weight of α_i. Thus, if α_j is a coset member with smallest weight, then $x^{(j)}$ must be a code word that is closest to x_t. In this case, $x^{(j)} = \bar{0} + x^{(j)} = x_t + x_t + x^{(j)} = x_t + \alpha_j$. An element α_j, having smallest weight, is called a **coset leader**. Note that a coset leader need not be unique.

If $e: B^m \to B^n$ is a group code, we now state the following procedure for obtaining a maximum likelihood decoding function associated with e.

Step 1 Determine all the left cosets of $N = e(B^m)$ in B^n.

Step 2 For each coset, find a coset leader (a word of least weight). Steps 1 and 2 can be carried out in a systematic tabular manner that will be described later.

Step 3 If the word x_t is received, determine the coset of N to which x_t belongs. Since N is a normal subgroup of B^n, the cosets of N form a partition of B^n, so each element of B^n belongs to one and only one coset of N in B^n. Moreover, there are $2^n/2^m$ or 2^r distinct cosets of N in B^n.

Step 4 Let α be a coset leader for the coset determined in Step 3. Compute $x = x_t + \alpha$. If $x = e(b)$, we let $d(x_t) = b$. That is, we decode x_t as b.

To implement the foregoing procedure, we must keep a complete list of all the cosets of N in B^n, usually in tabular form, with each row of the table containing one coset. We identify a coset leader in each row. Then, when a word x_t is received, we locate the row that contains it, find the coset leader for that row, and add it to x_t. This gives us the code word closest to x_t. We can eliminate the need for these additions if we construct a more systematic table.

Before illustrating with an example, we make several observations. Let

$$N = \{x^{(1)}, x^{(2)}, \ldots, x^{(2^m)}\},$$

where $x^{(1)}$ is $\bar{0}$, the identity of B^n.

Steps 1 and 2 in the preceding decoding algorithm are carried out as follows. First, list all the elements of N in a row, starting with the identity $\bar{0}$ at the left. Thus we have

$$\bar{0} \quad x^{(2)} \quad x^{(3)} \quad \cdots \quad x^{(2^m)}.$$

This row is the coset $[\bar{0}]$, and it has $\bar{0}$ as its coset leader. For this reason we will also refer to $\bar{0}$ as α_1. Now choose any word y in B^n that has not been listed in the first row. List the elements of the coset $y + N$ as the second row. This coset also

has 2^m elements. Thus we have the two rows

$$\begin{array}{cccccc} \bar{0} & & x^{(2)} & x^{(3)} & \cdots & x^{(2^m)} \\ y \oplus \bar{0} & & y \oplus x^{(2)} & y \oplus x^{(3)} & \cdots & y \oplus x^{(2^m)}. \end{array}$$

In the coset $y \oplus N$, pick an element of least weight, a coset leader, which we denote by $\alpha^{(2)}$. In case of ties, choose any element of least weight. Since $\alpha^{(2)} \in y \oplus N$, we have $y \oplus N = \alpha^{(2)} \oplus N$. This means that every word in the second row can be written as $\alpha^{(2)} \oplus \nu$, $\nu \in N$. Now rewrite the second row as follows:

$$\alpha^{(2)} \quad \alpha^{(2)} \oplus x^{(2)} \quad \alpha^{(2)} \oplus x^{(3)} \quad \cdots \quad \alpha^{(2)} \oplus x^{(2^m)}$$

with $\alpha^{(2)}$ in the leftmost position.

Next, choose another element z in B^n that has not yet been listed in either of the first two rows and form the third row $(z \oplus x^{(j)})$, $1 \le j \le 2^m$ (another coset of N in B^n). This row can be rewritten in the form

$$\alpha^{(3)} \quad \alpha^{(3)} \oplus x^{(2)} \quad \alpha^{(3)} \oplus x^{(3)} \quad \cdots \quad \alpha^{(3)} \oplus x^{(2^m)},$$

where $\alpha^{(3)}$ is a coset leader for the row.

Continue this process until all elements of B^n have been listed. The resulting Table 3 is called a **decoding table**. Notice that it contains 2^r rows, one for each coset of N. If we receive the word x_t, we locate it in the table. If x is the element of N that is at the top of the column containing x_t, then x is the code word closest to x_t. Thus, if $x = e(b)$, we let $d(x_t) = b$.

TABLE 3

$\bar{0}$	$x^{(2)}$	$x^{(3)}$	$\cdots$	$x^{(2^m-1)}$
$\alpha^{(2)}$	$\alpha^{(2)} \oplus x^{(2)}$	$\alpha^{(2)} \oplus x^{(3)}$	$\cdots$	$\alpha^{(2)} \oplus x^{(2^m-1)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
$\alpha^{(2^r)}$	$\alpha^{(2^r)} \oplus x^{(2)}$	$\alpha^{(2^r)} \oplus x^{(3)}$	$\cdots$	$\alpha^{(2^r)} \oplus x^{(2^m-1)}$

Example 4 Consider the $(3, 6)$ group code defined in Example 7 of Section 1. Here

$$\begin{aligned} N &= \{000000, 001100, 010011, 011111, 100101, 101001, 110110, 111010\} \\ &= \{x^{(1)}, x^{(2)}, \ldots, x^{(8)}\} \end{aligned}$$

as defined in Example 1. We now implement the decoding procedure for e as follows.

Steps 1 and 2: Determine all the left cosets of N in B^6, as rows of a table. For each row i, locate the coset leader α_i, and rewrite the row in the order

$$\alpha_i, \quad \alpha_i \oplus 001100, \quad \alpha_i \oplus 010011, \quad \ldots, \quad \alpha_i \oplus 111010.$$

The result is shown in Table 4.

Steps 3 and 4: If we receive the word 000101, we decode it by first locating it in the decoding table; it appears in the fifth column, where it is underlined. The word at the top of the fifth column is 100101. Since $e(100) = 100101$, we decode 000101 as 100. Similarly, if we receive the word 010101, we first locate it in the third column of the decoding table, where it is underlined twice. The word at the top of the third column is 010011. Since $e(010) = 010011$, we decode 010101 as 010.

TABLE 4

000000	001100	010011	011111	100101	101001	110110	111010
000001	001101	010010	011110	100100	101000	110111	111011
000010	001110	010001	011101	100111	101011	110100	111000
000100	001000	010111	011011	100001	101101	110010	111110
010000	011100	000011	001111	110101	111001	100110	101010
100000	101100	110011	111111	000101	001001	010110	011010
000110	001010	010101	011001	100011	101111	110000	111100
010100	011000	000111	001011	110001	111101	100010	101110

We make the following observations for this example. In determining the decoding table in Steps 1 and 2, there was more than one candidate for coset leader of the last two cosets. In row 7 we chose 00110 as coset leader. If we had chosen 001010 instead, row 7 would have appeared in the rearranged form

$$001010 \quad 001010 \square 001100 \quad \cdots \quad 001010 \square 111010$$

or

$$001010 \quad 000110 \quad 011001 \quad 010101 \quad 101111 \quad 100011 \quad 111100 \quad 110000.$$

The new decoding table is shown in Table 5.

TABLE 5

000000	001100	010011	011111	100101	101001	110110	111010
000001	001101	010010	011110	100100	101000	110111	111011
000010	001110	010001	011101	100111	101011	110100	111000
000100	001000	010111	011011	100001	101101	110010	111110
010000	011100	000011	001111	110101	111001	100110	101010
100000	101100	110011	111111	000101	001001	010110	011010
001010	000110	011001	010101	101111	100011	111100	110000
010100	011000	000111	001011	110001	111101	100010	101110

Now, if we receive the word 010101, we first locate it in the *fourth* column of Table 5. The word at the top of the fourth column is 011111. Since $e(011) = 011111$, we decode 010101 as 011. ◆

Suppose that the (m, n) group code is $e_H : B^m \to B^n$, where $\mathbf{H}$ is a given parity check matrix. In this case, the decoding technique above can be simplified. We now turn to a discussion of this situation.

Recall from Section 1 that $r = n - m$,

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

and the function $f_H : B^n \to B^r$ defined by

$$f_H(x) = x \cdot \mathbf{H}$$

is a homomorphism from the group B^n to the group B^r.

THEOREM 3 If m, n, r, $\mathbf{H}$, and f_H are as defined, then f_H is onto.

Proof
Let $b = b_1 b_2 \cdots b_r$ be any element in B^r. Letting

$$x = \underbrace{0 0 \cdots 0}_{m\ 0\text{'s}} b_1 b_2 \cdots b_r$$

we obtain $x \cdot \mathbf{H} = b$. Thus $f_H(x) = b$, so f_H is onto. ■

It follows that B^r and B^n/N are isomorphic, where $N = \ker(f_H) = e_H(B^m)$, under the isomorphism $g : B^n/N \to B^r$ defined by

$$g(xN) = f_H(x) = x \cdot \mathbf{H}.$$

The element $x \cdot \mathbf{H}$ is called the **syndrome** of x. We now have the following result.

THEOREM 4 Let x and y be elements in B^n. Then x and y lie in the same left coset of N in B^n if and only if $f_H(x) = f_H(y)$, that is, if and only if they have the same syndrome.

Proof
It follows that x and y lie in the same left coset of N in B^n if and only if $x - y = (-x) + y \in N$. Since $N = \ker(f_H)$, $x - y \in N$ if and only if

$$\begin{aligned} f_H(x - y) &= \bar{0}_{B^r} \\ f_H(x) - f_H(y) &= \bar{0}_{B^r} \\ f_H(x) &= f_H(y). \end{aligned}$$

■

In this case, the decoding procedure given previously can be modified as follows. Suppose that we compute the syndrome of each coset leader. If the word x_t is received, we also compute $f_H(x_t)$, the syndrome of x_t. By comparing $f_H(x_t)$ and the syndromes of the coset leaders, we find the coset in which x_t lies. Suppose that a coset leader of this coset is ℓ. We now compute $x = x_t - \ell$. If $x = e(b)$, we then decode x_t as b. Thus we need only the coset leaders and their syndromes in order to decode. We state the new procedure in detail.

Step 1 Determine all left cosets of $N = e_H(B^m)$ in B^n.

Step 2 For each coset, find a coset leader, and compute the syndrome of each leader.

Step 3 If x_t is received, compute the syndrome of x_t and find the coset leader ℓ having the same syndrome. Then $x_t - \ell = x$ is a code word $e_H(b)$, and $d(x_t) = b$.

For this procedure, we do not need to keep a table of cosets, and we can avoid the work of computing a decoding table. Simply list all cosets once, in any order, and select a coset leader from each coset. Then keep a table of these coset leaders and their syndromes. The foregoing procedure is easily implemented with such a table.

Example 5 Consider the parity check matrix

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and the $(3, 6)$ group $e_H : B^3 \to B^6$. Then

$$\begin{bmatrix} e(000) = 000000 \\ e(001) = 001011 \\ e(010) = 010101 \\ e(011) = 011110 \\ e(100) = 100110 \\ e(101) = 101101 \\ e(110) = 110011 \\ e(111) = 111000 \end{bmatrix}$$

code words.

TABLE 6

Syndrome of Coset Leader	Coset Leader
000	000000
001	000001
010	000010
011	001000
100	000100
101	010000
110	100000
111	001100

Thus

$$N = \{000000, 001011, 010101, 011110, 100110, 101101, 110011, 111000\}.$$

We now implement the decoding procedure as follows.

In Table 6 we give only the coset leaders together with their syndromes. Suppose now that we receive the word 001110. We compute the syndrome of $x_t = 001110$, obtaining $f_H(x_t) = x_t \cdot \mathbf{H} = 101$, which is the sixth entry in the first column of Table 6. This means that x_t lies in the coset whose leader is $\square = 010000$. We compute

$$x = x_t \square \square = 001110 \square 010000 = 011110.$$

Since $e(011) = 011110$, we decode 001110 as 011. ◆

2 Exercises

1. Let d be the $(4, 3)$ decoding function defined by letting m be 3 in Example 1. Determine $d(y)$ for the word y in B^4.

(a) $y = 0110$ **(b)** $y = 1011$

2. Let d be the $(6, 5)$ decoding function defined by letting m be 5 in Example 1. Determine $d(y)$ for the word y in B^6.

(a) $y = 001101$ **(b)** $y = 110100$

3. Let d be the $(6, 2)$ decoding function defined in Example 2. Determine $d(y)$ for the word y in B^6.

(a) $y = 111011$ **(b)** $y = 010100$

4. Let d be the $(9, 3)$ decoding function defined in the same way as the decoding function in Example 2. Determine $d(y)$ for the word y in B^9.

(a) $y = 101111101$ **(b)** $y = 100111100$

5. Let $e : B^2 \to B^4$ be the $(2, 4)$ encoding function defined by $e(00) = 0000$, $e(01) = 1011$, $e(10) = 0110$, $e(11) = 1101$. Construct a table of left cosets in B^4 for $N = \{0000, 1011, 0110, 1101\}$. Place the coset leader at the beginning of each row.

6. Let e be the $(2, 5)$ encoding function defined in Exercise 20 of Section 1. Construct a table of left cosets in B^4 for $N = e(B^2)$. Place the coset leader at the beginning of each row.

In Exercises 7 through 12, let e *be the indicated encoding function and let* d *be an associated maximum likelihood decoding function. Determine the number of errors that* (e, d) *will correct.*

7. e is the encoding function in Exercise 16 of Section 1.

8. e is the encoding function in Exercise 17 of Section 1.

9. e is the encoding function in Exercise 18 of Section 1.

10. e is the encoding function in Exercise 19 of Section 1.

11. e is the encoding function in Exercise 20 of Section 1.

12. e is the encoding function in Exercise 21 of Section 1.

13. Consider the group code de ned in Exercise 20 of Section 1. Decode the following words relative to a maximum likelihood decoding function.

(a) 11110 **(b)** 10011 **(c)** 10100

14. Consider the $(2, 4)$ group encoding function $e: B^2 \to B^4$ de ned by

$$e(00) = 0000 \qquad e(10) = 1001$$
$$e(01) = 0111 \qquad e(11) = 1111.$$

Decode the following words relative to a maximum likelihood decoding function.

(a) 0011 **(b)** 1011 **(c)** 1111

15. Consider the $(3, 5)$ group encoding function $e: B^3 \to B^5$ de ned by

$$e(000) = 00000 \qquad e(100) = 10011$$
$$e(001) = 00110 \qquad e(101) = 10101$$
$$e(010) = 01001 \qquad e(110) = 11010$$
$$e(011) = 01111 \qquad e(111) = 11100.$$

Decode the following words relative to a maximum likelihood decoding function.

(a) 11001 **(b)** 01010 **(c)** 00111

16. Consider the $(3, 6)$ group encoding function $e: B^3 \to B^6$ de ned by

$$e(000) = 000000 \qquad e(100) = 100101$$
$$e(001) = 000110 \qquad e(101) = 100011$$
$$e(010) = 010010 \qquad e(110) = 110111$$
$$e(011) = 010100 \qquad e(111) = 110001.$$

Decode the following words relative to a maximum likelihood decoding function.

(a) 011110 **(b)** 101011 **(c)** 110010

17. Let G be a group and H a subgroup of G.

(a) Prove that if g_1 and g_2 are elements of G, then either $g_1H = g_2H$ or $g_1H \cap g_2H = \{\ \}$.

(b) Use the result of part (a) to show that the left cosets of H form a partition of G.

18. Let $e: B^m \to B^n$ be a group encoding function.

(a) How many code words are there in B^n?

(b) Let $N = e(B^m)$. What is $|N|$?

(c) How many distinct left cosets of N are there in B^n?

19. Let e be as in Exercise 18. Give conditions on m and n so that coset leaders are of weight less than or equal to 1.

In Exercises 20 through 22, determine the coset leaders for $N = e_H(B^m)$ *for the given parity check matrix* $\mathbf{H}$.

20. $\mathbf{H} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$

21. $\mathbf{H} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

22. $\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

In Exercises 23 through 25, compute the syndrome for each coset leader found in the specified exercise.

23. Exercise 20 **24.** Exercise 21

25. Exercise 22

26. Let

$$\mathbf{H} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

be a parity check matrix. Decode the following words relative to a maximum likelihood decoding function.

(a) 0101 **(b)** 1010 **(c)** 1101

27. Let

$$\mathbf{H} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Decode the following words relative to a maximum likelihood decoding function associated with e_H.

(a) 10100 **(b)** 01101 **(c)** 11011

28. Let

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Decode the following words relative to a maximum likelihood decoding function associated with e_H.

(a) 011001 **(b)** 101011 **(c)** 111010

29. Let e be the $(2, 5)$ encoding function de ned in Exercise 20 of Section 1 and d an associated maximum likelihood decoding function.

(a) Give an example that veri es that (e, d) corrects one error.

(b) Give an example that veri es that (e, d) does not always correct two errors.

3 Public Key Cryptology

In 1978, Ronald Rivest, Adi Shamir, and Leonard Adelman published A Method for Obtaining Digital Signatures and Public Key Cryptosystems. In this paper, the authors describe a method of sending coded messages using a pair of publicly available integers. This method is widely called the RSA public key cryptosystem. We begin with a result on congruences that extends Fermat s Little Theorem.

THEOREM 1 Suppose that p and q are distinct primes and k is any integer. Then

(a) For any integer a with $\text{GCD}(a, pq) = 1$,

$$a^{k(p-1)(q-1)} \equiv 1 \pmod{pq} \tag{1}$$

(b) For any integer a,

$$a^{k(p-1)(q-1)+1} \equiv a \pmod{pq} \tag{2}$$

Proof

(a) If $\text{GCD}(a, pq) = 1$, then a is not divisible by p or q; it is relatively prime to both. Thus by Fermat s Little Theorem we have $a^{p-1} \equiv 1 \pmod{p}$, and so

$$a^{k(p-1)(q-1)} \equiv 1^{k(q-1)} = 1 \pmod{p}.$$

Similarly, $a^{k(p-1)(q-1)} \equiv 1 \pmod{q}$. Thus there exist integers r and s with

$$a^{k(p-1)(q-1)} = 1 + rp = 1 + sq.$$

It follows that $rp = sq$, and since q is not divisible by p, s must be, say, $s = pt$. Then

$$a^{k(p-1)(q-1)} = 1 + pqt$$

and

$$a^{k(p-1)(q-1)} \equiv 1 \pmod{pq}.$$

(b) If a is relatively prime to pq, the result follows from (1) by multiplying both sides by a. If not, then a is divisible by either p or q or both. If a is divisible by pq, then both sides of (2) are congruent to 0 mod pq and are therefore congruent to each other. In the remaining case, a is divisible by exactly one of the integers p or q, and without loss of generality, we may suppose that it is p. Then $a = bp^s$, with $s \geq 1$ and b relatively prime to pq. We note for later reference that b must satisfy (2).

Since p is relatively prime to q, we can show as in the proof of part (a) that for some integer r, $p^{k(p-1)(q-1)} = 1 + rq$. Multiplying by p then shows that

$$p^{k(p-1)(q-1)+1} \equiv p \pmod{pq},$$

and therefore

$$(p^s)^{k(p-1)(q-1)+1} = \left[p^{k(p-1)(q-1)+1}\right]^s \equiv p^s \pmod{pq}.$$

We see that both b and p^s satisfy (2), and therefore so does their product a. ■

Example 1 Let $p = 5$ and $q = 13$. Since 28 is relatively prime to $5 \times 13 = 65$ and $48 = 4 \times 12 = (5-1) \times (13-1)$, $28^{48} \equiv 1 \pmod{65}$. ◆

Example 2 Compute the remainder of 7^{293} after division by 65.

Solution

We use Theorem 1(a), with $p = 5$ and $q = 13$, so that $65 = pq$.

$$293 = (48 \times 6) + 5,$$

and since 7 is relatively prime to 65,

$$7^{293} = (7^{48})^6 \times 7^5 \equiv 7^5 \pmod{65}.$$

But $7^3 \equiv 343 \equiv 18 \pmod{65}$ and therefore

$$7^5 \equiv 18 \times 49 = 882 \equiv 37 \pmod{65}.$$

The remainder of 7^{293} after division by 65 is 37. ◆

We now construct a system in which we can make public a method of encoding messages to us (called a **public key**), but nevertheless be relatively sure that only we can decode these messages. Theorem 1 will play a major role in this effort. As a first step, we note that any message can be turned into a string of integers using a variety of methods. One way is to use the letters of the alphabet to represent a number base 26. Let A, B, . . . , Z stand for the integers 0, 1, . . . , 25. Then any pair of letters $\alpha\beta$ can be regarded as the base 26 representation of the number $(26\alpha) + \beta$. In this way numbers in the range 0 to 675 can be used in place of any two-letter pair, and any message, when divided into two-letter pairs, can be represented by a sequence of integers in this range.

Example 3 Consider the message ACT FIRST. Separate the letters into pairs, and replace each pair with the number it represents in base 26. The pairs AC, TF, IR, and ST become, respectively, the integers 2, 499, 225, and 487. If a message has an odd number of letters, we can add an agreed upon filler letter, say X, at the end. A variation of this method would be to replace triples of letters by the base 26 number they represent. Then we would use numbers in the range 0 to $25 \times 26^2 + 25 \times 26 + 25 = 17575$. ◆

We now describe a method of encoding messages. Select two primes, p and q, and let $m = pq$ and $n = (p-1)(q-1)$. Now choose any integer s that is relatively prime to n. We publish the integers m and s (that is, make them publicly available) and instruct anyone wishing to send us a secret message to proceed as follows: Divide the message into letter pairs $\alpha\beta$ and represent each pair as a number $x = (26\alpha) + \beta$ in the range 0 to 675. Then replace each of these numbers x by the unique integer y between 0 and $m - 1$ for which $y \equiv x^s \pmod{m}$, and send us the resulting number sequence. For this procedure to produce unique results, m must be at least 675.

Decoding

Since s is chosen to be relatively prime to n, $\bar{s}$, the remainder class of s mod n, has a multiplicative inverse $\bar{t}$ in the ring Z_n. Thus for some integer t we have $st \equiv 1 \pmod{n}$ or $st = 1 + k(p-1)(q-1)$ for some integer k. We can find t by using

the Euclidean algorithm. If we receive the integer $y = x^s \pmod m$, we compute $y^t \pmod m$ and apply Theorem 1. Since $m = pq$, Theorem 1(a) guarantees that

$$y^t = x^{st} = x^{1+k(p-1)(q-1)} \equiv x \pmod m.$$

Since x does not exceed m, we have $y^t \pmod m = x$, so we have recovered the original integer x. We do this to all received integers, and thus decode the message.

Example 4 Let $p = 19$ and $q = 37$. Since $m = pq = 703 > 675$, we can use the RSA method to encode messages in groups of two letters. Here $n = 18 \cdot 36 = 648$. Choose $s = 25$, which is relatively prime to 648. Now we publish the integers 703 and 25 as our public key. If someone wants to send us the message GO, she first computes $6 \times 26 + 14 = 170$ and then $170^{25} \pmod{703}$. Note that $170^2 = 28900 \equiv 77 \pmod{703}$. So

$$170^4 \equiv 77^2 = 305 \pmod{703}$$
$$170^8 \equiv 305^2 = 229 \pmod{703}$$
$$170^{16} \equiv 229^2 = 419 \pmod{703}$$

It follows that

$$170^{25} = 170^{16}170^8170 \equiv 419 \cdot 229 \cdot 170 = 16311670 \equiv 664 \pmod{703},$$

so she sends 664.

To decode the message, we first find t. Using the Euclidean algorithm, we compute

$$648 = 25 \times 25 + 23$$
$$25 = 1 \times 23 + 2$$
$$23 = 11 \times 2 + 1.$$

Thus

$$1 = 23 - 11 \cdot 2 = 23 - 11(25 - 23) = 12 \cdot 23 - 11 \cdot 25$$
$$= 12(648 - 25 \cdot 25) - 11 \cdot 25 = 12 \cdot 648 - 311 \cdot 25.$$

Thus $t = -311 \equiv 337 \pmod{648}$.

Now we compute $664^{337} \pmod{703}$. A series of computations such as those used previously to find 664 shows that $664^{337} \equiv 170 \pmod{703}$. Since $6 \times 26 + 14 = 170$, we can then recover the original message GO. ◆

Security

This method of coding is vulnerable to an attack based on an analysis of the frequency with which letters appear in ordinary language. By encoding pairs or triples of letters as we do with the public key method of this section, an attack by frequency analysis is much more difficult. But there are also other methods of attack on a public key cryptosystem.

In order to decode the message, someone must know t, which means that he must know n. This in turn requires him to know p and q. Thus the problem is to factor a number m, known to be the product of two primes. We can find the prime factors of m by trial and error if we divide n by all primes less than $\sqrt{m}$. In Example 4, $m = 703$, and the square root of 703 is less than 27. Thus we need only divide by 2, 3, 5, 7, 11, 13, 17, 19, and 23, at most 9 divisions, to find the prime

factors. In practice, one chooses p and q to have something like 100 digits, that is, to be of the order of magnitude 10^{100}, so that m is about 10^{200}. A famous theorem about prime numbers, called appropriately the Prime Number Theorem, states that the number of primes less than or equal to an integer m is approximately $\frac{m}{\ln(m)}$, and this approximation gets better as m gets larger. Thus, the number of primes less than $\sqrt{m} = 10^{100}$ is about

$$\frac{10^{100}}{\ln(10^{100})} = \frac{10^{100}}{100\ln(10)} \approx \frac{10^{100}}{230} > 10^{97}.$$

Presently, the fastest known computer has a speed of about 36,000 giga ops a second. With this computer it would take about 10^{83} seconds, or about 10^{66} billion years to do the required number of divisions. A similar enormous number of the world s largest hard drives would be required to just store these prime divisors, *if we even knew what they all were*.

The dif culty of factoring extremely large numbers provides some level of security, but even so, messages can be decoded if additional information leaks out. For example, the factorization can be found if $n = (p-1)(q-1)$ becomes known. This follows from the fact that p and q are roots of the quadratic equation

$$0 = (x-p)(x-q) = x^2 - (p+q)x + pq = x^2 - (p+q)x + m.$$

On the other hand,

$$n = (p-1)(q-1) = pq - (p+q) + 1 = m - (p+q) + 1,$$

so

$$(p+q) = m - n + 1.$$

We can therefore nd p and q by solving the equation $0 = x^2 + (n-m-1)x + m$.

Methods of coding for ef ciency, for error detection and correction, or for security are an active area of mathematical research. Here, we have presented only some of the basic ideas and procedures.

3 Exercises

1. Verify that $12^{704} \equiv 1 \pmod{391}$.

2. Verify that $10^{577} \equiv 10 \pmod{221}$.

In Exercises 3 through 6, compute the remainder when a^k is divided by c for the given values.

3. $a = 9, k = 199, c = 221$

4. $a = 17, k = 1123, c = 1189$

5. $a = 23, k = 3750, c = 3869$

6. $a = 12, k = 1540, c = 1649$

7. Let $p = 23$ and $q = 41$.

(a) Compute $m = pq$ and $n = (p-1)(q-1)$.

(b) Let $s = 41$. Find t such that $st \equiv 1 \pmod{n}$.

8. Using m and s from Exercise 7 and pairs of letters, apply the RSA method to encode the message BEAR.

9. Using m and s from Exercise 7 and pairs of letters, apply the RSA method to decode the message 371, 640.

In Exercises 10 through 12, use the RSA method, pairs of letters, and the public key $m = 779$, $s = 49$.

10. Encode the message STOP.

11. Encode the message NO.

12. Encode the message EXIT.

In Exercises 13 through 15, use the RSA method, triples of letters, and the public key $m = 19781$, $s = 19$.

13. Encode the message RUN.

14. Encode the message YES.

15. Encode the message END.

16. The public key $m = 779$, $s = 49$ has been published. Suppose you discover that n for this cryptosystem is 720. Find p and q.

17. The public key $m = 19781$, $s = 19$ has been published. Suppose you discover that n for this cryptosystem is 19500. Find p and q.

18. Use the information from Exercise 16 to decode the message 142, 525.

19. Use the information from Exercise 17 to decode the message 14032.

Tips for Proofs

The proofs in this chapter rely heavily on earlier results. Many of the concepts developed throughout the book are applied here to the problems of coding and decoding. In Section 1, we pointed out the similarity of proving two numbers equal to proving two sets are the same. Analogous proofs could be developed for any relation that has the antisymmetric property such as is less than and is a subset of.

In Section 2, Theorem 2 we use a one-to-one, onto function to match the elements of two sets in order to show that they have the same number of elements. This is also a technique that can be used in solving counting problems if the cardinality of one of the sets used is known.

Key Ideas for Review

- Message: nite sequence of characters from a nite alphabet
- Word: sequence of 0 s and 1 s
- (m, n) encoding function: one-to-one function $e: B^m \to B^n$, $m < n$
- Code word: element in Ran(e)
- Weight of x, $|x|$: number of 1 s in x
- Parity check code
- Hamming distance between x and y, $\delta(x, y)$: $|x \square y|$
- Theorem (Properties of the Distance Function): Let x, y, and z be elements of B^m. Then
  - **(a)** $\delta(x, y) = \delta(y, x)$.
  - **(b)** $\delta(x, y) \geq 0$.
  - **(c)** $\delta(x, y) = 0$ if and only if $x = y$.
  - **(d)** $\delta(x, y) \leq \delta(x, z) + \delta(z, y)$.
- Minimum distance of an (m, n) encoding function: minimum of the distances between all distinct pairs of code words
- Theorem: An (m, n) encoding function $e: B^m \to B^n$ can detect k or fewer errors if and only if its minimum distance is at least $k + 1$.
- Group code: (m, n) encoding function $e: B^m \to B^n$ such that $e(B^m) = \{e(b) \mid b \square B^m\}$ is a subgroup of B^n
- Theorem: The minimum distance of a group code is the minimum weight of a nonzero code word.
- Mod-2 sum of Boolean matrices **D** and **E**, **D** $\square$ **E**
- Mod-2 Boolean product of Boolean matrices **D** and **E**, **D** $\square$ **E**
- Theorem: Let m and n be nonnegative integers with $m < n$, $r = n - m$, and let **H** be an $n \times r$ Boolean matrix. Then the function $f_H : B^n \to B^r$ de ned by

$$f_H(x) = x \square \mathbf{H}, \qquad x \square B^n$$

  is a homomorphism from the group B^n to the group B^r.
- Group code e_H corresponding to parity check matrix **H**
- (n, m) decoding function
- Maximum likelihood decoding function associated with e
- Theorem: Suppose that e is an (m, n) encoding function and d is a maximum likelihood decoding function associated with e. Then (e, d) can correct k or fewer errors if and only if the minimum distance is at least $2k + 1$
- Decoding procedure for a group code
- Decoding procedure for a group code given by a parity check matrix
- RSA public key cryptosystem
- Theorem: Suppose that p and q are distinct primes and k is any integer. Then
  - **(a)** For any integer a with GCD$(a, pq) = 1$,

$$a^{k(p-1)(q-1)} \equiv 1 \pmod{pq}.$$

  - **(b)** For any integer a,

$$a^{k(p-1)(q-1)+1} \equiv a \pmod{pq}.$$

1 Languages

Consider the set S^* consisting of all finite strings of elements from the set S. There are many possible interpretations of the elements of S^*, depending on the nature of S. If we think of S as a set of "words," then S^* may be regarded as the collection of all possible "sentences" formed from words in S. Of course, such "sentences" do not have to be meaningful or even sensibly constructed. We may think of a language as a complete specification, at least in principle, of three things. First, there must be a set S consisting of all "words" that are to be regarded as being part of the language. Second, a subset of S^* must be designated as the set of "properly constructed sentences" in the language. The meaning of this term will depend very much on the language being constructed. Finally, it must be determined which of the properly constructed sentences have meaning and what the meaning is.

Suppose, for example, that S consists of all English words. The specification of a properly constructed sentence involves the complete rules of English grammar; the meaning of a sentence is determined by this construction and by the meaning of the words. The sentence

"Going to the store John George to sing."

is a string in S^*, but is not a properly constructed sentence. The arrangement of nouns and verb phrases is illegal. On the other hand, the sentence

"Noiseless blue sounds sit cross-legged under the mountaintop."

is properly constructed, but completely meaningless.

For another example, S may consist of the integers, the symbols $+$, $-$, $\times$, and $\div$, and left and right parentheses. We will obtain a language if we designate as proper those strings in S^* that represent unambiguously parenthesized algebraic expressions. Thus

$$((3 - 2) + (4 \times 7)) \div 9 \quad \text{and} \quad (7 - (8 - (9 - 10)))$$

are properly constructed "sentences" in this language. On the other hand, the three strings $(2 - 3)) + 4$, $4 - 3 - 2$, and $)2 + (3 -) \times 4$ are not properly constructed. The first has too many parentheses, the second has too few (we do not know which subtraction to perform first), and the third has parentheses and numbers completely out of place. All properly constructed expressions have meaning except those involving division by zero. The meaning of an expression is the rational number it represents. Thus the meaning of $((2 - 1) \div 3) + (4 \times 6)$ is 73/3, while $2 + (3 \div 0)$ and $(4 + 2) - (0 \div 0)$ are not meaningful.

The specification of the proper construction of sentences is called the **syntax** of a language. The specification of the meaning of sentences is called the **semantics** of a language. Among the languages that are of fundamental importance in computer science are the programming languages. These include BASIC, FORTRAN, JAVA, PASCAL, C++, LISP, ADA, FORTH, and many other general and special-purpose languages. When they are taught to program in some programming language, people are actually taught the syntax of the language. In a compiled language such as FORTRAN, most mistakes in syntax are detected by the compiler, and appropriate error messages are generated. The semantics of a programming language forms a much more difficult and advanced topic of study. The meaning of a line of programming is taken to be the entire sequence of events that takes place inside the computer as a result of executing or interpreting that line.

We will not deal with semantics at all. We will study the syntax of a class of languages called phrase structure grammars. Although these are not nearly complex enough to include natural languages such as English, they are general enough to encompass many languages of importance in computer science. This includes most aspects of programming languages, although the complete specification of some higher-level programming languages exceeds the scope of these grammars. On the other hand, phrase structure grammars are simple enough to be studied precisely, since the syntax is determined by substitution rules. The grammars that will occupy most of our attention lead to interesting examples of labeled trees.

Grammars

A **phrase structure grammar** G is defined to be a 4-tuple $(V, S, v_0, \rightarrow)$, where V is a finite set, S is a subset of V, $v_0 \in V - S$, and $\rightarrow$ is a finite relation on V^+. The idea here is that S is, as discussed previously, the set of all allowed "words" in the language, and V consists of S together with some other symbols. The element v_0 of V is a starting point for the substitutions, which will shortly be discussed. Finally, the relation $\rightarrow$ on V^+ specifies allowable replacements, in the sense that, if $w \rightarrow w'$, we may replace w by w' whenever the string w occurs, either alone or as a substring of some other string. Traditionally, the statement $w \rightarrow w'$ is called a **production** of G. Then w and w' are termed the **left** and **right** sides of the production, respectively. We assume that no production of G has the empty string λ as its left side. We will call $\rightarrow$ the **production relation** of G.

With these ingredients, we can introduce a substitution relation, denoted by $\Rightarrow$, on V^*. We let $x \Rightarrow y$ mean that $x = l \cdot w \cdot r$, $y = l \cdot w' \cdot r$, and $w \rightarrow w'$, where l and r are completely arbitrary strings in V^*. In other words, $x \Rightarrow y$ means that y results from x by using one of the allowed productions to replace part or all of x. The relation $\Rightarrow$ is usually called **direct derivability**. Finally, we let $\Rightarrow^\infty$ be the transitive closure of $\Rightarrow$, and we say that a string w in S^* is a syntactically correct sentence if and only if $v_0 \Rightarrow^\infty w$. In more detail, this says that a string w is a properly constructed sentence if w is in S^*, not just in V^*, and if we can get from v_0 to w by making a finite number of substitutions. This may seem complicated, but it is really a simple idea, as the following examples will show.

If $G = (V, S, v_0, \rightarrow)$ is a phrase structure grammar, we will call S the set of **terminal symbols** and $N = V - S$ the set of **nonterminal symbols**. Note that $V = S \cup N$.

The reader should be warned that other texts have slight variations of the definitions and notations that we have used for phrase structure grammars.

Example 1 Let S = {John, Jill, drives, jogs, carelessly, rapidly, frequently}, N = {sentence, noun, verbphrase, verb, adverb}, and let $V = S \cup N$. Let v_0 = sentence, and suppose that the relation $\rightarrow$ on V^* is described by

sentence $\rightarrow$ noun verbphrase
noun $\rightarrow$ John
noun $\rightarrow$ Jill
verbphrase $\rightarrow$ verb adverb
verb $\rightarrow$ drives
verb $\rightarrow$ jogs
adverb $\rightarrow$ carelessly
adverb $\rightarrow$ rapidly
adverb $\rightarrow$ frequently

The set S contains all the allowed words in the language; N consists of words that describe parts of sentences but that are not actually contained in the language.

We claim that the sentence "Jill drives frequently," which we will denote by w, is an allowable or syntactically correct sentence in this language. To prove this, we consider the following sequence of strings in $V^{\square}$.

sentence		
noun	verbphrase	
Jill	verbphrase	
Jill	verb	adverb
Jill	drives	adverb
Jill	drives	frequently

Now each of these strings follows from the preceding one by using a production to make a partial or complete substitution. In other words, each string is related to the following string by the relation $\square$, so sentence $\square^{\infty}$ w. By definition then, w is syntactically correct since, for this example, v_0 is sentence. In phrase structure grammars, correct syntax simply refers to the process by which a sentence is formed, nothing else. ◆

It should be noted that the sequence of substitutions that produces a valid sentence, a sequence that will be called a **derivation** of the sentence, is not unique. The following derivation produces the sentence w of Example 1 but is not identical with the derivation given there.

sentence		
noun	verbphrase	
noun	verb	adverb
noun	verb	frequently
noun	drives	frequently
Jill	drives	frequently

The set of all properly constructed sentences that can be produced using a grammar G is called the **language** of G and is denoted by L(G). The language of the grammar given in Example 1 is a somewhat simple-minded sublanguage of English, and it contains exactly 12 sentences. The reader can verify that "John jogs carelessly" is in the language L(G) of this grammar, while "Jill frequently jogs" is not in L(G).

It is also true that many different phrase structure grammars may produce the same language; that is, they have exactly the same set of syntactically correct sentences. Thus a grammar cannot be reconstructed from its language. In Section 2 we will give examples in which different grammars are used to construct the same language.

Example 1 illustrates the process of derivation of a sentence in a phrase structure grammar. Another method that may sometimes be used to show the derivation process is the construction of a **derivation tree** for the sentence. The starting symbol, v_0, is taken as the label for the root of this tree. The level-1 vertices correspond to and are labeled in order by the various words involved in the first substitution for v_0. Then the offspring of each vertex, at every succeeding level, are labeled by the various words (if any) that are substituted for that vertex the next time it is subjected to substitution. Consider, for example, the first derivation of sentence w in Example 1. Its derivation tree begins with "sentence," and the next-level vertices correspond to "noun" and "verbphrase" since the first substitution replaces the word "sentence" with the string "noun verbphrase." This part of the

tree is shown in Figure 1(a). Next, we substitute "Jill" for "noun," and the tree becomes as shown in Figure 1(b). The next two substitutions, "verb adverb" for "verbphrase" and "drives" for "verb," extend the tree as shown in Figures 1(c) and (d). Finally, the tree is completed with the substitution of "frequently" for "adverb." The finished derivation tree is shown in Figure 1(e).

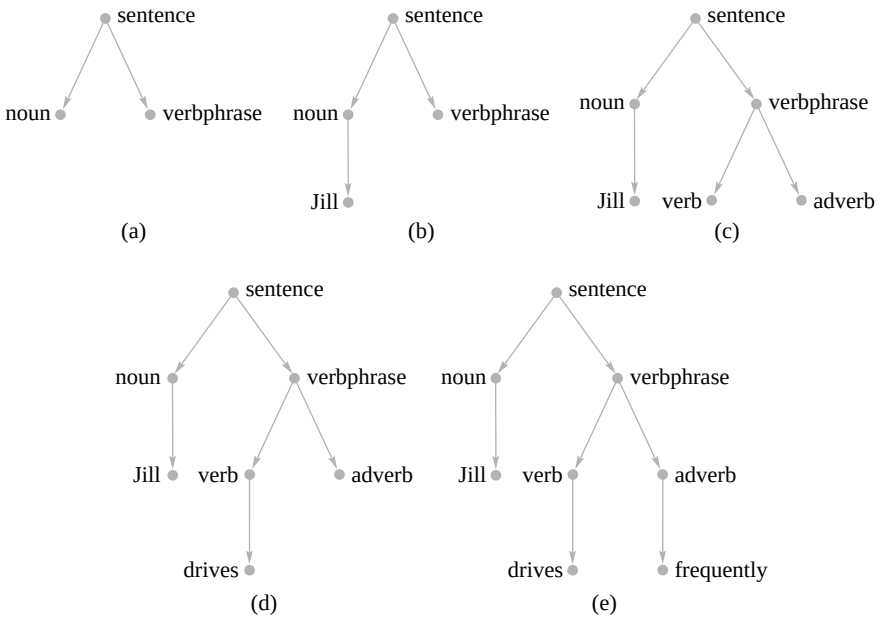


Figure 1

The second derivation sequence, following Example 1 for the sentence w, yields a derivation tree in the stages shown in Figure 2. Notice that the same tree results in both figures. Thus these two derivations yield the same tree, and the differing orders of substitution simply create the tree in different ways. The sentence being derived labels the leaves of the resulting tree.

Example 2 Let $V = \{v_0, w, a, b, c\}$, $S = \{a, b, c\}$, and let $\Rightarrow$ be the relation on V^* given by

$$1.\ v_0 \Rightarrow aw. \qquad 2.\ w \Rightarrow bbw. \qquad 3.\ w \Rightarrow c.$$

Consider the phrase structure grammar $G = (V, S, v_0, \Rightarrow)$. To derive a sentence of $L(G)$, it is necessary to perform successive substitutions, using (1), (2), and (3), until all symbols are eliminated other than the terminal symbols a, b, and c. Since we begin with the symbol v_0, we must first use production (1), or we could never eliminate v_0. This first substitution results in the string aw. We may now use (2) or (3) to substitute for w. If we use production (2), the result will contain a w. Thus one application of (2) to aw produces the string ab^2w (here b^n means n consecutive b's). If we use (2) again, we will have the string ab^4w. We may use production (2) any number of times, but we will finally have to use production 3 to eliminate the symbol w. Once we use (3), only terminal symbols remain, so the process ends. We may summarize this analysis by saying that $L(G)$ is the subset of S^* corresponding to the regular expression $a(bb)^*c$. Thus the word ab^6c is in the language of G, and its derivation tree is shown in Figure 3. Note that, unlike the tree of Example 1, the derivation tree for ab^6c is not a binary tree. ◆

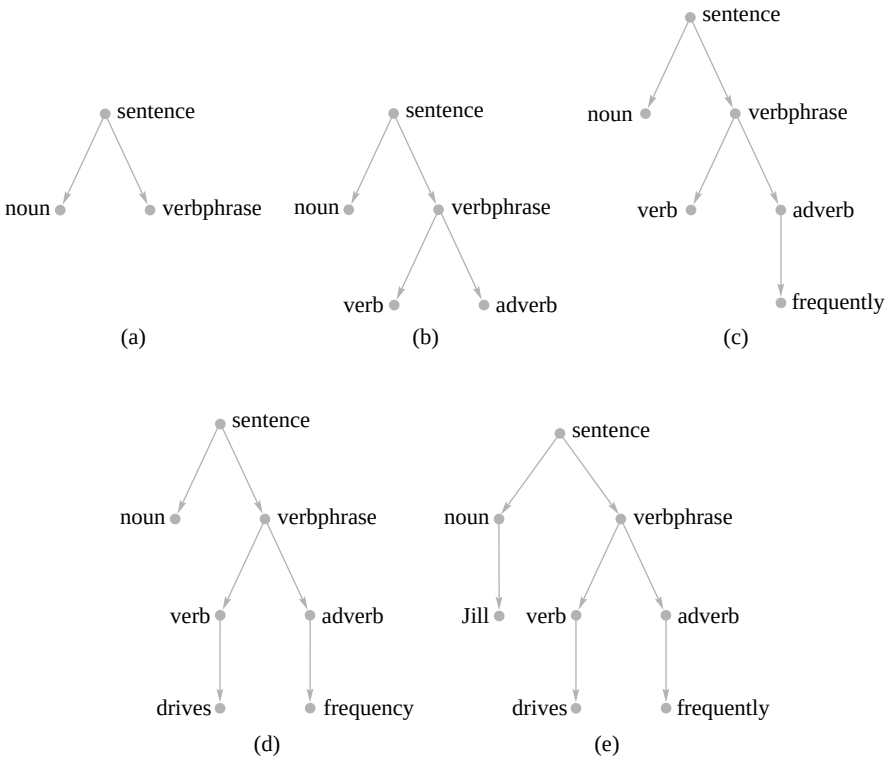


Figure 2

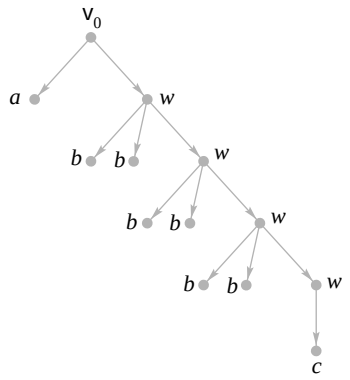


Figure 3

Example 3 Let $V = \{v_0, w, a, b, c\}$, $S = \{a, b, c\}$, and let $\Rightarrow$ be a relation on V^* given by

1. $v_0 \Rightarrow av_0b$. 2. $v_0b \Rightarrow bw$. 3. $abw \Rightarrow c$.

Let $G = (V, S, v_0, \Rightarrow)$ be the corresponding phrase structure grammar. As we did in Example 2, we determine the form of allowable sentences in $L(G)$.

Since we must begin with the symbol v_0 alone, we must use production (1) first. We may continue to use (1) any number of times, but we must eventually use production (2) to eliminate v_0. Repeated use of (1) will result in a string of the form $a^nv_0b^n$, $n \geq 1$; that is, there are equal numbers of a's and b's. When (2) is used, the result is a string of the form $a^m(abw)b^m$ with $m \geq 0$. At this point the only production that can be used is (3), and we must use it to remove the nonterminal symbol w. The use of (3) finishes the substitution process and produces a string in S^*. Thus the allowable sentences $L(G)$ of the grammar G all have the form $w = a^ncb^n$, where $n \geq 0$. In this case it can be shown that $L(G)$ does not correspond to a regular expression over S. ◆

Another interesting feature of the grammar in Example 3 is that the derivations of the sentences cannot be expressed as trees. Our construction of derivation trees works only when the left-hand sides of all productions used consist of single, nonterminal symbols. The left-hand sides of the productions in Example 3 do not have this simple form. Although it is possible to construct a graphical representation of these derivations, the resulting digraph would not be a tree. Many other problems can arise if no restrictions are placed on the productions. For this reason, a classification of phrase structure grammars has been devised.

Let $G = (V, S, v_0, \rightarrow)$ be a phrase structure grammar. Then we say that G is

TYPE 0: if no restrictions are placed on the productions of G

TYPE 1: if for any production $w_1 \rightarrow w_2$, the length of w_1 is less than or equal to the length of w_2 (where the **length** of a string is the number of words in that string)

TYPE 2: if the left-hand side of each production is a single, nonterminal symbol and the right-hand side consists of one or more symbols

TYPE 3: if the left-hand side of each production is a single, nonterminal symbol and the right-hand side has one or more symbols, including at most one nonterminal symbol, which must be at the extreme right of the string

In each of the preceding types, we permit the inclusion of the trivial production $v_0 \rightarrow \lambda$, where λ represents the empty string. This is an exception to the defining rule for types 1, 2, and 3, but it is included so that the empty string can be made part of the language. This avoids constant consideration of unimportant special cases.

It follows from the definition that each type of grammar is a special case of the type preceding it. Example 1 is a type 2 grammar, Example 2 is type 3, and Example 3 is type 0. Grammars of types 0 or 1 are quite difficult to study and little is known about them. They include many pathological examples that are of no known practical use. We will restrict further consideration of grammars to types 2 and 3. These types have derivation trees for the sentences of their languages, and they are sufficiently complex to describe many aspects of actual programming languages. Type 2 grammars are sometimes called **context-free grammars**, since the symbols on the left of the productions are substituted for wherever they occur. On the other hand, a production of the type $l \cdot w \cdot r \rightarrow l \cdot w' \cdot r$ (which could not occur in a type 2 grammar) is called **context sensitive**, since w' is substituted for w only in the context where it is surrounded by the strings l and r. Type 3 grammars have a very close relationship with finite-state machines. (See Section 3.) Type 3 grammars are also called **regular grammars**.

A language will be called **type 2** or **type 3** if there is a grammar of type 2 or type 3 that produces it. This concept can cause problems. Even if a language is produced by a non-type-2 grammar, it is possible that some type 2 grammar also produces this same language. In this case, the language is type 2. The same situation may arise in the case of type 3 grammars.

The process we have considered in this section, namely deriving a sentence within a grammar, has a converse process. The converse process involves taking a sentence and verifying that it is syntactically correct in some grammar G by constructing a derivation tree that will produce it. This process is called **parsing** the sentence, and the resulting derivation tree is often called the **parse tree** of the sentence. Parsing is of fundamental importance for compilers and other forms of language translation. A sentence in one language is parsed to show its structure, and a tree is constructed. The tree is then searched and, at each step, corresponding sentences are generated in another language. In this way a C++ program, for example, is compiled into a machine-language program. The contents of this section and the next two sections are essential to the compiling process, but the complete details must be left to a more advanced course.

1 Exercises

In Exercises 1 through 7, a grammar G *is specified. In each case describe precisely the language,* $L(G)$*, produced by this grammar; that is, describe all syntactically correct "sentences."*

1. $G = (V, S, v_0, \Rightarrow)$
$V = \{v_0, v_1, x, y, z\}$, $S = \{x, y, z\}$
$\Rightarrow$: $v_0 \Rightarrow xv_0$
$v_0 \Rightarrow yv_1$
$v_1 \Rightarrow yv_1$
$v_1 \Rightarrow z$

2. $G = (V, S, v_0, \Rightarrow)$
$V = \{v_0, a\}$, $S = \{a\}$
$\Rightarrow$: $v_0 \Rightarrow aav_0$
$v_0 \Rightarrow aa$

3. $G = (V, S, v_0, \Rightarrow)$
$V = \{v_0, a, b\}$, $S = \{a, b\}$
$\Rightarrow$: $v_0 \Rightarrow aav_0$
$v_0 \Rightarrow a$
$v_0 \Rightarrow b$

4. $G = (V, S, v_0, \Rightarrow)$
$V = \{v_0, x, y, z\}$, $S = \{x, y, z\}$
$\Rightarrow$: $v_0 \Rightarrow xv_0$
$v_0 \Rightarrow yv_0$
$v_0 \Rightarrow z$

5. $G = (V, S, v_0, \Rightarrow)$
$V = \{v_0, v_1, v_2, a, +, (,)\}$,
$S = \{(,), a, +\}$
$\Rightarrow$: $v_0 \Rightarrow (v_0)$ (where left and right parentheses are symbols from S)
$v_0 \Rightarrow a + v_1$
$v_1 \Rightarrow a + v_2$
$v_2 \Rightarrow a + v_2$
$v_2 \Rightarrow a$

6. $G = (V, S, v_0, \Rightarrow)$
$V = \{v_0, v_1, a, b\}$, $S = \{a, b\}$
$\Rightarrow$: $v_0 \Rightarrow av_1$
$v_1 \Rightarrow bv_0$
$v_1 \Rightarrow a$

7. $G = (V, S, v_0, \Rightarrow)$
$V = \{v_0, v_1, v_2, x, y, z\}$, $S = \{x, y, z\}$
$\Rightarrow$: $v_0 \Rightarrow v_0v_1$
$v_0v_1 \Rightarrow v_2v_0$
$v_2v_0 \Rightarrow xy$
$v_2 \Rightarrow x$
$v_1 \Rightarrow z$

8. For each grammar in Exercises 1 through 7, state whether the grammar is type 1, 2, or 3.

9. Let $G = (V, S, I, \Rightarrow)$, where

$V = \{I, L, D, W, a, b, c, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$S = \{a, b, c, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$\Rightarrow$ is given by

1. $I \Rightarrow L$
2. $I \Rightarrow LW$
3. $W \Rightarrow LW$
4. $W \Rightarrow DW$
5. $W \Rightarrow L$
6. $W \Rightarrow D$
7. $L \Rightarrow a$
8. $L \Rightarrow b$
9. $L \Rightarrow c$
10. $D \Rightarrow 0$
11. $D \Rightarrow 1$

⋮

19. $D \Rightarrow 9$

Which of the following statements are true for this grammar?

(a) $ab092 \in L(G)$ **(b)** $2a3b \in L(G)$
(c) $aaaa \in L(G)$ **(d)** $I \in a$
(e) $I \in^* ab$ **(f)** $DW \in 2$
(g) $DW \in^* 2$ **(h)** $W \in^* 2abc$
(i) $W \in^* ba2c$

10. Draw a derivation tree for $ab3$ in the grammar of Exercise 9.

11. If G is the grammar of Exercise 9, describe $L(G)$.

12. Let $G = (V, S, v_0, \Rightarrow)$, where
$V = \{v_0, v_1, v_2, a, b, c\}$, $S = \{a, b, c\}$
$\Rightarrow$: $v_0 \Rightarrow aav_0$
$v_0 \Rightarrow bv_1$
$v_1 \Rightarrow cv_2b$
$v_1 \Rightarrow cb$
$v_2 \Rightarrow bbv_2$
$v_2 \Rightarrow bb$.
State which of the following are in $L(G)$.

(a) $aabcb$ **(b)** $abbcb$
(c) $aaaabcbb$ **(d)** $aaaabcbbb$
(e) $abcbbbbb$

13. If G is the grammar of Exercise 12, describe $L(G)$.

14. Draw a derivation tree for the string a^8 in the grammar of Exercise 2.

15. Give two distinct derivations (sequences of substitutions that start at v_0) for the string $xyz \in L(G)$, where G is the grammar of Exercise 7.

16. Let G be the grammar of Exercise 5. Can you give two distinct derivations (see Exercise 15) for the string $((a + a + a))$?

17. Let G be the grammar of Exercise 9. Give two distinct derivations (see Exercise 15) of the string $a100$.

18. Let $G = (V, S, v_0, \Rightarrow)$, where $V = \{v_0, v_1\}$, $S = \{a, b\}$, and
$\Rightarrow$: $v_0 \Rightarrow bv_0$
$v_0 \Rightarrow av_1$
$v_0 \Rightarrow b$
$v_1 \Rightarrow av_0$
$v_1 \Rightarrow bv_1$
$v_1 \Rightarrow a$.
Describe $L(G)$.

19. Let $G = (V, S, v_0, \rightarrow)$, where $V = \{v_0, v_1, v_2\}$, $S = \{a, b\}$, and

$\rightarrow$: $v_0 \rightarrow v_1v_1v_0$
$v_0 \rightarrow v_2v_2v_1$
$v_1 \rightarrow a$
$v_2 \rightarrow b$.

Prove or disprove that $L(G)$ is the same as the language described in Exercise 18.

In Exercises 20 through 26, construct a phrase structure grammar G *such that the language,* $L(G)$*, of* G *is equal to the language* L.

20. $L = \{a^n b^n \mid n \geq 1\}$

21. $L = \{$strings of 0's and 1's with an equal number $n \geq 0$ of 0's and 1's$\}$

22. $L = \{a^n b^m \mid n \geq 1, m \geq 1\}$

23. $L = \{a^n b^n \mid n \geq 3\}$

24. $L = \{a^n b^m \mid n \geq 1, m \geq 3\}$

25. $L = \{x^n y^m \mid n \geq 2,$ m nonnegative and even$\}$

26. $L = \{x^n y^m \mid n$ even, m positive and odd$\}$

27. A **palindrome** is a string that reads the same from right to left as it does from left to right. Construct a phrase structure grammar G such that $L(G)$ is the set of palindromes using $\{a, b\}$.

28. Construct a phrase structure grammar G such that $L(G)$ is the set of Boolean polynomials in $x_1, x_2, \ldots, x_n$.

In Exercises 29 and 30, let $V = \{v_0, v_1, v_2, v_3\}$, $S = \{a\}$*, and let* $\rightarrow$ *be the relation on* V^* *given by*

1. $v_0 \rightarrow av_1$ 2. $v_1 \rightarrow av_2$ 3. $v_2 \rightarrow av_3$
4. $v_2 \rightarrow a$ 5. $v_3 \rightarrow av_1$

Let $G = (V, S, v_0, \rightarrow)$.

29. Complete the following proof that each $(aaa)^n$, $n \geq 1$ belongs to $L(G)$.

Proof: ***Basis Step*:** For $n = 1$, aaa is produced by _____. Hence $aaa \in L(G)$
***Induction Step*:** We use $P(k)$: $(aaa)^k$ is in $L(G)$ to show $P(k+1)$: $(aaa)^{k+1}$ is in $L(G)$. If $(aaa)^k = a^{3k}$ is in $L(G)$, then we must have applied _____ to _____. Instead we may use production rule _____ to produce $a^{3k}av_3$. (Complete the derivation of aaa^{3k+3}.) Hence $(aaa)^{3(k+1)}$ is in $L(G)$.

30. Complete the following proof that any string in $L(G)$ is of the form $(aaa)^n$.

Proof: Clearly all strings in $L(G)$ are of the form a^i, $1 \leq i$, because _____. Let $P(n)$: If a^i is in $L(G)$ and $3n < i \leq 3(n+1)$, then $i = 3(n+1)$.
***Basis Step*:** For $n = 0$, suppose a^i is in $L(G)$ and _____. It is easy to see that the smallest possible i with these production rules is 3. Hence $P(0)$ is true.
***Induction Step*:** Let $k \geq 1$. We use $P(k)$: _____ to show $P(k+1)$: _____. Suppose $3(k+1) < i \leq 3(k+2)$. We know that a^i must have been produced from _____ by production rule _____. But at the step just before that we must have used production rule _____ on $a^{i-2}v_1$. The string $a^{i-2}v_1$ could not have been produced using production rule 1, because _____, but was produced from _____. Since $3(k+1) < i \leq 3(k+2)$, we have $3k < i \leq 3(k+1)$. By $P(k)$, _____.

31. Let G_1 and G_2 be regular grammars with languages L_1 and L_2 respectively. Define a new language $L_3 = \{w_1 \cdot w_2 \mid w_1 \in L_1$ and $w_2 \in L_2\}$. Describe how to create a regular grammar for L_3 from G_1 and G_2.

32. Complete the following proof that no regular grammar G can produce the language of Exercise 20.

Proof: Suppose that there is a regular grammar $G = (V, S, v_0, \rightarrow)$ with n nonterminals. $L(G)$ must contain exactly the strings of the form a^ib^i. But since G is regular, in the derivation of a^ib^i we must have _____ with some nonterminal v_j. But with only n nonterminals, there must be at least two strings $\underline{\quad(1)\quad}$ and $\underline{\quad(2)\quad}$ with the same nonterminals. (Why?) There is no way to guarantee that using the production rules on (1) and (2) will produce exactly _____b's from (1) and exactly _____b's from (2). Hence there is no such regular grammar G.

33. Show that $\{a^m b a^n, m \geq 0, n \geq 1\}$ is a context-free language.

2 Representations of Special Grammars and Languages

BNF Notation

For type 2 grammars (which include type 3 grammars), there are some useful, alternative methods of displaying the productions. A commonly encountered alternative is called the **BNF notation** (for Backus–Naur form). We know that the left-hand sides of all productions in a type 2 grammar are single, nonterminal symbols. For any such symbol w, we combine all productions having w as the left-hand side. The symbol w remains on the left, and all right-hand sides associated with w are listed together, separated by the symbol |. The relational symbol

$\Rightarrow$ is replaced by the symbol ::=. Finally, the nonterminal symbols, wherever they occur, are enclosed in pointed brackets $\langle\ \rangle$. This has the additional advantage that nonterminal symbols may be permitted to have embedded spaces. Thus ⟨word1 word2⟩ shows that the string between the brackets is to be treated as one "word," not as two words. That is, we may use the space as a convenient and legitimate "letter" in a word, as long as we use pointed brackets to delimit the words.

Example 1 In BNF notation, the productions of Example 1 of Section 1 appear as follows.

⟨sentence⟩ ::= ⟨noun⟩⟨verbphrase⟩
⟨noun⟩ ::= John | Jill
⟨verbphrase⟩ ::= ⟨verb⟩⟨adverb⟩
⟨verb⟩ ::= drives | jogs
⟨adverb⟩ ::= carelessly | rapidly | frequently ◆

Example 2 In BNF notation, the productions of Example 2 of Section 1 appear as follows.

$\langle v_0 \rangle$ ::= a⟨w⟩
⟨w⟩ ::= bb⟨w⟩ | c ◆

Note that the left-hand side of a production may also appear in one of the strings on the right-hand side. Thus, in the second line of Example 2, ⟨w⟩ appears on the left, and it appears in the string bb⟨w⟩ on the right. When this happens, we say that the corresponding production $w \Rightarrow bbw$ is **recursive**. If a recursive production has w as left-hand side, we will say that the production is **normal** if w appears only once on the right-hand side and is the rightmost symbol. Other nonterminal symbols may also appear on the right side. The recursive production $w \Rightarrow bbw$ given in Example 2 is normal. Note that any recursive production that appears in a type 3 (regular) grammar is normal, by the definition of type 3.

Example 3 BNF notation is often used to specify actual programming languages. PASCAL and many other languages had their grammars given in BNF initially. In this example, we consider a small subset of PASCAL's grammar. This subset describes the syntax of decimal numbers and can be viewed as a mini-grammar whose corresponding language consists precisely of all properly formed decimal numbers.

Let $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, .\}$. Let V be the union of S with the set

$$N = \{\text{decimal-number, decimal-fraction, unsigned-integer, digit}\}.$$

Then let G be the grammar with symbol sets V and S, with starting symbol "decimal-number" and with productions given in BNF form as follows:

1. ⟨decimal-number⟩ ::= ⟨unsigned-integer⟩ | ⟨decimal-fraction⟩ | ⟨unsigned-integer⟩⟨decimal-fraction⟩
2. ⟨decimal-fraction⟩ ::= .⟨unsigned-integer⟩
3. ⟨unsigned-integer⟩ ::= ⟨digit⟩ | ⟨digit⟩⟨unsigned-integer⟩
4. ⟨digit⟩ ::= 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

Figure 4 shows a derivation tree, in this grammar, for the decimal number 23.14. Notice that the BNF statement numbered 3 is recursive in the second part of its right-hand side. That is, the production "unsigned-integer $\Rightarrow$ digit unsigned-integer" is recursive, and it is also normal. In general, we know that many different grammars may produce the same language. If the line numbered 3 were replaced by the line

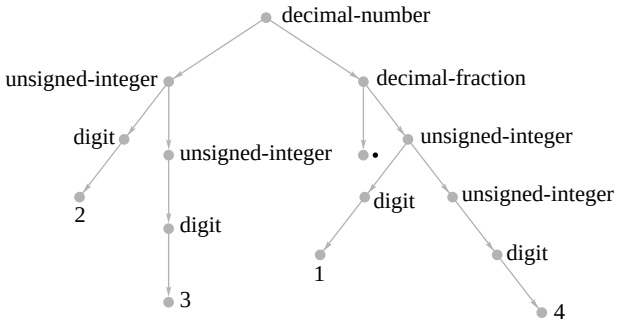


Figure 4

3°. ⟨unsigned-integer⟩ ::= ⟨digit⟩ | ⟨unsigned-integer⟩⟨digit⟩

we would have a different grammar that produced exactly the same language, namely the correctly formed decimal numbers. However, this grammar contains a production that is recursive but not normal. ◆

Example 4 As in Example 3, we give a grammar that specifies a piece of several actual programming languages. In these languages, an identifier (a name for a variable, function, subroutine, and so on) must be composed of letters and digits and must begin with a letter. The following grammar, with productions given in BNF, has precisely these identifiers as its language.

$$G = (V, S, \text{identifier}, \Rightarrow)$$
$$N = \{\text{identifier, remaining, digit, letter}\}$$
$$S = \{a, b, c, \ldots, z, 0, 1, 2, 3, \ldots, 9\},$$
$$V = N \cup S$$

1. ⟨identifier⟩ ::= ⟨letter⟩ | ⟨letter⟩⟨remaining⟩
2. ⟨remaining⟩ ::= ⟨letter⟩ | ⟨digit⟩ | ⟨letter⟩⟨remaining⟩ | ⟨digit⟩⟨remaining⟩
3. ⟨letter⟩ ::= a | b | c · · · | z
4. ⟨digit⟩ ::= 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

Again we see that the productions "remaining ⇒ letter remaining" and "remaining ⇒ digit remaining," occurring in BNF statement 2, are recursive and normal. ◆

Syntax Diagrams

A second alternative method for displaying the productions in some type 2 grammars is the **syntax diagram**. This is a pictorial display of the productions that allows the user to view the substitutions dynamically, that is, to view them as movement through the diagram. We will illustrate, in Figure 5, the diagrams that result from translating typical sets of productions, usually all the productions appearing on the right-hand side of some BNF statement.

A BNF statement that involves just a single production, such as ⟨w⟩ ::= ⟨w_1⟩⟨w_2⟩⟨w_3⟩, will result in the diagram shown in Figure 5(a). The symbols (words) that make up the right-hand side of the production are drawn in sequence from left to right. The arrows indicate the direction in which to move to accomplish a substitution, while the label w indicates that we are substituting for the symbol w. Finally, the rectangles enclosing w_1, w_2, and w_3 denote the fact that

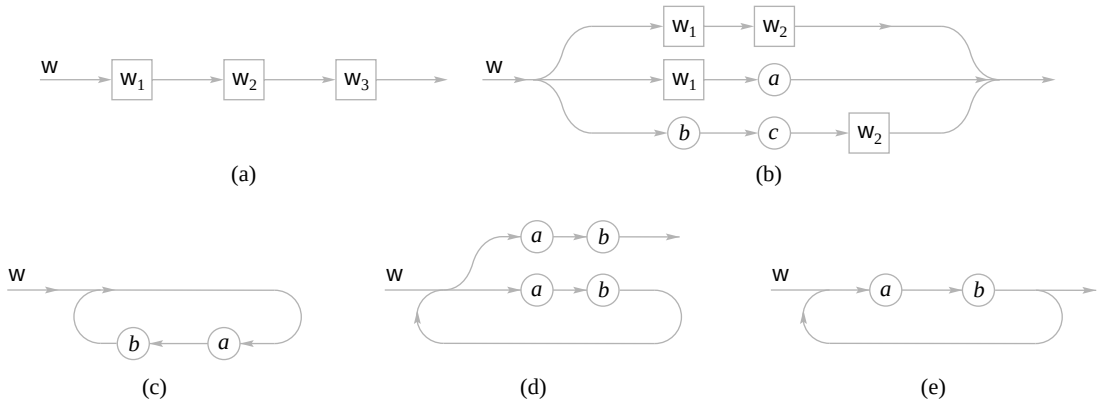


Figure 5

these are nonterminal symbols. If terminal symbols were present, they would instead be enclosed in circles or ellipses. Figure 5(b) shows the situation when there are several productions with the same left-hand side. This figure is a syntax diagram translation of the following BNF specification:

$$\langle w \rangle ::= \langle w_1 \rangle\langle w_2 \rangle \mid \langle w_1 \rangle a \mid bc\langle w_2 \rangle$$

(where a, b, and c are terminal symbols). Here the diagram shows that when we substitute for w, by moving through the figure in the direction of the arrows, we may take any one of three paths. This corresponds to the three alternative substitutions for the symbol w. Now consider the following normal, recursive production, in BNF form:

$$\langle w \rangle ::= ab\langle w \rangle.$$

The syntax diagram for this production is shown in Figure 5(c). If we go through the loop once, we encounter a, then b, and we then return to the starting point designated by w. This represents the recursive substitution of abw for w. Several trips around the diagram represent several successive substitutions. Thus, if we traverse the diagram three times and return to the starting point, we see that w will be replaced by ababab w in three successive substitutions. This is typical of the way in which movement through a syntax diagram represents the substitution process.

The preceding remarks show how to construct a syntax diagram for a normal recursive production. Nonnormal recursive productions do not lead to the simple diagrams discussed, but we may sometimes replace nonnormal, recursive productions by normal recursive productions and obtain a grammar that produces the same language. Since recursive productions in regular grammars must be normal, syntax diagrams can always be used to represent regular grammars.

We also note that syntax diagrams for a language are by no means unique. They will not only change when different, equivalent productions are used, but they may be combined and simplified in a variety of ways. Consider the following BNF specification:

$$\langle w \rangle ::= ab \mid ab\langle w \rangle.$$

If we construct the syntax diagram for w using exactly the rules presented, we will obtain the diagram of Figure 5(d). This shows that we can “escape” from w, that is, eliminate w entirely, only by passing through the upper path. On the other hand, we may first traverse the lower loop any number of times. Thus any movement through the diagram that eventually results in the complete elimination

of w by successive substitutions will produce a string of terminal symbols of the form $(ab)^n$, $n \geq 1$.

It is easily seen that in the simpler diagram of Figure 5(e), produced by combining the paths of Figure 5(d) in an obvious way, is an entirely equivalent syntax diagram. These types of simplifications are performed whenever possible.

Example 5 The syntax diagrams of Figure 6(a) represent the BNF statements of Example 2, constructed with our original rules for drawing syntax diagrams. A slightly more aesthetic version is shown in Figure 6(b). ◆

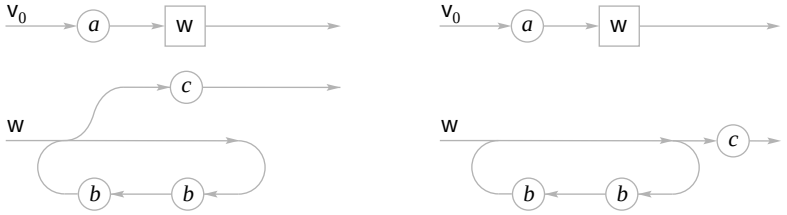


Figure 6

Example 6 Consider the BNF statements 1, 2, 3, and 4 of Example 4. The direct translation into syntax diagrams is shown in Figure 7. In Figure 8 we combine the first two diagrams of Figure 7 and simplify the result. We thus eliminate the symbol "remaining," and we arrive at the customary syntax diagrams for identifiers. ◆

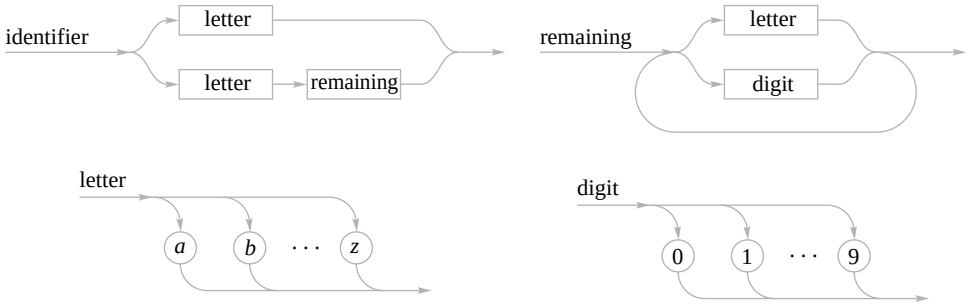


Figure 7

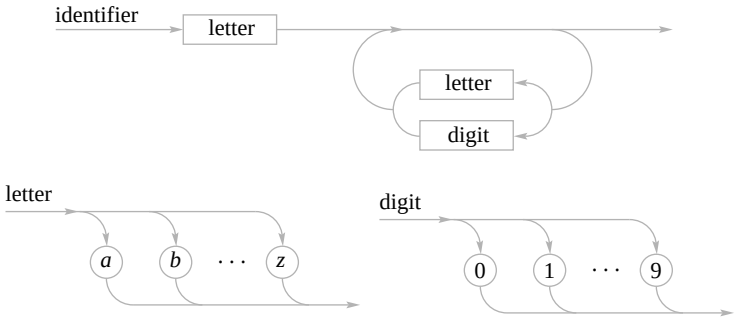


Figure 8

Machine Congruence and Quotient Machines

Let $M = (S, I, F)$ be a finite-state machine, and suppose that R is an equivalence relation on S. We say that R is a **machine congruence** on M if, for any $s, t \in S$, $s \, R \, t$ implies that $f_x(s) \, R \, f_x(t)$ for all $x \in I$. In other words, R is a machine congruence if R-equivalent pairs of states are always taken into R-equivalent pairs of states by every input in I. If R is a machine congruence on $M = (S, I, F)$, we let $\overline{S} = S/R$ be the partition of S corresponding to R. Then $\overline{S} = \{[s] \mid s \in S\}$.

For any input $x \in I$, consider the relation $\overline{f}_x$ on $\overline{S}$ defined by

$$\overline{f}_x = \{([s], [f_x(s)])\}.$$

If $[s] = [t]$, then $s \, R \, t$; therefore, $f_x(s) \, R \, f_x(t)$, so $[f_x(s)] = [f_x(t)]$. This shows that the relation $\overline{f}_x$ is a function from $\overline{S}$ to $\overline{S}$, and we may write $\overline{f}_x([s]) = [f_x(s)]$ for all equivalence classes $[s]$ in $\overline{S}$. If we let $\overline{F} = \{\overline{f}_x \mid x \in I\}$, then the triple $\overline{M} = (\overline{S}, I, \overline{F})$ is a finite-state machine called the **quotient of *M* corresponding to *R***. We will also denote $\overline{M}$ by M/R.

Generally, a quotient machine will be simpler than the original machine. We will show in Section 6 that it is often possible to find a simpler quotient machine that will replace the original machine for certain purposes.

Example 6 Let M be the finite-state machine whose state transition table is shown in Figure 27. Then $S = \{s_0, s_1, s_2, s_3, s_4, s_5\}$. Let R be the equivalence relation on S whose matrix is

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}.$$

	a	*b*
s_0	s_0	s_4
s_1	s_1	s_0
s_2	s_2	s_4
s_3	s_5	s_2
s_4	s_4	s_3
s_5	s_3	s_2

Figure 27

Then we have $S/R = \{[s_0], [s_1], [s_4]\}$, where

$$[s_0] = \{s_0, s_2\} = [s_2],$$
$$[s_1] = \{s_1, s_3, s_5\} = [s_3] = [s_5],$$

and

$$[s_4] = \{s_4\}.$$

We check that R is a machine congruence. The state transition table in Figure 27 shows that f_a takes each element of $[s_i]$ to an element of $[s_i]$ for $i = 0, 1, 4$. Also, f_b takes each element of $[s_0]$ to an element of $[s_4]$, each element of $[s_1]$ to an element of $[s_0]$, and each element of $[s_4]$ to an element of $[s_1]$. These observations show that R is a machine congruence; the state transition table of the quotient machine M/R is shown in Figure 28. ◆

	a	*b*
$[s_0]$	$[s_0]$	$[s_4]$
$[s_1]$	$[s_1]$	$[s_0]$
$[s_4]$	$[s_4]$	$[s_2]$

Figure 28

Example 7 Let $I = \{0, 1\}$, $S = \{s_0, s_1, s_2, s_3, s_4, s_5, s_6, s_7\}$, and $M = (S, I, F)$, the finite-state machine whose digraph is shown in Figure 29.

Suppose that R is the equivalence relation whose corresponding partition of S, S/R, is $\{\{s_0, s_4\}, \{s_1, s_2, s_5\}, \{s_6\}, \{s_3, s_7\}\}$. Then it is easily checked, from the digraph of Figure 29, that R is a machine congruence. To obtain the digraph of the quotient machine $\overline{M}$, draw a vertex for each equivalence class, $[s_0] = \{s_0, s_4\}$, $[s_1] = \{s_1, s_2, s_5\}$, $[s_6] = \{s_6\}$, $[s_3] = \{s_3, s_7\}$, and construct an edge from $[s_i]$ to

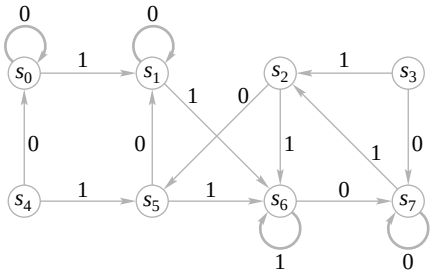


Figure 29

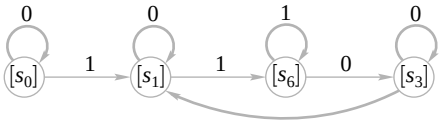


Figure 30

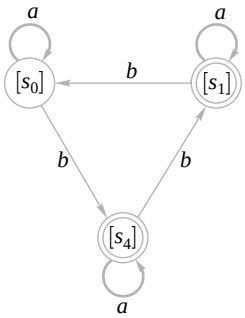


Figure 31

$[s_j]$ if there is, in the original digraph, an edge from some vertex in $[s_i]$ to some vertex in $[s_j]$. In this case, the constructed edge is labeled with all inputs that take some vertex in $[s_i]$ to some vertex in $[s_j]$. Figure 30 shows the result. The procedure illustrated in this example works in general. ◆

If $M = (S, I, F, s_0, T)$ is a Moore machine, and R is a machine congruence on M, then we may let $\overline{T} = \{[t] \mid t \in T\}$. Here, the sequence $\overline{M} = (\overline{S}, I, \overline{F}, [s_0], \overline{T})$ is a Moore machine. In other words, we compute the usual quotient machine M/R; then we designate $[s_0]$ as a starting state, and let $\overline{T}$ be the set of equivalence classes of acceptance states. The resulting Moore machine $\overline{M}$, constructed this way, will be called the **quotient Moore machine** of M.

Example 8 Consider the Moore machine (S, I, F, s_0, T), where (S, I, F) is the finite-state machine of Example 6 and T is the set $\{s_1, s_3, s_4\}$. The digraph of the resulting quotient Moore machine is shown in Figure 31. ◆

3 Exercises

In Exercises 1 through 6, draw the digraph of the machine whose state transition table is shown. Remember to label the edges with the appropriate inputs.

1.

	0	1
s_0	s_0	s_1
s_1	s_1	s_2
s_2	s_2	s_0

2.

	0	1	2
s_0	s_1	s_0	s_2
s_1	s_0	s_0	s_1
s_2	s_2	s_0	s_2

3.

	a	b
s_0	s_1	s_0
s_1	s_2	s_0
s_2	s_2	s_0

4.

	a	b
s_0	s_1	s_0
s_1	s_2	s_1
s_2	s_3	s_2
s_3	s_3	s_3

5.

	a	b	c
s_0	s_0	s_1	s_2
s_1	s_2	s_1	s_1
s_2	s_1	s_1	s_2
s_3	s_2	s_0	s_1

6.

	0	1	2
s_0	s_0	s_2	s_1
s_1	s_1	s_3	s_2
s_2	s_2	s_1	s_3
s_3	s_3	s_3	s_2

In Exercises 7 through 12 (Figures 32 through 37), construct the state transition table of the finite-state machine whose digraph is shown.

7.

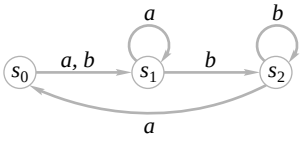


Figure 32

8.

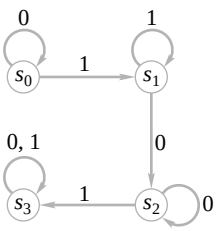


Figure 33

9.

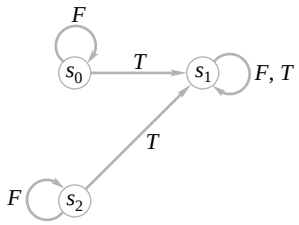


Figure 34

10.

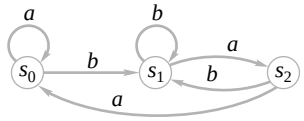


Figure 35

11.

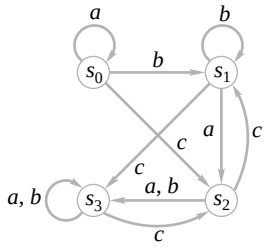


Figure 36

12.

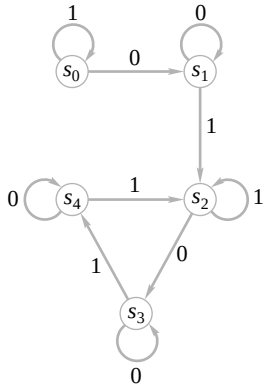


Figure 37

13. Let $M = (S, I, F)$ be a finite-state machine. Define a relation R on I as follows: $x_1 \, R \, x_2$ if and only if $f_{x_1}(s) = f_{x_2}(s)$ for every s in S. Show that R is an equivalence relation on I.

14. Let $(S, \Box)$ be a finite semigroup. Then we may consider the machine (S, S, F), where $F = \{f_x \mid x \in S\}$, and $f_x(y) = x \Box y$ for all $x, y \in S$. Thus we have a finite-state machine in which the state set and the input are the same. Define a relation R on S as follows: $x \, R \, y$ if and only if there is some $z \in S$ such that $f_z(x) = y$. Show that R is transitive.

15. Consider a finite group $(S, \Box)$ and let (S, S, F) be the finite-state machine constructed in Exercise 14. Show that if R is the relation defined in Exercise 14, then R is an equivalence relation.

16. Let $I = \{0, 1\}$ and $S = \{a, b\}$. Construct all possible state transition tables of finite-state machines that have S as state set and I as input set.

17. Consider the machine whose state transition table is

	0	1
1	1	4
2	3	2
3	2	3
4	4	1

Here $S = \{1, 2, 3, 4\}$.

(a) Show that $R = \{(1, 1), (1, 4), (4, 1), (4, 4), (2, 2), (2, 3), (3, 2), (3, 3)\}$ is a machine congruence.

(b) Construct the state transition table for the corresponding quotient machine.

18. Consider the machine whose state transition table is

	a	b	c
s_0	s_0	s_1	s_3
s_1	s_0	s_1	s_2
s_2	s_2	s_3	s_0
s_3	s_2	s_2	s_0

Let $R = \{(s_0, s_1), (s_0, s_0), (s_1, s_1), (s_1, s_0), (s_3, s_2), (s_2, s_2), (s_3, s_3), (s_2, s_3)\}$.

(a) Show that R is a machine congruence.

(b) Construct the digraph for the corresponding quotient machine.

19. Consider the machine whose state transition table is

	0	1
s_0	s_1	s_2
s_1	s_0	s_2
s_2	s_3	s_0
s_3	s_1	s_3

Let $R = \{(s_0, s_0), (s_1, s_1), (s_2, s_2), (s_3, s_3), (s_0, s_1), (s_1, s_0)\}$.

(a) Show that R is a machine congruence.

(b) Construct the digraph for the corresponding quotient machine.

20. Consider the Moore machine whose digraph is shown in Figure 38.

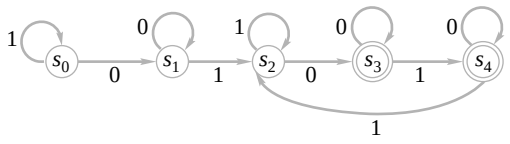


Figure 38

(a) Show that the relation R on S whose matrix is

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

is a machine congruence.

(b) Draw the digraph of the corresponding quotient machine.

21. Consider the Moore machine whose digraph is shown in Figure 39. Show that the relation R on S whose matrix is

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

is a machine congruence. Draw the digraph of the corresponding quotient Moore machine.

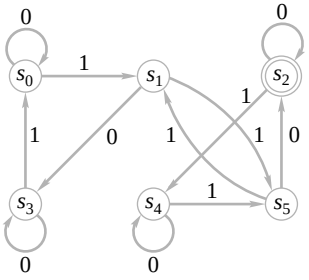


Figure 39

22. Consider the Moore machine whose digraph is shown in Figure 40. Show that the relation R on S whose matrix is

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

is a machine congruence. Draw the digraph of the corresponding quotient Moore machine.

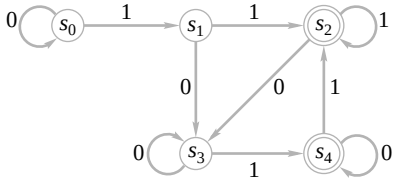


Figure 40

23. Let $M' = (S', I, F', s_0', T')$ and $M'' = (S'', I, F'', s_0'', T'')$ be two Moore machines with the same input set. Define $S = S' \times S''$, $s_0 = (s_0', s_0'')$, $T = T' \times T''$, and F by $f_x(s', s'') = (f'_x(s'), f''_x(s''))$. Show that $M = (S, I, F, s_0, T)$ is a Moore machine.

24. Let $M' = (S', I, F', s_0', T')$ and $M'' = (S'', I, F'', s_0'', T'')$ be two Moore machines with the same input set. Define $S = S' \times S''$, $s_0 = (s_0', s_0'')$, $T = \{(t', t'') \mid t' \in T' \text{ or } t'' \in T''\}$, and F by $f_x(s', s'') = (f'_x(s'), f''_x(s''))$. Show that $M = (S, I, F, s_0, T)$ is a Moore machine.

25. Using the definition of M in Exercise 23, give its state transition table if M' is the machine in Exercise 1 and M'' is the machine in Exercise 8.

4 Monoids, Machines, and Languages

Let $M = (S, I, F)$ be a finite-state machine with state set $S = \{s_0, s_1, \ldots, s_n\}$, input set I, and state transition functions $F = \{f_x \mid x \in I\}$.

We will associate with M two monoids. First, there is the free monoid I^* on the input set I. This monoid consists of all finite sequences (or "strings" or "words") from I, with catenation as its binary operation. The identity is the empty string λ. Second, we have the monoid S^S, which consists of all functions from S to S and which has function composition as its binary operation. The identity in S^S is the function 1_S defined by $1_S(s) = s$, for all s in S.

If $w = x_1x_2 \cdots x_n \in I^*$, we let $f_w = f_{x_n} \circ f_{x_{n-1}} \circ \cdots \circ f_{x_1}$, the composition of the functions $f_{x_n}, f_{x_{n-1}}, \ldots, f_{x_1}$. Also, we define f_λ to be 1_S. In this way we assign an element f_w of S^S to each element w of I^*. If we think of each f_x as the effect of the input x on the states of the machine M, then f_w represents the combined effect of all the input letters in the word w, received in the sequence specified by w. We call f_w the **state transition function corresponding to *w***.

Example 1 Let $M = (S, I, F)$, where $S = \{s_0, s_1, s_2\}$, $I = \{0, 1\}$, and F is given by the following state transition table.

	0	1
s_0	s_0	s_1
s_1	s_2	s_2
s_2	s_1	s_0

Let $w = 011 \in I^*$. Then

$$f_w(s_0) = (f_1 \circ f_1 \circ f_0)(s_0) = f_1(f_1(f_0(s_0))) = f_1(f_1(s_0)) = f_1(s_1) = s_2.$$

Similarly,

$$f_w(s_1) = f_1(f_1(f_0(s_1))) = f_1(f_1(s_2)) = f_1(s_0) = s_1$$

and

$$f_w(s_2) = f_1(f_1(f_0(s_2))) = f_1(f_1(s_1)) = f_1(s_2) = s_0.$$ ◆

Example 2 Let us consider the same machine M as in Example 1 and examine the problem of computing f_w a little differently. In Example 1 we used the definition directly, and for a large machine we would program an algorithm to compute the values of f_w in just that way. However, if the machine is of moderate size, we may find another procedure to be preferable.

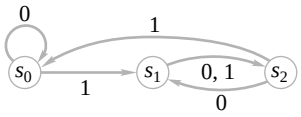


Figure 41

We begin by drawing the digraph of the machine M as shown in Figure 41. We may use this digraph to compute word transition functions by just following the edges corresponding to successive input letter transitions. Thus, to compute $f_w(s_0)$, we start at state s_0 and see that input 0 takes us to state s_0. The input 1 that follows takes us on to state s_1, and the final input of 1 takes us to s_2. Thus $f_w(s_0) = s_2$, as before.

Let us compute $f_{w'}$, where $w' = 01011$. The successive transitions of s_0 are

$$s_0 \xrightarrow{0} s_0 \xrightarrow{1} s_1 \xrightarrow{0} s_2 \xrightarrow{1} s_0 \xrightarrow{1} s_1,$$

so $f_{w'}(s_0) = s_1$. Similar displays show that $f_{w'}(s_1) = s_2$ and $f_{w'}(s_2) = s_0$. ◆

This method of interpreting word transition functions such as f_w and $f_{w'}$ is useful in designing machines that have word transitions possessing certain desired properties. This is a crucial step in the practical application of the theory and we will consider it in the next section.

Let $M = (S, I, F)$ be a finite-state machine. We define a function T from I^* to S^S. If w is a string in I^*, let $T(w) = f_w$ as defined previously. Then we have the following result.

THEOREM 1

(a) If w_1 and w_2 are in I^*, then $T(w_1 \cdot w_2) = T(w_2) \circ T(w_1)$.

(b) If $M = T(I^*)$, then M is a submonoid of S^S.

Proof

(a) Let $w_1 = x_1x_2 \cdots x_k$ and $w_2 = y_1y_2 \cdots y_m$ be two strings in I^*. Then

$$\begin{aligned} T(w_1 \cdot w_2) &= T(x_1x_2 \cdots x_k y_1y_2 \cdots y_m) \\ &= (f_{y_m} \circ f_{y_{m-1}} \circ \cdots \circ f_{y_1}) \circ (f_{x_k} \circ f_{x_{k-1}} \circ \cdots \circ f_{x_1}) \\ &= T(w_2) \circ T(w_1). \end{aligned}$$

Also, $T(\lambda) = 1_S$ by definition. Thus T is a monoid homomorphism.

(b) Part (a) shows that if f and g are in M, then $f \circ g$ and $g \circ f$ are in M. Thus M is a subsemigroup of S^S. Since $1_S = T(\square)$, $1_S \in M$. Thus M is a submonoid of S^S. The monoid M is called the **monoid of the machine M**. ■

Example 3

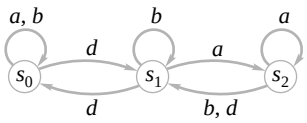


Figure 42

Let $S = \{s_0, s_1, s_2\}$ and $I = \{a, b, d\}$. Consider the finite-state machine $M = (S, I, F)$ defined by the digraph shown in Figure 42. Compute the functions f_{bad}, f_{add}, and f_{badadd}, and verify that

$$f_{add} \circ f_{bad} = f_{badadd}.$$

Solution

f_{bad} is computed by the following sequence of transitions:

$$s_0 \xrightarrow{b} s_0 \xrightarrow{a} s_0 \xrightarrow{d} s_1$$
$$s_1 \xrightarrow{b} s_1 \xrightarrow{a} s_2 \xrightarrow{d} s_1$$
$$s_2 \xrightarrow{b} s_1 \xrightarrow{a} s_2 \xrightarrow{d} s_1.$$

Thus $f_{bad}(s_0) = s_1$, $f_{bad}(s_1) = s_1$, and $f_{bad}(s_2) = s_1$.

Similarly, for f_{add},

$$s_0 \xrightarrow{a} s_0 \xrightarrow{d} s_1 \xrightarrow{d} s_0$$
$$s_1 \xrightarrow{a} s_2 \xrightarrow{d} s_1 \xrightarrow{d} s_0$$
$$s_2 \xrightarrow{a} s_2 \xrightarrow{d} s_1 \xrightarrow{d} s_0,$$

so $f_{add}(s_i) = s_0$ for $i = 0, 1, 2$. A similar computation shows that

$$f_{badadd}(s_0) = s_0, \qquad f_{badadd}(s_1) = s_0, \qquad f_{badadd}(s_2) = s_0$$

and the same results hold for $f_{add} \circ f_{bad}$. In fact,

$$(f_{add} \circ f_{bad})(s_0) = f_{add}(f_{bad}(s_0)) = f_{add}(s_1) = s_0$$
$$(f_{add} \circ f_{bad})(s_1) = f_{add}(f_{bad}(s_1)) = f_{add}(s_1) = s_0$$
$$(f_{add} \circ f_{bad})(s_2) = f_{add}(f_{bad}(s_2)) = f_{add}(s_1) = s_0.$$ ◆

Example 4

Consider the machine whose graph is shown in Figure 43. Show that $f_w(s_0) = s_0$ if and only if w has $3n$ 1's for some $n \geq 0$.

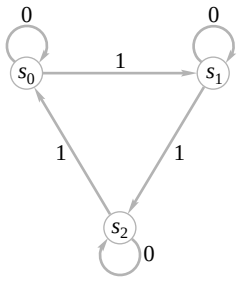


Figure 43

Solution

From Figure 43 we see that $f_0 = 1_S$, so the 0's in a string $w \in I^*$ have no effect on f_w. Thus, if $\overline{w}$ is w with all 0's removed, then $f_w = f_{\overline{w}}$. Let $l(w)$ denote the **length** of w, that is, the number of digits in w. Then $l(\overline{w})$ is the number of 1's in w, for all $w \in I^*$. For each $n \geq 0$, consider the statement

$P(n)$: Let $w \in I^*$ and let $l(\overline{w}) = m$.
(a) If $m = 3n$, then $f_w(s_0) = s_0$.
(b) If $m = 3n + 1$, then $f_w(s_0) = s_1$.
(c) If $m = 3n + 2$, then $f_w(s_0) = s_2$.

We prove by mathematical induction that $P(n)$ is true for all $n \geq 0$.

Basis Step

Suppose that $n = 0$. In case (a), $m = 0$; therefore, w has no 1's and $f_w(s_0) = 1_S(s_0) = s_0$. In case (b), $m = 1$, so $\overline{w} = 1$ and $f_w(s_0) = f_{\overline{w}}(s_0) = f_1(s_0) = s_1$. Finally, in case (c), $m = 2$, so $\overline{w} = 11$, and $f_w(s_0) = f_{\overline{w}}(s_0) = f_{11}(s_0) = f_1(s_1) = s_2$.

Induction Step

We must use $P(k)$ to show $P(k + 1)$. Let $w \in I^*$, and denote $l(\overline{w})$ by m. In case (a), $m = 3(k + 1) = 3k + 3$; therefore, $\overline{w} = w' \cdot 111$, where $l(w') = 3k$. Then $f_{w'}(s_0) = s_0$ by $P(k)$, part (a), and $f_{111}(s_0) = s_0$ by direct computation, so $f_{\overline{w}}(s_0) = f_{w'}(f_{111}(s_0)) = f_{w'}(s_0) = s_0$. Cases (b) and (c) are handled in the same way. Thus $P(k + 1)$ is true.

By mathematical induction, $P(n)$ is true for all $n \geq 0$, so $f_w(s_0) = s_0$ if and only if the number of 1's in w is a multiple of 3. ◆

Suppose now that (S, I, F, s_0, T) is a Moore machine. As in Section 1, we may think of certain subsets of I^* as "languages" with "words" from I. Using M, we can define such a subset, which we will denote by $L(M)$, and call the **language of the machine** M. Define $L(M)$ to be the set of all $w \in I^*$ such that $f_w(s_0) \in T$. In other words, $L(M)$ consists of all strings that, when used as input to the machine, cause the starting state s_0 to move to an acceptance state in T. Thus, in this sense, M accepts the string. It is for this reason that the states in T were named acceptance states in Section 3.

Example 5

Let $M = (S, I, F, s_0, T)$ be the Moore machine in which (S, I, F) is the finite-state machine whose digraph is shown in Figure 43, and $T = \{s_1\}$. The discussion of Example 4 shows that $f_w(s_0) = s_1$ if and only if the number of 1's in w is of the form $3n + 1$ for some $n \geq 0$. Thus $L(M)$ is exactly the set of all strings with $3n + 1$ 1's for some $n \geq 0$. ◆

Example 6

Consider the Moore machine M whose digraph is shown in Figure 44. Here state s_0 is the starting state, and $T = \{s_2\}$. What is $L(M)$? Clearly, the input set is $I = \{a, b\}$. Observe that, in order for a string w to cause a transition from s_0 to s_2, w must contain at least two b's. After reaching s_2, any additional letters have no effect. Thus $L(M)$ is the set of all strings having two or more b's. We see, for example, that $f_{aabaa}(s_0) = s_1$, so *aabaa* is rejected. On the other hand, $f_{abaab}(s_0) = s_2$, so *abaab* is accepted. ◆

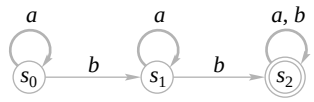


Figure 44

4 Exercises

In Exercises 1 through 5, we refer to the finite-state machine whose state transition table is

	0	1
s_0	s_0	s_1
s_1	s_1	s_2
s_2	s_2	s_3
s_3	s_3	s_0

1. List the values of the transition function f_w for $w = 01001$.

2. List the values of the transition function f_w for $w = 11100$.

3. Describe the set of binary words (sequences of 0's and 1's) w having the property that $f_w(s_0) = s_0$.

4. Describe the set of binary words w having the property that $f_w = f_{010}$.

5. Describe the set of binary words w having the property that $f_w(s_0) = s_2$.

In Exercises 6 through 10, we refer to the finite-state machine whose digraph is shown in Figure 45.

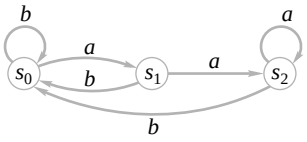


Figure 45

6. List the values of the transition function f_w for $w = abba$.

7. List the values of the transition function f_w for $w = babab$.

8. Describe the set of words w having the property that $f_w(s_0) = s_2$.

9. Describe the set of words w having the property that $f_w(s_0) = s_0$.

10. Describe the set of words w having the property that $f_w = f_{aba}$.

In Exercises 11 through 15, describe (in words) the language accepted by the Moore machines whose digraphs are given in Figures 46 through 50.

11.

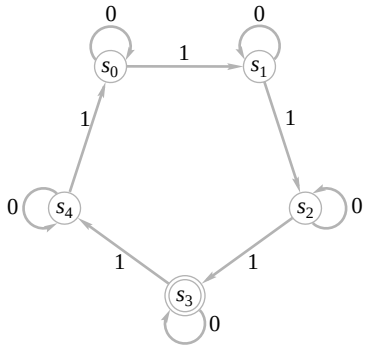


Figure 46

12.

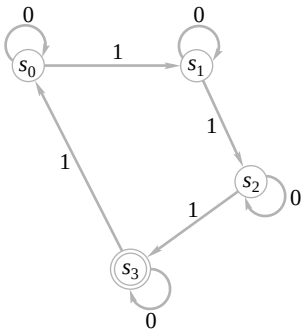


Figure 47

13.

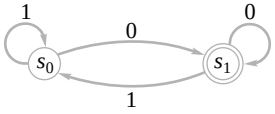


Figure 48

14.

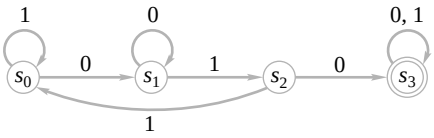


Figure 49

15.

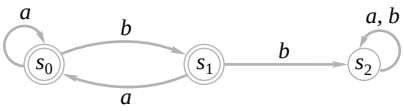


Figure 50

In Exercises 16 through 22, describe (in words) the language accepted by the Moore machine whose state table is given. The starting state is s_0, and the set T of acceptance states is shown.

16.

	0	1
s_0	s_1	s_2
s_1	s_1	s_2
s_2	s_2	s_1

$T = \{s_2\}$

17.

	0	1
s_0	s_1	s_0
s_1	s_1	s_2
s_2	s_1	s_0

$T = \{s_2\}$

18.

	0	1
s_0	s_0	s_1
s_1	s_0	s_1

$T = \{s_1\}$

19.

	x	y	z
s_0	s_1	s_3	s_4
s_1	s_4	s_2	s_4
s_2	s_4	s_4	s_4
s_3	s_4	s_4	s_2
s_4	s_4	s_4	s_4

$T = \{s_2\}$

20.

	x	y
s_0	s_1	s_2
s_1	s_3	s_2
s_2	s_1	s_2
s_3	s_3	s_3

$T = \{s_1, s_2\}$

21.

	0	1
s_0	s_2	s_1
s_1	s_1	s_1
s_2	s_1	s_2

$T = \{s_0, s_1\}$

22.

	0	1
s_0	s_0	s_1
s_1	s_2	s_1
s_2	s_2	s_2

$T = \{s_1\}$

23. Let $M = \{S, I, F, s_0, T\}$ be a Moore machine. Suppose that if $s \in T$ and $w \in I^*$, then $f_w(s) \in T$. Prove that $L(M)$ is a subsemigroup of $(I^*, \cdot)$, where $\cdot$ is catenation.

24. Let M be as constructed in Exercise 23 of Section 3. Show that $L(M) = L(M') \cap L(M'')$.

25. Let M be as constructed in Exercise 24 of Section 3. Show that $L(M) = L(M') \cup L(M'')$.

26. Let M be a Moore machine such that $L(M)$ is finite. Describe a Moore machine M' that accepts $I^* - L(M)$.

5 Machines and Regular Languages

Let $M = (S, I, F, s_0, T)$ be a Moore machine. In Section 4 we defined the language $L(M)$ of the machine M. It is natural to ask if there is a connection between such a language and the languages of phrase structure grammars, discussed in Section 1. The following theorem, due to S. Kleene, describes the connection.

THEOREM 1 Let I be a set and let $L \subseteq I^*$. Then L is a type 3 language; that is, $L = L(G)$, where G is a type 3 grammar having I as its set of terminal symbols, if and only if $L = L(M)$ for some Moore machine $M = (S, I, F, s_0, T)$. ■

We stated in Section 2 that a set $L \subseteq I^*$ is a type 3 language if and only if L is a regular set, that is, if and only if L corresponds to some regular expression over I. This leads to the following corollary of Theorem 1.

Corollary 1 Let I be a set and let $L \subseteq I^*$. Then $L = L(M)$ for some Moore machine $M = (S, I, F, s_0, T)$ if and only if L is a regular set. ■

We will not give a complete and detailed proof of Theorem 1. However, it is easy to give a construction that produces a type 3 grammar from a given Moore machine. This is done in such a way that the grammar and the machine have the same language. Let $M = (S, I, F, s_0, T)$ be a given Moore machine. We construct a type 3 grammar $G = (V, I, s_0, \to)$ as follows. Let $V = I \cup S$; that is, I will be the set of terminal symbols for G, while S will be the set of nonterminal symbols. Let s_i and s_j be in S, and $x \in I$. We write $s_i \to xs_j$, if $f_x(s_i) = s_j$, that is, if the input x takes state s_i to s_j. We also write $s_i \to x$ if $f_x(s_i) \in T$, that is, if the input x takes the state s_i to some acceptance state. Now let $\to$ be the relation determined by the preceding two conditions and take this relation as the production relation of G.

The grammar G constructed previously has the same language as M. Suppose, for example, that $w = x_1x_2x_3 \in I^*$. The string w is in $L(M)$ if and only if $f_w(s_0) = f_{x_3}(f_{x_2}(f_{x_1}(s_0))) \in T$. Let $a = f_{x_1}(s_0)$, $b = f_{x_2}(a)$, and $c = f_{x_3}(b)$, where $c = f_w(s_0)$ is in T. Then the rules given for constructing $\to$ tell us that

1. $s_0 \to x_1a$
2. $a \to x_2b$
3. $b \to x_3$

are all productions in G. The last one occurs because $c \in T$. If we begin with s_0 and substitute, using (1), (2), and (3) in succession, we see that $s_0 \Rightarrow^* x_1x_2x_3 = w$ (see Section 1), so $w \in L(G)$. A similar argument works for any string in $L(M)$, so $L(M) \subseteq L(G)$. If we reverse the argument, we can see that we also have $L(G) \subseteq L(M)$. Thus M and G have the same language.

Example 1 Consider the Moore machine M shown in Figure 44. Construct a type 3 grammar G such that $L(G) = L(M)$. Also, find a regular expression over $I = \{a, b\}$ that corresponds to $L(M)$.

Solution

Let $I = \{a, b\}$, $S = \{s_0, s_1, s_2\}$, and $V = I \cup S$. We construct the grammar $(V, I, s_0, \rightarrow)$, where $\rightarrow$ is described as follows:

$$\begin{array}{lll} \rightarrow: & s_0 \rightarrow as_0 & s_2 \rightarrow bs_2 \\ & s_0 \rightarrow bs_1 & s_1 \rightarrow b \\ & s_1 \rightarrow as_1 & s_2 \rightarrow a \\ & s_1 \rightarrow bs_2 & s_2 \rightarrow b \\ & s_2 \rightarrow as_2. & \end{array}$$

The production relation $\rightarrow$ is constructed as we indicated previously; therefore, $L(M) = L(G)$.

If we consult Figure 44, we see that a string $w \in L(M)$ has the following properties. Any number $n \geq 0$ of a's can occur at the beginning of w. At some point, a b must occur in order to cause the transition from s_0 to s_1. After this b, any number $k \geq 0$ of a's may occur, followed by another b to cause transition to s_2. The remainder of w, if any, is completely arbitrary, since the machine cannot leave s_2 after once entering this state. From this description we can readily see that $L(M)$ corresponds to the regular expression

$$a^*ba^*b(a \cup b)^*.$$ ◆

Example 2 Consider the Moore machine whose digraph is shown in Figure 51. Describe in words the language $L(M)$. Then construct the regular expression that corresponds to $L(M)$ and describe the production of the corresponding grammar G in BNF form.

Solution

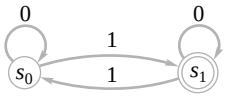


Figure 51

It is clear that 0's in the input string have no effect on the states. If an input string w has an odd number of 1's, then $f_w(s_0) = s_1$. If w has an even number of 1's, then $f_w(s_0) = s_0$. Since $T = \{s_1\}$, we see that $L(M)$ consists of all w in I^* that have an odd number of 1's.

We now find the regular expression corresponding to $L(M)$. Any input string corresponding to the expression 0^*10^* will be accepted, since it will have exactly one 1. If an input w begins in this way, but has more 1's, then the additional ones must come in pairs, with any number of 0's allowed between, or after each pair of 1's. The previous sentence describes the set of strings corresponding to the expression $(10^*10^*)^*$. Thus $L(M)$ corresponds to the regular expression

$$0^*10^*(10^*10^*)^*.$$

Finally, the type 3 grammar constructed from M is $G = (V, I, s_0, \rightarrow)$ with $V = I \cup S$. The BNF of the relation $\rightarrow$ is

$$\begin{aligned} \langle s_0 \rangle &::= 0\langle s_0 \rangle \mid 1\langle s_1 \rangle \mid 1 \\ \langle s_1 \rangle &::= 0\langle s_1 \rangle \mid 1\langle s_0 \rangle \mid 0. \end{aligned}$$ ◆

Occasionally, we may need to determine the function performed by a given Moore machine, as we did in the preceding examples. More commonly, however, it is necessary to construct a machine that will perform a given task. This task may be defined by giving a verbal description, a regular expression, or an equivalent type 3 grammar, perhaps in BNF or with a syntax diagram. There are systematic, almost mechanical ways to construct such a machine. Most of these use the concept of *nondeterministic machines* and employ a tedious translation process from such machines to the Moore machines that we have discussed. If the task

of the machine is not too complex, we may use simple reasoning to construct the machine in steps, usually in the form of its digraph. Whichever method is used, the resulting machine may be quite inefficient; for example, it may have unneeded states. In Section 6, we will give a procedure for constructing an equivalent machine that may be much more efficient.

Example 3 Construct a Moore machine M that will accept exactly the string 001 from input strings of 0's and 1's. In other words, $I = \{0, 1\}$ and $L(M) = \{001\}$.

Solution

We must begin with a starting state s_0. If w is an input string of 0's and 1's and if w begins with a 0, then w *may* be accepted (depending on the remainder of its components). Thus one step toward acceptance has been taken, and there needs to be a state s_1 that corresponds to this step. We therefore begin as in Figure 52(a). If we next receive another 0, we have progressed one more step toward acceptance. We therefore construct another state s_2 and let 0 give a transition from s_1 to s_2. State s_1 represents the condition "first symbol of input is a 0," whereas state s_2 represents the condition "first two symbols of the input are 00." This situation is shown in Figure 52(b). Finally, if the third input symbol is a 1, we move to an acceptance state, as shown in Figure 52(c). Any other beginning sequence of input digits or any additional digits will move us to a "failure state" s_4 from which there is no escape. Thus Figure 52(d) shows the completed machine. ◆

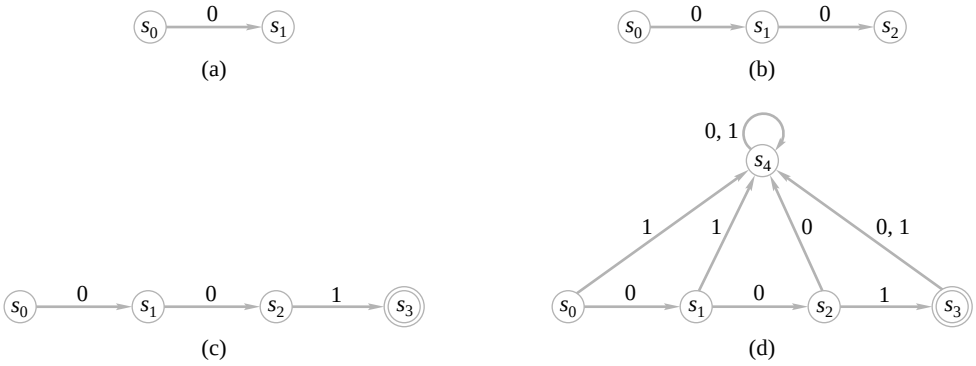


Figure 52

The process illustrated in Example 3 is difficult to describe precisely or to generalize. We try to construct states representing each successive stage of input complexity leading up to an acceptable string. There must also be states indicating the ways in which a promising input pattern may be destroyed when a certain symbol is received. If the machine is to recognize several, essentially different types of input, then we will need to construct separate branches corresponding to each type of input. This process may result is some redundancy, but the machine can be simplified later.

Example 4 Let $I = \{0, 1\}$. Construct a Moore machine that accepts those input sequences w that contain the string 01 or the string 10 anywhere within them. In other words, we are to accept exactly those strings that do not consist entirely of 0's or entirely of 1's.

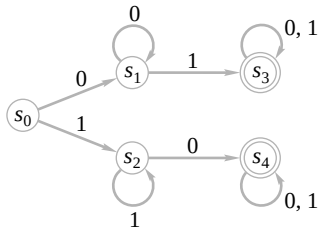


Figure 53

Solution

This is a simple example in which, whatever input digit is received first, a string will be accepted if and only if the other digit is eventually received. There must be a starting state s_0, states s_1 and s_2 corresponding respectively to first receiving a 0 or 1, and (acceptance) states s_3 and s_4, which will be reached if and when the other digit is received. Having once reached an acceptance state, the machine stays in that state. Thus we construct the digraph of this machine as shown in Figure 53. ◆

In Example 3, once an acceptance state is reached, any additional input will cause a permanent transition to a nonaccepting state. In Example 4, once an acceptance state is reached, any additional input will have no effect. Sometimes the situation is between these two extremes. As input is received, the machine may repeatedly enter and leave acceptance states. Consider the Moore machine M whose digraph is shown in Figure 54. This machine is a slight modification of the finite-state machine given in Example 4 of Section 4. We know from that example that $w \in L(M)$ if and only if the number of 1's in w is of the form $3n$, $n \geq 0$. As input symbols are received, M may enter and leave s_0 repeatedly. The conceptual states "one 1 has been received" and "four 1's have been received" may both be represented by s_1. When constructing machines, we should keep in mind the fact that a state, previously defined to represent one conceptual input condition, may be used for a new input condition if these two conditions represent the same degree of progress of the input stream toward acceptance. The next example illustrates this fact.

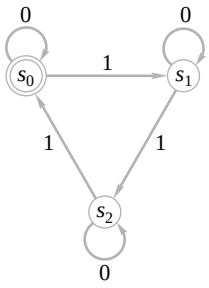
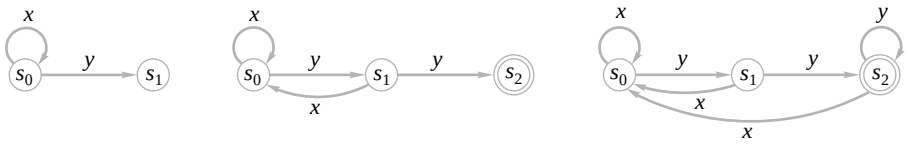


Figure 54

Example 5 Construct a Moore machine that accepts exactly those input strings of x's and y's that end in *yy*.

Solution

Again we need a starting state s_0. If the input string begins with a y, we progress one step to a new state s_1 ("last input component received is a y"). On the other hand, if the input begins with an x, we have made no progress toward acceptance. Thus we may suppose that M is again in state s_0. This situation is shown in Figure 55(a). If, while in state s_1, a y is received, we progress to an acceptance state s_2 ("last two symbols of input received were y's"). If instead the input received is an x, we must again receive two y's in order to be in an acceptance state. Thus we may again regard this as a return to state s_0. The situation at this point is shown in Figure 55(b). Having reached state s_2, an additional input of y will have no effect, but an input of x will necessitate two more y's for acceptance. Thus we can again regard M as being in state s_0. The final Moore machine is shown in Figure 55(c). ◆



(a) (b) (c)

Figure 55

We have not mentioned the question of implementation of finite-state machines. Indeed, many such machines, including all digital computers, are implemented as hardware devices, that is, as electronic circuitry. There are, however,

many occasions when finite-state machines are simulated in software. This is frequently seen in compilers and interpreters, for which Moore machines may be programmed to retrieve and interpret words and symbols in an input string. We provide just a hint of the techniques available by simulating the machine of Example 2 in pseudocode. The reader should refer back to Example 2 and Figure 51 for the details of the machine. The following subroutine gives a pseudocode program for this machine.

This program uses a subroutine INPUT to get the next 0 or 1 in variable X and assumes that a logical variable EOI will be set true when no further input is available. The variable RESULT will be true if the input string contains an odd number of 1's; otherwise, it will be false.

```
SUBROUTINE ODDONES (RESULT)
1. EOI ← F
2. RESULT ← F
3. STATE ← 0
4. UNTIL (EOI)
    a. CALL INPUT (X, EOI)
        1. IF (EOI = F) THEN
            a. IF (STATE = 0) THEN
                1. IF (X = 1) THEN
                    a. RESULT ← T
                    b. STATE ← 1
    b. ELSE
        1. IF (X = 1) THEN
            a. RESULT ← F
            b. STATE ← 0
5. RETURN
END OF SUBROUTINE ODDONES
```

In this particular coding technique, a state is denoted by a variable that may be assigned different values depending on input and whose values then determine other effects of the input. An alternative procedure is to represent a state by a particular location in code. This location then determines the effect of input and the branch to a new location (subsequent state). The following subroutine shows the same subroutine ODDONES coded in this alternative way.

```
    SUBROUTINE ODDONES (RESULT) version 2
    1. RESULT ← F
S0: 2. CALL INPUT (X, EOI)
    3. IF (EOI) THEN
        a. RETURN
    4. ELSE
        a. IF (X = 1) THEN
            1. RESULT ← T
            2. GO TO S1
        b. ELSE
            1. GO TO S0
```

```
S1: 5. CALL INPUT (X, EOI)
    6. IF (EOI) THEN
        a. RETURN
    7. ELSE
        a. IF (X = 1) THEN
             1. RESULT ← F
             2. GO TO S0
        b. ELSE
             1. GO TO S1
END OF SUBROUTINE ODDONES version 2
```

It is awkward to avoid **GO TO** statements in this approach, and we have used them. In languages with multiple **GO TO** statements, such as FORTRAN's indexed **GO TO** or PASCAL's CASE statement, this method may be particularly efficient for finite-state machines with a fairly large number of states. In such cases, the first method may become quite cumbersome.

5 Exercises

1. Let M be the Moore machine of Figure 56. Construct a type 3 grammar $G = (V, I, s_0, \rightarrow)$, such that $L(M) = L(G)$.

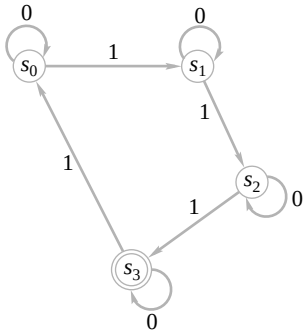


Figure 56

2. Let M be the Moore machine of Figure 57. Give a regular expression over $I = \{0, 1\}$, which corresponds to the language $L(M)$.

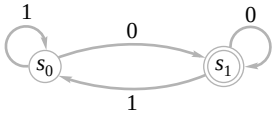


Figure 57

3. Let M be the Moore machine of Exercise 18, Section 4. Give a regular expression over $I = \{0, 1\}$, which corresponds to the language $L(M)$.

4. Let M be the Moore machine of Figure 58. Construct a type 3 grammar $G = (V, I, s_0, \rightarrow)$, such that $L(M) = L(G)$. Describe $\rightarrow$ in BNF.

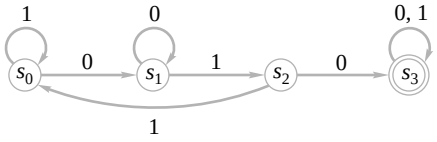


Figure 58

5. Let M be the Moore machine of Figure 59. Construct a type 3 grammar $G = (V, I, s_0, \rightarrow)$, such that $L(M) = L(G)$. Describe $\rightarrow$ in BNF.

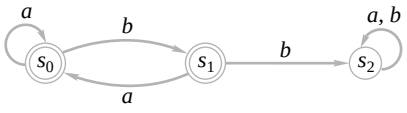


Figure 59

In Exercises 6 through 20, construct the digraph of a Moore machine that accepts the input strings described and no others.

6. Inputs a, b: strings that end in *ab*

7. Inputs a, b: strings where the number of b's is divisible by 3

8. Inputs a, b: strings where the number of a's is even and the number of b's is a multiple of 3

9. Inputs x, y: strings that have an even number of y's

10. Inputs 0, 1: strings that contain 0011

11. Inputs 0, 1: strings that end with 0011

12. Inputs , □: strings that contain □ or □

13. Inputs +, ×: strings that contain + × × or × + +

14. Inputs w, z: strings that contain *wz* or *zzw*

15. Inputs a, b: strings that contain *ab* and end in *bbb*

18.

	0	1
s_0	s_5	s_2
s_1	s_6	s_2
s_2	s_0	s_4
s_3	s_3	s_5
s_4	s_6	s_2
s_5	s_3	s_0
s_6	s_3	s_1

$T = \{s_2\}$

19.

	0	1
a	a	c
b	g	d
c	f	e
d	a	d
e	a	d
f	g	f
g	g	c

$s_0 = a$
$T = \{d, e\}$

20.

	0	1
s_0	s_1	s_2
s_1	s_2	s_3
s_2	s_2	s_4
s_3	s_2	s_3
s_4	s_5	s_4
s_5	s_5	s_5

$T = \{s_5\}$

21.

	0	1
s_0	s_1	s_4
s_1	s_4	s_2
s_2	s_3	s_2
s_3	s_3	s_2
s_4	s_4	s_4

$T = \{s_3\}$

22. Draw the digraph of the Moore machine whose state table is given in Exercise 18. Also, draw the digraph of the quotient machine $\overline{M}$.

23. Draw the digraph of the Moore machine whose state table is given in Exercise 19. Also, draw the digraph of the quotient machine $\overline{M}$.

24. Draw the digraph of the Moore machine whose state table is given in Exercise 20. Also, draw the digraph of the quotient machine $\overline{M}$.

25. Draw the digraph of the Moore machine whose state table is given in Exercise 21. Also, draw the digraph of the quotient machine $\overline{M}$.

26. Describe $L(\overline{M})$ for $\overline{M}$ in Exercise 25.

Tips for Proofs

Direct proofs using definitions and previously proven results are usually the first strategy to try. Example 4, Section 4, uses an induction proof, because the statement to be proved is of the form $P(n)$, $n \geq 0$. The form of the statement is often a clue to what proof strategy to choose. As another example, you should recognize that phrases such as "at least two" in the theorem or statement may make it a good candidate for a pigeonhole proof. (See Exercise 32, Section 1.) Even when you have a good idea which proof technique will be useful, there is still likely to be some trial and error involved in developing the proof. Just how to use a definition or which previous theorem is applicable is not always obvious, but persistence is also a powerful tool for producing a proof.

Key Ideas for Review

- Phrase structure grammar $(V, S, v_0, \rightarrow)$
- Production: a statement $w \rightarrow w'$, where $(w, w') \in \rightarrow$
- Direct derivability
- Terminal symbols: the elements of S
- Nonterminal symbols: the elements of $V - S$
- Derivation of a sentence: substitution process that produces a valid sentence
- Language of a grammar G: set of all properly constructed sentences that can be produced from G
- Derivation tree for a sentence
- Types 0, 1, 2, 3 phrase structure grammars
- Context-free grammar: type 2 grammar
- Regular grammar: type 3 grammar
- Parsing: process of obtaining a derivation tree that will produce a given sentence
- BNF notation
- Syntax diagram
- Theorem: Let S be a finite set, and $L \subseteq S^*$. Then L is a regular set if and only if $L = L(G)$ for some regular grammar $G = (V, S, v_0, \rightarrow)$.

- Finite-state machine: (S, I, F), where S is a finite set of states, I is a set of inputs, and $F = \{f_x \mid x \in I\}$
- State transition tables
- R_M: $s_i \; R_M \; s_j$, if there is an input x so that $f_x(s_i) = s_j$
- Moore machine: $M = (S, I, F, s_0, T)$, where $s_0 \in S$ is the starting state and $T \subseteq S$ is the set of acceptance states
- Machine congruence R on M: For any $s, t \in S$, $s \; R \; t$ implies that $f_x(s) \; R \; f_x(t)$ for all $x \in I$.
- Quotient of M corresponding to R
- State transition function f_w, $w = x_1x_2 \cdots x_n$: $f_w = f_{x_n} \circ f_{x_{n-1}} \circ \cdots \circ f_{x_1}$, $f_\lambda = 1_S$
- Theorem: Let $M = (S, I, F)$ be a finite-state machine. Define $T : I^* \to S^S$ by $T(w) = f_w$, $w \in I^*$, and $T(\lambda) = 1_S$. Then
  (a) If w_1 and w_2 are in I^*, then $T(w_1 \cdot w_2) = T(w_2) \circ T(w_1)$.
  (b) If $M = T(I^*)$, then M is a submonoid of S^S.
- Monoid of a machine: M in the preceding theorem
- w-compatible
- Equivalent machines M and N: $L(M) = L(N)$
- $l(w)$: length of the string w
- Language accepted by M: $L(M) = \{w \in I^* \mid f_w(s_0) \in T\}$
- Theorem: Let I be a set and $L \subseteq I^*$. Then L is a type 3 language, that is, $L = L(G)$ if and only if $L = L(M)$ for some Moore machine M.

Chapter Self-Test

1. Can every phrase structure grammar be given in BNF?

2. What types of phrase structure grammars can be represented by a syntax diagram?

3. What is the type of the language $\{0^k1^k, k \geq 1\}$?

4. What does it mean for a machine to accept or recognize a language?

5. What is the connection between regular languages and Moore machines?

6. Let $G = (V, S, v_0, \Rightarrow)$ be a phrase structure grammar with $V = \{v_0, v_1, a, b, c, d\}$, $S = \{a, b, c, d\}$, and

$$\Rightarrow : \; v_0 \Rightarrow av_0b \quad v_0 \Rightarrow v_1 \quad v_1 \Rightarrow cv_1 \quad v_1 \Rightarrow d.$$

Tell whether each of the following is true or false.

(a) $abcd \in L(G)$

(b) $aaav_1bbb \Rightarrow^* aaaccv_1bbb$

(c) $ccccd \in L(G)$

(d) $av_1b \Rightarrow acdb$

7. Describe precisely $L(G)$, the language of G as given in Problem 1.

8. Let $G = (V, S, v_0, \Rightarrow)$ be a phrase structure grammar with $V = \{v_0, v_1, 1, 2, 3, 4, 5\}$, $S = \{1, 2, 3, 4, 5\}$ and

$$\begin{aligned} \Rightarrow : \quad & v_0 \Rightarrow 1v_1 \quad v_0 \Rightarrow 3v_1 \quad v_0 \Rightarrow 5v_1 \\ & v_1 \Rightarrow 1v_1 \quad v_1 \Rightarrow 2v_1 \quad v_1 \Rightarrow 3v_1 \\ & v_1 \Rightarrow 4v_1 \quad v_1 \Rightarrow 5v_1 \quad v_1 \Rightarrow 2. \end{aligned}$$

(a) Draw the syntax diagram for the productions of G.

(b) Give the BNF for the productions of G.

9. Describe the language of G as given in Problem 8.

10. Construct a phrase structure grammar G such that $L(G) = \{0^n10^m \mid n \geq 0, m \geq 1\}$.

11. Consider the finite-state machine whose state transition table is

	a	b
s_0	s_0	s_1
s_1	s_1	s_2
s_2	s_2	s_3
s_3	s_3	s_0

Construct the digraph of this machine.

12. Describe the language accepted by the Moore machine whose digraph is given in Figure 69.

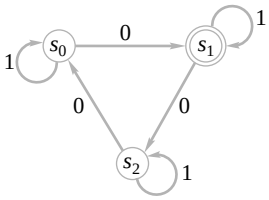


Figure 69

13. Consider the Moore machine M whose digraph is given in Figure 70.

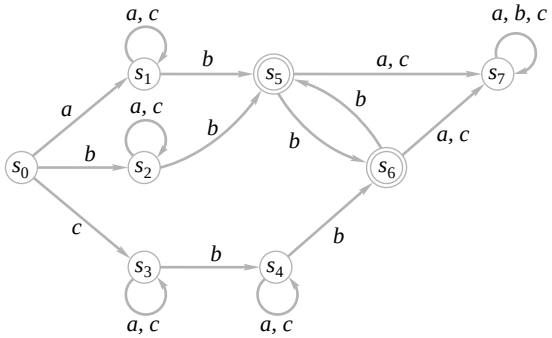


Figure 70

(a) Show that $R = \{(s_0, s_0), (s_1, s_1), (s_2, s_2), (s_3, s_3), (s_4, s_4), (s_5, s_5), (s_6, s_6), (s_7, s_7), (s_1, s_2), (s_1, s_4), (s_2, s_1), (s_2, s_4), (s_4, s_1), (s_4, s_2), (s_5, s_6), (s_6, s_5)\}$ is a machine congruence.

(b) Draw the digraph of the corresponding quotient Moore machine.

14. For the machine described in Problem 11, describe all words w in $\{a, b\}^*$ such that $f_w(s_0) = s_0$.

15. Construct a Moore machine that accepts a string of 0's and 1's if and only if the string has exactly two 1's.

16. Consider the Moore machine M whose digraph is given in Figure 71. Define R_k, $k = 0, 1, 2, \ldots$, as follows: $s\ R_k\ t$ if and only if s and t are w-compatible for all $w \in \{0, 1\}^*$ with length of $w \leq k$.

(a) Give the matrix of R_0.

(b) List the elements of R_1.

17. Using the machine M and the definitions in Problem 16,

(a) Find the smallest k such that $R_k = R_{k+1}$.

(b) Let $R = R_k$ for the k found in part (a) and draw the digraph of M/R.

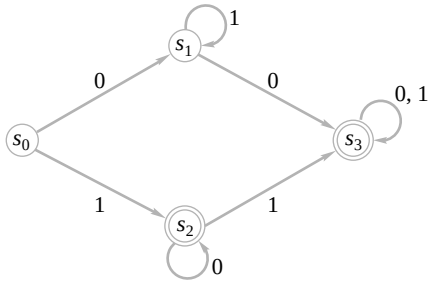


Figure 71

Experiment 10

Moore machines (Section 3) are examples of finite-state machines that recognize regular languages. Many computer languages, however, are not regular (type 3), but are context free (type 2). For example, a computer language may include expressions using balanced parentheses (a right parenthesis for every left parenthesis). A Moore machine has no way to keep track of how many left parentheses have been read to determine if the same number of right parentheses have also been read. A finite-state machine that includes a feature to do this is called a pushdown automaton.

A **pushdown automaton** is a sequence (S, I, F, s_0, T) in which S is a set of states, T is a subset of S and is the set of final states, $s_0 \in S$ is the start state, I is the input set, and F is a function from $S \times I \times I^*$ to $S \times I^*$. Roughly speaking, the finite-state machine can create a string of elements from the input set to serve as its memory. This string may be the empty string λ. The transition function F uses the current state, the input, and the string to determine the next state and the next string. For example, $F(s_3, a, w) = (s_2, w')$ means that if the machine is in state s_3 with current memory string w and a is read, the machine will move to state s_2 with new string w'. In actual practice there are only two ways to change the memory string:

(1) From w to bw for some b in I; this is called pushing b on the stack.

(2) From bw to w for some b in I; this is called popping b off the stack.

A pushdown automaton accepts a string v if this input causes the machine to move from s_0 with memory string λ to a final state s_j with memory string λ.

1. Construct a Moore machine that will accept strings of the form 0^m1^n, $m \geq 0$, $n \geq 0$, and no others.

2. Explain why the Moore machine in Question 1 cannot be modified to accept strings of the form 0^n1^n, $n \geq 0$, and no others.

3. Let $P = (S, I, F, s_0, T)$ with $S = \{s_0, s_1\}$, $I = \{0, 1\}$, $T = \{s_1\}$, and

$$F(s_0, 0, w) = (s_0, 0w), \quad F(s_0, 1, 0w) = (s_1, w),$$
$$F(s_1, 1, 0w) = (s_1, w),$$

where w is any string in I^*. Show that P accepts strings of the form 0^n1^n, $n \geq 0$, and no others.

4. Let $I = \{a, b, c\}$ amd $w \in \{a, b\}^*$. We define w^R to be the string formed by the elements of w in reverse order. For example, if w is *aabab*, then w^R is *babaa*. Design a pushdown automaton that will accept strings of the form wcw^R, and no others.

5. Let $G = (V, S, s_0, \rightarrow)$ be a phrase structure grammar with

$$V = \{v_0, w, a, b, c\}, \quad S = \{a, b, c\},$$

and

$$\rightarrow: \quad v_0 \rightarrow av_0b, \quad v_0b \rightarrow bw, \quad abw \rightarrow c.$$

(a) Describe the language $L(G)$.

(b) Design a pushdown automaton whose language is $L(G)$. That is, it only accepts strings in $L(G)$.

Coding Exercises

1. Let $M = (S, I, F)$ be a finite state machine where $S = \{s_0, s_1\}$, $I = \{0, 1\}$, and F is given by the following state transition table:

	0	1
s_0	s_0	s_1
s_1	s_1	s_0

Write a subroutine that given a state and an input returns the next state of the machine.

2. Write a function ST_TRANS that takes a word w, a string of 0's and 1's, and a state s and returns $f_w(s)$, the state transition function corresponding to w evaluated at s.

3. Let $M = (S, I, F)$ be a Moore machine where $S = \{s_0, s_1, s_2\}$, $I = \{0, 1\}$, $T = \{s_2\}$ and F is given by the following state transition table:

	0	1
s_0	s_0	s_1
s_1	s_2	s_2
s_2	s_1	s_0

Write a program that determines if a given word w is in $L(M)$.

4. Write a subroutine that simulates the Moore machine given in Exercise 2, Section 5.

5. Write a subroutine that simulates the Moore machine given in Exercise 4, Section 5.

Answers to Odd-Numbered Exercises

Exercise Set 1

1. $\{x^m y^n z, m \geq 0, n \geq 1\}$.

3. $\{a^{2^{n+1}}, n \geq 0\} \cup \{a^{2n}b, n \geq 0\}$.

5. $\{(\underbrace{\ldots(}_{k}\underbrace{a + a + \cdots + a}_{n\ a\text{'s}}\underbrace{)\ldots)}_{k}, k \geq 0, n \geq 3\}$.

7. $\{x^m y z^n, m \geq 1, n \geq 0\}$.

9. (a), (c), (e), (h), (i).

11. $L(G)$ is the set of strings from $\{a, b, c, 1, 2, \ldots, 9, 0\}^*$ that begin with a, b, or c.

13. $L(G) = \{(aa)^n bc^k (bb)^j b^k, n \geq 0, k \geq 1, j \geq 0\}$.

15.

v_0	v_0
v_0v_1	v_0v_1
$v_0v_1v_1$	$v_2v_0v_1$
v_2v_0z	xyv_1
xyz	xyz

17.

I	I
LW	LW
aW	LDW
aDW	LDDW
a1W	LDDD
a1DW	aDDD
a1DD	a1DD
a10D	a10D
a100	a100

19. The languages are not the same; here L(G) contains *aabba*.

21. G = (V, S, v_0, $\Rightarrow$), V = {v_0, v_1, 0, 1}, S = {0, 1}
$\Rightarrow$: $v_0 \Rightarrow 0v_11$, $v_0 \Rightarrow 1v_10$, $v_1 \Rightarrow 0v_11$, $v_1 \Rightarrow 1v_10$, $v_1 \Rightarrow 01$, $v_1 \Rightarrow 10$.

23. G = (V, S, v_0, $\Rightarrow$), V = {v_0, v_1, a, b}, S = {a, b}
$\Rightarrow$: $v_0 \Rightarrow aav_1bb$, $v_1 \Rightarrow av_1b$, $v_1 \Rightarrow ab$.

25. G = (V, S, v_0, $\Rightarrow$), V = {v_0, x, y}, S = {x, y}
$\Rightarrow$: $v_0 \Rightarrow v_0yy$, $v_0 \Rightarrow xv_0$, $v_0 \Rightarrow xx$.

27. $v_0 \Rightarrow av_0a$ $v_0 \Rightarrow bv_0b$ $v_0 \Rightarrow a$
$v_0 \Rightarrow aav_0 \Rightarrow b$ $v_0 \Rightarrow bb$.

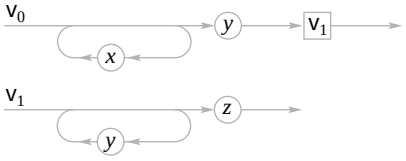
29. By using production rules 1, 2, and 4; production rule 4 to $a^{3k-1}v_2$; 3; $a^{3k}av_3 \Rightarrow a^{3k+1}v_1 \Rightarrow a^{3k+2}v_2 \Rightarrow a^{3(k+1)}$.

31. Let $G_1 = (V_1, S_1, v_0, \Rightarrow_1)$ and $G_2 = (V_2, S_2, v_0', \Rightarrow_2)$. Define $G = (V_1 \cup V_2, S_1 \cup S_2, v_0, \Rightarrow)$ as follows. If $v_i \Rightarrow_1 wv_k$, then $v_i \Rightarrow wv_k$. If $v_i \Rightarrow_1 w$ where w consists of terminal symbols, then $v_i \Rightarrow wv_0'$. All productions in $\Rightarrow_2$ become productions in $\Rightarrow$.

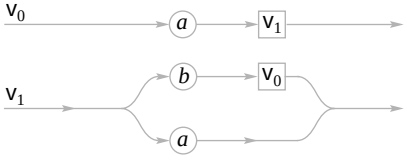
33. $v_0 \Rightarrow av_0$ $v_1 \Rightarrow av_1$
$v_0 \Rightarrow bv_1$ $v_1 \Rightarrow a$
The language is in fact regular.

Exercise 2

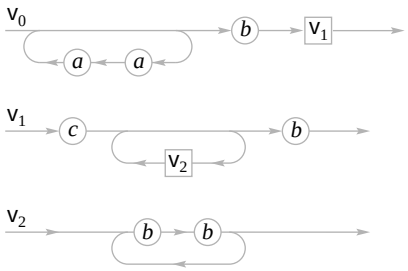
1. ⟨v_0⟩ ::= x⟨v_0⟩ | y⟨v_1⟩
⟨v_1⟩ ::= y⟨v_1⟩ | z



3. ⟨v_0⟩ ::= a⟨v_1⟩
⟨v_1⟩ ::= b⟨v_0⟩ | a



5. ⟨v_0⟩ ::= aa⟨v_0⟩ | b⟨v_1⟩
⟨v_1⟩ ::= c⟨v_2⟩b | cb
⟨v_2⟩ ::= bb⟨v_2⟩ | bb



7. ⟨v_0⟩ ::= x⟨v_0⟩ | y⟨v_0⟩ | z

9. ⟨v_0⟩ ::= a⟨v_1⟩
⟨v_1⟩ ::= b⟨v_0⟩ | a

11. ⟨v_0⟩ ::= ab⟨v_1⟩
⟨v_1⟩ ::= c⟨v_1⟩ | ⟨v_2⟩
⟨v_2⟩ ::= dd⟨v_2⟩ | d

13. ⟨v_0⟩ ::= a⟨v_1⟩
⟨v_1⟩ ::= a⟨v_2⟩
⟨v_2⟩ ::= a⟨v_1⟩ | a

15. ⟨v_0⟩ ::= b⟨v_0⟩ | a⟨v_1⟩ | b
⟨v_1⟩ ::= a⟨v_0⟩ | b⟨v_1⟩ | a

17. $(aa)^+aa$.

19. $((()^+(a + a + (a +)^+a())^+$. Note: Right and left parentheses must be matched.

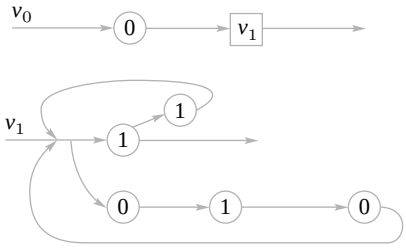
21. $(a \cup b \cup c)(a \cup b \cup c \cup 0 \cup 1 \cup \cdots \cup 9)^*$.

23. $ab(d \cup (d(c \cup d)d))^*$.

25. $ab(abc)^nb$, $n \geq 1$.

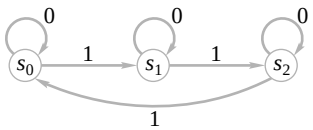
27. $(aab \cup ab)^*$.

29.

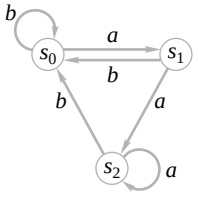


Exercise Set 3

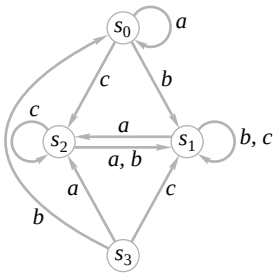
1.



3.



5.



7.

	a	b
s_0	s_1	s_1
s_1	s_1	s_2
s_2	s_0	s_2

9.

	T	F
s_0	s_1	s_0
s_1	s_1	s_1
s_2	s_1	s_2

11.

	a	b	c
s_0	s_0	s_1	s_2
s_1	s_2	s_1	s_3
s_2	s_3	s_3	s_1
s_3	s_3	s_3	s_2

13. Let $x \in I$. Certainly $f_x(s) = f_x(s)$ for all $s \in S$. Thus $x \mathrel{R} x$ and R is reflexive.
Suppose $x \mathrel{R} y$. Then $f_x(s) = f_y(s)\ \forall s \in S$. But then $y \mathrel{R} x$ and R is symmetric.
Suppose $x \mathrel{R} y$, $y \mathrel{R} z$. Then $f_x(s) = f_y(s) = f_z(s), \forall s \in S$. Hence $x \mathrel{R} z$ and R is transitive.

15. Using Exercise 14, we need only show that R is reflexive and symmetric. Let $s \in S$. $s = e \cdot s$ so $f_e(s) = s$ and $s \mathrel{R} s$. Suppose $x \mathrel{R} y$. Then $f_z(x) = y$ for some $z \in S$. $y = z \cdot x \Rightarrow z^{-1} \cdot y = x$ and thus $f_{z^{-1}}(y) = x$. Hence $y \mathrel{R} x$ and R is symmetric.

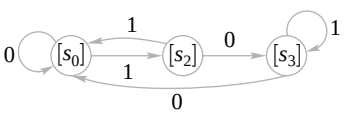
17. (a) Inspection of $\mathbf{M}_R$ shows that R is reflexive and symmetric. Since $\mathbf{M}_R \odot \mathbf{M}_R = \mathbf{M}_R$, R is transitive. Thus R is an equivalence relation. The table below shows that it is a machine congruence.

(b)

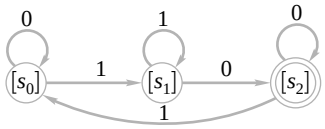
	0	1
[1]	[1]	[1]
[2]	[2]	[2]

19. (a) R is clearly an equivalence relation. $s_0 \mathrel{R} s_1$ and $f_0(s_0) = s_1$, $f_0(s_1) = s_0$, $f_1(s_0) = s_2 = f(s_1)$ so $f_x(s_0) \mathrel{R} f_x(s_1), \forall x \in I$.

(b)



21. Inspection of $\mathbf{M}_R$ shows that R is reflexive and symmetric. Since $\mathbf{M}_R \odot \mathbf{M}_R = \mathbf{M}_R$, R is transitive. Thus R is an equivalence relation. The digraph below shows that it is a machine congruence.



23. f_x, $x \in I$, is a function on S so f_x is a state transition function. $T \subseteq S$ so the conditions for a Moore machine are met.

25.

	0	1
(s_0, s_0)	(s_0, s_0)	(s_1, s_1)
(s_0, s_1)	(s_0, s_2)	(s_1, s_1)
(s_0, s_2)	(s_0, s_2)	(s_1, s_3)
(s_0, s_3)	(s_0, s_3)	(s_1, s_3)
(s_1, s_0)	(s_1, s_0)	(s_2, s_1)
(s_1, s_1)	(s_1, s_2)	(s_2, s_1)
(s_1, s_2)	(s_1, s_2)	(s_2, s_3)
(s_1, s_3)	(s_1, s_3)	(s_2, s_3)
(s_2, s_0)	(s_2, s_0)	(s_0, s_1)
(s_2, s_1)	(s_2, s_2)	(s_0, s_1)
(s_2, s_2)	(s_2, s_2)	(s_0, s_3)
(s_2, s_3)	(s_2, s_3)	(s_0, s_3)

Exercise Set 4

1. $f_w(s_0) = s_2$, $f_w(s_1) = s_3$, $f_w(s_2) = s_0$, $f_w(s_3) = s_1$.

3. The number of 1's in w is divisible by 4.

5. The number of 1's in w is $2 + 4k$, $k \geq 0$.

7. $f_w(s_0) = s_0$, $f_w(s_1) = s_0$, $f_w(s_2) = s_0$.

9. All words ending in b.

11. Strings of 0's and 1's with $3 + 5k$ 1's, $k \geq 0$.

13. Strings of 0's and 1's that end in 0.

15. Strings of a's and b's that do not contain bb.

17. Strings of 0's and 1's that end in 01.

19. Strings xy and yz.

21. Strings that begin 1 or 00.

23. Let $w, u \in L(M)$. Then $f_w(s_0) = s \in T$, $f_u(s) \in T$. Hence $f_{w \cdot u}(s_0) \in T$, so $w \cdot u \in L(M)$.

25. $w \in L(M_1) \cup L(M_2)$ if and only if $w \in L(M_1)$ or $w \in L(M_2)$. Say $w \in L(M_2)$. But $w \in L(M_2)$ if and only if $f_w(s_0') \in T'$ if and only if $f_w(s_0, s_0') = (f_w(s_0), f_w'(s_0')) \in T$.

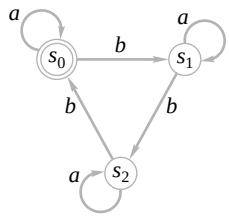
Exercise Set 5

1. $G = (V, I, s_0, \rightarrow)$, $V = \{s_0, s_1, s_2, s_3, 0, 1\}$, $I = \{0, 1\}$
$\rightarrow$: $s_0 \rightarrow 0s_0$, $s_0 \rightarrow 1s_1$, $s_1 \rightarrow 0s_1$, $s_1 \rightarrow 1s_2$, $s_2 \rightarrow 0s_2$, $s_2 \rightarrow 1s_3$, $s_2 \rightarrow 1$, $s_3 \rightarrow 0s_3$, $s_3 \rightarrow 0$, $s_3 \rightarrow 1s_0$.

3. $(0 \cup 1)^*1$.

5. $G = (V, I, s_0, \rightarrow)$, $V = \{s_0, s_1, s_2, a, b\}$, $I = \{a, b\}$.
$\langle s_0 \rangle ::= a\langle s_0 \rangle \mid b\langle s_1 \rangle \mid a \mid b$
$\langle s_1 \rangle ::= a\langle s_0 \rangle \mid b\langle s_2 \rangle \mid a$
$\langle s_2 \rangle ::= a\langle s_2 \rangle \mid b\langle s_2 \rangle$

7.



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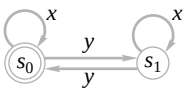

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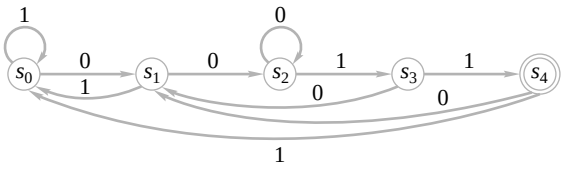
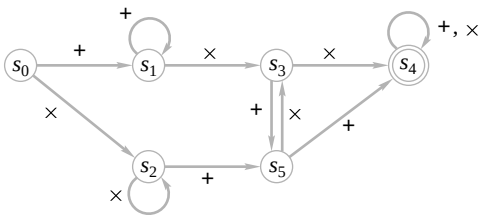


13.



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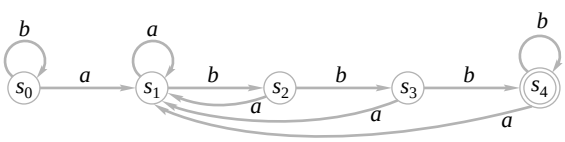
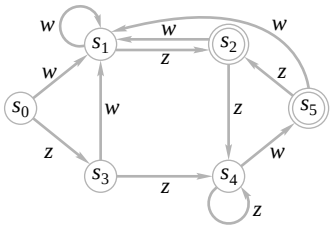
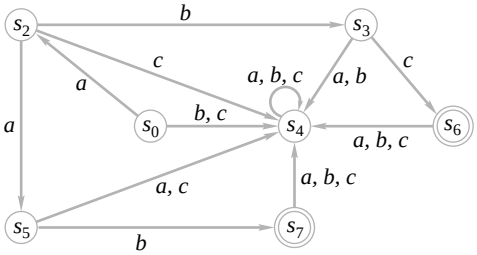


17.



19.



21.

	0	1
s_0	s_1	s_0
s_1	s_1	s_2
s_2	s_3	s_0
s_3	s_1	s_4
s_4	s_3	s_0

$T = \{s_4\}$

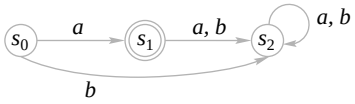
23.

	x	y
s_0	s_1	s_0
s_1	s_2	s_1
s_2	s_3	s_2
s_3	s_3	s_3

$T = \{s_2\}$

25. R is reflexive because $f_w(x) = f_w(x)$. R is symmetric because if $f_w(s_i)$, $f_w(s_j)$ are both (not) in T, then $f_w(s_j)$, $f_w(s_i)$ are both (not) in T. R is transitive because $s_i \, R \, s_j$, $s_j \, R \, s_k$ if and only if $f_w(s_i)$, $f_w(s_j)$, $f_w(s_k)$ are all in (or not in) T.

27.



29. The construction of Exercise 24 of Section 3 gives a machine that accepts $\alpha \square \beta$.

Exercise Set 6

1. $R_0 = \{(s_0, s_0), (s_0, s_1), (s_1, s_0), (s_1, s_1), (s_2, s_2)\}$.

3. $R_1 = \{(s_0, s_0), (s_1, s_1), (s_2, s_2), (s_3, s_3), (s_4, s_4), (s_0, s_3), (s_3, s_0), (s_1, s_2), (s_2, s_1)\}$.

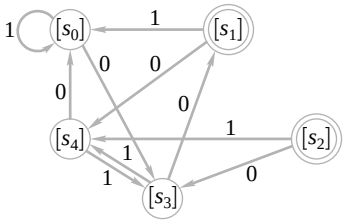
5. $R_{127} = R_1$.

7. $R_2 = R_1$.

9. $R = \{(s_0, s_0), (s_1, s_1), (s_2, s_2)\}$.

11. $R = R_1$ as given in Exercise 6.

13. $R = \{(s_0, s_0), (s_1, s_1), (s_2, s_2), (s_3, s_3), (s_4, s_4), (s_4, s_5), (s_5, s_4), (s_5, s_5), (s_6, s_6), (s_3, s_6), (s_6, s_3)\}$.



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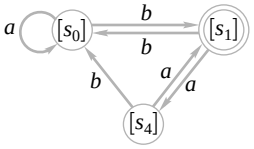

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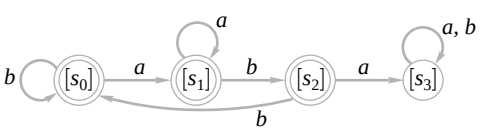
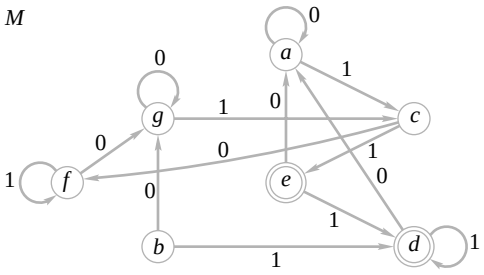


19. $P = \{\{a, g\}, \{f\}, \{b\}, \{c\}, \{d, e\}\}$.

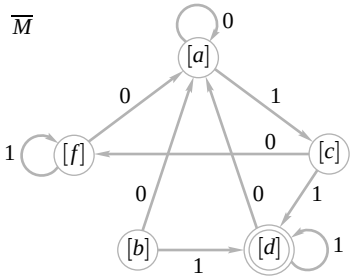
	0	1
[a]	[a]	[c]
[b]	[a]	[d]
[c]	[f]	[d]
[d]	[a]	[d]
[f]	[a]	[f]

21. The equivalence classes each contain a single state, so the table is essentially that given in the statement of the exercise.

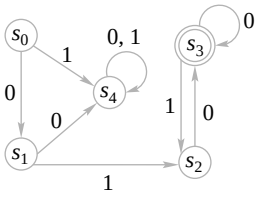
23. M



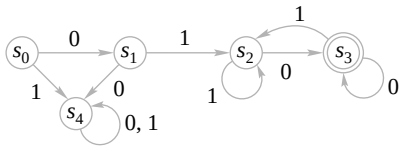
$\overline{M}$



25. M



$\overline{M}$



Answers to Chapter Self-Tests

1. No, type 0 and type 1 languages cannot be given in BNF.

2. Some type 2 and type 3 grammars can be represented by a syntax diagram.

3. This language is type 2, context-sensitive, but importantly it is not regular.

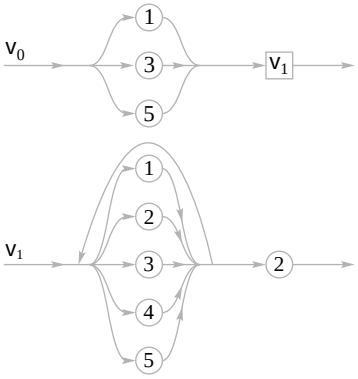
4. A machine accepts or recognizes a language if and only if every string in the language leads from the starting state to an acceptance state; that is, $f_w(s_0)$ is an acceptance state.

5. A language L is regular if and only if $L = L(M)$ for some Moore machine M.

6. **(a)** False. **(b)** True. **(c)** True. **(d)** False.

7. $L(G) = \{a^n c^m d b^n, n \geq 0, m \geq 0\}$.

8. **(a)**

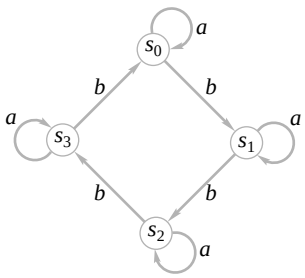


(b) $\langle v_0 \rangle ::= 1\langle v_1 \rangle \mid 3\langle v_1 \rangle \mid 5\langle v_1 \rangle$
$\langle v_1 \rangle ::= 1\langle v_1 \rangle \mid 2\langle v_1 \rangle \mid 3\langle v_1 \rangle \mid 4\langle v_1 \rangle \mid 5\langle v_1 \rangle \mid 2$

9. $L(G) = \{(1 \square 3 \square 5)(1 \square 2 \square 3 \square 4 \square 5)^n 2, n \geq 0\}$.

10. $G = (V, S, v_0, \Rightarrow)$ with $V = \{v_0, v_1, 0, 1\}$, $S = \{0, 1\}$, and $\Rightarrow : v_0 \Rightarrow 0v_0$, $v_0 \Rightarrow 1v_1$, $v_1 \Rightarrow 0v_1$, $v_1 \Rightarrow 0$.

11.

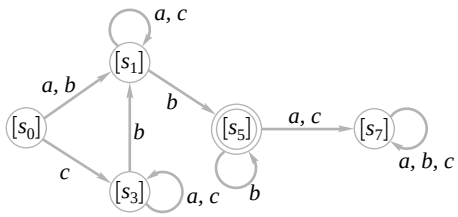


12. Strings of 0's and 1's with $3k$ zeros, $k \geq 0$.

13. **(a)** R is easily seen to be an equivalence relation with equivalence classes $\{s_0\}$, $\{s_3\}$, $\{s_1, s_2, s_4\}$, $\{s_5, s_6\}$, and $\{s_7\}$.

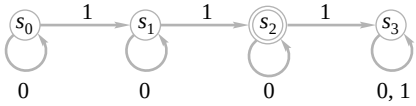
	a	b	c
$[s_0]$	$[s_1]$	$[s_1]$	$[s_3]$
$[s_1]$	$[s_1]$	$[s_5]$	$[s_1]$
$[s_3]$	$[s_3]$	$[s_1]$	$[s_3]$
$[s_5]$	$[s_7]$	$[s_5]$	$[s_7]$
$[s_7]$	$[s_7]$	$[s_7]$	$[s_7]$

(b)



14. Strings with exactly $4k$ b's, $k \geq 0$.

15.

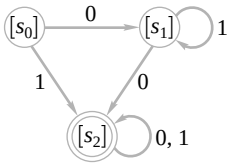


16. $\mathbf{M}_{R_0} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$.

$R_1 = \{(s_0, s_0), (s_1, s_1), (s_2, s_2), (s_3, s_3), (s_2, s_3), (s_3, s_2)\}$.

17. (a) $k = 1$.

(b) $R = \{(s_0, s_0), (s_1, s_1), (s_2, s_2), (s_3, s_3), (s_2, s_3), (s_3, s_2)\}$.



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Appendix

Algorithms and Pseudocode

Algorithms

An **algorithm** is a complete list of the steps necessary to perform a task or computation. The steps in an algorithm may be general descriptions, leaving much detail to be filled in, or they may be totally precise descriptions of every detail.

Example 1 A recipe for baking a cake can be viewed as an algorithm. It might be written as follows.

```
1. ADD MILK TO CAKE MIX.
2. ADD EGG TO CAKE MIX AND MILK.
3. BEAT MIXTURE FOR 2 MINUTES.
4. POUR MIXTURE INTO PAN AND COOK IN OVEN FOR 40
   MINUTES AT 350°F.
```

●

The preceding algorithm is fairly general and assumes that the user understands how to pour milk, break an egg, set controls on an oven, and perform a host of other unspecified actions. If these steps were all included, the algorithm would be much more detailed, but long and unwieldy. One possible solution, if the added detail is necessary, is to group collections of related steps into other algorithms that we call **subroutines** and simply refer to these subroutines at appropriate points in the main algorithm. We hasten to point out that we are using the term "subroutine" in the general sense of an algorithm whose primary purpose is to form part of a more general algorithm. We do not give the term the precise meaning that it would have in a computer programming language. Subroutines are given names, and when an algorithm wishes the steps in a subroutine to be performed, it signifies this by *calling* the subroutine. We will specify this by a statement **CALL** NAME, where NAME is the name of the subroutine. ◆

Example 2 Consider the following version of Example 1, which uses subroutines to add detail. Let us title this algorithm BAKECAKE.

Algorithm BAKECAKE

```
1. CALL ADDMILK.
2. CALL ADDEGG.
3. CALL BEAT(2).
4. CALL COOK(OVEN, 40, 350).
```

●

The subroutines of this example will give the details of each step. For example, subroutine ADDEGG might consist of the following general steps.

From *Discrete Mathematical Structures*, Sixth Edition, Bernard Kolman, Robert C. Busby, Sharon Cutler Ross.
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```
SUBROUTINE ADDEGG
1. Remove egg from carton.
2. Break egg on edge of bowl.
3. Drop egg, without shell, into bowl.
4. RETURN
END OF SUBROUTINE ADDEGG
```

Of course, these steps could be broken into substeps, which themselves could be implemented as subroutines. The purpose of Step 4, the "return" statement, is to signify that one should continue with the original algorithm that "called" the subroutine. ◆

Our primary concern is with algorithms to implement mathematical computations, investigate mathematical questions, manipulate strings or sequences of symbols and numbers, move data from place to place in arrays, and so on. Sometimes the algorithms will be of a general nature, suitable for human use, and sometimes they will be stated in a formal, detailed way suitable for programming in a computer language. Later in this appendix we will describe a reasonable language for stating algorithms.

It often happens that a test is performed at some point in an algorithm, and the result of this test determines which of two sets of steps will be performed next. Such a test and the resulting decision to begin performing a certain set of instructions will be called a **branch**.

Example 3 Consider the following algorithm for deciding whether to study for a "discrete structures" test.

Algorithm FLIP

```
1. Toss a coin.
2. IF the result is "heads," GO TO 5.
3. Study for test.
4. GO TO 6.
5. See a show.
6. Take test next day.
```

●

Note that the branching is accomplished by **GO TO** statements, which direct the user to the next instruction to be performed, in case it is not the next instruction in sequence. In the past, especially for algorithms written in computer programming languages such as FORTRAN, the **GO TO** statement was universally used to describe branches. Since then there have been many advances in the art of algorithm and computer program design. Out of this experience has come the view that the indiscriminate use of **GO TO** statements to branch from one instruction to any other instruction leads to algorithms (and computer programs) that are difficult to understand, hard to modify, and prone to error. Also, recent techniques for actually proving that an algorithm or program does what it is supposed to do will not work in the presence of unrestricted **GO TO** statements.

In light of the foregoing remarks, it is a widely held view that algorithms should be **structured**. This term refers to a variety of restrictions on branching, which help to overcome difficulties posed by the **GO TO** statement. In a structured branch, the test condition follows an **IF** statement. When the test is true, the instructions following a **THEN** statement are performed. Otherwise, the instructions following an **ELSE** statement are performed.

Example 4 Consider again the algorithm FLIP described in Example 3. The following is a structured version of FLIP.

Algorithm FLIP

```
1. Toss a coin.
2. IF (heads) THEN
     a. Study for test.
3. ELSE
     a. See a show.
4. Take test next day.
```

This algorithm is easy to read and is formulated without **GO TO** statements. In fact, it does not require numbering or lettering of the steps, but we keep these to set off and emphasize the instructions. Of course, the algorithm FLIP of Example 3 is not very different from that of Example 4. The point is that the **GO TO** statement has the potential for abuse, which is eliminated in the structured form.

Another commonly encountered situation that calls for a branch is the **loop**, in which a set of instructions is repeatedly followed either for a definite number of times or until some condition is encountered. In structured algorithms, a loop may be formulated as shown in the following example.

Example 5 The following algorithm describes the process of mailing 50 invitations.

Algorithm INVITATIONS

```
1. COUNT ← 50
2. WHILE (COUNT > 0)
     a. Address envelope.
     b. Insert invitation in envelope.
     c. Place stamp on envelope.
     d. COUNT ← COUNT − 1
3. Place envelopes in mailbox.
```

In this algorithm, the variable COUNT is first assigned the value 50. The symbol ← may be read "is assigned." The loop is handled by the **WHILE** statement. The condition COUNT > 0 is checked, and as long as it is true, statements **a** through **d** are performed. When COUNT = 0 (after 50 steps), the looping stops.

Later in this appendix we will give the details of this and other methods of looping, which are generally considered to be structured. In structured algorithms, the only deviations permitted from a normal, sequential execution of steps are those given by loops or iterations and those resulting from the use of the **IF-THEN-ELSE** construction. Use of the latter construction for branching is called **selection**.

We need to describe numerous algorithms, many of which are highly technical. Descriptions of these algorithms in ordinary English may be feasible, and in many cases we will give such descriptions. However, it is often easier to get an overview of an algorithm if it is presented in a concise, symbolic form. Some authors use diagrammatic representations called **flow charts** for this purpose. Figure 1 shows a flow chart for the algorithm given in Example 4. These diagrams

have a certain appeal and are still used in the computer programming field, but many believe that they are undesirable since they are more in accord with older programming practice than with structured programming ideas.

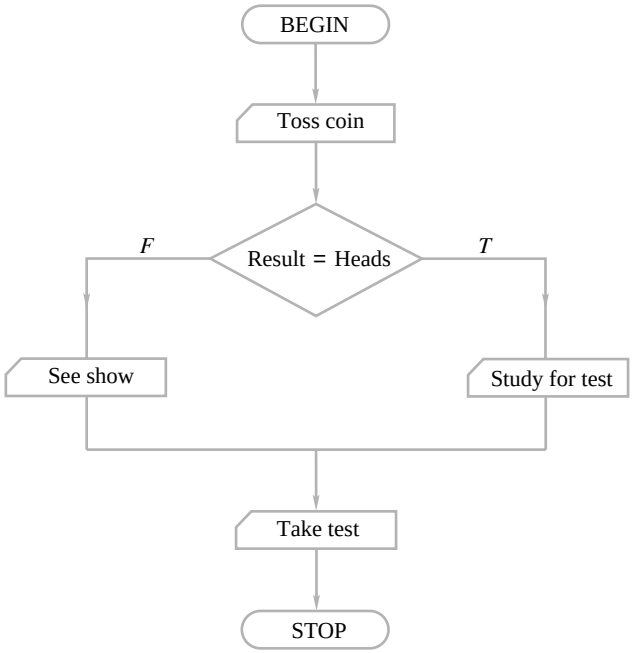

Figure 1

The other alternative is to express algorithms in a way that resembles a computer programming language or to use an actual programming language such as PASCAL. We choose to use a **pseudocode** language rather than an actual programming language, and the earlier examples of this section provide a hint as to the structure of this pseudocode form. There are several reasons for making this choice. First, knowledge of a programming language is not necessary for the understanding of the contents here. The fine details of a programming language are necessary for communication with a computer, but may serve only to obscure the description of an algorithm. Moreover, we feel that the algorithms should be expressed in such a way that an easy translation to any desired computer programming language is possible. Pseudocode is very simple to learn and easy to use, and it in no way interferes with one's learning of an actual programming language.

The second reason for using pseudocode is the fact that many professional programmers believe that developing and maintaining pseudocode versions of a program, before and after translation to an actual programming language, encourage good programming practice and aid in developing and modifying programs. We feel that the student should see a pseudocode in use for this reason. The pseudocode described is largely taken from Rader (*Advanced Software Design Techniques*, Petrocelli, New York, 1968) and has seen service in a practical programming environment. We have made certain cosmetic changes in the interest of pedagogy.

One warning is in order. An algorithm written in pseudocode may, if it is finely detailed, be very reminiscent of a computer program. This is deliberate, even to the use of terms like **SUBROUTINE** and the statement **RETURN** at the end of a subroutine to signify that we should return to the steps of the main algorithm. Also, the actual programming of algorithms is facilitated by the similarity of pseudocode to a programming language. However, always remember that a pseudocode algorithm is *not* a computer program. It is meant for humans, not machines, and we are only obliged to include sufficient detail to make the algorithm clear to human readers.

Pseudocode

In pseudocode, successive steps are usually labeled with consecutive numbers. If a step begins a selection or a loop, several succeeding steps may be considered subordinate to this step (for example, the body of a loop). Subordinate lines are indented several spaces and labeled with consecutive letters instead of numbers. If these steps had subordinates, they in turn would be indented and labeled with numbers. We use only consecutive numbers or letters as labels, and we alternate them in succeeding levels of subordination. A typical structuring of steps with subordinate steps is illustrated in the following.

```
1. line 1
    a. line 2
    b. line 3
        1. line 4
        2. line 5
    c. line 6
2. line 7
3. line 8
    a. line 9
        1. line 10
    b. line 11
4. line 12
```

Steps that have the same degree of indentation will be said to be at the **same level**. Thus the next line at the level of line 1 is line 7, while the next line at the level of line 3 is line 6, and so on.

Selection in pseudocode is expressed with the form **IF-THEN-ELSE**, as follows:

```
1. IF (CONDITION) THEN
   true-block
2. ELSE
   false-block
```

The true- and false-blocks (to be executed respectively when CONDITION is true and CONDITION is false) may contain any legitimate pseudocode including selections or iterations. Sometimes we will omit statement 2, the **ELSE** statement, and the false-block. In this case, the true-block is executed only when CONDITION is true and then, whether CONDITION is true or false, control passes to the next statement that is at the same level as statement 1.

Example 6 Consider the following statements in pseudocode. Assume that X is a rational number.

```
1. IF (X > 13,000) THEN
    a. Y ← X + 0.02(X − 13,000)
2. ELSE
    a. Y ← X + 0.03X
```

In statement 1, CONDITION is: $X > 13{,}000$. If X is greater than 13,000, then Y is computed by the formula

$$X + 0.02(X - 13{,}000),$$

while if $X \leq 13{,}000$, then Y is computed by the formula

$$X + 0.03X.$$ ◆

We will use the ordinary symbols of mathematics to express algebraic relationships and conditions in pseudocode. The symbols $+$, $-$, $\times$, and $/$ will be used for the basic arithmetic operations, and the symbols $<$, $>$, $\leq$, $\geq$, $=$, and $\neq$ will be used for testing conditions. The number X raised to the power Y will be denoted by X^Y, the square of a number A will be denoted by A^2, and so on. Moreover, products such as $3 \times A$ will be denoted by $3A$, and so on, when no confusion is possible.

We will use a left arrow, $\leftarrow$, rather than the equal sign, for assignments of values to variables. Thus, as in Example 6, the expression $Y \leftarrow X + 0.03X$ means that Y is assigned the value specified by the right-hand side. The use of $=$ for this purpose conflicts with the use of this symbol for testing conditions. Thus $X = X + 1$ could either be an assignment or a question about the number X. The use of $\leftarrow$ avoids this problem.

A fundamental way to express iteration expressions in pseudocode is the **WHILE** form:

```
1. WHILE (CONDITION)
   repeat-block
```

Here CONDITION is tested and, if true, the block of pseudocode following it is executed. This process is repeated until CONDITION becomes false, after which control passes to the next statement that is at the same level as statement 1.

Example 7 Consider the following algorithm in pseudocode; N is assumed to be a positive integer.

```
1. X ← 0
2. Y ← 0
3. WHILE (X < N)
    a. X ← X + 1
    b. Y ← Y + X
4. Y ← Y/2
```

In this algorithm, CONDITION is $X < N$. As long as CONDITION is true, that is, as long as $X < N$, statements a and b will be executed repeatedly. As soon as CONDITION is false, that is, as soon as $X = N$, statement 4 will be executed. This means that the **WHILE** loop is executed N times and the algorithm computes

$$\frac{1 + 2 + \cdots + N}{2},$$

which is the value of variable Y at the completion of the algorithm. ◆

A simple modification of the **WHILE** form called the **UNTIL** form is useful and we include it, although it could be replaced by completely equivalent statements using **WHILE**. This construction is

```
1. UNTIL (CONDITION)
   repeat-block
```

Here the loop continues to be executed *until* the condition is true; that is, continues only as long as the condition is false. Also, CONDITION is tested *after* the repeat-block rather than *before,* so the block must be repeated at least once.

Example 8 The algorithm given in Example 7 could also be written with an **UNTIL** statement as follows:

```
1. X ← 0
2. Y ← 0
3. UNTIL (X ≥ N)
    a. X ← X + 1
    b. Y ← Y + X
4. Y ← Y/2
```

In this algorithm, the CONDITION X ≥ N is tested at the completion of Step 3. If it is false, the body of Step 3 is repeated. This process continues until the test reveals that CONDITION is true (when X = N). At that time Step 4 is immediately executed. ◆

The **UNTIL** form of iteration is a convenience and could be formulated with a **WHILE** statement. The form

```
1. UNTIL (CONDITION)
   block 1
```

is actually equivalent to the form

```
1. block 1
2. WHILE (CONDITION = FALSE)
   block 1
```

In each case, the instructions in block 1 are followed once, regardless of CONDITION. After this, CONDITION is checked, and, if it is true, the process stops; otherwise, block 1 instructions are followed again. This procedure of checking CONDITION and then repeating instructions in block 1 if CONDITION is false is continued until CONDITION is true. Since both forms produce the same results, they are equivalent.

The other form of iteration is the one most like a traditional DO loop, and we express it as a **FOR** statement:

```
1. FOR VAR = X THRU Y [BY Z]
   repeat-block
```

In this form, VAR is an integer variable used to count the number of times the instructions in repeat-block have been followed. X, Y, and Z, if desired, are either integers or expressions whose computed values are integers. The variable VAR begins at X and increases Z units at a time (Z is 1 if not specified). After each increase in X, the repeat-block is executed as long as the new value of X is not

greater than Y. The conditions on VAR, specified by X, Y, and Z, are checked before each repetition of the instructions in the block. The block is repeated only if those conditions are true. The brackets around **BY** Z are not part of the statement, but simply mean that this part is optional and may be omitted. Note that the repeat-block is always executed at least once, since no check is made until X is changed.

Example 9 The pseudocode statement

```
FOR VAR = 2 THRU 10 BY 3
```

will cause the repeat-block to be executed three times, corresponding to VAR = 2, 5, 8. The process ends then, since the next value of VAR would be 11, which is greater than 10. ◆

We will use lines of pseudocode by themselves to illustrate different parts of a computation. However, when the code represents a complete thought, we may choose to designate it as an algorithm, a subroutine, or a function.

A set of instructions that will primarily be used at various places by other algorithms is often designated as a **subroutine**. A subroutine is given a name for reference, a list of input variables, which it will receive from other algorithms, and output variables, which it will pass on to the algorithms that use it. A typical title of a subroutine is

```
SUBROUTINE NAME (A, B, ...; X, Y, ...)
```

The values of the input variables are assumed to be supplied to the subroutine when it is used. Here NAME is a name generally chosen as a memory aid for the task performed by the subroutine; A, B, and so on, are input variables; and X, Y, and so on, are output variables. The semicolon is used to separate input variables from output variables.

A subroutine will end with the statement **RETURN**. As we remarked earlier in this section, this simply reminds us to return to the algorithm (if any) that is using the subroutine.

An algorithm uses a subroutine by including the statement

```
CALL NAME (A, B, ...; X, Y, ...)
```

where NAME is a subroutine and the input variables A, B, and so on have all been assigned values. This process was also illustrated in earlier examples.

Example 10 The following subroutine computes the square of a positive integer N by successive additions.

```
SUBROUTINE SQR(N; X)
1. X ← N
2. Y ← 1
3. WHILE (Y ≠ N)
    a. X ← X + N
    b. Y ← Y + 1
4. RETURN
END OF SUBROUTINE SQR
```

◆

If the result of the steps performed by a subroutine is a single number, we may call the subroutine **FUNCTION**. In this case, we title such a program as follows:

```
FUNCTION NAME (A, B, C, ...)
```

G

Games, 12, 97
Geometry, 19, 97, 308
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Graphical information, 350
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